

Structural Relations among Physical Constants

Action, Geometric Reciprocity,
and Memory of Transformations

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Abstract

This work establishes compatibility, reciprocity, and reconstruction relations among the structures of action, mass, and constitutive response obtained through Triadic Modular Holography and Dimensional Mechanics. The development starts from the composition of the arithmetic path structure *All Possible Paths* (APP), the oriented ternary signature TRIT, and the *Topological Phase Kernel* (TPK), which updates and transports the enriched state. This composition retains orientation, carry, memory, and return. The constants enter as evaluations of previously constructed structures; composing them determines the properties they share and the information recovered by joint observation. Within the normalized constitutive collection, the propagation speed and the Barbero–Immirzi functional are proved to determine the labelled eigenvalues of the two channels. The logarithms of speed and impedance form the gradient of a strictly concave spectral potential. Its Hessian characterizes the sensitivity of inversion and provides explicit stability bounds. The potential shares a depth-two spectral moment with the oriented construction whose extremal value is Catalan’s constant; the constitutive response uniquely recovers that collection with its boundary condition at the origin.

Composing quarter-turns associated with conjugate elliptic forms produces a unimodular relative transport whose reciprocal dilations reconstruct the dimensionless action coefficient, with the closure coordinate and orientation retained. The dodecaphase lift and transport iteration characterize the memory of the order of operations. Chronological analysis distinguishes information growth from the entropy rate per position, and memory re-entry determines its contribution to the operator response. The inscribed return and its polar composition determine a length L_α . Joint preservation of action and area fixes $G = c^3 L_\alpha^2 / \hbar$ in the radial realization, including under noncommutative variations and compatible memory reduction.

Electronic memory is reconstructed through linear and spectral invariants that retain the order of the record; transport between multisections admits an explicit criterion for mass preservation. The thermal composition combines capacities, multiplicities, and spectral memory in an exact trace and constructs its thermometric response for a marked readout with a fixed reference. This realization relates energy, information, action, and area while retaining the normalization and observation conditions of each quantity. Transported composition of responses and Higgs–impedance reconstruction connect the constitutive channels with the electroweak sector. Electronic substitution retains its scale in Fermi, cancels it in the angular ratios, and transports the inverse length by operator congruence, including in the noncommutative case.

Finally, under the specified circular and thermal conditions, a single mass coordinate determines the proper frequency, two reciprocal lengths with invariant product, and the thermal weighting of the cycle. The results thus bring production, compatibility, and reconstruction into a common mathematical description, with explicit proofs and a distinction between change of representation, relative transport, and physical realization.

The uniform dodecaphasic reader also determines a sequence of first roots of critical return. Its separation ratio is proved to converge to the Feigenbaum constant, preserving the original selection, orientation and memory of the construction.

The same composition extends to a total action of geometry, gauge and matter: it jointly incorporates gravitation and the strong, weak and electromagnetic interactions, preserving the chiral representations and the previously generated couplings. Its spin current, total metric source and torsional cross-terms between species are obtained. The Legendre transformation and symmetry identities produce joint first-class constraints and their propagation. On their regular surface, the curvature of the total canonical connection is calculated and shown to vanish in the characteristic deformation directions. The result preserves the boundary conditions and the possible global holonomy of the enriched state.

Keywords: action; geometric reciprocity; operator memory; physical constants; Triadic Modular Holography; Dimensional Mechanics.

Contents

1	Introduction	9
2	Formal core of Triadic Modular Holography	11
2.1	APP: positional support and arithmetic evaluations	11
2.2	TRIT: orientation and quadratic regimes	14
2.3	TPK: topological phase kernel	19
2.4	Accumulated memory and transport resolvents	26
3	Topological Phase Kernel and generation of the seed catalogue	29
3.1	Phase selection, transport and enriched state	29
3.2	Initial conditions and reversible enumeration	30
3.3	Sheet evaluation and arithmetic record	31
3.4	Six windows and decimal block emission	32
3.5	Complete calculation of an emission beginning with 501	33
3.6	Factorisation of the finite census	34
3.7	Oriented ternary reading and preservation of fibres	36
3.8	Dihedral action and the forty-three orbits	37
3.9	Self-contained enumeration and verification algorithm	38
3.10	Preservation of the record and the function of the seeds	38
4	Development of the formal kernel: initial conditions, geometry, and refinement	39
4.1	Initial states and the emission recurrence	39
4.2	Calendar closure and memory advancement	40
4.3	Observable compression and orthogonal conservation	42
4.4	Local development: commutation, Lorentzian signature, and discrete curvature	42
4.5	Cubic completion and two notions of boundary	43
4.6	Cubic incidence and Lorentzian realization	45
4.7	Compatibility between modular and positional reductions	46
4.8	Number as a compatible section and value as a readout	47
4.9	Composition of histories and memory multiplier	47
4.10	Double variance of refinement and preservation of the unit	48
4.11	Phase return, Weyl representation, and two readouts	49
5	Coinductive generation of π, φ, e to arbitrary depth	49
5.1	Orbits, regions, and characters of closure, propagation, and self-scaling	49
5.2	Finite transport and the prolongation boundary	50
5.3	Five closure regions and angular compensation	51
5.4	Propagation readout and normalization of the unit	52
5.5	Modal incidence and generation of self-scaling	52
5.6	Exponential realizations of the generated coordinates	55
5.7	Compatible cylinders and arbitrary depth	57
5.8	Specular involution and synchronization of the regions	57
5.9	Sturmian helical suspension	58
5.10	Nonadic connection, holonomy and monodromic representation	61
5.11	Block clock and capacity clock	62
5.12	Symbolic suspension and aperiodic time crystal	63
5.13	Mirror symmetry of the regions and dual realization	64
5.14	Temporal transport and tritic exponential composition	66
5.15	Multiplicative irreducibility and return periods	67
6	Regional generation and refinement	69
6.1	Positional emission through rational separation of cells	69
6.2	Regional blocks of six trits at arbitrary depth	70
6.3	Transition registers and reconstruction of linear lifts	71
6.4	Simultaneous refinement in ternary and decimal bases	72
6.5	Selection of the next block by the regional reader	74
6.6	Ternary signature, carry, Witt incidence, and reading clock	74
6.7	Composition theorem for the block, cylinder, and signature	75
6.8	Calendar census and simultaneous compatibility of the regional biographies	76
7	Regional determination of the dodecaphasic descriptor and selection of the coupling register	77
7.1	Regional bands, encoding, and integer signatures	77
7.2	Conversion clock and first zero advance	78
7.3	Oriented construction of the twenty-four-trit descriptor	79

7.4	Functional panel and terminal repertoire	80
7.5	Comparative readers, affine reader, and temporal incidence	84
7.6	Enumeration of cylinders and decimal candidates	85
7.7	Exceptional incidence and heterotypy condition	87
7.8	Regional composition and continuity of the realizations	89
7.9	Executable certification and traceability of the operations	90
7.10	Complete census of cylinders and selection of the terminal boundary	91
8	Dodecaphase record and the rational phase constant	94
8.1	Cyclic extension and memory grading	95
8.2	The oriented record and the integral $1 + 5 + 6$ chart	96
8.3	Evaluation of incidences and determination of the transverse coordinates	97
8.4	Integral image and inversion of the transverse coordinates	98
8.5	Conjugation, covariance, and memory of the incidence register	99
8.6	Two readouts of the terminal state	101
8.7	From the channel record to the signed vector	102
8.8	Orbitwise Hadamard transformation and integer reconstruction	103
8.9	Cyclic differences and the uniform component	104
8.10	Two rational readouts: κ_{36} and κ_{per}	105
8.11	Reversibility of the rational constant and phase change	105
8.12	Periodicity and the cokernel of the transport operator	106
9	Derivation of the fine-structure constant	107
9.1	First route: positional determination through integer closure	107
9.2	Second route: analytic determination through vacancies	113
9.3	Identification criterion for compatible readouts	115
9.4	Compatibility between the dodecaphasic and analytic representations of α	116
10	Exceptional prolongation: from dodecaphasic incidence to Moonshine	122
10.1	Compatibility of the Archimedean and incidence realizations	122
10.2	The finite realization and the ternary code	123
10.3	Dodecaphasic flag associated with K and α	124
10.4	Isometric intertwining of the incidence representations	127
10.5	The Witt star and the generation of M_{12}	128
10.6	The binary residue and the five regions of π	129
10.7	From the ternary code to the Niemeier lattice	131
10.8	Incidence selection of the rootless unimodular neighbor	132
10.9	The vertex operator algebra and the Monster	133
10.10	Graded Moonshine traces and the Monster representation	134
11	Planck constants and the determinantal structure of action	135
11.1	Determinantal action scale and decimal valuation	135
11.2	Action sections and dimensionless return quotient	136
11.3	Comparison of the reduced Planck constant	139
11.4	Angular coordinates, normalization of the turn, and units of action	139
11.5	Covariance of the channels and the incidence functional	140
12	Action ellipse and modular reciprocity	141
12.1	Semiaxes, foci, and interior radii	142
12.2	Parametric phase, polar angle, and action area	142
12.3	Shape reciprocity and interior distance	143
12.4	Bohr radius: a consequence of the action scale	144
12.5	Inductive–capacitive realization and constitutive continuity	144
13	Sheet return, areal–volumetric incidence and constitutive orientation	145
13.1	Projected return and restoration of the oriented cover	145
13.2	Descent of incidence from the tritic calendar	145
13.3	Boundary marking and areal–volumetric ratio	146
13.4	Transport of orientation to the constitutive channels	146
14	Hexad–octad incidence and the oriented functional	147
14.1	Dodecaphasic incidence and return functional	147
15	Constitutive operator and reconstruction of the vacuum channels	148
15.1	Positive response on two sheets	148
15.2	Four coordinates of a common response	148
15.3	Cubic inversion and angular recovery	149
15.4	Transported composition of constitutive responses	150

15.5	Logarithmic coordinates and refinement	152
15.6	Dimensional types of the four evaluations	152
15.7	Reconstruction of the action shape from the constitutive response	152
16	Joint reconstruction of gauge couplings, Higgs and the constitutive response	154
16.1	Angular provenance and route signatures	154
16.2	Rationalized tree-level realization	155
16.3	Joint inverse and exact domain	155
16.4	Higgs-impedance reconstruction	156
16.5	Retained information and transport of orientation	157
17	Cyclic decomposition, quarter-turn, and oriented Catalan moment	157
17.1	Two subspaces of the dodecaphase cycle	157
17.2	Faithful representation of the quarter-turn	158
17.3	Resolvent and unlimited depth	158
17.4	Composition with the constitutive response	159
17.5	Return traces and moments of the constitutive quotient	159
17.6	Residual composition and products of moments	161
17.7	Factorization through the constitutive operator	161
17.8	Reconstruction from vacuum coordinates	163
18	Constitutive evaluation and dimensional realization	163
18.1	Evaluation of the internal chain	163
18.2	Unit charts and covariance	164
19	Generated coordinates, action, and electronic normalisation	165
19.1	Regional readers and compatibility of evaluation	165
19.2	Signed dodecaphase chart of the full TPK state	166
19.3	From temporal incidences to the dodecaphase record	167
19.4	Periodic representation of K and normalisation of α	171
19.5	Three-dimensional construction of the electronic prefactor	172
19.6	Boundary reading and basal electronic composition	173
19.7	Determinantal scale and the reduced Planck constant	174
19.8	Angular coordinates and the clock of the absolute realisation	176
19.9	Two readers of the electronic trajectory and beta evaluation	178
20	The particle as a class of persistent sections	179
20.1	Extension, observable history, and compatible section	179
20.2	Identity under refinement and equivalence of representations	180
20.3	The species as a fibre: scope of the finite atlas	181
20.4	From the material record to the mass observable	182
21	Conjugation, orientation, and return of material sections	182
21.1	Dodecaphase involution and parity of observables	182
21.2	Lifting to antisections and preservation of mass	183
21.3	Sign holonomy and the persistence band	184
21.4	Bidirectional lattice and channel conjugation	184
21.5	Spinor representation of the electronic block	185
22	From the trajectory record to the mass operator	186
22.1	Chronological reading and oriented events	186
22.2	Torsional memory and signature	187
22.3	Character and internal specialisation	187
22.4	Bilayer hierarchy and preservation of the two genealogies	188
22.5	Displacement and autocorrelation readers	189
22.6	Operator on a fibre and dimensional section	189
22.7	The null propagation sector	190
23	Action, gravity, and thermal information	190
23.1	Dodecaphasic period and circular realization of action	190
23.2	Structural definition and function of the gravitational constant	191
24	Reduced information, energy, and thermal coordinates	191
24.1	Reader, memory, and state recovery	191
24.2	Reversible update, tritic decision, and erasure	192
24.3	Action, frequency, and decision energy	193
24.4	Dimensional definition of Boltzmann and coordinate selection	193
24.5	Dimensional entropy and information of the reduced state	194

25 Algebraic construction of quantum completions over APP	194
25.1 Linear state space associated with a particle realization	194
25.2 Symmetric and exterior completions	195
25.3 Sheet grading and exchange algebra	195
25.4 Canonical operators and idempotence of occupation	195
26 Bose–Einstein and Fermi–Dirac distributions	196
26.1 Constitutive speed and kinematic realization of the clock	196
27 Oriented radial recovery and symplectic transport	198
27.1 Recovery of Planck’s action with orientation preserved	198
27.2 Intertwining phase and anisotropy	199
28 Positive domain of the elliptic chart	199
29 Primitive returns and lifted frequency	200
29.1 Local extension automaton	200
29.2 Oriented primitive counts	200
29.3 Convergence of primitive refinement	200
29.4 Control identities	201
30 Route clock and absolute normalization	201
30.1 Integer returns do not suffice to distinguish species	201
30.2 Temporal holonomy	201
31 Antecedents of operator reconstruction and memory	202
31.1 The full orbit as a reference for its carrier	202
31.2 Operator identification and relative holonomy	203
31.3 Reversible coupling and nonstationary transport	205
31.4 Canonical covariance and reciprocity of the action ellipse	208
31.5 Covariance, normal form, and Gaussian spectrum	209
31.6 Prolongation of memory transport: quantum realization	211
32 Operational memory and invariants of an evolving structure	214
32.1 Observable value, capacity to act, and compatibility	214
32.2 Retained memory and operationally visible memory	215
32.3 Regularity of operations and prolongation of the state	215
32.4 Action character and oriented radial reconstruction	216
32.5 Action, information, and temperature	216
32.6 Comparison programme and hierarchy of checks	216
32.7 Scientific comparison and scope of interpretation	217
33 Chronological memory, balance, and vacancy response	217
33.1 Object, measure, and two readout operators	218
33.2 Increasing information and zero entropy rate	218
33.3 Chronological information not determined by balance	218
33.4 Centred current: bounded balance and persistent activity	219
33.5 Current spectrum and orientation	219
33.6 Electronic operator link	220
33.7 Retained information and energy composition	221
34 Visible repetition and genealogical differentiation within a region	221
35 Genealogical regularity, operator reconstruction, and the geometry of action	222
35.1 Production, compatibility, and reconstruction	222
35.2 Vacancy regimes and return of the capacity phase	223
35.3 The contribution of memory to operator composition	224
35.4 Action character and complete reciprocity coordinate	225
35.5 Phase circuits, memory, and conservativity	227
35.6 Dynamical compatibility between observations	227
36 Operator antecedents of the criticality family	227
36.1 Arithmetic cells, orientation and transport	228
36.2 Cylinder refinement and ordered readout	235
36.3 Joint construction of the continuum through compatible histories and prolongations	236
36.4 Extension of states and transfinite semantic extension	249
36.5 Ordered readout of block cylinders	254
36.6 Descent along fibers and preservation of the selector	255

36.7	Dodecaphase readout and quadratic normalization	255
36.8	Transport of iterates, roots and minima	256
36.9	Finite spectral representation of the uniform reader	256
37	Dodecaphase profile and quadratic return	258
37.1	Genealogy and object of the proof	258
37.2	Exact profile and quadratic criticality	258
37.3	Structural robustness with uniform bounds	259
37.4	Global quadratic holomorphic coordinate on a strip	259
37.5	Return, rescaling and memory	260
37.6	Accumulated scales and derivative of the operator	261
37.7	Certified roots and itineraries through period 256	261
37.8	Analytic persistence of the deformed roots	262
37.9	Analytic tails and error propagation	262
38	Nonadic depth and persistence of the itinerary	263
38.1	Outer growth and inner resolution	263
38.2	Explicit depth budget for each return	264
38.3	Doubling word and nonadic reader of chronology	264
38.4	Compatible bilateral memory at infinite depth	265
38.5	Existence and separation of asymptotic obligations	266
39	Compatibility of readers and recovery of the parameter	268
39.1	Compatibility of readers, minima and recovery of history	268
40	Ordered transport and Jacobi representation	270
40.1	Complete transport from the continuum and spectral realization of the profile	270
41	Classification and continuation of the first-root selector	273
41.1	Object and provenance	273
41.2	Classification before the infinite word	273
41.3	First parameter with infinite itinerary	275
41.4	Existence of each first new root	275
41.5	Convergence of the global selector to the first infinite itinerary	276
41.6	Logical checks and falsifiers	277
42	Transverse entry and local scaling	277
42.1	Provenance and result	277
42.2	Analytic chart and effective return	278
42.3	External bounds used as subsequent recognition	278
42.4	Rational entry and direction certificate	279
42.5	Quantitative construction of the stable leaf	279
42.6	Existence, uniqueness and transversality of the parameter	280
42.7	Real admissibility of all returns	280
42.8	Scaling of the local superstable cascade	281
42.9	Certified continuation of the eight initial roots	282
43	Isolation of the limit and scaling of the global selector	283
43.1	Indefinite continuation of the first root and exclusion of the intermediate interval	283
43.2	First-exit isolation and coincidence of the limits	284
43.3	Identification of the original centers and global scaling	286
44	Analytic identification of the original centers	288
44.1	Degree-two realization and metric identification of the centers	288
45	Return, memory and universality of the dodecaphase cascade	292
45.1	The generating structure and the critical reading	292
45.2	Twelve phases, moments and spectral representation	293
45.3	Reading depth and recovery of histories	293
45.4	Nonadic chronology and doubling combinatorics	294
45.5	The ordered chart and global root selection	294
45.6	The stable sheet and identification of the selected limit	295
45.7	Universal scaling and the meaning of precision	296
46	Articulation of the critical, spectral and constitutive readers	297
46.1	Critical separation as an intrinsic asymptotic reading	297
46.2	The finite measure and spectral recovery of the profile	298
46.3	Constitutive channels and their labeled inversion	298

46.4	The Catalan moment and functional recovery	299
46.5	Composition of the recoveries and joint scope	300
47	Joint normalization of action, length, and gravitational coupling	302
47.1	Inscribed return and longitudinal section	302
47.2	Polar transport and reciprocity of the same character	303
47.3	Planck scales and specialization of the unit character	304
47.4	Area and reconstruction of the memory sectors	304
47.5	Joint variation and compatible memory reduction	305
47.6	Temporal rigidity and boundary memory	306
47.7	Geometric normalization and evaluation	306
48	Geometric and material background of the joint action	307
48.1	Incidence, screen and soldering	307
48.2	Transport representation and geometric realization	308
48.3	Transport differential and material current	309
48.4	Spinorial representation and affine current	310
48.5	Torsional contraction and cross-terms between species	311
48.6	Normal ordering of the total current	312
49	Chromatic residue and internal representation of the joint action	312
49.1	Residual support and preserved orientation	313
49.2	Qutrit chart, Weyl pair and colour algebra	313
49.3	Chromatic transport and independent internal factors	314
50	Gravity, energy and self-inertia of the joint system	315
50.1	Energy, mass and radius of the same character	315
50.2	Static and dynamic memory, and spectral domain	316
50.3	Backreaction in the same transport	317
50.4	The same action and its currents	319
50.5	Constraint propagation with the total source	321
50.6	Boundary and self-inertial operator	321
50.7	Self-inertia of the state and acceleration of its projection	323
50.8	Result and scope checks	323
51	Joint action, backreaction and canonical closure of the constraints	323
51.1	Starting structure and action with the full source	324
51.2	A proof of compatibility, propagation and closure	325
51.3	Domain of the result and comparison of realizations	328
52	Structural compatibility of action, mass and operational information	329
52.1	Action in the angular composition and the electronic structure	329
52.2	Proper frequency and structural normalization of mass	330
52.3	Reciprocal dilation of the spatial readouts of mass	330
52.4	Channel orientation and the constitutive response of the vacuum	331
52.5	Boltzmann, structural frequency and ternary thermal weighting	332
52.6	The action ellipse and transport of geometric shape	332
52.7	Joint compatibility and relational testing	333
52.8	Phase closure and relative holonomy of the state	333
52.9	Reconstruction of the action coefficient from relative transport	334
52.10	Holonomy iteration and operational information of the history	334
52.11	Spectral potential and differential reciprocity of the responses	335
52.12	The Catalan moment and contractive response of the incidences	336
52.13	Exact reconstruction and stability under finite precision	336
52.14	Synthesis: coordinates of the section and memory of the operations	336
53	Composition and reconstruction of the angular, electronic and constitutive readouts	337
53.1	Action coefficient and construction of the counterangle	337
53.2	Angular charts and conservation of action	338
53.3	Complete electronic readout and action return	338
53.4	Barbero functional and the hexad–octad transition	339
53.5	Oriented Catalan moment and constitutive response	340
53.6	Reconstruction from normalized speed and Barbero	341
53.7	Reconstruction on the generated image	341
53.8	Structural definitions of the composite readouts	342
53.9	Transport of the nilpotent electronic block	342

54 Action, mass and propagation: structural compatibility and transport	343
54.1 Shared action section and mass law	343
54.2 Spatial reciprocity and gravitational composition	345
54.3 Constitutive response, orientation and magnitude	345
54.4 Action, clock and thermal information	346
54.5 Energy identity of the ellipse and exact transport	347
54.6 Momentum, tests and scope of the composition	349
55 Marked Reading and the Thermometric Conjugacy of Action	350
55.1 Capacity, Spectral Reference and Retained Origin	350
55.2 Tensor Identity and Specification of the Readers	351
55.3 Variational Selection and Information Balance	352
55.4 Composition with Action, Area and Gravitational Reciprocity	353
56 Horizon composition and thermal-geometric recovery	354
56.1 Fixed area, fixed mass, and preservation of the thermal product	354
56.2 Angular separation and recovery of the action section	354
57 Electronic History and Mass Conservation under Transport	355
57.1 Separation and Reconstruction of the Two Central Histories	355
57.2 Spectral Form of the Return Reader	356
57.3 Criterion for Transported Mass and Its Associated Readings	357
57.4 Transport of the electronic section to the electroweak sector	357
57.5 Preservation of route refinements	358
57.6 Mass congruence and spectral inversion without commutation	359
58 Constitutive compatibility, relative holonomy and operational information	359
58.1 Differential reciprocity of speed and impedance	359
58.2 Spectral potential and the depth-two connection with Catalan	360
58.3 Exact reconstruction and finite sensitivity	361
58.4 The quarter-turn arises from the dodecaphasic cycle	362
58.5 Relative loop of two transports	363
58.6 Distinguishing a chart change from a relative effect	365
58.7 History information versus occupation statistics	366
58.8 Consequences and proposed new definitions	367
59 Conclusions	368
A Compensated transport, geometric memory and energy tests	372
A.1 Operator origin and domain of the realization	372
A.2 Compensation of operations and area conservation	372
A.3 Positive readout of the residual and path reversal	373
A.4 Reversible coupling and reduced entanglement readout	373
A.5 Second moment of the vacancy current	375
A.6 Conditions for a material test	375
B Action, reduced memory and recoverable work	376
B.1 Composition of action with an energy realization	376
B.2 Preparation and readout operators for the energy result	376
B.3 Identity between visible energy and purity	377
B.4 Recoverable work under local operations	377
B.5 Exact cases and transport orientation	378
B.6 Boltzmann and the equivalent temperature of the passive state	379
B.7 Specificity and testing conditions	379
C Exhaustive enumeration and finite determination of transport operators	379
C.1 Complete domain of initial conditions	380
C.2 Census of signatures and emissions	380
C.3 Transition register and reconstruction of the lifts	382
C.4 Exhaustiveness of the calendar and stability of reconstruction	383
C.5 Reproduction certificate and location of the data	384
D Sources, preservation, and reproduction	384

1 Introduction

Reading a constant as a scalar value represents the final stage of an operation. Retaining the structure that produces this readout makes further questions accessible: which relations it shares with other readouts, which transformation preserves its orientation, and which part of its history can be reconstructed from joint observations. The present development situates these questions within the continuity between the discrete HMT structure and its realizations in Dimensional Mechanics.

The common origin retains the operations of APP, the oriented regime of TRIT, and the effective composition of TPK. Phase return is distinguished from the evolution of the enriched state. The regional coordinates and their dodecaphase closure are subsequently transmitted to the action, incidence, and mass readout operators. Each transmission preserves the objects and conditions it uses; an output can enter the next construction without losing its provenance. The opening core presents this architecture with its operations, prolongations, and proofs. The sectoral antecedents that follow explicitly supply the dependencies used by the compositions developed here.

The central question can be formulated through three complementary terms: *production*, *compatibility*, and *reconstruction*. Production fixes the object and its genealogy. Compatibility determines the relations that its different readouts must satisfy. Reconstruction establishes which data of the object can be recovered from them. This inverse procedure is a property of already constructed readouts and retains their bases, orientations, and domains; it does not replace their generative order.

The first line of development links action to the angular pair. The dimensionless coefficient of the action section participates in the counter-angle; together with the angle, the latter determines the channels that enter the constitutive response and the Barbero–Immirzi functional. The same section enters the return quotient of the electronic readout. Substituting the structures into their equations therefore reveals shared dependencies that a list of separate constants leaves implicit. The distinction between a coefficient and a dimensionful quantity is essential: reconstructing a shape ratio and reconstructing a dimensionful action require different reference data.

The second line of development examines the information retained in the response. Normalized speed and impedance retain different aspects of the distribution between channels. Their composition proves both a global inversion within the stated collection and a differential relation between the two responses. A spectral potential organizes this relation, and its Hessian controls the sensitivity of inversion. Two properties are thus distinguished: exact data uniquely determine the object, whereas finite precision permits reconstruction with a bounded error. The oriented collection whose extremal value is Catalan’s constant enters through its spectral operation, with its arguments and depth retained.

Transported composition of the responses provides their additive coordinate; the angular gauge and Higgs signatures reconstruct the same channel pair. The moments of the cyclic defect complete this reading with convergence bounds and a Mellin representation. Subsequently, the electronic section is substituted into Fermi and the Higgs scale: the proof retains the factors that remain, those that cancel, and the transformation of the inverse mass operator without assuming commutation.

The third line of development addresses the order of operations. The quarter-turn plane of the dodecaphase cycle and the conjugate forms of the ellipse yield an explicit commutator. The balance of phase increments closes while the action on the state retains a reciprocal dilation. Two programs with the same multiplicities of operations can produce different responses because of their order. This difference admits defined detectors and reconstruction maps: the trace records certain invariants, whereas responses along oriented axes distinguish directions to which the trace assigns the same value.

The physical interpretation is presented together with the conditions of each realization. In the circular gravitational section, the reduced Compton wavelength and the reduced gravitational radius undergo reciprocal dilations and retain a common area. In the thermal realization of the cycle, the same structural coordinate enters as the exponent of an equilibrium weight. In elliptic transport, preservation of symplectic area and preservation of a common quadratic energy are examined through their respective operators. Any prospective experimental realization of the relative program requires a joint specification of the system,

reference, available operations, and controller. These conditions are part of the meaning of the result and remain adjacent to its proofs.

Gravitational normalization is developed before its scalar consequences are used. The Hadamard return, the inscription of incidences, and the local norm of α produce a length section. Its polar transport and the same route character produce two reciprocal readout operators. Energy in turn preserves the action length. Combining $RA = L_\alpha^2 I$ with $H\Lambda = \hbar c I$ fixes the gravitational constant as the common coefficient of all positive modes of the realization. The circular clock is then expressed in terms of length and speed. This precedence explains the normalization without using the experimental value of G or a tabulated Planck length.

The electronic and thermal compositions specify how memory enters these relations. In the former, the dodecaphase record is reconstructed from its sum, differences, and orientation; the spectrum distinguishes orders having the same multiplicities. In the latter, the geometric spectrum already constructed by memory transport supplies the ratio 1/10. Chronological capacities and occupation sectors determine a marked trace that is composed with a Gibbs state. Conditional energy, marked information, and the resulting temperature are specified separately; this choice of observations makes it possible to prove the thermometric transducer and follow its relation to the action scale and area. The corresponding sections retain the preparation data, gradings, and variation hypotheses used.

The exposition first develops the antecedents, then preserves the explanatory narrative of the investigation and presents the complete compositions. The chronological study of vacancies, the regional example of visible recurrence, and memory re-entry make it possible to follow the origin of the compatibility questions. The energetic and Gaussian developments are retained in the appendices as auxiliary realizations, with their full hypotheses and proofs. The conclusions bring together the structural characterizations obtained. The digital supplement preserves the verbatim working sources, programs, and results; the manuscript contains the proofs used and distinguishes their mathematical interpretation from the accompanying finite checks.

This organization gives a precise formulation of the work's transversal contribution. A constant retains its numerical value, while its structural definition also specifies the operation producing that value, the domain on which it acts, and the transformations it preserves. Two quantities sharing a dependency can therefore constrain one another: knowing a response and its orientation reconstructs a distribution that an isolated scalar readout does not distinguish. The inverse problem is posed on the already generated image and retains its reference data. Generation and reconstruction are thereby connected without reversing the precedence of the construction.

The same distinction organizes the relation between memory and information. The present state, the accumulated balance, and the operator that will respond to a continuation are different objects. The return of a phase can coexist with a persistent transformation of the state; two sequences with identical counts can differ through their order of composition. The proofs develop this distinction through operators, invariants, and reconstructive observations. Their physical relevance lies in specifying what each description preserves and which operation provides access to the retained information.

The passage to dimensionful quantities preserves the corresponding bases and normalizations. Reconstruction of an action coefficient, preservation of an area, and characterization of a temperature are related results of different types. The exposition brings them together under their conditions of validity and shows how they compose. The conclusions make explicit both the characterization obtained for each object and the relation it establishes with the other structures.

The Feigenbaum constant acquires a precise structural definition: it is the asymptotic separation invariant of the parameters selected by the critical returns of the uniform dodecaphasic reader. The construction preserves its APP-TRIT-TPK origin, the joint continuum and the first-root order. Identifying the selected centres with the analytic centres makes the scaling theorem applicable to that same sequence; the proofs in 43.1–43.3 and (46.2) establish the parametric result. The Jacobi representation recovers the profile from its twelve-node measure, while constitutive inversion recovers the oriented channels and the moment whose endpoint is Catalan's constant. These operations exhibit relations between readers of the same antecedent, preserving the domain and information of each.

The composition also extends to a common action of geometry, gauge fields and matter. Gravitation and the strong, weak and electromagnetic interactions enter through realizations of the same enriched state, with the couplings inherited from the preceding construction. Joint variation determines the sources and torsional response; the Legendre transformation produces a common algebra of first-class constraints. Their propagation and the closure of the canonical connection are proved on the regular constraint surface, preserving boundary conditions and global holonomy. Variational and canonical unification thus provides an explicit dynamical composition for the compatibility and reconstruction relations organizing the manuscript.

2 Formal core of Triadic Modular Holography

Triadic Modular Holography, abbreviated HMT, begins with a finite structure of positions and operations. The order of construction matters: positions receive arithmetic evaluations, transitions acquire a ternary signature, and the dynamics preserves the information required to prolong them. Only subsequently are numerical realizations formed. A constant recognized at the end of this process does not replace either the rule that produced its region or the history that determines its evaluation.

The designation *All Possible Paths*, APP, expresses the role of paths on the support. It is adopted as the name of the construction, with a conceptual reference to the path formulation of quantum mechanics; it does not imply that its arithmetic law is the Feynman path integral. The *Topological Phase Kernel*, TPK,¹ denotes the composition of selection, transport, memory updating, and prolongation. The acronyms are retained only after the objects they represent have been defined.

2.1 APP: positional support and arithmetic evaluations

Consider the nine interior marks of a line graduated between zero and ten,

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Their horizontal and vertical arrangement produces the eighty-one positions of D_9^2 . The interior marks must not be confused with the ten unit intervals of the reference line. This arrangement fixes an alphabet and two directions; it does not yet use a real expansion to determine a trajectory.

Cyclic identification of the positions yields the support

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The chart of positive representatives retains the symbol nine for the zero class. For every positive integer, define

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n). \quad (2.1)$$

The positive digital root is therefore a choice of representative of the residue class. By itself, it does not retain the integer prior to reduction; the coordinate q_9 supplies precisely the missing information.

At each position (i, j) , accumulating i a total of j times produces ij . The two integer evaluations and their projections are

$$S(i, j) = i + j, \quad P(i, j) = ij, \quad \Sigma(i, j) = \rho_9(i + j), \quad \Pi(i, j) = \rho_9(ij).$$

The symmetry $(i, j) \mapsto (j, i)$ interchanges the directions of accumulation and preserves both evaluations. This is a property of a single support, not a subsequent identification between two independent matrices.

Definition 2.1 (Lifted arithmetic evaluation). The APP evaluation that preserves the two sheets is

$$\mathcal{A}(i, j) = ((R_{ij}^+, q_{ij}^+), (R_{ij}^\times, q_{ij}^\times)) = ((\rho_9(i + j), q_9(i + j)), (\rho_9(ij), q_9(ij))). \quad (2.2)$$

The term *sheet* here denotes an evaluation on the same support: additive or multiplicative. The two quotient coordinates belong to the arithmetic state and are not discarded when the residues are reduced.

Proposition 2.2 (Reconstruction of the evaluations). *The two pairs in (2.2) determine $i + j$ and ij exactly. The residues alone determine neither these integer evaluations nor their quotients.*

Proof. The first assertion is (2.1) applied to each sheet. For the second, $10 = 1 + 9$ and $19 = 1 + 2 \cdot 9$ have the same residue and different quotients. On the two-dimensional support, the positions $(5, 5)$ and $(8, 2)$ give the same sum and the same product residue, but products $25 = 7 + 18$ and $16 = 7 + 9$. The residual projection identifies data that the lifted evaluation distinguishes. $\square$

¹The authors also dedicate the initials TPK to Tan Pengkiang. This dedication is independent of the mathematical definition.

	1	2	3	4	5	6	7	8	9	
1	1	2	3	4	5	6	7	8	9	One support: 81 positions.
2	2	4	6	8	1	3	5	7	9	Two evaluations: $S(i, j) = i + j, P(i, j) = ij.$
3	3	6	9	3	6	9	3	6	9	Each row is generated by accumulation: $P(i, j + 1) - P(i, j) = i.$
4	4	8	3	7	2	6	1	5	9	
5	5	1	6	2	7	3	8	4	9	
6	6	3	9	6	3	9	6	3	9	The chart represents 0 (mod 9) by 9.
7	7	5	3	1	8	6	4	2	9	The marked block is
8	8	7	6	5	4	3	2	1	9	$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix},$
9	9	9	9	9	9	9	9	9	9	with eigenvalues 9 and 5.

Figure 1: APP multiplication table in positive representatives. Nonunit rows and columns are distinguished by grey shading; the frame identifies the adjacent diagonal block with equal diagonals. The displayed table is an observable of the arithmetic torus; its representational edges do not constitute an intrinsic topological boundary of the torus.

2.1.1 Cayley graph, toroidal topology, and winding number

The cardinal translations are given, in row and column coordinates, by

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \quad T_d(x) = x + \nu_d.$$

The first coordinate increases southward and the second eastward. The graph

$$\mathcal{G}_9 = \text{Cay}(X_9, \{\nu_N, \nu_E, \nu_S, \nu_O\})$$

is the two-dimensional cyclic lattice: it has eighty-one vertices and four outgoing oriented edges per vertex. It is a Cayley graph for the *additive* structure of the support. Multiplication of residues has zero divisors and does not turn all APP rows into a multiplicative group.

A cardinal word $w = d_1 \cdots d_r$ and an origin x_0 determine

$$x_k = x_0 + \sum_{j=1}^k \nu_{d_j} \pmod{9}.$$

There are $81 \cdot 4^r$ oriented paths of length r , including returns and backtracking. When the path closes, its integer displacement prior to reduction defines

$$\text{wind}(w) = \frac{1}{9}(\#S(w) - \#N(w), \#E(w) - \#O(w)) \in \mathbb{Z}^2. \quad (2.3)$$

Path reversal changes the sign of this vector. Return to the same residual position therefore does not require the winding to vanish. This is the first instance of a distinction that will recur in phase return: one coordinate may close while another records a nontrivial evolution.

2.1.2 Transport quotient and cocycle identity

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \rightarrow D_9$ be the section of positive representatives. The transport quotient associated with additive composition, also called the *carry* in positional arithmetic, is

$$c_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9}.$$

The compositional character of this memory is expressed by an exact identity.

Lemma 2.3 (Cocycle identity). *For $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a, b) + c_9(a + b, c) = c_9(b, c) + c_9(a, b + c).$$

Proof. Both sums equal

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$

Associativity of the sum of lifted representatives requires precisely this additional term when composing in the quotient. $\square$

With positive representatives, this cocycle is not normalized at the identity: $c_9(0, a) = 1$. If s_0 takes representatives in $\{0, \dots, 8\}$, its normalized carry c_0 satisfies

$$c_9(a, b) = c_0(a, b) + \mathbf{1}_{a=0} + \mathbf{1}_{b=0} - \mathbf{1}_{a+b=0}.$$

The difference is the change of section; it does not alter the cocycle identity. The subsequent decomposition of the calendar into phase and window will use representatives from zero to eight.

The totals of the two evaluations distinguish the visible contribution from that retained in the quotient:

$$\begin{aligned} \sum S = 810, \quad \sum P = 2025, \quad \sum \Sigma = 405, \quad \sum \Pi = 459, \\ \sum q^+ = 45, \quad \sum q^\times = 174. \end{aligned}$$

It follows that

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \quad (2.4)$$

The sum-product difference is not exhausted by the visible defect 54. The identity also preserves the quotient defect 129. Suppressing it would change the arithmetic being transported, even if the residual table remained identical.

2.1.3 Ternary reduction and multiplicative radical

The ring $R = \mathbb{Z}/9\mathbb{Z}$ has unit group $\{1, 2, 4, 5, 7, 8\}$ and maximal ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\}, \quad \mathfrak{m}^2 = 0.$$

Every additive row is a permutation of the nine representatives. A multiplicative row is such a permutation exactly when its index is a unit. Rows three and six contain three repeated values; row nine contains only the representative of the zero class. Thus multiplicative folding already distinguishes the units from the nilpotent sector before any continuous geometry is introduced.

The map $x \mapsto 3x$ has kernel and image both equal to $\mathfrak{m}$. It induces an isomorphism of vector spaces over $\mathbb{F}_3$,

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \quad [x] \longmapsto 3x,$$

but not a ring isomorphism. In particular, the visible sequence 3, 6, 9 admits two different readouts: its direct reduction modulo three is 0, 0, 0, whereas extraction of the factor three followed by reduction of its internal coordinate gives 1, 2, 0. This second operation preserves a coordinate of the ideal; it is not the same projection as direct reduction of the original number.

2.1.4 Latin realization of the additive evaluation

The distinction between the two evaluations can also be expressed through operators. Let $(e_r)_{r \in \mathbb{Z}/9\mathbb{Z}}$ be the orthonormal basis of $\mathbb{C}^9$. The assignment $v_{ab} = e_{a+b}$ produces an orthonormal basis in each row and column. The projectors $P_{ab} = v_{ab}v_{ab}^*$ satisfy

$$\sum_b P_{ab} = I_9 = \sum_a P_{ab}, \quad P_{ab}^2 = P_{ab}.$$

Indeed, each translation $b \mapsto a + b$ permutes the nine residues. This yields a rank-one-projector realization of a Latin square, in the terminology of quantum Latin squares [1, 3]. This construction starts from the additive sheet already defined and makes explicit the map to its operator representation.

The multiplicative evaluation preserves different information: for $a = 3$, the vectors e_{3b} repeat the residues 0, 3, 6, and

$$\sum_{b \in \mathbb{Z}/9\mathbb{Z}} |e_{3b}\rangle\langle e_{3b}| = 3(|e_0\rangle\langle e_0| + |e_3\rangle\langle e_3| + |e_6\rangle\langle e_6|).$$

The multiplicity is recorded in the operator and characterizes the multiplicative radical. Accordingly, the toroidal support, the Latin property, and the operator resolution of the identity are properties requiring different definitions. The quantum magic squares studied by De las Cuevas, Drescher, and Netzer require positive entries and row and column sums equal to the identity; their theory also distinguishes dilations with commuting entries[2]. This reference provides precise terminology for comparing APP realizations.

2.2 TRIT: orientation and quadratic regimes

Definition 2.4 (Ternary signature and local lift). The TRIT signature uses the balanced identification

$$\mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1.$$

For $n \in \mathbb{Z}$, its lift preserves $n = r + 3c$, where $r \in \{-1, 0, +1\}$ and $c \in \mathbb{Z}$. In a transition of bounded local amplitude, the three states $(0, -1)$, $(0, 0)$, $(0, +1)$ may exhibit the same zero residue and different carries.

The signature does not replace the trajectory. A residual zero does not mean the absence of information about orientation, phase, or quotient. Nor does it by itself introduce a physical quantity. Its function is to distinguish algebraically the transport classes to be realized next.

2.2.1 Three quadratic regimes and a common exponential

Once the ternary signature has been obtained, its quadratic realization is defined by

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \quad \tau \in \{-1, 0, +1\}. \quad (2.5)$$

If J_τ is the class of X , then $J_\tau^2 = -\tau I$. The structure distinguishes the elliptic type $\tau = +1$, the parabolic type $\tau = 0$, and the hyperbolic type $\tau = -1$. The real realization does not generate the signature: it expresses the types already produced by oriented reduction.

Proposition 2.5 (Exponentiation of the three types). *In each algebra (2.5),*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad (2.6)$$

where

$$C_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n}}{(2n)!}, \quad S_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n+1}}{(2n+1)!}.$$

The three realizations are

τ	J_τ^2	$\exp(tJ_\tau)$
+1	$-I$	$\cos tI + \sin tJ_\tau$
0	0	$I + tJ_\tau$
-1	I	$\cosh tI + \sinh tJ_\tau$

Proof. Separating the even and odd terms of the exponential series and substituting $J_\tau^{2n} = (-\tau)^n I$ and $J_\tau^{2n+1} = (-\tau)^n J_\tau$ gives the formulas. For $\tau = 0$, every term of degree at least two vanishes. In the other two cases, the trigonometric or hyperbolic series are recovered. $\square$

The exponential is common to the three regimes: it is not assigned exclusively to the parabolic branch. The invariants

$$C_\tau(t)^2 + \tau S_\tau(t)^2 = 1$$

follow from $\exp(tJ_\tau)\exp(-tJ_\tau) = I$. The same algebraic principle preserves unity across realizations that differ in their quadratic type.

In the hyperbolic regime, $P_\pm = (I \pm J_{-1})/2$ are complementary projectors, and

$$\exp(tJ_{-1}) = e^t P_+ + e^{-t} P_-.$$

Growth and contraction occur jointly in two eigendirections. In the elliptic regime, the parameter describes a rotation; in the parabolic regime, a nilpotent translation. The temporal interpretation adopted by HMT associates +1 with return with memory, 0 with the threshold, and -1 with bidirectional opening. This interpretation requires an ordering of the parameterization; it does not add a fourth algebraic type.

2.2.2 Balanced division and composition of lifts

Ternary reduction makes it possible to express an integer APP evaluation in a visible coordinate and a prolongation coordinate. Its function is not merely to abbreviate the writing of an integer: it provides a composition rule determining how the quotient changes when evaluations are added or multiplied. The balanced choice also makes explicit the compatibility of that rule with sign reversal.

Definition 2.6 (Balanced ternary coordinates). For $n \in \mathbb{Z}$, let

$$q_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \quad r_3(n) = n - 3q_3(n).$$

The map

$$\mathcal{B} : \mathbb{Z} \longrightarrow \{-1, 0, +1\} \times \mathbb{Z}, \quad n \longmapsto (r_3(n), q_3(n))$$

is called the balanced ternary decomposition. Its inverse is $\mathcal{L}(r, c) = r + 3c$.

Indeed, $r_3(n) \in \{-1, 0, +1\}$. If two pairs represent the same integer, $r + 3c = s + 3d$, then $r - s$ is a multiple of 3 lying in the interval $[-2, 2]$; necessarily $r = s$ and $c = d$. The representation is therefore unique. Division can be repeated on $q_3(n)$, so that each level preserves the datum needed to reconstruct the preceding one. The quotient is not a perturbation of the residue: it constitutes its complementary coordinate.

The signature obtained from an evaluation F , with $F = S$ or $F = P$ in APP, is $r_3 \circ F$. When reduction acts on the phase cursor, the same balanced section produces the classes $\{1, 4, 7\}$, $\{2, 5, 8\}$, and $\{3, 6, 9\}$, with values +1, -1, 0. The TPK selector defined next uses these classes to determine the operational mode. Reading an evaluation and selecting a mode thus have the same ternary reduction, but different domains and functions.

Proposition 2.7 (Arithmetic of balanced pairs). Let $n = r + 3c$ and $m = s + 3d$, with $r, s \in \{-1, 0, +1\}$. Define

$$\chi(r, s) = \frac{r + s - r_3(r + s)}{3}.$$

Integer operations are transported to pairs by

$$(r, c) \boxplus (s, d) = (r_3(r + s), c + d + \chi(r, s)), \quad (2.7)$$

$$(r, c) \boxtimes (s, d) = (rs, rd + sc + 3cd). \quad (2.8)$$

The map $\mathcal{L}$ preserves both operations.

Proof. The equality $r + s = r_3(r + s) + 3\chi(r, s)$ gives

$$n + m = r_3(r + s) + 3(c + d + \chi(r, s)).$$

On the other hand,

$$nm = rs + 3(rd + sc + 3cd).$$

Since the product of two balanced representatives again belongs to $\{-1, 0, +1\}$, it requires no additional residue reduction. Uniqueness of the decomposition proves both formulas. $\square$

For example, $2 = (-1) + 3$ and $4 = 1 + 3$. Their sum and product are reconstructed as

$$(-1, 1) \boxplus (1, 1) = (0, 2), \quad (-1, 1) \boxtimes (1, 1) = (-1, 3).$$

The resulting evaluations are 6 and 8. Adding quotients suffices in this additive example; the product involves the cross terms rd , sc , and $3cd$. The discrepancy between the two operations also resides in the prolongation coordinate, not only in the residue.

The term χ is a cocycle of the balanced section. If $r \oplus s = r_3(r + s)$, it satisfies

$$\chi(r, s) + \chi(r \oplus s, t) = \chi(s, t) + \chi(r, s \oplus t).$$

Both sides represent the quotient of the same sum of three integers. This identity guarantees that regrouping the operations does not change the reconstructed evaluation. At the same time, projection onto the first component removes that accounting: $1 + 1 = -1 + 3$, whereas the residue readout retains only -1 . The discarded information is determined, in this case, by $\chi(1, 1) = 1$.

The relation to modulus 9 preserves a second discrete scale. The map

$$\beta : \{-1, 0, +1\} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad (r, \bar{c}) \longmapsto r + 3c \pmod{9}$$

is well-defined and bijective. Changing the representative of $\bar{c}$ adds a multiple of 9; conversely, equality of images first fixes r by reduction modulo 3, and then $\bar{c}$. The addition transported by this bijection contains the term $\chi(r, s)$ from (2.7), now reduced modulo 3. Thus the two ternary levels are not composed by independent coordinate addition: normalizing the first level modifies the second.

In particular, the classes 3, 6, and 0 in $\mathbb{Z}/9\mathbb{Z}$ have first component 0, but second components 1, 2, and 0, respectively. The distinction already observed between direct reduction and extraction of the ideal coordinate appears as a property of β . Preserving that coordinate prevents the zero-residue sector from being artificially turned into a single position. The modulus 1000, used later to organize decimal blocks, serves another positional-representation function; its quotient is preserved through the corresponding division. A change of base preserves the integer when it retains residue and quotient, whereas agreement of residues in different bases does not by itself identify their transport states.

2.2.3 Reversal, residue class, and nonvisible coordinates

Additive inversion lifts unambiguously:

$$\mathcal{B}(-n) = (-r_3(n), -q_3(n)).$$

In particular, 3, 0, and -3 have coordinates $(0, 1)$, $(0, 0)$, and $(0, -1)$. They share a zero signature but represent different evaluations. In a transition restricted to a quotient of maximum magnitude 1, they are the three possibilities in the visible fiber over zero; in a general integer lift, that fiber contains all pairs $(0, c)$, with $c \in \mathbb{Z}$.

The quotient coordinate recovers the evaluated integer, but does not by itself recover the trajectory that produced it. Two APP paths may share evaluation, residue, and quotient while differing in orientation, mode sequence, or winding. Thus the decomposition $\mathcal{B}$ is integrated into the transport state instead of replacing it. The arithmetic memory of a division and the memory of a trajectory are related coordinates, not synonyms.

For a signature sequence of length n , oriented reversal is

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1).$$

Applying it twice restores the sequence. The operation reverses the order of positions and the sign of each signature; it preserves the length and the zero positions in reverse order. In the adopted parametrization, it interchanges return with memory, associated with $+1$, and bidirectional opening, associated with -1 , while maintaining the threshold 0. Transport of the remaining trajectory data is specified through the corresponding TPK involution.

This signature reversal must be distinguished from conjugation within a quadratic algebra. The former may change the type τ ; the latter, constructed next, acts within the type already fixed. The distinction prevents attributing to sign reversal an algebraic equivalence between different regimes.

2.2.4 Quadratic product, conjugation, and algebraic norm

An element of $\mathbb{A}_\tau$ has a unique expression $aI + bJ_\tau$. The relation $J_\tau^2 = -\tau I$ determines its product:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \quad (2.9)$$

This is a real realization of the type already selected. The coordinates a, b are not APP residues, and the product (2.9) does not replace (2.8); it expresses another operation in the declared quadratic codomain.

Definition 2.8 (Conjugation and algebraic norm). Conjugation of type τ and its algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \quad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 2.9 (Multiplicativity and invertibility). For $z, w \in \mathbb{A}_\tau$,

$$z\bar{z} = N_\tau(z)I, \quad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

Moreover, z is invertible if and only if $N_\tau(z) \neq 0$, in which case

$$z^{-1} = \frac{\bar{z}}{N_\tau(z)}.$$

Proof. The product of $aI + bJ_\tau$ and $aI - bJ_\tau$ is $(a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so $(zw)\bar{z}\bar{w} = (z\bar{z})(w\bar{w})$. If the norm is nonzero, the stated formula provides the inverse. Conversely, if $zw = I$, multiplicativity implies $N_\tau(z)N_\tau(w) = 1$. $\square$

The quadratic form N_τ is positive definite only in the elliptic type. For $\tau = 0$, it equals a^2 , and the element J_0 is nonzero although its square and norm are zero. For $\tau = -1$, it equals $a^2 - b^2$; the nonzero elements $I + J_{-1}$ and $I - J_{-1}$ have zero norm and their product vanishes. The algebraic norm here is a multiplicative quadratic form: it does not presuppose a positive vector-space norm.

All three types can be faithfully realized by

$$aI + bJ_\tau \mapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}.$$

The determinant of this matrix is $N_\tau(aI + bJ_\tau)$. Thus the complex structure, the dual-number structure, and the split-real structure are represented by the same matrix expression. Each preserves its law and zero divisors. The form of signature $(1, 1)$ in the last case has a Lorentzian interpretation in dimension $1 + 1$; identifying a physical metric additionally requires a specific space, field, and transport rule.

2.2.5 Multiplicative conservation and composition of evolutions

The exponential already obtained belongs to the set of norm-one elements:

$$N_\tau(\exp(tJ_\tau)) = 1.$$

This conservation persists under composition of parameters. Indeed, all factors are functions of the same generator, and

$$\exp(tJ_\tau)\exp(sJ_\tau) = \exp((t+s)J_\tau).$$

Comparing the two components yields the addition laws

$$C_\tau(t+s) = C_\tau(t)C_\tau(s) - \tau S_\tau(t)S_\tau(s), \quad (2.10)$$

$$S_\tau(t+s) = S_\tau(t)C_\tau(s) + C_\tau(t)S_\tau(s). \quad (2.11)$$

Conjugation changes t to $-t$ and coincides with inversion within this subgroup. It does not change τ : it reverses an evolution in the chosen regime.

In the elliptic type, the preserved form is $a^2 + b^2$, and the exponential trajectory traverses the norm-one circle. In the hyperbolic type, the eigen-coordinates are $u = a + b$, $v = a - b$, with $uv = N_{-1}(z)$. The exponential gives

$$(u, v) = (e^t, e^{-t}), \quad uv = 1.$$

Growth of one coordinate is accompanied by contraction of the other. Conservation expresses a relation between the two coordinates; it does not assert that each remains constant.

In the parabolic type,

$$(I + tJ_0)(I + sJ_0) = I + (t + s)J_0, \quad (I + tJ_0)^{-1} = I - tJ_0.$$

Evolution does not stop because $J_0^2 = 0$. Nilpotence eliminates terms of order at least two, but preserves the linear term transporting the parameter. The visible symbol 0, the zero element of the algebra, and the nonzero generator J_0 are distinct. This distinction is necessary to describe the threshold without identifying it with the absence of a state or dynamics.

Proposition 2.10 (Composition of affine coordinates). *In the chart where the scalar component is nonzero, represent a direction by $I + uJ_\tau$. If $1 - \tau uv \neq 0$, the direction of its product with $I + vJ_\tau$ has coordinate*

$$u \star_\tau v = \frac{u + v}{1 - \tau uv}.$$

Proof. We have

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u + v)J_\tau.$$

Dividing by the scalar component gives the stated expression. $\square$

The affine chart preserves the ratio between components, not the scalar factor divided out. When $1 - \tau uv = 0$, the product must be examined before changing charts: it may have a defined direction with zero scalar component, or be the zero element in the presence of zero divisors. The composition law still belongs to the complete algebra; a coordinate singularity does not automatically amount to a singularity of the object. This formula will reappear in the elliptic regional readout. Its use requires retaining the denominator and, where applicable, the normalization factor.

2.2.6 Transport of type and subsequent realizations

The algebras $\mathbb{A}_{+1}$, $\mathbb{A}_0$, and $\mathbb{A}_{-1}$ are not isomorphic as unital real algebras. The first is a field; the second contains a nonzero nilpotent; the third has nontrivial idempotents and is isomorphic to $\mathbb{R} \times \mathbb{R}$. Thus the signature change organizing a route cannot be interpreted as an undifferentiated isomorphism between the three realizations. The TPK preserves the regime index, domains, and effective operator acting at each transition.

In the electron chapter, the pair of projections of the APP sheets produces a normalized commutator whose square is $-I$. The same block contains an involution squaring to I and two nilpotent directions. The joint realization will be proved on that concrete pair of projections; the preceding classification explains why the three quadratic relations can appear in the same matrix algebra without being identified with one another.

In the exceptional prolongation, the Paley matrix reduced modulo 3 satisfies $A_W^2 = -I_6$. The relation equips the ternary space with a structure of dimension 3 over $\mathbb{F}_9$. Its link to a real operator root is expressed through the integral presentation

$$\mathbb{Z}[u]/(u^2 + 1).$$

Base changes to $\mathbb{F}_3$ and $\mathbb{R}$ produce representations of the same relation. Each preserves its characteristic, dimension, and incidence information. The development of the respective operators establishes the link, not equality of their cardinalities.

Preservation of norm one therefore describes the quadratic realizations of evolutions. Preservation of quotients, orientation, and trajectory belongs to the transported arithmetic state. Both properties enter the construction, with different functions: one determines the identities of the representations; the other makes it possible to compose, reconstruct, and prolong the operations giving rise to them.

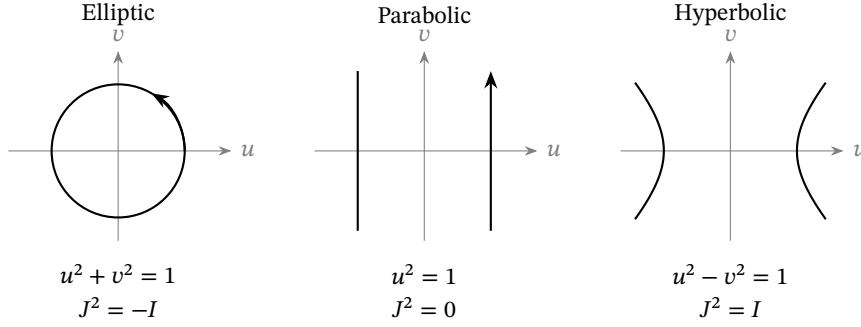


Figure 2: Quadratic realizations of TRIT. For $J_\tau = \begin{pmatrix} 0 & -\tau \\ 1 & 0 \end{pmatrix}$, the exponential preserves $u^2 + \tau v^2$. The parabolic degeneration preserves u and transports v linearly. The three diagrams represent orbits of quadratic forms; their geometric type corresponds to the algebraic relation of the generator.

2.3 TPK: topological phase kernel

The dynamics uses the phase selector

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Its value determines which evaluation acts. Cardinal transport displaces the corresponding cursor; the record preserves the previous value, the new quotient, the orientation, and the transition. Selection and transport are therefore distinct functions: a ternary sign is not a displacement on the torus.

We denote by $\hat{X}$ the space of arithmetic states with trajectory memory. An element contains the positions and directions of the two cursors, the active mode, the TRIT signature, the phase, the residues and quotients of both evaluations, the carries, the constructed prefix, and the boundary data. When a refinement cylinder is introduced, it forms part of the state. This notation avoids replacing the object with a reduced version that retains only the visible symbol.

Definition 2.11 (TPK update). Let Sel_t be mode selection by M_{ph} , Tra_t the lifted cardinal transport, and Mem_t the record update. The transition is

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (2.12)$$

Each application preserves the fields it does not modify and updates the remaining fields through Euclidean division and trajectory concatenation.

The notation (2.12) expresses the order of composition; it does not replace the effective rules. The next section fully specifies its finite two-cursor realization. Prolongation will then be obtained not by erasing its memory and repeating the first emission, but by transporting the prefixes and their compatible boundaries. Thus APP supplies the evaluations, TRIT distinguishes their regimes, and TPK composes their evolution.

2.3.1 Dynamic state, observables, and dependence of the operations

The Topological Phase Kernel organizes the evolution of the two APP sheets with the orientation determined by TRIT. Its definition comprises the state on which it acts, the transition rules, and the relations that make prolongations compatible. Equation (2.12) acquires operational content by specifying which coordinates each of its factors receives and modifies. Position belongs to the toroidal support; phase belongs to the calendar; orientation determines transport; quotients record the arithmetic evaluation before its reduction. These coordinates are related through the transition and preserve different mathematical functions.

The observable projection of a state is

$$q = (x_+, d_+, x_-, d_-, \theta) \in \mathcal{Q}_{\text{obs}} := (X_9 \times \mathcal{D})^2 \times \Theta_{108}, \quad \mathcal{D} = \{N, E, S, O\}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z}. \quad (2.13)$$

The subscripts distinguish the additive and multiplicative cursors. The fibers of complete states over q preserve the ordered trajectory, the integer evaluations of both sheets, their residues and quotients, and the orientation and boundary registers. A precision prolongation adds the prefix and the compatible cylinder. Thus two states with the same projection q may preserve different registers.

The dependence of the operations has three levels. At the local level, the selector activates a sheet, transport modifies its position, and the update calculates the new arithmetic coordinates. At the trajectory level, these transitions are composed and preserve the registers of earlier steps. At the refinement level, extension relations determine which continuation preserves the prefix, signature, and boundary of its predecessor. The finite calendar describes the phase of these operations; holonomy describes their composition on the history.

2.3.2 Trit selection and lifted cardinal transport

Let $\bar{\theta} \in \{0, \dots, 107\}$ be the calendar representative. The operational phase and dodecaphasic sector are

$$r(\theta) = 1 + (\bar{\theta} \bmod 9), \quad \beta(\theta) = \left[\left[\frac{\bar{\theta}}{9} \right] \right]_{12}. \quad (2.14)$$

The selector already defined can be written $\varepsilon_9 = M_{\text{ph}}$. Its value vector, in the ordered chart of D_9 , is

$$m_{\text{ph}} = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The function $M_{\text{ph}} : D_9 \rightarrow \{-1, 0, +1\}$ and this vector are two presentations of the same selector, with their respective domains.

Define the activation indicators

$$a_+(\theta) = \mathbf{1}_{\{1,4,7\}}(r(\theta)), \quad a_-(\theta) = \mathbf{1}_{\{2,5,8\}}(r(\theta)).$$

Cursor transport is then

$$x'_+ = x_+ + a_+(\theta)v_{d_+}, \quad x'_- = x_- + a_-(\theta)v_{d_-} \quad \text{in } X_9. \quad (2.15)$$

In phases 3, 6, 9, both positions remain unchanged; the neutral event is part of the chronology and its register. Cardinal directions determine the vectors v_d , while the phase signature determines which of those vectors is applied.

The integer lift preserves boundary crossing. If $s : X_9 \rightarrow D_9^2$ chooses positive representatives and $x' = x + av_d$, with $a \in \{0, 1\}$, define

$$b_d(x, a) = \frac{s(x) + av_d - s(x')}{9} \in \mathbb{Z}^2. \quad (2.16)$$

Divisibility by 9 follows from equality in X_9 . If $z \in \mathbb{Z}^2$ records earlier crossings, the update is $z' = z + b_d(x, a)$. Thus

$$s(x') + 9z' = s(x) + 9z + av_d.$$

The integer transport datum remains reconstructible after reducing the position on the torus.

Proposition 2.12 (Preservation of lifted displacement). *For a cardinal trajectory of n steps, whose activation indicators are a_t , we have*

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t v_{d_t}.$$

The same equality is preserved when two compatible trajectories are concatenated.

Proof. The one-step identity is (2.16). Summing it over the n steps cancels the intermediate coordinates. Dividing the summation interval into two segments produces exactly the concatenation law. For a closed trajectory, the difference $z_n - z_0$ is the winding number defined earlier. $\square$

The same principle governs APP evaluations. If a sheet takes the integer value N_t , its register simultaneously preserves $N_t = R_t + 9q_t$. The difference between two steps satisfies

$$N_{t+1} - N_t = (R_{t+1} - R_t) + 9(q_{t+1} - q_t).$$

Spatial-crossing memory z_t and the arithmetic quotient q_t have different definitions; their joint preservation maintains the geometric and arithmetic provenance of each transition.

2.3.3 Two-cursor chronology and composite returns

The cardinal involution j interchanges N with S and E with O . After (2.15) is applied, both directions are reversed whenever a block of 27 steps ends. Writing

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete rule on the observable projection is

$$F_{\text{obs}}(q) = (x'_+, j^{h(\theta)} d_+, x'_\times, j^{h(\theta)} d_\times, \theta + 1). \quad (2.17)$$

The readout forming the emission uses the positions before transport, according to the explicit recurrence in Section 4. The observable update and emission are therefore composed in a fixed order.

Proposition 2.13 (Reversibility and period of the observable calendar). *The map F_{obs} is a permutation of $\mathcal{Q}_{\text{obs}}$ and satisfies $F_{\text{obs}}^{108} = I$. In a block of 54 steps, positions and directions are restored; the coordinate in Θ_{108} advances by 54.*

Proof. Given the final state, one first recovers the previous phase by subtracting one in Θ_{108} . That phase determines whether the directions were reversed and which cursor moved. Applying the corresponding involution and subtracting the cardinal vector uniquely recovers the initial state.

Each interval of 27 steps contains nine activations of each cursor. The next block preserves the activations and reverses the directions, so the integer displacements cancel in 54 steps. The rule has period 54 in its position and orientation increments; the same cancellation holds for any calendar origin. After 108 steps, θ is also restored. $\square$

We reserve

$$\mathcal{K}_{27} = F_{\text{obs}}^{27} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}$$

for the iterate of 27 transitions. Its fourth iterate is the identity on the observable state defined in (2.13). Projecting history onto the calendar makes this finite check possible. The trajectory register, its evaluations, and new prefixes are preserved in the dynamic state on which $\mathcal{U}_t$ acts; the observable equality does not reset them.

2.3.4 Affine memory transport and state composition

The observable projection admits fibers of arithmetic and geometric data. Over a configuration q , write a state in the form

$$x = (q, o, h, c, \sigma, C, b, \lambda).$$

Here o preserves orientation and sheet; h , residues and quotients; c , additive transport memory; σ , the boundary signature; C , the precision cylinder; b , its boundary label; and λ , the recorded trajectory. The signature is determined by a map

$$\text{Sig}_q : H_q \times C_q \times \text{Hist}_q \longrightarrow S_q, \quad \sigma = \text{Sig}_q(h, c, \lambda).$$

The symbols H_q , C_q , Hist_q and S_q denote, respectively, the sets of residue–quotient data, the abelian group of transport memory, the recorded trajectories, and the boundary signatures. Realizations of this signature are specified with their components in the regional and terminal constructions. This dependence prevents treating the signature as a free parameter once the data producing it are fixed.

The elementary residue-and-quotient chart is explicit. For $n \in \mathbb{Z}$, let

$$r = 1 + ((n - 1) \bmod 9), \quad u = \frac{n - r}{9}.$$

Then $n = r + 9u$, with $r \in D_9$ and $u \in \mathbb{Z}$. Uniqueness follows because two representatives in D_9 congruent modulo 9 are equal. This chart provides local integer reconstruction. The precision towers additionally incorporate truncation maps and the cylinders developed in 4 and 5.

Let $\gamma : q \rightarrow q'$ be an observable trajectory. Fiber transport is expressed by maps

$$H(\gamma) : H_q \rightarrow H_{q'}, \quad C(\gamma) : C_q \rightarrow C_{q'}, \quad \text{Hist}(\gamma) : \text{Hist}_q \rightarrow \text{Hist}_{q'}.$$

The maps respect identities and composition, and $C(\gamma)$ is an abelian-group homomorphism. The increment $k_{\text{tr}}(\gamma) \in C_{q'}$ satisfies

$$\begin{aligned} k_{\text{tr}}(\text{id}_q) &= 0, \\ k_{\text{tr}}(\gamma_2 \circ \gamma_1) &= k_{\text{tr}}(\gamma_2) + C(\gamma_2)k_{\text{tr}}(\gamma_1). \end{aligned} \tag{2.18}$$

The symbol k_{tr} here denotes the transport cocycle; the rational coordinate κ of the register K retains its notation in the chapter on dodecaphasic closure.

The transportable coordinates are updated by

$$\begin{aligned} h' &= H(\gamma)h, & c' &= C(\gamma)c + k_{\text{tr}}(\gamma), \\ \lambda' &= \gamma \circ \lambda, & \sigma' &= \text{Sig}_{q'}(h', c', \lambda'). \end{aligned} \tag{2.19}$$

The second equation is an affine action: the linear term transports the previous memory, and the cocycle adds the increment produced by the trajectory. In the spatial-crossing realization for each cursor separately, $C(\gamma) = I$, and $k_{\text{tr}}(\gamma)$ is the sum of the vectors in (2.16). This concrete instance links the abstract law to the cardinal operations already defined.

Proposition 2.14 (Compatibility of the update with concatenation). *Updating (2.19) by γ_1 , followed by updating by γ_2 , coincides with updating by $\gamma_2 \circ \gamma_1$.*

Proof. For the affine coordinate, two updates give

$$c'' = C(\gamma_2)C(\gamma_1)c + C(\gamma_2)k_{\text{tr}}(\gamma_1) + k_{\text{tr}}(\gamma_2).$$

Functoriality of C and (2.18) turn this expression into $C(\gamma_2 \circ \gamma_1)c + k_{\text{tr}}(\gamma_2 \circ \gamma_1)$. The assertion for h follows from composition of H , and that for λ from associativity of trajectories. The final signature agrees because it is evaluated on the same three reconstructed coordinates. $\square$

The cylinder and its boundary complete this law through the relations Ext_γ , defined in 2.3.7. Their elements determine which of the preceding updates preserve an admissible prolongation. States, together with the triples (γ, x, x') admitted by these relations, form a category over observable trajectories:

$$(\gamma_2, x', x'') \circ (\gamma_1, x, x') = (\gamma_2 \circ \gamma_1, x, x'').$$

The identity of x is (id_q, x, x) . Composition preserves both arithmetic transport and boundary prolongability. This is the structure in which returns with memory and successive refinements take place.

2.3.5 Dodecaphasic register, phase, and accumulated memory

The 108 transitions are distributed over the ordered windows

$$E_m = \{9m - 8, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The register preserves the phase origin, orientation, and data produced during each window. Its map is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12},$$

where τ_m is the window subroute, σ_m its orientation, κ_m the integer boundary increment, and Λ_m the local incidence register. We adopt the chart to be developed in 8.2; its window indices correspond to $m = \beta + 1$. The domain comprises compatible TPK histories, and the codomain their oriented temporal registers. The composition of segments and their increments is governed by (2.19). The group C_{12} , by contrast, describes translation of the window index; Θ_{108} preserves the step calendar. This distinction makes it possible to specify what is transformed and what is observed when a cycle is completed.

In coordinates $\theta = 9b + r$, with $0 \leq r < 9$, advancement by 9 steps preserves r and increases b by one. Phase addition introduces the quotient

$$c_0(r, r') = \left\lfloor \frac{r + r'}{9} \right\rfloor,$$

whose cocycle identity ensures associativity of composition. The finite calendar retains $[b]_{12}$; a winding register retains $b \in \mathbb{Z}$, or $b \in \mathbb{N}_0$ for a history with fixed origin. Section 4 proves the nonsplit exact sequence (4.5). Its role is to describe the relation between these coordinates before they are used in constructing the signed state.

The signed register receives phase and memory data through the composition developed in Section 8:

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \quad (2.20)$$

Its domain is the terminal state with memory; its output belongs to $\mathbb{Z}^{12}$. The reversible transformation between that signed state and K , as well as the rational readout of K , are proved in the corresponding chapter. The registration operations precede those transformations. Reconstructing a register from its cyclic differences verifies recoverability; its dynamic provenance must be preserved through the events that produced it.

For the encoded trace (d_t) preserved by the register, the event readout counts the codes $\{2, 4, 6, 8\}$ in each window with orientation $(+, +, +, -, -, -, +, +, +, -, -, -)$. These codes and the cardinal symbols of $\mathcal{D}$ are different coordinates of the register. Chapter 8 develops this readout, the integral decomposition $1 + 5 + 6$, the congruences characterizing its image, and its exact inverse. The inversion proof establishes which data that chart preserves; the definition of $\mathcal{R}_{12}$ specifies the oriented history on which it acts.

2.3.6 Phase incidence and internal construction of the channels

The selector determines three subsets of D_9 :

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}.$$

Let $H_{\text{act}} = H_+ \sqcup H_-$ be the set of active phases, and let $O_{\text{ph}} = D_9 \setminus \{9\}$ be the chart preceding the marked return. Denote by $P_2(H_{\text{act}})$ the set of unordered pairs of active phases. Internal incidence is given by

$$\mathcal{F}_6^{\text{ph}} = H_{\text{act}} \times P_2(H_{\text{act}}), \quad \mathcal{F}_8^{\text{ph}} = O_{\text{ph}} \times P_2(H_{\text{act}}), \quad \Theta_{\text{ph}} = D_9 \times P_2(H_{\text{act}}). \quad (2.21)$$

The indices describe the number of positions in the first component. The construction uses the phases and calendar origin already fixed; subsequent exceptional realizations will transport these incidences to their own domains.

Proposition 2.15 (Cardinalities and oriented phase incidence). *The sets in (2.21) have cardinalities 90, 120, 135, respectively. If*

$$\partial_{\text{ph}} = \Theta_{\text{ph}} \times \{-1, +1\}, \quad E_{\text{ph}} = \partial_{\text{ph}} \sqcup \{\infty\}, \quad L_{\text{ph}} = H_0 \times P_2(H_{\text{act}}),$$

then

$$|\partial_{\text{ph}}| = 270, \quad |E_{\text{ph}}| = 271, \quad |L_{\text{ph}}| = 45, \quad |\partial_{\text{ph}} \sqcup L_{\text{ph}}| = 315.$$

Simultaneously translating the phase origin and marked return preserves all these cardinalities and their inclusions.

Proof. There are six active phases, so $|P_2(H_{\text{act}})| = \binom{6}{2} = 15$. Products by 6, 8, 9 give 90, 120, 135. Orientation doubles 135; the return mark adds one element; the three neutral phases give $3 \cdot 15 = 45$. A calendar translation transports each factor by a bijection and carries the marked point to its image. Products and disjoint unions thus preserve incidence. $\square$

The cardinality 271 here has a realization through phase incidence. The cubic completion in Section 4 has its own construction of the shell $1000 - 729$. The correspondence between both realizations must preserve the maps and markings that identify them, in addition to cardinality. Likewise, the selector ε_9 , the incidence (90, 120), and the extractor $\varepsilon_{108}^{90,120}$ preserve the domains and operations just distinguished.

2.3.7 Trajectory reversal and conjugate prolongations

Bidirectionality begins with an operation on trajectories, before scalar values are assigned to them. For $\gamma = (x_0; d_1, \dots, d_n)$, with endpoint x_n , define

$$\text{rev } \gamma = (x_n; jd_n, \dots, jd_1).$$

The origin of the reverse trajectory is the endpoint of the original. Composition satisfies

$$\text{rev}^2 \gamma = \gamma, \quad \text{rev}(\gamma_2 \circ \gamma_1) = \text{rev } \gamma_1 \circ \text{rev } \gamma_2,$$

and the integer displacement changes sign. These identities are obtained by reversing the order of steps and replacing each cardinal vector by its opposite. For a loop it follows that $\text{wind}(\text{rev } \gamma) = -\text{wind}(\gamma)$.

On registers, reversal also requires transporting the order of their components and the orientation of their events. Route reversal and signature conjugation are distinguishable operations: their composition is specified when applied to a concrete register. In particular, the regional involution interchanging the closure and propagation components preserves the self-scaling axis; its formula is presented in Section 5.13. This regional operation acts on a different domain from reversal of a cardinal trajectory.

Extensions over a route $r : q \rightarrow q'$ are expressed through relations between fibers of compatible states. If $\mathcal{F}_q$ denotes the fiber of complete states over q , one requires

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \Delta_{\mathcal{F}_q} \subseteq \text{Ext}_{\text{id}_q},$$

$$\text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

Their composition preserves the evaluations of both sheets, orientation, quotients, prefix, and boundary. Route reversal does not by itself imply that every extension relation is a bijection. Conjugate prolongation is defined on the domain where transport of these data satisfies the corresponding compatibilities.

2.3.8 Nonadic holonomy and continuity of refinement

One phase winding composes transitions on the state with memory:

$$\Gamma_{9,k} = \mathcal{U}_{k+8} \circ \cdots \circ \mathcal{U}_k. \quad (2.22)$$

On an individualized history, this functional composition realizes the holonomy relation; on the complete domain one retains $\Gamma_{9,k} \subseteq \hat{X}_k \times \hat{X}_{k+9}$, with the admissible prolongations developed below. The cyclic base has nine phases, whereas the fiber preserves the trajectory, quotients, and boundary. Monodromy is the action of one winding of that base on the fibers; holonomy $\Gamma_{9,k}$ expresses its realization by the effective TPK transitions.

If g observes the phase and M the number of retained windings,

$$g(\Gamma_{9,k}x) = g(x), \quad M(\Gamma_{9,k}x) = M(x) + 1.$$

The first identity expresses phase closure, and the second identifies the new temporal depth. Coinductive prolongation additionally preserves its cylinders and prefixes. Restrictions between depths compose an inverse system; a compatible section gathers its predecessors without replacing them by the last observation. The development of this arithmetic of histories and its unit occupies 4.8–4.10; the construction of regions and their readouts is carried out in Section 5.

Phase transfer on emissions already constructed has a different domain. Let $\mathcal{A}_+$ and $\mathcal{A}_\times$ be the alphabets of additive and multiplicative signatures, with cardinalities 18 and 26. Their product identifies the catalogue $\mathcal{U}_6$. For $f : \mathcal{A}_+ \times \mathcal{A}_\times \rightarrow \mathbb{R}$, the sheet averages are

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s, e').$$

These emissions and their cardinalities are constructed through the recurrence in Section 4. Once they have been obtained, the transfer operator is defined on $\mathcal{U}_6 \times C_9$ by

$$P_+ = E_+, \quad P_- = E_\times, \quad P_0 = I, \quad \mathcal{K}_{\text{ph}}((u, r), (v, r+1)) = P_{M_{\text{ph}}(r)}(u, v),$$

with zero weight for the remaining phase transitions. The index r uses the positive representative of the cyclic class. The phase order $(+1, -1, 0)^3$ then produces the renewal

$$\mathcal{K}_{\text{ph}}^9|_r = E_+ E_\times.$$

Indeed, both averages are idempotent, commute, and average different factors; their composition assigns weight $1/(18 \cdot 26) = 1/468$ to each emission. This equality characterizes a subsequent transfer observable. Transport of histories remains given by (2.22), with its memory and boundary.

The resulting hierarchy establishes the role of the following chapters. Finite emission constructs the domain and regions; extension relations preserve their prefixes; the nonadic connection prolongs them; the dodecaphasic register determines the data entering K and the closure of α . The numerical and incidence realizations receive that prior structure. This order makes it possible to present each consequence in its chapter while preserving, from the formal kernel, the objects that make it possible.

2.4 Accumulated memory and transport resolvents

Calendar return does not eliminate the increments produced during the traversal. To express this difference in operator terms, an accumulative register of trajectory events is constructed first, followed by its generating series. Memory variables will be reduced on that transport, while preserving the source and output readout as well.

2.4.1 Oriented events and register update

Let (d_t) be the sequence of integer direction codes preserved by a TPK trajectory. The code d_t and the cursor's cardinal direction are different coordinates; no identification between their alphabets is assumed. Number the windows from zero:

$$I_m = \{9m + 1, \dots, 9m + 9\}, \quad m \geq 0.$$

In the first cycle, I_m is window E_{m+1} of the dodecaphasic register. Its orientation is $\sigma = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1)$. For a prolonged history, $\sigma_m \in \{-1, 1\}$ is the sign preserved in the corresponding window. The signed event is

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9. \quad (2.23)$$

Each summand is incorporated when its transition occurs. This definition counts the events in the trace; it does not choose directions through a value to be obtained later.

Let $\mathbf{e}_0, \dots, \mathbf{e}_{11}$ be the basis of $\mathbb{Z}^{12}$, and $S\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$. Write $R = S^{-1}$. The accumulated register in fixed coordinates satisfies

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}.$$

A prefix with no accumulated history starts at $z_0 = 0$; at another origin, z_0 preserves the previous register. The frame accompanying the calendar is $y_m = S^{-m} z_m$.

Proposition 2.16 (Update and return with memory). *The register in the moving frame satisfies*

$$y_{m+1} = R y_m + \eta_m, \quad \eta_m = a_m R \mathbf{e}_0. \quad (2.24)$$

For every $n \geq 1$,

$$y_n = R^n y_0 + \sum_{j=0}^{n-1} R^{n-1-j} \eta_j. \quad (2.25)$$

In particular,

$$y_{12} = y_0 + \sum_{j=0}^{11} a_j \mathbf{e}_j. \quad (2.26)$$

Proof. Multiplying the update of z_m by $S^{-(m+1)}$ gives $y_{m+1} = S^{-1} y_m + a_m S^{-1} \mathbf{e}_0$, because $S^{-m} \mathbf{e}_{m \bmod 12} = \mathbf{e}_0$. Formula (2.25) follows by induction. For $n = 12$, $R^{12} = I$ and $R^{11-j} \eta_j = a_j R^{12-j} \mathbf{e}_0 = a_j \mathbf{e}_j$, proving (2.26). $\square$

Thus twelve windows restore the frame and preserve the event vector. The count alone does not recover the order of all steps within each window; that order remains in the recorded trajectory.

To specify the balance, define

$$D_3 = I - S^3, \quad D_4 = I - S^4, \quad B = D_3^T D_3 + D_4^T D_4,$$

$$\mathcal{M}(y) = \frac{1}{2} y^T B y, \quad q(y) = \sum_{j=0}^{11} y_j.$$

The first is a quadratic form of the register; the second records the sum of its coordinates. They are not identified here with physical energy or charge.

Proposition 2.17 (Balance produced by the event). *The update (2.24) satisfies*

$$\begin{aligned}\mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) &= a_m(B y_m)_0 + 2a_m^2, \\ q(y_{m+1}) - q(y_m) &= a_m.\end{aligned}\tag{2.27}$$

Proof. Expanding the products gives $B = 4I - S^3 - S^{-3} - S^4 - S^{-4}$; hence $R^\top B R = B$ and $B_{00} = 4$. Since $y_{m+1} = R(y_m + a_m \mathbf{e}_0)$, the quadratic difference is $a_m \mathbf{e}_0^\top B y_m + \frac{1}{2} a_m^2 B_{00}$. The sum of coordinates is invariant under the permutation R , giving the second equality. $\square$

2.4.2 Generating series and uniform control of memory

The series preserves all increments, not only their sum at the end of a cycle. With a formal variable u , let

$$Y(u) = \sum_{m \geq 0} y_m u^m, \quad H_\eta(u) = \sum_{m \geq 0} \eta_m u^m.$$

Proposition 2.18 (Resolvent and tail bound). *The formal identity*

$$Y(u) = (I - uR)^{-1}(y_0 + uH_\eta(u)).\tag{2.28}$$

holds. Both series converge for $|u| < 1$. If $\ell : \mathbb{R}^{12} \rightarrow \mathbb{R}$ is linear, $L = \|\ell\|_{1 \rightarrow \mathbb{R}}$, $M_0 = \|y_0\|_1$, and $0 < r < 1$, then, for every integer $N \geq 0$ and $|u| \leq r$,

$$\left| \sum_{m > N} \ell(y_m) u^m \right| \leq L r^{N+1} \left(\frac{M_0 + 9(N+1)}{1-r} + \frac{9r}{(1-r)^2} \right).\tag{2.29}$$

The derivative series converges uniformly on each disk $|u| \leq r < 1$.

Proof. Multiplying (2.24) by u^{m+1} and summing gives $Y - y_0 = uRY + uH_\eta$. The constant term of $I - uR$ is invertible, so its formal inverse is $\sum_{k \geq 0} u^k R^k$. Moreover, R preserves the ℓ^1 norm and $\|\eta_m\|_1 \leq 9$, whence $\|y_m\|_1 \leq M_0 + 9m$. For the tail, sum the majorants $L(M_0 + 9m)r^m$, using

$$\sum_{m > N} r^m = \frac{r^{N+1}}{1-r}, \quad \sum_{m > N} m r^m = r^{N+1} \left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2} \right).$$

Finally, $\sum_{m \geq 1} Lm(M_0 + 9m)r^{m-1} < \infty$ majorizes the derivative series and proves uniform convergence. $\square$

The variable u counts windows of nine transitions. Identifying it with the variable of another readout requires preserving that clock and the map relating the two readouts.

2.4.3 Elimination of memory: operator, source, and readout

On a domain where the update is fixed as $U : X \rightarrow X$, the free vector space $V = \mathbb{Q}^{(X)}$ on the states admits the operator $T\delta_x = \delta_{U(x)}$. The clock is included in x if the transition depends on it. This linear extension preserves distinct states even when they share an observable phase.

Consider a decomposition $V = V_0 \oplus V_1$, where V_1 is the auxiliary sector to be eliminated, and write

$$T = \begin{pmatrix} A & B_1 \\ C & D \end{pmatrix}, \quad v = \begin{pmatrix} v_0 \\ v_1 \end{pmatrix}, \quad \ell = (\ell_0, \ell_1).$$

Here $A : V_0 \rightarrow V_0$, $B_1 : V_1 \rightarrow V_0$, $C : V_0 \rightarrow V_1$, and $D : V_1 \rightarrow V_1$. The following identities are formal; they also hold wherever the resolvents converge.

Proposition 2.19 (Schur reduction with source and readout). *Let*

$$D_u = (I - uD)^{-1}, \quad \mathcal{F}(u) = I - uA - u^2 B_1 D_u C.$$

The retained component of $(I - uT)^{-1}v$ is

$$w_0 = \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1), \quad w_1 = D_u(v_1 + uCw_0). \quad (2.30)$$

Its complete readout satisfies

$$\begin{aligned} \ell(I - uT)^{-1}v &= (\ell_0 + u\ell_1 D_u C)\mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1) \\ &\quad + \ell_1 D_u v_1. \end{aligned} \quad (2.31)$$

Proof. The two equations of $(I - uT)w = v$ are $w_0 = v_0 + uAw_0 + uB_1 w_1$ and $w_1 = v_1 + uCw_0 + uDw_1$. Solving the second and substituting into the first gives (2.30); applying ℓ_0 and ℓ_1 to both components gives (2.31). All formal inverses exist because their constant terms are identities. $\square$

The term $u^2 B_1 D_u C = \sum_{k \geq 0} u^{k+2} B_1 D^k C$ describes an exit into memory, k steps within it, and a return. By contrast, $u\ell_1 D_u C$ reads memory before that return. Thus reducing only the operator does not necessarily preserve the observation: source and readout must also be transformed.

The difference is visible in the cycle already constructed. If P retains $\mathbf{e}_0$, then $PRP = 0$, but

$$\langle \mathbf{e}_0, (I - uR)^{-1} \mathbf{e}_0 \rangle = \frac{1}{1 - u^{12}}, \quad \langle \mathbf{e}_{11}, (I - uR)^{-1} \mathbf{e}_0 \rangle = \frac{u}{1 - u^{12}}. \quad (2.32)$$

Indeed, $R^n \mathbf{e}_0 = \mathbf{e}_{-n \bmod 12}$: the two series collect, respectively, $n \equiv 0$ and $n \equiv 1 \pmod{12}$. The resolvent of the compression is I , whereas compression of the resolvent preserves the omitted returns. This example corresponds to the twelve-window register, not to the entire TPK state.

2.4.4 Composition of segments and coherence of reduction

Let there be three composable affine transports $F_i(y) = R_i y + b_i$, with compatible domains and codomains. The memory produced by the complete traversal is

$$F_3 F_2 F_1(y) = R_3 R_2 R_1 y + b_3 + R_3 b_2 + R_3 R_2 b_1. \quad (2.33)$$

The proof is successive substitution of the three affine formulas. Grouping either the first two segments or the last two first produces the same expression; the factors transport each increment from the point where it originated. In (2.24), $R_i = R$ and b_i is the event in its window.

Eliminating several layers preserves a similar coherence. To make it explicit, now decompose $V = V_0 \oplus V_1 \oplus V_2$ and write

$$T = \begin{pmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{pmatrix}, \quad Q_2 = (I - uD_{22})^{-1}.$$

Eliminating the third component leaves the four blocks

$$\begin{aligned} A' &= A + uB_2 Q_2 C_2, & B'_1 &= B_1 + uB_2 Q_2 D_{21}, \\ C'_1 &= C_1 + uD_{12} Q_2 C_2, & D'_{11} &= D_{11} + uD_{12} Q_2 D_{21}. \end{aligned} \quad (2.34)$$

Proposition 2.20 (Transitivity of elimination). *If $D_* = (D_{ij})_{i,j=1}^2$, the complement obtained in two stages coincides with that obtained by eliminating both memories together:*

$$\begin{aligned} I - uA' - u^2 B'_1 (I - uD'_{11})^{-1} C'_1 &= I - uA \\ &\quad - u^2 (B_1 \ B_2) (I - uD_*)^{-1} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \end{aligned} \quad (2.35)$$

Proof. Solving the third equation of $(I - uT)w = v$ and substituting into the first two produces (2.34). Subsequently eliminating the second component gives the left-hand side of (2.35). Solving both memory equations simultaneously gives the right-hand side, by Proposition 2.19. The formal solution is unique because $I - uT$ has constant term I . In both procedures, source and readout are also transformed according to (2.30)–(2.31). $\square$

The balance (2.27) describes register increments. The preceding identities allow them to be transported when memory is reused in dodecaphasic closure; they do not by themselves determine additional analytic coefficients.

3 Topological Phase Kernel and generation of the seed catalogue

The two APP evaluations acquire generative efficacy once their activation, evaluation position, positional transport and the information preserved by each transition have been specified. The *Topological Phase Kernel*, abbreviated TPK, combines these operations into a dynamics on enriched states. Its definition comprises selection, transport, arithmetic updating and preservation of the record. The finite construction developed in this section provides the seeds on which the regional lifts and coinductive prolongation subsequently act.

The principal result of this stage is a generated and classified catalogue: the initial conditions of the two cursors produce integer emissions; their ternary reading determines oriented words; and the dihedral action organises these words into orbits. Each step preserves its preimages and the relations that permit recovery of the preceding description. The chain of cardinalities

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729, \quad 243 \longrightarrow 43,$$

abbreviates distinct objects. Each object is constructed and each count justified below, before any of them is used as data for regional selection.

3.1 Phase selection, transport and enriched state

3.1.1 Tritic selector of operative phase

The nonadic phase takes its values in $D_9 = \{1, \dots, 9\}$. The tritic selector is the function

$$M_{\text{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \quad M_{\text{ph}}(g) = \begin{cases} +1, & g \in \{1, 4, 7\}, \\ -1, & g \in \{2, 5, 8\}, \\ 0, & g \in \{3, 6, 9\}. \end{cases} \quad (3.1)$$

In the emitter realisation, the value $+1$ activates the additive evaluation, -1 activates the multiplicative evaluation, and 0 records the neutral event. The cardinal orientation of the cursor is another state coordinate. This distinction specifies both the operation being read and the direction in which its evaluation point is transported.

For transitions indexed by $t \geq 1$, the phase preceding the reading is

$$g_t = 1 + [(t - 1)]_9, \quad \epsilon_t = M_{\text{ph}}(g_t), \quad (3.2)$$

where $[n]_b$ denotes the representative of n modulo b in $\{0, \dots, b - 1\}$. A window of nine transitions contains the sequence $(+1, -1, 0, +1, -1, 0, +1, -1, 0)$.

3.1.2 Oriented cursors and cardinal transport

The indices $I_9 = \{0, \dots, 8\}$ are used for positions, and the positive labels $(a, b) = (i + 1, j + 1)$ for APP evaluations. A finite cursor is

$$c = (i, j; d) \in \mathcal{S} := I_9^2 \times \mathcal{D}, \quad \mathcal{D} = \{N, E, S, O\}.$$

The four directions act by

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The cardinal map corresponding to d is

$$T_d(i, j) = ([i + (v_d)_1]_9, [j + (v_d)_2]_9).$$

Thus M_{ph} and T_d have distinct domains, codomains and actions. Tritic activation selects one of the two cursors; cardinal transport changes its position in the direction already retained by that cursor.

The integer lift of the cursor is $\tilde{c} = (x, y; d) \in \mathbb{Z}^2 \times \mathcal{D}$, with finite position $([x]_9, [y]_9)$. Its winding counts are recovered from

$$x = 9[x/9] + [x]_9, \quad y = 9[y/9] + [y]_9. \quad (3.3)$$

These two divisions concern spatial transport. The division of an additive or multiplicative evaluation constitutes a distinct arithmetic record, which is preserved together with them.

3.1.3 Composition of the transition

The state of the finite realisation includes the two lifted cursors, their directions, the phase, the accumulators of the active window, the ordered list of blocks already emitted and the transition record. Each local record contains the preceding positions, the integer evaluations of both sheets, their residues and quotients, the active regime and the orientations. Prefix and boundary fields arising in a prolongation are incorporated into the same state with their compatibility rules.

The transition is ordered as

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (3.4)$$

Selection determines the regime and preserves the sample taken before any displacement. Transport applies the active movement and the cardinal reversal prescribed by the schedule. Updating incorporates that sample into the accumulators and the record; at the end of a window, it emits the corresponding block. The composition therefore uses a sample taken before transport, although its definitive storage occurs at the end of the transition.

3.2 Initial conditions and reversible enumeration

The additive and multiplicative sheets have independent cursors. The domain of initial conditions is the ordered product

$$\Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}| = 9^2 \cdot 4 = 324.$$

Consequently,

$$|\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (3.5)$$

The number 81 counts positions; 324 counts the oriented positions of one cursor; 104 976 counts ordered pairs of cursors. Choosing a position in APP retains only part of an initial condition of the emitter.

Fix the numbering

$$\text{ind}(N) = 0, \quad \text{ind}(E) = 1, \quad \text{ind}(S) = 2, \quad \text{ind}(O) = 3,$$

and define

$$\nu(i, j; d) = 4(9i + j) + \text{ind}(d). \quad (3.6)$$

Proposition 3.1 (Exact enumeration of the initial conditions). *The map ν is a bijection from S onto $\{0, \dots, 323\}$. The pair of indices is reversibly encoded by $324\nu_+ + \nu_\times \in \{0, \dots, 104\,975\}$.*

Proof. Division by four recovers

$$\text{ind}(d) = [\nu]_4, \quad j = [\lfloor \nu/4 \rfloor]_9, \quad i = \lfloor \nu/36 \rfloor.$$

The bounds on the domain ensure that these three quantities belong to their respective index sets and reconstruct ν through (3.6). For the pair, Euclidean division by 324 recovers the quotient ν_+ and the residue $\nu_\times$. $\square$

The enumeration traverses the complete domain before any regional grouping. The indices locate the preimages of an emission while keeping explicit the condition that produced each trajectory.

3.3 Sheet evaluation and arithmetic record

At a positive position (a, b) , the integer evaluations are $A_+(a, b) = a + b$ and $A_\times(a, b) = ab$. For $n > 0$, retain

$$\rho_9(n) = 1 + [n - 1]_9, \quad q_9(n) = \lfloor (n - 1)/9 \rfloor, \quad n = \rho_9(n) + 9q_9(n). \quad (3.7)$$

Additive activation incorporates $\rho_9(A_+)$ into the corresponding accumulator. Multiplicative activation uses an observable of the residue $\rho_9(A_\times)$, defined next.

3.3.1 Discrete exponent in the units modulo nine

The powers of 2 modulo 9 traverse

$$2^0, 2^1, \dots, 2^5 \equiv 1, 2, 4, 8, 7, 5 \pmod{9}, \quad 2^6 \equiv 1 \pmod{9}.$$

The discrete logarithm in this group of six units is represented by

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5. \quad (3.8)$$

The emission law extends the observable by $\ell_9(3) = \ell_9(6) = \ell_9(9) = 0$. To preserve its provenance, the recorded sample includes the original residue and the indicator of membership in the group of units. An observed value of zero can therefore originate from the unit 1 or from one of the three non-invertible residues, with its origin remaining recoverable.

3.3.2 Order of reading and movement

Evaluations are performed at the positions preceding the transition. If $\epsilon_t = +1$, the additive reading is accumulated and its cursor is displaced. If $\epsilon_t = -1$, the multiplicative discrete exponent is accumulated and the second cursor is displaced. If $\epsilon_t = 0$, the neutral counter is incremented and both positions are preserved.

At the end of steps 27 and 54, both directions are replaced by their opposites. The windows are concatenated with positions and orientations preserved. At the start of each window, only its three local accumulators are reset to zero; the preceding record remains available.

Proposition 3.2 (Inversion of the lifted transport). *Let $a, b \in \{0, 1\}$ be the activation and reversal indicators of a cursor. If opp interchanges $N \leftrightarrow S$ and $E \leftrightarrow O$, the transformation*

$$F_{a,b}(z, d) = (z + av_d, \text{opp}^b(d))$$

has inverse

$$F_{a,b}^{-1}(z', d') = (z' - av_{\text{opp}^b(d')}, \text{opp}^b(d')).$$

Spatial reduction modulo nine intertwines this transformation with the finite transport.

Proof. Applying opposition twice recovers the initial direction. Once that direction has been recovered, subtracting av_d reverses the displacement. For the spatial projection, it suffices to use $[z + av_d]_9 = [[z]_9 + av_d]_9$ componentwise. Directional reversal preserves this equality. $\square$

The phase determines the indicators used by the inverse. The spatial and arithmetic records can therefore be preserved jointly, including when the boundary of the representative I_g^2 is crossed.

3.4 Six windows and decimal block emission

3.4.1 Accumulation within a nonadic window

Window m , with $1 \leq m \leq 6$, comprises steps $9m - 8, \dots, 9m$. Let S_m be the sum of its active positive additive residues, E_m the sum of its active discrete exponents, and Z_m its number of neutral events.

Lemma 3.3 (Activation multiplicities). *Each window contains three additive activations, three multiplicative activations and three neutral activations. In particular, $Z_m = 3$.*

Proof. The phases of a window traverse exactly $1, \dots, 9$. Each of the three sets in (3.1) has cardinality three. $\square$

The six windows comprise 54 transitions and produce six blocks. This length belongs to the initial emitter defined here. Continuation in depth subsequently acts on the words and their compatible records.

3.4.2 Decimal encoding rule

The emission law specifies the digits

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \tag{3.9}$$

The weights 100, 10, 1 are the hundreds, tens and units positions. The subscript 10 denotes the decimal residue, and each block satisfies $0 \leq U_m \leq 999$. The emitted word is the ordered sequence $\mathbf{U} = (U_1, \dots, U_6)$, with a fixed width of three digits per block.

The coefficients 3, 5, 7 constitute the explicit weights of this emission law. Once the rule has been fixed, each digit is calculated from the accumulators, and the results of this section follow from that rule. The number 1 appearing in its reduced form has a further specific provenance: $7Z_m = 21 \equiv 1 \pmod{10}$.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, one obtains

$$G(s_m, e_m) = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10} = U_m. \tag{3.10}$$

Here the letter e_m denotes a component of the discrete-exponent signature. Its definition belongs to the finite emitter.

Proposition 3.4 (Recovery of the two signatures). *The map $G : \{0, \dots, 9\}^2 \rightarrow \{0, \dots, 999\}$ is injective. For a block $U = 100a + 10b + c$ in its image, the inverse is*

$$s = a, \quad e = [b - a]_{10}, \quad c = [3s + 5e + 1]_{10}.$$

Consequently, the word $\mathbf{U}$ determines the two six-component signatures $\mathbf{s}$ and $\mathbf{e}$.

Proof. The last two summands of (3.10) belong to $\{0, \dots, 99\}$, so the hundreds digit recovers s . The tens digit is $[s + e]_{10}$; subtracting s and reducing modulo ten recovers e . The units digit verifies the third congruence. The reconstruction applies independently to the six positions of the word. $\square$

The integer totals S_m, E_m are recovered from the recorded readings; the hundreds and tens digits recover their residues. For example, a total $S_m = 15$ produces $s_m = 5$. This distinction identifies precisely which information each level of encoding preserves.

3.4.3 Positional encoding of the six-block word

The six blocks admit a fixed-width integer encoding,

$$\mathcal{N}(\mathbf{U}) = \sum_{m=1}^6 U_m 1000^{6-m}, \quad 0 \leq \mathcal{N}(\mathbf{U}) < 1000^6. \quad (3.11)$$

The first block occupies the position of greatest weight, and the sixth the base-thousand units position. A block 058 occupies a complete position with integer value 58; its three-digit width belongs to the decimal presentation of that position.

Proposition 3.5 (Positional recovery of the emissions). *The encoding (3.11) determines the complete word by*

$$U_m = \left\lfloor \left\lfloor \frac{\mathcal{N}(\mathbf{U})}{1000^{6-m}} \right\rfloor \right\rfloor_{1000}, \quad 1 \leq m \leq 6.$$

Proof. On division by 1000^{6-m} , the preceding blocks yield integer multiples of 1000, block m yields U_m , and the contribution of the subsequent blocks belongs to $[0, 1)$. Taking the integer part eliminates the latter contribution, and the residue modulo 1000 eliminates the former. $\square$

This encoding preserves the six emissions. Its domain has ordered base-thousand positions, whereas the APP domain has spatial positions modulo nine. The composition retains both indices: m identifies an emission window, and (i, j) identifies the cell visited during a transition in that window.

3.5 Complete calculation of an emission beginning with 501

Fix the initial conditions

$$c_{+,0} = (0, 4; N), \quad c_{\times,0} = (2, 3; N), \quad \nu_+ = 16, \quad \nu_{\times} = 84.$$

The initial positive positions are $(1, 5)$ and $(3, 4)$. The preceding chronological index is $t = 0$, the first phase read is $g_1 = 1$, and the record is empty. These coordinates are the inputs to the calculation; the blocks are obtained by executing the recurrence.

Table 1: First window: preceding readings and active accumulation.

t	ϵ_t	position +	position $\times$	ΔS	ΔE	(S, E, Z)
1	+1	(1, 5)	(3, 4)	6	0	(6, 0, 0)
2	-1	(9, 5)	(3, 4)	0	0	(6, 0, 0)
3	0	(9, 5)	(2, 4)	0	0	(6, 0, 1)
4	+1	(9, 5)	(2, 4)	5	0	(11, 0, 1)
5	-1	(8, 5)	(2, 4)	0	3	(11, 3, 1)
6	0	(8, 5)	(1, 4)	0	0	(11, 3, 2)
7	+1	(8, 5)	(1, 4)	4	0	(15, 3, 2)
8	-1	(7, 5)	(1, 4)	0	2	(15, 5, 2)
9	0	(7, 5)	(9, 4)	0	0	(15, 5, 3)

Note. Positions are positive labels of the cells preceding movement. Neutral events increment only Z .

The three active additive evaluations are

$$1 + 5 = 6, \quad 9 + 5 = 14 = 5 + 9, \quad 8 + 5 = 13 = 4 + 9,$$

so $S_1 = 6 + 5 + 4 = 15$. The three active multiplicative evaluations are $3 \cdot 4 = 12$, $2 \cdot 4 = 8$ and $1 \cdot 4 = 4$. Their positive residues are 3, 8, 4, with observables 0, 3, 2; hence $E_1 = 5$. Finally, $Z_1 = 3$. The decimal rule gives

$$d_{1,1} = [15]_{10} = 5, \quad d_{1,2} = [15 + 5]_{10} = 0, \quad d_{1,3} = [45 + 25 + 21]_{10} = 1,$$

and one obtains $U_1 = 501$.

The second additive reading above has arithmetic quotient 1, whereas its lifted cursor is at $(-1, 4; N)$, with vertical winding count -1 . These are distinct data in the same record. Both participate in the complete recovery of the evaluation and its position.

Table 2: The six windows of initial condition (16, 84).

m	steps	S_m	E_m	Z_m	s_m	U_m	$[-U_m]_3$
1	1–9	15	5	3	5	501	0
2	10–18	6	5	3	6	614	1
3	19–27	24	5	3	4	498	0
4	28–36	21	5	3	1	169	2
5	37–45	12	5	3	2	272	1
6	46–54	12	5	3	2	272	1

The resulting word and its signatures are

$$\mathbf{U} = (501, 614, 498, 169, 272, 272),$$

$$\mathbf{s} = (5, 6, 4, 1, 2, 2),$$

$$\mathbf{e} = (5, 5, 5, 5, 5, 5).$$

Cardinal reversal after step 27 changes the trajectories of the final three windows. The repeated block 272 corresponds to two successive windows with their own positions and chronological record.

As a second case, the minimal initial condition $(\nu_+, \nu_-) = (0, 0)$ produces

$$\mathbf{U} = (252, 109, 252, 903, 850, 850), \quad \mathbf{s} = (2, 1, 2, 9, 8, 8), \quad \mathbf{e} = (3, 9, 3, 1, 7, 7).$$

In this case, both cursors return to their initial positions and orientations after 54 transitions. The nonadic phase completes six cycles, the chronological counter advances by 54 units, and the record contains 54 samples. The positional reading of the return and the recorded history describe distinct properties of the execution.

3.6 Factorisation of the finite census

The signatures $f_+(\nu) = \mathbf{s}$ and $f_-(\nu) = \mathbf{e}$ are calculated by executing the corresponding cursor through the six windows. The independence of positions and directions between the two cursors gives

$$T_{\text{em}}(\nu_+, \nu_-) = G^{(6)}(f_+(\nu_+), f_-(\nu_-)), \quad (3.12)$$

where $G^{(6)}$ applies (3.10) at each position. This factorisation first permits enumeration of $324 + 324$ trajectories and then combination of the distinct signatures, preserving the multiplicities of their preimages.

3.6.1 Images of the two signatures

The following tables include every signature. Its multiplicity is the number of initial conditions producing it; the minimal index locates one such condition. The complete preimage is defined by

$$f_{\sigma}^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_{\sigma}(\nu) = a\}.$$

The recurrence of the preceding sections determines every element of this set by a finite integer calculation.

Table 3: The eighteen additive signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
122564	18	17	564122	18	16
122898	18	24	645212	18	4
212645	18	5	654889	18	33
212988	18	0	889221	18	13
221456	18	29	889654	18	32
221889	18	12	898122	18	25
456221	18	28	898465	18	20
465898	18	21	988212	18	1
546988	18	9	988546	18	8

Table 4: The twenty-six multiplicative signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
000000	108	8	555555	24	84
177393	8	1	555717	8	14
177555	8	54	555771	8	18
177771	8	11	555933	8	24
339555	8	6	717555	8	12
339771	8	15	717717	8	21
339933	8	59	717933	8	19
393177	8	0	771177	8	9
393393	8	45	771339	8	13
393555	8	40	771555	8	16
555177	8	52	933339	8	57
555339	8	4	933555	8	26
555393	8	41	933717	8	17

Proposition 3.6 (Image and multiplicities of the emitter). *The image $\mathcal{U}_6 = \text{im } T_{\text{em}}$ contains 468 distinct words. Their preimages are distributed into 432 fibres of cardinality 144, 18 of cardinality 432 and 18 of cardinality 1944.*

Proof. Exhaustive evaluation of the algorithm in Section 3.9 on the indices $0, \dots, 323$ gives the two tables above. The first partition has 18 classes of cardinality 18; the second has 24 classes of cardinality 8, one of cardinality 24 and one of cardinality 108. The sums $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$ verify the total mass of both partitions. Proposition 3.4 makes the coupling of their images injective; there are therefore $18 \cdot 26 = 468$ emissions. By (3.12), the preimage of a pair of signatures is the product of their two preimages. This gives $18 \cdot 24 = 432$ products of cardinality $18 \cdot 8 = 144$, 18 of cardinality $18 \cdot 24 = 432$, and 18 of cardinality $18 \cdot 108 = 1944$. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The enumerative part is an exact finite check of the recurrence; the factorisation explains why this count covers every pair. $\square$

The emission in Example 3.5 has a fibre of cardinality $18 \cdot 24 = 432$. The example with indices $(0, 0)$ has a fibre of cardinality $18 \cdot 8 = 144$. In both cases, the emitted word recovers the signatures, whereas identification of a particular initial condition uses its index or its trajectory record.

3.7 Oriented ternary reading and preservation of fibres

Oriented reduction acts separately on each block:

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \quad \Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (3.13)$$

Its sign fixes an orientation of the catalogue. The balanced identification $0 \mapsto 0, 1 \mapsto +1, 2 \mapsto -1$ is then applied componentwise.

Modulo nine organises APP evaluations and positions; residues modulo ten form each digit; base thousand combines three decimal positions; and modulo three determines the ternary signature of each block. These four uses have defined objects and functions. In particular, Ψ reads six blocks and produces six trits; a positional conversion of the integer formed by eighteen digits would be a different operation.

For the two conditions already calculated, one obtains

$$\begin{aligned} \Psi(501, 614, 498, 169, 272, 272) &= (0, 1, 0, 2, 1, 1), \\ \Psi(252, 109, 252, 903, 850, 850) &= (0, 2, 0, 0, 2, 2). \end{aligned}$$

The decimal emitter is defined before this reduction. The function of Ψ is to produce the ternary object on which orbital symmetry and regional selectors act. This function explains its position in the genealogy: it connects an already calculated emission with a structural classification.

The image $\mathcal{W}_6 := \Psi(\mathcal{U}_6)$ contains exactly 243 words within the ambient set of $3^6 = 729$ words. Enumeration by the final algorithm gives the histogram

$$\begin{array}{c|cccccc} |\Psi^{-1}(w)| & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline \text{number of words } w & 132 & 66 & 18 & 3 & 6 & 18. \end{array} \quad (3.14)$$

The sums for the two readings are

$$132 + 66 + 18 + 3 + 6 + 18 = 243, \quad 132 + 2 \cdot 66 + 3 \cdot 18 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 18 = 468.$$

The first counts ternary words; the second recovers the number of emissions distributed among their fibres.

Example 3.7 (Two emissions with the same ternary signature). The emissions

$$(498, 501, 614, 252, 252, 109), \quad (870, 810, 923, 252, 252, 109)$$

are distinct and both have image $(0, 0, 1, 0, 0, 2)$ under Ψ . Each has 144 initial conditions. The ternary word preserves a reading class; the emissions and their preimages distinguish the data that converge in that class.

Preservation of fibres is expressed concretely by

$$\mathcal{F}_w = \{\mathbf{U} \in \mathcal{U}_6 : \Psi(\mathbf{U}) = w\}, \quad \Omega_{\mathbf{U}} = T_{\text{em}}^{-1}(\mathbf{U}).$$

The complete preimage of w in the initial domain is therefore the disjoint union of the sets $\Omega_{\mathbf{U}}$, with $\mathbf{U} \in \mathcal{F}_w$. The enriched record can retain the emission, its signatures and the initial condition together with its ternary reading. The exact inverse then concerns the recorded datum with its type; the isolated six-trit word retains the precise scope of (3.13).

3.8 Dihedral action and the forty-three orbits

Define the following permutations on six coordinates:

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4), \quad (3.15)$$

$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3). \quad (3.16)$$

The first rotates the two groups of three coordinates in conjugate directions; the second interchanges the groups. Their composition satisfies $r^3 = s^2 = 1$ and $srs = r^{-1}$. They therefore generate an action of the six-element dihedral group D_3 .

Proposition 3.8 (Equivariance and stability of the catalogue). *Reduction by Ψ commutes with the action of D_3 . The sets $\mathcal{U}_6$ and $\mathcal{W}_6$ are stable under its generators.*

Proof. The equality $\Psi(g\mathbf{U}) = g\Psi(\mathbf{U})$ holds at every coordinate because g permutes positions and Ψ applies the same reduction at each position. Stability of the catalogue is a further check: generating its 468 elements by (3.12), one verifies that $r\mathbf{U}$ and $s\mathbf{U}$ again belong to that set. The algorithm in Section 3.9 performs these membership checks. Equivariance then transfers stability to the ternary image. $\square$

Proposition 3.9 (Complete orbital decomposition). *The set $\mathcal{W}_6$ decomposes into 43 orbits: 38 of cardinality six and 5 of cardinality three.*

Proof. For each $w \in \mathcal{W}_6$, form

$$\mathcal{O}(w) = \{w, rw, r^2w, sw, srw, sr^2w\}.$$

Choose the lexicographically least word in this set as representative. Two words have the same representative exactly when they belong to the same orbit. The algorithm generates the list of representatives in Table 5; their cardinalities sum to $38 \cdot 6 + 5 \cdot 3 = 243$. Since every word in the catalogue participates in this operation and the catalogue is stable, the list covers the entire image. $\square$

Table 5: Minimal representatives and cardinalities of the forty-three orbits.

representative	cardinality	representative	cardinality	representative	cardinality
001002	6	002112	6	012222	6
001011	6	002211	6	021022	6
001012	6	002220	6	021121	6
001021	6	011021	6	021202	6
001022	6	011112	6	022022	3
001100	3	011120	6	022112	6
001101	6	011201	6	022121	6
001110	6	011211	6	022202	3
001112	6	011220	6	022211	6
001202	6	011222	6	022222	6
001210	6	012112	6	112112	3
001211	6	012121	6	112211	3
001220	6	012202	6	112222	6
001222	6	012210	6		
002012	6	012220	6		

The orbits inherit integer invariants from their fibres. For each word, $n(w) = |\mathcal{F}_w|$ counts emissions, whereas

$$M(w) = \sum_{\mathbf{U} \in \mathcal{F}_w} |\Omega_{\mathbf{U}}|$$

counts initial conditions. The symbol census $c(w) = (n_0, n_1, n_2)$ records the number of occurrences of each ternary class. Together with the orientation and the support of the trajectories, these data provide the predicates for subsequent regional selection. The classification thus preserves more information than the form of a representative word.

3.9 Self-contained enumeration and verification algorithm

The enumeration uses integers, finite lists and the operations already defined. The following procedure specifies its execution; σ takes the values $+$ and $\times$.

1. For each $v = 0, \dots, 323$, recover $(i, j; d)$ using the inverse of (3.6). Initialise an empty signature list.
2. For each window $m = 1, \dots, 6$, initialise the accumulator $A = 0$. Traverse $t = 9m - 8, \dots, 9m$, calculating ϵ_t by (3.2).
3. If $\sigma = +$ and $\epsilon_t = +1$, add $\rho_9(i + j + 2)$ to A and update $(i, j) \leftarrow T_d(i, j)$. If $\sigma = \times$ and $\epsilon_t = -1$, add $\ell_9(\rho_9((i + 1)(j + 1)))$ and perform the same transport.
4. At the end of $t = 27$ or $t = 54$, replace d by its opposite. At the end of the window, append $[A]_{10}$ to the signature list. Preserve the position and direction for the next window.
5. Group the 324 indices according to equality of their six-component lists. These two groupings give f_+ , $f_\times$ and all their preimages.
6. For each pair of distinct signatures, apply $G^{(6)}$. Assign to the emission the product multiplicity of the two preimages. Verify signature recovery in each block and the sum of multiplicities 104 976.
7. Apply Ψ to each emission and group by equality of words. Count the 243 values and the multiplicities in (3.14).
8. Apply r, s to each emission and verify membership in the catalogue. For each ternary word, form its six dihedral images, remove repetitions and record its lexicographic minimum. Count the 43 orbits and the two sizes in Table 5.

The execution terminates: the first stage evaluates 648 trajectories of 54 transitions; the second combines 468 pairs of signatures; the third groups a set of 243 words. An additional direct check can execute the 104 976 pairs and compare the result with the factorisation (3.12); its scope is the same finite domain.

3.10 Preservation of the record and the function of the seeds

Let $\text{Sample}(q_t)$ be the sample preceding the transition, and $\mathcal{M}_t$ the accumulated record. The memory update is

$$\mathcal{M}_{t+1} = \mathcal{M}_t \parallel \text{Sample}(q_t), \quad (3.17)$$

where $\parallel$ denotes ordered concatenation. Induction gives

$$\mathcal{M}_n = \mathcal{M}_0 \parallel (\text{Sample}(q_0), \dots, \text{Sample}(q_{n-1})).$$

The accumulators are reconstructed by summing the active contributions of each segment of nine records. Their quotients, the winding counts of the cursors and the orientations remain associated with the same transitions. Execution determines that record through the recurrence; storage preserves the specific provenance of the readings.

At the end of this stage, the complete domain of initial conditions, decimal emission, its two recoverable signatures, the ternary image, its fibres and its orbital classification have been constructed. The counts 104 976, 468, 243, 729 and 43 now have a domain and a mechanism of derivation. Regional selections receive these structures with their invariants, and the lifts act on oriented words with preserved provenance. This continuity makes the finite catalogue a constructive antecedent of the coinductive development.

4 Development of the formal kernel: initial conditions, geometry, and refinement

The eighty-one-position support does not coincide with the domain of initial conditions of the dynamics. An initial condition must also specify an orientation; and the additive and multiplicative sheets have independent cursors. This distinction makes it possible to reconstruct the emission exhaustively without selecting a numerical region in advance.

4.1 Initial states and the emission recurrence

For enumeration we use indices $I_9 = \{0, \dots, 8\}$, related to the positive labels by $i \mapsto i + 1$. Let

$$\mathcal{S} = I_9^2 \times \{N, E, S, O\}, \quad \Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times.$$

Each initial condition $\omega = (s_+, s_\times)$ determines two positions and two directions. The domain comprises all ordered pairs, so

$$|\mathcal{S}| = 324, \quad |\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (4.1)$$

The independent annex enumerates all these conditions; the main text presents the law that transforms them. The rows of an emission table are fibers of this domain of initial conditions.

The finite realization uses six windows of nine transitions. At each time t , the phase signature is $M_{\text{ph}}(1 + ((t-1) \bmod 9))$. If it equals $+1$, the positive residue $\Sigma(i+1, j+1) = \rho_9(i+j+2)$ of the first cursor is read, and that cursor is then moved; if it equals -1 , the positive residue $\Pi(i+1, j+1) = \rho_9((i+1)(j+1))$ of the second cursor is read, and the second cursor is then moved. The value zero increments the neutral counter. At the end of transitions twenty-seven and fifty-four, the cardinal orientations are reversed. Integer evaluations and their quotients are retained in the register, but the emitter uses the residue readouts specified here.

In the unit sector, the multiplicative readout uses the base-two discrete logarithm in $(\mathbb{Z}/9\mathbb{Z})^\times$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

For the finite emission it is extended by zero to the nonunit sector. This extension is an observable, not an identification of the elements 3, 6, 9 with the multiplicative identity. The sheet and its membership in the radical sector remain in the trajectory register.

Let S_m be the sum of the three additive positive residues Σ in window m , E_m the sum of the three discrete exponents of the multiplicative residues Π , and $Z_m = 3$ the number of neutral events. Thus S_m does not sum the integer evaluations $S = i + j$, whose quotients remain in the register. With $[a]_{10} \in \{0, \dots, 9\}$, the decimal emission rule is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (4.2)$$

The integer coefficients and calendar of this recurrence are an explicit part of the declared emission law. No values of π, φ, e or α are received. The distinction between an emission rule and a constant recognized from its output prevents the genealogy from being hidden in a table of decimals.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, the same rule reads

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \quad (4.3)$$

The final one is the residue of $7Z_m = 21$ modulo ten. The oriented ternary reduction of the catalogue adopts the convention

$$\Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (4.4)$$

The sign of this chart is explicit: changing it modifies the words and requires transporting subsequent orientations. The balanced identification of each component of $\mathbb{F}_3$ with the TRIT is made after (4.4).

Proposition 4.1 (Injective factorization of the emission). *The word $U = (U_1, \dots, U_6)$ determines the two signatures $s = (s_1, \dots, s_6)$ and $e = (e_1, \dots, e_6)$. The emission image has $18 \cdot 26 = 468$ elements. The ternary projection has 243 distinct values in the ambient space of 729 words.*

Proof. The hundreds digits of U_m recover s_m . If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit checks the third congruence in (4.3). Thus the coupling is injective on the product of signatures.

Enumerating the 324 conditions of a cursor using the preceding recurrence produces eighteen additive signatures, each with eighteen preimages; and twenty-six multiplicative signatures, of which twenty-four have eight preimages, one has twenty-four, and one has one hundred and eight. The annex preserves these signatures and their initial conditions, allowing the finite enumeration to be repeated without a reference numerical catalogue. Injectivity of the coupling gives $18 \cdot 26$ emissions and multiplicities

$$\underbrace{432}_{18 \cdot 24} \text{ fibers of } 144, \quad 18 \text{ fibers of } 432, \quad 18 \text{ fibers of } 1944.$$

Their sum is $432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976$. Applying (4.4) to the 468 emissions produces exactly 243 words. This last number is the cardinality of the image, not of the space $\mathbb{F}_3^6$. $\square$

This factorization explains the difference between finite exhaustiveness and unlimited depth. The domain (4.1) is complete for the declared support and emitter; it does not already contain every prefix of every prolongation. Depth belongs to the system of compatible extensions constructed over those conditions.

4.2 Calendar closure and memory advancement

The calendar of one hundred and eight transitions consists of twelve nonadic windows. Its group structure is more precise than an informal product of two periods:

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \quad \iota([b]) = [9b], \quad p([t]) = [t]_9. \quad (4.5)$$

If $t = 9b + r$ with $0 \leq r < 9$, adding two phases produces the carry

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor, [r + r']_9 \right),$$

where the first coordinate is reduced modulo twelve. Composition preserves the term linking the two coordinates.

Proposition 4.2 (Nonsplitting of the calendar). *The sequence (4.5) is exact and admits no section that is a group homomorphism.*

Proof. The kernel of reduction modulo nine is the subgroup of multiples of nine, the image of ι . If the extension split, commutativity would give $C_{108} \cong C_{12} \times C_9$. The first group has exponent 108, whereas the second has exponent $\text{mcm}(12, 9) = 36$, a contradiction. $\square$

Nine transitions restore the nonadic phase and advance one window; one hundred and eight restore the finite calendar. Neither fact requires erasing the trajectory or restoring the prefix of a prolongation. The nonadic holonomy Γ_9 denotes the transport of one winding on states with memory, so that

$$g(\Gamma_9 x) = g(x), \quad M(\Gamma_9 x) = M(x) + 1. \quad (4.6)$$

Equality of phase and increment of memory are compatible and constitute two observations of the same transport.

4.2.1 Temporal parametrization of compatibility and tritic orientation

The temporal interpretation of the TRIT is established on transitions of the complete state. Let $x \xrightarrow{\text{adm}} y$ be the admissibility relation determined by the TPK operations on $\hat{X}$. A trajectory parametrized by a discretely ordered set $\mathcal{T}$ is a map

$$\gamma : \mathcal{T} \longrightarrow \hat{X}, \quad \gamma(t) \xrightarrow{\text{adm}} \gamma(t^+),$$

where t^+ is the successor of t . The compatibility relation precedes the choice of parameter: the order expresses its transitions as an evolution. When the term section is used, the projection $p : E \rightarrow \mathcal{T}$ and condition $p \circ \gamma = \text{id}_{\mathcal{T}}$ are also declared. This specifies the state space, temporal order and information transported.

The signature τ selects a quadratic regime and admits a temporal readout relative to that parametrization. Type +1, with $J_{+1}^2 = -I$, represents return and memory; type 0, with $J_0^2 = 0$, represents the transition threshold; type -1, with $J_{-1}^2 = I$, represents opening in two eigendirections. In the model's temporal terminology, these functions correspond, respectively, to past, present and future. The correspondence concerns the operational position of a transition within a history: preserved antecedent, update and admissible continuation.

This interpretation acquires content through three distinct operations. Restricting a trajectory recovers the prefix already constructed. The update $\mathcal{U}_t$ composes selection, transport and registration in the observed transition. Prolongation considers the continuations that preserve the boundary conditions of that prefix. If $\mathcal{P}_n(x)$ denotes the set of admissible continuations of length n from x , deletion of the last step defines

$$r_n : \mathcal{P}_{n+1}(x) \longrightarrow \mathcal{P}_n(x).$$

An unbounded continuation is a collection $(\gamma_n)_{n \geq 1}$ satisfying $r_n(\gamma_{n+1}) = \gamma_n$. The inverse system thus preserves the restrictions that each depth imposes on the preceding ones. Existence of a compatible collection is established in the corresponding prolongation domain; merely setting up the sets $\mathcal{P}_n(x)$ does not replace that proof.

Bidirectionality therefore combines reconstruction of antecedents with compatibility of continuations. On the signature sequence, the involution C_{rev} reverses the order and changes the sign of each component. The trajectory is also acted on by the transformations of position, sheet and boundary specified by transport. The reversed signature is an observation of the conjugate trajectory, not a replacement for its remaining coordinates. The threshold remains distinguished: the visible value 0 retains its transition function even when its quotient or historical register is nonzero.

The relationship with exponential geometries is equally precise. In each fixed regime, $\exp(tJ_\tau)$ composes parameters and preserves the algebraic norm. Elliptic closure belongs to this local representation. The phase return of Γ_9 , by contrast, acts on the trajectory with memory, according to (4.6). A trajectory may recover its phase while retaining a new antecedent: its historical data have changed even though the phase observation is the same. The parabolic threshold retains a linear displacement; hyperbolic opening separates two eigendirections of expansion and contraction. These three realizations provide composition laws, while the TPK determines their selection and effective concatenation.

The aperiodic suspension of Section 5 will make this distinction explicit through a finite phase factor, an irrational rotation and a memory degree. The connection between TRIT and temporality is thus expressed through operations on trajectories and their quadratic representations. Its content goes beyond a classification of signs: it preserves the regime, direction of transport and relationship between antecedent, transition and continuation.

The monodromic representation of this return is developed in Section 5.10. There, the holonomy of the state is distinguished from its representation on phase and counter. The symbolic suspension in Section 5.12 defines the aperiodic time-crystal model, and Section 5.14 composes this temporality with the three Euler realizations.

4.3 Observable compression and orthogonal conservation

An isometric realization makes it possible to quantify what a reduced observation loses. Let $U : \mathcal{H} \rightarrow \mathcal{H}$ be an isometric realization of the transport of one winding, $U^*U = I$, and $J : \mathcal{H}_{\text{vis}} \rightarrow \mathcal{H}$ an isometric inclusion. Define

$$T = J^*UJ, \quad D = (I - JJ^*)UJ.$$

Then

$$UJ = JT + D, \quad J^*D = 0, \quad I - T^*T = D^*D. \quad (4.7)$$

The last equality follows by taking the adjoint product of the first and using orthogonality. Contraction of the visible component does not by itself constitute destruction of information in the total state.

Proposition 4.3 (Finite-horizon conservation). *If $I - T^*T = D^*D$, then, for every $n \geq 1$,*

$$I = \sum_{r=0}^{n-1} T^{*r}D^*DT^r + T^{*n}T^n. \quad (4.8)$$

Proof. Multiplying $D^*D = I - T^*T$ by T^{*r} and T^r and summing produces a telescoping sum. The interior terms cancel, leaving $I - T^{*n}T^n$. $\square$

The last summand preserves the terminal state. Removing it before justifying a limit would alter the equality. This distinction is the operational content of preservation of the unit under compression: the norm is distributed among the visible component, memory, and terminal, without being replaced by a single residue.

4.4 Local development: commutation, Lorentzian signature, and discrete curvature

The adjacent diagonal block with equal diagonal entries modulo nine is unique. Indeed,

$$k^2 \equiv (k+1)^2 \pmod{9} \iff 2k+1 \equiv 0 \pmod{9}$$

gives $k = 4$. Its visible matrix is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I.$$

With $P_{\pm} = (I \pm R)/2$, one obtains $B_c = 9P_+ + 5P_-$. The unit of stochastic normalization and preservation of an indefinite form are two realizations of this block:

$$\frac{B_c}{9} = P_+ + \frac{5}{9}P_-, \quad \frac{B_c}{\sqrt{45}} = \exp(\eta R), \quad \eta = \frac{1}{2} \log \frac{9}{5}.$$

For $Q = \text{diag}(1, -1)$, the second matrix satisfies $B_c^T Q B_c = 45Q$. This gives a local realization in $SO^+(1, 1)$; no prior introduction of a four-dimensional spacetime is needed to prove it.

Uniform averaging over the classes $C_1 = \{1, 4, 7\}$, $C_2 = \{2, 5, 8\}$, $C_0 = \{3, 6, 9\}$ gives

$$M_{\Pi} = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \quad M_{\Sigma} = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

The characteristic polynomial of M_{Π} is

$$(\lambda + 1)((\lambda - 9)^2 - 72),$$

so its quadratic form has signature $(2, 1)$. The local spectral gap $9 - 5 = 4$, the coefficient 2, the aggregate difference $51 - 45 = 6$, and the integrated defect $9 \cdot 6 = 54$ describe different levels of the same arithmetic comparison. They are not interchangeable indices of the same operator.

Discrete curvature can be described directly before this spectral realization. In residues, let $S(x, y) = x + y$, $P(x, y) = xy$, and Δ_x, Δ_y be the forward differences. The link variables are potential differences, $A_x^T = \Delta_x T$, $A_y^T = \Delta_y T$, and we set

$$F(T_x, T_y) = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Since $\Delta_x S = \Delta_y S = 1$, $\Delta_x P = y$, and $\Delta_y P = x$,

$$F(S, S) = F(P, P) = 0, \quad F(S, P) = +1, \quad F(P, S) = -1 \pmod{9}. \quad (4.9)$$

Sum and product, taken separately, induce flat connections; their ordered mixing changes the sign of plaquette curvature. Summing over a rectangle with sides a, b , the interior edges cancel and the oriented boundary sum equals $\pm ab$ modulo nine. An exact relation thus appears between composition order and boundary orientation.

4.5 Cubic completion and two notions of boundary

Enlarging the cubic support of side nine to that of side ten admits an exact positional stratification:

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1. \quad (4.10)$$

The first term corresponds to three interior coordinates; the second to one new coordinate; the third to two; and the last to the vertex with three new coordinates. The enlargement shell contains 271 positions. By contrast, the internal boundary of the discrete cube of side nine contains $9^3 - 7^3 = 386$. The difference between these counts is a difference of domain, not a conflict of values.

Dividing (4.10) by 1000 gives a normalized partition of unity. Cube incidence distinguishes six faces, twelve edges, and eight vertices. The subsequent link with Steiner hexads and octads requires the corresponding combinatorial maps; the cardinalities of the cube do not replace them. This completion here provides the relation between the bases 729 and 1000, while the exceptional prolongation will be developed after numerical generation and the electron.

4.5.1 Stratification of the completion and restoration of the apex

The completion admits a set-theoretic description that makes its operation explicit. Let E_9 be the set of the nine residue classes, represented positively by $1, \dots, 9$, and let $*$ be an additional element. The symbol $*$ differs from the zero class, already represented by 9. Consequently, enlarging E_9 to $\hat{E} = E_9 \sqcup \{*\}$ increases the cardinality from nine to ten without modifying the internal modular identification. The cubic completion is $\hat{C} = \hat{E}^3$, and the initial cube is $C = E_9^3$.

Definition 4.4 (Strata of combinatorial codimension). For $0 \leq r \leq 3$, define

$$S_r = \{(x_1, x_2, x_3) \in \hat{E}^3 : \#\{j : x_j = *\} = r\}.$$

The apex of the completion is $a_* = (*, *, *)$, and the punctured completion is $\hat{C}^\circ = \hat{C} \setminus \{a_*\}$.

Proposition 4.5 (Exact decomposition of the completion). *The strata are disjoint and satisfy*

$$\hat{C} = S_0 \sqcup S_1 \sqcup S_2 \sqcup S_3, \quad |S_r| = \binom{3}{r} 9^{3-r}, \quad (4.11)$$

$$|\hat{C}^\circ| = 729 + 243 + 27 = 999, \quad |\hat{C}| = 999 + 1 = 1000. \quad (4.12)$$

The punctured enlargement contains 270 elements in addition to the initial cube; the full enlargement contains 271.

Proof. An element of S_r is obtained by choosing the r coordinates containing $*$, in $\binom{3}{r}$ ways, and assigning to each remaining coordinate one of the nine classes of E_9 . Classification by the number of additional coordinates is exhaustive, and its classes are disjoint. The only element of S_3 is a_* . Removing it subtracts one from the total cardinality; restoring it recovers the complete Cartesian product. $\square$

The expression *holographic extreme and mean ratio*, abbreviated EMRH in the corpus, here denotes this articulation between the initial body, the enlargement strata, and unit restoration. Its immediate content is the exact partition of a finite product and its normalization. The number 999 represents a definite object, the punctured completion; 1000 represents its closure by restoration of the apex. The added unit therefore has an explicit incidence location.

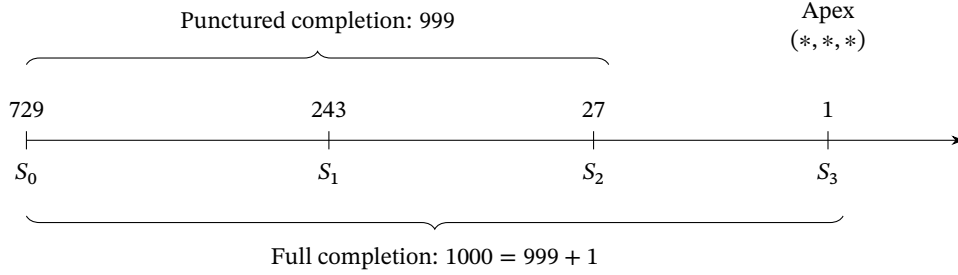


Figure 3: Strata of the cubic completion. The layout represents the set-theoretic decomposition, not a metric scale. The 243 elements of S_1 form three coordinate strata of 81 elements; the 27 elements of S_2 form three strata of 9. The apex completes the partition.

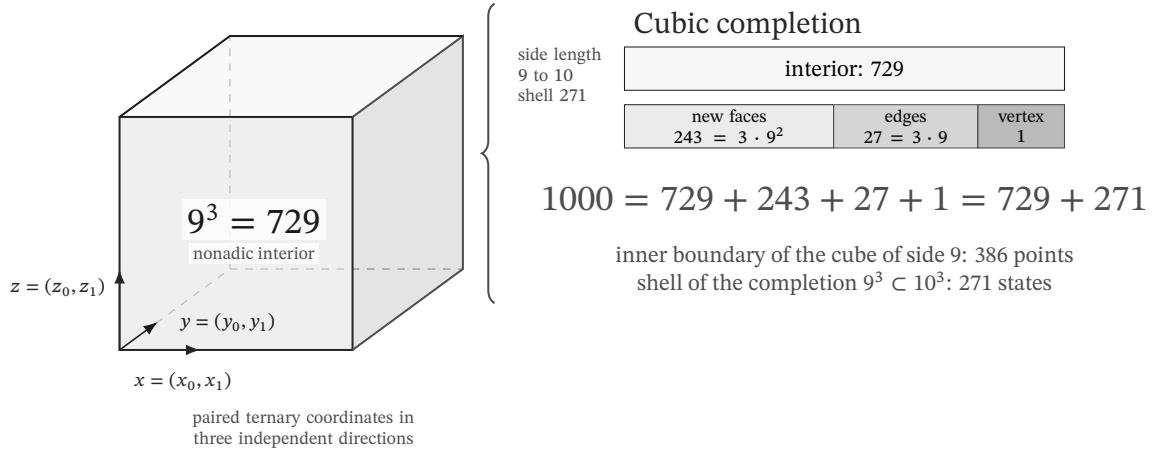


Figure 4: Cubic representation of the expansion from nine to ten classes per coordinate. Strata S_1, S_2, S_3 locate the added incidences on faces, edges, and the apex; their cardinalities form the 271-element shell. The 386-element boundary belongs to the initial cube. This geometric perspective complements the partition in Figure 3; graphical spacing is schematic.

4.5.2 Preservation of unity and dimensional compatibility

Normalization is not limited to dividing an isolated identity. In dimension d , the product $\hat{E}^d$ carries the uniform measure $\mu_d(\{x\}) = 10^{-d}$. The coordinate projection $p_d : \hat{E}^{d+1} \rightarrow \hat{E}^d$ removes the last coordinate.

Proposition 4.6 (Projective compatibility of the normalized measures). *For every $d \geq 1$,*

$$(p_d)_* \mu_{d+1} = \mu_d, \quad \mu_d(\hat{E}^d) = 1.$$

The mass of the stratum with r additional coordinates is $\binom{d}{r} 9^{d-r} / 10^d$, and the sum of the masses of all strata is one.

Proof. Each point of $\hat{E}^d$ has ten preimages. Their total mass is $10 \cdot 10^{-(d+1)} = 10^{-d}$. Equality for any subset follows by summing over its points. The stratified formula follows from the same count as (4.11); the binomial $(9 + 1)^d$ gives the total normalization. $\square$

In dimension three, removing the apex leaves mass 999/1000, and restoring it adds 1/1000. Renormalizing before restoration would define another measure: the distinction is operationally relevant. In particular, preservation of probability mass, preservation of transport registers, and thermodynamic dissipation are statements about different objects. The proposition proves the first and provides the compatible support for studying the second; the third requires a specified dynamical and energetic realization.

The internal geometric boundary also remains distinct. On the Cartesian representative $\{1, \dots, 9\}^3$, removing $\{2, \dots, 8\}^3$ leaves 386 points. This boundary belongs to the initial cube, whereas $\hat{C} \setminus C$ belongs to its enlargement. The cubic completion, the positional base-thousand expansion, and the exceptional incidence readout share support data; their respective maps determine which information from those data each representation uses.

4.6 Cubic incidence and Lorentzian realization

Cubic completion provides a relation between interior and boundary. Its geometry can be developed through the incidence of the six faces with the eight vertices. Label the faces by (i, ε) , $i \in \{1, 2, 3\}$, $\varepsilon \in \{+1, -1\}$, and the vertices by $\nu \in \{+1, -1\}^3$. The incidence matrix is

$$M_{(i,\varepsilon),\nu} = \mathbf{1}_{\{\nu_i=\varepsilon\}}.$$

Let $u = (1, \dots, 1)^\top \in \mathbb{R}^6$, let R_F be face opposition, and let $P_u = uu^\top/6$, $P_- = (I - R_F)/2$.

Proposition 4.7 (Four-dimensional incidence quotient). *The incidence Gram matrix satisfies*

$$MM^\top = 12P_u + 4P_-.$$

Its eigenvalues 12, 4, 0 have multiplicities 1, 3, 2. Consequently,

$$Q_\square = \mathbb{R}^6 / \ker(MM^\top) \simeq \mathbb{R}u \oplus E_-, \quad E_- = \text{im } P_-.$$

Proof. A face contains four vertices; two opposite faces have empty intersection, and two distinct nonopposite faces share two vertices. The matrix $12P_u + 4P_-$ has precisely these entries. The uniform space is one-dimensional; differences between opposite faces generate a three-dimensional space orthogonal to the uniform space. Its complement is two-dimensional and belongs to the kernel. $\square$

The decomposition $1 + 3$ comes from incidence. The temporal choice is declared through the bilinear form

$$B_\square([x], [y]) = \langle x, (P_- - P_u)y \rangle.$$

It is well defined on the quotient because both projectors annihilate the kernel. In the basis

$$t = [u]/\sqrt{6}, \quad d_i = [e_{i,+} - e_{i,-}]/\sqrt{2},$$

its matrix is $\text{diag}(-1, 1, 1, 1)$. This makes explicit the domain, decomposition and sign convention producing the Minkowski realization. The positive Gram matrix determines the quotient; the temporal assignment determines the indefinite form on it.

The operator

$$W_\square = \sqrt{6}P_u + \sqrt{2}P_-, \quad W_\square e_{i,\varepsilon} = t + \varepsilon d_i$$

sends the six face labels to null directions, since $B_\square(t + \varepsilon d_i, t + \varepsilon d_i) = -1 + 1 = 0$. Reversal of a direction and face opposition thus have a concrete geometric realization.

4.6.1 Soldering, reversal and subdivision

The cellular realization uses an invertible transport $U_m(a)$, preserving the Lorentzian form, between the fibres at the endpoints of an oriented dual edge a , a face label $\lambda_m(a)$ and an orientation $\sigma_m(a)$. With $\ell_m = \ell_0/9^m$, the soldering map is

$$\beta_m(a) = \ell_m \sigma_m(a) W_{\square, s(a)} e_{\lambda_m(a)}.$$

Compatible reversal satisfies

$$\beta_m(\bar{a}) = -U_m(a) \beta_m(a).$$

Identify the source fibres at two consecutive levels through the declared refinement isometry. If the nine subsegments $a_1, \dots, a_9$, at level $m+1$, preserve the label and orientation when transported to that common fibre, then

$$\beta_m(a) = \sum_{j=1}^9 U_{m+1}(a_1 \cdots a_{j-1})^{-1} \beta_{m+1}(a_j). \quad (4.13)$$

Each transported term equals $\ell_m/9$ times the same null direction, proving the identity. The compatibility hypothesis determines which subdivisions realize this refinement. The equation brings together direction, scale and transport, instead of limiting the geometric correspondence to a dimension count.

With orientation and the Lorentzian form also fixed, Hodge duality on $\Lambda^2 Q_{\square}^*$ satisfies $\star^2 = -I$. In dimension four the sign is $(-1)^{2(4-2)+1} = -1$. This yields a real complex structure on two-forms and, after complexification, eigenspaces with eigenvalues $+i$ and $-i$. The electromagnetic realization will use this domain after constructing the electronic section and declaring its coupling.

4.7 Compatibility between modular and positional reductions

The blockwise decimal base enters through a Euclidean division that simultaneously preserves remainder and quotient:

$$n = 1000q + r, \quad 0 \leq r < 1000.$$

Since $1000 \equiv 1 \pmod{9}$ and $1000 \equiv 1 \pmod{3}$,

$$n \equiv q + r \pmod{9}, \quad n \equiv q + r \pmod{3}.$$

Thus, reducing a block requires the quotient's contribution to recover the integer's class. The numbers 0 and 1000 have the same remainder modulo 1000, but different classes modulo 9 and modulo 3. This is an operational reason to preserve memory during a change of resolution.

Ternary refinement and determination of a decimal prefix can be coordinated through a single integer residual. If $N/3^L$ is a cylinder endpoint and $D/1000^r$ a decimal endpoint, define

$$A = 1000^r N - D 3^L.$$

On adjoining six ternary symbols, $N' = 729N + \nu$, $L' = L + 6$, with $0 \leq \nu < 729$. If the decimal prefix remains fixed, one obtains

$$A' = 729A + \nu 1000^r.$$

If a decimal block δ is also incorporated, then $D' = 1000D + \delta$, $r' = r + 1$, and

$$A' = 729000A + \nu 1000^{r+1} - 729\delta 3^L. \quad (4.14)$$

Both identities follow by substituting the definitions. Endpoint comparison is thus reduced to integer operations that retain each prolongation's contribution. This compatibility allows a region to be refined before its real coordinate is evaluated.

4.8 Number as a compatible section and value as a readout

The distinction between state and observation makes it possible to specify which object is prolonged as resolution increases. At depth n , let X_n be the set of admissible states produced by APP-TRIT-TPK. Their data include position, sum and product sheets, residues and quotients, regime, orientation, carry, phase, route, boundary, memory, and survival. They are not independent coordinates: arithmetic evaluations and previous transports impose their relations. A prolongation preserves these relations in every predecessor, even as it adds new steps and new memory data. This is the meaning of the arithmetic of histories developed in (see 4.9; 4.10).

Let $r_{m,n} : X_m \rightarrow X_n$, for $m \geq n$, be the depth restrictions, with $r_{n,n} = \text{id}$ and $r_{\ell,n} = r_{m,n} \circ r_{\ell,m}$. Each restriction preserves the prefix and the information already determined within it.

Definition 4.8 (Compatible numerical section). An HMT number is a compatible history of the state system:

$$x = (x_n)_{n \geq 0} \in X_\infty := \varprojlim_n X_n, \quad r_{m,n}(x_m) = x_n. \quad (4.15)$$

A readout is an additional operation on a specified domain of these histories. A digit is a finite observation; a scalar value is an evaluation of the readout. The pair consisting of the section and its readout constitutes a read presentation, not a redefinition of the section's identity.

On the Archimedean domain X_∞^{Arch} , the readout associates a closed cylinder with each prefix, preserves their nesting, and makes their diameters tend to zero. It thus determines

$$\text{ev}_\mathbb{R} : X_\infty^{\text{Arch}} \longrightarrow \mathbb{R}, \quad \{\text{ev}_\mathbb{R}(x)\} = \bigcap_n I_n(x). \quad (4.16)$$

The concrete construction of these cylinders and the decision of their boundaries are developed in Section 5. Definition (4.15) does not by itself assert that the image of (4.16) is the whole line: a coverage assertion requires its own theorem. Nor does equality of values establish, without an injectivity result, equality of their histories. This separation allows precision to be understood as depth while preserving the difference between the arithmetic predecessor and its scalar realization.

4.9 Composition of histories and memory multiplier

Fix a sector of states at depth n and a category $\mathcal{C}$ whose arrows are their admissible histories. Composition is written $vu = v \circ u$: first traverse u , then v . Elementary steps arise from TPK selection, transport, and update. A local trit fiber admits the notation $\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x)$, where $\tau(x) \in \{+1, 0, -1\}$. Regime and orientation belong to the transported state; a transition between regimes does not amount to identifying their quadratic algebras.

The following formulation specifies an algebraic sector of that composition. For each object x , a unital associative real algebra A_x is given, possibly noncommutative. For each $u : x \rightarrow y$, a unital homomorphism $\rho_u : A_x \rightarrow A_y$ is given. Assume a strict action:

$$\rho_{\text{id}_x} = \text{id}_{A_x}, \quad \rho_v \rho_u = \rho_{vu}. \quad (4.17)$$

For each composable pair $u : x \rightarrow y, v : y \rightarrow z$, fix an invertible central multiplier $\Omega(v, u) \in Z(A_z)^\times$. Require the normalization $\Omega(\text{id}_y, u) = \Omega(u, \text{id}_x) = 1_y$ and, for each composable triple, the identity

$$\rho_w(\Omega(v, u))\Omega(w, vu) = \Omega(w, v)\Omega(wv, u). \quad (4.18)$$

These are explicit hypotheses on the transports and their coefficients, not a consequence of calling the dynamics holonomic. When this presentation is applied to a TPK sector, its ρ_u and $\Omega(v, u)$ must be identified. Transitions expressed through bimodules require preserving that structure and are not replaced by (4.17).

The space of finite sums

$$\mathcal{H}(\mathcal{C}, A, \rho, \Omega) = \bigoplus_{u \in \text{Mor}(\mathcal{C})} A_{I(u)} T_u$$

receives the bilinear product

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v, u)T_{vu}, \quad (4.19)$$

if the arrows are composable, and zero otherwise. Here $t(u)$ denotes the terminal state. The sum gathers histories with coefficients; the product concatenates histories while preserving the register multiplier.

Proposition 4.9 (Sectorial associativity with a central multiplier). *Under (4.17) and (4.18), the product (4.19) is associative. If the set of objects is finite, its unit is $\sum_x 1_x T_{\text{id}_x}$.*

Proof. For $u : x \rightarrow y, v : y \rightarrow z, w : z \rightarrow t$, let $a \in A_y, b \in A_z, c \in A_t$. The two coefficients of T_{wvu} are

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: c\rho_w(b)\Omega(w, v)\rho_{wv}(a)\Omega(wv, u), \\ (cT_w)((bT_v)(aT_u)) &: c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v, u))\Omega(w, vu). \end{aligned}$$

The strict action identifies $\rho_w\rho_v(a)$ with $\rho_{wv}(a)$. Centrality of $\Omega(w, v)$ allows it to be moved to the right of that factor; (4.18) then equates the remaining products. The coefficients a, b, c are not permuted. If the triple is not composable, both parenthesizations are zero by their initial and terminal states. Bilinearity completes the proof. Local identities and the normalization of Ω verify the stated unit. $\square$

Calendar carry provides an effective example. In the category with one object and arrows C_9 , take coefficients $A = \mathbb{R}[z, z^{-1}]$, the identity action, and

$$\Omega(b, a) = z^{c_0(a, b)}, \quad c_0(a, b) = \left\lfloor \frac{a+b}{9} \right\rfloor, \quad 0 \leq a, b < 9.$$

Both carry sums of a triple equal $\lfloor (a+b+c)/9 \rfloor$; hence (4.18) holds. The formal variable z records the winding without assigning it a numerical value. Subsequently imposing $z^{12} = 1$ recovers the dodecaphasic register of the calendar (4.5); retaining z without that relation distinguishes successive windings. The example realizes the memory of this calendar without identifying it with all TPK memory.

4.10 Double variance of refinement and preservation of the unit

Restricting states prolongs observables in the opposite direction. Consider the function algebras, with pointwise product, $D_n = \text{Fun}(X_n, \mathbb{R})$. Precomposition defines

$$r_{m,n}^* : D_n \longrightarrow D_m, \quad r_{m,n}^* f = f \circ r_{m,n}. \quad (4.20)$$

These are unital morphisms; they are also injective when the corresponding restrictions are surjective. The direct limit $D_{\text{cyl}} = \varinjlim_n (D_n, r_{n+1,n}^*)$ gathers finite-depth observations. This algebra is not identified here with the entire route algebra: prolonging transport operators also requires their compatibilities with refinement.

Lemma 4.10 (Unit and naturality of evaluation). *If e_x is the indicator of $x \in X_n$, then*

$$r_{m,n}^* e_x = \sum_{y: r_{m,n}(y)=x} e_y, \quad r_{m,n}^* 1_n = 1_m, \quad (4.21)$$

and, for every $f \in D_n, y \in X_m$,

$$\langle r_{m,n}^* f, y \rangle = \langle f, r_{m,n} y \rangle.$$

Each section $x \in X_\infty$ determines a well-defined evaluation $D_{\text{cyl}} \rightarrow \mathbb{R}$, given by $[f] \mapsto f(x_n)$ for $f \in D_n$.

Proof. The fibers of $r_{m,n}$ partition X_m . Their indicators are orthogonal and sum to 1_m , proving (4.21). Naturality is the definition of precomposition. Finally, $(r_{m,n}^* f)(x_m) = f(x_n)$, so two compatible representatives give the same evaluation in the direct limit. $\square$

The global unit is the constant-one function on all possibilities; the individual section selects a compatible history. Preserving the former does not mean immobilizing the latter. Refinement can increase the number of extensions and the depth of each history while maintaining the unit and all evaluations of its predecessors.

4.11 Phase return, Weyl representation, and two readouts

The law (4.6) acquires an operator realization in the bilateral suspension to be constructed in Section 5. There, T is one-step advancement, V_9 observes the nonadic phase, W_δ the rotation phase, and D_M the memory degree. Writing $\hat{\Gamma}_9 = T^9$ for the winding in this representation, the Weyl relations (5.17) imply, on finitely supported vectors,

$$\begin{aligned} V_9 \hat{\Gamma}_9 &= \hat{\Gamma}_9 V_9, & W_\delta \hat{\Gamma}_9 &= e^{2\pi i 9\delta} \hat{\Gamma}_9 W_\delta, \\ [D_M, \hat{\Gamma}_9] &= \hat{\Gamma}_9. \end{aligned} \quad (4.22)$$

The first equality follows from $\zeta_9^9 = 1$; the second from nine applications of the covariance of W_δ ; the third expresses the unit memory increment. The finite phase returns, while irrational rotation and memory distinguish the next state. The characters and parameter δ are subsequently realized from the numerical construction specified there: they do not retrospectively select the seeds. This representation describes observables and translations of the suspension; it does not replace all coordinates or all operators of the TPK.

The double readout prolongs the same distinction between history and observable. On its Archimedean domain, (4.16) is applied; on the calibrated incidence domain, the maps Q_n in Section 10.1 preserve words and boundary frames. The equality proved there, $r_{n,m} Q_n = Q_m p_{n,m}$, determines their prolongation Q_∞ . On the common domain of both readouts one may jointly consider $\text{ev}_\mathbb{R}(x)$ and $Q_\infty(x)$: the former retains the value; the latter, the transported incidence that its codomain preserves. The section precedes both. This articulation permits passing from the arithmetic of histories to constants and the exceptional readout without turning their realizations into new initial data.

5 Coinductive generation of π, φ, e to arbitrary depth

The six-block emission is a finite stage of the construction, not a complete decimal expansion. Its role is to produce regions with multiplicity, orientation, and transport rules. Prolongation must preserve these data and determine a compatible collection of prefixes of arbitrary length. The term *generation* denotes this operation; *genealogy* denotes the reconstruction of its dependencies. Both notions are involved, but they serve different purposes.

5.1 Orbits, regions, and characters of closure, propagation, and self-scaling

For a word $w = (w_1, \dots, w_6)$, define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \quad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

We have $r^3 = s^2 = I$ and $srs = r^{-1}$; these permutations therefore realize the dihedral group D_3 . The action preserves the catalogue and the multiplicities and commutes with Ψ . The quotient of the 243 visible words has 43 orbits: thirty-eight of cardinality six and five of cardinality three.

The closure class is recognized by a fibre predicate: its six orientations have five emissions per word, total multiplicity 1008, and distribution $432 + 4 \cdot 144$. There is exactly one orbit with these properties in the catalogue. To specify the other two selections, write $f(w) = \#\Psi^{-1}(w)$, $m(w) = \sum_{U \in \Psi^{-1}(w)} \text{mult}(U)$, and $\nu(w) = (n_0, n_1, n_2)$. The additive diagonal class of an emission is $d(U) = [i + j]_9$ in indices from zero to eight; its additive signature retains this class and the orientation *ES* or *NO*. Let $\mathcal{D}(\mathcal{O})$ be the set of these classes for all emissions over an orbit.

Among the orbits of size six, the joint predicate

$$f(w) = 1, \quad m(w) = 144 \quad (w \in \mathcal{O}), \quad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\}$$

leaves exactly six orbits. In this sector, the census $\nu = (2, 3, 1)$, the same as for closure, uniquely selects propagation; the census $\nu = (2, 2, 2)$ uniquely selects self-scaling. The diagonal support cannot be omitted: without it there are four balanced single-preimage orbits of multiplicity 144, not one. Finally, the existence of an emission with $d = 6$ and orientation ES selects one representative of each orbit and yields

$$w^{\text{cl}} = 010211, \quad w^{\text{pr}} = 201101, \quad w^{\text{au}} = 121200. \quad (5.1)$$

These symbols first designate regions of the catalogue. Scalar identification will be established through their readouts; they are not chosen because they coincide with digits of a known constant.

The closure word 010211 has the following five preimages:

Six-block emission	Multiplicative signature	Role	Multiplicity
501, 614, 498, 169, 272, 272	555555	transverse	432
810, 923, 870, 169, 272, 272	339555	positive orientation	144
810, 983, 810, 169, 272, 272	393555	positive orientation	144
870, 923, 810, 169, 272, 272	933555	positive orientation	144
870, 923, 810, 418, 674, 521	933717	oriented return	144

The multiplicity 432 and the constant signature 555555 characterize the transverse region. They do not identify the central position (5, 5) of APP. Confusing a six-window signature with a position in the support would erase the distinction between numerical region and electronic structure.

The microfibre also has a precise bipartite morphology. Each emission is a product of an additive cursor class and a multiplicative class. The three positive regions share the same additive class of eighteen conditions; together they contain three multiplicative classes of eight conditions each. The transverse region has a product 18×24 ; the three positive regions taken together, another 18×24 ; and the return, a product 18×8 . The decomposition by emissions has five parts, whereas the decomposition into connected components of the bipartite graph has three. Both preserve the same 1008 seeds without reducing form to a cardinality.

5.2 Finite transport and the prolongation boundary

The endomorphisms L_0, L_1 of the ternary word space act before the Archimedean realization. Their determination begins with the transitions of the arithmetic state, not with the prescription of two matrices. If U_t is the composite step of selection, transport, and update, the observed block satisfies

$$b_j^\chi = \Pi_6(x_j^\chi), \quad x_{j+1}^\chi = U_{9j+9} \cdots U_{9j+1}(x_j^\chi), \quad \chi \in \{\text{cl}, \text{pr}, \text{au}\}.$$

Six transitions are formed for each regime: two consecutive transitions in each of the three regions. The pairs of source and target matrices obtained in the record are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

Each string of six digits represents a row in $\mathbb{F}_3^6$. Since $\det X_0 = 1$ and $\det X_1 = -1$ in $\mathbb{F}_3$, the equations $X_a L_a = Y_a$ uniquely determine $L_a = X_a^{-1} Y_a$. In this row-vector chart, they yield

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix} \pmod{3}.$$

The schedule (L_0, L_0, L_1, L_1) produces four new blocks of six components. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate accumulated length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace the production of those transitions by U_t . The provenance of their record and the examination of the sixteen binary schedules are documented in the technical supplement. The indicated schedule is the only one of these sixteen that jointly reproduces the three sequences of five blocks.

5.3 Five closure regions and angular compensation

The closure region contains one transverse component and four accompanying components. The symmetric readout acts on the space of its five positions through

$$P_5 = \frac{1}{5} \mathbf{1} \mathbf{1}^T, \quad \mathbf{1} = (1, 1, 1, 1, 1)^T.$$

Invariance $P_5 \lambda = \lambda$ and conservation $\mathbf{1}^T \lambda = 1$ force $\lambda = \mathbf{1}/5$. This normalization distributes the unit among positions of the readout; it does not replace the seed multiplicities 432, 144, 144, 144, 144 with uniform frequencies. The elliptic chart of the readout realizes each coordinate through $1 + \lambda_j J$; its slope is λ_j , so the primitive slope is $1/5$. This makes explicit the rule taking a normalized coordinate to a slope, rather than identifying the two notions without a map. The readout composes the four accompanying positions by the law

$$a \oplus b = \frac{a + b}{1 - ab}. \quad (5.2)$$

This law is obtained by multiplying $(1 + aJ)(1 + bJ)$ in the regime $J^2 = -I$ and dividing by its scalar component. The composition of slopes is therefore a realization of the quadratic structure of TRIT, not an ordinary sum of four fractions.

Proposition 5.1 (Closure compensator). *The composition of four slopes $1/5$ is $120/119$. The condition of return to slope one determines a unique positive compensator of the form $1/q$, with $q > 1$, and yields $q = 239$.*

Proof. The first doubling gives $(1/5) \oplus (1/5) = 5/12$; the second, $(5/12) \oplus (5/12) = 120/119$. The condition

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1$$

is equivalent to $(120 - 119)q = 120 + 119$, hence $q = 239$. $\square$

The resulting closure character has the realization

$$\Pi_{\text{cl}} = 4 \left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \right). \quad (5.3)$$

Machin's angular identity recognizes $\Pi_{\text{cl}} = \pi$. The direction of the construction is precise: the slope and the number of compositions are specified by the penta fibre readout; the compensation equation forces 239. Once this character has been produced, its evaluation by alternating series does not select the seed orbit anew.

For $0 < x < 1$, the alternating sums

$$A_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{2j+1}$$

enclose $\arctan x$ between two consecutive truncations, with difference $x^{2n+3}/(2n+3)$. Applied to the two slopes in (5.3), they provide rational endpoints with explicit error bounds. The numerical value is thus calculated as a consequence of the exact character, without introducing a target prefix.

5.4 Propagation readout and normalization of the unit

The propagation region is realized by a formal collection $P_t \in \mathcal{A}[[t]]$, where $\mathcal{A}$ is a unital commutative algebra over $\mathbb{Q}$. Its composition satisfies $P_{t+s} = P_t P_s$, with unit $P_0 = 1$, and the oriented elementary transport fixes the scalar generator $P'_0 = +1$. The condition fixes its sign, not merely its magnitude. Writing

$$P_t = \sum_{n \geq 0} a_n t^n,$$

the normalization determines $a_0 = a_1 = 1$. Comparing the coefficient of $s^n t$ in $P_{s+t} = P_s P_t$ gives $(n+1)a_{n+1} = a_n a_1$; equivalently, $P'_t = P_t$. Thus,

$$(n+1)a_{n+1} = a_n, \quad a_n = \frac{1}{n!}. \quad (5.4)$$

The character at one unit of the parameter is $\Pi_{\text{pr}} = P_1$, subsequently recognized as e .

Normalization of the generator matters. Without this datum, the semigroup law would admit $P_t = e^{ct}$ for different c ; the condition fixing the unit of the parameter in the readout is therefore not omitted. This condition precedes decimal comparison and belongs to the exposition of the operational rule, not to the choice of an experimental value.

Proposition 5.2 (Rational propagation bounds). *For $n \geq 1$, let $E_n = \sum_{j=0}^n 1/j!$. Then*

$$E_n < \Pi_{\text{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

Proof. The first omitted term is $1/(n+1)!$. Each subsequent ratio is at most $1/(n+2)$. The geometric upper bound for the tail is

$$\frac{1}{(n+1)!} \frac{1}{1 - 1/(n+2)} = \frac{n+2}{(n+1)(n+1)!}.$$

Strict inequality follows from the decrease of the successive ratios. $\square$

5.5 Modal incidence and generation of self-scaling

The tritic selector produces the cycle $(+1, -1, 0)$. On the modes Σ, Π , the rule is $+1 : m \mapsto \Sigma$, $-1 : m \mapsto \Pi$, $0 : m \mapsto m$. With cyclic closure, the observed mode is (Σ, Π, Π) . Its edges are $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Its incidence, in this ordering of the modes, is therefore

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (5.5)$$

The four entries are obtained from the schedule, not from an equation that has already received the value of φ . Repeating the schedule gives counts $3F_{\text{av}}$, $9F_{\text{av}}$, $36F_{\text{av}}$ at lengths nine, twenty-seven, and one hundred and eight. These counts are not powers of the matrix.

Theorem 5.3 (Positive self-scaling character). *The incidence (5.5) has a unique strictly positive eigenray. When normalized as $(1, x)^\top$, its coordinate x satisfies $x^2 - x - 1 = 0$, $x > 0$, and equals $\varphi = (1 + \sqrt{5})/2$.*

Proof. The eigenvalue equation gives $x = \lambda$ and $1 + x = \lambda x$. Eliminating λ yields the polynomial. Only one of its two roots is positive. Since all entries of F_{av}^2 are positive, the positive ray is unique and simple. The radical expression recognizes the coordinate obtained; it does not participate in the production of F_{av} . $\square$

With $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, the incidence induces, for $n \geq 1$,

$$F_{av}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

The consecutive ratios F_{n+1}/F_n alternate around φ . The determinant identity

$$F_{n+1}^2 - F_n F_{n+2} = (-1)^n$$

gives an exact width $1/(F_n F_{n+1})$ between two consecutive ratios. Their growth makes this rational bracket arbitrarily narrow.

The measure on all compatible histories does not coincide with the mode frequency of the three-event schedule. The chronological frequency is $(1/3, 2/3)$. The measure of maximal entropy of the symbolic envelope defined by F_{av} is constructed from its eigenvectors and has weight ratio φ^{-2} . This latter measure enters the areal–volumetric readout; it is not obtained by simply replacing one counting frequency with another.

5.5.1 Trajectory space and invariant measure between sheets

The toroidal topology of APP fixes the support of the cardinal translations and their winding classes. The correspondence between sheets arises from another operation on the same support: temporal selection of the additive or multiplicative evaluation. Its geometric typing is

$$\eta_{av}(\Sigma) = \mathcal{H}_{ar}, \quad \eta_{av}(\Pi) = \mathcal{H}_{vol}.$$

The first sheet corresponds to the additive boundary readout; the second, to the multiplicative interior readout. This map transports the mode labels and preserves the incidence (5.5). The square of the self-scaling ratio is determined through the trajectories of that incidence.

Definition 5.4 (One-step-memory symbolic envelope). Let

$$X_{av} = \{(s_k)_{k \in \mathbb{Z}} \in \{\mathcal{H}_{ar}, \mathcal{H}_{vol}\}^{\mathbb{Z}} : (F_{av})_{s_k, s_{k+1}} = 1 \text{ for all } k\}.$$

The index shift acts on this space. It is the smallest one-step-memory symbolic system containing the modal sequences and preserving their consecutive pairs: it admits both cross-transitions and volumetric self-retention. Phase, cardinal orientation, and trajectory records remain in the TPK state from which this envelope is obtained.

Proposition 5.5 (Weighting of trajectories between sheets). *In areal–volumetric order, the invariant measure of maximal entropy on X_{av} has transition matrix and stationary distribution*

$$P_{av} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}, \quad (\mu_{ar}, \mu_{vol}) = \frac{(1, \varphi^2)}{1 + \varphi^2}. \quad (5.6)$$

In particular, $\mu_{ar}/\mu_{vol} = \varphi^{-2}$.

Proof. The already constructed incidence has positive eigenvector $r = (1, \varphi)^\top$ and eigenvalue φ . The normalization

$$(P_{av})_{ij} = \frac{(F_{av})_{ij} r_j}{\varphi r_i}$$

gives the stated matrix. Its rows sum to one because $\varphi^{-2} + \varphi^{-1} = 1$. Symmetry of F_{av} implies that weights proportional to r_i^2 satisfy detailed balance; hence $\mu P_{\text{av}} = \mu$. Irreducibility and volumetric self-retention determine a unique stationary distribution.

To verify maximality, every allowed transition satisfies $\log(P_{\text{av}})_{ij} = \log r_j - \log r_i - \log \varphi$. Under any invariant measure, the endpoint terms cancel on average. If v_i are its vertex weights and $q_i(j) = v(s_1 = j \mid s_0 = i)$, for $v_i > 0$, then

$$h_v \leq H_v(s_1 \mid s_0) = \log \varphi - \sum_{i: v_i > 0} v_i D(q_i \parallel R_i) \leq \log \varphi,$$

where $D(q \parallel p) = \sum_j q_j \log(q_j/p_j)$ is the relative entropy on the allowed pairs and the convention $0 \log 0 = 0$ is adopted. The Markov measure of (5.6) attains the bound. Equality in the first inequality requires the Markov property; equality in the second requires $q_i = R_i$. Stationarity and irreducibility then fix μ , establishing uniqueness. This is the Parry measure of the defined envelope. $\square$

The frequency $(1/3, 2/3)$ counts the instants of the primitive calendar; μ weights the trajectories of the envelope. Its quadratic ratio expresses the product of incoming and outgoing incidence amplitudes. TPK memory and phase restrictions retain their own domain; the entropy of the envelope pertains to the symbolic system specified in this definition.

5.5.2 Closure operator and limit of marked returns

The closure coordinate π , generated by the regional readout, enters after construction of the modal incidence. Let $e_{\text{ar}} = (1, 0)^\top$, $e_{\text{vol}} = (0, 1)^\top$, and $R_i = e_i e_i^\top$ be the sheet projectors. The readout conditions are preservation of both sheets, unit volumetric normalization, and an areal mark equal to the closure coordinate. These conditions determine the positive operator

$$D_\pi = \pi P_{\text{ar}} + P_{\text{vol}} = \begin{pmatrix} \pi & 0 \\ 0 & 1 \end{pmatrix}. \quad (5.7)$$

Diagonality expresses sheet preservation, and the two entries express their normalizations; this makes explicit the uniqueness of the operator for the fixed readout.

Proposition 5.6 (Boundary-return ratio). *For $n \geq 1$, the marked evaluation*

$$\Theta_n = \frac{\langle e_{\text{ar}}, D_\pi F_{\text{av}}^n e_{\text{ar}} \rangle}{\langle e_{\text{vol}}, F_{\text{av}}^n e_{\text{vol}} \rangle} = \pi \frac{F_{n-1}}{F_{n+1}} \quad (5.8)$$

satisfies

$$\lim_{n \rightarrow \infty} \Theta_n = \pi \frac{\mu_{\text{ar}}}{\mu_{\text{vol}}} = \frac{\pi}{\varphi^2}. \quad (5.9)$$

Proof. The diagonal entries of F_{av}^n count paths of length n returning to their initial sheet. The power formula already proved gives F_{n-1} and F_{n+1} . The operator D_π multiplies the first count by π . Finally, F_{n-1}/F_{n+1} is the product of two consecutive quotients converging to φ^{-1} ; its limit is φ^{-2} . The identity with the weight ratio follows from (5.6). $\square$

The closure operator appears exactly once, in the terminal evaluation of the return. The transfer F_{av} remains determined by the calendar. Information from two readouts of the same state is thus composed: modal incidence determines the quadratic weighting, and the closure region contributes its coordinate to the boundary.

The composition also preserves refinement. For positive cylinders $I = [a, b]$ and $J = [c, d]$, the image of the readout $\mathcal{R}(x, y) = x/y^2$ is the interval

$$\mathcal{Q}(I, J) = \left[\frac{a}{d^2}, \frac{b}{c^2} \right].$$

Monotonicity in each argument proves that nested closure and self-scaling cylinders produce nested intervals of the ratio; the positivity of their limits and contraction of their diameters determine π/φ^2 . The scalar expression thus retains a chain of operations compatible with depth, applied next in its two orientations.

Once the characters have been generated, the two orientations of the readout

$$R(x, y) = \frac{x}{y^2}$$

provide

$$\rho_A = \frac{\pi}{\varphi^2}, \quad \rho_E = \frac{\varphi}{\pi^2}, \quad \varphi^3 \rho_A^2 \rho_E = 1. \quad (5.10)$$

The second expression will appear in the electronic exponent. The first is also obtained as the limit of the marked readout $\pi F_{n-1}/F_{n+1}$. Interchanging the arguments relates the two, but does not turn a reparametrization of the electronic equation into a second independent derivation of all its coefficients.

5.6 Exponential realizations of the generated coordinates

The regional coordinates already determined allow recognition of the concrete realizations of the TRIT quadratic collection. The order is substantive: construction of the regions precedes evaluation of an exponential at the parameters π or $2 \log \varphi$. These expressions identify the role of the coordinates in operators constructed from APP and TPK; they do not enter the selection of the prefixes that produce them.

5.6.1 Ternary cycle, nilpotent transport, and sheet incidence

Multiplication $a(r) = 4r$ in $\mathbb{Z}/9\mathbb{Z}$ has order 3. On the units it determines the orbits $\{1, 4, 7\}$ and $\{2, 8, 5\}$. In the augmentation module of either orbit, formed by the integer combinations whose coefficients sum to zero, the basis $(e_r - e_{4r}, e_{4r} - e_{7r})$ gives

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^2 + C + I = 0.$$

The restricted form has Gram matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. In residual evaluations modulo 9, the action distinguishes the sheets: $S(ax, ay) = aS(x, y)$, whereas $P(ax, ay) = a^{-1}P(x, y)$. The cyclic realization therefore preserves a contragredient orientation between sum and product. The quotients of the integer lift are preserved in the corresponding arithmetic register.

Elementary parabolic transport is represented by

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The areal–volumetric incidence already constructed in (5.5) provides

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = F_{av}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The three operators act on declared real spaces: the augmentation module tensored with $\mathbb{R}$, the nilpotent transport plane, and the modal incidence plane. Sharing a matrix dimension does not identify these spaces or their coordinates.

Proposition 5.7 (Concrete generators of the three types). *The operators*

$$J_e = \frac{2C + I}{\sqrt{3}}, \quad J_p = N, \quad J_h = \frac{2A - 3I}{\sqrt{5}}$$

satisfy, respectively,

$$J_e^2 = -I, \quad J_p^2 = 0, \quad J_h^2 = I.$$

Proof. The relation for C implies $(2C + I)^2 = -3I$. Direct multiplication gives $N^2 = 0$. Finally, A has trace 3 and determinant 1, so $A^2 - 3A + I = 0$ and $(2A - 3I)^2 = 5I$. $\square$

Proposition 5.8 (Three evaluations of the exponential identity). *In the preceding realizations,*

$$\exp(\pi J_e) + I = 0, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2. \quad (5.11)$$

Proof. The elliptic specialization of the tritic exponential gives $\cos \pi I + \sin \pi J_e = -I$. Nilpotence removes exactly the terms of degree at least 2 in the second expression. For the third, $\varphi^2 = \varphi + 1$ implies

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2}.$$

The exponential therefore equals $\frac{3}{2}I + \frac{\sqrt{5}}{2}J_h = A$. $\square$

The first equality represents the elliptic half-turn; the complete turn satisfies $\exp(2\pi J_e) = I$. The second retains a nonzero linear term: the parabolic regime expresses transport, not absence of evolution. The third realizes a unimodular hyperbolic advance. Its eigenvalues are φ^2 and φ^{-2} ; one direction expands and the other contracts while preserving the determinant. The three evaluations use different parameters of a common exponential collection.

In that collection, e enters through the scalar evaluation $\exp(I) = eI$ and the law $\exp(sI) \exp(tI) = \exp((s + t)I)$. Its role is not restricted to the parabolic type. The coordinate e , obtained previously from its region, is recognized here as the unit-parameter evaluation of the exponential; this identity does not replace its coinductive genealogy.

5.6.2 Polynomial resolution of the regimes

The three realizations admit a compact presentation on the direct sum of their spaces. Define

$$\mathbb{J} = J_e \oplus J_p \oplus J_h, \quad Q = \mathbb{J}^2.$$

The preceding relations imply

$$\mathbb{J}^6 = \mathbb{J}^2, \quad Q^3 = Q.$$

The use of Q keeps this operator square separate from the dodecaphasic register K .

Proposition 5.9 (Polynomial regime projectors). *The maps*

$$p_e = \frac{Q^2 - Q}{2}, \quad p_p = I - Q^2, \quad p_h = \frac{Q^2 + Q}{2}$$

are mutually orthogonal idempotents, sum to I , and recover the three summands of the representation.

Proof. The three polynomials take values $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ at the points $-1, 0, +1$, respectively. The differences between their squares and themselves, as well as their cross-products, are multiples of $X^3 - X$. Substituting Q , which annihilates that polynomial, proves the identities. On the three blocks, Q acts as $-I, 0, I$. $\square$

This yields

$$\begin{aligned} \exp(t\mathbb{J}) &= p_e(\cos tI + \sin t\mathbb{J}) + p_p(I + t\mathbb{J}) \\ &\quad + p_h(\cosh tI + \sinh t\mathbb{J}). \end{aligned} \quad (5.12)$$

Each summand preserves its quadratic relation, and conjugation reverses its evolution. The total representation brings together the three regimes, but does not by itself define a transition operator between them. Such a transition requires the TPK transport maps with their domains, orientation, and memory. The exponential articulation allows their types to be recognized without replacing that dynamics by an identification of the fibres.

5.7 Compatible cylinders and arbitrary depth

The realization of a ternary block $b = (b_1, \dots, b_6)$ is the integer

$$\nu(b) = \sum_{j=1}^6 b_j 3^{6-j} \in \{0, \dots, 728\}.$$

A sequence of blocks defines integer prefixes by $N_{n+1} = 729N_n + \nu(b_{n+1})$, starting from $N_0 = m$, where m is the integer part determined by the readout. The preceding brackets place the closure, propagation, and self-scaling characters in (3, 4), (2, 3), and (1, 2), respectively; their initial values are therefore $m = 3, 2, 1$. Subsequent blocks refine the fractional part. The corresponding cylinder is

$$I_n = \left[\frac{N_n}{729^n}, \frac{N_n + 1}{729^n} \right], \quad |I_n| = 729^{-n}. \quad (5.13)$$

The half-open convention is used to select a representative when a value lies on a boundary. Passage to the limit uses the closures of these intervals and identifies the two equivalent boundary expansions. This qualification prevents the assumption that every intersection of nested half-open intervals is automatically nonempty.

Theorem 5.10 (Determination of the scalar realization). *A compatible block history determines a unique limit of the left endpoints of (5.13). If an exact character has rational brackets of arbitrarily small width and is separated from the boundary of every decimal cylinder examined, each decimal prefix of finite length is determined after a finite computation. The prefixes thus obtained are mutually compatible.*

Proof. Extending a block preserves the previous prefix by Euclidean division. The closed cylinders are nested and their diameter tends to zero; their left endpoints form a Cauchy sequence and have a unique limit. For a fixed decimal length, a value separated from the boundaries of the partition has positive distance from them. A sufficiently narrow bracket lies within a single cell and determines its prefix. Uniqueness of the cell and nesting of the partitions prove compatibility under truncation. At points with a terminating expansion, the exact boundary decision and its representation convention must be added; this decision does not follow from numerical width alone. $\square$

Corollary 5.11 (Finite determination and compatibility of regional prefixes). *For each of the realizations π, φ, e and every finite decimal length, the constructed brackets determine the corresponding prefix by a finite computation. The prefixes obtained at different depths are compatible under truncation.*

Proof. The closure, propagation, and self-scaling characters have the preceding brackets. After their recognition, the irrationality of π, e, φ ensures their separation from the rational boundaries of every finite decimal partition. The preceding theorem applies to each character. $\square$

Arbitrary depth expresses the quantifier in the corollary: for every finite horizon there is a computation determining its prefix. The mathematical object is the complete compatible collection; each execution obtains a finite projection of it.

5.8 Specular involution and synchronization of the regions

The joint nonadic connection transports the three characters together. In the ninth-phase chart, the self-scaling axis and the ternary sum of closure and propagation remain fixed under specular interchange. Helical inversion changes the direction of transport; the mirror of the two channels changes their relative orientation. These are different involutions, although both act on the same trajectory system.

The matching census values illustrate this coordination:

Phase	N_π	N_e	N_φ	Equality of counts
4	207	207	122	$N_\pi = N_e$
5	39	151	151	$N_e = N_\varphi$
6	110	110	69	$N_\pi = N_e$
7	81	29	81	$N_\varphi = N_\pi$
8	59	59	59	triple synchronization
9	43	43	43	triple synchronization

These integers count compatible coordinates of the regions. They do not assert equality between π , e , and φ as real numbers. If

$$\partial(a, b, c) = (a - b, b - c, c - a),$$

its image belongs to the lattice $A_2 = \{(u, v, w) \in \mathbb{Z}^3 : u + v + w = 0\}$. In phases eight and nine the relative component vanishes, but the diagonal component changes from $(59, 59, 59)$ to $(43, 43, 43)$. Relative synchronization coexists with a shift of -16 per channel. The closure that will lead to α must preserve both pieces of information: relative coincidence and memory evolution.

5.9 Sturmian helical suspension

The relation between bases nine and ten of the completion defines the dimensionless parameter

$$\delta = \frac{\log(10/9)}{\log 10}. \quad (5.14)$$

It is not obtained from a target physical constant. It is the relative logarithmic scale of two bases already present in the construction. Its use as a symbolic readout of refinement has an exact consequence.

Lemma 5.12 (Irrationality of the slope). *The number δ is irrational.*

Proof. If $\delta = p/q$, $q > 0$, then $(10/9)^q = 10^p$ and $9^q = 10^{q-p}$. The factorization of the left-hand side contains the prime three, whereas that of the right-hand side contains only the primes two and five. Equality is impossible. $\square$

The lower mechanical word, with intercept θ , is

$$z_n(\theta) = \lfloor (n+1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor \in \{0, 1\}.$$

It is obtained by observing the rotation $\theta \mapsto \theta + \delta \pmod{1}$ through the interval $[1 - \delta, 1)$. The event sequence has density δ , and its sum over any window of length L equals $\lfloor L\delta \rfloor$ or $\lceil L\delta \rceil$. Since $21 < 1/\delta < 22$, the gaps between events are twenty-one or twenty-two. The word indicating which intervals are short is another mechanical word, of slope $\sigma = 22 - 1/\delta$; its events are separated by six or seven primary intervals. The second scale is induced by the first and is not a second free parameter.

5.9.1 Capacity index, vacancies, and induction of the tritic regime

The vacancy sequence is linked to refinement through an increment identity. To exhibit it, fix the phase origin $\theta = 0$. Let

$$\rho = \log_{10} 9 = 1 - \delta, \quad \mathcal{C}_t = \lfloor (t+1)\rho \rfloor \quad (t \geq 0).$$

The index $\mathcal{C}_t$ expresses the decimal capacity of refinement in the normalization comparing a ternary block of 6 positions, with 729 possibilities, and a decimal block of 3 positions, with 1000 possibilities. Indeed, $\log_{1000} 729 = \log_{10} 9 = \rho$. The letter K is reserved for the dodecaphase record to be constructed later.

Proposition 5.13 (Complementarity of capacity and vacancy). *For $t \geq 1$,*

$$\mathcal{C}_t - \mathcal{C}_{t-1} = 1 - z_t(0). \quad (5.15)$$

Consequently, if $0 \leq a < b$,

$$\mathcal{C}_b - \mathcal{C}_a + \sum_{t=a+1}^b z_t(0) = b - a.$$

Proof. The irrationality of δ gives

$$\mathcal{C}_t = t + 1 - [(t+1)\delta] = t - [(t+1)\delta].$$

Subtracting two consecutive terms yields (5.15). The sum over the interval telescopes. $\square$

Each transition increases the block counter by one. If $z_t = 0$, capacity also increases; if $z_t = 1$, capacity remains unchanged while a transition is added to the history. A vacancy is here an event of capacity retention, with a determined position and antecedent. The preceding balance preserves the length exactly, partitioned into acquired capacity and recorded vacancies. The observed phase $g_t^{\text{cap}} = [\mathcal{C}_t]_9$ satisfies

$$g_t^{\text{cap}} - g_{t-1}^{\text{cap}} \equiv 1 - z_t \pmod{9}.$$

This capacity phase and the chronological phase $[t]_9$ are two different observations. Local capacity retention and a complete holonomy turn have distinct mechanisms, although both allow the phase observation to remain unchanged while the state changes.

The relationship with TRIT can be followed at the vacancy positions themselves. For $j \geq 1$, define

$$p_j = \left\lfloor \frac{j}{\delta} \right\rfloor, \quad v_j = \mathcal{C}_{p_j}, \quad h_j = p_{j+1} - p_j.$$

The inequality $p_j\delta < j < (p_j + 1)\delta$ identifies p_j as the position of event j . Since $0 < \delta < 1$, it also gives $[(p_j + 1)\delta] = j$, and hence

$$v_j = p_j - j, \quad v_{j+1} - v_j = h_j - 1.$$

If $s_j = 22 - h_j$, then $s_j \in \{0, 1\}$, and

$$[v_{j+1}]_3 = [v_j - s_j]_3. \quad (5.16)$$

An interval of length 22 preserves the class; one of length 21 shifts it by one unit in the negative direction. The canonical selector is applied to the positive phase $r_j = 1 + [v_j]_9$:

$$\tau_j = M_{\text{ph}}(r_j).$$

A tritic regime is thus determined by the capacity index retained at each event, in addition to the binary vacancy coding.

The positions of the short intervals form the induced mechanical coding of slope $22 - 1/\delta$. Their spacings 6 and 7 determine the plateau lengths of the class $[v_j]_3$, once the origin and endpoint convention are fixed. The first classes are 2 for 6 events, 1 for 7, and 0 for 7; the first segment of 6 is the initial restriction of a plateau. Their signatures traverse 0, -1 , $+1$. Equations (5.15) and (5.16) make explicit the link between refinement, vacancies, and regime selection. The quadratic TRIT readout subsequently realizes these types through their respective generators; the temporal event and its geometric representation therefore retain an identifiable dependency.

Inverting the monotone index provides another useful observation:

$$\mathcal{T}(k) = \min\{t : \mathcal{C}_t = k\} = \left\lceil \frac{k}{\rho} \right\rceil - 1 \quad (k \geq 1).$$

If $\sigma = \rho^{-1} - 1$, the number of blocks required to gain m units of capacity is

$$\mathcal{T}(k+m) - \mathcal{T}(k) = m + [(k+m)\sigma] - [k\sigma] \in \{m + [m\sigma], m + \lceil m\sigma \rceil\}.$$

In particular, 9 units of capacity require 9 or 10 blocks. The period of a finite phase remains exact, while the return time measured by the other index admits two values. This distinction belongs to the dynamics of refinement and is not an a posteriori correction of its numerical values.

Theorem 5.14 (Complexity of the phase readouts). *For $m \geq 1$, the mechanical word has factor complexity $p(m) = m + 1$. When the phase $n \bmod 9$ is retained, the complexity is $p_9(m) = 9(m + 1)$; when only its ternary class is retained, it is $p_3(m) = 3(m + 1)$. All three topological entropies vanish.*

Proof. For factors of length m , the predecessors of the two endpoints of the coding interval form $m + 1$ distinct points on the circle: coincidence between shifted endpoints removes duplicates, and irrationality prevents further coincidences. These points determine $m + 1$ distinct intervals on each of which the itinerary is constant. Every irrational orbit visits each interval, hence $p(m) = m + 1$.

For a fixed phase modulo nine, the positions of that phase follow a rotation with step 9δ , which is also irrational. Density implies that every factor occurs in every phase, and the first coordinate distinguishes its nine copies. The same argument with step 3δ gives three copies after ternary reduction. Finally, $m^{-1} \log(c(m + 1)) \rightarrow 0$ for $c = 1, 3, 9$. $\square$

The phase product $C_9 \times \mathbb{T}$, translated by $(1, \delta)$, has frequency characters

$$\frac{a}{9} + k\delta \pmod{1}, \quad a \in \mathbb{Z}/9\mathbb{Z}, k \in \mathbb{Z}.$$

The spectral module is not $\log 10$: it is the additive module generated by $1/9$ and δ , modulo the integers. The factor $\log 10$ belongs to the normalization in (5.14).

An operator presentation makes memory explicit. On $\ell^2(\mathbb{Z})$, let $T|n\rangle = |n + 1\rangle$, $V_9|n\rangle = \zeta_9^n|n\rangle$, $W_\delta|n\rangle = e^{2\pi i n \delta}|n\rangle$, and $D_M|n\rangle = \lfloor n/9 \rfloor |n\rangle$. On finitely supported vectors,

$$V_9 T = \zeta_9 T V_9, \quad W_\delta T = e^{2\pi i \delta} T W_\delta, \quad [D_M, T^9] = T^9. \quad (5.17)$$

The first two relations are Weyl covariance relations; the third expresses that a nonadic turn increases the memory degree. The base rotation is abelian, whereas the algebra of observables and translations is not. The inversion $|n\rangle \mapsto |-n\rangle$ conjugates forward and backward motion. This realization provides an aperiodic helical suspension with finite phase return, not a periodic word of length nine.

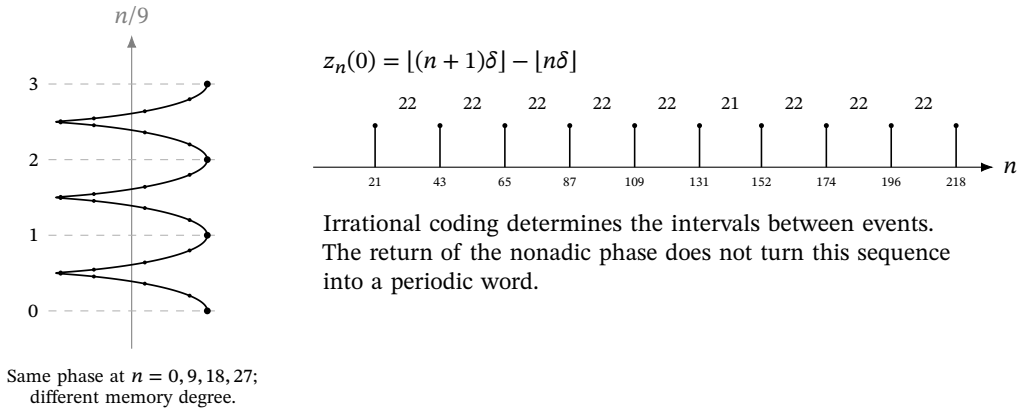


Figure 5: Complementary representations of transport. Left: projection of the helix $(\cos(2\pi n/9), \sin(2\pi n/9), n/9)$: advancing nine steps recovers the phase and increases the height. Right: the first ten events of the lower word for $\delta = \log_{10}(10/9)$, with exact gaps 21 or 22. The helical representation introduces no additional physical metric.

For this count we fix the intercept $\theta = 0$ and the cumulative endpoints $(N_0, \dots, N_4) = (0, 729, 972, 999, 1000)$, corresponding to the cubic ordering $(729, 243, 27, 1)$. The lower counts are $\lfloor N_j \delta \rfloor - \lfloor N_{j-1} \delta \rfloor$, giving $(33, 11, 1, 0)$. The upper counts are $\lfloor N_j \delta \rfloor - \lfloor N_{j-1} \delta \rfloor$, with the same initial boundary zero, giving $(34, 11, 1, 0)$. These values depend on the intercept and the ordering of the strata; they are not asserted for an arbitrary initial phase. Their totals are forty-five and forty-six. The increment corresponds to a single boundary unit, located in the first stratum of this ordered partition. The numbers thirty-four and eleven thus appear in a single count preceding any chart of

units; identifying them with physical exponents additionally requires the corresponding dimensional derivation.

The designation *aperiodic crystal* is used here for this symbolic order, with its complexity, spectrum, and return specified. Vanishing topological entropy does not by itself imply vanishing physical dissipation: a thermodynamic realization requires a defined energy dynamics and erasure operation. Likewise, this Sturmian readout neither replaces all TPK operators nor proves by itself periodicity or aperiodicity of the complete output of α . Its role is to make explicit an order structure that a mere return of the schedule does not describe.

5.10 Nonadic connection, holonomy and monodromic representation

The regional connection already constructed is now studied through three objects: transport of the complete state, representation of its windings and suspension of its itineraries. The APP-TRIT-TPK operations and the initial catalogue retain the preceding definitions. The necessary distinction here concerns what each representation retains of the same history (see 4.8).

At prolongation index k , a state chart preserves

$$\hat{x}_k = (B_k, C_k, p_k, c_k, \Sigma_k, W_k, g_k, m_k).$$

Here B_k collects the three regional blocks, C_k is the cylinder, p_k the prefix, c_k the carry register, Σ_k the boundary signature and W_k the incidence information. The phase is $g_k = 1 + [(k-1)]_9$; m_k counts complete windings. The reduction $\beta_k = [m_k]_{12}$ preserves the dodecaphasic position, while the integer m_k distinguishes its successive windings.

Definition 5.15 (Transport of a nonadic winding). Let $\mathcal{R}_k \subseteq \hat{X}_k \times \hat{X}_{k+1}$ be the TPK admissible-extension relations. The realization of holonomy at level k is the relational composition

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \cdots \circ \mathcal{R}_k.$$

For $s \geq 1$, its prolongation to s windings is

$$\Gamma_{9s,k} = \Gamma_{9,k+9(s-1)} \circ \cdots \circ \Gamma_{9,k}.$$

The relational formulation preserves admissible continuations. On an individually specified history, each arrow leads to the state actually prolonged; the isolated visible block may still admit more than one continuation. This difference is precisely the role of memory and boundary conditions in determining the trajectory.

Proposition 5.16 (Iteration of return and accumulated memory). *In every admissible composition of s windings,*

$$g_{k+9s} = g_k, \quad m_{k+9s} = m_k + s, \quad \beta_{k+9s} = \beta_k + s \pmod{12}.$$

Proof. The one-winding rule preserves phase and increments the counter by one. Applying it to the already transported state repeats exactly these two operations. Induction gives the first two equalities; reducing the second modulo 12 gives the third. $\square$

The counter thus provides an integer witness of the distinction between windings. After 12 windings the finite phase pair is restored, but the counter increases by 12. The nonsplit extension (4.5) expresses composition of the calendar through its cocycle; the trajectory also preserves the integer lift of this datum. Preservation here means transport of previous contributions, not immobility of every coordinate.

5.10.1 Bilateral representation of the return index

The shift T of (5.17) admits an expression in phase and winding coordinates. Number phases from 0 to 8, by shifting the origin relative to g_k , and write $k = 9m + r$, with $0 \leq r < 9$, on the integer line covering the phase cycle. On $\mathcal{H}_{\text{ind}} = \ell^2(\mathbb{Z})$, the basis $|k\rangle = |r, m\rangle$ expresses that same unitary shift:

$$T|r, m\rangle = \begin{cases} |r+1, m\rangle, & r < 8, \\ |0, m+1\rangle, & r = 8. \end{cases}$$

The inverse decreases r by one when $r > 0$, and takes $|0, m\rangle$ to $|8, m-1\rangle$. Passage through the phase origin thus incorporates the exact quotient unit.

Proposition 5.17 (Representation of cycle monodromy). *Let $S = T^9$. Then*

$$S|r, m\rangle = |r, m+1\rangle, \quad \mathbb{Z} \longrightarrow \mathcal{U}(\mathcal{H}_{\text{ind}}), \quad s \longmapsto S^s$$

is a faithful unitary representation of the cycle's winding group.

Proof. Nine shifts preserve the remainder upon division by 9 and increment the quotient. Powers satisfy $S^{s+t} = S^s S^t$, and $S^s|r, m\rangle = |r, m+s\rangle$ distinguishes every integer s . A permutation of an orthonormal basis is unitary. $\square$

This representation corresponds to the cover of the cycle, whose winding group is $\mathbb{Z}$, and represents phase together with its counter. The TPK state additionally contains the trajectory coordinates already described. The bilateral extension admits negative indices as coordinates of the cover; a trajectory beginning at an initial condition uses its admissible half-line. This provides an operator inverse of the index without turning every extension relation into a bijection of the complete state.

With $D_M|k\rangle = \lfloor k/9 \rfloor |k\rangle$, the identity $[D_M, S] = S$ expresses memory increment on the dense domain of finitely supported vectors. If $\mathcal{J}|k\rangle = |-k\rangle$ and P_0 projects onto indices divisible by 9, one also obtains

$$\mathcal{J}T\mathcal{J} = T^{-1}, \quad \mathcal{J}D_M\mathcal{J} = -D_M - (I - P_0). \quad (5.18)$$

Indeed, for $k = 9m + r$, $\lfloor -k/9 \rfloor = -m$ if $r = 0$, and equals $-m - 1$ if $r > 0$. Inversion thus preserves the boundary term of Euclidean division.

5.11 Block clock and capacity clock

The index of a refinement map and the number of available decimal positions obey related but different laws. To express them, reserve n for the ternary-block index and define

$$\rho = \log_{1000} 729 = \log_{10} 9, \quad \delta = 1 - \rho, \quad \mathcal{C}_n = \lfloor (n+1)\rho \rfloor, \quad n \geq 0.$$

The symbol $\mathcal{C}_n$ denotes capacity here, not the dodecaphasic register K . At the canonical phase origin, the preceding mechanical word gives

$$Z_n = \sum_{j=0}^n z_j(0) = \lfloor (n+1)\delta \rfloor.$$

Proposition 5.18 (Exact balance between capacity and vacancies). *For $n \geq 0$ and, in the second identity, $n \geq 1$,*

$$\mathcal{C}_n = n - Z_n, \quad \mathcal{C}_n - \mathcal{C}_{n-1} = 1 - z_n(0).$$

Proof. Irrationality of δ implies $\lfloor (n+1)\delta \rfloor = \lfloor (n+1)\delta \rfloor + 1$. Then

$$\lfloor (n+1)(1-\delta) \rfloor = n+1 - \lfloor (n+1)\delta \rfloor = n - Z_n.$$

Subtracting two consecutive identities completes the proof. $\square$

An event $z_n = 1$ adds a block to the history while maintaining capacity $\mathcal{C}_n$; an event $z_n = 0$ increments both counters. In any interval, capacity increments and vacancies sum exactly to its length. Reducing modulo q , for $q = 9$ or $q = 108$, gives

$$[\mathcal{C}_n]_q = [n]_q - [Z_n]_q.$$

Uniform block phase and capacity phase are related through integrated vacancy memory. In particular, a capacity hold and a complete winding of the connection have distinct mechanisms, although both may preserve a phase observation.

Monotone inversion of the clock gives a first-arrival time:

$$T_{\text{cap}}(a) = \min\{n : \mathcal{C}_n = a\} = \left\lceil \frac{a}{\rho} \right\rceil - 1, \quad a \geq 1.$$

With $\sigma = \rho^{-1} - 1$, the time required to gain b units of capacity is

$$T_{\text{cap}}(a + b) - T_{\text{cap}}(a) = b + [(a + b)\sigma] - [a\sigma].$$

The difference of ceilings takes one of the two integers adjacent to $b\sigma$. Thus, 9 or 10 blocks are required to gain 9 capacity units, and 113 or 114 to gain 108. These are inverse-clock times measured from a first arrival; the winding counter is always interpreted in the index to which it belongs.

5.12 Symbolic suspension and aperiodic time crystal

Rotation coding brings together phase periodicity, local recurrence and global aperiodicity. Let $\mathcal{X}_\delta \subseteq \{0, 1\}^{\mathbb{Z}}$ be the closure, in the product topology, of the translates of the bilateral word $z_n(\theta)$. Denote the shift of its sequences by σ_{sh} . Preserving a nonadic phase, the step is

$$F(r, z) = ([r + 1]_9, \sigma_{\text{sh}} z), \quad (r, z) \in C_9 \times \mathcal{X}_\delta.$$

Definition 5.19 (Symbolic aperiodic time crystal). The unit-roof suspension of the preceding system is the space

$$\mathcal{S}_\delta = ((C_9 \times \mathcal{X}_\delta) \times \mathbb{R}) / \sim, \quad (y, t + 1) \sim (Fy, t),$$

with flow $\Phi_s[y, t] = [y, t + s]$. This model of temporal order, together with its vacancy coding and winding-counter representation, is called a *symbolic aperiodic time crystal*.

The adjective temporal designates the action of the flow and its return section; aperiodicity characterizes its itineraries. The frequency of ones in each sequence is δ : the uniform discrepancy bound strictly less than one, obtained by telescoping in any window, persists under closure. A periodic sequence would have rational frequency. Thus, $\mathcal{X}_\delta$ contains no periodic orbits. The suspension has no closed orbits either: a flow return would require an integer time and periodicity of the itinerary.

Finite words do recur. Length- m itineraries arise from a partition of the circle by $m + 1$ points; each interval is visited recurrently by the irrational rotation. Even when times are restricted to a fixed phase, the step 9δ remains irrational. This is why every word appears in all 9 phases, and explains the previously established complexities

$$p(m) = m + 1, \quad p_9(m) = 9(m + 1), \quad p_3(m) = 3(m + 1).$$

Recurrence of local configurations coexists with the absence of a global period. Its topological entropy is zero because these complexities grow linearly.

Realization through the observables V_δ, W_δ and shift T expresses the same order through the covariance relations (5.17). Uniform-clock frequencies belong to $\frac{1}{9}\mathbb{Z} + \delta\mathbb{Z}$, modulo 1. The capacity clock additionally preserves the relation $[\mathcal{C}_n]_9 = [n]_9 - [Z_n]_9$; it is not replaced by uniform phase when interpreting returns. The term *time crystal* is thus defined through a mathematical dynamical system. Its physical realization requires specifying energy dynamics, its temporal observables and stability, data distinct from the present symbolic construction.

5.13 Mirror symmetry of the regions and dual realization

Mirror interchange acts on the regional block $B = (u^\pi, u^e, u^\varphi) \in (\mathbb{F}_3^6)^3$ by

$$\mathfrak{c}B = (u^e, u^\pi, u^\varphi).$$

Its fixed axis and aggregate are, respectively, u^φ and $s = u^\pi + u^e$. This fixedness concerns the involution in a fibre; both coordinates may evolve during nonadic transport. With $q = u^\pi + u^e + u^\varphi$, $a = u^\pi + u^e - u^\varphi$ and $d = u^\pi - u^e$, one recovers

$$u^\varphi = 2(q - a), \quad s = q - u^\varphi, \quad u^\pi = 2(s + d), \quad u^e = 2(s - d).$$

The equalities hold componentwise in $\mathbb{F}_3$, where $2^{-1} = 2$. The mirror preserves q, a, s and reverses d ; this antisymmetric coordinate individualizes the allocation that the aggregate preserves without distinguishing. Time-index inversion $\mathcal{I}$ and channel involution $\mathfrak{c}$ therefore act on different data.

Phase return has an explicit witness:

$$B_1 = (012222 \mid 121221 \mid 112202),$$

$$B_{10} = (101222 \mid 112221 \mid 112002).$$

Both positions have phase 1; the block, axis and memory have changed. Likewise, synchronization of the censuses at phases 8 and 9 annihilates their relative differences, while the diagonal component changes from $(59, 59, 59)$ to $(43, 43, 43)$. Equality of a relative observable does not exhaust the state determining it.

The dual realization preserves that architecture. In Section 10.1, the same prefix determines nested cylinders on one side and a sequence of encoded words with their frames on the other. The first map passes to the limit by contraction of diameters; the second satisfies $r_{n,m}Q_n = Q_m p_{n,m}$ and defines Q_∞ . Figure 6 anticipates these two maps. The subsequent incidence route constructs codes, lattices and the Moonshine realization in their own domains; the automorphism group acts on that realization, not on the digits through an implicit identification.

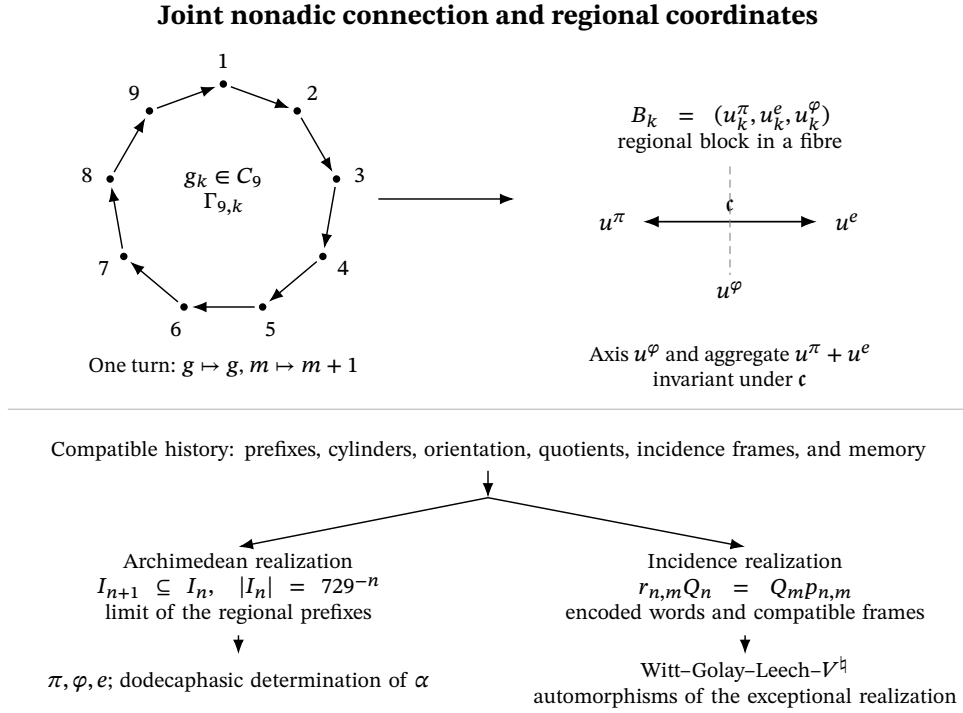


Figure 6: Nine phases of a connection and two realizations of the compatible history. The cycle represents the phase; each turn increments the counter. The regional diagram represents the involution c , which interchanges closure and propagation and fixes the self-scale axis in a fibre. This invariance does not require the axis to remain constant during transport. The two lower maps use the same prefix: one determines its numerical limit, and the other preserves words and incidence frames. The maps of this second realization are specified in Section 10.1; the exceptional route is developed afterwards.

5.13.1 Rectilinear representation of phases and return

The representation on the segment $[0, 10]$ preserves the positional origin of APP and orders its nine interior marks. In this diagram, the integers $1, \dots, 9$ are phase representatives. The horizontal position indicates the ordering of the charts; the upper labels specify regional operations. Figure 7 thus brings together the initial line, fixation of the self-scaling axis, the specular distribution of the two channels, and return with memory (see 5.10).

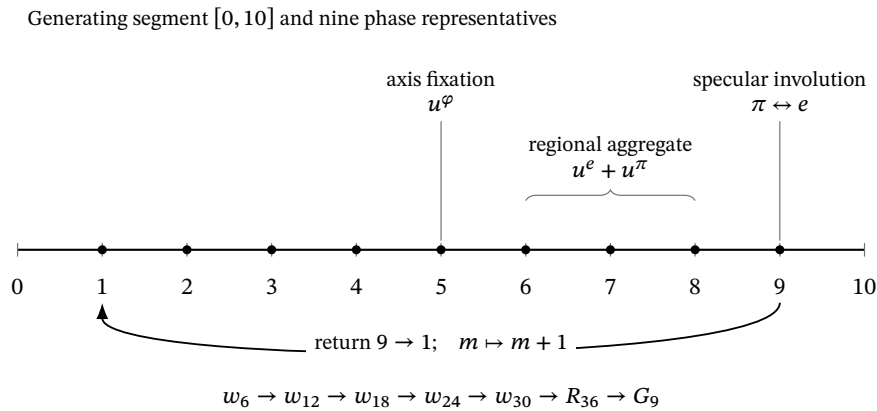


Figure 7: Rectilinear representation of the joint nonadic connection. The mark 5 locates the first strong saturation and the regional self-scaling axis; the brace 6–8 identifies the ternary aggregate fixed in each fibre; phase 9 selects the closure orientation. Return preserves phase and advances memory. The abscissae number operational positions, whereas the labels u^π, u^e, u^φ designate blocks of $\mathbb{F}_3^6$.

Axis fixation is formulated in the fibre of a given boundary. If $q = u^\pi + u^e + u^\varphi$ and $a = u^\pi + u^e - u^\varphi$, then

$$u^\varphi = 2(q - a), \quad s = u^\pi + u^e = q - u^\varphi \quad \text{in } \mathbb{F}_3^6.$$

The data (q, a) determine the axis and the aggregate; the ordered distribution of s retains the specular freedom resolved by prolongation. The fifth phase has one local solution and one extension, with boundary datum $c = 11111$ and residual signature $\rho_W = 000$. This is the meaning of its first strong saturation. The symbol φ above the mark 5 identifies the regional axis involved in that operation.

The intermediate phases preserve this decomposition within each fibre, while their boundary data evolve. The complete census of local multiplicities and extensions of the distinguished branch is

phase	1	2	3	4	5	6	7	8	9
N_{loc}	3	2	5	4	1	1	3	3	2
N_{ext}	2	5	4	1	1	3	3	2	1

Phase 6 combines local uniqueness with three prolongations; phase 7 resolves a triple specular fibre; phase 8 synchronizes the three regional censuses. Phase 9 retains two local distributions with the same axis and aggregate, one of which prolongs. These counts express different functions within the same connection and justify the marks on the line.

In particular, for phases 6, 7, 8, the regional sums are 012100, 101001, and 112000, respectively. The conservation represented by the brace is invariance under specular distribution within each fibre. Subsequent real evaluation applies the readouts of compatible cylinders to complete histories; the internal sum represented in the figure belongs to the ternary domain.

Finally, the arc $9 \rightarrow 1$ represents passage from one turn to the next. The formulas in Section 5.10 preserve the phase remainder and increment the turn counter. The rectilinear diagram and the cyclic diagram in Figure 6 therefore express two complementary aspects of the same holonomy: the ordering of positions and composition of return on the state with memory.

5.14 Temporal transport and tritic exponential composition

The Euler relations in Section 5.6 describe local realizations of the three regimes. For a fixed signature τ , the generator satisfies $J_\tau^2 = -\tau I$; algebraic conjugation takes J_τ to $-J_\tau$. Consequently,

$$\exp(sJ_\tau) \exp(tJ_\tau) = \exp((s+t)J_\tau), \quad \overline{\exp(tJ_\tau)} = \exp(-tJ_\tau),$$

and the product of an evolution with its conjugate is the unit. In the elliptic type this is realized by $\cos^2 t + \sin^2 t = 1$; in the hyperbolic type, by $\cosh^2 t - \sinh^2 t = 1$; in the parabolic type, $(I+tN)(I-tN) = I$, since $N^2 = 0$. The same conservation law admits three quadratic expressions.

The elliptic realization closes a winding; the nilpotent one preserves a linear displacement; the hyperbolic one separates expansion and contraction with unit determinant. At the regional parameters already generated,

$$\exp(\pi J_e) = -I, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{\text{av}}^2.$$

The evaluation $\exp(I) = eI$ recognizes the unit of the common exponential law. The number e is not assigned exclusively to the parabolic regime.

Transport of a local realization by an isomorphism U preserves these relations: if $J' = UJU^{-1}$, the exponential series gives $U \exp(tJ) = \exp(tJ')U$. This identity transports the regime and its norm. A transition between different quadratic types belongs, by contrast, to the TPK selection of sheets and regimes; it is not a conjugation changing the value of J^2 . The temporal product therefore preserves the order of transitions and information about their endpoints.

This articulation distinguishes closure of a local representation from holonomy of a history. One may have $\exp(2\pi J_e) = I$ at the same time as $S|r, m\rangle = |r, m+1\rangle$. The first equality restores regime orientation; the second records another winding. The discrete structure preserves both: local return and the accumulated provenance of the state realizing it.

5.15 Multiplicative irreducibility and return periods

The distinction between structure, evaluation, and phase also appears in the multiplicative sector. The local APP product contains both the residue and the quotient; its iteration makes it possible to construct normal forms and factorizations before examining their decimal periods. This development places the expression *prime mode* in a precise domain and relates irreducibility to the operations already defined, not to an isolated coincidence of periods (see 5.15).

Let

$$\mathcal{W}_9 = \{\emptyset\} \sqcup \bigsqcup_{\ell \geq 1} \{1, \dots, 9\}^\ell.$$

Sequences are ordered from the least significant digit. The evaluation $\text{ev}_9(u) = \sum_j u_j 9^j$, with $\text{ev}_9(\emptyset) = 0$, is a bijection onto the nonnegative integers: for a positive integer, dividing $n - 1$ by 9 uniquely determines the first digit and the continuation. This is bijective nonadic numeration; it preserves the boundary representative 9.

The sum $\boxplus_\#$ is obtained by applying the additive APP sheet to the digits at each position and propagating their quotients. If there is only one digit, it constitutes the provisional residue; a nonzero incoming quotient is normalized through a second additive operation. The sum of the quotients is transported to the next position. To multiply a sequence by a digit d , the multiplicative cell produces $u_j d = 9q_j + r_j$, and the incoming quotient c_j is incorporated through

$$(w_j, c_{j+1}) = \begin{cases} (r_j, q_j), & c_j = 0, \\ (R^+(r_j, c_j), q_j + q^+(r_j, c_j)), & c_j > 0. \end{cases}$$

The procedure begins with $c_0 = 0$ and incorporates the final quotient when it is nonzero. During addition, $c_j \leq 2$; during multiplication by a digit, $c_j \leq 9$. The local operations thus receive arguments within their domain. Denote this second procedure by $d \odot_\# u$, with $d \odot_\# \emptyset = \emptyset$.

For $v = (v_0, \dots, v_{m-1})$, the full product is constructed by the recurrence

$$a_m = \emptyset, \quad a_j = (9 \odot_\# a_{j+1}) \boxplus_\# (v_j \odot_\# u), \quad u \boxtimes_\# v = a_0.$$

The rule composes the two local sheets. Its evaluation satisfies

$$\text{ev}_9(u \boxtimes_\# v) = \text{ev}_9(u) \text{ev}_9(v). \quad (5.19)$$

Indeed, each transition preserves the partial sum plus the pending quotient weighted by the power of 9. The recurrence then gives $\text{ev}_9(a_j) = 9 \text{ev}_9(a_{j+1}) + v_j \text{ev}_9(u)$, whose iteration proves (5.19). Uniqueness of the normal form transports associativity and commutativity to the constructed product; its unit is (1).

Definition 5.20 (Reduced multiplicative coproduct). On the vector space of finite support generated by the nonempty normal forms, define

$$\tilde{\Delta}_\# e_w = \sum_{\substack{u \boxtimes_\# v = w \\ u, v \neq (1)}} e_u \otimes e_v.$$

A nonunit normal form is irreducible when its image under $\tilde{\Delta}_\#$ is zero.

The factorization fibres are finite by (5.19). In the completion $\mathcal{H}_\# = \ell^2(\mathcal{W}_9 \setminus \{\emptyset\})$, the images of distinct forms have disjoint supports. If $d_\#(w)$ counts the ordered nonunit factorizations, the domain of the closure is

$$\text{Dom}(\overline{\tilde{\Delta}_\#}) = \left\{ x \in \mathcal{H}_\# : \sum_w d_\#(w) |x_w|^2 < \infty \right\}.$$

Indeed, the squared norm of the image is that weighted sum. Therefore, the operator

$$A_\# = (\overline{\tilde{\Delta}_\#})^* \overline{\tilde{\Delta}_\#}$$

is diagonal, positive, and self-adjoint: its eigenvalue at e_w is the number of ordered nonunit factorizations of w . The projection

$$P_{\text{irr}} = (I - |e_{(1)}\rangle\langle e_{(1)}|)\mathbf{1}_{\{0\}}(A_{\#})$$

selects exactly the irreducible forms. Through multiplicative evaluation, these correspond to the ordinary primes. Selection depends on the factorization structure and keeps the unit state separate.

For a value $n \geq 2$ coprime to 10, decimal division induces the residue permutation $r \mapsto 10r \pmod{n}$. Return of the residue 1 has period

$$\tau_n = \text{ord}_n(10).$$

If $n = \prod_p p^{a_p}$, the Chinese remainder theorem gives

$$\tau_n = \text{mcm}_{p^{a_p} \parallel n} \text{ord}_{p^{a_p}}(10).$$

The composite return synchronizes the returns of the prime powers. For a prime $p \notin \{2, 3, 5\}$, we have

$$\tau_p \mid p - 1, \quad M_p = \frac{10^{\tau_p} - 1}{p} \in \mathbb{N}, \quad M_p \equiv 0 \pmod{9}.$$

The last congruence expresses the nonadic closure of the decimal repetition: $10^{\tau_p} - 1$ is a multiple of 9, and p is invertible modulo 9. The same congruence holds for any n coprime to 90. In particular, 7, 13, 77, 91, 143 have decimal period 6, although only the first two are irreducible. The coproduct and the period therefore record different properties of the same constructed value.

This distinction makes precise the relationship with aperiodic refinement. The vacancy sequence determines the increments of the capacity index; factorization determines the multiplicative modes; decimal division determines their return times. The nonadic representative preserves phase closure, while the quotient and factorization preserve information that this closure does not distinguish. The link with the spectral constructions of the treatise resides in those operators and their intertwining. The proofs of their subsequent analytic results retain their own development; they are not replaced by the periodicity of a residual observation.

5.15.1 Vacancy discrepancy and Dirichlet readouts

The link also admits an exact analytic expression. The same encoding $z_n = z_n(0)$, for $n \geq 1$, is preserved, and two subsequent observables are considered: the series over all indices and the product over the indices selected by irreducibility. For $\text{Re } s > 1$,

$$Z_{\text{vac}}^+(s) = \sum_{n \geq 1} \frac{z_n}{n^s}, \quad Z_{\text{vac}}^\times(s) = \prod_p (1 - p^{-s})^{-z_p}.$$

Both converge absolutely in that half-plane. Their common factor is the density δ , but their discrepancies differ.

Proposition 5.21 (Separation of vacancy density). *We have*

$$Z_{\text{vac}}^+(s) = \delta \zeta(s) + H_{\text{vac}}(s), \quad H_{\text{vac}} \in \mathcal{O}(\text{Re } s > 0).$$

In $\text{Re } s > 1$, with logarithms defined by the absolutely convergent Euler series,

$$Z_{\text{vac}}^\times(s) = \zeta(s)^\delta G_{\text{vac}}(s), \quad \log G_{\text{vac}}(s) = \sum_p (z_p - \delta) \log(1 - p^{-s})^{-1}.$$

Proof. The centred partial sums satisfy

$$A_N := \sum_{n=1}^N (z_n - \delta) = \lfloor (N+1)\delta \rfloor - N\delta, \quad |A_N| < 1.$$

Abel summation represents the centred term by $s \int_1^\infty A_{\lfloor x \rfloor} x^{-s-1} dx$. The integral converges locally uniformly in $\text{Re } s > 0$, proving holomorphy. The second identity follows by separating $z_p = \delta + (z_p - \delta)$ in the logarithm of the product, within its domain of absolute convergence. $\square$

The decomposition makes visible what the readout on primes adds: it examines the same vacancy sequence on a subset determined by factorization. Control of the discrepancy over all integers gives the first holomorphic extension; the second observable preserves a discrepancy of its own on primes. These are two applications of the same discrete structure, with explicit analytic domains. This articulation explains their relevance to the arithmetic core without transferring to this article the proofs of the terminal results of the treatise.

6 Integral compatibility of regional generation, cylindrical refinement, and incidence

Regional prolongation produces ordered sequences of blocks together with their reading depths. The Archimedean projection and ternary incidence are calculated on these same sequences. The connection between the two realizations can be formulated entirely through integer operations: selection by rational intervals, Euclidean division, lifting of six trits, carry reading, and the Witt transformation. This formulation makes it possible to trace each coefficient from its generating block to its subsequent use.

The construction uses the three regional readers already obtained from the APP-TRIT-TPK state. We denote them by the indices $c \in \{P, E, F\}$, corresponding, respectively, to the closure, propagation, and autoscaling regions. Their real realization will be written as v_c . The subsequent identifications $v_P = \pi_{\text{HMT}}$, $v_E = e_{\text{HMT}}$, and $v_F = \varphi_{\text{HMT}}$ preserve this causal order. The operation that emits a block inspects the rational intervals of the regional reader; the real value enters the proof of correctness of that operation.

6.1 Positional emission through rational separation of cells

For each region, two rational sequences $\ell_c(j), u_c(j)$, constructed by its reader, are available and satisfy

$$\ell_c(j) \leq v_c \leq u_c(j).$$

The cell-separation property proved for these readers states that, for each integer $s > 0$, there exists J such that, for every $j \geq J$,

$$\lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1 = \lfloor sv_c \rfloor. \quad (6.1)$$

Avoidance of rational boundaries by the regional values is the hypothesis of the preceding theorems that guarantees this separation. Here we use their operational conclusion together with the readers and intervals that produce it.

Definition 6.1 (Separation index and emitted prefix). For $s > 0$, let

$$j_c(s) = \min\{j \in \mathbb{N} : \lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1\}.$$

For an integer base $b \geq 2$ and a depth $n \geq 0$, define

$$P_c(b, n) = \lfloor b^n \ell_c(j_c(b^n)) \rfloor. \quad (6.2)$$

The prefix contains the integer part and the first n positions in base b ; consequently, $P_c(b, 0)$ is the regional integer part.

Theorem 6.2 (Termination and correctness of emission). *The index $j_c(s)$ exists for every $s > 0$. The emitted prefixes satisfy*

$$P_c(b, n) = \lfloor b^n v_c \rfloor, \quad P_c(b, n) \leq b^n v_c < P_c(b, n) + 1. \quad (6.3)$$

For each scale s , a single sufficiency index $J(s)$ can also be chosen that is valid simultaneously for all three regions.

Proof. The eventual equality in (6.1) makes the set defining $j_c(s)$ nonempty. Its minimum exists by the well-ordering of $\mathbb{N}$. At any index where the equality holds, the rational lower bound gives $\lfloor s\ell_c(j) \rfloor \leq \lfloor sv_c \rfloor$. The upper bound, together with avoidance of a cell boundary, gives $\lfloor sv_c \rfloor \leq \lfloor su_c(j) \rfloor - 1$. When the two integer endpoints coincide, the intervening integer coincides with them. This proves (6.3). For the joint statement, it suffices to take the maximum of the three sufficiency indices obtained at the same scale. The joint index controls all three emissions simultaneously, while each region retains its own intervals. $\square$

Proposition 6.3 (Exact compatibility of prefixes). For $n, m \geq 0$,

$$\left\lfloor \frac{P_c(b, n+m)}{b^m} \right\rfloor = P_c(b, n). \quad (6.4)$$

In particular, the next block $d_c(b, n) = P_c(b, n+1) \bmod b$ satisfies

$$P_c(b, n+1) = bP_c(b, n) + d_c(b, n), \quad 0 \leq d_c(b, n) < b. \quad (6.5)$$

Proof. Write $b^n v_c = N + \theta$, with $N = P_c(b, n)$ and $0 \leq \theta < 1$. Then $P_c(b, n+m) = b^m N + \lfloor b^m \theta \rfloor$, and the second summand belongs to $\{0, \dots, b^m - 1\}$. Euclidean division by b^m recovers N . The specialization $m = 1$ yields (6.5) and its residual bound. $\square$

6.2 Regional blocks of six trits at arbitrary depth

The integer equality $729 = 3^6$ determines the grouping of six ternary positions into a block. For $t \geq 0$, define

$$u_c(t) = P_c(729, t+1) \bmod 729, \quad B_{c,j}(t) = \left\lfloor \frac{u_c(t)}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6. \quad (6.6)$$

Thus $B_c(t)$ is an ordered word of six trits, and

$$u_c(t) = \sum_{j=0}^5 B_{c,j}(t) 3^{5-j}. \quad (6.7)$$

The word and its integer lift are mutually recoverable representations of the same emitted block.

Theorem 6.4 (Stability of extraction at any depth). For every n ,

$$P_c(729, n) = P_c(3, 6n).$$

If $t < n$, then

$$u_c(t) = \left\lfloor \frac{P_c(729, n)}{729^{n-t-1}} \right\rfloor \bmod 729. \quad (6.8)$$

Consequently, every extension of the prefix preserves all previously emitted blocks and all signatures calculated from them.

Proof. The first equality follows because both prefixes are separated at the same scale, $729^n = 3^{6n}$, and from Theorem 6.2. For the second, apply (6.4) at depths n and $t + 1$, and then take the residue modulo 729. The ternary extraction in (6.6) is deterministic. Any explicit function of those eighteen coordinates therefore retains the same result when the prefix is extended. $\square$

The construction is quantified over $t \in \mathbb{N}$. A table materializing one hundred blocks is a finite evaluation of this construction; formula (6.8) determines its compatibility with every subsequent extension exactly.

Example 6.5 (First joint block). Evaluation of the selected regional readers gives the words

$$B_P(0) = 010211, \quad B_E(0) = 201101, \quad B_F(0) = 121200.$$

Their lifting by (6.7) produces

$$u_P(0) = 103, \quad u_E(0) = 523, \quad u_F(0) = 450.$$

For example, $201101_3 = 2 \cdot 243 + 27 + 9 + 1 = 523$. All six positions, including leading zeros, are retained in the band representation. The leading zero in 010211 has a defined position even though the decimal notation for 103 does not contain it.

6.3 Transition registers and reconstruction of linear lifts

Over $\mathbb{F}_3$, collect two consecutive emissions from each region in the matrix

$$R(t) = \begin{pmatrix} B_P(t) \\ B_P(t+1) \\ B_E(t) \\ B_E(t+1) \\ B_F(t) \\ B_F(t+1) \end{pmatrix}. \quad (6.9)$$

This definition fixes both the regional order and the temporal order. The first four registers are

$$X_0 = R(0), \quad Y_0 = R(1), \quad X_1 = R(2), \quad Y_1 = R(3).$$

Their values, obtained by extraction from the regional prefixes, are

$$\begin{aligned} X_0 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, & Y_0 &= \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Theorem 6.6 (Unique reconstruction of the two initial links). *The equations $X_0L_0 = Y_0$ and $X_1L_1 = Y_1$ each have a unique solution in $M_6(\mathbb{F}_3)$. These solutions are*

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}.$$

Both matrices are invertible.

Proof. The matrices

$$A_0 = \begin{pmatrix} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 & 2 & 2 \\ 2 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}, \quad A_1 = \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

satisfy $A_iX_i = X_iA_i = I$, by multiplication in $\mathbb{F}_3$. The solutions are therefore forced by $L_i = A_iY_i$; multiplication produces the two displayed matrices. Their inverses are

$$L_0^{-1} = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}, \quad L_1^{-1} = \begin{pmatrix} 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

The bilateral verification $L_iL_i^{-1} = L_i^{-1}L_i = I$ proves exact recovery of each transformed word. All products use residues 0, 1, 2 with their interpretation in $\mathbb{F}_3$. $\square$

The causal reading of this result is precise: the regional emissions produce $R(t)$; the registers $R(0), \dots, R(3)$ determine the two lifts above; their inverses recover the transformed words. Continuity for every t belongs to the coinductive readers and Theorem 6.4. The matrix reconstruction proved here has the two links specified in its statement as its domain.

6.4 Simultaneous refinement in ternary and decimal bases

A ternary emission of six trits and a decimal emission of three digits are readings in different bases. To preserve their relationship, record the state

$$s = (n, k, N, D),$$

where N is a prefix at depth n in base 729, and D is a prefix at depth k in base 1000. Their Archimedean cylinders are

$$I_{729}(s) = \left[\frac{N}{729^n}, \frac{N+1}{729^n} \right), \quad I_{1000}(s) = \left[\frac{D}{1000^k}, \frac{D+1}{1000^k} \right). \quad (6.10)$$

The half-open upper endpoint assigns each value to a unique cell at a given base and depth.

Definition 6.7 (Integer compatibility defects). Define

$$A(s) = N1000^k - D729^n, \quad (6.11)$$

$$B(s) = (N+1)1000^k - D729^n = A(s) + 1000^k. \quad (6.12)$$

The state is compatible when

$$A(s) < 729^n, \quad B(s) > 0. \quad (6.13)$$

Lemma 6.8 (Integer criterion for intersection). *The cylinders in (6.10) have a nonempty intersection if and only if (6.13) holds.*

Proof. Two half-open intervals $[a, b)$ and $[c, d)$, with $a < b$ and $c < d$, intersect exactly when $a < d$ and $c < b$. Applying this condition to (6.10) and multiplying by the positive denominators gives $N1000^k < (D + 1)729^n$ and $D729^n < (N + 1)1000^k$. These are the two inequalities in (6.13). $\square$

Definition 6.9 (Update of depths and prefixes). For $0 \leq u < 729$ and $0 \leq d < 1000$, define

$$T_u(n, k, N, D) = (n + 1, k, 729N + u, D),$$

$$T_{u,d}(n, k, N, D) = (n + 1, k + 1, 729N + u, 1000D + d).$$

The first operation refines only the ternary cylinder. The second refines both readings and preserves both depths in the state.

Proposition 6.10 (Exact boundary recurrences). *The updates above satisfy*

$$A(T_u s) = 729A(s) + u1000^k, \quad (6.14)$$

$$A(T_{u,d} s) = 729000A(s) + u1000^{k+1} - d729^{n+1}. \quad (6.15)$$

Proof. For the first identity,

$$(729N + u)1000^k - D729^{n+1} = 729(N1000^k - D729^n) + u1000^k.$$

For the second,

$$\begin{aligned} (729N + u)1000^{k+1} - (1000D + d)729^{n+1} \\ = 729000(N1000^k - D729^n) + u1000^{k+1} - d729^{n+1}. \end{aligned}$$

Both are integer identities preceding any real approximation. $\square$

Theorem 6.11 (Refinement of the generated regional cylinders). *Let*

$$s_c(n, k) = (n, k, P_c(729, n), P_c(1000, k)).$$

Then $s_c(n, k)$ is compatible for all n, k . Moreover,

$$T_{u_c(n)} s_c(n, k) = s_c(n + 1, k),$$

$$T_{u_c(n), d_c(1000, k)} s_c(n, k) = s_c(n + 1, k + 1).$$

Euclidean division of the refined prefixes recovers the preceding prefixes exactly. For every $a, b \geq 0$,

$$\frac{P_c(729, n + a) - P_c(729, n + a) \bmod 729^a}{729^a} = P_c(729, n),$$

and the analogous identity holds in base 1000.

Proof. Theorem 6.2 places v_c simultaneously in both intervals in (6.10); the intersection criterion gives compatibility. The update identities are (6.5) in each base. Recovery follows from (6.4). Thus the defect recurrence and prefix recovery are consequences of the same update. $\square$

Example 6.12 (First pair of regional cylinders). For $n = k = 1$, evaluation of the three regions gives

c	N	D	A	B
P	2290	3141	211	1211
E	1981	2718	-422	578
F	1179	1618	-522	478

In all three rows, $A < 729$ and $B > 0$. The first prefix column decomposes as $2290 = 3 \cdot 729 + 103$, $1981 = 2 \cdot 729 + 523$, and $1179 = 729 + 450$: exactly the previously emitted six-trit blocks reappear. The different signs of A express the relative positions of the two lower endpoints without altering the regional value's membership in both intervals.

6.5 Selection of the next block by the regional reader

Compatibility of two intervals expresses a relationship between readings. Determining the next block additionally uses the rational interval produced by the region. This operation can be formulated without discarding the depth history.

Definition 6.13 (Admissibility of regional prolongation). For fixed c, n , let $s = 729^{n+1}$ and $j = j_c(s)$. A residue $u \in \{0, \dots, 728\}$ is admissible when

$$\lfloor s\ell_c(j) \rfloor \leq 729P_c(729, n) + u \leq \lfloor su_c(j) \rfloor - 1. \quad (6.16)$$

Theorem 6.14 (Uniqueness of the prolongation compatible with the reader). *Condition (6.16) is equivalent to $u = u_c(n)$. There is therefore exactly one admissible prolongation of each region at each depth.*

Proof. At the separation index, both integer endpoints in (6.16) coincide with $P_c(729, n+1)$. The condition is therefore equivalent to $729P_c(729, n) + u = P_c(729, n+1)$. By (6.5), the right-hand side is $729P_c(729, n) + u_c(n)$. Cancelling the common term gives equality of the residues. Conversely, this equality satisfies the condition. $\square$

The effective dependence of this selection is fully specified: selected region, rational reader, depth, separation index, preceding prefix, and next residue. The defect A summarizes a boundary relationship; the state (n, k, N, D) and the regional intervals retain the data used by the update.

6.6 Ternary signature, carry, Witt incidence, and reading clock

At a fixed depth t , write $B_{c,j} = B_{c,j}(t)$. For each column j , calculate the following integers and residues:

$$S_j = B_{P,j} + B_{E,j} + B_{F,j}, \quad q_j = S_j \bmod 3, \quad c_j = \lfloor S_j/3 \rfloor, \quad (6.17)$$

$$a_j = (B_{P,j} + B_{E,j} - B_{F,j}) \bmod 3, \quad w_j = \sum_c \mathbf{1}_{B_{c,j} \neq 0}, \quad \bar{w}_j = w_j \bmod 3. \quad (6.18)$$

The residue of the axial expression is taken in the integers. An implementation using natural numbers calculates it as $(B_{P,j} + B_{E,j} + 3 - B_{F,j}) \bmod 3$, which preserves that residue exactly.

The ternary incidence matrix used by the reader is

$$W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \quad (6.19)$$

Two residual charges are preserved for each region:

$$v_c^{\text{res}} = \sum_j B_{c,j} \bmod 3, \quad v_c^W = \sum_j (B_c W)_j \bmod 3. \quad (6.20)$$

The reader's complete signature is $\mathcal{S}(B) = (q, a, c, w, \bar{w}, v^{\text{res}}, v^W)$. Its structure separately preserves the column residue, carry, axial contrast, support weight, and the two charge readings.

Theorem 6.15 (Exact recovery and bounds of the signature). *For each column and each region,*

$$S_j = q_j + 3c_j, \quad 0 \leq q_j, a_j < 3, \quad 0 \leq c_j \leq 2, \quad (6.21)$$

$$B_{F,j} = 2(q_j + 3 - a_j) \bmod 3, \quad 0 \leq w_j \leq 3, \quad 0 \leq \bar{w}_j < 3. \quad (6.22)$$

Moreover,

$$v_c^W = \left(\sum_j ((B_c W)_j \bmod 3) \right) \bmod 3. \quad (6.23)$$

The stated recoveries determine each column's total and its autoscaling coordinate; the regional register additionally preserves the three ordered bands.

Proof. Identity (6.21) is Euclidean division of S_j by 3. Each summand of S_j lies between 0 and 2, so $0 \leq S_j \leq 6$ and $c_j \leq 2$. In $\mathbb{F}_3$, subtracting the two congruences defining q_j and a_j gives $q_j - a_j = 2B_{F,j}$. Since the inverse of 2 is 2, this yields (6.22); the summand 3 permits an expression using natural numbers. The bounds on w_j follow by summing three indicators. Finally, reducing an integer sum modulo 3 is equivalent to reducing each summand and then reducing their sum, which proves (6.23). $\square$

Example 6.16 (Signature of the first joint block). The three initial bands produce

$$\begin{aligned} q &= (0, 0, 2, 2, 1, 2), & a &= (1, 2, 0, 1, 1, 2), \\ c &= (1, 1, 0, 1, 0, 0), & w &= (2, 2, 2, 3, 1, 2), \\ v^{\text{res}} &= (2, 2, 0), & v^{\text{W}} &= (2, 1, 1). \end{aligned}$$

The fourth column has total $2 + 1 + 2 = 5$; its signature records $q_3 = 2$ and $c_3 = 1$, so $q_3 + 3c_3 = 5$. Axial recovery gives $2(2 + 3 - 1) \bmod 3 = 2$, precisely the autoscaling coordinate. The Witt images, reduced componentwise, are

$$(2, 0, 2, 1, 1, 2), \quad (0, 0, 0, 2, 0, 2), \quad (2, 1, 1, 2, 1, 0).$$

Their residual sums by row yield the three charges $(2, 1, 1)$.

Definition 6.17 (Integer clock of decimal reading). The transition index t and decimal representation depth are related by

$$\kappa(t) = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\}. \quad (6.24)$$

Proposition 6.18 (Characterization and monotonicity of the clock). The integer $\kappa(t)$ is uniquely determined by

$$1000^{\kappa(t)} \leq 729^{t+1} < 1000^{\kappa(t)+1}. \quad (6.25)$$

The function κ is monotone and satisfies $\kappa(20) = \kappa(21) = 20$.

Proof. The powers of 1000 are strictly increasing and unbounded; there is therefore a unique interval between consecutive powers containing 729^{t+1} . Monotonicity of the latter expression proves that of κ . The final two evaluations are the integer comparisons

$$1000^{20} \leq 729^{21} < 1000^{21}, \quad 1000^{20} \leq 729^{22} < 1000^{21}.$$

These comparisons use exact integer powers. $\square$

Equality of decimal depth at two distinct transitions explains why both indices must be retained: the reading clock can remain constant while the ternary band advances and its signature is updated. Chronological information resides in the ordered register of transitions.

6.7 Composition theorem for the block, cylinder, and signature

Theorem 6.19 (Joint compatibility of regional readings). For each depth n , there exists a unique triple of blocks (u_P, u_E, u_F) satisfying regional admissibility (6.16) in all three regions. This triple is $(u_P(n), u_E(n), u_F(n))$. Simultaneously:

1. each block updates its cylinder through recurrences (6.14) and (6.15);
2. its six trits produce the signature $\mathcal{S}(B_P(n), B_E(n), B_F(n))$;
3. the column totals are recovered as $q + 3c$, and the Witt charges are obtained from the same bands through (6.23);
4. every deeper prefix returns exactly these blocks, truncated cylinders, and signatures.

Proof. Apply Theorem 6.14 to each of the three regional indices. Coordinatewise uniqueness gives joint uniqueness. Substitute those same residues into the operations $T_u, T_{u,d}$, which allows Theorem 6.11 to be applied. The deterministic application of (6.6) produces the bands used by (6.17)–(6.20). The recovery identities are those of Theorem 6.15. Finally, Theorem 6.4 and truncation in each base prove stability at greater depth. At every step the identity of the emitted block has been preserved; neither reading requires the selection of a different block. $\square$

The conclusion is an operational compatibility between cylindrical contraction and preservation of incidence. Both use the same generated register, retain the depths, and admit integer recovery. This result is the explicit form of the connection

$$\text{emitted regional block} \longrightarrow \begin{cases} \text{cylinder and integer boundary,} \\ \text{band, signature, carry, and Witt charge.} \end{cases}$$

The subsequent development of incidence and fields uses the second reading while preserving its common provenance with the first.

6.8 Calendar census and simultaneous compatibility of the regional biographies

The preceding rows are assembled, without yet applying any decimal reader, into the three block biographies produced by the transitions:

$$\begin{aligned} B^{\text{cl}} &= 010211|012222|010211|002111|110221, \\ B^{\text{pr}} &= 201101|121221|102011|012222|102011, \\ B^{\text{au}} &= 121200|112202|121020|010210|010200. \end{aligned} \tag{6.26}$$

Indeed, the consecutive pairs of the first two segments are the rows of X_0, Y_0 , and those of the last two are the rows of X_1, Y_1 . Thus, the biographies are another representation of the record of twelve transitions, not a collection of conventional prefixes.

To verify the operative word of the calendar, let $c = c_1c_2c_3c_4 \in \{0, 1\}^4$ and, for each regime $\chi \in \{\text{cl}, \text{pr}, \text{au}\}$, define

$$u_0^\chi = b_0^\chi, \quad u_j^\chi = u_{j-1}^\chi L_{c_j} \quad (1 \leq j \leq 4).$$

To condense the census, denote by

$$N_{\text{ar}}(c) := \#\{(j, \chi) : u_j^\chi = b_j^\chi, 1 \leq j \leq 4\}, \quad N_{\text{bio}}(c) := \#\{\chi : (u_0^\chi, \dots, u_4^\chi) = B^\chi\}.$$

Exact multiplication in $\mathbb{F}_3$ for the sixteen calendars gives:

c	$N_{\text{ar}}(c)$	$N_{\text{bio}}(c)$
0000, 0001	6	0
0010	9	0
0011	12	3
0100, 0101, 0110, 0111	3	0
1000, 1001, 1010, 1011	0	0
1100, 1101, 1110, 1111	0	0

(D)

The table exhausts the $2^4 = 16$ possibilities, and each of its entries is obtained by four multiplications of a row by one of the two displayed matrices. Consequently, 0011 is the unique calendar that simultaneously reconstructs the three complete biographies. In operator notation,

$$(L_0, L_0, L_1, L_1) \tag{6.27}$$

is not an independent prescription: it is the unique binary reading compatible with the twelve edges already produced by U_t .

The calendar (6.27) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate cumulative length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace their production by U_t . Conversely, (D) proves within the article the uniqueness of the ordering of the four transports. The causal chain is thereby closed: the APP census produces the initial regions, TRIT fixes regime and orientation, TPK produces the transition record, and that record determines L_0, L_1 and 0011 before any Archimedean realization.

7 Regional determination of the dodecaphasic descriptor and selection of the coupling register

The determination of the coupling register combines two constructions that should be developed in their actual order. The first produces the inputs to the selector from the regional representations of the TPK: six-trit bands, column signatures, a conversion calendar, an oriented descriptor, and a functional panel. The second operates on these inputs and on the terminal eight-block repertoire, enumerates its admissible orderings, and selects a unique register through two simultaneous conditions: exceptional incidence and reading heterotypy. The value of the register appears at the end of this composition.

The present development presupposes the regional representations obtained in the preceding chapters from APP–TRIT–TPK. Its purpose is to make explicit every intermediate operation leading from those representations to the finite selector. The realization of the same register through signed incidences, Hadamard inversion, and the representation of the fine-structure constant are then incorporated. The three constructions retain their respective domains: selection of the register, recovery from its channels, and arithmetic evaluation of its representations.

7.1 Regional bands, encoding, and integer signatures

7.1.1 Domains and encoding of the regional bands

The channel order is fixed as

$$\chi \in \{\pi, e, \varphi\}, \quad B_\chi(t) \in \mathbb{F}_3^6, \quad t \in \mathbb{N}.$$

The channel symbols identify the regions already generated. Each $B_\chi(t)$ retains its place in its sequence of representations; the index t counts ternary blocks. The natural-number evaluation of a band $b = (b_0, \dots, b_5)$ is

$$\text{enc}_6(b) = 243b_0 + 81b_1 + 27b_2 + 9b_3 + 3b_4 + b_5, \quad 0 \leq \text{enc}_6(b) < 729. \quad (7.1)$$

Its fixed-width inverse is

$$\text{dig}_j(n) = \left\lfloor \frac{n}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6.$$

Leading zeros are thus retained: the words 002001 and 2001 could share an abbreviated integer notation, but the type of the former contains six positions. All subsequent concatenations use this width.

Within the hundred-block window, the six-hundred-trit representation of channel χ determines an integer $P_\chi^{(600)}$. Extraction is performed by

$$b_\chi(t) = \left\lfloor \frac{P_\chi^{(600)}}{729^{99-t}} \right\rfloor \bmod 729, \quad 0 \leq t < 100, \quad B_\chi(t) = \text{enc}_6^{-1}(b_\chi(t)). \quad (7.2)$$

The six-hundred-trit integer is a representation of the regional producer; the hundred-row table is the evaluation of (7.2). This separation allows a historical table to be replaced by its producer without changing the selector that consumes it.

7.1.2 Residues, quotients, occupancy, and orientation

For each column j and time t , introduce

$$T_j(t) = B_\pi(t)_j + B_e(t)_j + B_\varphi(t)_j, \quad (7.3)$$

$$q_j(t) = T_j(t) \bmod 3, \quad c_j(t) = \lfloor T_j(t)/3 \rfloor, \quad (7.4)$$

$$a_j(t) = (B_\pi(t)_j + B_e(t)_j - B_\varphi(t)_j) \bmod 3, \quad (7.5)$$

$$w_j(t) = \mathbf{1}_{B_\pi(t)_j \neq 0} + \mathbf{1}_{B_e(t)_j \neq 0} + \mathbf{1}_{B_\varphi(t)_j \neq 0}, \quad \bar{w}_j(t) = w_j(t) \bmod 3. \quad (7.6)$$

The subtraction in the third expression is performed in the integers and then reduced. In an implementation over natural numbers, the equivalent expression is $(B_\pi + B_e + 3 - B_\varphi) \bmod 3$; truncated subtraction would change the reader.

Proposition 7.1 (Reconstruction of the column sum). *For all admissible bands,*

$$T_j(t) = q_j(t) + 3c_j(t), \quad 0 \leq q_j(t) < 3, \quad 0 \leq c_j(t) \leq 2, \quad 0 \leq w_j(t) \leq 3.$$

Proof. Each column sums three representatives from $\{0, 1, 2\}$, so that $0 \leq T_j(t) \leq 6$. Euclidean division by three yields the stated residue and quotient. Occupancy counts three binary conditions; its bound follows directly from that definition. $\square$

This identity preserves the quotient of the local sum. Reconstructing the three individual bands requires the bands themselves or an additional reader that distinguishes them: a column sum has fewer coordinates than the triple that produces it. Accordingly, the temporal row used below jointly retains the three bands and the four fields $q, a, c, \bar{w}$.

7.2 Conversion clock and first zero advance

7.2.1 Ternary capacity and decimal depth

A representation of $t + 1$ bands contains $6(t + 1)$ trits. The associated conversion calendar is defined by the integer

$$d_t = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\} = \max\{k \in \mathbb{N} : 1000^k \leq 729^{t+1}\}. \quad (7.7)$$

The definition compares integer capacities. The counter t and the depth d_t record different quantities: a transition adds one six-trit band; the clock counts complete powers of the decimal base one thousand contained in the accumulated ternary capacity.

Lemma 7.2 (Integer characterization of the clock). *For $t, k \in \mathbb{N}$,*

$$d_t = k \iff 1000^k \leq 729^{t+1} < 1000^{k+1}.$$

Moreover, $d_t \leq d_{t+1}$.

Proof. The powers of one thousand form a strictly increasing, unbounded sequence. The power 729^{t+1} lies between two consecutive terms; the lower exponent is precisely the maximum in (7.7). The inequality $729^{t+1} < 729^{t+2}$ implies monotonicity of the maximum. $\square$

Proposition 7.3 (First zero advance). *The least time satisfying $d_{t+1} = d_t$ is*

$$t_* = 20.$$

In particular,

$$d_0 = 0, \quad d_{19} = 19, \quad d_{20} = 20, \quad d_{21} = 20, \quad d_{22} = 21.$$

Proof. The exact integer comparisons

$$1000^{20} \leq 729^{21}, \quad 729^{22} < 1000^{21}, \quad 1000^{21} \leq 729^{23} < 1000^{22}$$

give the values at times twenty, twenty-one, and twenty-two. For $0 \leq t \leq 20$, the expression $729^{t+1}/1000^t = 729(729/1000)^t$ is decreasing and its value at twenty is greater than or equal to one. Hence, $1000^t \leq 729^{t+1} < 1000^{t+1}$ and $d_t = t$. No time less than twenty can satisfy $d_{t+1} = d_t$; at twenty it does hold. All comparisons used involve integer powers. $\square$

The number twenty is therefore an output of the minimality criterion. The rule selecting the instant is “the first zero advance of the clock”; the algorithm and its proof can use this rule without receiving twenty as an external parameter.

7.2.2 Nonadic return and temporal antecedent

The length of the phase return determines

$$r = t_* - 9 = 11. \quad (7.8)$$

The descriptor queries are made at $t_* + 1$, $r + 1$, and r . This distribution preserves the relation between the pause instant, the nonadic translation, and the immediately associated phase reading. Times eleven, twelve, and twenty-one designate positions in the regional calendar; their temporal difference remains recorded even when two times produce the same decimal depth.

7.3 Oriented construction of the twenty-four-trit descriptor

7.3.1 Opposite half-band defects

Take the oriented defects of the regional chart

$$\delta_+ = (0, 0, 0, 1, 1, 1), \quad \delta_- = (0, 0, 0, 2, 2, 2) \in \mathbb{F}_3^6.$$

Their opposition is verified componentwise:

$$\delta_+ + \delta_- = 0 \quad \text{in } \mathbb{F}_3^6.$$

The representative 2 realizes -1 in the ternary chart. Concatenating the three-position tails in negative-positive order produces the word 222111. Adding the defects and concatenating their tails are different operations: the former has codomain $\mathbb{F}_3^6$, whereas the latter preserves the order of two segments of length three.

7.3.2 Dual charge and phase selection

In the regional chart, the six-coordinate transformation uses

$$A_{\text{reg}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

With row vectors, the dual charge of $b \in \mathbb{F}_3^6$ is the sum of the coordinates of bA_{reg} . The integer row sums of A_{reg} are (5, 7, 7, 7, 7, 7); this gives the local expression

$$\eta^\vee(b) = 2b_0 + b_1 + b_2 + b_3 + b_4 + b_5 \pmod{3}. \quad (7.9)$$

The reading phase at instant r is $\theta_r = (r - 1) \bmod 3$. For each channel, define

$$\varepsilon_\chi(r) = \begin{cases} 1, & \eta^\vee(B_\chi(r)) = \theta_r, \\ 2, & \eta^\vee(B_\chi(r)) \neq \theta_r. \end{cases}$$

The phase–charge representation is

$$D(r) = \varepsilon_\pi(r)B_\pi(r) + \varepsilon_e(r)B_e(r) + \varepsilon_\varphi(r)B_\varphi(r) \quad \text{in } \mathbb{F}_3^6. \quad (7.10)$$

The coefficient is determined by an equality of charges before evaluating the word that will result from the combination.

7.3.3 Composition of the descriptor

Let $\parallel$ denote the concatenation of fixed-width words. The complete construction is

$$\begin{aligned} \Omega_{24} = & (B_e(t_* + 1) + \delta_-) \parallel (\delta_-|_{3,4,5} \parallel \delta_+|_{3,4,5}) \\ & \parallel (B_\pi(r + 1) + B_e(r + 1)) \parallel D(r). \end{aligned} \quad (7.11)$$

Each length-six summand is evaluated in $\mathbb{F}_3^6$; the final concatenation has length $6 + 6 + 6 + 6 = 24$. This construction distinguishes the descriptor Ω_{24} from the regional stages w_{24}^χ : the two objects share a length, but their producers and their places in the composition differ.

Proposition 7.4 (Regional evaluation of the descriptor). *The regional representations and the first-zero-advance criterion determine*

$$\Omega_{24} = 020022 \, 222111 \, 211201 \, 021101, \quad [\Omega_{24}]_3 = 66241910521.$$

Proof. The required bands, preserving the order (π, e, φ) , are

t	$B_\pi(t)$	$B_e(t)$	$B_\varphi(t)$
11	022212	110001	100021
12	012012	202222	000120
21	221022	020100	211021

and come from extraction (7.2). At time twenty-one, $020100 + 000222 = 020022$ in $\mathbb{F}_3^6$. The defect tails provide 222111. At time twelve, $012012 + 202222 = 211201$.

At time eleven, the dual charges are 0, 1, 2, respectively; the phase is $(11 - 1) \bmod 3 = 1$. The resulting coefficients are (2, 1, 2) and

$$2(022212) + (110001) + 2(100021) = 021101 \quad \text{in } \mathbb{F}_3^6.$$

Concatenating these four results gives the stated word. Its integer evaluation is obtained by Horner's method in base three, or by three successive concatenations in base 729. $\square$

The preceding table serves a verification purpose: each of its entries is obtained from the regional producer. Construction rule (7.11) makes it possible to reproduce the descriptor without introducing the final twenty-four-trit string as an input.

7.4 Functional panel and terminal repertoire

7.4.1 Nine regional positions

The functional panel retains the first three bands of each of the three channels, in the declared regional order:

$$\mathcal{P}_9 = (b_\pi(0), b_\pi(1), b_\pi(2), b_e(0), b_e(1), b_e(2), b_\varphi(0), b_\varphi(1), b_\varphi(2)). \quad (7.12)$$

Its evaluation is

$$\mathcal{P}_9 = (103, 161, 103, 523, 457, 301, 450, 398, 438).$$

The nine positions retain their regional provenance, although the value 103 occurs twice. The panel has nine ordered incidences and eight distinct values. The subsequent enumeration uses the incidences; candidate deduplication takes place at a later stage.

7.4.2 Unordered eight-block repertoire

The terminal repertoire involved in this construction is

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\} \subset \mathbb{F}_3^6. \quad (7.13)$$

Its eight elements are distinct. The integer encoding, sorted by value solely to fix a computational representation, is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

The documentary provenance of this repertoire is the terminal segmentation of the seventy-two-trit word and its position-wise recovery through incidences of the phase-charge and affine readers. The selector developed below receives the repertoire without its historical ordering and tests its 8! orderings.

The exact formulation of the reassembled implementation maintains an explicit interface here: the regional producers materially replace the descriptor, the panel, and the temporal rows; repertoire (7.13) retains its earlier documentary producer. This distinction identifies where each input enters. Membership of eight words in a temporal reader repertoire, selection of those eight words, and determination of their terminal order are three different statements. The terminal-selection proof establishes the third for the declared repertoire.

Documentary recovery by position can be expressed exactly. Let $\mathcal{I}$ be the phase-charge incidence register and let $\mathcal{J}$ be the register of the affine extension. Each incidence retains the position ℓ in the seventy-two-trit word and the emitted word u . For $5 \leq \ell \leq 12$, form

$$\mathcal{S}_\ell = \{u : (\ell, u) \in \mathcal{I} \cup \mathcal{J}\}.$$

The recovery procedure checks that all eight positions occur and that each $\mathcal{S}_\ell$ is a singleton. If s_ℓ is its unique element, the output delivered to the selector is

$$\mathcal{S}_{\text{term}} = \{s_5, s_6, \dots, s_{12}\}.$$

The singleton proof for each position preserves the coherence of the historical witnesses; the subsequent export retains the eight words and removes only the historical ordering. The selector reconstructs the order through complete enumeration and the terminal conditions, instead of receiving it from the provenance register.

7.4.3 Temporal incidences of the terminal repertoire and recovery by position

The representation of the terminal repertoire uses the same regional blocks, the same carries, and the same calendar that produce the initial twenty-four-trit descriptor. Two operations in this composition should be made explicit separately. The first recovers eight words from recorded temporal incidences. The second presents those eight words to the dodecaphasic selector as an unordered set. The selector re-establishes the order through its incidence, cylinder, and closure conditions. The distinction preserves

the provenance witnesses without turning the order of a historical segmentation into an input to the selector.

Let $B_t = (b_{t,0}, b_{t,1}, b_{t,2}) \in (\mathbb{F}_3^6)^3$ be the regional boundary already generated at time t , with $0 \leq t < 100$. In the chart used by the historical register, the incidence matrix is

$$A_{\text{hist}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

The words are regarded as row vectors. Their two charges and the chronological phase are

$$r_{t,i} = \sum_{j=1}^6 b_{t,i,j}, \quad d_{t,i} = \sum_{j=1}^6 (b_{t,i} A_{\text{hist}})_j, \quad \vartheta_t = t - 1 \pmod{3}.$$

All operations in these three expressions take place in $\mathbb{F}_3$. The historical subscript identifies a specific chart of the Witt incidence; it keeps the coordinate transformation explicit when another presentation of that same incidence is used.

For $c \in \{r_t, d_t\}$, $\delta \in \mathbb{F}_3$ and $\epsilon \in \{1, -1\}$, define

$$\eta_i(c, \delta, \epsilon) = \begin{cases} \epsilon, & c_i = \vartheta_t + \delta, \\ -\epsilon, & c_i \neq \vartheta_t + \delta, \end{cases} \quad L_{t,c,\delta,\epsilon} = \sum_{i=0}^2 \eta_i(c, \delta, \epsilon) b_{t,i}.$$

The orientation -1 is written as 2 when printing the ternary coefficients. Alongside these twelve comparative readers per boundary, the register has the affine reader

$$L_t^{\text{af}} = \sum_{i=0}^2 (d_{t,i} + \vartheta_t) b_{t,i}.$$

Adding the phase in the affine reader and comparing charges in the preceding reader are different operations, both acting on data already produced by the TPK boundary.

t	$b_{t,0}$	$b_{t,1}$	$b_{t,2}$	r_t	d_t
8	200121	212212	110110	011	202
11	022212	110001	100021	001	012
30	210112	110202	211110	100	012
34	121022	000011	112110	220	021
37	120121	011202	112221	100	201
49	110212	200212	100122	110	201
58	111121	022121	122201	122	220
67	010012	001220	011120	122	122
75	220000	020111	100002	120	021
81	122201	020202	201122	202	001
90	022112	022110	121010	202	200

Table 6: Temporal boundaries realizing the printed incidences of the terminal repertoire. Each six-digit entry preserves the order of the six coordinates of the regional boundary.

The following table displays all comparative incidences of the register on the twelve blocks of the segmentation under consideration. The letters r and d denote the visible charge and the dual charge, respectively. Column j retains the position index of the historical witness; the operation delivering the data to the selector forgets this index after forming the set of words.

The tenth block has an affine witness. At $t = 30$, $d_{30} = (0, 1, 2)$ and $\vartheta_{30} = 2$, whence

$$(d_{30,0} + \vartheta_{30}, d_{30,1} + \vartheta_{30}, d_{30,2} + \vartheta_{30}) = (2, 0, 1)$$

j	t	charge	δ	ϵ	$(\eta_0\eta_1\eta_2)$	$L_{t,c,\delta,\epsilon}$
4	11	d	0	1	212	021101
5	67	d	1	-1	211	002001
5	67	d	2	1	211	002001
5	67	r	1	-1	211	002001
5	67	r	2	1	211	002001
6	81	d	0	1	222	020111
6	81	r	2	1	222	020111
7	34	r	1	-1	111	200110
7	75	d	1	-1	211	200110
7	75	r	2	-1	211	200110
8	90	r	0	1	121	121012
8	90	r	1	-1	121	121012
9	49	d	0	-1	121	010122
11	11	d	2	-1	211	221110
11	37	d	0	-1	121	221110
12	8	d	0	-1	111	222110
12	8	r	1	-1	111	222110
12	58	d	1	-1	111	222110
12	58	r	0	-1	111	222110

Table 7: Complete phase-charge incidences on the terminal segmentation within the hundred boundaries of the register. Word matches between rows retain their times, orientations, and charge types.

and, using the bands in Table 6,

$$L_{30}^{\text{af}} = 2(210112) + 0(110202) + (211110) = (001001) \quad \text{in } \mathbb{F}_3^6.$$

The equality holds coordinatewise: before reduction modulo three, the sum vector is $(6, 3, 1, 3, 3, 4)$.

Proposition 7.5 (Complete recovery of the repertoire through incidences). *Consider the comparative-incidence register in Table 7, augmented by the affine witness $(j, t, L) = (10, 30, 001001)$. For each position $5 \leq j \leq 12$, the set of output words of its incidences is a singleton. Extracting these eight words and subsequently forgetting their order produces exactly*

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\}.$$

Their encoding as base-three integers is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

Proof. The incidence multiplicities at positions $j = 5, 6, 7, 8, 9, 10, 11, 12$ are, respectively,

$$(4, 2, 3, 2, 1, 1, 2, 4).$$

At each position, all rows print the same word. Each row is verified by substituting its three bands and three coefficients into the formula for L ; the tenth case has been calculated explicitly through L_{30}^{af} . For example, at $t = 67$ and with coefficients $(2, 1, 1)$, one obtains

$$2(010012) + (001220) + (011120) = (002001) \pmod{3}.$$

The eight results are distinct as words of length six. Their encoding $\sum_{k=1}^6 w_k 3^{6-k}$ gives, in position order, $(55, 175, 498, 437, 98, 28, 714, 687)$. Forgetting the order yields the stated set of integers. Thus, extraction preserves each complete word, including its zero positions, and delivers eight distinct elements to the selector. $\square$

The preceding proof is an exact recovery from identified incidences. Its input comprises the incidence register with its positions and witnesses. The subsequent orbital selection receives only $\mathcal{S}_{\text{term}}$ and the initial regional descriptor; it traverses the $8!$ orderings and determines their compatibility with the

functional boundaries and the oriented incidences of the dodecaphasic wheel. Both operations are thus presented contiguously, with explicit domains: temporal provenance of the eight words and constructive determination of their order.

The comparative table contains the nineteen incidences that its formula produces on the twelve-block segmentation over the hundred declared times. This finite exhaustiveness is checked by traversing $100 \cdot 2 \cdot 3 \cdot 2 = 1200$ evaluations. The affine reader extends coverage of the terminal positions from seven to eight. This count characterizes the local reader repertoire just defined; the other TPK operations retain the domains and constructive functions established in their respective derivations.

7.5 Comparative readers, affine reader, and temporal incidence

7.5.1 Witt transformation of the selector

The selector uses the six-coordinate chart

$$A_{\text{ter}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Define $W(b) = bA_{\text{ter}}$ in $\mathbb{F}_3^6$, together with the charges

$$\eta(b) = \sum_{j=0}^5 b_j \pmod{3}, \quad \eta_{\text{ter}}^{\vee}(b) = \eta(W(b)).$$

The descriptor chart and the selector chart are related by the permutation $\sigma = (0, 1, 2, 4, 5, 3)$, written as a list of images:

$$(A_{\text{ter}})_{ij} = (A_{\text{reg}})_{\sigma(i), \sigma(j)}. \quad (7.14)$$

Lemma 7.6 (Compatibility of the charge functional). *For every band b ,*

$$\eta_{\text{ter}}^{\vee}(b) = \eta^{\vee}(b).$$

Proof. The integer sum of each corresponding row of the two matrices agrees: it is five in the first row and seven in the other five. By distributivity,

$$\sum_j \sum_i b_i (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{reg}})_{ij}.$$

Reduction modulo three gives the equality of charges. Matrix identity (7.14) is verified by substituting its thirty-six entries and also preserves the coordinate change. $\square$

Equality of the total charge functional allows its representations to be compared. The vector transformations remain identified by their matrices and the explicit permutation; the preceding proof preserves that coordinate information.

7.5.2 Twelve comparative readers per instant

Define the temporal phase

$$\theta(t) = (t + 2) \pmod{3}.$$

This expression realizes $(t - 1) \pmod{3}$ also at $t = 0$, where the phase equals two. For each charge type $\nu \in \{\eta, \eta_{\text{ter}}^{\vee}\}$, offset $o \in \{0, 1, 2\}$ and orientation $s \in \{1, 2\}$, take

$$\epsilon_{\chi}^{\nu, o, s}(t) = s \begin{cases} 1, & \nu(B_{\chi}(t)) = \theta(t) + o \pmod{3}, \\ 2, & \nu(B_{\chi}(t)) \neq \theta(t) + o \pmod{3}. \end{cases}$$

The corresponding comparative word is

$$C_{\nu,o,s}(t) = \sum_{\chi \in \{\pi,e,\varphi\}} \epsilon_{\chi}^{\nu,o,s}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (7.15)$$

The parameters generate twelve labelled representations per row. Two labels may produce the same word; the temporal witnesses and labels remain available in the complete calendar.

7.5.3 Affine phase–charge reader

The affine reader uses coefficients

$$a_{\chi}^{\text{af}}(t) = \eta_{\text{ter}}^{\vee}(B_{\chi}(t)) + \theta(t) \pmod{3},$$

and produces

$$F(t) = \sum_{\chi \in \{\pi,e,\varphi\}} a_{\chi}^{\text{af}}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (7.16)$$

The comparative rule distinguishes agreement and difference of charges; the affine rule adds charge and phase before combining the bands. This operational difference is what heterotypic selection uses.

7.5.4 Word–residue incidence

Let R be the cyclic rotation of the six positions of a band. In the row at time t , form the residual repertoire

$$\mathcal{R}(t) = \{R^j v : v \in \{q(t), a(t), c(t), \bar{w}(t)\}, 0 \leq j < 6\}.$$

The temporal incidence relations are

$$\mathcal{A}_{\text{cmp}} = \{(C_{\nu,o,s}(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t), \nu, o, s \text{ admissible}\}, \quad (7.17)$$

$$\mathcal{A}_{\text{af}} = \{(F(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t)\}. \quad (7.18)$$

Each pair relates a read word to a residue of the same row. Time is quantified jointly across both coordinates of the pair. In the calendar used, times $0, \dots, 99$ are distinct; identifying the same row and identifying the same time are therefore equivalent operations.

Proposition 7.7 (Existential compression of the temporal repertoire). *The function*

$$M(v, u) = \mathbf{1}_{(v,u) \in \mathcal{A}_{\text{cmp}}} + 2\mathbf{1}_{(v,u) \in \mathcal{A}_{\text{af}}} \in \{0, 1, 2, 3\}$$

preserves exactly the existence of each reading type for the pair (v, u) .

Proof. The first bit equals one exactly when there exist a time, a residual field, a rotation, and a comparative label that produce the pair. The second bit expresses the corresponding condition for the affine reader. Union by bitwise disjunction preserves each existence condition as all rows are traversed. A pair may admit both types, in which case the mask equals three. $\square$

The function M is an existence reader. The witness times, their multiplicities, and the individual labels remain in (7.18) and in the calendar that produces them. The finite implementation uses M to decide heterotypy, whereas the genealogical archive retains the richer provenance data.

7.6 Enumeration of cylinders and decimal candidates

7.6.1 Concatenation of seventy-two and seventy-eight trits

For an ordering $\tau = (s_1, \dots, s_8)$ of $\mathcal{S}_{\text{term}}$, define

$$P_{72}(\tau) = [\Omega_{24} \| s_1 \| \dots \| s_8 \|]_3.$$

Recursively, start from $p_0 = [\Omega_{24}]_3$ and compute $p_i = 729p_{i-1} + [s_i]_3$ for $1 \leq i \leq 8$. This recurrence preserves the leading zeros of each block through explicit multiplication by 729. For each position b of the functional panel,

$$P_{78}(\tau, b) = 729P_{72}(\tau) + b.$$

The lengths are $24 + 8 \cdot 6 = 72$ and $72 + 6 = 78$, respectively. The corresponding ternary cylinder is

$$I_{\tau, b} = \left[\frac{P_{78}(\tau, b)}{3^{78}}, \frac{P_{78}(\tau, b) + 1}{3^{78}} \right). \quad (7.19)$$

7.6.2 Exact intersection with the twelve-triad grid

The decimal reading uses twelve triads, that is, the grid $N/10^{36}$. Write $D = 10^{36}$ and $Q = 3^{78}$ for brevity.

Lemma 7.8 (Integer endpoints of the cylinder). *For an integer prefix p of length seventy-eight, the numerators of the grid points belonging to its cylinder are*

$$\left\{ N \in \mathbb{Z} : \left\lfloor \frac{pD}{Q} \right\rfloor \leq N < \left\lfloor \frac{(p+1)D}{Q} \right\rfloor \right\}. \quad (7.20)$$

Each cylinder contains at most one point of this grid.

Proof. The condition $N/D \in [p/Q, (p+1)/Q]$ is equivalent to $pD/Q \leq N < (p+1)D/Q$. For integer N , the first inequality is equivalent to $\lfloor pD/Q \rfloor \leq N$; the second is equivalent to $N < \lfloor (p+1)D/Q \rfloor$. The interval length in coordinate N is $D/Q < 1$, since $10^{36} < 3^{78}$. It can therefore contain at most one integer. $\square$

Ceilings are evaluated by integer division: $\lceil a/Q \rceil = \lfloor (a+Q-1)/Q \rfloor$ for $a \geq 0$. The entire enumeration uses exact integers; the choice of boundary points is fixed by the half-open right endpoint.

First retain the indexed list of incidences

$$\mathcal{C}_{\text{ind}} = \{(\tau, j, N) : N/D \in I_{\tau, \mathcal{P}_9(j)}\}, \quad 1 \leq j \leq 9,$$

and then obtain the set of numerators

$$\mathcal{C} = \{N : \exists \tau, j (\tau, j, N) \in \mathcal{C}_{\text{ind}}\}.$$

Passing to $\mathcal{C}$ removes repetitions of a numerator, retained in $\mathcal{C}_{\text{ind}}$ together with their provenance.

Lemma 7.9 (Recovery of the ordering from the cylinder point). *If $(\tau, j, N) \in \mathcal{C}_{\text{ind}}$, block number i of the seventy-eight-trit word is recovered by*

$$\left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13.$$

In particular, blocks five through twelve recover τ .

Proof. Membership in the cylinder means $p \leq QN/D < p+1$, where $p = P_{78}(\tau, \mathcal{P}_9(j))$. Hence, $\lfloor QN/D \rfloor = p$. Extraction of the earlier blocks agrees with successive division of the prefix p by powers of 729: truncating a positional expansion before extracting a block preserves that block. The displayed formula performs exactly this extraction. The first four blocks form Ω_{24} and the next eight are the ordering τ . $\square$

Proposition 7.10 (Exact candidate count). *For Ω_{24} , $\mathcal{S}_{\text{term}}$ and $\mathcal{B}$ defined above,*

$$\# \text{Ord}(\mathcal{S}_{\text{term}}) = 40320, \quad \# \mathcal{C}_{\text{ind}} = 21902, \quad \# \mathcal{C} = 19446.$$

Proof. The eight blocks are distinct, so their orderings are the $8! = 40320$ permutations. For each permutation and each of the nine panel positions, compute $p = P_{78}(\tau, b)$, apply endpoints (7.20), and add the unique integer when the integer interval is nonempty. The sum of the cardinalities of these intervals is 21902. The union of their integers contains 19446 distinct elements. The exhaustive evaluation of these operations is certified by the theorems `indexed_cylinder_count` and `candidate_count` of the finite selector. The algorithm performs exactly the enumeration described: permutations, nine panel incidences, bounds by integer division, and duplicate removal. $\square$

The numbers 21902 and 19446 count different objects. The former counts indexed incidences containing a decimal point; the latter counts distinct numerators. The total number of cylinder queries is $40320 \cdot 9 = 362880$, including cylinders whose intersection with the grid is empty.

7.7 Exceptional incidence and heterotypy condition

7.7.1 Exceptional face produced by the panel

A band is lifted to twelve coordinates by

$$\Gamma(b) = (b, W(b)) \in \mathbb{F}_3^{12}.$$

The support of a word is the set of positions of its nonzero coordinates, with human-readable indices $1, \dots, 12$. The first regional bands of the panel produce

$$\mathcal{B} = \text{supp } \Gamma(B_\pi(0)) \cap \text{supp } \Gamma(B_e(0)) = \{4, 6, 9, 10\}. \quad (7.21)$$

The operation uses the regional words 103 and 523 and the selector's Witt matrix. The face is computed before querying the candidate coordinates.

For $N \in \mathcal{C}$, its twelve triads are

$$K_i(N) = \left\lfloor \frac{N}{1000^{12-i}} \right\rfloor \bmod 1000, \quad 1 \leq i \leq 12. \quad (7.22)$$

The crown support is

$$\mathcal{H}(N) = \{i : K_i(N) \geq 729\}, \quad \mathcal{C}_{\mathcal{B}} = \{N \in \mathcal{C} : \mathcal{H}(N) = \mathcal{B}\}.$$

The boundary $729 = 3^6$ relates the six-trit band to the decimal triad. Exhaustive evaluation of the equality of supports gives

$$\# \mathcal{C}_{\mathcal{B}} = 79. \quad (7.23)$$

The support preserves positions, while the complete register also preserves the twelve magnitudes. The next stage uses this positional information together with relations between magnitudes.

7.7.2 Ternary words and oriented differences of the candidate

The i th ternary band of the point N/D is obtained by

$$T_i(N) = \left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13. \quad (7.24)$$

The oriented edge $a \rightarrow b$ has residue

$$R_{a,b}(N) = (K_a(N) - K_b(N)) \bmod 729. \quad (7.25)$$

Subtraction is performed in $\mathbb{Z}$ before reduction. For this edge, the pair

$$(\text{enc}_6^{-1}(T_b(N)), \text{enc}_6^{-1}(R_{a,b}(N))).$$

is queried in relations (7.18). It is therefore a joint query of the destination word and the oriented difference of the triads.

Write $\text{Cmp}(a, b; N)$ when the pair belongs to $\mathcal{A}_{\text{cmp}}$ and $\text{Af}(a, b; N)$ when it belongs to $\mathcal{A}_{\text{af}}$.

7.7.3 Determination of the transverse orbit

The twenty-four-trit descriptor determines the first three decimal triads before the terminal filtrations. Indeed, its cylinder satisfies the exact inclusion

$$\left[\frac{66241910521}{3^{24}}, \frac{66241910522}{3^{24}} \right) \subseteq \left[\frac{234543140}{10^9}, \frac{234543141}{10^9} \right).$$

The two endpoint inequalities are verified by integer cross-products. Thus, every candidate shares the three reference positions 1, 2, 3 and their representations (234, 543, 140).

The step-four translation on $\mathbb{Z}/12\mathbb{Z}$ has the four orbits

$$(1, 5, 9), \quad (2, 6, 10), \quad (3, 7, 11), \quad (4, 8, 12).$$

The first three each contain one of the reference positions; the fourth is the only one disjoint from $\{1, 2, 3\}$. The oriented forest of steps three and four is fixed by the edges

$$\begin{aligned} 1 &\rightarrow 4, & 1 &\rightarrow 5, & 2 &\rightarrow 6, \\ 3 &\rightarrow 7, & 4 &\rightarrow 8, & 5 &\rightarrow 9, \\ 6 &\rightarrow 10, & 7 &\rightarrow 11, & 8 &\rightarrow 12. \end{aligned}$$

Its first edge has step three and the remaining edges have step four. The two internal edges of the distinguished orbit are precisely $4 \rightarrow 8$ and $8 \rightarrow 12$. The location of the reading condition is thus fixed by the geometry of the positions and by the initial prefix, before examining the candidates of the exceptional face.

7.7.4 Terminal heterotypy condition

The terminal condition is

$$\begin{aligned} \mathcal{J}(N) &= [\text{Cmp}(4, 8; N) \wedge \text{Af}(8, 12; N)] \\ &\quad \vee [\text{Af}(4, 8; N) \wedge \text{Cmp}(8, 12; N)]. \end{aligned} \tag{7.26}$$

Both assignments of types to the two edges are admitted. The selection requires the existence of a realization with distinct types; the individual labels and their times remain available in the incidence witnesses.

Theorem 7.11 (Terminal selection of the coupling register). *On the candidate domain produced by Ω_{24} , $\mathcal{S}_{\text{term}}$, $\mathcal{P}_3$ and the hundred-row regional calendar, there exists a unique numerator simultaneously satisfying exceptional incidence and heterotypy:*

$$\{N \in \mathcal{C}_{\mathcal{B}} : \mathcal{J}(N)\} = \{234543140729659824621058914794146601\}.$$

Its twelve-triad register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \tag{7.27}$$

Proof. The preceding count produces the 79 elements of $\mathcal{C}_{\mathcal{B}}$. For each one, its twelve coordinates are extracted by (7.22); the destination bands T_8 , T_{12} and residues $R_{4,8}$, $R_{8,12}$ are computed; the two masks of the temporal repertoire are queried; and disjunction (7.26) is applied. The resulting list contains exactly

the stated integer, according to the exhaustive evaluation certified by `terminal_selection`. Equality of the list with a singleton implies existence and uniqueness in the declared domain. Base-one-thousand extraction from that integer gives (7.27).

The selection procedure receives the enumerated producers and rules. The final integer appears in the statement of the verified equality after selection has been executed. This logical position allows the result to be checked without using it as a selection criterion. $\square$

7.7.5 Arithmetic witness of membership in a cylinder

One of the incidences of the selected numerator uses the ordering

$$(002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110)$$

and the boundary $438 = 121020_3$. For this incidence,

$$P_{72} = 5283881584882673136514102926090957,$$

$$P_{78} = 3851949675379468716518781033120308091.$$

The integer-endpoint formula yields

$$\left\lceil \frac{P_{78} 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146601,$$

$$\left\lceil \frac{(P_{78} + 1) 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146602.$$

The unique integer in the interval is therefore N_K . The thirteen ternary bands of its decimal point are

$$\begin{array}{cccc} 020022 & 222111 & 211201 & 021101 \\ 002001 & 020111 & 200110 & 121012 \\ 010122 & 001001 & 221110 & 222110 \\ 121020 & & & \end{array}$$

This calculation exhibits a concrete witness of membership in the enumerated domain. Uniqueness of the register follows from the exhaustive examination of all candidates, in addition to this particular witness.

7.8 Regional composition and continuity of the realizations

7.8.1 Composition of the regional generators

Let $\text{Sel}(\Omega, \mathcal{S}, \mathcal{P}, \mathcal{N})$ be the selector defined by the enumeration and the two preceding filters. The regional producers satisfy the equalities

$$\Omega_{\text{reg}} = \Omega_{24}, \quad \mathcal{P}_{\text{reg}} = \mathcal{P}_9, \quad \mathcal{N}_{\text{reg}} = \mathcal{N}_{100},$$

where $\mathcal{N}_{\text{reg}}$ is computed by band extraction and signature reading. By congruence of functions,

$$\begin{aligned} \text{Sel}(\Omega_{\text{reg}}, \mathcal{S}_{\text{term}}, \mathcal{P}_{\text{reg}}, \mathcal{N}_{\text{reg}}) \\ = \text{Sel}(\Omega_{24}, \mathcal{S}_{\text{term}}, \mathcal{P}_9, \mathcal{N}_{100}) = \{N_K\}. \end{aligned} \tag{7.28}$$

The equality replaces three documentary interfaces with their actual regional construction. The terminal repertoire retains the provenance specified in (7.13).

The register is then formed by the twelve Euclidean divisions in (7.22). Each coordinate belongs to $\{0, \dots, 999\}$ and the length is twelve by construction. The implementation that takes the first element of the selected list is justified by equality with the singleton: the default value of a computational operation on empty lists never enters this result.

7.8.2 Relation to signed incidences and inversion

The present construction provides a prospective selection in an explicit finite domain. The subsequent development of the dodecaphasic register produces its oriented channels from the enriched terminal state and proves an integral inversion through Hadamard. That inversion recovers a register from compatible channels; here a register is selected from its regional and incidence-based inputs. Equality of their results allows the two realizations to be composed while preserving the origin of each operation.

Uniqueness of selection and uniqueness of inversion are distinct properties. The former quantifies over the candidates constructed in this chapter. The latter quantifies over the registers sharing a compatible signature. Their conjunction justifies transporting the selected register to the signed realization and recovering it from that realization, without turning a retrospective recovery into the producer of the register.

7.8.3 Relation between the terminal counts

The sequence

$$40320 \longrightarrow 21902 \longrightarrow 19446 \longrightarrow 79 \longrightarrow 1$$

records, successively, orderings, indexed incidences with the grid, distinct numerators, candidates of the exceptional face, and heterotypic survivors. The arrow notation indicates the order of the operations; the second number counts a different class of objects from the first.

The regional-catalogue count $196 \cdot 197 \cdot 196 \longrightarrow 331 \longrightarrow 2 \longrightarrow 1$, developed in its corresponding place, starts from another population: regional blocks from three catalogues and terminal conditions of Gram, occupancy, distance, and orientation. Both counts retain their domains. Their convergence to the same representation is expressed through the identification of their readers and outputs; the intermediate numbers describe different operations and remain documented separately.

7.8.4 Regional composition of the fine-structure constant

The output K subsequently enters the ordered regional composition and the base-one-thousand carry. Its later use preserves the twelve positions and the reading boundary. Coupling to the analytic root and representations at arbitrary depth uses that same output, together with the regional coordinates already generated and the rules of the analytic chart. This dependency prepares the chapter devoted to the two correlated realizations of the fine-structure constant.

The scope of this chapter is precise: production of the descriptor and its regional inputs, determination of the finite domain, unique selection on that domain, and material composition with the producers. The analytic recurrence at all orders and the compatibility of infinite carries are developed subsequently with their own proofs.

7.9 Executable certification and traceability of the operations

The formal development preserves the correspondence between the mathematical definitions and their executable realizations. The signature and clock proofs are found in `N69RegionalSignature.lean`; band extraction and reproduction of the hundred rows, in `GeneratedN69Rows.lean`; the minimality criterion and construction of Ω_{24} , in `RegionalW24.lean`. Compatibility of the two Witt matrices is assembled in `WittReaderCompatibility.lean`.

The temporal readers, rotations, oriented residues, and heterotypic condition are formalized in `TerminalReaders.lean`. Enumeration and counts are formalized in `TerminalOrbitalSelection.lean`. The regional panel is linked through `RegionalPanelBridge.lean`, and composition (7.28) is expressed in `SelectedFromRegionalInputs.lean`.

The finite enumeration checks use Lean’s native evaluation; their reports record `Lean.ofReduceBool`, in addition to the library’s customary logical axioms. The regional reconstruction of the descriptor uses kernel proofs and ordinary reduction. This distinction describes the

verification mechanisms actually employed. The compilation receipts attest to their declared modules and dependencies; the mathematical proofs retain the domains, producers, and hypotheses made explicit in the body of the chapter.

7.10 Complete census of cylinders and selection of the terminal boundary

The terminal readout jointly uses two kinds of information: compatibility of the regional cylinders and an incidence profile with labelled channels and positions. This section constructs the complete catalogues, enumerates their triples, and proves the effect of each reader condition. The starting point consists of the regional outputs already produced in Section 5, not values introduced to define the APP-TRIT-TPK transport. Reduction to a matrix belongs to a subsequent readout of these outputs and preserves the order of the three channels.

The result is a census theorem with an explicit starting structure. The profile of Gram products, weights, distances, and marginals is declared as a datum of the terminal reader. The exhaustiveness of the enumeration proves what this profile selects on the specified catalogues; it does not, by itself, prove that the profile is a universal law or that it can be reconstructed from the unique triple that ultimately satisfies it. This distinction allows the complete finite proof to be incorporated without reversing its genealogy.

7.10.1 From regional outputs to ternary catalogues

Let $x_\pi, x_e, x_\varphi \in [0, 1)$ be the fractional parts of the three regional coordinates already constructed. We use their expressed cylinders of one thousand decimal places,

$$J_\chi^{(1000)} = \left[\frac{d_\chi}{10^{1000}}, \frac{d_\chi + 1}{10^{1000}} \right), \quad \chi \in \{\pi, e, \varphi\}.$$

The integers d_χ are finite output readouts. A length of one thousand is sufficient for the integers that follow, as verified from the two bounds of each interval; no infinite depth is assumed. We define

$$p = 30, \quad r = 1050, \quad L = p + r = 1080, \quad k = 171, \quad D = 3^L, \quad Q = 1000^k, \quad (7.29)$$

and

$$P_\chi = \lfloor 3^{30} x_\chi \rfloor, \quad T_\chi = \lfloor Q x_\chi \rfloor.$$

Here k counts decimal triads and does not denote the dodecaphasic register K . We have $1000^{171} \leq 3^{1080} < 1000^{172}$.

Lemma 7.12 (Stable extraction of an integer). *For $d, m, h \in \mathbb{Z}$, with $m, h > 0$, the value of $\lfloor mx \rfloor$ is constant on $[d/h, (d+1)/h)$ if and only if*

$$\left\lfloor \frac{md}{h} \right\rfloor = \left\lfloor \frac{m(d+1)-1}{h} \right\rfloor. \quad (7.30)$$

When this holds, either side gives that value.

Proof. The minimum of mx is attained at the left endpoint. Its supremum is $m(d+1)/h$ and is not attained. The largest integer value that $\lfloor mx \rfloor$ can take is $\lceil m(d+1)/h \rceil - 1$, which equals the second side because the numerator and denominator are integers. Equality of the endpoints of the range of values proves the assertion. $\square$

The stability test holds in all three channels for $m = 3^{30}, Q, 3^{1080}$. The last choice will be used solely for a subsequent comparison, not to select the profile. The first thirty recovered positions are precisely the five six-trit stages of the regional transport:

Channel	Thirty-trit prefix	P_χ
π	010211012222010211002111110221	29152671743887
e	201101121221102011012222102011	147887858824447
φ	121200112202121020010210010200	127247717616687

The agreement preserves the prolongation already constructed; the decimal readout is not used to replace its transport matrices retrospectively.

For a fixed channel, a tail of r trits is represented by $0 \leq R < 3^r$. Its cylinder and the cylinder of the first k triads are

$$I_{\chi,R} = \left[\frac{P_\chi 3^r + R}{D}, \frac{P_\chi 3^r + R + 1}{D} \right), \quad J_{\chi,T} = \left[\frac{T_\chi}{Q}, \frac{T_\chi + 1}{Q} \right). \quad (7.31)$$

The catalogue includes all tails for which these two cylinders have a nonempty intersection.

Proposition 7.13 (Catalogue by exact intersection). *The compatible tails are the integers in the interval $[a_\chi, b_\chi]$, where*

$$a_\chi = \max \left(0, \left\lfloor \frac{T_\chi D}{Q} \right\rfloor - P_\chi 3^r \right), \quad (7.32)$$

$$b_\chi = \min \left(3^r - 1, \left\lfloor \frac{(T_\chi + 1)D - 1}{Q} \right\rfloor - P_\chi 3^r \right). \quad (7.33)$$

For the specified regional cylinders, no clipping applies and we obtain

χ	$a_\chi \bmod 729$	$b_\chi \bmod 729$	$b_\chi - a_\chi + 1$
π	67	262	196
e	584	51	197
φ	117	312	196

(7.34)

Proof. Write $N = P_\chi 3^r + R$. The two half-open intervals in (7.31) intersect exactly when

$$NQ < (T_\chi + 1)D, \quad (N + 1)Q > T_\chi D.$$

The second inequality is equivalent to $N \geq \lfloor T_\chi D / Q \rfloor$; the first, to $N \leq \lfloor ((T_\chi + 1)D - 1) / Q \rfloor$. Subtracting $P_\chi 3^r$ and retaining $0 \leq R < 3^r$ proves the formulae. Evaluation by integer division of the T_χ extracted using (7.30) gives the table. Since $r \geq 6$, $P_\chi 3^r \equiv 0 \pmod{729}$, so the residue of the first endpoint can also be obtained directly from $\lfloor T_\chi D / Q \rfloor \bmod 729$. $\square$

The last six trits of a tail form the word $\nu_6(R \bmod 729)$, where ν_6 denotes the ternary representation of length six, including its leading zeros. By the table, each catalogue contains fewer than 729 consecutive elements, so this reduction is injective on it. Thus, the three boundary catalogues are completely described, without choosing the final triples:

$$\begin{aligned} \mathcal{B}_\pi &= \{\nu_6(67 + i) : 0 \leq i < 196\}, \\ \mathcal{B}_e &= \{\nu_6((584 + j) \bmod 729) : 0 \leq j < 197\}, \\ \mathcal{B}_\varphi &= \{\nu_6(117 + \ell) : 0 \leq \ell < 196\}. \end{aligned} \quad (7.35)$$

Requiring the left endpoint of $I_{\chi,R}$ to belong to $J_{\chi,T}$ would impose a different condition: it would lose the first cylinder that intersects the boundary and would give 195, 196, 195. The census uses the intersection of the complete cylinders, in accordance with its definition.

7.10.2 Declared profile and compatibility indicators

For $u, v \in \mathbb{F}_3^6$, let $g(u, v) = \sum u_i v_i \in \mathbb{F}_3$, $n(u) = g(u, u)$, $w(u) = \#\{i : u_i \neq 0\}$, and $h(u, v) = \#\{i : u_i \neq v_i\}$. The last two functions take integer values; the first two take values in $\mathbb{F}_3$. The reader's local profile, in the order (π, e, φ) , is

$$\begin{aligned} (g_{\pi e}, g_{\pi \varphi}, g_{e \varphi}) &= (2, 2, 0), & (n_\pi, n_e, n_\varphi) &= (0, 0, 0), \\ (w_\pi, w_e, w_\varphi) &= (3, 3, 3), & (h_{\pi e}, h_{\pi \varphi}, h_{e \varphi}) &= (2, 5, 3). \end{aligned} \quad (7.36)$$

The words are ordered as in (7.35). We use $[E] \in \{0, 1\}$ for the indicator of a condition E . For each pair of channels χ, ψ , the rectangular matrix $A_{\chi\psi}$ has entry $[g(u, v) = g_{\chi\psi}]$. The matrix $A_{\chi\psi}^h$ multiplies that entry by $[h(u, v) = h_{\chi\psi}]$. Let D_χ^n be the diagonal matrix of $[n(u) = 0]$, and D_χ^{nw} that of $[n(u) = 0][w(u) = 3]$.

Theorem 7.14 (Exhaustive enumeration by finite sums). *The product of the three catalogues contains 7 567 952 triples. The successive conditions in (7.36) leave*

$$7\,567\,952 \longrightarrow 275\,794 \longrightarrow 6235 \longrightarrow 3127 \longrightarrow 331. \quad (7.37)$$

These four counts are, respectively,

$$\begin{aligned} N_g &= \text{tr}(A_{\pi e} A_{e \varphi} A_{\varphi \pi}), \\ N_{gn} &= \text{tr}(D_\pi^n A_{\pi e} D_e^n A_{e \varphi} D_\varphi^n A_{\varphi \pi}), \\ N_{gnw} &= \text{tr}(D_\pi^{nw} A_{\pi e} D_e^{nw} A_{e \varphi} D_\varphi^{nw} A_{\varphi \pi}), \\ N_{gnwh} &= \text{tr}(D_\pi^{nw} A_{\pi e}^h D_e^{nw} A_{e \varphi}^h D_\varphi^{nw} A_{\varphi \pi}^h). \end{aligned} \quad (7.38)$$

All products and traces in this formula are calculated over the integers, not over the field with three elements.

Proof. The size of the product is $196 \cdot 197 \cdot 196$. When a trace in (7.38) is expanded, its indices run through exactly one triple of words. The summand equals one if it satisfies all the represented conditions and zero otherwise. Each triple is counted once. The diagonal matrices add the individual conditions; the rectangular factors add the pairwise conditions. This proves that the traces are precisely the stated cardinalities.

To make their evaluation explicit, fix the indices i, j of the first two channels and form the sets of indices ℓ of the third that satisfy each pairwise condition. Their indicators are intersected, the relevant individual indicators are added, and the cardinality is summed. The final sum runs over $0 \leq i < 196, 0 \leq j < 197$. Substitution of the words in (7.35) and the profile (7.36) gives, in that order, 275794, 6235, 3127, 331. The supplement records the four contributions of each of the 196 indices i and the complete list of 331 triples; its program evaluates these sums with exact binary indicators and checks each survivor by scalar evaluation. No sample or tolerance is involved. $\square$

7.10.3 Labelled incidence and orientation charge

For a surviving triple, let $B \in \mathbb{F}_3^{3 \times 6}$ be its matrix of ordered rows. The second profile requires

$$\text{rank}_{\mathbb{F}_3} B = 3, \quad w_{\text{col}}(B) = (1, 1, 2, 2, 3, 0), \quad q_\partial(B) = (1, 2, 2, 1, 2, 0), \quad (7.39)$$

where w_{col} counts the nonzero elements in each column and q_∂ sums its entries modulo three. This is a profile of labelled columns; permuting them without transporting the profile defines a different reader. Denote by $\mathcal{F}$ the set of 331 triples from the preceding theorem.

Proposition 7.15 (Reduction of the census to the terminal pair). *The sums of indicators over $\mathcal{F}$, successively imposing rank, column weights, and column sums, are*

$$\sum_{B \in \mathcal{F}} [\text{rank } B = 3] = 331, \quad N_{\text{rango,pesos}} = 18, \quad N_{\text{rango,pesos,sumas}} = 2. \quad (7.40)$$

With catalogue indices starting at one, the two final triples are (120, 197, 157) and (129, 197, 148), and their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

Proof. For each triple in the already enumerated set, the rank is computed by elimination over $\mathbb{F}_3$, the nonzero elements of the six columns are counted, and their entries are summed modulo three. The products of the indicators in (7.39) give (7.40). Substituting the two final indices into (7.35) gives the displayed matrices. The complete list of survivors and the exact filters in the supplement allow the elimination to be reproduced without presupposing these two matrices as the domain. $\square$

The orientation $\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3$ selects the line $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$. The row charges of B_0, B_1 are (0, 2, 0) and (2, 2, 1); only the latter belongs to this line. This recovers the last step of the selector in Section 7.15, now preceded by the construction and enumeration of its complete domain. The stable readout of $[3^{1080}x_\chi]$, performed after the filters, gives the boundaries 021020, 001220, 100210, with the same ranks (129, 197, 148). This subsequent comparison does not define the profile or enter the sums of indicators.

7.10.4 Preserved information and scope of the result

The procedure preserves the thirty-trit prefix, the compatible tail, the regional order, the position of each trit, and the orientation. Reduction modulo 729 preserves the identity of each candidate here because the interval of tails has length less than 729; this injectivity is not asserted for an arbitrary history. The census of 331 also proves that the local profile alone does not select a single matrix: the labelled marginals and the charge supply additional effective conditions.

Thus, a complete finite enumeration and its selector have been recovered on output cylinders, with an explicit readout profile. Construction of the signed register additionally uses its memory reader and its channels, as specified below; the visible matrix is not identified with that register by equality of the number of components. Nor does this count prove the universality of the profile, generate the regional coordinates anew, or decide a claim about alpha or about the complete continuum.

8 Dodecaphase record and the rational phase constant

Generation of the regional coordinates does not exhaust the information in the APP-TRIT-TPK state. One Archimedean readout determines a value through compatible cylinders; another readout preserves the opposition of the sheets, the orientation of the route, and the record of the incidences traversed. The dodecaphase record belongs to this second readout before entering the closure of alpha. It is therefore not defined by subtracting a physical constant from a combination of previously known values. It is reconstructed from a signed coordinate of the same arithmetic state with trajectory memory that produces the regional coordinates.

The distinction is necessary even when two objects have twelve coordinates. A twelve-block catalogue word, a list of twelve oriented events, a signed integer vector, and a base-thousand dodecaphase record are different objects. Comparing them requires the maps that relate them. Equality of length does not provide these maps, and loss of signs is not merely a simplification of notation.

This chapter preserves the causal cutoff established in the preceding chapters: the nine APP positions on the horizontal and vertical axes produce the arithmetic sheets; TRIT types regime and orientation; TPK selects, transports, updates, and prolongs, preserving residue, quotient, carry, sheet, route, boundary, and memory. The coordinates used here are obtained from the joint discrete structure of the continuum. The chain $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ is not replaced by a finite schedule. Nor is the order $U_t \rightarrow \Gamma_9 \rightarrow K_{\text{ph}}$ reversed: the last map is a reduced-memory readout, not the generator of the signed record.

Reversible coordinatization and incidence determination. The result organizing this chapter is the equivalence of three coordinatizations of a register previously produced by APP–TRIT–TPK. The Hadamard transformation relates the signed register to the integer register; the difference extractor relates the latter to its transverse variations and its uniform component:

$$U \xrightleftharpoons[\mathcal{H}_{12}]{\mathcal{H}_{12}/4} K \xrightleftharpoons{(D_3, D_4, Q)} (D_3 K, D_4 K, Q(K)).$$

The first equivalence is established on the integral sublattice $\mathcal{L}_H$; the second, on the image of the extractor. With base 1000, the 12 positions and the phase origin fixed, the evaluation $K \mapsto \kappa_{\text{per}}$ adds a fourth reversible coordinatization: multiplying by $1000^{12} - 1$ and performing the Euclidean divisions recovers the full register. The proofs in §8.8, §8.9 and §8.10 establish these equivalences in their respective domains.

Two compositions subsequently act on this register. The positional composition combines its components with the closure, propagation and self-scaling coordinates, and its normalization determines α_A . The incidence composition applies the cubic threshold and yields

$$K \longmapsto B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The full register allows both compositions to be reconstructed. The support B_K specifically retains membership in the class selected by the threshold. The identity $B_K = H_e \cap P_\pi$ in (10.12) connects this integer extraction to the incidence of the regional codes. The negative support of their balance with K completes the flag $B_K \subset H_\alpha^-$ developed in the exceptional prolongation. Thus the role of K is defined by an invertible transformation and by precise subsequent maps; the additional information on depth and transport remains in the state from which the register originates.

8.1 Cyclic extension and memory grading

The observable schedule contains one hundred and eight positions, organized into twelve nonadic windows. With positive indices, let

$$E_m = \{9(m-1) + 1, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The ordered support $E_1 \sqcup \dots \sqcup E_{12}$ preserves position and window membership. The group $C_{108} = \mathbb{Z}/108\mathbb{Z}$, by contrast, preserves the law of cyclic translation. For the representative $0 \leq \tilde{\theta} < 108$, the coordinates

$$\beta(\theta) = \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad r(\theta) = 1 + (\tilde{\theta} \bmod 9)$$

separate the dodecaphase sector from the interior nonadic position. Neither by itself preserves the total number of turns.

Proposition 8.1 (The carry of the schedule). *The maps*

$$\iota : C_{12} \longrightarrow C_{108}, \quad [b] \longmapsto [9b], \quad p : C_{108} \longrightarrow C_9, \quad [\theta] \longmapsto [\theta]_9$$

form a nonsplit exact sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0.$$

For the set-theoretic section choosing the representatives $0, \dots, 8$, the failure of additivity is the carry

$$c(r, r') = \left[\left[\frac{\tilde{r} + \tilde{r}'}{9} \right] \right]_{12}.$$

Proof. The kernel of p consists of the multiples of nine and coincides with the image of ι . Reduction is surjective, and multiplication by nine is injective on the stated domain. If the sequence split, the middle group would be isomorphic to $C_{12} \times C_9$, whose exponent is $\text{mcm}(12, 9) = 36$; the exponent of C_{108} is 108. This contradiction proves nonsplitting. Finally, Euclidean division of $\tilde{r} + \tilde{r}'$ by nine gives the stated carry and verifies the identity for the section. $\square$

The result explains a distinction that will reappear in alpha: returning to the phase is not equivalent to returning to the state. If q_{mem} counts nonadic turns, the schedule preserves only $\beta = [q_{\text{mem}}]_{12}$. After one hundred and eight steps, β returns, but q_{mem} increases by twelve. The record also preserves the cylinder, the incidence ledger, and the vertical memory. Closure of a finite chart does not turn this history into a memoryless cycle.

8.2 The oriented record and the integral 1 + 5 + 6 chart

For a history produced by TPK, let τ_m be the subroute over E_m , σ_m its orientation, κ_m the boundary carry, and Λ_m the local incidence ledger. The record is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}.$$

Its domain is the space of arithmetic histories with trajectory memory, not the set of phase labels. Projection onto the schedule forgets precisely some of the data required by the signed readout.

An initial chart of the record can be written in terms of oriented events. The reference construction (Subsection 8.2) fixes the event code set $\{2, 4, 6, 8\}$ and the sector orientation

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

If d_t is the direction code at step t , this gives

$$e_i = \Sigma_{12}(i+1) \# \{t \in E_{i+1} : d_t \in \{2, 4, 6, 8\}\}, \quad 0 \leq i < 12.$$

The formula keeps the nonnegative event count separate from orientation. It does not identify a cardinal direction with a TRIT sign. The vector e , whose components are signed local counts, is also not yet the vector U that will appear below.

Opposite positions determine, for $0 \leq i < 6$,

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}.$$

With

$$s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5),$$

the integral chart is defined by

$$\mathcal{L}_{12}(e) = (s_C, M_0, \dots, M_4, a_0, \dots, a_5).$$

The decomposition 1 + 5 + 6 is not an arbitrary allocation of coordinates: one coordinate preserves the charge, five compare the pairs with a reference pair, and six preserve the opposition of the half-crowns.

Lemma 8.2 (Inversion of the integral chart). *An integer tuple $(s_C, M_0, \dots, M_4, a_0, \dots, a_5)$ belongs to the image of $\mathcal{L}_{12}$ if and only if*

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6),$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad p_i = p_5 + M_i \quad (i < 5).$$

In that case its preimage is unique and is given by

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}.$$

Proof. The sum of the five margins is $s_C - 6p_5$, which gives the division by six. Once the p_i are reconstructed, each pair of equations $p_i = e_i + e_{i+6}$, $a_i = e_i - e_{i+6}$ has the indicated solution. This solution is integral exactly when the parity congruence holds. All operations are inverted, so no residual freedom remains. $\square$

This reversibility shows what information is preserved; it does not authorize discarding the rest of the ledger. The complete record has a larger domain than this count chart. Extraction of the channels feeding U uses that complete record and cannot be deduced from the list (e_i) by identifying names.

8.3 Evaluation of incidences and determination of the transverse coordinates

The dodecaphasic register preserves a history of visits and oriented traces. Its transverse coordinates are obtained through a composition of operations: arithmetic evaluation of the visits, counting by windows, decimal normalization, and cyclic differences. Reconstructing the register from those differences is another operation, whose existence and uniqueness will be proved below. This separation makes it possible to specify both the data used by the extractor and the information it transmits to the harmonic readout.

Let $\mathcal{L}$ be an incidence register produced by the selection, transport, and updating of an APP-TRIT-TPK history. Each visit preserves its route, APP position, sheet, time, multiplicity, and boundary class; each trace preserves its members, orientation, multiplicity, and length in the unit declared by its readout. The windows are

$$E_m = \{9(m-1) + 1, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

Evaluation of the sum or product sheet determines the integer $n(v)$ of each visit and its positive division

$$n(v) = r(v) + 9q(v), \quad 1 \leq r(v) \leq 9.$$

The residue is calculated from that evaluation. The boundary class of a visit with residue 9 is preserved by $\epsilon_{\partial}(v) = 1$ on the corona and -1 in the interior.

The incidence presentation of the register uses the three counts

$$A_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{1,3,5,7\}}(r(v)), \quad (8.1)$$

$$C_m = \sum_{j \geq 0} (\nu_{120,m}^-(j) - \nu_{120,m}^+(j)), \quad (8.2)$$

$$V_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{9\}}(r(v)) \epsilon_{\partial}(v). \quad (8.3)$$

Here $w(v)$ is the incidence multiplicity and $\nu_{120,m}^{\pm}(j)$ counts the oriented traces of the indicated channel with reduced length $120(1 + 3j)$, according to the enumeration of the register. The collection of traces and its multiplicity are data of the evaluation, together with their corresponding production rule. The integer sum requires finite support or a construction that determines its meaning. The reduced length of a trace and the number of times in the window are differently typed quantities; the readout unit specifies their relationship. The symbols A_m, C_m, V_m here denote integer counts, distinct from the angular pair and the constitutive operator.

Let $[x]_{10} \in \{0, \dots, 9\}$ be the Euclidean representative. Normalization also preserves the quotient in

$$x = [x]_{10} + 10[x/10].$$

The block is formed as

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}, \quad 0 \leq z_m \leq 999. \quad (8.4)$$

With cyclic indices modulo 12, let

$$(D_a z)_m = z_m - z_{m+a}, \quad a \in \{3, 4\}, \quad Q(z) = \sum_{m=1}^{12} z_m.$$

The incidence composition is expressed by

$$\mathcal{E}_{\text{cnt}}(\mathcal{L}) = \mathcal{D}(z) := (D_3 z, D_4 z, Q(z)) \in \mathbb{Z}^{25}. \quad (8.5)$$

Its arguments are the visits and traces with their counting rules. The evaluation does not receive K , U , α , or the expected transverse coordinates. To identify it with $\mathcal{E}_{108}^{90,120}$, the composition with the actual collection and incidences of the terminal state is also preserved. The invertibility of $\mathcal{D}$ determines a subsequent reconstruction; by itself, it does not select that collection of incidences.

Proposition 8.3 (Residual information determined by the extractor). *Let $N, N' \in \mathbb{Z}^{12 \times 3}$ be two count registers, ordered by (A, C, V) . For the operations (8.4)–(8.5),*

$$\mathcal{E}_{\text{cnt}}(N) = \mathcal{E}_{\text{cnt}}(N') \iff N - N' \in 10\mathbb{Z}^{36}.$$

The 36 sectoral residues determine exactly the output of 25 coordinates. The quotients and provenance constitute additional information in the register.

Proof. Equality modulo 10 produces the same blocks and the same transverse coordinates. Conversely, equality of the coordinates gives $D_3(z - z') = D_4(z - z') = 0$. Steps 3 and 4 generate the cycle of length 12, so $z - z'$ is constant. Equality of Q makes that constant zero. The expansion of each integer from 0 to 999 into three decimal digits is unique; hence the three residues of each window are recovered. $\square$

The proposition is valid on the entire indicated integer domain. It does not postulate that the 36 classes are independent on the subset of admissible registers produced by TPK.

8.4 Integral image and inversion of the transverse coordinates

To express the inversion, the blocks are reindexed by $i \in \mathbb{Z}/12\mathbb{Z}$, from 0 to 11. Let $(Sz)_i = z_{i+1}$, $D_3 = I - S^3$, and $D_4 = I - S^4$. Given an integer triple (b, c, q) , define

$$w_i = c_i - b_{i+1}, \quad P_0 = 0, \quad P_i = \sum_{j=0}^{i-1} w_j \quad (1 \leq i \leq 12), \quad T(w) = \sum_{i=0}^{11} P_i. \quad (8.6)$$

Theorem 8.4 (Characterization of the integral image). *A triple $(b, c, q) \in \mathbb{Z}^{25}$ belongs to $\text{im } \mathcal{D}$ if and only if*

$$b_i = w_i + w_{i+1} + w_{i+2}, \quad 0 \leq i < 12, \quad (8.7)$$

$$\sum_{i=0}^{11} w_i = 0, \quad (8.8)$$

$$q + T(w) \equiv 0 \pmod{12}. \quad (8.9)$$

Its unique integer preimage is

$$z_i = \frac{q + T(w)}{12} - P_i, \quad 0 \leq i < 12. \quad (8.10)$$

The twelve relations (8.7) and relation (8.8) are linearly independent over $\mathbb{Q}$.

Proof. For an output of $\mathcal{D}$,

$$c_i - b_{i+1} = (z_i - z_{i+4}) - (z_{i+1} - z_{i+4}) = z_i - z_{i+1}.$$

Three consecutive increments sum to b_i , and their total cyclic sum is zero. Moreover, $P_i = z_0 - z_i$, whence $q = 12z_0 - T(w)$. This gives the conditions and the inverse formula. Conversely, the congruence makes the coordinates of (8.10) integers, and closure allows them to be extended cyclically. Their adjacent differences are w_i , so $D_3z = b$ and

$$(D_4z)_i = w_i + w_{i+1} + w_{i+2} + w_{i+3} = w_i + b_{i+1} = c_i.$$

Summation gives $Q(z) = q$. The formula also determines uniqueness. Finally, the invertible integer change $(b, c) \mapsto (b, w = c - Sb)$ transforms the first twelve relations into $b = (I + S + S^2)w$: each solves for a different coordinate of b . Closure $\sum_i w_i = 0$ adds an independent relation. $\square$

The rational image has dimension 12. The integral condition adds the congruence of order 12 in (8.9). The eleven increments $w_0, \dots, w_{10}$ and the charge q are sufficient coordinates, with $w_{11} = -\sum_{i=0}^{10} w_i$. The 25 coordinates preserve those 12 degrees of freedom through 13 linear relations. If the triple also comes from (8.4), the inverse belongs to $\{0, \dots, 999\}^{12}$ and recovers its three residues per window.

8.4.1 Agreement with the harmonic transform

The transform $\mathcal{H}_{12}$ acts through H_4 on the orbits $(r, r+3, r+6, r+9)$, $r = 0, 1, 2$, where

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I_4.$$

For a compatible triple, let

$$B_r = \sum_{j=0}^3 c_{r+3j}, \quad a_0 = \frac{q + 2B_0 + B_1}{3}, \quad a_1 = a_0 - B_0, \quad a_2 = a_1 - B_1.$$

The harmonic readout has blocks

$$(\Pi_H(b, c, q))^{(r)} = (a_r, b_{r+3} - b_{r+9}, b_r + b_{r+6}, b_r - b_{r+6}).$$

Proposition 8.5 (Agreement of reconstructions). *On $\text{im } \mathcal{D}$,*

$$\Pi_H \mathcal{D} = \mathcal{H}_{12}, \quad \mathcal{D}^{-1} = \frac{1}{4} \mathcal{H}_{12} \Pi_H.$$

The image of Π_H restricted to that domain is

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}.$$

Proof. For $z = \mathcal{D}^{-1}(b, c, q)$, let $h_r = \sum_{j=0}^3 z_{r+3j}$. Shifting by four positions takes each orbit to the next; therefore $B_r = h_r - h_{r+1}$. The identity $q = h_0 + h_1 + h_2$ gives $a_r = h_r$. Within an orbit, write (x_0, x_1, x_2, x_3) for its coordinates and $d_j = x_j - x_{j+1}$. Then

$$\begin{aligned} d_1 - d_3 &= x_0 + x_1 - x_2 - x_3, \\ d_0 + d_2 &= x_0 - x_1 + x_2 - x_3, \\ d_0 - d_2 &= x_0 - x_1 - x_2 + x_3. \end{aligned}$$

These are the remaining three rows of $H_4 x$. Inversion and the integral characterization follow from $\mathcal{H}_{12}^2 = 4I$. $\square$

The harmonic congruence condition controls the output of Π_H ; membership of the channels in $\text{im } \mathcal{D}$ also requires (8.7)–(8.9). An exact negative control is obtained by taking $b = 0$, $q = 0$, and $c = e_0 - e_3$. The three orbital sums of c vanish, so $\Pi_H(b, c, q) = 0$, but the image relations fail. The harmonic readout does not replace the compatibility check for the channels.

8.5 Conjugation, covariance, and memory of the incidence register

Let $\rho(m) = 13 - m$. If incidence conjugation reverses the order of the windows, preserves the arithmetic and boundary classes of the visits, and interchanges the trace channels, then

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\rho(m)}, -C_{\rho(m)}, V_{\rho(m)}).$$

With $c_m = [C_m]_{10}$, decimal normalization gives

$$z_m^\dagger = z_{\rho(m)} + 100\mathbf{1}_{\{c_{\rho(m)} \neq 0\}} - 20c_{\rho(m)}. \quad (8.11)$$

Indeed, $[-C]_{10} = 0$ if $[C]_{10} = 0$, and is $10 - [C]_{10}$ otherwise. The channel negation is applied before reconstructing the decimal block.

Application to a specific involution also preserves the window endpoints. For a route $\gamma = (x_0, \dots, x_n)$ read before advancement,

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0). \quad (8.12)$$

Both sums contain the same interior terms; the difference is the final endpoint minus the initial one. On closed routes this correction vanishes. On open windows it is incorporated before reduction modulo 10.

Proposition 8.6 (Consequence of a periodic reduction). *If the observable of a fixed collection of routes satisfies $o(t + 54) = o(t)$, its multiplicities are preserved under that return, and the same local readout acts on windows of nine times, then*

$$(A, C, V)_{m+6} = (A, C, V)_m, \quad z_{m+6} = z_m.$$

Proof. Translation $t \mapsto t + 54$ takes each window bijectively to the one six positions later and preserves each counted incidence and its multiplicity. Equality is transmitted to the counts and their decimal normalizations. $\square$

A presentation with twelve distinct blocks therefore preserves information that does not factor through that observable of period 54: incidence selection, orientation, endpoints, multiplicity, or memory. The proposition delimits the effect of that reduction; it does not assign period 54 to the complete history.

8.5.1 Determination on orbits and accumulation of events

Fixing only $Q(z) = q$ leaves an affine class of the lattice

$$\mathcal{Z}_0 = \{v \in \mathbb{Z}^{12} : \sum_i v_i = 0\} = \bigoplus_{i=0}^{10} \mathbb{Z}(e_i - e_{11}).$$

For example, $1001 + e_0$ and $1001 + e_1$ have charge 1201 and satisfy the image checks through their respective channels. They are witnesses to the compatibility equations, without any assertion that they belong to HMT terminal histories.

Proposition 8.7 (Cyclic covariance). *Let ρ be the window translation of a register R , of minimal period $p \mid 12$, and let S be the translation of its coordinates. A covariant readout on that orbit is determined by $z \in \text{Fix}(S^p)$. The condition $Q(z) = q$ admits integer solutions if and only if $12/p$ divides q ; when they exist, they form an affine class of rank $p - 1$.*

Proof. The rule $F(\rho^j R) = S^j z$ is well defined exactly when $S^p z = z$. These vectors consist of $12/p$ repetitions of a block of length p , and their charge is $\frac{12}{p} \sum_{i=0}^{p-1} z_i$. The formula proves the divisibility and rank. Covariance of the channels follows from $D_a S = S D_a$ and $Q S = Q$. $\square$

In the harmonic chart, the corresponding action is $\hat{S}_H = \mathcal{H}_{12} S \mathcal{H}_{12} / 4$. If the origin is marked, the action also transports that mark. The period of the orbit indicated here is not identified with that of the enriched history. For an initial readout F_{108} already defined, naturality under truncation is expressed by $F_N = F_{108} \circ \tau_{N,108}$; this identity preserves the initial evaluation without selecting its production rule.

Proposition 8.8 (Additive composition of incidence contributions). *Suppose that the initial state and each event a_i produce, through previously fixed rules on the register, contributions $(\delta w_t, \delta q_t)$ such that*

$$\sum_i \delta w_{t,i} = 0, \quad \delta q_t + T(\delta w_t) \equiv 0 \pmod{12}.$$

With compatible initial conditions $(w^{(0)}, q^{(0)})$, updating by summation determines a unique register at each stage. Its increment is

$$\delta z_{t,i} = \frac{\delta q_t + T(\delta w_t)}{12} - P_i(\delta w_t).$$

The construction respects the concatenation of segments and preserves the readout of a prefix under subsequent extensions.

Proof. The compatibility conditions are preserved by summation, and the inverse (8.10) is additive in (w, q) . Induction fixes each state. Summation over concatenated segments is the sum of their contributions; a prefix receives the same summands before and after extension. $\square$

This proposition gives a sufficient criterion for additive accumulation, not a requirement that every TPK readout be linear. When events are transported by nontrivial actions, their contributions are expressed in the terminal chart through the corresponding affine cocycle. In both cases, the event rules and the initial state preserve their incidence provenance.

The preceding development determines the passage

$$\mathcal{L} \mapsto (A, C, V) \mapsto z \mapsto (D_3 z, D_4 z, Q(z)) \mapsto \mathcal{H}_{12} z$$

with the conditions of each map. Its domain differs from the regional register in Appendix C: there, the equations $X_a L_a = Y_a$ determine the lifts from ternary transitions. Here, the incidences of visits and traces determine integer blocks and their channels. The uniqueness of one reconstruction does not replace the production of the register used by the other. Both preserve their primitives, operations, and proofs in the APP-TRIT-TPK chain.

8.6 Two readouts of the terminal state

Let x_{term} denote the state produced by the composition of selection, transport, and update up to the terminal cutoff. The operator expression is

$$x_{\text{term}} = (\mathcal{U}_{108} \circ \cdots \circ \mathcal{U}_1)(x_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Its two relevant readouts are

$$x_{\text{term}} \xrightarrow{(\pi_{\text{term}}, R_{\text{sgn}})} (B_{\text{term}}, U).$$

The matrix B_{term} retains the visible ternary readout; the vector U preserves another coordinate of the same history. No arrow $B_{\text{term}} \rightarrow U$ is inserted between them.

The terminal selector of the reference construction (Subsection 8.6) acts on catalogues of sizes 196, 197, and 196. The local signatures reduce their $196 \cdot 197 \cdot 196 = 7\,567\,952$ triples to 331, and the column margin $q_{\partial} = (1, 2, 2, 1, 2, 0)$ leaves two matrices:

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row orientation is $\sigma = (1, 1, -1) = (1, 1, 2)$ in $\mathbb{F}_3^3$. The row charges of the candidates are $(0, 2, 0)$ and $(2, 2, 1)$. Only the second belongs to $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$; hence $B_{\text{term}} = B_1$. This check proves the final step of the selector from the stated census. It does not turn the column margin into a consequence of the row charge, nor replace the preceding enumeration with inspection of these two matrices.

The signed readout has the typed composition

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\text{TPK}}^{\text{term,mem}} \longrightarrow \mathbb{Z}^{12}. \quad (8.13)$$

The channel record determines

$$\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\text{term}})) = (b^{90}, b^{120}, Q_{\text{TPK}}),$$

with coordinates

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, \\ &\quad -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, \\ &\quad 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned}$$

These quantities are used as coordinates of the preceding record, not as parameters fitted through alpha. The reconstruction proved in the following subsections begins exactly at this typed output. The material trace of the preceding extractor and the scope of its self-contained transcription are preserved in the technical provenance note; the subsequent arithmetic is not presented as a substitute for that trace.

8.7 From the channel record to the signed vector

The harmonic readout Π_H can be written without using either K or alpha. To avoid confusing its orbit sums with the alpha blocks, we denote them by s_1, s_2, s_3 . Let

$$B_r = \sum_{j=0}^3 b_{r+3j}^{120}, \quad d_{r,j} = b_{r+3j}^{90}, \quad 1 \leq r \leq 3,$$

and

$$s_1 = \frac{Q_{\text{TPK}} + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2. \quad (8.14)$$

Division by three belongs to the readout of the three orbits. Its integrality is checked on compatible outputs of the record; it is not assumed for an arbitrary element of $\mathbb{Z}^{25}$.

The three signed blocks are

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \quad (8.15)$$

Integer substitution gives

$$(B_1, B_2, B_3) = (972, -1073, 101), \quad (s_1, s_2, s_3) = (2378, 1406, 2479),$$

and therefore

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns yields

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (8.16)$$

The presence of negative values and components greater than 999 shows that U is not a base-thousand digit word. Nor is it $U_6 \parallel U_6$, a catalogue concatenation of period six. The distinction concerns domain and preserved information, not merely values.

The constructive order thus far is

$$x_{\text{term}} \longrightarrow \mathcal{R}_{12} \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U.$$

The inversion that follows produces the dodecaphase record. The fact that the channels can subsequently be recovered from the dodecaphase record is a reversibility check; it does not reverse this causal order.

8.8 Orbitwise Hadamard transformation and integer reconstruction

In the twelve-position cycle, step three determines three orbits:

$$(1, 4, 7, 10), \quad (2, 5, 8, 11), \quad (3, 6, 9, 12).$$

The separation of four sectors in each orbit is what, once origin and orientation have been fixed, admits the ninety-degree angular readout. The transformation does not act on three consecutive tetrads in the natural ordering.

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

If P_{orb} reorders by the three indicated orbits, define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

Lemma 8.9 (Hadamard inversion and integrality). *We have*

$$H_4^2 = 4I_4, \quad H_4^{-1} = \frac{1}{4}H_4, \quad \det H_4 = -16.$$

For $u \in \mathbb{Z}^4$, the preimage $H_4^{-1}u$ is integral if and only if $H_4u \equiv 0 \pmod{4}$.

Proof. The rows are orthogonal and have squared norm four; the matrix is symmetric. This gives $H_4^2 = 4I_4$ and the inverse. Integer elimination, or expansion of the determinant, gives the negative sign and the value -16 . The integrality condition is exactly divisibility by four of all four coordinates of H_4u . $\square$

Definition 8.10 (Integral dodecaphase transformation). The sublattice of signed records and its integral coordinatization are

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}, \quad \mathcal{K}(u) = \frac{1}{4}\mathcal{H}_{12}u.$$

The map $\mathcal{K} : \mathcal{L}_H \rightarrow \mathbb{Z}^{12}$ is an isomorphism of abelian groups, with inverse $k \mapsto \mathcal{H}_{12}k$. The canonical record K is the image of the signed record generated by the preceding operations, with phase origin and orientation fixed.

Proof. The congruence defines precisely the integrality of $\mathcal{K}(u)$. The identity $\mathcal{H}_{12}^2 = 4I$ proves that both compositions are inverse. In particular, every $k \in \mathbb{Z}^{12}$ has preimage $\mathcal{H}_{12}k \in \mathcal{L}_H$. $\square$

Proposition 8.11 (Canonical integral record). *The inversion $K^{(r)} = H_4U^{(r)}/4$ of the blocks in (8.15) produces*

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 58, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

In the natural ordering, the dodecaphase record is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (8.17)$$

It is the unique rational preimage of U and belongs to $\{0, \dots, 999\}^{12}$.

Proof. The three products before division are

$$H_4 U^{(1)} = (936, 2916, 2484, 3176),$$

$$H_4 U^{(2)} = (2172, 2636, 232, 584),$$

$$H_4 U^{(3)} = (560, 3296, 3656, 2404).$$

All coordinates are divisible by four. Dividing and undoing the orbit permutation gives (8.17). Uniqueness follows from rational invertibility, not from a search among candidates close to a physical value. $\square$

Integrality can also be formulated in lattice terms. The determinantal divisors of H_4 are 1, 2, 4, 16: the entries have greatest common divisor one; the minors of order two are even and one has absolute value two; those of order three are multiples of four and one has absolute value four. Hence

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4),$$

and for the global transformation,

$$\text{coker } \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3.$$

The index of the image is $16^3 = 4096$. Thus, not every signed integer vector admits an integer preimage; the vector obtained satisfies the precise congruences required by inversion. The factor $1/4$ is forced by the matrix and does not have the status of a free coefficient.

8.9 Cyclic differences and the uniform component

With cyclic indices modulo twelve, let

$$(D_3 z)_m = z_m - z_{m+3}, \quad (D_4 z)_m = z_m - z_{m+4}, \quad Q(z) = \sum_{m=1}^{12} z_m.$$

The first operator compares sectors separated by three positions; the second, by four. The channel names 90 and 120 refer here to these separations in the oriented cycle, not to identification with a physical operator from another domain.

Proposition 8.12 (Completeness of the transverse system). *The operator*

$$\mathcal{D} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and nonzero invariant factors

$$\text{diag}(1, \dots, 1, 12),$$

with eleven units. In particular, its outputs determine a unique dodecaphase record.

Proof. If $D_3 z = D_4 z = 0$, the vector is constant along both steps. Since 3 and 4 generate the cycle modulo twelve, $\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle$. The total charge eliminates this last freedom because $Q(\mathbf{1}) = 12$.

For the integral statement, the rows of (D_3, D_4) are oriented incidences of a connected graph. A spanning tree provides eleven unit invariant factors. Every minor of order twelve must contain the row Q . Adding the first eleven columns to the last makes the latter zero in the incidence rows and twelve in row Q . All maximal minors are divisible by twelve, and the tree produces one of absolute value exactly twelve. The final factor is therefore twelve. $\square$

Checking (8.17) gives

$$D_3 K = b^{90}, \quad D_4 K = b^{120}, \quad Q(K) = 6263.$$

Formulas (8.14) and (8.15), in turn, reconstruct U from those differences. This closes a triangle of exact representations of the same object: transverse record, signed vector, and dodecaphase record. Equality of results does not turn their codomains into a single one; it proves that the specified maps recover the relevant information.

8.10 Two rational readouts: κ_{36} and κ_{per}

Once K has been constructed, the base-thousand chart allows two different operations. The first reads a single turn; the second prolongs the dodecaphase record periodically. Let $B = 1000$ and define its concatenation integer

$$N_K = \sum_{m=1}^{12} K_m B^{12-m} = 234543140729659824621058914794146601. \quad (8.18)$$

The leading zeros of a block, as in 058, are part of its three-digit representation. They do not change its integer value, but they make it possible to concatenate the twelve blocks unambiguously.

Definition 8.13 (Finite and periodic readouts of the dodecaphase record). The one-turn readout is

$$\kappa_{36} = \sum_{m=1}^{12} K_m B^{-m} = \frac{N_K}{B^{12}}.$$

The periodic prolongation is

$$K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}, \quad \kappa_{\text{per}} = \sum_{j \geq 1} K_j^{\text{per}} B^{-j}.$$

Proposition 8.14 (Exact value and period of the prolonged dodecaphase record). *We have*

$$\kappa_{\text{per}} = \frac{N_K}{B^{12} - 1}, \quad \kappa_{\text{per}} - \kappa_{36} = \frac{N_K}{B^{12}(B^{12} - 1)} = B^{-12} \kappa_{\text{per}}. \quad (8.19)$$

The infinite word of the dodecaphase record has minimum period twelve in base thousand and minimum period thirty-six in base ten.

Proof. The sum of the first turn is $N_K B^{-12}$; each new turn multiplies its contribution by B^{-12} . The geometric series gives $N_K/(B^{12} - 1)$, and subtraction gives the second identity.

The twelve components of K are distinct. A proper period of the cycle would identify two of them; the minimum block period is therefore twelve. The decimal expansion is canonical, since the dodecaphase record has no tail of 999 blocks. If its minimum decimal period were d , it would divide 36. Grouping digits in threes would produce a block period of $d/\gcd(d, 3)$. No proper divisor of 36 gives twelve under this operation. Hence the minimum decimal period is 36. $\square$

Both readouts are rational, but they are not the same rational number. The first terminates after twelve blocks; the second repeats the turn. Moreover, $0 < \kappa_{\text{per}} - \kappa_{36} < B^{-12}$. Their proximity does not allow one to be substituted for the other in an exact identity. In particular, the finite closure of alpha uses κ_{36} , whereas the sum of the prolonged section uses κ_{per} .

8.11 Reversibility of the rational constant and phase change

The rational constant κ_{per} determines the register K completely when the base, length, and readout origin are preserved. Its numerical and structural functions are related through an exact inversion.

Proposition 8.15 (Integral recovery and cyclic transformation). *Let $I(K) = \sum_{j=1}^{12} K_j 1000^{12-j}$. Then*

$$I(K) = (1000^{12} - 1) \kappa_{\text{per}}.$$

The integer expansion of $I(K)$ into twelve base-1000 blocks recovers every K_j , including the leading zeros of each block. If $\rho(K) = (K_2, \dots, K_{12}, K_1)$, then

$$\kappa(\rho K) = 1000 \kappa(K) - K_1. \quad (8.20)$$

Proof. The first identity inverts the definition of periodic evaluation. Iterated Euclidean division by 1000, with twelve fixed positions, recovers the vector. On the other hand,

$$I(\rho K) = 1000I(K) - K_1(1000^{12} - 1).$$

Dividing by $1000^{12} - 1$ gives (8.20). $\square$

Recovery therefore composes $\kappa \mapsto K \mapsto U$ and the transformations D_3K, D_4K, Q already constructed. Rational evaluation preserves the signed register and the cyclic differences that depend on it. The total depth and transport operators of a history remain in their own coordinates: the domain of the preceding inversion is the integral dodecaphasic register.

Write $\kappa_j = \kappa(\rho^j K)$, with indices modulo 12. The same identity gives

$$K_{j+1} = 1000\kappa_j - \kappa_{j+1}, \quad \sum_{j=0}^{11} \kappa_j = \frac{\sum_{j=1}^{12} K_j}{999} = \frac{6263}{999}.$$

Phase acts exactly on the constant: its value is fixed in the canonical readout, while its rotations follow an affine law. The orbit sum is invariant under the choice of origin.

Subsequent incidence also has an expression in these coordinates. The cubic threshold determines

$$B_K = \{j + 1 : 1000\kappa_j - \kappa_{j+1} \geq 729\} = \{4, 6, 9, 10\}.$$

The tetrad is recovered from the rational orbit because the latter first recovers the integer components. The signed support associated with α additionally uses its regional registers and boundary normalization. The relationship between coordinate and incidence is thus expressed through identifiable operations.

This is an operational definition of the role of K : it provides a reversible integral coordinatization of the oriented register, whose periodic evaluation and incidence functions participate in subsequent constructions. Value, transformation, and domain are preserved jointly.

8.12 Periodicity and the cokernel of the transport operator

The periodicity just proved belongs to the dodecaphase record. It does not imply periodicity of the regional coordinates, of the closure carries, or of alpha. It is useful to make this separation precise through a second, purely integral quotient.

Let S be the cyclic shift of twelve coordinates, with fixed orientation, and let $b \geq 2$ be an integer base. The periodic carry operator is $\partial_b = S - bI$. If a periodic identity is written

$$A = P + E - \Phi - K + \partial_b C,$$

then the transformations

$$K \mapsto K + \partial_b h, \quad C \mapsto C + h$$

preserve its right-hand side. This invariance compares representatives of a carry class; it is not authorization to modify the dodecaphase record already selected by the record.

Proposition 8.16 (Periodic carry quotient). *With the preceding orientation,*

$$\text{coker}(S - bI) \cong \mathbb{Z}/(b^{12} - 1)\mathbb{Z}.$$

Proof. The cyclic module is presented as $\mathbb{Z}[t]/(t^{12} - 1)$. Imposing $S = bI$ amounts to imposing $t = b$, leaving the relation $b^{12} - 1 = 0$. Evaluating the coefficients at b , with the chosen concatenation order, represents the resulting class. $\square$

The denominator $b^{12} - 1$ of κ_{per} and this quotient come from the same cyclic length, but their objects remain different: one is a rational evaluation and the other a quotient of a carry module. The finite

closure of alpha will be an oriented chain with a right boundary. Its prolongation will transport a compatible remote boundary, not a carry identified by decree between the first and thirteenth positions.

This prepares the next chapter. The dodecaphase record has been fixed before solving the alpha recurrence; its Hadamard reconstruction is integral and unique; its transverse differences recover the record; and its two rational evaluations have separate domains and uses. None of these facts uses alpha as an input, and none yet determines the arithmetic type of the output of the complete closure.

9 Derivation of the fine-structure constant

In this development, the fine-structure constant receives an arithmetic determination that brings together the three regional coordinates and the integral dodecaphase record. The operation combines their blocks with fixed orientation and transports the quotients of positional normalization. The resulting parameter expresses compatibility of that oriented difference with cylinder prolongation. Its value and the incidence data used in its construction will remain related throughout the exceptional development.

The internal order is important. The coordinates π , e , and φ , already constructed in the preceding section, are prior coordinates of the same arithmetic state with trajectory memory and may legitimately act as inputs to a subsequent reader. They are not conventional constants injected to select alpha. Correlated generation of the four coordinates shares origin and coinductive compatibility without requiring that all four components be evaluated in a single step.

The two routes express different mathematical roles of the same parameter. The first realizes an integer closure by Euclidean division in base 1000. The second studies the analytic equilibrium of vacancies through the resolvent $E_{21/22}$ and its graded prolongations. One route determines blocks and quotients; the other determines a root through an equation and a uniqueness domain. Their identification is formulated afterward in the common cylinders at every depth. The exposition distinguishes these three results: positional construction, analytic determination, and compatibility of the two.

9.1 First route: positional determination through integer closure

9.1.1 Domain and orientation of integer closure

Let $P_j, E_j, \Phi_j \in \{0, \dots, 999\}$ be the canonical blocks of the three regional coordinates, and prolong the dodecaphase record by

$$K_j = K_{1+((j-1) \bmod 12)}.$$

The channel orientation is $(1, 1, -1)$. The contribution of the dodecaphase record is subtracted according to the already typed dodecaphase closure. For $B = 1000$, the local operation distinguishes the signed balance

$$Z_j := P_j + E_j - \Phi_j - K_j$$

from the sum incorporating the incoming quotient. Normalization is

$$s_j = Z_j + c_{j+1}, \quad A_j = s_j - B \left\lfloor \frac{s_j}{B} \right\rfloor, \quad c_j = \left\lfloor \frac{s_j}{B} \right\rfloor. \quad (9.1)$$

Thus $A_j \in \{0, \dots, B-1\}$, even when s_j is negative. The carry entering from the right belongs to the domain of the operation; it cannot be omitted because the table is printed from left to right.

Definition 9.1 (Finite-depth closure). For a depth N and an integer boundary t , the map $\mathcal{C}_N$ receives $(P_{1:N}, E_{1:N}, \Phi_{1:N}, K_{1:N}, c_{N+1} = t)$ and applies (9.1) in the order $N, N-1, \dots, 1$. Its output is the pair of words

$$(A_{1:N}, c_{1:N+1}).$$

Euclidean division has no residual sign choice: the remainder is fixed in $\{0, \dots, 999\}$, and the quotient is determined. For example, $-431 = 569 - 1000$ and $-1200 = 800 - 2 \cdot 1000$. The negative carries of closure are not anomalies; they are the integer information needed to maintain the oriented equality.

9.1.2 Dodecaphase window and carry conditions

The first twelve triads obtained are

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884), \\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497), \\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117). \end{aligned}$$

In the closure channel, the five regions determining coordinate P preserve their distinct antecedents. Sharing this reading does not identify them as regions or histories.

With boundary $c_{13} = 0$, the recurrence produces the following table. Each input, incoming carry, total before normalization, obtained block, and outgoing carry is shown. The table is a reproducible integer computation, not a list of blocks selected for their resemblance to a target.

j	P_j	E_j	Φ_j	K_j	c_{j+1}	s_j	A_j	c_j
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	-1	352	352	0
4	589	459	749	729	-1	-431	569	-1
5	793	045	894	659	-2	-717	283	-1
6	238	235	848	824	-1	-1200	800	-2
7	462	360	204	621	0	-3	997	-1
8	643	287	586	058	-1	285	285	0
9	383	471	834	914	-1	-895	105	-1
10	279	352	365	794	0	-528	472	-1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

Table 8: Initial closure with zero right boundary. The dependency propagates from the last row to the first.

The carry word is

$$(c_1, \dots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0),$$

and the window reading is the rational number

$$\alpha_{12}^{(0)} = \frac{7297352569283800997285105472380663}{10^{36}} = 0.007297352569283800997285105472380663. \quad (9.2)$$

The superscript recalls the chosen boundary. This window is exact in its finite domain; it is not simply identified with the truncation of the infinite section. The difference between the two readings will become visible when the carry of the remote tail is transported.

9.1.3 Telescoping and reversibility in a boundary fiber

The local form of the equality is

$$P_j + E_j - \Phi_j - K_j - A_j = Bc_j - c_{j+1}. \quad (9.3)$$

The carry term is a link between adjacent positions. Eliminating it coordinate by coordinate gives a false equality. Only an evaluation preserving its endpoints allows it to telescope.

Proposition 9.2 (Integer identity at arbitrary depth). *For $\iota_N(z) = \sum_{j=1}^N z_j B^{N-j}$, every execution of $\mathcal{C}_N$ satisfies*

$$\iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(K) - \iota_N(A) = B^N c_1 - c_{N+1}. \quad (9.4)$$

Proof. Multiplying (9.3) by B^{N-j} and summing produces on the right-hand side

$$\sum_{j=1}^N (Bc_j - c_{j+1}) B^{N-j} = B^N c_1 - c_{N+1}.$$

Each interior carry appears twice with opposite signs; only the two endpoints survive. $\square$

For Table 8, both endpoints are zero. If p_{12}, e_{12}, f_{12} are the three truncated fractional sums,

$$\alpha_{12}^{(0)} = p_{12} + e_{12} - f_{12} - \kappa_{36}.$$

The dodecaphase-record constant appearing here is κ_{36} , not κ_{per} . The domain of this identity contains truncations and a finite boundary; its form does not authorize erasing both conditions when passing to the complete section.

Local inversion is exact as well:

$$K_j = P_j + E_j - \Phi_j + c_{j+1} - A_j - Bc_j.$$

Given P, E, Φ , blocks A , and the complete carries, it recovers the dodecaphase record. Concatenation additionally recovers

$$\iota_N(K) = \iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(A) - B^N c_1 + c_{N+1}.$$

This reversibility is a property of the fiber with fixed data and boundaries. It must not be turned into a new causal recipe starting from alpha and manufacturing the record afterward. The direction of construction remains signed record, dodecaphase record, closure, and output.

9.1.4 Prolongation and canonical normalization of the complete section

Prolongation preserves the periodic dodecaphase record and receives the complete regional sequences. Let

$$p = \sum_{j \geq 1} P_j B^{-j}, \quad e_0 = \sum_{j \geq 1} E_j B^{-j}, \quad f = \sum_{j \geq 1} \Phi_j B^{-j}.$$

Each series is a realization of the corresponding internal character. The integer parts already obtained are 3, 2, and 1, so $p = \pi - 3$, $e_0 = e - 2$, and $f = \varphi - 1$.

Proposition 9.3 (Existence and uniqueness of complete normalization). *The series*

$$y = p + e_0 - f - \kappa_{\text{per}} = \sum_{j \geq 1} (P_j + E_j - \Phi_j - K_j) B^{-j} \quad (9.5)$$

converges absolutely. On the domain of the obtained blocks, $0 < y < 10^{-2}$. There is a unique canonical expansion (A_j) of y , with no eventually constant 999 tail, and a unique bounded integer sequence (c_j) satisfying (9.3) with $c_1 = 0$. These carries belong to the invariant set $\mathcal{C} = \{-2, -1, 0, 1, 2\}$.

Proof. The coefficients of the series are bounded in absolute value by $2(B - 1)$, so a geometric series dominates it. If $Z_j = P_j + E_j - \Phi_j - K_j$ and $R_j(Z) = \sum_{r \geq 0} Z_{j+r} B^{-r-1}$, each tail is the difference of two sums of two canonical tails and belongs to $(-2, 2)$. The first coefficient is $Z_1 = 141 + 718 - 618 - 234 = 7$; hence $y = (7 + R_2(Z))/1000 \in (0.005, 0.009)$.

Let (A_j) be the canonical expansion of y . Define

$$c_j = R_j(Z) - R_j(A).$$

Multiplying equality of the two series by B^{j-1} and separating their first $j - 1$ terms gives

$$c_j = \sum_{r=1}^{j-1} (A_r - Z_r) B^{j-1-r} \in \mathbb{Z},$$

and $c_1 = 0$. Furthermore, $R_j(A) \in [0, 1)$, so $-3 < c_j < 2$, placing all carries in $\mathcal{C}$. The tail-shift identities give $Bc_j = Z_j - A_j + c_{j+1}$, which is the required recurrence.

Conversely, any bounded solution with canonical output and $c_1 = 0$ telescopes in the limit to the same series y . The canonical expansion is unique, and once it is fixed, the tail formula determines all carries. Finally, if a finite execution is applied with $c_{j+1} \in \mathcal{C}$, then $-2000 \leq s_j \leq 2000$, so its Euclidean quotient again belongs to $\mathcal{C}$. $\square$

The proposition makes explicit the normalization of the subsequent reader: it does not replace generation of the three sequences with a conventional series introduced as data. Once those sequences have been produced by the coinductive state, their combination and the normalization boundary are determined without consulting alpha.

The finite procedure preserved in the corpus propagates the five boundaries of $\mathcal{C}$ and retains only the prefix common to all of them. If two depths have coalesced before a boundary, the deterministic right-to-left recurrence gives the same blocks to its left. This is compatibility with truncation. A boundary is not chosen because it gives favorable digits, nor is all-depth compatibility inferred from observing a long run.

The initial comparison shows why this precaution matters. Stabilized prolongation gives $c_{13}^\infty = -1$, whereas the isolated window imposed $c_{13} = 0$. Thus,

$$884 + 497 - 117 - 601 - 1 = 662$$

in the last block of the first turn. The first eleven triads do not change, but the prolongation begins

$$(007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \dots).$$

An occurrence of block 999 does not violate the canonical convention: what is excluded is an eventually constant 999 tail. The zero-boundary window ends in 663; the truncation of the complete section ends in 662. Both statements describe different and exact operations. They must not be merged into a single projection notation.

9.1.5 Operative definition and complete additive equation

Definition 9.4 (Alpha coordinate of closure). The integer route determines the coordinate

$$\alpha_A = \sum_{j \geq 1} A_j B^{-j} = p + e_0 - f - \kappa_{\text{per}}, \quad (9.6)$$

where (A_j, c_j) is the compatible canonical normalization of (9.1). In the scalar realization of the HMT coordinates,

$$\boxed{\alpha_A = \pi + e - \varphi - 4 - \kappa_{\text{per}}.} \quad (9.7)$$

The integer four is not calibrated: it is $3 + 2 - 1$, the combination of the integer parts of the three coordinates. Base thousand arises from the cubic chart, and period twelve from the dodecaphase record. The signs are those of the oriented channel and closure, and the remote boundary is that imposed by the compatible section. These are the effective origins of the coefficients of the equation.

At this level, alpha can be described as the scalar coordinate of the closure defect between the three regional realizations and the rational dodecaphase return, computed with compatible carry. The term *defect* denotes a determinate algebraic difference, not an arbitrary loss or an imperfection of the system. This description does not suppress the genealogy: without the regions, signed record, orientation, and carry law, the scalar equality would not be a sufficient exposition of the construction.

Three objects that alpha notation may accompany in subsequent readers must also be kept separate: A is a word of blocks; the negative pre-carry support is obtained from the signs of the totals before normalization; and an exceptional radial vector belongs to an incidence codomain. None is interchangeable with the real value α_A .

Signed balance and sum with incoming quotient. Positional division consists of two consecutive stages. With $K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}$, denoting the periodic prolongation also written K_j in (9.1), the balance before quotient transport is

$$Z_j := R_j + E_j - \Phi_j - K_j^{\text{per}}.$$

The variable s_j in (9.1) denotes the sum already including the incoming contribution, so that

$$s_j = Z_j + c_{j+1}, \quad A_j = Z_j + c_{j+1} - 1000c_j.$$

The incidence configuration H_{α}^- uses the signs of the balance Z_j in the initial window; the determination of the digits also uses the linking quotients c_{j+1} and c_j . This distinction jointly preserves the signed support and the normalization of the expansion.

Exact dependence on the truncation boundary. The regional coordinates and the dodecaphasic register come from the common APP-TRIT-TPK generation. Once produced, their positional composition has an explicit dependence on readout depth. Let $B = 1000$,

$$Z_j = P_j + E_j - \Phi_j - K_j^{\text{per}}, \quad y_N = \sum_{j=1}^N Z_j B^{-j}, \quad R_{N+1}(Z) = \sum_{r \geq 0} Z_{N+1+r} B^{-r-1}.$$

The complete series in (9.5) satisfies exactly

$$\alpha_A = y_N + B^{-N} R_{N+1}(Z), \quad -2 < R_{N+1}(Z) < 2.$$

This identity combines the contribution of the window and that of its extension, both expressed in the same coordinates.

Proposition 9.5 (Boundary transport and canonical cylinder). *Let $A_j^{(N,t)}$ be the output of finite positional division with $c_{N+1} = t \in \mathcal{C} = \{-2, -1, 0, 1, 2\}$, and let $a_N^{(t)} = \sum_{j=1}^N A_j^{(N,t)} B^{-j}$. For the registers in the domain considered, we have*

$$\alpha_A - a_N^{(t)} = B^{-N}(R_{N+1}(Z) - t), \quad |\alpha_A - a_N^{(t)}| < 4B^{-N}. \quad (9.8)$$

The quotient of the complete extension is $c_{N+1}^{\infty} = \lfloor R_{N+1}(Z) \rfloor$. With this boundary,

$$0 \leq \alpha_A - a_N^{(c_{N+1}^{\infty})} < B^{-N},$$

so the resulting window is the canonical prefix of order N .

Proof. Telescoping (9.3) gives $a_N^{(t)} = y_N + tB^{-N} - c_1$. Since $Z_1 = 7$, every incoming state belongs to $\mathcal{C}$, and the first dividend lies between 5 and 9, we have $c_1 = 0$. Substituting this equality into the series decomposition proves (9.8). The bound follows from $R_{N+1}(Z) \in (-2, 2)$ and $t \in \mathcal{C}$.

For complete canonical normalization, $c_{N+1}^{\infty} = R_{N+1}(Z) - R_{N+1}(A)$ and $R_{N+1}(A) \in [0, 1)$. Integrality of the quotient implies $c_{N+1}^{\infty} = \lfloor R_{N+1}(Z) \rfloor$. The error formula then becomes $\alpha_A - a_N^{(c_{N+1}^{\infty})} = B^{-N} R_{N+1}(A)$, proving membership in the canonical cylinder. $\square$

The proposition identifies the meaning of the right boundary: it is the integer part of the signed tail expressed at the scale of the next position. Calculation over the five boundaries preserves this information as a finite collection of possibilities until the blocks to the left stabilize. The quantitative contraction comes from depth and tail transport; its object is the normalization of coordinates already generated.

Graded domain of the analytic resolvent. The analytic composition displayed in §9.2.2 acts on $\text{span}\{e_7, e_8, e_9\}$ and satisfies $Ne_9 = 0$. Its evaluation produces exactly $T_{7:9}$; degree 9 is the last component of that domain. The identity $N^3 = 0$ explains why the resolvent is expressed through three summands. Describing an extension to higher degrees additionally requires specification of the transport between successive graded domains and its action on the initial component of each domain. Local nilpotence and global compatibility thus serve distinct mathematical functions.

9.1.6 Constructive determination and role of the dodecaphasic register

The preceding definition specifies what it means to determine α from structure. The operation's domain contains four registers of known provenance: the three regional sequences and the integral phase register. Normalization also incorporates a boundary condition compatible with prolongation. Once these data are fixed, Euclidean division determines each remainder and its quotient. Proposition 9.3 extends this determination to the complete system. Independence from a target physical value is therefore a property of the construction order: the value compared at the end does not enter the selection of any argument of this map.

This causal independence must be distinguished from algebraic independence between numbers. The former is established by identifying the domain, operations and provenance of coefficients; the latter would be a claim about polynomial relations. The result used here is the former. The construction determines a coordinate through registers produced by APP-TRIT-TPK and allows subsequent verification of the relations it satisfies. That order makes it possible to use an already generated coordinate in a later operation without converting it into an external parameter.

The register K has a more precise function than that of a numerical correction added to a formula. Its twelve positions come from the reversible transformation of the signed register. The equation $U = \mathcal{H}_{12}K$ preserves the possibility of returning to that register, while the transverse differences and total sum permit another reconstruction. In periodic evaluation, the order of positions determines the positional weight of each component. The same set of components, arranged in a different order, generally produces a different rational coordinate. The phase origin and orientation therefore belong to the definition of the number entering (9.7).

Normalization of α composes this information with the regions. Its local identity distinguishes two successive operations. First, the oriented block balance is formed; that balance is then represented by a canonical remainder and a quotient linking adjacent positions. The first operation preserves the sign of the balance and allows subsequent extraction of a support. The second allows prefixes of the real coordinate to be formed. Both readouts remain related because the state preserves the data before and after normalization. The signed support and positional expansion are thus different realizations of an explicit arithmetic composition.

The telescoping identity expresses the global content of this composition. Interior quotients cancel in pairs when a prefix is evaluated, while their endpoints remain in the equality. This cancellation is a property of the weighted sum; it neither removes quotients from the state nor allows them to be suppressed before summation. The right boundary of a window receives the contribution of the continuation. For this reason, evaluation of an isolated window and restriction of a prolonged section are distinct operations. The change of the final block from 663 to 662, proved above, gives an explicit realization of that difference without altering the blocks already stabilized to its left.

In this sense, bidirectionality has a concrete arithmetic formulation. Generation provides compatible prolongations; normalization transmits their boundary's integer information from lower-weight to higher-weight positions. The equation preserves the contribution of both directions through its indices and boundary conditions. Reversibility in the fixed fibre is checked by reconstructing K from the remaining registers and the complete quotients. This inversion certifies consistency of the original operation; the genealogical direction remains the one that first produces the register and then the α coordinate.

The second determination has complementary content. There, the immediate object is a vacancy balance, and the parameter appears as its root in an isolating domain. The positional route provides a canonical normalization of registers; the analytic route studies the vanishing of an operator. Presenting both requires preserving precisely this difference of functions. Their relations with the common state must be expressed through refinement maps and comparison of their complete realizations, as developed in the corresponding subsection. Agreement of a finite approximation constitutes a localized check; identification of the complete constructions has the domain and scope of its own statement.

Finally, this organization explains alpha's position in the article. Its equation appears after regional generation and the construction of K , because it uses both results. The electronic section subsequently uses that coordinate as a factor of its composition. The exceptional prolongation receives the registers and their supports, where incidence information is preserved. These two continuations take up the same provenance and develop distinct mathematical functions: evaluation of a central section and construction of a boundary configuration. The development of each makes it possible to follow the structure through the equation, instead of limiting comparison to its scalar value.

9.2 Second route: analytic determination through vacancies

9.2.1 Transport invariants and graded resolvent

The analytic route receives invariants of the oriented state, not the output α_A . Its sixth-order truncation is

$$P_6(x) = 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \quad (9.9)$$

and the vacancy condition uses

$$\delta = \log_{10} \frac{10}{9}.$$

The ratio 10/9 compares the decimal chart with the nine APP positions of the nonadic construction. Its logarithm belongs to this subsequent compactification reading; it is not a target value of alpha.

The genealogy of the coefficients must be preserved in full, but with its exact logical strength. Recent developments bring together a marked signature of the state. The synchronization matrix and associated census are

$$D_{\text{syn}} = \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}, \quad N_9 = 43.$$

From its determinant one extracts

$$a = -\frac{\det D_{\text{syn}}}{N_9} = \frac{172}{43} = 4.$$

The clock induced on the short gaps has secondary spectrum $\{6, 7\}$; its upper endpoint gives $b = 7$. The tritic cardinality $m = 3$ determines

$$t_* = mb = 21, \quad k_* = mb - 1 = 20.$$

The vacancy reading in the thousand chart gives

$$v_- = \lfloor 1000\delta \rfloor = 45, \quad v_+ = \lceil 1000\delta \rceil = 46.$$

These integers are typed antecedents of the truncation's evaluation. They are not obtained from its root or from a metrological approximation to it.

The lower branch additionally coincides with the determinant of the APP block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \det B_c = 45.$$

The transverse rank of the preceding chapter's record gives $r_{\text{dod}} = 11$, and the phase semicrown provides $f_{\text{ph}} = 8\binom{6}{2} = 120$. In particular, $45r_{\text{dod}} = 495$.

With $\omega = 2\pi$, the signature

$$\mathcal{I}_9 = (\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$$

feeds the graded reader

$$\begin{aligned} \mathfrak{G}_9(\mathcal{I}_9)(x) = & \omega x - \frac{b}{a}x^2 + \omega^{-1}x^3 + \frac{x^4}{k_*} - \frac{2x^5}{t_*} - \frac{x^6}{v_+} \\ & - \frac{x^7}{f_{\text{ph}}} + \frac{x^8}{v_-} + \frac{2x^9}{v_-r_{\text{dod}}}. \end{aligned} \quad (9.10)$$

Substitution recovers the coefficients of P_6 and its ninth-order prolongation. The pair 45/46 crosses the separation between the two: the upper branch occupies degree six, and the lower reappears in degrees eight and nine.

The factorization of (9.10) is formulated for a state preserving origin, channel order, and orientation. On that marked domain, each coefficient is fixed by $(\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$. Two readouts preserving the signature and applying the same graded evaluation agree term by term through degree nine. This uniqueness does not extend to readouts that forget those data.

9.2.2 Nilpotent resolvent in degrees 7, 8, and 9

The first prolongation admits an especially explicit finite operator construction. On the space with basis e_7, e_8, e_9 , let

$$Ne_7 = -\frac{8}{3}e_8, \quad Ne_8 = \frac{2}{11}e_9, \quad Ne_9 = 0, \quad v_7 = -\frac{1}{120}e_7.$$

The symbol N here denotes a nilpotent operator, not a truncation depth. Its domain and grading are fixed before the polynomial is evaluated at a number.

Lemma 9.6 (Exact nilpotent prolongation). *One has $N^3 = 0$ and*

$$(I - N)^{-1}v_7 = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}.$$

Evaluation $e_n \mapsto x^n$ produces

$$T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}. \quad (9.11)$$

Proof. The operator raises degree and vanishes on the last basis vector, so its cube is zero. The inverse is the finite resolvent $I + N + N^2$. Moreover,

$$Nv_7 = \frac{1}{45}e_8, \quad N^2v_7 = \frac{2}{495}e_9.$$

Summing the three terms and evaluating gives the formula. $\square$

Here the denominators 120, 45, and 495 are not decimal fits: they arise from the semicrown, APP determinant, and its product with the dodecaphase rank. The transfers $-8/3$ and $2/11$, together with the sign and degree of the seed, belong to the typed reader. Preserving only cycle symmetry and replacing this operator by another symmetric operator does not necessarily preserve the coefficients.

9.2.3 Multiplicity-one root of the ninth-order truncation

Define

$$R_9 = P_6 + T_{7:9}, \quad E_{21/22}^{(9)}(x) = R_9(x) - \delta, \quad I_\alpha = (0, 10^{-2}).$$

This finite operator has its own existence and uniqueness result.

As a subsequent check of the characters already constructed, the rational isolation can be reproduced without a decimal table. Let

$$A_n(t) = \sum_{j=0}^n \frac{(-1)^j t^{2j+1}}{2j+1}, \quad L_n(t) = 2 \sum_{j=0}^n \frac{t^{2j+1}}{2j+1} \quad (0 < t < 1).$$

The alternating-series criterion and the geometric tail sum give

$$A_{2n+1}(t) < \arctan t < A_{2n}(t), \quad (9.12)$$

$$L_n(t) < \log \frac{1+t}{1-t} < L_n(t) + \frac{2t^{2n+3}}{(2n+3)(1-t^2)}. \quad (9.13)$$

Apply (9.12) to Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, and (9.13) to

$$\log(10/9) = \log \frac{1 + 1/19}{1 - 1/19}, \quad \log 10 = \log(10/9) + 2 \log \frac{1 + 1/2}{1 - 1/2}.$$

Write

$$\delta := \frac{\log(10/9)}{\log 10}.$$

Truncation orders 80, 20, 40, 120, applied respectively to the four preceding series, give rational intervals for π and δ , both of width less than 10^{-70} . These bounds check a subsequent realization; they neither select the APP-TRIT-TPK state nor receive a value of alpha.

Proposition 9.7 (Multiplicity-one root of the truncation). *The operator $E_{21/22}^{(9)}$ has a unique root ξ_9 in I_α , and that root is simple. It also has the rational isolation*

$$\frac{729735256928380099}{10^{20}} < \xi_9 < \frac{729735256928380100}{10^{20}}. \quad (9.14)$$

Proof. The internal-coordinate bounds $3.14 < \pi < 3.15$ and $0.045 < \delta < 0.046$ suffice for existence. At zero, $E_{21/22}^{(9)}(0) = -\delta < 0$. At 10^{-2} , the linear contribution minus the negative terms exceeds 0.0626, so the operator value is positive.

For $0 \leq x \leq 10^{-2}$, discarding the positive terms of the derivative gives

$$(E_{21/22}^{(9)})'(x) \geq 2\pi - \frac{7}{2} \cdot 10^{-2} - \frac{10}{21} \cdot 10^{-8} - \frac{6}{46} \cdot 10^{-10} - \frac{7}{120} \cdot 10^{-12} > 6.24.$$

Continuity gives existence; the derivative bound gives uniqueness and simplicity. For the left endpoint l and the right endpoint u of (9.14), rational substitution of the bounds (9.12)–(9.13) in the polynomial gives, with directed rounding,

$$-4.6 \cdot 10^{-20} < E_{21/22}^{(9)}(l) < -4.5 \cdot 10^{-20} < 0, \quad 0 < 1.6 \cdot 10^{-20} < E_{21/22}^{(9)}(u) < 1.8 \cdot 10^{-20}.$$

The monotonicity already proved places the unique root between these endpoints. The interval is checked after constructing the operator and does not enter the selection of its coefficients. $\square$

The result concerns ξ_9 . It does not authorize writing $\xi_9 = \alpha_A$ as an exact equality of complete numbers. A truncation root may share a prefix with the root of the prolonged operator and differ at a later position. Simplicity allows control of root movement under perturbations; it does not annul those perturbations.

9.3 Identification criterion for compatible readouts

Identification of two complete readouts requires their cylinders and restriction maps to be preserved. For a compatible graded collection,

$$E_{21/22}^{(6)} \longleftarrow E_{21/22}^{(9)} \longleftarrow E_{21/22}^{(12)} \longleftarrow \cdots, \quad E_{21/22} = \varprojlim_m E_{21/22}^{(6+3m)},$$

the limit denotes the readout together with its transports, not a freely selected tail. The following criterion separates the consequence of compatibility from the effective construction establishing it below.

Proposition 9.8 (Identification of compatible limits). *If the values of the integer and analytic routes belong to the same cylinders C_N^α for every N , and $\text{diam } C_N^\alpha \rightarrow 0$, then*

$$\alpha_A = \alpha_B = \alpha.$$

Proof. For every N , the distance between the two values is bounded by $\text{diam } C_N^\alpha$. Letting N tend to infinity gives zero distance. Existence of compatible sections ensures that this is not the empty intersection of an approximation system. $\square$

The difference can be expressed algebraically. Let $F = \mathbb{Q}(\pi, \delta)$, and let $j_9 : F[[x]] \rightarrow F[x]/(x^{10})$ be truncation. Then

$$\ker j_9 = x^{10}F[[x]].$$

All operators

$$E_h(x) = E_{21/22}^{(9)}(x) + x^{10}h(x)$$

have the same ninth-order truncation. However, taking $h = 0$ and $h = 1/729$,

$$E_{1/729}(\xi_9) = \frac{\xi_9^{10}}{729} > 0,$$

so they do not share that root. The observation does not permit choosing a tail foreign to HMT transport; it demonstrates precisely why the generated tail and its compatibility must be preserved. Nor does an arbitrary formal series suffice for speaking of analytic roots without its convergence domain or the realization of the cylinder system.

9.4 Compatibility between the dodecaphasic and analytic representations of alpha

Compatibility is established at every depth. The reversible dodecaphasic representation and the analytic representation of the alpha coordinate $E_{21/22}$ are two internal representations with distinct codomains, produced from the same enriched APP-TRIT-TPK state. The jet $E_{21/22}^{(9)}$ is an exact finite witness, but does not by itself determine the tail. The completion constructed below is not deduced from the bare jet: it receives the shared coordinate of the state and expresses its analytic image through an explicit recurrence. It therefore does not constitute an independent causal selection; it constitutes a typed analytic representation of the same generated point.

This separation preserves the causal order

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \mathcal{X}_\alpha^{\text{enr}} \longrightarrow (P, E, \Phi, U_{12}^{\text{sgn}}, K) \longrightarrow \alpha_{\text{HMT}}.$$

The coordinates P, E, Φ and the register K are earlier outputs of the same enriched state. Their use in the closure of alpha is therefore a subsequent internal composition, not an injection of conventional constants.

For $r \in 6\mathbb{N}$, $r \geq 36$, let $\hat{\mathcal{X}}_r^{\text{enr}, \alpha}$ be the sector of truncated states that extend the internal alpha seed fixed at R_{36} . Write

$$x_r^\alpha = (\tilde{x}_r, \chi_\alpha, \varepsilon_r^\alpha, N_r^\alpha),$$

where χ_α is the internal character of the sector, not the expressed scalar value. Each state preserves remainder, quotient, carry, sheet, orientation, phase, route, cylinder, boundary, signature, register, memory, survival, provenance and depth. If

$$d_r^\alpha = \text{Dig}_{729}(\varepsilon_r^\alpha), \quad \varepsilon_{r+6}^\alpha = 729\varepsilon_r^\alpha - d_r^\alpha, \quad N_{r+6}^\alpha = 729N_r^\alpha + d_r^\alpha,$$

the effective transition is

$$\mathcal{U}_r^\alpha = \text{Upd}_r^\alpha \circ \text{Tra}_r^\alpha \circ \text{Sel}_r^\alpha, \quad \text{Ext}_{r+6, r}^{\alpha, \rightarrow} = \text{graph}(\mathcal{U}_r^\alpha).$$

The single-valuedness of these extensions, with the seed R_{36} fixed, produces a unique compatible section: the phase returns while memory and depth advance. The global domain of both determinations is

$$\mathcal{X}_\alpha^{\text{enr}} := \hat{\Omega}_\alpha := \varprojlim_{r \geq 36} (\hat{\mathcal{X}}_r^{\text{enr}, \alpha}, \text{Ext}_{r+6, r}^{\alpha, \rightarrow}). \quad (9.15)$$

The notation κ_{per} retains the periodic readout of the register already constructed in the base-one-thousand chart fixed in (9.7).

9.4.1 The dodecaphasic closure as a complete section

For $N \geq 1$, set

$$\mathcal{W}_N := \{0, \dots, 999\}^N, \quad \mathsf{T}_{\alpha, N}(\mathbf{s}) := (A_1, \dots, A_N), \quad \mathbf{s} \in \mathcal{X}_\alpha^{\text{enr}},$$

where the blocks A_j are obtained through (9.1) with the compatible boundary $c_{N+1}^\infty = \lfloor R_{N+1}(Z) \rfloor$. If $N' \geq N$, let $\rho_{N', N} : \mathcal{W}_{N'} \rightarrow \mathcal{W}_N$ be the restriction to the first N blocks.

Theorem 9.9 (Complete dodecaphasic output and projective compatibility). *The collection $(\mathsf{T}_{\alpha, N})_{N \geq 1}$ is compatible:*

$$\rho_{N', N} \circ \mathsf{T}_{\alpha, N'} = \mathsf{T}_{\alpha, N} \quad (N' \geq N). \quad (9.16)$$

Its canonical cylinders

$$C_N^A(\mathbf{s}) := \left[\sum_{j=1}^N A_j 1000^{-j}, \sum_{j=1}^N A_j 1000^{-j} + 1000^{-N} \right) \cap I_\alpha \quad (9.17)$$

are nested, have diameter at most 1000^{-N} and satisfy

$$\bigcap_{N \geq 1} C_N^A(\mathbf{s}) = \{\alpha_A(\mathbf{s})\}, \quad \alpha_A = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}. \quad (9.18)$$

Consequently, the first route expresses a unique infinite section, which we denote by $\alpha_{\text{HMT}} := \alpha_A$.

Proof. Proposition 9.3 provides the unique canonical expansion and the unique bounded sequence of carries. The local identity (9.3) and its telescoping (9.4) preserve the remote boundary; Proposition 9.5 proves that restricting an extended execution recovers exactly the preceding prefix. This gives (9.16). Membership of the full sum in each cylinder follows from the tail identity, and $\text{diam } C_N^A \leq 1000^{-N} \rightarrow 0$; the intersection contains a single point. Finally, (9.7) identifies that point with the expression in (9.18). All its coefficients and signs are fixed before the representation of alpha. $\square$

The window $\alpha_{12}^{(0)}$, closed with $c_{13} = 0$, remains an exact finite calculation; it must not be confused with the restriction of the complete section, whose compatible boundary satisfies $c_{13}^\infty = -1$. In particular,

$$\rho_{36, 12}(\alpha_{A, 36}) = \alpha_{A, 12} \quad (9.19)$$

is an equality of compatible words, not an equality between words of different lengths or an identification of the isolated window with the limit.

9.4.2 Analytic representation of the alpha coordinate

The analytic extension is specified through all its coefficients. Let $\mathbf{s} \in \mathcal{X}_\alpha^{\text{enr}}$. Before decimal normalization, the same state already expresses the pre-coordinate

$$a(\mathbf{s}) := p(\mathbf{s}) + e_0(\mathbf{s}) - f(\mathbf{s}) - \kappa_{\text{per}}(\mathbf{s}) \in I_\alpha, \quad (9.20)$$

with $p = \pi_{\text{HMT}} - 3$, $e_0 = e_{\text{HMT}} - 2$ and $f = \varphi_{\text{HMT}} - 1$. Equivalently, $a = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}$. No conventional α is introduced here: p, e_0, f are the three regional characters already generated, and κ_{per} is the rational reading of the terminal register. The integer closure applies to (9.20) the normalization in (9.1); the chart below expresses a compatible analytic representation on the same pre-coordinate. Its dependency graph receives $(p, e_0, f, \kappa_{\text{per}})$ directly: it does not contain an edge $\alpha_A \rightarrow \lambda$, although the equality proved below allows both representation names to be identified subsequently.

Write $q = 1/729$, let F_s be the ordered field generated by the coordinates of $\mathbf{s}$, and set $E_{9, s}(X) = E_{21/22}^{(9)}(\mathbf{s}; X)$. We define the linking coefficient

$$\lambda(\mathbf{s}) := -\frac{E_{9, s}(a(\mathbf{s}))(1 - q a(\mathbf{s})^3)}{a(\mathbf{s})^{10}}. \quad (9.21)$$

All its arguments belong to the state before metrological recognition. In particular, λ is not obtained by fitting a target decimal string. Equation (9.21) must be read with its exact type: once the first representation of the state has produced $a(s)$, it fixes the unique extension within the geometric class defined below. It is not a second independent selection of $a(s)$, but a correlated analytic representation of the same enriched state.

Indeed, for $\mu \in F_s$ set

$$E_{s,\mu}(X) := E_{9,s}(X) + \mu \frac{X^{10}}{1 - qX^3}.$$

Since $0 < a(s) < 10^{-2}$ and $qa(s)^3 < 1$, we have

$$E_{s,\mu}(a(s)) = 0 \iff \mu = \lambda(s). \quad (9.22)$$

Thus, the coefficient seed contains no freedom once the state and the class of q -geometric extensions have been fixed. This uniqueness is relative to the declared class; it does not assert that the ninth-order jet by itself selects a tail among all possible analytic series.

The local action on blocks of three coefficients is

$$\mathcal{C}_\alpha : F_s^3 \longrightarrow F_s^3, \quad \mathcal{C}_\alpha(u, v, w) = q(u, v, w), \quad c^{(0)} = (\lambda(s), 0, 0). \quad (9.23)$$

Therefore, for every $n \geq 0$,

$$b_{10+3n} = q^n \lambda(s), \quad b_{11+3n} = b_{12+3n} = 0. \quad (9.24)$$

This is the rule that computes each coefficient beyond degree nine. No free parameter remains at each extension.

For $M \geq 0$, let

$$E_{s,M}(X) := E_{9,s}(X) + \lambda(s) \sum_{n=0}^M q^n X^{10+3n}. \quad (9.25)$$

The complete function is

$$E_{s,\infty}(X) := E_{9,s}(X) + \lambda(s) \frac{X^{10}}{1 - X^3/729}. \quad (9.26)$$

To formulate the projective system without implicit types, we fix $d_M = 12 + 3M$ and define the jet fibration

$$\mathfrak{F}_M := \coprod_{s \in \mathcal{X}_\alpha^{\text{nr}}} \{s\} \times F_s[X]/(X^{d_M+1}). \quad (9.27)$$

If $M' \geq M$, the restriction is

$$\tau_{M',M}(s, [P]_{d_{M'}}) := (s, [P]_{d_M}). \quad (9.28)$$

Proposition 9.10 (Naturality of the analytic recurrence). *Let*

$$\Gamma_M(s) = (s, [E_{s,M}]_{d_M}).$$

If $M' \geq M$, the truncation $\tau_{M',M}$ satisfies

$$\tau_{M',M} \circ \Gamma_{M'} = \Gamma_M, \quad \Gamma_{E21}(s) = \varprojlim_M \Gamma_M(s). \quad (9.29)$$

The projection onto the three new degrees is precisely $\mathcal{C}_\alpha^{M+1}(c^{(0)})$.

Proof. Equation (9.24) fixes the block of degrees $10 + 3n, 11 + 3n, 12 + 3n$. Passing from M to $M + 1$ adds only the next block; truncating it recovers the preceding polynomial literally. Compatibility for $M' \geq M$ is obtained by composition. Thus, the inverse limit is not a section chosen from the bare jet, but the unique orbit of $c^{(0)}$ under $\mathcal{C}_\alpha$. $\square$

9.4.3 Uniform convergence, simple root and rational intervals

The regional cylinders and the periodic register provide bounds through rational interval arithmetic. For each coordinate

$$c \in \{\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}\},$$

let P_c be its certified thousand-digit nonadic prefix. The datum used is not the rational number P_c treated as though it were the complete value, but the half-open cylinder and its outer closure

$$\mathcal{C}_c^{(1000)} = [P_c, P_c + \varepsilon), \quad \overline{\mathcal{C}}_c^{(1000)} = [P_c, P_c + \varepsilon], \quad \varepsilon = 10^{-1000}.$$

Directed arithmetic operates conservatively with these closures. If P_π, P_e, P_φ denote the three prefixes, we obtain

$$\begin{aligned} a_- &:= P_\pi + P_e - (P_\varphi + \varepsilon) - 4 - \kappa_{\text{per}}, \\ a_+ &:= (P_\pi + \varepsilon) + (P_e + \varepsilon) - P_\varphi - 4 - \kappa_{\text{per}}, \\ A_{1000} &:= [a_-, a_+], \quad a(s) \in [a_-, a_+] = A_{1000}, \quad \text{diam } A_{1000} = 3\varepsilon. \end{aligned} \quad (9.30)$$

The natural interval extension of (9.21) then transports A_{1000} , the cylinder of π_{HMT} and the directed rational bound for δ to a closed interval $\Lambda_{1000} \ni \lambda(s)$; none of the three tails is frozen. In particular,

$$\frac{7297352569283800997285105472380662}{10^{36}} < a(s) < \frac{7297352569283800997285105472380663}{10^{36}}, \quad (9.31)$$

$$-8.711 \times 10^{-6} < \lambda(s) < -8.709 \times 10^{-6}.$$

These inequalities use certified prefixes with their rational tail bound; they do not receive a table of alpha values. The package verifier reproduces them from the three nonadic outputs and from the terminal chart.

Let

$$r := \frac{10^{-6}}{729} < 1.372 \times 10^{-9}.$$

For $0 \leq x \leq 10^{-2}$, the tail beyond M satisfies

$$|E_{s,\infty}(x) - E_{s,M}(x)| \leq 8.711 \times 10^{-26} \frac{r^{M+1}}{1-r}. \quad (9.32)$$

Moreover,

$$\left| \frac{d}{dx} \left(\lambda \frac{x^{10}}{1 - x^3/729} \right) \right| \leq 8.711 \times 10^{-6} \left(\frac{10 \cdot 10^{-18}}{1-r} + \frac{3 \cdot 10^{-24}/729}{(1-r)^2} \right) < 8.712 \times 10^{-23}. \quad (9.33)$$

The tail of the derivatives beyond block M also admits the depth-dependent estimate

$$\sup_{0 \leq x \leq 10^{-2}} |E'_{s,\infty}(x) - E'_{s,M}(x)| \leq 8.711 \times 10^{-24} r^{M+1} \left(\frac{13+3M}{1-r} + \frac{3r}{(1-r)^2} \right) \xrightarrow{M \rightarrow \infty} 0. \quad (9.34)$$

Theorem 9.11 (Limit function and unique simple root of the correlated chart). *The sequence $E_{s,M}$ converges uniformly, together with its derivatives, to $E_{s,\infty}$ on $\bar{I}_\alpha = [0, 10^{-2}]$. We have*

$$E_{s,\infty}(a(s)) = 0, \quad \inf_{x \in \bar{I}_\alpha} E'_{s,\infty}(x) > 6.239. \quad (9.35)$$

Consequently, $a(s)$ is the unique simple root of the complete function in I_α .

Proof. The ratio of two consecutive terms of the tail is bounded by $r < 1$, which gives (9.32); term-by-term differentiation and summation of the two geometric series give (9.33) and (9.34). The proof for the jet establishes $E'_{9,s} > 6.24$ on $\bar{I}_\alpha$; hence, the complete perturbation leaves the derivative above 6.239. Finally, substituting (9.21) into (9.26), the two summands evaluated at $a(s)$ cancel exactly. The function is strictly increasing; its unique root is therefore simple and coincides with that pre-coordinate. $\square$

To make the identification effective at every depth, directed endpoint evaluations first certify

$$E_{s,\infty}(0) < 0 < E_{s,\infty}(10^{-2}), \quad E'_{s,\infty} \geq \mu := 6239/1000 > c := 6 \quad \text{on } [0, 10^{-2}].$$

By continuity there is a root in the interval and, by the second inequality, it is unique. Only after this existence, monotonicity and the invariant that the bracket contains the root have been established is the mean-value-theorem fallback applied. We start from $D_0 = [0, 10^{-2}]$. If $D_j = [r_j, s_j]$, let $m_j = (r_j + s_j)/2$, and let $[F_j^-, F_j^+]$ be a directed rational enclosure of $E_{s,\infty}(m_j)$, obtained by propagating the cylinders of $\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}, \delta$ and λ , with

$$0 \leq F_j^+ - F_j^- \leq \frac{c}{4} (s_j - r_j). \quad (9.36)$$

When $F_j^+ < 0$, the right half is retained; when $F_j^- > 0$, the left half is retained. If $F_j^- \leq 0 \leq F_j^+$, no attempt is made to decide whether the midpoint is exactly the root. Let $w_j = s_j - r_j$. The uniform bound $E'_{s,\infty} \geq c = 6$ and the mean value theorem imply that any root contained in D_j belongs to

$$D_{j+1} = D_j \cap \left[m_j - \frac{w_j}{4}, m_j + \frac{w_j}{4} \right]. \quad (9.37)$$

Indeed, let ξ_j be the root and let $y_j = E_{s,\infty}(m_j) \in [F_j^-, F_j^+]$. Since the enclosure contains zero and satisfies (9.36), we have $|y_j| \leq cw_j/4$. The mean value theorem gives $|m_j - \xi_j| \leq |y_j|/c \leq w_j/4$, proving (9.37). The argument includes the case $y_j = 0$: the root is then at the midpoint, but the algorithm does not need to recognize this equality in order to retain it.

If the available enclosure does not satisfy (9.36), the source cylinders are first refined and the evaluation is repeated; no sign is declared and no insufficient contraction is accepted. In each of the three branches the new interval retains the root and has diameter at most $w_j/2$. For each finite target depth, directed refinement allows the width bound to be met without an uncertified comparison of a real number with zero. Monotonicity proves inductively that

$$a(s) \in D_{j+1} \subseteq D_j, \quad \text{diam } D_{j+1} \leq \frac{1}{2} \text{diam } D_j, \quad \text{diam } D_j \longrightarrow 0, \quad (9.38)$$

where the inequality is a contraction bound, not an imposed equality. The executable certificate applies this rule to twenty-four base-one-thousand levels and checks that the complete outer closure A_{1000} , not only its lower endpoint, lies inside each expressed cylinder. Its negative tests reject an excessively wide enclosure, an incorrectly ordered enclosure, a point other than the midpoint, a domain without a sign change, and a degenerate bracket. Coinductive extension allows one thousand to be replaced by any required depth.

Specifically, for $S_N = 1000^N$ the verifier computes

$$j_N^- := \lfloor S_N a_- \rfloor, \quad j_N^+ := \lfloor S_N a_+ \rfloor \quad (9.39)$$

and expresses the level only if $j_N^- = j_N^+$. This equality, which includes the upper endpoint of the outer closure, proves $A_{1000} \subset [j_N^-/S_N, (j_N^- + 1)/S_N)$ and avoids assigning to the left cell an endpoint lying exactly on its right boundary.

Let $C_N^A(s)$ be the integer cylinder in (9.17). We recursively choose $m(N) > m(N-1)$ so that $\text{diam } D_{m(N)} < 1000^{-N}/4$, and define

$$I_{B,N} := D_{m(N)} \cap C_N^A(s). \quad (9.40)$$

Define $\ell_N := \inf I_{B,N}$ and $u_N := \sup I_{B,N}$. Both are rational; the interval is nonempty because both factors contain $a(s)$. Since the cylinder is half-open, u_N may not belong to $I_{B,N}$. Evaluation at that endpoint is understood through the continuous extension of $E_{s,\infty}$ to the outer closure. Thus,

$$I_{B,N+1} \subseteq I_{B,N} \subseteq C_N^A(s), \quad \text{diam } I_{B,N} \longrightarrow 0, \quad E_{s,\infty}(\ell_N) \leq 0 \leq E_{s,\infty}(u_N). \quad (9.41)$$

The bound (9.32) allows a truncation to be chosen that preserves the sign at each endpoint where the complete function does not vanish. If an endpoint coincides with the root, the identity of the

complete function and the membership already proved are used; that same sign is not attributed to its truncations. Indeed, the exact remainder is

$$E_{s,\infty}(x) - E_{s,M}(x) = \lambda(s)x^{10} \frac{(qx^3)^{M+1}}{1 - qx^3}.$$

In the positive domain under consideration, where $\lambda(s) < 0$, we obtain at the root

$$E_{s,M}(a(s)) = -\lambda(s)a(s)^{10} \frac{(qa(s)^3)^{M+1}}{1 - qa(s)^3} > 0 \quad (M < \infty).$$

Therefore, the certification of the weak signs in (9.41) refers to $E_{s,\infty}$. Its evaluation uses the complete function or the truncation accompanied by the signed remainder. This covers both the included infimum and the supremum that may be excluded from the cylinder.

Proposition 9.12 (Compatibility squares and common cylinders). *The restrictions of words and of analytic representations commute:*

$$\begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{T_{\alpha,N'}} & \mathcal{W}_{N'} \\ \parallel & & \downarrow \rho_{N',N} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{T_{\alpha,N}} & \mathcal{W}_N \end{array} \quad \begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_{M'}} & \mathfrak{F}_{M'} \\ \parallel & & \downarrow \tau_{M',M} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_M} & \mathfrak{F}_M. \end{array}$$

The collection $I_{B,N}$ provides the effective connection between the two squares and satisfies concretely $I_{B,N} \subseteq C_N^A$ for every N .

Proof. The first square is the compatibility of integer normalization; the second is Proposition 9.10. The inclusion follows from the definition (9.40), and nonemptiness and the signs follow from Theorem 9.11. No step identifies the root of a truncation with that of an arbitrary tail. $\square$

We now define, without leaving the prefix reader implicit,

$$\alpha_B(s) := \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x), \quad \text{pref}_N(x) := \frac{\lfloor 1000^N x \rfloor}{1000^N}, \quad \alpha_{B,N} := \text{pref}_N(\alpha_B). \quad (9.42)$$

Theorem 9.13 (Projective equality of the two correlated representations). *For every N , the integer and analytic routes express the same prefix:*

$$\boxed{\alpha_{A,N} = \text{pref}_N(\alpha_A) = \text{pref}_N(\alpha_B) = \alpha_{B,N}.} \quad (9.43)$$

In the limit,

$$\boxed{\alpha_A = \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x) = \alpha_B = \alpha_{\text{HMT}}.} \quad (9.44)$$

Proof. Integer normalization expresses exactly $a(s)$. Theorem 9.11 proves that the analytic chart represents that same point as its unique simple root. The half-open cylinders C_N^A fix a canonical prefix, and $I_{B,N} \subseteq C_N^A$ contains $\alpha_B = \alpha_A$. Therefore, their prefixes coincide for each N ; nesting and diameters tending to zero give equality of the limits. Metrological recognition occurs afterwards. $\square$

The proved identity connects two representations of the same state through the coefficient recurrence and compatible cylinders. The analytic function is a correlated representation of the already generated precoordinate, not a second causally independent selection. The jet, geometric class, and state coordinates retain their respective roles; conventional recognition comes afterwards.

10 Exceptional prolongation: from dodecaphasic incidence to Moonshine

The numerical expression does not exhaust the state that generated it. In the preceding construction, APP preserves both sheets and division with remainder and quotient; the TRIT determines the local regime and its orientation; the TPK composes selection, transport, carry, memory, and return. The Archimedean readout contracts compatible cylinders. The readout that we now study instead preserves the incidence of positions and its transport between frames. Both start from the same arithmetic state with trajectory memory. No exceptional geometry is reconstructed from the final decimals of π , φ , e , or α .

This point allows us to specify the meaning of the prolongation $\alpha/K \longrightarrow$ exceptional incidence. The arrow does not assert that an isolated real number determines a code, a lattice, or a group. It transports the dodecaphasic register and the signed data that entered the determination of α , together with the words and frames arising from APP–TRIT–TPK. In particular, the ordered register K can be recovered from its rational constant κ with the base, length, and origin conventions already fixed. Projection onto a support, by contrast, preserves only the selected incidences. Meanwhile, K_{ph} denotes another readout, with reduced memory, subsequent to the holonomy Γ_9 .

The exposition separates three mathematical operations: constructing the finite objects and their maps; prolonging them to a lattice and a vertex operator algebra; and subsequently recognizing the objects obtained by the relevant classical theorems. The names Paley, Witt, Mathieu, Golay, Leech, and Monster designate these recognitions and the explicit realizations that make them verifiable. They are not retrospective selectors of the seeds or coefficients of the generation.

The argument begins with compatibility of prefixes and frames, constructs the ternary code, and determines the flag through K and the signed record of α . The marked incidence then selects the lattice origin and the Leech neighbour. The final step realizes the vertex algebra and its automorphisms. At each stage, the structure preserved will be specified: linear combinations, supports, discriminant classes, bilinear form, or graded product. Explanations of the role of K , the normalization, and the origin of norm 54 accompany their respective constructions in Sections 10.3 and 10.8.

10.1 Compatibility of the Archimedean and incidence realizations

Let X_n^{cal} be the space of admissible prefixes of length n of the arithmetic state with trajectory memory, with truncations $p_{n,m} : X_n^{\text{cal}} \rightarrow X_m^{\text{cal}}$ for $m \leq n$. The superscript recalls that the declared initial frame and its transport are preserved, not that a parameter is fitted to an experimental value. We write (w_j, f_j) for the visible six-trit word and the frame at step j . The frames satisfy a causal law

$$f_{j+1} = g_j(x)f_j,$$

where $g_j(x)$ depends on the prefix already constructed. Consequently, a prolongation does not alter earlier frames.

The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is preserved in this readout. The return of the phase resets neither the frame nor the state memory. Likewise, the census of 468 decimal emissions, the 243 visible six-trit words, and the ambient space $\mathbb{F}_3^6$, with 729 words, remain distinct. The ambient space will be used to construct a linear code; it does not replace the admissible domain of emissions.

For a block b of six trits, the Archimedean readout takes its integer address $\nu(b) \in \{0, \dots, 728\}$. A prefix determines

$$q_n = m + \sum_{j=0}^{n-1} \frac{\nu(b_j)}{729^{j+1}}, \quad I_n = [q_n, q_n + 729^{-n}].$$

Prefix compatibility gives nested cylinders whose diameter tends to zero. Their Cauchy class first belongs to the internal completion $\mathbb{R}_{\text{HMT}}$; its subsequent identification with the usual real line does not enter the choice of blocks.

For the incidence readout we introduce, on the same prefix, the graph map

$$\gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad w \longmapsto (w, wA). \quad (10.1)$$

Words are written as row vectors. If $A^f = fAf^{-1}$, the change of frame satisfies the concrete identity

$$\gamma_{A^f}(wf^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}). \quad (10.2)$$

The calibrated codomain retains f together with the encoded word. We then define

$$Q_n(x) = ((f_j, \gamma_{f_j A_w f_j^{-1}}(w_j)))_{0 \leq j < n}. \quad (10.3)$$

Proposition 10.1 (Compatibility and prolongation of the incidence readout). *If $r_{n,m}$ truncates the encoded sequences, then*

$$r_{n,m} Q_n = Q_m p_{n,m}.$$

Thus the Q_n determine a unique map Q_∞ between the respective inverse limits. In each frame, the map preserves exactly the visible word, since $\text{pr}_1 \gamma_A = \text{id}$.

Proof. Two compatible prefixes have the same words and the same transport operators through index $m - 1$. By induction, they have the same frames at those indices. The first m components of (10.3) coincide, proving the square. The universal property of the inverse limit gives Q_∞ . The recovery assertion is projection onto the first six coordinates of each graph. $\square$

This local injectivity has a precise domain. It does not prove that Q_∞ recovers all the deep memory of the TPK state if only the sequence of visible words and frames has been retained. Still less does it prove injectivity after passing from words to supports. Moreover, a general change of basis in $\text{GL}_6(\mathbb{F}_3)$ does not itself preserve ordinary Hamming weight. In a general chart, the incidence and metric of the initial chart are transported as well. The support calculations below are performed in the declared Paley chart; monomial relabelings directly preserve their coordinate interpretation.

10.2 The finite realization and the ternary code

The elliptic operator of the TRIT admits the following finite realization in the six-position chart. These positions coordinatize the six units $\{1, 2, 4, 5, 7, 8\}$ modulo nine, preserved by modular transport. On $\mathbb{P}^1(\mathbb{F}_5)$, after declaring the order $\infty, 0, 1, 2, 3, 4$, let C be the matrix with zero diagonal, $C_{\infty,a} = C_{a,\infty} = 1$, and $C_{a,b} = \chi(b - a)$ in the five finite positions. Here $\chi(1) = \chi(4) = 1$, $\chi(2) = \chi(3) = -1$, and $\chi(0) = 0$. This is a coordinate realization of the quadratic structure already transported; identifying the six positions with the projective line is a chart datum, not a new numerical input to the generator.

The quadratic-sum identities give $C^2 = 5I_6$. Reducing modulo three, we obtain

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \quad A_W^2 = 2I_6 = -I_6. \quad (10.4)$$

The quadratic equality realizes the elliptic regime; it does not identify the field $\mathbb{F}_3$ with an Archimedean realization of the imaginary unit. These are different realizations of a common operator relation, with their respective domains.

Definition 10.2 (Code of the dodecaphasic chart). The ternary code associated with (10.4) is

$$C_W = \{(w, wA_W) : w \in \mathbb{F}_3^6\} \subset \mathbb{F}_3^{12}.$$

Its set of hexads is

$$\mathcal{H} = \{\text{supp}(c) : c \in C_W, \text{wt}(c) = 6\}.$$

Proposition 10.3 (Self-duality, distance, and Witt design). *The code C_W is self-dual with parameters $[12, 6, 6]_3$. Its weight enumerator is*

$$\sum_{c \in C_W} z^{\text{wt}(c)} = 1 + 264z^6 + 440z^9 + 24z^{12}. \quad (10.5)$$

The set $\mathcal{H}$ has 132 elements and forms an $S(5, 6, 12)$ system: every set of five positions is contained in a unique hexad.

Proof. The generator matrix $[I_6 \mid A_W]$ has rank six and, by symmetry of A_W , satisfies

$$[I_6 \mid A_W][I_6 \mid A_W]^T = I_6 + A_W^2 = 0.$$

The code is self-orthogonal with dimension half its length and is therefore self-dual. Finite evaluation of the 3^6 words gives (10.5); this is an enumeration over the explicit matrix, without real values or external targets. In particular, the minimum distance is six.

If two words of weight six have the same support, among the six ratios of their nonzero coordinates one of the signs 1, -1 occurs at least three times. Subtracting the corresponding multiple would yield a nonzero word of weight at most three, which is impossible. Hence a support corresponds to exactly the pair $\{c, -c\}$, and there are $264/2 = 132$ supports. If two different supports shared five positions, the same argument would cancel at least three coordinates in a union of size seven, producing weight at most four. Thus a five-element set belongs to at most one hexad. Finally,

$$132 \binom{6}{5} = 792 = \binom{12}{5},$$

so it belongs to one and only one. □

The subsequent recognition is the extended ternary Golay code and the small Witt design [7]. The preceding proof explains which object is recognized. It does not replace the APP-TRIT-TPK genealogy with a search among known codes.

10.3 Dodecaphasic flag associated with K and α

Articulation of regional coordinates and closure incidence. The function of this prolongation is understood by distinguishing three operations: preserving a record, evaluating a coordinate, and extracting an incidence relation. All three act on information produced by APP-TRIT-TPK. Their articulation makes explicit the structural thesis of the article: a numerical realization has an operator provenance, and certain data of that provenance additionally admit a combinatorial and lattice realization. The equations allow both paths to be followed, together with their domains and the information retained in each.

The starting point is the terminal state with its compatible prolongations. It retains the residues and quotients of the two APP sheets, the TRIT regime and orientation, and TPK transport with its memory. The closure, propagation, and self-scaling regions are determined in that domain before their readouts establish the coordinates π, e, φ . At the same time, the oriented record of the twelve windows determines the signed vector U . The subsequent appearance of K comes from the integral transformation proved in §8.8:

$$U = \mathcal{H}_{12}K, \quad K = \frac{1}{4}\mathcal{H}_{12}U, \quad \mathcal{H}_{12}^2 = 4I_{12}. \quad (10.6)$$

The congruence defining $\mathcal{L}_H$ guarantees the integrality of the inverse. Thus K is a reversible coordinatization of the signed record in the relevant sublattice. Its positions retain the phase order; its differences D_3K, D_4K retain the transverse relations; the sum $Q(K)$ determines the uniform component. This structure explains why the record participates both in numerical composition and in the selection of a boundary configuration.

The periodic evaluation

$$K \mapsto \kappa_{\text{per}} = \frac{\sum_{m=1}^{12} K_m 1000^{12-m}}{1000^{12} - 1}$$

keeps the twelve blocks recoverable when the base, length, and phase origin are fixed. Extracting a support performs a different operation: it preserves the membership of each position in a class defined by a threshold and groups records having the same support. The distinction between these two maps is decisive. The scalar precision of κ_{per} and the incidence of B_K are two specific mathematical contents, related through the complete record from which they arise.

In the frame inherited from the dodecaphasic register, the data are

$$\begin{aligned} K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601), & (10.7) \\ w_\pi &= 010211, & w_e &= 201101, \\ w_\varphi &= 121200, & w_{\text{APP}} &= 022020. \end{aligned}$$

Applying γ_{A_w} yields the following incidences:

$$\begin{aligned} P_\pi &= \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, \\ H_e &= \{1, 3, 4, 6, 9, 10\}, \\ H_\varphi &= \{1, 2, 3, 4, 7, 9\}, \\ H_{\text{APP}} &= \{2, 3, 5, 10, 11, 12\}. \end{aligned}$$

The first is the support of a word of weight nine; the other three are hexads. Through its high-block readout, the register provides

$$B_K = \{i : K_i \geq 729\} = \{4, 6, 9, 10\} = H_e \cap P_\pi. \quad (10.8)$$

The integer 729 here marks a boundary of the declared block format. It is not identified merely by cardinality with the ternary ambient space used in (10.2). In this specific register, all integer thresholds between 660 and 729 produce the same support: the support recovers neither the threshold nor the twelve blocks.

Before carry, the arithmetic route to α preserves the signed vector

$$Z_0 = (7, 297, 353, -430, -715, -1199, -3, 286, -894, -528, 380, 663). \quad (10.9)$$

Its negative support is

$$H_\alpha^- = \{4, 5, 6, 7, 9, 10\}. \quad (10.10)$$

These signs are not extracted from the decimal expansion of the scalar α ; they are preserved in the state preceding its numerical expression.

The normalization of alpha and its signed configuration. The coordinate of α is determined along the positional route by composing the regional blocks and the dodecaphasic register. Before positional division, the oriented combination has components $Z_j = B_j + E_j - \Phi_j - K_j^{\text{per}}$, and the balance with the incoming quotient is $s_j = Z_j + c_{j+1}$. The normalization satisfies

$$A_j = s_j - 1000c_j = Z_j + c_{j+1} - 1000c_j, \quad 0 \leq A_j < 1000. \quad (10.11)$$

This identity preserves both the normalized block and the contribution of the transported quotient. The limit of the compatible prefixes determines the scalar coordinate of the closure. The signed

configuration, in turn, retains which positions of the combination before normalization exhibit a negative defect. In the initial window, this configuration is the vector Z_0 in (10.9), and its negative support is H_α^- .

This defines two kinds of content of a single operation. Positional normalization determines a real coordinate through its compatible remainders and quotients; the negative support determines a configuration of positions in the twelve-component cycle. The sign refers to the register balance before normalization. Its incidence readout remains available because the state retains that balance together with the normalized result. This articulation belongs to the proved positional route; the comparison with the analytic vacancy operators retains the precise scope established in the chapter on α .

Proposition 10.4 (Agreement of the two flag selectors). *Within $\mathcal{H}$, the support (10.10) is the unique hexad H satisfying*

$$B_K \subset H \subset P_\pi, \quad (|H \cap H_e|, |H \cap H_\varphi|, |H \cap H_{\text{APP}}|) = (4, 3, 2).$$

Thus the signed readout and the incidence readout select the same flag $B_K \subset H_\alpha^-$.

Proof. The four hexads containing B_K are obtained by adding, respectively, the pairs

$$\{1, 3\}, \quad \{2, 11\}, \quad \{5, 7\}, \quad \{8, 12\}.$$

Only the second and third pairs are contained in P_π . For the second, the intersection with H_φ has size three and that with H_{APP} has size three; it fails the last condition. For the third, the three intersections have sizes 4, 3, 2. The latter produces precisely the negative support of (10.9). $\square$

It is useful to preserve both the flag and its marked frame:

$$\mathcal{J}_* = (B_K \subset H_\alpha^-), \quad \tilde{\mathcal{J}}_* = (\mathcal{J}_*; P_\pi, H_e, H_\varphi, H_{\text{APP}}).$$

Union, intersection, and difference are equivariant under simultaneous relabeling of the design. The marked class, not the position numbers taken individually, is the transported object.

Lemma 10.5 (Origin selected by marked incidence). *For the preceding frame,*

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

The rule is equivariant under every automorphism applied simultaneously to the data of $\tilde{\mathcal{J}}_$.*

Proof. The first intersection is $\{1, 3, 4, 9\}$. The union being subtracted contains 3, 4, 9 and does not contain 1. Equivariance follows because a bijection commutes with set operations. $\square$

This origin will select a direction of the lattice neighbor when the A_2 metric and the positive chamber are declared. Incidence alone does not contain that metric. The lattice step will explicitly preserve both data.

From ternary coding to the marked flag. The regions also provide the codes w_π, w_e, w_φ and w_{APP} in the inherited frame. The map

$$\gamma_{A_W} : w \mapsto (w, wA_W)$$

preserves the six-component word as its first coordinate and adds its transform by A_W . The relation $A_W^2 = -I_6$ fixes the quadratic compatibility of this realization. The next step, $c \mapsto \text{supp}(c)$, retains the nonzero positions. For words of weight six, the support identifies exactly the pair $\{c, -c\}$, as proved by the minimum distance of the code. Thus, the quotient by sign produces the hexads, while the complete code remains available for operations requiring its symbols.

In this chart, the two constructions of the tetrad coincide:

$$\{i : K_i \geq 729\} = H_e \cap P_\pi = B_K = \{4, 6, 9, 10\}. \quad (10.12)$$

The left-hand side comes from an integer comparison of the components of K ; the right-hand side, from the intersection of the coded propagation and closure supports. The threshold belongs to the declared block chart and remains explicit in the definition of the readout. The complete record preserves the values on which it acts, even when different thresholds produce the same subset. The equality therefore expresses compatibility between specified maps.

The support H_α^- completes this tetrad to a hexad. The condition of containment in the support P_π , together with intersection sizes $(4, 3, 2)$ relative to $H_e, H_\varphi, H_{\text{APP}}$, selects a unique hexad. The following proofs bring together the two determinations: that obtained from the sign of Z_0 and that obtained from incidence. The result is organized as

$$\tilde{J}_* = (B_K \subset H_\alpha^-; P_\pi, H_e, H_\varphi, H_{\text{APP}}). \quad (10.13)$$

The inclusion is the flag; the four additional supports constitute its marked frame. Simultaneous permutations transport the frame and preserve its inclusions and intersections. The relevant information resides in this transported configuration and its isomorphism class.

The five regions of π recover their internal differentiation here. They share w_π as their visible ternary code, but retain different emissions, signatures, and multiplicities. In the binary realization on a marked dodecad, the transverse region corresponds to the octad whose intersection with the dodecad is the tetrad; the other four regions correspond to lifts of the four hexads of its star. The negative region is distinguished by H_α^- , and the chart ordering individuates the three positive regions. This correspondence preserves the regional roles and the distribution $432 + 4 \cdot 144 = 1008$, as well as realizing the incidence partition $1 + 4$.

10.4 Isometric intertwining of the incidence representations

Incidence defines a linear map whose action can be checked position by position:

$$I : \mathbb{R}^{12} \longrightarrow \mathbb{R}^{\mathcal{H}}, \quad (Iv)(H) = \sum_{p \in H} v_p. \quad (10.14)$$

Let $V_{11} = \mathbf{1}^\perp \subset \mathbb{R}^{12}$. The two Euclidean realizations are here subsequent languages for the same finite design.

Proposition 10.6 (Isometry of the dodecaphasic screen). *We have*

$$I^\top I = 36I_{12} + 30J_{12},$$

where J_{12} is the all-ones matrix. Consequently,

$$U_W = \frac{1}{6}I|_{V_{11}} : V_{11} \xrightarrow{\sim} E_{\text{inc}} := I(V_{11})$$

is an equivariant isometry for the permutation actions of the design automorphisms.

Proof. Each point belongs to $r = \binom{11}{4}/\binom{5}{4} = 66$ hexads, and each pair to $\lambda_2 = \binom{10}{3}/\binom{4}{3} = 30$. These are, respectively, the diagonal and off-diagonal entries of $I^\top I$. On $\mathbf{1}^\perp$, the J_{12} term vanishes, leaving $36I$. Finally, if g is an automorphism, $p \in H$ is equivalent to $g(p) \in g(H)$; hence $I\rho_{12}(g) = \rho_{\mathcal{H}}(g)I$. $\square$

The incidence information also has a concrete spectral readout. Define

$$G_{H,H'} = \binom{|H \cap H'|}{4}.$$

Lemma 10.7 (Action of the intersection kernel). *If $J_{132,12}$ is the rectangular all-ones matrix,*

$$GI = 30I + 15J_{132,12}.$$

Thus G acts by the scalar 30 on E_{inc} .

Proof. The (H, p) entry of GI counts pairs (T, H') with $T \subset H \cap H'$, $|T| = 4$, and $p \in H'$. If $p \in H$, there are ten four-element sets T containing p , each contained in four hexads, and five not containing it, each with a unique extension of the five-element set $T \cup \{p\}$. The total is $10 \cdot 4 + 5 = 45$. If $p \notin H$, the fifteen four-element subsets of H each give one extension: the total is 15. This proves the identity. The rectangular term vanishes on V_{11} . $\square$

Here a material intertwiner does exist: its domain, codomain, entries, and normalization are determined. For example, any orthogonal projector P already constructed on V_{11} is transported to $Q = U_W P U_W^*$ on E_{inc} , with $U_W P = Q U_W$. This assertion does not identify all operators acting on those spaces solely by their trace.

10.5 The Witt star and the generation of M_{12}

Definition 10.8 (Star over a four-element set). For $B \in \binom{[12]}{4}$, the Witt star is the collection of the four hexads containing B . Its petals are the pairs $H \setminus B$. The permutation σ_B fixes B and interchanges the two points of each petal.

The existence of four hexads follows from

$$\lambda_4 = \frac{\binom{8}{1}}{\binom{2}{1}} = 4.$$

Two petals cannot share a point, since two hexads would then contain the same five-element set. They are therefore four disjoint pairs partitioning $[12] \setminus B$. This description is neither a five-pointed star nor a geometric metaphor: it is a four-to-one incidence of the design.

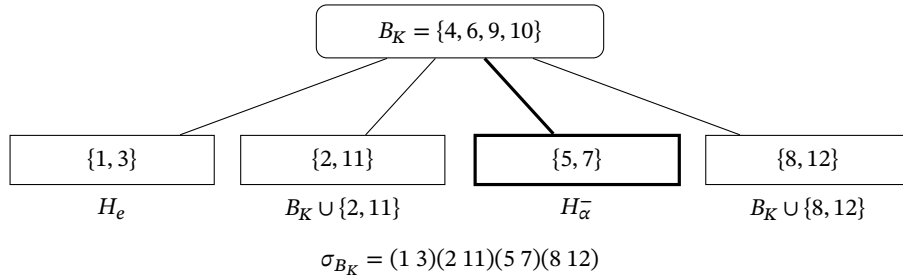


Figure 8: The concrete star of the dodecaphasic flag. Each line indicates the union of the central four-element set with the pair at its endpoint; the thick line marks the hexad also selected by the signs preceding the carry of α . The four pairs exhaust the eight outer positions.

Proposition 10.9 (The collection of stars). *The $\binom{[12]}{4} = 495$ permutations σ_B preserve $\mathcal{H}$. The group they generate has order 95040 and is the Mathieu group M_{12} in its action on the small Witt design.*

Proof. Preservation is a finite check on the 132 hexads of (10.2): for each four-element set, the four petals are constructed and one verifies $\{\sigma_B(H) : H \in \mathcal{H}\} = \mathcal{H}$. Closure under composition of these permutations produces 95040 elements. In lexicographic enumeration, it suffices to successively add six stars not yet contained in the group; the intermediate orders are 2, 6, 18, 360, 7920, 95040, and the 495 stars belong to the final group. The procedure and the six generators are made explicit in the provenance technical note. Finally, one applies the recognition of the automorphism group of $S(5, 6, 12)$ as M_{12} , whose order is $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95040$. $\square$

An individual star generates a cyclic group of order two. It does not by itself generate M_{12} , much less the Monster. Nor does the existence of 495 generators of a permutation group prove that each is the transport of a specific TPK loop. That assertion requires the corresponding loop map; the result established here is the finite incidence action.

Remark 10.10 (Equality of traces does not identify operators). On V_{11} , a star has spectrum $+1$ with multiplicity seven and -1 with multiplicity four. Its normalized trace is $3/11$. A rank-three projector on the same space also has normalized trace $3/11$, but spectrum $1^3, 0^8$. They are not conjugate: their minimal polynomials and spectra differ. The intertwiner (10.14) transports defined operators; it does not turn this scalar coincidence into an equivalence of operators.

10.6 The binary residue and the five regions of π

The next recognition uses the extended binary Golay code G_{24} , with parameters $[24, 12, 8]_2$. Its weights are 0, 8, 12, 16, 24, and the weight-eight supports form the design $S(5, 8, 24)$. Fix a dodecad D , the support of a word of weight twelve. This datum and the incidence isomorphism to be declared below belong to the realization chart; they are not used to generate or select the decimals of π .

Proposition 10.11 (Residue of a dodecad and lifting of hexads). *The set*

$$\mathcal{H}(D) = \{O \cap D : O \text{ is an octad, } |O \cap D| = 6\}$$

is an $S(5, 6, 12)$ system on D . Each of its hexads has a unique lift to an octad. For any four-element set $B \subset D$, the five octads containing it split into four residual lifts and a unique octad $O_B^\perp$ with $O_B^\perp \cap D = B$.

Proof. A five-element set $A \subset D$ belongs to a unique octad O . If $j = |O \cap D|$, the binary word given by the sum of O and D has weight $20 - 2j$. Among the allowed weights, and with $0 \leq j \leq 8$, only $j = 2, 4, 6$ are possible. Since $A \subset O \cap D$, we have $j \geq 5$ and therefore $j = 6$. This gives existence and uniqueness of the residual hexad, as well as uniqueness of its lift by choosing five of its points. Moreover,

$$\lambda_4(S(5, 8, 24)) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5, \quad \lambda_4(S(5, 6, 12)) = 4.$$

The four residual hexads over B give four distinct octads. The fifth cannot meet D in six points, since it would give another residual hexad; because it contains B , its intersection has size four and equals B . $\square$

A chart $\ell : [12] \rightarrow D$ preserving hexads identifies $\mathcal{H}$ with $\mathcal{H}(D)$. Its existence is the recognition of the design already constructed as the small Witt design; a concrete choice of ℓ fixes the relabeling. The lift

$$\iota_D : \mathcal{H} \longrightarrow \{\text{octads of } G_{24}\}, \quad H \longmapsto O \text{ such that } O \cap D = \ell(H),$$

is injective because intersection with D recovers $\ell(H)$. This injectivity has hexads as its domain, not the register K .

The relation to the five regions of π preserves richer data than the number five. The emissions and their signatures are:

Region	Multiplicity	Emission U_6	Signature
$\perp$	432	(501, 614, 498, 169, 272, 272)	555555
$++_1$	144	(810, 923, 870, 169, 272, 272)	339555
$++_2$	144	(810, 983, 810, 169, 272, 272)	393555
$++_3$	144	(870, 923, 810, 169, 272, 272)	933555
$--$	144	(870, 923, 810, 418, 674, 521)	933717

In each three-digit decimal block, the signature uses the modular difference of the first two digits. The five emissions have the same visible trit word $w_\pi = 010211$, but not the same regional genealogy. The

three positive regions preserve the tail 169|272|272; the negative region changes it to 418|674|521; the transverse region has constant signature and greater multiplicity.

The marked extension of the regional chart sends the transverse region to $O_{\ell(B_K)}^\perp$, the negative one to $\iota_D(H_\alpha^-)$, and the three positive ones to the other three lifts of the star. This realizes the $4 + 1$ partition of the five octads over $\ell(B_K)$. The individual correspondence between the three positive names and the three remaining petals requires a chart ordering; incidence does not determine it without that datum. Thus the precise assertion is a marked correspondence of regions, compatible with their roles and the flag, not a canonical bijection deduced from cardinality five.

The identity $432 + 4 \cdot 144 = 1008$ preserves the regional multiplicities. Also, $1008 = 36 \cdot 28$, and an octad has $\binom{8}{2} = 28$ pairs. These equalities are arithmetic catalogue checks, not substitutes for the map ι_D or the regional chart.

10.6.1 Pentadic representation of the microfibre and its incidence

The arrangement of the five regions displays simultaneously the unity of the ternary quotient and the differentiation retained by the catalogue. The relevant map is the restriction

$$\Psi|_{\mathcal{R}_\pi^{\text{micro}}} : \mathcal{R}_\pi^{\text{micro}} \longrightarrow \{010211\}, \quad \Psi(U) = (-U) \bmod 3.$$

Its five preimages are the U_6 vectors in the preceding table. Figure 9 recovers their radial organization, preserving the chart order, signatures and multiplicities of the catalogue (see 9).

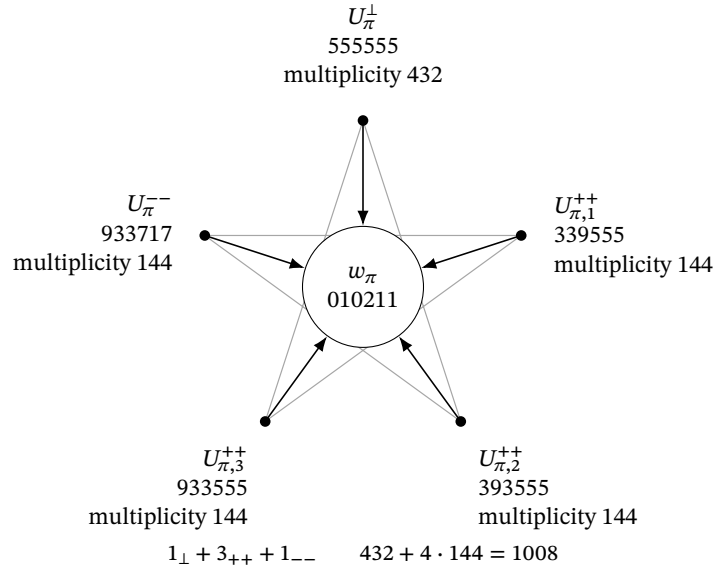


Figure 9: The five regions of π in a pentadic arrangement. Each position preserves its multiplicative signature and multiplicity; the radial arrows represent the common projection $\Psi(U) = 010211$. The star-shaped outline organizes the five positions of the chart. Their realization in the five Witt octads is effected through the regional alignment and the $4 + 1$ decomposition in the text.

Projection to the centre identifies the five ternary images; the register of outer positions retains the regional information that makes the incidence realization possible. The distribution $1_\perp + 3_{++} + 1_{--}$ specifies one transverse region, three positive regions and one negative region. Their normalized weights are $3/7$ and $1/7$ for each of the other four: the transverse multiplicity is threefold, even though the five images under Ψ coincide.

The connection with the Witt star is formulated through the flag $B_K \subset H_\alpha^-$, the alignment ℓ and the lift ι_D already constructed. The five regional positions are realized in the five octads over $\ell(B_K)$:

$$\begin{aligned} U_\pi^\perp &\longmapsto O_{\ell(B_K)}^\perp, \\ U_\pi^{--} &\longmapsto \iota_D(H_\alpha^-), \\ \{U_{\pi,1}^{++}, U_{\pi,2}^{++}, U_{\pi,3}^{++}\} &\longmapsto \{\iota_D(B_K \cup P) : P \in \mathcal{P}_+\}, \\ \mathcal{P}_+ &= \{\{1, 3\}, \{2, 11\}, \{8, 12\}\}. \end{aligned}$$

The three positive positions are assigned to the three pairs in an ordered chart. The transverse and negative rows are distinguished by their regional predicates and by the flag.

The two figures are thereby articulated: Figure 8 displays the four hexads of the small Witt design over B_K ; Figure 9 displays the five regions that, under the alignment, occupy the four residual octads and the transverse octad of the binary design. The $4 + 1$ law, proved in Proposition 10.11, specifies the change of incidence. In this composition, K determines the marked tetrad and the signature of α distinguishes one of its hexads. In addition to their common ternary value, the regions retain the structure required for this transport to the exceptional realization.

10.7 From the ternary code to the Niemeier lattice

The passage to dimension twenty-four need not first go through the binary code. It is performed directly from C_W by discriminant gluing. Two compatible prolongations are thus distinguished: one preserves the residue $W_{12} \subset W_{24}$; the other constructs the lattice on which the vertex operator algebra will be realized.

Let $A_2 = \mathbb{Z}a_1 \oplus \mathbb{Z}a_2$ have Gram matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \rho = a_1 + a_2, \quad \rho^2 = 2.$$

We represent the three classes of A_2^*/A_2 by

$$\varpi_0 = 0, \quad \varpi_1 = \frac{2a_1 + a_2}{3}, \quad \varpi_2 = \frac{a_1 + 2a_2}{3}.$$

The map $j \mapsto \varpi_j + A_2$ additively identifies $\mathbb{F}_3$ with that discriminant group. For $c \in C_W$, write $\phi(c) = (\varpi_{c_1}, \dots, \varpi_{c_{12}})$ and define

$$N(C_W) = \bigcup_{c \in C_W} (A_2^{12} + \phi(c)). \quad (10.15)$$

Theorem 10.12 (Ternary gluing). *The object (10.15) is a positive, even, unimodular lattice of rank 24. Its root system is exactly A_2^{12} ; it is therefore recognized as the Niemeier lattice with that root system.*

Proof. Linearity of the code makes the union of classes closed under addition. Self-orthogonality annihilates the discriminant pairing: this is a multiple of $c \cdot d/3$ modulo $\mathbb{Z}$. Thus the lattice is integral. The norm of a glue class is $2 \text{ wt}(c)/3$ modulo $2\mathbb{Z}$. The weights in (10.5) are multiples of three, so it is even.

The index of A_2^{12} in $N(C_W)$ is $|C_W| = 3^6$. Consequently,

$$\det N(C_W) = \frac{3^{12}}{(3^6)^2} = 1.$$

Positivity comes from the orthogonal sum of the twelve A_2 metrics. In a nonzero class of A_2^*/A_2 , the minimum norm is $2/3$; every nonzero word has at least six nonzero positions. Thus a vector in a nontrivial glue class has norm at least $6 \cdot 2/3 = 4$. No roots of norm two occur outside A_2^{12} . Within it, the roots are precisely those of its twelve components. $\square$

10.8 Incidence selection of the rootless unimodular neighbor

From the incidence origin to lattice modification. The marked flag determines a position through an equivariant set-theoretic operation:

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

Its function is to distinguish one component among the twelve involved in the lattice realization. The complete ternary code acts as a glue code in the discriminant module $(A_2^*/A_2)^{12}$. Its preimage defines $N(C_W)$, and the self-duality, weights, and distance of the code translate, respectively, into integrality and determinant properties, parity, and norm bounds for the glue classes. Here the symbolic information of the code serves a function that the isolated support no longer performs.

The A_2 metric and the positive radial chamber are fixed in the lattice development. Within that chamber, the congruence required for an even index-three neighbour has twelve realizations of minimum norm 54, permuted by changing the distinguished component. The origin o_* selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 2(4^2 + 11) = 54.$$

The coefficient 4 belongs to that solution of the congruence in the stated chamber. The subscript α records the incidence provenance of the selection: its content is the frame that participated in the closure and now singles out a lattice direction.

Pairing with v_α determines the index-three sublattice $N_{\alpha,0}$, and adjoining the class $v_\alpha/3$ produces the neighbour N_α of (10.17). This modification preserves rank and recovers unimodularity and evenness; congruence selection eliminates the previous roots, and local bounds exclude roots in the new classes. Identification with Leech follows these verifications. The binary realization of the five regions and this lattice construction are two prolongations of the common frame: the latter directly receives the ternary code as glue data.

The flag label $o_* = 1$ is now realized as one of the twelve A_2 components. This operation declares the preceding metric and the positive radial chamber

$$v(a) = \sum_{i=1}^{12} a_i \rho_i, \quad a_i = 1 + 3b_i, \quad b_i \in \mathbb{Z}_{\geq 0}.$$

The evenness requirement for the index-three neighbor is $v(a)^2 \equiv 0 \pmod{18}$. Since

$$v(a)^2 = 2 \sum_i (1 + 3b_i)^2,$$

this congruence is equivalent to $\sum_i b_i \equiv 1 \pmod{3}$. The smallest norm in the chamber is attained with a single $b_i = 1$ and all others zero; it is 54. The marked origin selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 54. \tag{10.16}$$

The word “smallest” refers to this positive chamber. It does not assert minimality throughout the residue class: the vector $(a_1 - 2a_2, \rho, \dots, \rho)$ represents the same class modulo three and has norm $14 + 11 \cdot 2 = 36$. The chamber restriction is part of the construction data, not a conclusion that can be omitted.

Let $N = N(C_W)$. Define

$$N_{\alpha,0} = \{x \in N : \langle x, v_\alpha \rangle \equiv 0 \pmod{3}\}, \quad N_\alpha = N_{\alpha,0} + \mathbb{Z} \frac{v_\alpha}{3}. \tag{10.17}$$

Theorem 10.13 (Leech neighbor determined by the marked flag). *With the declared metric and chamber, N_α is a positive, even, unimodular lattice of rank 24, with no vectors of norm two. By the corresponding classification theorem, it is isometric to the Leech lattice Λ_{24} .*

Proof. The character $x \mapsto \langle x, v_\alpha \rangle \bmod 3$ is nonzero: a simple root of a component pairs with $a_i \rho_i$ to give $a_i \equiv 1 \pmod{3}$. Hence $[N : N_{\alpha,0}] = 3$ and $\det N_{\alpha,0} = 9$. Moreover, $v_\alpha/3 \in N_{\alpha,0}^*$ and $v_\alpha \in N_{\alpha,0}$. If $v_\alpha/3$ belonged to N , integrality of N would make all pairings with v_α divisible by three, contradicting the nonzero character. The class of $v_\alpha/3$ therefore has order three. It follows that

$$[N_\alpha : N_{\alpha,0}] = 3, \quad \det N_\alpha = 9/3^2 = 1.$$

The extension is integral, and for $x \in N_{\alpha,0}$,

$$\left(x + m \frac{v_\alpha}{3}\right)^2 = x^2 + 2m \frac{\langle x, v_\alpha \rangle}{3} + 6m^2 \in 2\mathbb{Z}.$$

It is therefore even.

It remains to exclude roots, which is the decisive step. The roots of N belong to A_2^{12} . Their pairings with v_α are congruent to ± 1 or ± 2 modulo three; none belongs to $N_{\alpha,0}$. For the other two classes of the neighbor, each component belongs to one of the local classes

$$A_2 + \varpi_j \pm \rho/3, \quad j = 0, 1, 2.$$

Since $4\rho/3 \equiv \rho/3 \pmod{A_2}$, this description also holds in the distinguished component. Writing a local vector as $(ua_1 + va_2)/3$, its norm is $2(u^2 - uv + v^2)/9$. The six preceding classes have nonzero residues and minimum $2/9$: representatives minimizing the quadratic form are $(\pm 1, 0)$, $(0, \pm 1)$, $(1, 1)$, $(-1, -1)$ in the appropriate residues. A vector in a new class therefore has norm at least $12 \cdot 2/9 = 8/3 > 2$. There are no roots in any of the three classes. The classification of positive even unimodular lattices of rank twenty-four identifies the unique rootless case with Λ_{24} . $\square$

The argument does not recognize Leech from a coincidence of dimension. It constructs the gluing, proves integrality, evenness, and determinant, and eliminates roots by a local bound. Only then does it apply the classification theorem. Dependence on α is preserved in the flag selecting o_* and in the vector (10.16); the scalar value of the constant is not converted into a lattice length.

10.9 The vertex operator algebra and the Monster

Let $\Lambda = N_\alpha$, already recognized as Leech. The next step uses its integral lattice, not a decimal expansion. The lattice vertex operator algebra construction is written

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

Here $M(1)$ is the rank-twenty-four Heisenberg oscillator space; $\mathbb{C}_\varepsilon[\Lambda]$ is the group algebra twisted by the cocycle of the lattice central extension. The cocycle and the lifting of isometries are part of the construction: a bare vector-space sum does not determine the vertex products.

Negation $\lambda \mapsto -\lambda$ admits the standard lift θ . With its twisted module and the compatible parity choice, the Frenkel–Lepowsky–Meurman construction produces

$$V_{\text{HMT}}^\natural = V_\Lambda^+ \oplus (V_\Lambda^T)^+. \quad (10.18)$$

The sign $+$ denotes the even part of the lifted action. The expression includes the orbifold vertex product; it is not merely a sum of graded spaces.

Theorem 10.14 (Prolongation to the Moonshine module). *The algebra (10.18), constructed on the neighbor (10.17), is a realization of the Moonshine algebra $V^\natural$. It has central charge 24, zero weight-one component, and automorphism group isomorphic to the Monster $\mathbb{M}$. Its graded character is*

$$\text{Tr}_{V_{\text{HMT}}^\natural} q^{L_0-1} = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots, \quad q = e^{2\pi i \tau}. \quad (10.19)$$

Proof. Theorem 10.13 verifies the lattice hypotheses: positivity, evenness, unimodularity, rank twenty-four, and absence of roots. One then applies the Frenkel–Lepowsky–Meurman construction and theorem for the Leech lattice. These identify the orbifold with $V^\natural$ and establish its character and automorphism group. Composing the flag, the neighbor, and the VOA construction functor gives the realization stated here. The group is not deduced from the coefficient 196884. $\square$

The theorem used is a subsequent classical recognition with verified hypotheses, not a new proof in this article of the FLM construction [5]. Furthermore, the notation $q = e^{2\pi i\tau}$ is the conventional Fourier variable used to recognize the character. It introduces neither π nor e as inputs to the coinductive generator that has already determined them.

Part of the trace mechanism can be seen before the orbifold. Only the zero lattice vector is fixed by negation; each oscillator contributes alternating sign. Hence

$$\mathrm{Tr}_{V_\Lambda} \theta q^{L_0-1} = t_2(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}.$$

This trace in the lattice algebra must not be confused with the character without insertion of $V^\natural$. Nor does it by itself identify a Monster conjugacy class: the space, insertion, and lift must be specified before functions are compared.

Grading, unit and automorphisms. The Frenkel–Lepowsky–Meurman construction is applied to the lattice N_α once its properties have been verified. It incorporates the oscillator space, the central extension of the lattice, the vertex product and the sector twisted by the lift of negation. The result is the algebra $V_{\mathrm{HMT}}^\natural$ of (10.18), whose automorphism group is isomorphic to the Monster. Its domain of action is thus fixed by an algebraic structure equipped with a grading and a product. Borcherds’ Moonshine theorem subsequently determines the graded traces with insertions of these automorphisms [5, 6].

At this level the unit acquires a precise realization: the weight-zero component is the vacuum line and supplies the coefficient 1 of q^{-1} in the graded character. At the cylindrical level, refinement maps preserve the constant function by precomposition. Each preservation statement therefore has its own object and map. The development of the proofs makes this continuity of construction explicit, from arithmetic operations and memory to incidence, the metric and vertex products.

This articulation places K and the closure of α within the main argument. K reversibly preserves an integral coordinate of the history; the normalization of α composes the regional registers with that coordinate; the signed configuration determines a flag; and the marked flag selects the lattice modification on which Moonshine is realized. The following proofs develop these maps, distinguishing reversible equivalences, information quotients and realizations that add a metric or an algebraic structure. This expresses a demonstrable relationship between structure, equation and value, preserving the attribution of each construction and the scope of its theorems.

10.10 Graded Moonshine traces and the Monster representation

For $g \in \mathrm{Aut}(V_{\mathrm{HMT}}^\natural)$, the trace with insertion is

$$T_g(\tau) = \mathrm{Tr}_{V_{\mathrm{HMT}}^\natural} g q^{L_0-1}.$$

The monstrous Moonshine theorem identifies these series as Hauptmoduln of the corresponding genus-zero groups [6, Theorem 1.1]. The character J , the group $\mathbb{M}$, and the collection $g \mapsto T_g$ are realizations of the algebra already constructed. They do not form a causal sequence in which J is first guessed and the Monster then deduced.

The unit has a precise algebraic location here. The weight-zero subspace is the vacuum line, $V_0^\natural = \mathbb{C}\mathbf{1}$, and the coefficient of q^{-1} in J is therefore 1. In weight one, the dimension is zero, accounting for the absence of the constant term in $J = j - 744$. A vertex algebra morphism preserves the vacuum vector. This property supplies the technical meaning of preservation of the unit at this level; the unit of the cylindrical observables is preserved by the precomposition proved earlier. The relation between the two levels must follow the construction maps with their units.

In weight two, the equality

$$196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

appears. The first decomposition corresponds to the trivial conformal line and the irreducible component of dimension 196883 of the Monster representation. The other expressions have lattice or arithmetic readouts. Their equality does not automatically identify their summands with regions, words, or TPK routes. Doing so would require presenting the maps between those spaces and proving that they respect the relevant actions.

In particular, an isometry $N_\alpha \simeq \Lambda_{24}$ induces a realization of (10.18), and a choice of isomorphism with $V^\natural$ transports its automorphism group. This has not constructed a Monster action on the digits of π , on the blocks of K , or on all TPK histories. Such an additional assertion would require a representation ρ_{TPK} and a map F with an explicit square

$$F\rho_{\text{TPK}}(g) = \rho_{V^\natural}(g)F,$$

including domain, codomain, and preserved information. The absence of that step from the present assertion does not withdraw the incidence intertwiner U_W , the flag, or the neighbor already constructed; it only delimits a different strengthening, concerning action on the source state.

The chain established in this section can be summarized, keeping its transport data visible, as

$$\begin{aligned} &\text{APP-TRIT-TPK and arithmetic state with trajectory memory} \longrightarrow (w_j, f_j; K, Z_0) \\ &\longrightarrow (C_W, \mathcal{H}; B_K \subset H_\alpha^-, P_\pi, H_e, H_\varphi, H_{\text{APP}}) \\ &\longrightarrow (N(C_W), o_*, \text{metric and chamber}) \longrightarrow N_\alpha \longrightarrow V_{\text{HMT}}^\natural. \end{aligned}$$

At the incidence level, the stars generate M_{12} and the residual chart realizes the regional $4 + 1$ structure in W_{24} . At the lattice and VOA level, the construction gives Leech and the Moonshine module. These are prolongations of the same marked source, with different objects and maps. Causal continuity does not require identifying their codomains or erasing the explicit choices of frame, chamber, and lift.

The result of the section is therefore a consolidated derivation: the finite and lattice proofs make effective the passage from dodecaphasic incidence to the hypotheses of the recognition theorems. This result concerns the specified constructions and hypotheses; it does not add experimental validation to the comparison of constants.

11 Planck constants and the determinantal structure of action

11.1 Determinantal action scale and decimal valuation

The absolute scale and the relative corrector originate in distinct objects. The former uses the local block of the step-three dodecaphase record,

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad G_H = \mathbb{Z}^4 / H_4 \mathbb{Z}^4. \quad (11.1)$$

The operation uses a local quotient. The three blocks of the full record are not replaced by a tensorization that would multiply the number of sheets of this reader once again.

Lemma 11.1 (Number of sheets of the local quotient). *The Smith normal form of H_4 is $\text{diag}(1, 2, 2, 4)$. Consequently,*

$$G_H \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \quad |G_H| = 16.$$

Proof. The greatest common divisor of the entries is one. The minors of order two are even, and one has value -2 , so their greatest common divisor is two. Since $H_4 H_4^\top = 4I$ and $\det H_4 = -16$, the cofactors are ± 4 ; the third determinantal divisor is four. The fourth is 16. The successive quotients of the divisors 1, 2, 4, 16 are 1, 2, 2, 4. Their product gives the order of the quotient. $\square$

Definition 11.2 (Determinantal action reader). The multiplicative norm of the scalar section α , constant over the sheets of G_H , and one application of the self-scale φ produce

$$\Sigma_H = \varphi N_{G_H}(\alpha), \quad N_{G_H}(\alpha) = \prod_{g \in G_H} \alpha = \alpha^{16}. \quad (11.2)$$

The associated decimal chart determines $\text{ord}_{10}(\Sigma_H) = \lfloor \log_{10} \Sigma_H \rfloor$.

The sixteenth power thus arises from the number of sheets of the quotient, once this local reader has been fixed. The factor φ acts once on the basis of the action line. The Smith computation alone would not select any possible functional on the quotient: the multiplicative norm and the self-scale form part of the declared operation.

Theorem 11.3 (Decade of the action section). *Reader (11.2), applied to the already generated HMT coordinates, satisfies*

$$10^{-34} < \Sigma_H < 10^{-33}, \quad \text{ord}_{10}(\Sigma_H) = -34. \quad (11.3)$$

Proof. The following rational intervals suffice; they are much coarser than the available cylinders:

$$\frac{1618}{1000} < \varphi < \frac{1619}{1000}, \quad \frac{7297}{10^6} < \alpha < \frac{7298}{10^6}.$$

The function $(x, a) \mapsto xa^{16}$ is increasing in both positive arguments. The two integer inequalities

$$1618 \cdot 7297^{16} > 10^{65}, \quad 1619 \cdot 7298^{16} < 10^{66}$$

imply

$$10^{-34} < \frac{1618}{1000} \left(\frac{7297}{10^6} \right)^{16} < \Sigma_H < \frac{1619}{1000} \left(\frac{7298}{10^6} \right)^{16} < 10^{-33}.$$

The definition of decimal order completes the proof. No SI unit or reference mass enters these inequalities. $\square$

The normalized section begins with $\Sigma_H = 1.0462438381 \dots \times 10^{-34}$. Its mantissa is not identified with the action coordinate before or after return. The determinantal reader fixes the decimal order; the two return coordinates are obtained through the following readers.

11.2 Action sections and dimensionless return quotient

Let $\mathcal{L}_5$ be an oriented action line, and let $\mathcal{U}_5$ be its positive basis. The fifth-order character and the regional return term have the expressions

$$H_5 = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \quad (11.4)$$

$$D_A = \exp\left(-\frac{100\pi\alpha}{9}\right), \quad (11.5)$$

$$\mathcal{C}_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \quad (11.6)$$

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi} \mathcal{C}_\pi(\alpha). \quad (11.7)$$

These are subsequent readings of the dodecaphase closure and its regional incidence. The integer 169 is retained as the regional selector already constructed, and the coefficients of H_5 as those of its local character; they are not replaced by an interpolation of the Planck constant. The formulas make explicit exactly which readings are needed for the electronic stage. It is not necessary to introduce other functionals of mass theory here.

For $0 < \alpha < 1$, one has $0 < D_A < 1$, so $D_A \alpha < 1$. The rational term in (11.6) preserves the complete sum of the geometric tail

$$\frac{D_A \alpha^7}{1 - D_A \alpha} = \sum_{n=0}^{\infty} D_A^{n+1} \alpha^{n+7}. \quad (11.8)$$

It is not a truncation that may be silently modified when the precision changes. After the term of index N , the remaining tail is $D_A^{N+2} \alpha^{N+8} / (1 - D_A \alpha)$, allowing evaluation to be controlled without recourse to a target quantity.

Definition 11.4 (Action sections and return reader). The pre-return and post-return sections are

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S. \quad (11.9)$$

Whenever $\eta_{\text{ret}} > 0$, define

$$\delta_{\text{ret}} = H_5 - \eta_{\text{ret}}, \quad R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5}{\eta_{\text{ret}}}. \quad (11.10)$$

Proposition 11.5 (Positivity and invariance of the ratio). *In the intervals used in the proof of (11.3), $0 < \eta_{\text{ret}} < H_5$, and therefore $R_{\text{act}} > 1$. The ratio does not change when the basis $\mathcal{U}_S$ is replaced by any other common positive basis.*

Proof. The terms of $\mathcal{C}_\pi$ are positive. For an elementary bound, it suffices to use $\alpha < 0.008$, $\varphi < 1.619$, $\alpha > 0.007$, $D_A < 1$, and $\pi > 3$. The root in (11.4) is greater than one, whereas the sum of the negative terms is less than 0.008001; in particular, $H_5 > 0.99$. Likewise,

$$0 < \frac{90}{\pi} \mathcal{C}_\pi < 30 \left(169(0.008)^6 + \frac{(0.008)^7}{1 - 0.008} \right) < 10^{-6}.$$

Thus $\eta_{\text{ret}} > 0.989999 > 0$ and $\eta_{\text{ret}} < H_5$. Finally, the basis change divides both action coordinates by the same positive factor, which cancels in their quotient. $\square$

Subsequent evaluation gives

$$\begin{aligned} H_5 &= 1.0545718176461544696 \dots, \\ \eta_{\text{ret}} &= 1.0545718169150390611 \dots, \\ R_{\text{act}} &= 1.0000000006932817630 \dots. \end{aligned}$$

The difference and the quotient are two readings of the same sections, one additive and the other multiplicative. They do not represent two independent Planck constants. The passage from angular action to action per complete turn is expressed, in each section, by $h = 2\pi\hbar$. The subsequent chart $\mathcal{U}_S \mapsto \mathcal{J}_s$ assigns a unit but does not act retroactively on the decimal order already proved.

Regional return can also be written in an angular chart. This reparametrization is not an additional indispensable dependency: (11.4)–(11.7) directly supply H_5 , η_{ret} , and R_{act} . If angle and counter-angle are introduced, both must be transported in the same chart; degrees and radians are not mixed within a difference.

11.2.1 Action character and provenance of its coefficients

The reduced Planck constant, $\hbar$, is the physical object that motivates realization of the angular-action section. Its position in the electronic dependency must be specified at two levels. The determinantal norm Σ_H determines the decade of the scale. The character H_5 and the regional return then determine the coordinates of the action sections. This separation allows the result needed for the electron to be presented without anticipating the full developments of other sectors of dimensional mechanics.

The provenance of the fifth-order character can be made explicit through invariants already fixed in the support. In the central chart, consider

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \lambda_+ = 9, \quad \lambda_- = 5.$$

The calendar has length 12, the step-3 orbit has length 4, and the relevant census quotient is $104976/1944 = 54$. The character coefficients satisfy

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \quad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \quad \frac{7}{48} = \frac{\text{tr}(B_c)/2}{4 \cdot 12}, \quad \frac{1}{54} = \frac{1944}{104976}. \quad (11.11)$$

The assignment of these invariants to degrees 2, 3, 4, 5, with the alternating signs of (11.4), belongs to the definition of the analytic character used. Making it explicit is essential: the collection of cardinalities, considered without the character and without the ordering of its degrees, would admit other expressions. The constructive content lies in the composition of the support, orientation, and declared analytic rule.

11.2.2 Regional continuation and renewal equation

The selector 169 determines the degree-six impulse. Subsequent prolongation is organized by the transfer D_A and a unit normalization of the determinant line. These rules admit a formulation in formal series before substituting $z = \alpha$, with D_A already constructed. Write

$$\mathcal{C}(z) = 169z^6 + R(z), \quad R(z) \in z^7\mathbb{R}[[z]].$$

Concatenating a prolongation increases the degree by one and multiplies its weight by D_A . The first strict prolongation has weight D_A . Its renewal equation is therefore

$$R(z) = D_A z^7 + D_A z R(z). \quad (11.12)$$

Proposition 11.6 (Formal uniqueness of the normalized continuation). *Equation (11.12) determines a unique series, and*

$$\mathcal{C}(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z} = 169z^6 + \sum_{n=7}^{\infty} D_A^{n-6} z^n.$$

The realization is analytic for $|D_A z| < 1$.

Proof. The element $1 - D_A z$ has constant term one and is invertible in the ring of formal series. Solving the equation produces the displayed series. Equivalently, the degree-seven coefficient is D_A , and each subsequent coefficient is D_A times the preceding one. The geometric series converges absolutely in the indicated disk. $\square$

Normalization of the first prolongation is an indispensable operative datum. If only the degree-six impulse and the concatenation ratio were retained, the series

$$169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}$$

would still share both data for every λ . The unit rule fixes $\lambda = 1$ before evaluation is performed. This locates the contribution of each rule to the uniqueness of the reader, and the rational expression preserves the full continuation.

11.2.3 Reduced Planck constant and return of the action section

In the dimensional chart of (11.9), the pre-return section is the model construction intended for comparison with the reduced Planck constant:

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S.$$

The post-return section additionally preserves the regional compensation $90\mathcal{C}(\alpha)/\pi$. Both belong to the same action line, and their quotient R_{act} measures the multiplicative effect of return. The passage to h through 2π uses the relation between angular parametrization and a complete turn.

This construction enters effectively into $R_{\text{act}}^{80-54\alpha+6\Delta_4}$. The electron preserves its central coordinate and its trajectory; the chronological scale does not replace that structure with an undifferentiated

quantum. Nor is a mass already expressed in MeV multiplied again by 10^{-34} : the decade belongs to the action section, cancels in its dimensionless quotient, and reappears only where the corresponding dimensional realization acts.

The numerical comparison of H_5 , the returned section, and $h/(2\pi)$ is presented in Section 11.3. This distinguishes the name of the physical object, the equation assigned to it by the model, and the precision with which that equation reproduces it.

11.3 Comparison of the reduced Planck constant

The International System in force since 20 May 2019 fixes $h = 6.62607015 \times 10^{-34}$ J s exactly [8]. Its reduced constant is therefore

$$\hbar_{\text{SI}} = \frac{6.62607015}{2\pi} 10^{-34} \text{ J s} = 1.0545718176461563912 \dots \times 10^{-34} \text{ J s}.$$

The ellipsis denotes the expansion of an exact expression, not an experimental uncertainty in h in the current SI. This reference value is used here after evaluation of the HMT readers; its function is comparative.

The article's two return coordinates are

$$\begin{aligned} H_5 &= 1.0545718176461544696 \dots, \\ \eta_{\text{ret}} &= 1.0545718169150390611 \dots. \end{aligned}$$

Under the identification of $\mathcal{U}_5$ with J s, the relative difference between the preceding section and $\hbar_{\text{SI}}$ is approximately -1.82×10^{-15} . For the subsequent section, it is approximately -6.93×10^{-10} . The second difference contains the return effect used by the electronic corrector; it does not equal the evaluation error of the first section.

The precision of these comparisons allows the quantitative proximity of the construction of H_5 to be recognized while distinguishing proximity from exact identity. In the present manuscript, the displayed expressions do not provide an exact equality with $h/(2\pi)$. The result proved for Σ_H is its decade -34 ; the analytic result for H_5 is the evaluation of the stated character. The physical identification is examined in this chart and with the explicit differences just given. The SI value does not participate in the definition of the determinantal norm or in the regional recurrence.

11.4 Angular coordinates, normalization of the turn, and units of action

The quantity of action and the angular coordinate belong to distinct spaces. The section $\hbar_{\text{ret}}$ occupies the action line $\mathcal{L}_S$; η_{ret} is its coefficient in the declared basis. Angular representation acts on the dimensionless pair $(\alpha, \eta_{\text{ret}})$, after its construction:

$$\mathcal{P}(\alpha, \eta_{\text{ret}}) = ([1000\alpha]_{360}, [2(\eta_{\text{ret}} + \alpha)]_{360}). \quad (11.13)$$

Nonadic completion determines the factor 1000; bidirectional closure determines the factor two. The circular chart of 360 units makes the partitions $12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120$ visible. On the positive branch of the article, the representatives are

$$A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha).$$

Proposition 11.7 (Shift of the base-thousand representation). *Let $\alpha = \sum_{n \geq 1} b_n 1000^{-n}$, with $b_n \in \{0, \dots, 999\}$, be the compatible expansion already generated. The lifted coordinate $A_{\text{deg}} = 1000\alpha$ satisfies*

$$A_{\text{deg}} = b_1 + \sum_{n \geq 1} b_{n+1} 1000^{-n}.$$

The representation preserves the block sequence from the second block onward; the initial block becomes the integer part.

Proof. The series converges absolutely. Multiplying each term by 1000 and separating the initial term gives the identity. The operation acts on the constructed expansion, with its blocks and memory; the subsequent circular representation takes its class modulo 360. $\square$

Definition 11.8 (Charts of the angular coordinate). The degree chart expresses a complete turn in 360 units. The radian chart expresses it as 2π . The coordinate changes and their inverses are

$$x = \frac{\pi}{180} A_{\text{deg}}, \quad y = \frac{\pi}{180} C_{\text{deg}}^*, \quad A_{\text{deg}} = \frac{180}{\pi} x, \quad C_{\text{deg}}^* = \frac{180}{\pi} y.$$

On a path with preserved winding number, the lifted coordinates are used; the circular quotient expresses its class modulo the corresponding turn.

The definition characterizes degrees and radians through their exact relation to the turn. The record also retains orientation, turns, and transport memory. Representation of its angular class preserves phase, whereas the lift distinguishes paths with the same final phase. The physical reading of action requires specifying which of these coordinates enters the integral.

Proposition 11.9 (Covariance of the action reader). *For $h_a = 2\pi\hbar_a$, the action of a lifted coordinate satisfies*

$$\mathcal{S}_a(\theta) = \hbar_a \theta_{\text{rad}} = h_a \frac{\theta_{\text{deg}}}{360}. \quad (11.14)$$

A positive turn expresses h_a .

Proof. Substituting $\theta_{\text{rad}} = \pi\theta_{\text{deg}}/180$ gives $\hbar_a \pi\theta_{\text{deg}}/180 = (2\pi\hbar_a)\theta_{\text{deg}}/360$. In one turn $\theta_{\text{deg}} = 360$, yielding h_a . $\square$

The periodicity of a representation is declared separately. If the transported representation satisfies $U(2\pi) = -I$, its operator restoration occurs at 4π . This property of the lift coexists with the per-turn normalization of (11.14). Each preserves its own object: integrated quantity, circular phase, and represented state.

11.5 Covariance of the channels and the incidence functional

The channels are written equivalently as

$$q_{\pm} = \exp[-(x \pm y)] = \exp\left[-\frac{\pi}{180}(A_{\text{deg}} \pm C_{\text{deg}}^*)\right]. \quad (11.15)$$

Proposition 11.10 (Recovery of the even and odd coordinates). *In the lifted real chart, the positive channels uniquely determine the two coordinates:*

$$A_{\text{deg}} = -\frac{90}{\pi} \log(q_+ q_-), \quad C_{\text{deg}}^* = \frac{90}{\pi} \log \frac{q_-}{q_+}.$$

The exchange $(q_+, q_-) \mapsto (q_-, q_+)$ preserves A_{deg} and reverses C_{deg}^ .*

Proof. The sum and difference of the logarithms of (11.15) are, respectively, $-\pi A_{\text{deg}}/90$ and $-\pi C_{\text{deg}}^*/90$. The real logarithm is single-valued on the positive channels. The product is symmetric, and the quotient is inverted when they are exchanged. $\square$

In the twelve-position record, the antipodal involution $r \mapsto r + 6 \pmod{12}$ forms six pairs. The uniform distribution of the odd coordinate over these pairs is $C_{\text{deg}}^*/6$: summing their six contributions recovers C_{deg}^* . This sectorial coordinate is distinguished from the Planck constant h , which belongs to the action line.

The conversion is applied only once to each argument. The quantity α remains dimensionless, and the quantity $\hbar$ continues to belong to the action line; their preceding coordinates enter the map $\mathcal{P}$.

In addition to the channels, the Barbero–Immirzi functional contains a component bilinear in the angular representatives. Its transport is

$$\frac{\pi}{180} A_{\text{deg}} C_{\text{deg}}^* = \frac{180}{\pi} xy. \quad (11.16)$$

The identity follows by substituting both degree representatives with their expressions in radians. The complete formula in analytic coordinates is therefore

$$\gamma_{\text{HMT}} = \frac{180}{\pi} xy + 12[S_{90}(e^{-x+y}) - S_{90}(e^{-x-y})] - [S_{120}(e^{-x+y}) - S_{120}(e^{-x-y})].$$

Equivalence of the two charts transports the bilinear component and the exponential functions simultaneously. This transformation makes explicit the precise role of the degree in the construction and allows the evaluation to be reproduced without ambiguity in angular units.

12 Action ellipse and modular reciprocity

With the outputs $A, C^*, \hbar > 0$ fixed in the chart $|C^*/A| < 1$, we define

$$\eta_{\text{el}} = \text{artanh}(C^*/A), \quad a_S = \sqrt{2\hbar} e^{\eta_{\text{el}}/2}, \quad b_S = \sqrt{2\hbar} e^{-\eta_{\text{el}}/2}.$$

The balanced canonical coordinates have units of the square root of action; position and momentum are not added here without a dimensional chart. The following hold:

$$a_S b_S = 2\hbar, \quad \tau_{\text{el}} = i \frac{b_S}{a_S}, \quad R_{\text{el}} = \sqrt{\frac{b_S}{a_S}} = \left(\frac{A - C^*}{A + C^*} \right)^{1/4}.$$

The positive root determines the composition $\tau_{\text{el}} = iR_{\text{el}}^2$, and makes it possible to recover the angular ratio:

$$\frac{C^*}{A} = \frac{1 - R_{\text{el}}^4}{1 + R_{\text{el}}^4}.$$

Shape inversion sends R_{el} to R_{el}^{-1} and τ_{el} to $-1/\tau_{\text{el}}$.

Proposition 12.1 (Coincident energy, distinguishable action). *Let*

$$D(\eta) = \begin{pmatrix} e^\eta & 0 \\ 0 & e^{-\eta} \end{pmatrix}, \quad S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Both maps satisfy $T^\top D(-\eta)T = D(\eta)$. However, for $\omega = dQ \wedge dP$,

$$S_2^* \omega = -\omega, \quad J_2^* \omega = \omega.$$

If $\mathcal{I}(\gamma) = \oint_\gamma P dQ$ on a closed curve,

$$\mathcal{I}(S_2 \gamma) = -\mathcal{I}(\gamma), \quad \mathcal{I}(J_2 \gamma) = \mathcal{I}(\gamma).$$

Proof. Conjugation exchanges the diagonal entries of D . The determinants of S_2, J_2 are $-1, 1$, respectively, which proves the transformation of ω . For $\lambda = P dQ$,

$$S_2^* \lambda = d(PQ) - \lambda, \quad J_2^* \lambda = \lambda - d(PQ).$$

Integrating the exact differential over the closed curve completes the proof. $\square$

The quadratic energy and the modulus do not suffice to distinguish the two transports. The orientation of action does distinguish them, even in the circular shape. The geometric marking is thus part of the recoverable reading, not an embellishment added to the value.

12.1 Semiaxes, foci, and interior radii

The figure is obtained from the angular and action outputs of the preceding sections. APP preserves the sheets and their residues and quotients; TRIT fixes orientation; TPK transports boundary, carry, and memory through to the representations of the same enriched state and the joint discrete structure of the continuum. The pair (A, C^*) and the positive section $\hbar$ are received here after that construction. The geometry makes explicit what information their readers preserve, without using an observed length to select the source state.

Let $\xi = C^*/A \in (-1, 1)$, with $A \neq 0$ and both angles in the same unit. Write $\eta = \operatorname{artanh} \xi$ and

$$a_S = \sqrt{2\hbar} e^{\eta/2}, \quad b_S = \sqrt{2\hbar} e^{-\eta/2}, \quad \gamma(\theta) = (a_S \cos \theta, b_S \sin \theta).$$

The semiaxes are labeled by their coordinates: if $\eta < 0$, the vertical axis is the larger. Both belong to the square-root-of-action line of the balanced chart, not to the line of spatial lengths.

Proposition 12.2 (Focal geometry and radial interior). *Let $a = \max(a_S, b_S)$, $b = \min(a_S, b_S)$. The focal half-distance and eccentricity are*

$$f_S = \sqrt{a^2 - b^2} = \sqrt{4\hbar \sinh |\eta|}, \quad e_f = \frac{f_S}{a} = \sqrt{1 - e^{-2|\eta|}}. \quad (12.1)$$

For $\eta > 0$, the foci are $(\pm f_S, 0)$; for $\eta < 0$, they are $(0, \pm f_S)$. At $\eta = 0$, they coincide with the center. The boundary radius in polar direction ϕ is

$$\rho_\partial(\phi) = \left(\frac{\cos^2 \phi}{a_S^2} + \frac{\sin^2 \phi}{b_S^2} \right)^{-1/2}. \quad (12.2)$$

The interior segment in that direction consists of $r(\cos \phi, \sin \phi)$, with $0 \leq r < \rho_\partial(\phi)$. In particular, $b \leq \rho_\partial(\phi) \leq a$.

Proof. The squares of the major and minor semiaxes are $2\hbar e^{|\eta|}$ and $2\hbar e^{-|\eta|}$. Their difference gives (12.1). To verify the focus, suppose $\eta > 0$. At $X = \gamma(\theta)$, the distances to $F_+ = (f_S, 0)$ and $F_- = (-f_S, 0)$ satisfy

$$|X - F_+|^2 = (a_S - f_S \cos \theta)^2, \quad |X - F_-|^2 = (a_S + f_S \cos \theta)^2.$$

Since $a_S > f_S$, their positive roots sum to $2a_S$. For $\eta < 0$, exchanging coordinates gives the same proof. In the circular case, $f_S = 0$. Finally, substituting $(Q, P) = r(\cos \phi, \sin \phi)$ into $Q^2/a_S^2 + P^2/b_S^2 = 1$ gives (12.2). The coefficient of r^2 lies between a^{-2} and b^{-2} , proving the bounds and the characterization of the interior. $\square$

12.2 Parametric phase, polar angle, and action area

Proposition 12.3 (Areal law and two angular coordinates). *The phase θ of γ and its continuous polar angle ϕ , lifted from zero, satisfy*

$$\phi(\theta) = \arg(a_S \cos \theta + ib_S \sin \theta), \quad \frac{d\phi}{d\theta} = \frac{a_S b_S}{a_S^2 \cos^2 \theta + b_S^2 \sin^2 \theta} > 0. \quad (12.3)$$

In a chart where both tangents are defined, $\tan \phi = (b_S/a_S) \tan \theta$. The oriented swept area is

$$\mathcal{A}(\theta) = \frac{1}{2} \int_0^\theta (QP' - PQ') dt = \hbar \theta, \quad \mathcal{A}(1) = \hbar, \quad \mathcal{A}(2\pi) = h. \quad (12.4)$$

Proof. The derivative of the argument of a nonzero vector is $(QP' - PQ')/(Q^2 + P^2)$. Here the numerator is $a_S b_S = 2\hbar$, proving the derivative and monotonicity. The quotient P/Q gives the tangent relation; the continuous lift fixes the quadrants and the number of turns. The areal integral is $a_S b_S \theta/2 = \hbar \theta$, and $h = 2\pi \hbar$ gives the final identity. $\square$

Parametric phase does not generally coincide with polar angle. One radian of phase sweeps $\hbar$, but is not thereby identified with a circular arc or a sector of unit polar angle. In polar coordinates, the same law is written $d\mathcal{A} = \rho_\partial(\phi)^2 d\phi/2$. With the orientation of γ used here, $\oint P dQ = -h$: positive area is h , whereas the integral of that one-form preserves its sign. This agrees with the distinction between symplectic and antisymplectic transports in the preceding section.

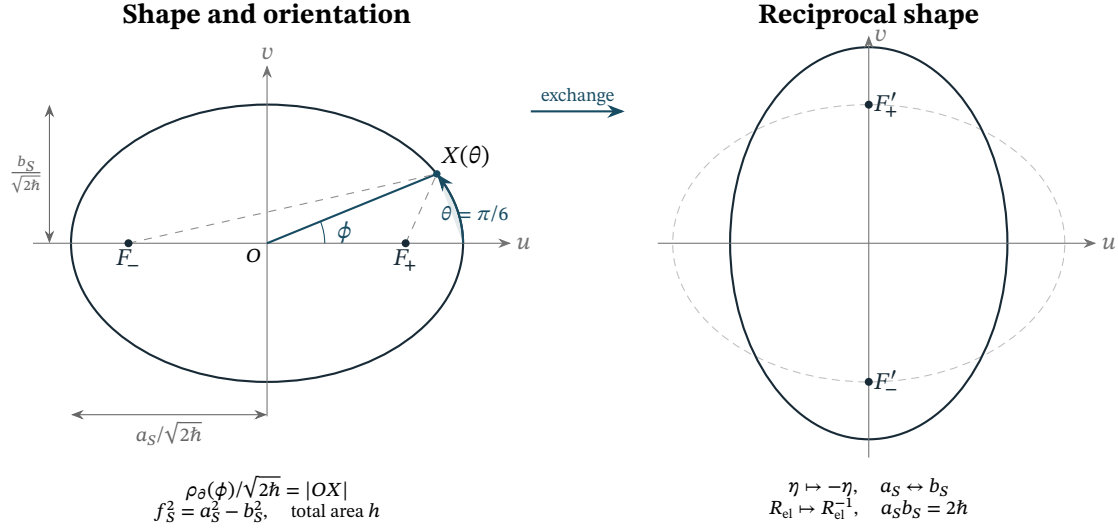


Figure 10: Semiaxes, foci, and reciprocity of the action ellipse in the normalized coordinates $u = Q/\sqrt{2\hbar}$, $v = P/\sqrt{2\hbar}$. The marked sector has parametric phase $\theta = \pi/6$, different from its polar angle ϕ ; its action area is $\hbar\pi/6$. The figure uses the algebraic case $\xi = 1/3$ to make the relations legible: it does not assign that value to the HMT angular output. The foci are computed from the semiaxes. On the right, exchanging axes preserves total area h and inverts the dimensionless modular radius.

For the control case of the figure, $\xi = 1/3$, the normalized coordinates are exactly

$$\eta = \frac{1}{2} \log 2, \quad \frac{a_S}{\sqrt{2\hbar}} = 2^{1/4} = 1.1892071150 \dots, \quad \frac{b_S}{\sqrt{2\hbar}} = \frac{f_S}{\sqrt{2\hbar}} = R_{\text{el}} = 2^{-1/4} = 0.8408964153 \dots$$

These are regression numbers for the drawing, not values assigned to the angular generator. The local certificate also checks the circular shape and the orientations $\xi = \pm 1/3, \pm 3/5$.

12.3 Shape reciprocity and interior distance

The rectangular representation recovers

$$R_{\text{el}} = \sqrt{b_S/a_S} = e^{-\eta/2}, \quad R_{\text{el}}^4 = \frac{1-\xi}{1+\xi}, \quad \xi = \frac{1-R_{\text{el}}^4}{1+R_{\text{el}}^4}. \quad (12.5)$$

The proof consists in substituting $2\eta = \log((1+\xi)/(1-\xi))$. Exchanging axes sends η to $-\eta$, R_{el} to R_{el}^{-1} , and $\tau = iR_{\text{el}}^2$ to $-1/\tau$. The product $a_S b_S$ is preserved. The radius R_{el} is dimensionless; the polar radius ρ_∂ has square-root-of-action units.

The interior also admits the Hilbert projective distance. If z_-, x, y, z_+ are ordered on a chord of the ellipse, define

$$d_H(x, y) = \frac{1}{2} \log \frac{|y - z_-| |x - z_+|}{|x - z_-| |y - z_+|}.$$

It is a dimensionless distance: the units cancel in the cross-ratio. The normalization $(Q, P) \mapsto (Q/a_S, P/b_S)$ takes the ellipse to the unit disk and preserves that ratio, because lengths on a single line are multiplied by a common factor. It thus transports the metric to the Beltrami–Klein model. Neither this interior distance nor the distance between foci is identified with an atomic length.

12.4 Bohr radius: a consequence of the action scale

In the electrodynamic realization, the Bohr scale is expressed as $a_B = \hbar/(m_e c \alpha)$ [4]. In the chain of the present article, this reading follows the constructions of action, vacuum, fine structure, and electronic mass. Here $\hbar, m_e, c, \alpha > 0$ are received in the same dimensional realization.

Proposition 12.4 (Areal expression of the Bohr scale). *In that realization,*

$$a_B = \frac{a_S b_S}{2m_e c \alpha} = \frac{\mathcal{A}(2\pi)}{2\pi m_e c \alpha} = \frac{\bar{\lambda}_C(m_e)}{\alpha}, \quad \bar{\lambda}_C(m_e) = \frac{\hbar}{m_e c}. \quad (12.6)$$

Proof. The identities $a_S b_S = 2\hbar$ and $\mathcal{A}(2\pi) = 2\pi\hbar$, substituted into the Bohr expression, give the first two members. The definition of $\bar{\lambda}_C(m_e)$ gives the third. $\square$

The numerator has action dimension, and $m_e c$ has momentum dimension; the quotient is a length. This composition uses the product of the semiaxes and the other three quantities: it does not assert that Bohr is a semiaxis, a focus, a Hilbert distance, or the modular radius. Nor does it equate electronic mass with the circular quantum of order 108. The notation a_B avoids confusing this length with the normalization a_0 of the area operator.

12.5 Inductive–capacitive realization and constitutive continuity

The electrodynamic bridge of the ellipse preserves an additional chart: a clock $\omega > 0$ and a reference impedance $Z_* > 0$. The construction recovered from the corpus defines

$$L_\eta = \frac{Z_*}{\omega} e^{-\eta}, \quad C_\eta = \frac{1}{Z_* \omega} e^\eta, \quad Z_\eta = \sqrt{L_\eta / C_\eta}. \quad (12.7)$$

Here C_η is capacitance, distinct from the counter-angle C^* .

Proposition 12.5 (Dynamical product and constitutive quotient). *In the preceding chart,*

$$L_\eta C_\eta = \omega^{-2}, \quad \frac{Z_\eta}{Z_*} = e^{-\eta} = \frac{b_S}{a_S} = R_{\text{el}}^2. \quad (12.8)$$

With $Q = \sqrt{Z_*} q$, $P = \Phi / \sqrt{Z_*}$, the energy $H_{LC} = q^2 / (2C_\eta) + \Phi^2 / (2L_\eta)$ takes the form

$$H_{LC} = \frac{\omega}{2} (e^{-\eta} Q^2 + e^\eta P^2).$$

On $\gamma(\theta)$, its electric and magnetic components are $U_E = \hbar\omega \cos^2 \theta$ and $U_B = \hbar\omega \sin^2 \theta$; their sum is $\hbar\omega$.

Proof. The product in (12.7) cancels the exponents and Z_* ; the quotient is $Z_*^2 e^{-2\eta}$. Taking the positive root and using (12.5) gives (12.8). Substituting $q = Q / \sqrt{Z_*}$ and $\Phi = \sqrt{Z_*} P$ into H_{LC} gives its balanced form. Finally, the semiaxes of γ cancel the factors $e^{\mp\eta}$, and $\cos^2 \theta + \sin^2 \theta = 1$. $\square$

This bridge distinguishes shape, clock, and transducer. The choice of dynamical direction must agree with the symplectic convention: for the equations $\dot{Q} = \partial_P H$, $\dot{P} = -\partial_Q H$, the preceding parametrization satisfies $\dot{\theta} = -\omega$. The energy identities do not depend on reversing that path.

The vacuum resolvent reader acts on the already produced channels $q_\pm = e^{-(A_{\text{rad}} \pm C_{\text{rad}}^*)}$ and incidence 90/120. Its responses $r_\pm = (1 - q_\pm^{90}) / (1 - q_\pm^{120})$ express $\hat{Z} = r_- / r_+$ and $\hat{c} = (r_+ r_-)^{-1}$. These are spectral responses, not geometric radii. Identity (12.8) fixes the LC realization here; identifying it with the constitutive vacuum impedance preserves the map between the two readers and their charts. Their values are not equated merely because they share the angular antecedent.

13 Sheet return, areal–volumetric incidence and constitutive orientation

The sheet structure enters before electromagnetic quantities are evaluated. Its function is to distinguish closure of a projection, restoration of an orientation and evolution of the transported register. These operations act on related objects of different types: a projected trajectory may have closed while its lift remains on the opposite sheet; orientation may have been restored while memory preserves the two windings performed. The constitutive construction must receive this information together with the structure determining it.

13.1 Projected return and restoration of the oriented cover

Let ℓ be a return loop of a surviving extension, and let $\varepsilon : \langle \ell \rangle \rightarrow \{+1, -1\}$ be the character of its lift, with $\varepsilon(\ell) = -1$. The associated interval bundle has the presentation

$$\mathcal{M} = ([0, 1] \times [-1, 1]) / (0, u) \sim (1, -u). \quad (13.1)$$

The decisive datum is the sign of transport around the specified loop. The identification of endpoints geometrically realizes that character.

Proposition 13.1 (Two closure orders). *One winding of the loop closes the base and reverses the fibre orientation. Two windings restore orientation. In the angular parametrization of the boundary readout, these returns correspond, respectively, to the areal closure 2π and the volumetric closure 4π of the oriented cover.*

Proof. The equivalence relation of (13.1) identifies the terminal fibre with the initial one by $u \mapsto -u$. Composing two transports gives $u \mapsto -(-u) = u$. Parametrizing one winding of the base by 2π assigns 4π to its square. The statement concerns this lift and this readout. $\square$

TPK memory also preserves the trajectory traversed and the register updates. The square of the sign character is the identity, but it does not impose vanishing of these updates. The spinorial representation, when used, must transport this datum through its own map.

13.2 Descent of incidence from the tritic calendar

In the primitive block $(+1, -1, 0)$, the TPK updates the arithmetic mode through $+1 : m \mapsto \Sigma$, $-1 : m \mapsto \Pi$ and $0 : m \mapsto m$. Its cyclic orbit is (Σ, Π, Π) . The sheet readout assigns Σ to the areal sheet and Π to the volumetric sheet. In the basis order $(e_{\text{ar}}, e_{\text{vol}})$, edge incidence is represented by

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (13.2)$$

The three consecutive pairs are $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$ and $\Pi \rightarrow \Sigma$, each with multiplicity one. The calendars of lengths 9, 27 and 108 contain 3, 9 and 36 copies of the primitive block. Their incidence matrices are therefore these multiples of F_{av} .

This construction preserves the distinction between chronological frequency and the measure of the continuation space. The former assigns weights $(1/3, 2/3)$ to the mode cycle. The latter is defined on the minimal memory-one symbolic envelope admitting exactly the preceding pairs. Its adjacency matrix is F_{av} ; trajectory counts belong to that envelope.

Proposition 13.2 (Return weights of the two sheets). *Let $F_0 = 0$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$. For $n \geq 1$,*

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}. \quad (13.3)$$

The Parry measure of this envelope assigns the sheets weights proportional to $(1, \varphi^2)$, and the ratio of diagonal counts converges to φ^{-2} .

Proof. The power formula follows by induction using the recurrence. The square of F_{av} is strictly positive. Its positive Perron vector is $(1, \varphi)$, with eigenvalue φ , since $\varphi^2 = \varphi + 1$. As the matrix is symmetric, the Parry weights are proportional to the squares of that vector's components. The other root of the characteristic polynomial is $-\varphi^{-1}$; its modulus is strictly less than φ , giving the limit of the diagonal ratios. $\square$

Spectral identification of this incidence follows its construction from the calendar. The regional coordinate φ retains its earlier coinductive genealogy: recognition of the eigenvalue does not select the calendar.

13.3 Boundary marking and areal–volumetric ratio

Let $\Pi_{\text{HMT}}(X_\infty)$ be the value of the regional closure readout already constructed. With the diagonal sheet projectors, define

$$D_\pi = \Pi_{\text{HMT}}(X_\infty)P_{\text{ar}} + P_{\text{vol}}, \quad \Theta_n = \frac{\langle e_{\text{ar}}, D_\pi F_{\text{av}}^n e_{\text{ar}} \rangle}{\langle e_{\text{vol}}, F_{\text{av}}^n e_{\text{vol}} \rangle}. \quad (13.4)$$

Volumetric normalization is unity; regional closure marks the areal return only once. Applying (13.3),

$$\Theta_n = \Pi_{\text{HMT}}(X_\infty) \frac{F_{n-1}}{F_{n+1}} \longrightarrow \frac{\pi}{\varphi^2}. \quad (13.5)$$

The return ratio is determined by incidence, measure and marking. The quotient $2\pi/4\pi$, by contrast, describes the cover's two closure orders. Both relations participate in the sheet readout without replacing each other.

13.4 Transport of orientation to the constitutive channels

In a two-channel representation, let $\mathcal{R}$ be a self-adjoint involution and $\mathcal{S}$ a unitary interchange operator satisfying $\mathcal{S}\mathcal{R}\mathcal{S}^{-1} = -\mathcal{R}$. Write the angular coordinates already constructed as $x = A_{\text{rad}}$ and $y = C_{\text{rad}}^*$, with $x > |y|$. Then

$$B = xI + y\mathcal{R}, \quad T = e^{-B}, \quad \mathcal{S}T(y)\mathcal{S}^{-1} = T(-y). \quad (13.6)$$

Indeed, conjugation transforms $B(y)$ into $B(-y)$ and commutes with its convergent exponential series. The eigenvalues $q_\pm = e^{-x \mp y}$ are interchanged. To fix the response order, the chamber $x > y > 0$ is used, where $0 < q_+ < q_- < 1$. The map $(x, y) \mapsto (x, -y)$ transports it to the opposite chamber $x > -y > 0$, where $0 < q_- < q_+ < 1$. Both belong to the contractive domain $x > |y|$. On the boundary $y = 0$ the two channels coincide. Conjugation interchanges the chambers; it does not preserve the strict ordering of their labels. It acts on the projectors through $\mathcal{S}P_\pm\mathcal{S}^{-1} = P_\mp$, with $P_\pm = (I \pm \mathcal{R})/2$. This concrete map transmits orientation to the constitutive operator developed next.

14 Hexad–octad incidence and the oriented functional

14.1 Dodecaphasic incidence and return functional

The Barbero–Immirzi contribution uses angular transport and hexad–octad incidence. Let $H \subset O$ be a hexad and an octad with the hexad’s marked pair preserved. The relevant flag sets are

$$\mathcal{F}_6 = H \times \binom{H}{2} \hookrightarrow \mathcal{F}_8 = O \times \binom{H}{2}, \quad |\mathcal{F}_6| = 90, \quad |\mathcal{F}_8| = 120. \quad (14.1)$$

The cardinalities are $6\binom{6}{2}$ and $8\binom{6}{2}$; the same incidence preserves the normalized defect $(90 - 120)/(24\binom{6}{2}) = -1/12$. This count identifies the two sectors used by return, while the calendar determines their relative multiplicities.

For $m \in \{90, 120\}$, the tritic return on a contractive channel is

$$S_m(q) = \sum_{j \geq 0} q^{m+3mj} = \frac{q^m}{1 - q^{3m}}, \quad 0 < q < 1. \quad (14.2)$$

Each term corresponds to the initial height m and j returns of length $3m$. The geometric series thus preserves both the initial height and return period. The dodecaphasic calendar $\mathbb{Z}/12\mathbb{Z}$ supplies twelve sectors and a unique oriented boundary incidence, denoted ∂_{ret} . In the free module of events, its fundamental chain is

$$c_{12}^+ = \sum_{j \in \mathbb{Z}/12\mathbb{Z}} [j, \mathcal{F}_6] - [\partial_{\text{ret}}, \mathcal{F}_8]. \quad (14.3)$$

The linear readout $\mathcal{L}_q$ assigns $S_{90}(q)$ to each generator $[j, \mathcal{F}_6]$ and $S_{120}(q)$ to the boundary generator. This yields

$$\mathcal{L}_q(c_{12}^+) = 12S_{90}(q) - S_{120}(q). \quad (14.4)$$

The weights 12 and -1 are the chain’s oriented multiplicities. The conjugate orientation satisfies $c_{12}^- = -c_{12}^+$, with the same incidence types.

Proposition 14.1 (Singular coefficient of the fundamental return). *The coefficient of $(1 - q)^{-1}$ in $\mathcal{L}_q(c_{12}^+)$ is $1/24$.*

Proof. For every integer $m > 0$, the factorization $1 - q^{3m} = (1 - q) \sum_{j=0}^{3m-1} q^j$ gives

$$\lim_{q \uparrow 1} (1 - q) S_m(q) = \frac{1}{3m}.$$

Applying this identity to the functional already determined in (14.4) gives

$$\lim_{q \uparrow 1} (1 - q) \mathcal{L}_q(c_{12}^+) = \frac{12}{270} - \frac{1}{360} = \frac{1}{24}. \quad (14.5)$$

In the complex coordinate $q - 1$, the residue is $-1/24$. □

Singular evaluation verifies a consequence of incidence and tritic return. The proof order is fixed by (14.1)–(14.4); the singular coefficient is computed after the chain has been constructed.

15 Constitutive operator and reconstruction of the vacuum channels

The constitutive quantities are obtained by an operation on angular transport. The starting information comprises the channels, their projectors, and the orientation distinguishing them. This organization makes it possible to follow separately the construction of the operator, the evaluation of its spectral functions, and its subsequent dimensional realization.

15.1 Positive response on two sheets

In the oriented chamber $x > y > 0$ of Section 13, let $P_{\pm} = (I \pm \mathcal{R})/2$ and $T = q_+P_+ + q_-P_-$, with $0 < q_+ < q_- < 1$. The incidences constructed by the calendar fix the sectors 90 and 120. The constitutive response is

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1}. \quad (15.1)$$

This definition receives the transport arising from the TPK and preserves the sheet assignments. The response rule and its domain are an explicit part of the construction.

The quotient arises from comparing the two calendar incidences on the same transport, not from fitting four quantities separately. If $S = T^{30}$ and $\Sigma_j(S) = I + S + \dots + S^{j-1}$, factoring $I - S^j$ gives

$$\mathcal{V}\Sigma_4(S) = \Sigma_3(S), \quad \mathcal{V} = \Sigma_3(S)\Sigma_4(S)^{-1}. \quad (15.2)$$

The factors Σ_3 and Σ_4 express, respectively, the sectors $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$. Since the spectrum of S is contained in $(0, 1)$, $\Sigma_4(S)$ is invertible: the completion equation determines a unique response for the fixed transport. This uniqueness concerns the declared constitutive rule and its incidences. Electromagnetic identification adds the oriented assignment of sheets and the dimensional lines specified below.

Proposition 15.1 (Decomposition and positivity). *The response is well-defined and*

$$\mathcal{V} = r_+P_+ + r_-P_-, \quad r_{\pm} = \frac{1 - q_{\pm}^{90}}{1 - q_{\pm}^{120}}. \quad (15.3)$$

Its eigenvalues satisfy $3/4 < r_- < r_+ < 1$ in this chamber.

Proof. Each coefficient $1 - q_{\pm}^{120}$ is positive. The inverse of $I - T^{120}$ is the sum of the projectors divided by those coefficients, proving (15.3). With $s = q^{30}$, one obtains

$$f(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}, \quad f'(s) = -\frac{s^2(3 + 2s + s^2)}{(1 + s + s^2 + s^3)^2} < 0. \quad (15.4)$$

The limits at 0 and 1 are 1 and 3/4, respectively. Monotonicity and the channel ordering complete the proof. $\square$

15.2 Four coordinates of a common response

Define the normalized evaluations

$$\hat{\varepsilon} = r_+^2, \quad \hat{\mu} = r_-^2, \quad \hat{Z} = \frac{r_-}{r_+}, \quad \hat{c} = \frac{1}{r_+r_-}. \quad (15.5)$$

The first pair preserves the quadratic responses assigned to each sheet. The product captures the common mode; the quotient distinguishes their relative orientation. The following hold exactly:

$$\hat{\mu}\hat{\varepsilon} = \hat{c}^{-2}, \quad \frac{\hat{\mu}}{\hat{\varepsilon}} = \hat{Z}^2, \quad r_+ = \frac{1}{\sqrt{\hat{Z}\hat{c}}}, \quad r_- = \sqrt{\frac{\hat{Z}}{\hat{c}}}. \quad (15.6)$$

The positive roots fix the inversion. Under sheet conjugation, read relative to the fixed markings $P_{\pm}$, $\hat{\varepsilon}$ and $\hat{\mu}$ are interchanged, $\hat{Z}$ is inverted, and $\hat{c}$ is preserved. The response is then transported to

the chamber $x > -y > 0$: the formulas remain valid, with the orderings $q_- < q_+$ and $r_+ < r_-$ reversed. Consequently, common mode and orientation are data recoverable through complementary observables.

Proposition 15.2 (Determination of the quadratic evaluations). *Let $W = r_+P_+ + r_-P_-$ be a positive response on the two sheets. With the product and quotient readouts fixed,*

$$\hat{c}^{-1} = \det W = r_+r_-, \quad \hat{Z} = r_-/r_+,$$

there exists a unique positive pair $(\hat{\varepsilon}, \hat{\mu})$ satisfying $\hat{\mu}\hat{\varepsilon} = \hat{c}^{-2}$ and $\hat{\mu}/\hat{\varepsilon} = \hat{Z}^2$. It is precisely (r_+^2, r_-^2) .

Proof. Positivity permits taking roots unambiguously:

$$\hat{\mu} = \frac{\hat{Z}}{\hat{c}} = \frac{r_-}{r_+}r_+r_- = r_-^2, \quad \hat{\varepsilon} = \frac{1}{\hat{Z}\hat{c}} = \frac{r_+}{r_-}r_+r_- = r_+^2.$$

These expressions satisfy both relations and are their only positive roots. $\square$

Thus the squares are not two independent choices added to the channels: they are determined by the multiplicative and oriented readouts once the constitutive relations are fixed. This proposition establishes uniqueness within that system of rules. Interpreting the rules as an electromagnetic response retains its additional physical-representation content.

The determinantal readout of transport and the constitutive response must also preserve their order. For $T = e^{-xI-yR}$, one has $\det T = e^{-2x}$, whereas

$$\det \mathcal{V} = f(q_+^{30})f(q_-^{30}) = \hat{c}^{-1}.$$

Both arise from the same two-sheet operator but evaluate different functions of its spectrum. In general, applying the determinant first and then a rational function is not equivalent to taking the determinant of that function of the operator. Keeping both eigenvalues until the response is completed preserves orientation information that an isolated determinant no longer distinguishes.

15.3 Cubic inversion and angular recovery

Theorem 15.3 (Constitutive inversion in the contractive domain). *For each $r \in (3/4, 1)$, there exists a unique $s \in (0, 1)$ such that $f(s) = r$. This s is the unique root in that interval of*

$$rs^3 + (r-1)(1+s+s^2) = 0. \quad (15.7)$$

The labeled response reconstructs $s_{\pm}$, $q_{\pm} = s_{\pm}^{1/30}$, and

$$x = -\frac{1}{2} \log(q_+q_-), \quad y = \frac{1}{2} \log \frac{q_-}{q_+}. \quad (15.8)$$

Proof. The derivative and limits in (15.4) prove that f is a decreasing bijection between the stated intervals. Multiplying $f(s) = r$ by its denominator produces (15.7). The positive root of order 30 recovers the channel. Finally, adding and subtracting $\log q_{\pm} = -x \mp y$ gives (15.8). $\square$

Reconstruction preserves the sheet labels. The complete operator $\mathcal{V}$ also preserves the projectors, and the inverse function is applied by spectral calculus. Recovering two angular coordinates is an operation defined on this representation; the complete historical TPK trace preserves additional data.

The interval $r \in (3/4, 1)$ and the inverse function of this theorem belong to the internal normalization of (15.5). If coordinates expressed in another unit chart are available, first recover this normalization through the transformation in Section 18; then apply cubic inversion.

15.4 Transported composition of constitutive responses

The angular transport and the incidences 90 and 120 originate in the same APP-TRIT-TPK structure: the sheets retain evaluations and their carries; the tritic regime retains orientation; the TPK calendar transports phase and memory to the angular channels. The constitutive response applies the function f of (15.4) to those channels. Its inversion expresses, in response coordinates, the composition of transports sharing a common frame. The constructive order is thereby preserved: transport, composition and spectral evaluation. The resulting operation acts on this representation of the enriched state; its inverse recovers the channels, while the complete memory history remains in the underlying structure.

Let $s = q^{30}$ and $\psi = f^{-1}$. The exponent 30 is the greatest common divisor of the incidences 90 and 120. Cubic inversion determines $\psi : (3/4, 1) \rightarrow (0, 1)$. Adjoining the boundary response $f(1) = 3/4$ extends the bijection to

$$\psi : D \rightarrow (0, 1], \quad D = [3/4, 1).$$

At this boundary the rational function f is used; the quotient $(1 - q^{90})/(1 - q^{120})$ has a removable singularity there.

Proposition 15.4 (Response monoid and additive coordinate). *The operation*

$$r \odot t = f(\psi(r)\psi(t)), \quad r, t \in D, \quad (15.9)$$

makes D a commutative cancellative monoid with identity $3/4$. The map

$$\ell(r) = -\frac{1}{30} \log \psi(r) \quad (15.10)$$

is an isomorphism from this monoid to $(\mathbb{R}_{\geq 0}, +)$. The interval $(3/4, 1)$ is closed under $\odot$; the identity belongs to the boundary extension. The identity is the only invertible element within D .

Proof. The product of two elements of $(0, 1]$ belongs to the same interval, and

$$\psi(r \odot t) = \psi(r)\psi(t).$$

For three responses, either association has image $\psi(r)\psi(t)\psi(u)$; injectivity of ψ proves associativity. Commutativity follows from scalar multiplication. If $r \odot t = r \odot u$, positivity of $\psi(r)$ permits cancellation, giving $\psi(t) = \psi(u)$ and hence $t = u$. The parameter 1 corresponds to $3/4$ and provides the identity. The logarithm transforms products into sums, and its range on $(0, 1]$ shows that ℓ is a bijection onto $\mathbb{R}_{\geq 0}$. In particular, the response of a channel q satisfies $\ell(r) = -\log q$. If both parameters are strictly less than 1, so is their product. Finally, two numbers in $(0, 1]$ have product 1 only when both are 1. $\square$

Proposition 15.5 (Admissible cancellation and inverse extension). *For $r, u \in D$, the equation $r \odot t = u$ has a solution $t \in D$ exactly when $u \geq r$. This solution is unique and is given by*

$$t = f\left(\frac{\psi(u)}{\psi(r)}\right). \quad (15.11)$$

The same function f is a bijection from $(0, \infty)$ onto $(0, 1)$. With this extension of ψ , the law (15.9) transports the multiplicative group of positive parameters to the interval $(0, 1)$. Group inversion satisfies

$$r^{\ominus} = \psi(r)r, \quad r \odot r^{\ominus} = \frac{3}{4}. \quad (15.12)$$

Proof. The equation is equivalent to $\psi(t) = \psi(u)/\psi(r)$. This quotient belongs to $(0, 1]$ exactly when $\psi(u) \leq \psi(r)$, which, by decreasing monotonicity, is equivalent to $u \geq r$. The bijection proves existence and uniqueness. The derivative in (15.4) is negative for every $s > 0$; the limits of f at 0 and infinity are 1 and 0, respectively. This establishes the extended bijection. Multiplying the numerator and denominator by s^3 gives

$$f(s^{-1}) = \frac{s^3 + s^2 + s}{s^3 + s^2 + s + 1} = sf(s).$$

The multiplicative inverse of $s = \psi(r)$ therefore has response $\psi(r)r$. Its composition with r corresponds to the parameter 1. $\square$

The extension to $s > 1$, equivalently to $q > 1$, is the algebraic extension of the reader. The chapter's physical contractive sector retains its declared domain. The coordinate ℓ extends to an isomorphism with $(\mathbb{R}, +)$ and changes sign under group inversion.

15.4.1 Operator realization in a common sheet frame

Proposition 15.6 (Composition with marked projectors). *Let $\mathcal{R} = \mathcal{R}^*$, $\mathcal{R}^2 = I$ and $P_{\pm} = (I \pm \mathcal{R})/2$. For $j = a, b$, let*

$$T_j = q_{j,+}P_+ + q_{j,-}P_-, \quad \mathcal{V}_j = f(T_j^{30}), \quad 0 < q_{j,\pm} < 1.$$

The composition $T_{a \circ b} = T_a T_b$ satisfies

$$\mathcal{V}_{a \circ b} = f(\psi(\mathcal{V}_a)\psi(\mathcal{V}_b)). \quad (15.13)$$

Its two responses are $r_{a,\pm} \odot r_{b,\pm}$. If $T_j = \exp[-(x_j I + y_j \mathcal{R})]$ with $x_j > |y_j|$, then

$$T_a T_b = \exp[-((x_a + x_b)I + (y_a + y_b)\mathcal{R})], \quad (15.14)$$

and $x_a + x_b > |y_a + y_b|$.

Proof. Orthogonality $P_+ P_- = 0$ gives

$$T_a T_b = q_{a,+}q_{b,+}P_+ + q_{a,-}q_{b,-}P_-.$$

In particular, T_a and T_b commute. Functional calculus on these projectors yields $\psi(\mathcal{V}_j) = T_j^{30}$ and $(T_a T_b)^{30} = T_a^{30} T_b^{30}$; applying f proves (15.13). More explicitly, for every function g defined on the two eigenvalues,

$$P_{\pm} g(T_j) = g(q_{j,\pm}) P_{\pm}.$$

Since $q_{j,\pm} = \exp[-(x_j \pm y_j)]$, their products prove (15.14). The triangle inequality concludes $x_a + x_b > |y_a| + |y_b| \geq |y_a + y_b|$. $\square$

The recovered coordinates can be written directly as

$$x = \frac{\ell(r_+) + \ell(r_-)}{2}, \quad y = \frac{\ell(r_+) - \ell(r_-)}{2}.$$

Composition adds these pairs. Simultaneous sheet exchange permutes the two channels of each transport, acts as $(x, y) \mapsto (x, -y)$ and is an automorphism of the componentwise law. Inversion of both transports in the algebraic extension acts as $(x, y) \mapsto (-x, -y)$. The two involutions commute and retain distinct functions: one exchanges orientation markings; the other inverts the transport.

To represent exchange by conjugation, use an operator L satisfying $L\mathcal{R}L^{-1} = -\mathcal{R}$. In the representation

$$\mathcal{R} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

direct multiplication gives $LP_{\pm}L^{-1} = P_{\mp}$, and hence $LT_jL^{-1} = q_{j,-}P_+ + q_{j,+}P_-$. Conjugation by $\mathcal{R}$ preserves its own projectors. A trace combining the channels likewise cannot replace the projectors in this functional identity: their labels contribute to the composition.

The common-frame hypothesis is substantive. For example, the positive definite operators

$$A_0 = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/3 \end{pmatrix}, \quad B_0 = \begin{pmatrix} 1/2 & 1/6 \\ 1/6 & 1/2 \end{pmatrix}$$

have positive eigenvalues, but

$$A_0 B_0 = \begin{pmatrix} 1/4 & 1/12 \\ 1/18 & 1/6 \end{pmatrix}$$

is not self-adjoint. The proposition specifies transport composition on common projectors; its domain does not encompass an arbitrary mixture of material media. Realizing a concatenation of HMT routes additionally retains their memory data: algebraic closure of the parameter cone describes the representation and does not by itself select new primary routes.

15.4.2 Channel product and transport composition

The normalized speed of one state remains determined by

$$\hat{c} = (r_+ r_-)^{-1}.$$

Here the ordinary product combines the two responses of that same state. The law $\odot$ evaluates a different operation: composition of two transports on each sheet. The following exact calculation distinguishes their arguments:

$$f(1/2) = \frac{14}{15}, \quad \frac{14}{15} \odot \frac{14}{15} = f(1/4) = \frac{84}{85} \neq \frac{196}{225} = \left(\frac{14}{15}\right)^2. \quad (15.15)$$

The fractions in this example are algebraic test parameters. The expression for $\hat{c}$ and the preceding constitutive identities remain unchanged. The added content is an explicit law transporting composition and recovering its additive angular coordinate.

15.5 Logarithmic coordinates and refinement

Let $\mathcal{E} = \log(r_+ r_-)$ and $\mathcal{B} = \log(r_-/r_+)$. Then

$$\log \mathcal{V} = \frac{1}{2}\mathcal{E}I - \frac{1}{2}\mathcal{B}\mathcal{R}. \quad (15.16)$$

This decomposition displays the invariant and oriented components. In the amplification $\mathcal{V}_m = \mathcal{V} \otimes I_{9m}$, normalized partial expectations satisfy $E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m$. For every polynomial p , $p(\mathcal{V}_m) = p(\mathcal{V}) \otimes I_{9m}$; hence $E_m(p(\mathcal{V}_{m+1})) = p(\mathcal{V}_m)$. Uniform continuity extends the identity to continuous functions on the spectrum. Thus projectors, powers, and the logarithm are preserved in the stated tower.

15.6 Dimensional types of the four evaluations

With the action, charge, and velocity lines, equipped with sections u_S, u_Q, u_C , the dimensional realization reads

$$Z_{\text{vac}} = \hat{Z}u_S u_Q^{-2}, \quad c_{\text{vac}} = \hat{c}u_C, \quad (15.17)$$

$$\mu_{\text{vac}} = \hat{\mu}u_S u_C^{-1} u_Q^{-2}, \quad \varepsilon_{\text{vac}} = \hat{\varepsilon}u_S^{-1} u_C^{-1} u_Q^2. \quad (15.18)$$

Multiplying and dividing these expressions preserves $\mu_{\text{vac}}\varepsilon_{\text{vac}} = c_{\text{vac}}^{-2}$ and $\mu_{\text{vac}}/\varepsilon_{\text{vac}} = Z_{\text{vac}}^2$. The SI chart subsequently evaluates the sections. Physical comparison requires presenting their construction and normalization alongside evaluation; the four normalized coordinates in (15.5) retain their dimensionless type.

15.7 Reconstruction of the action shape from the constitutive response

The constitutive response and the inductive–capacitive realization receive the same angular pair constructed from APP–TRIT–TPK. The former evaluates rational functions of the two channels; the latter uses the ratio of their exponents to determine the shape of the action ellipse. Cubic inversion allows the two readouts to be composed without identifying their numerical ratios.

Let $x = A_{\text{rad}}$ and $y = C_{\text{rad}}^*$, with $x > |y|$, be the lifted coordinates already constructed, and preserve their sheet labels. Write

$$q_{\pm} = e^{-(x \pm y)}, \quad s_{\pm} = q_{\pm}^{30}, \quad r_{\pm} = f(s_{\pm}), \quad f(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}. \quad (15.19)$$

The contractive domain gives $s_{\pm} \in (0, 1)$ and $r_{\pm} \in (3/4, 1)$. In the chamber $x > y > 0$, we have $s_+ < s_-$ and $r_+ > r_-$; sheet interchange transports this chamber to the opposite orientation.

The image of the two normalized constitutive coordinates is the domain

$$\mathcal{U} = \{(z, v) \in (0, \infty)^2 : (zv)^{-1/2} \in (3/4, 1), \quad (z/v)^{1/2} \in (3/4, 1)\}. \quad (15.20)$$

Here $(z, v) = (\hat{Z}, \hat{c})$. The restrictions express membership of the reconstructed channels in the domain of the inversion theorem; they do not constitute a subsequent choice of their values.

Proposition 15.7 (Constitutive composition of the shape and LC impedance). *For $(\hat{Z}, \hat{c}) \in \mathcal{U}$, let $s_{\pm} \in (0, 1)$ be the unique roots of*

$$r_{\pm} s_{\pm}^3 + (r_{\pm} - 1)(1 + s_{\pm} + s_{\pm}^2) = 0, \quad r_+ = \frac{1}{\sqrt{\hat{Z}\hat{c}}}, \quad r_- = \sqrt{\frac{\hat{Z}}{\hat{c}}}. \quad (15.21)$$

The angular ratio, the anisotropy of the action ellipse, and the relative LC impedance are reconstructed exactly by

$$\frac{y}{x} = \frac{\log s_+ - \log s_-}{\log s_+ + \log s_-}, \quad (15.22)$$

$$\eta_{\text{el}} = \frac{1}{2} \log \frac{\log s_+}{\log s_-}, \quad (15.23)$$

$$\frac{Z_{\text{LC}}}{Z_*} = e^{-\eta_{\text{el}}} = \sqrt{\frac{\log s_-}{\log s_+}}. \quad (15.24)$$

In the last identity, $Z_{\text{LC}} = Z_{\eta_{\text{el}}}$ is the impedance of the LC chart in (12.7).

Proof. The cubic inversion in (15.7) uniquely determines $s_{\pm}$, because f is a decreasing bijection from $(0, 1)$ onto $(3/4, 1)$. Their positive roots of order 30 recover $q_{\pm}$. The identities

$$\log s_+ = -30(x + y), \quad \log s_- = -30(x - y)$$

prove (15.22) by difference and sum. Both logarithms are negative; their ratio is positive and their sum is nonzero. Therefore,

$$\frac{1}{2} \log \frac{\log s_+}{\log s_-} = \frac{1}{2} \log \frac{x + y}{x - y} = \text{artanh}(y/x) = \eta_{\text{el}}.$$

The relation $Z_{\eta_{\text{el}}}/Z_* = e^{-\eta_{\text{el}}}$, proved in (12.8), gives (15.24). $\square$

The composition preserves the information needed for each reconstruction. With $\hbar$, one recovers $a_S = \sqrt{2\hbar} e^{\eta_{\text{el}}/2}$ and $b_S = \sqrt{2\hbar} e^{-\eta_{\text{el}}/2}$; with the chart (ω, Z_*) , one recovers the inductance and capacitance already defined. Action, clock, and dimensional reference accompany the shape as typed data. Inverting two dimensionless coordinates does not reconstruct their units or the entire memory of the state.

Normalization enters before the cubic equations are solved. If Z_{vac} and c_{vac} are received in a dimensional chart, relations (15.18) first determine

$$\hat{Z} = \frac{Z_{\text{vac}}}{u_S u_Q^{-2}}, \quad \hat{c} = \frac{c_{\text{vac}}}{u_C}.$$

These are the coordinates to which (15.21) applies. The reference Z_* belongs to the LC realization and is explicitly preserved in (15.24).

The two ratios are related through the channels, but evaluate different functions:

$$\hat{Z} = \frac{f(s_-)}{f(s_+)}, \quad \frac{Z_{\text{LC}}}{Z_*} = \sqrt{\frac{\log s_-}{\log s_+}}. \quad (15.25)$$

The preceding reconstruction provides the map between the two readouts. In particular, interchanging the sheets inverts both ratios, changes η_{el} to $-\eta_{\text{el}}$, and preserves $\hat{c}$. In the symmetric limit $y = 0$, the channels coincide and both ratios equal 1. Outside that case, a common origin does not impose equality of the two functions. The action shape is recoverable from the constitutive response while preserving orientation and normalization.

16 Joint reconstruction of gauge couplings, Higgs and the constitutive response

The vacuum response and the electroweak sector admit complementary readings of the angular transport already constructed. The former retains both marked channels; the determinant retains their common contraction; the angular Higgs component also retains the orientation that disappears from that determinant. Composing these readings reconstructs their spectral antecedent and expresses impedance through the scalar self-coupling and the gauge couplings within one and the same tree-level realization.

16.1 Angular provenance and route signatures

The APP-TRIT-TPK chain is preserved: the surviving prolongation determines the route, the route produces its oriented register, and the register reader expresses the angular signature. The coordinates $\pi = \pi_{\text{HMT}}$ and $\alpha = \alpha_{\text{HMT}}$ are preceding outputs of the same generation; A and C^* are their subsequent angular representations. Radians are used in this block:

$$x = \frac{\pi A_{\text{deg}}}{180}, \quad y = \frac{\pi C_{\text{deg}}^*}{180}, \quad A_{\text{deg}} = 1000\alpha, \quad 0 < y < x. \quad (16.1)$$

Thus $x = 50\pi\alpha/9$. The channels and sheet markings are those of Section 13:

$$\begin{aligned} q_+ &= e^{-x-y}, & q_- &= e^{-x+y}, & D &= q_+q_- = e^{-2x}, \\ P_{\pm} &= \frac{I \pm \mathcal{R}}{2}, & T &= q_+P_+ + q_-P_-. \end{aligned} \quad (16.2)$$

Here $\mathcal{R}$ is the marked self-adjoint involution, and each projector has rank one in the basic two-sheet realization.

The duration–persistence angular reader is

$$\chi_{\text{dur}}(n, s) = \exp\left(nx + \frac{s}{6}y\right). \quad (16.3)$$

To preserve the finite provenance of the coefficients used below, we assemble the nominal fingerprints of the three routes of the length-108 schedule. The cardinal-displacement columns are read as (N, E, S, W) ; the following pair retains the sheet and residue expressed by that route. For each identified route, the table preserves the fingerprint and angular signature expressed by the preceding chain.

Sector	Route and axis	(N, E, S, W)	Sheet/residue	(n, s)
W	15/37, E–W	(23, 44, 11, 30)	(41, 8)	(94, –1)
Z	20/23, E–W	(33, 45, 10, 20)	(57, 7)	(95, –1)
H	24/29, N–S	(40, 34, 22, 12)	(47, 10)	(97, 9)

The three typed fingerprints preserve the route identifier, orientation and provenance register of the signatures used. Angular evaluation follows extraction from the enriched register; the table assembles its outputs to compose the characters in this realization. The markings $W_{\pm}$ below denote winding coordinates, distinct from the species name W :

$$\begin{aligned} W_{\pm} &= 6n \pm s, & n &= \frac{W_+ + W_-}{12}, & s &= \frac{W_+ - W_-}{2}, \\ (n, s)_W &= (94, -1), & (W_+, W_-)_W &= (563, 565), \\ (n, s)_Z &= (95, -1), & (W_+, W_-)_Z &= (569, 571), \\ (n, s)_H &= (97, 9), & (W_+, W_-)_H &= (591, 573). \end{aligned} \quad (16.4)$$

The matrix of the first map is $\begin{pmatrix} 6 & 1 \\ 0 & -1 \end{pmatrix}$, with determinant -12 and image $\{(u, v) \in \mathbb{Z}^2 : u + v \in 12\mathbb{Z}\}$. Indeed, the inverse formulas above are integral precisely on this image. The character retains both orientations through

$$\chi_{\text{dur}}(n, s) = \left(q_+^{-W_+} q_-^{-W_-}\right)^{1/12}. \quad (16.5)$$

Taking logarithms proves the identity: the numerator is $W_+(x+y) + W_-(x-y) = 12nx + 2sy$. This identifies the twelfth root and the divisor 6 of the reader.

Proposition 16.1 (Angular ratios determined by the signatures). *On a common scale section, with the shared refinement factors that permit their cancellation in the ratios,*

$$\frac{m_W^<}{m_Z^<} = e^{-x}, \quad \frac{m_H^<}{m_W^<} = e^{3x+5y/3}, \quad \left(\frac{m_H^<}{m_W^<}\right)^2 = e^{6x+10y/3}. \quad (16.6)$$

Proof. Dividing the W and Z characters subtracts their signatures: $(94, -1) - (95, -1) = (-1, 0)$. For H/W , the difference is $(97, 9) - (94, -1) = (3, 10)$. Equation (16.3) gives $3x + (10/6)y$; squaring multiplies this exponent by two. The common scale and shared refinements cancel before these operations. $\square$

These ratios retain the identifier of the angular chart. Application to masses with distinct sectorial refinements retains their additional factors; the cancellation condition is part of the statement.

16.2 Rationalized tree-level realization

The electroweak reading of this chart defines $c_W^2 = D$, $s_W^2 = 1 - D$ and $e_g^2 = 4\pi\alpha$. The positive root fixes $g = e_g/s_W$ and $g' = e_g/c_W$. Hence

$$g^2 = \frac{4\pi\alpha}{1-D}, \quad g'^2 = \frac{4\pi\alpha}{D}. \quad (16.7)$$

The quantity e_g is a dimensionless coupling in the rationalized normalization, distinct from a dimensional charge section. The scalar-potential convention is

$$\begin{aligned} V(H) &= -\mu_H^2 H^\dagger H + \lambda_H (H^\dagger H)^2, \\ m_H^2 &= 2\lambda_H v^2, \quad m_W^2 = \frac{g^2 v^2}{4}. \end{aligned} \quad (16.8)$$

The same v occurs in both relations. Division cancels its scale and yields $m_H^2/m_W^2 = 8\lambda_H/g^2$. Together with (16.6), this gives

$$\boxed{\lambda_H = \frac{g^2}{8} \exp\left(6x + \frac{10}{3}y\right)}. \quad (16.9)$$

The factor 8 arises from the two quadratic normalizations in (16.8); the coefficients 6 and 10/3 arise from the signatures. This distinction preserves the operation producing each factor. The composition receives the preceding HMT outputs and uses this tree-level physical realization at a common scale; a change of field normalization, scale or scheme is also transported in its mass and coupling relations.

16.3 Joint inverse and exact domain

Theorem 16.2 (Angular reconstruction from gauge couplings and Higgs). *For fixed $\pi > 0$, the map defined by (16.2), (16.7) and (16.9), on $\alpha > 0$, $x > y > 0$, is injective and has the inverse*

$$\begin{aligned} D &= \frac{g^2}{g^2 + g'^2}, & x &= -\frac{1}{2} \log D, & \alpha &= \frac{g^2 g'^2}{4\pi(g^2 + g'^2)}, \\ y &= \frac{3}{10} \left[\log\left(\frac{8\lambda_H}{g^2}\right) - 6x \right]. \end{aligned} \quad (16.10)$$

Its exact image consists of the triples $g, g', \lambda_H > 0$ such that, with D and x reconstructed above,

$$\frac{g^2}{8} e^{6x} < \lambda_H < \frac{g^2}{8} e^{28x/3}. \quad (16.11)$$

Proof. The two gauge equations give $g^2/(g^2 + g'^2) = D$ and $g^2 g'^2/(g^2 + g'^2) = 4\pi\alpha$. The real logarithm recovers x , and the scalar equation recovers y . The conditions $g, g' > 0$ imply $0 < D < 1$ and $x > 0$. Under the strictly increasing logarithm, inequality (16.11) is equivalent to $0 < \log(8\lambda_H/g^2) - 6x < 10x/3$, namely $0 < y < x$. Conversely, the reconstructed expressions give $4\pi\alpha/(1 - D) = g^2$, $4\pi\alpha/D = g'^2$ and $g^2 e^{6x+10y/3}/8 = \lambda_H$. Positive roots recover the original couplings. This proves necessity, sufficiency and uniqueness. $\square$

The preceding inversion describes this map between readers exactly. The section generated by HMT also preserves the relation in (16.1), which becomes

$$-\frac{1}{2} \log\left(\frac{g^2}{g^2 + g'^2}\right) = \frac{25}{18} \frac{g^2 g'^2}{g^2 + g'^2}. \quad (16.12)$$

Substituting the inverse expression for α in $x = 50\pi\alpha/9$ proves the identity. The counter-angle likewise retains its preceding return equation. Thus the reader image and the section produced by the generator are distinct objects: the preceding compatibilities select the section inside the reconstructible domain. Inversion acts after generation and checks the outputs that have been retained.

16.4 Higgs-impedance reconstruction

To express the oriented content while preserving the channels, define

$$\rho_{\text{or}} = \frac{q_-}{q_+} = e^{2y}, \quad \Xi_H = \frac{8D^3 \lambda_H}{g^2}. \quad (16.13)$$

Equation (16.9) and $D^3 = e^{-6x}$ give

$$\begin{aligned} \Xi_H &= e^{10y/3} = \rho_{\text{or}}^{5/3}, & \rho_{\text{or}} &= \Xi_H^{3/5}, \\ q_- &= \sqrt{D} \Xi_H^{3/10}, & q_+ &= \sqrt{D} \Xi_H^{-3/10}. \end{aligned} \quad (16.14)$$

The powers are the positive real roots. The exact domain of the oriented chamber is

$$0 < D < 1, \quad 1 < \Xi_H < D^{-5/3}. \quad (16.15)$$

Indeed, $0 < y < x$ is equivalent to $1 < e^{10y/3} < e^{10x/3}$. The lower endpoint gives $q_+ = q_-$; the upper endpoint gives $q_- = 1$. For the two orientations together, $|y| < x$ is equivalent to $D^{5/3} < \Xi_H < D^{-5/3}$.

Let $F(q) = f(q^{30}) = (1 - q^{90})/(1 - q^{120})$ be the response already proved in (15.3)–(15.4).

Corollary 16.3 (Confluence of scalar and constitutive readings). *On the domain (16.15),*

$$\hat{Z} = \frac{F(\sqrt{D} \Xi_H^{3/10})}{F(\sqrt{D} \Xi_H^{-3/10})}. \quad (16.16)$$

The complete responses are

$$\begin{aligned} r_- &= F(\sqrt{D} \Xi_H^{3/10}), & r_+ &= F(\sqrt{D} \Xi_H^{-3/10}), \\ \hat{\mu} &= r_-^2, & \hat{\varepsilon} &= r_+^2, & \hat{c} &= (r_- r_+)^{-1}. \end{aligned} \quad (16.17)$$

The two reconstructions also satisfy

$$\frac{F^{-1}(\sqrt{\hat{\mu}})}{F^{-1}(\sqrt{\hat{\varepsilon}})} = \rho_{\text{or}} = \Xi_H^{3/5}. \quad (16.18)$$

Proof. Substituting the channels of (16.14) into the four readings of (15.5) proves the first identities. The constitutive inverse of (15.7) recovers q_- and q_+ ; their ratio is ρ_{or} . Its equality to $\Xi_H^{3/5}$ was proved in (16.14). $\square$

On the section compatible with $A_{\text{deg}} = 1000\alpha$, the coupling $Q(D) := -9 \log D/25$ satisfies $Q(D) = 4\pi\alpha$ and $g^2 = Q(D)/(1 - D)$. Thus the same Higgs coordinate takes the form

$$\Xi_H = \frac{8(1-D)D^3\lambda_H}{Q(D)}, \quad \frac{1}{g^2} + \frac{1}{g'^2} = \frac{1}{Q(D)}. \quad (16.19)$$

These expressions relate the total coupling, its rationalized partition and the scalar orientation on the same section.

16.5 Retained information and transport of orientation

The gauge pair recovers α and x ; the ratio λ_H/g^2 then recovers y . Together with the markings $P_{\pm}$, it reconstructs the complete operators T and $\mathcal{V} = r_+P_+ + r_-P_-$. The three scalars determine the eigenvalues; the marked projectors also determine their operator realization. The enriched route history remains preceding information of higher resolution.

Under $y \mapsto -y$ relative to the fixed markings, D, g, g' remain invariant and $\Xi_H \mapsto \Xi_H^{-1}$. The constitutive readings are exchanged, $\hat{Z} \mapsto \hat{Z}^{-1}$, and $\hat{c}$ is preserved. If λ_H^+ and λ_H^- denote the evaluations in the two conjugate orientations, then

$$\lambda_H^+ \lambda_H^- = \frac{g^4}{64} D^{-6}. \quad (16.20)$$

Multiplying (16.9) at y and $-y$ proves the identity: the oriented contributions cancel. This comparison retains the signature reader and changes its angular argument. A simultaneous conjugation of routes and reference markings must also transport the signatures.

For the amplification $T_m = T \otimes I_{9^m}$, with $\tau_m = \text{Tr} / (2 \cdot 9^m)$ and $\mathcal{R}_m = \mathcal{R} \otimes I_{9^m}$, the appropriate coordinates are

$$D = \exp(2\tau_m(\log T_m)), \quad \rho_{\text{or}} = \exp(-2\tau_m(\mathcal{R}_m \log T_m)). \quad (16.21)$$

Indeed, $\log T_m = (\log T) \otimes I_{9^m}$, so the two traces are $-x$ and $-y$, respectively. Normalization removes the amplification multiplicity. The gauge, Higgs and vacuum readings are therefore compatible with these refinements; the ordinary amplified determinant would be D^{9^m} and has a different role.

The composition has a precise conclusion: the even gauge invariant and the oriented scalar coordinate jointly reconstruct the spectral pair governing the vacuum. When λ_H is evaluated using the same formula, the confluence is a consistency identity. An independent reading of λ_H , transported to identical scheme and scale, makes the equality an additional test. Both uses retain the signatures, sheet markings and tree-level normalization explicitly.

17 Cyclic decomposition, quarter-turn, and oriented Catalan moment

The dodecaphase record relates the constitutive response to other transport invariants. The link is established on determined subspaces and maps: the rank defect measures an incidence difference, the determinant preserves a scalar response, and a coefficient of the resolvent retains orientation. The collection of moments culminating in Catalan's constant enters at this last level.

17.1 Two subspaces of the dodecaphase cycle

In $\mathbb{R}^{12}$, let $Ce_j = e_{j+1 \bmod 12}$ and let

$$P_3^{\text{cyc}} = \frac{I + C^3 + C^6 + C^9}{4}, \quad P_4^{\text{cyc}} = \frac{I + C^4 + C^8}{3}, \quad H_d = \text{Ran } P_d^{\text{cyc}}. \quad (17.1)$$

The averages over cyclic subgroups are orthogonal projectors. H_3 consists of the periodic sequences whose period divides 3, and H_4 of those whose period divides 4. Their dimensions are 3 and 4. The restriction of C to each space realizes the corresponding cycle. The superscript cyc identifies the average

over the dodecaphase cycle: P_3^{cyc} is the projector onto sequences whose period divides 3. Its domain and action differ from those of the three-dimensional isotypic projector of the icosahedral realization; equality of rank does not identify the two operators.

Proposition 17.1 (Rank defect and determinantal response). *The following identities hold:*

$$\frac{\text{Tr}(P_3^{\text{cyc}} - P_4^{\text{cyc}})}{12} = -\frac{1}{12}, \quad \frac{\det_{H_3}(I - sC)}{\det_{H_4}(I - sC)} = \frac{1 - s^3}{1 - s^4} = f(s), \quad 0 < s < 1. \quad (17.2)$$

Proof. The trace of an orthogonal projector is its rank, so the numerator of the first expression is $3 - 4$. The characteristic polynomial of a cycle of length d is $t^d - 1$; hence its determinant $\det(I - sC)$ is $1 - s^d$. Their quotient yields the second identity. $\square$

The substitution $s = q^{30}$ transports this representation to the $90/120$ quotient. At $s = 1$, the common constant mode is cancelled before taking the limit, which equals $3/4$. This realization of $-1/12$ as a trace defect has an explicit finite domain. The regularized and modular realizations of the same scalar retain their own operators and domains.

17.2 Faithful representation of the quarter-turn

Define the antipodal projector $P_{\text{ant}} = (I - C^6)/2$ and $Q = P_4^{\text{cyc}} P_{\text{ant}}$. This projector acts on the twelve-phase cycle and differs from the constitutive projector P_- in Section 15. The projectors P_4^{cyc} and P_{ant} commute, and Q has rank 2. In its image, fix the orthonormal basis

$$u = \frac{1}{\sqrt{6}}(1, 0, -1, 0, 1, 0, -1, 0, 1, 0, -1, 0)^T, \quad v = Cu. \quad (17.3)$$

Componentwise verification gives $Cu = v$ and $Cv = -u$. Consequently, the isometry $W(a, b) = au + bv$ satisfies

$$CW = WJ, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I. \quad (17.4)$$

The complex structure of the plane is realized by the effective action of the cycle. The sign depends on the chosen orientation: in the same basis, $C^3W = -WJ$. This precision allows the plane to be composed with the oriented readout.

17.3 Resolvent and unlimited depth

Since $C^2 = -I$ on $\text{Ran } Q$, for $0 \leq s < 1$ one has

$$(I - sC)^{-1}|_{\text{Ran } Q} = \frac{I + sC}{1 + s^2}, \quad k_4(s) := \langle v, (I - sC)^{-1}u \rangle = \frac{s}{1 + s^2}. \quad (17.5)$$

The sequence $\langle v, C^n u \rangle$ is $0, 1, 0, -1, \dots$. The depth readout preserves this character through the moment

$$\mathcal{C}_2(s) = \sum_{k \geq 0} \frac{(-1)^k s^{2k+1}}{(2k+1)^2}, \quad 0 \leq s \leq 1. \quad (17.6)$$

The series converges absolutely and uniformly on this interval by comparison with $\sum (2k+1)^{-2}$. For $0 < s < 1$, the differentiated series converge uniformly on every compact subinterval of the interior. With $\mathcal{D} = s d/ds$, one obtains

$$\mathcal{D}^2 \mathcal{C}_2(s) = k_4(s), \quad \mathcal{C}_2(s) = \int_0^s \frac{dt}{t} \int_0^t \frac{d\xi}{\xi} k_4(\xi). \quad (17.7)$$

The conditions $\mathcal{C}_2(0) = 0$ and $\lim_{s \downarrow 0} \mathcal{D} \mathcal{C}_2(s) = 0$ fix the two integration constants. Indeed, the difference between two solutions of $\mathcal{D}^2 F = k_4$ has the form $a \log s + b$; the second condition annuls a and the first annuls b . Uniform convergence permits evaluation at the endpoint $s = 1$: the result is the Catalan moment. The finite representation of the character preserves its orientation; the depth index remains unlimited.

17.4 Composition with the constitutive response

Theorem 17.2 (Constitutive differential relation). *For $0 < s < 1$, with the orientation of (17.3),*

$$f(s) = \frac{1 + s + s^2}{s(1 + s)} \mathcal{D}^2 \mathcal{C}_2(s). \quad (17.8)$$

Proof. Substituting (17.5) into (17.7) and using $(1 + s)(1 + s^2) = 1 + s + s^2 + s^3$ gives the identity. All denominators are positive on the domain. The intertwiner (17.4) identifies the operator coefficient used before its evaluation. $\square$

Upon evaluating $s_{\pm} = q_{\pm}^{30}$, the same collection produces the two channel responses of (15.3). The Catalan limit, the trace defect, and the constitutive eigenvalues retain their distinct functions within this composition. Reversing the cycle with the markings u, v fixed changes the sign of k_4 and preserves the determinant $1 + s^2$. This identifies orientation information that the determinant alone does not distinguish.

17.5 Return traces and moments of the constitutive quotient

The register update and the cycle response share the dodecaphasic transport and preserve different kinds of information. The signed event is obtained from the direction and orientation of each window of the APP-TRIT-TPK history. Its vector increment and balance (2.27) retain the interaction with the previous register. The construction below extends the spectral reading of the same cycle, once its incidence subspaces have been fixed. This order allows return, determinant and moments to be composed while preserving the history that precedes them.

Retain the operator C and projectors of (17.1), and define

$$E_{\text{cyc}} = P_3^{\text{cyc}} - P_4^{\text{cyc}}, \quad b_n = \text{Tr}(C^n E_{\text{cyc}}), \quad n \geq 0. \quad (17.9)$$

The symbol b_n distinguishes this spectral trace from the signed event a_m of a window. The trace uses the transport operator; the event also uses the trajectory actually traversed.

Proposition 17.3 (Trace sequence produced by incidence). *One has*

$$b_n = 3\mathbf{1}_{3|n} - 4\mathbf{1}_{4|n}, \quad (b_1, \dots, b_{12}) = (0, 0, 3, -4, 0, 3, 0, -4, 3, 0, 0, -1). \quad (17.10)$$

The least period is twelve and the sum over a period is zero. For $B_N = \sum_{n=1}^N b_n$,

$$B_N = 3\lfloor N/3 \rfloor - 4\lfloor N/4 \rfloor, \quad |B_N| \leq 3. \quad (17.11)$$

Proof. On H_d , the operator C cyclically permutes the d coordinate classes. Its n th power has d fixed points if d divides n , and none otherwise. Since P_d^{cyc} commutes with C , $\text{Tr}(C^n P_d^{\text{cyc}})$ is the trace of this restriction. Subtracting the cases $d = 3, 4$ proves the formula. The least positive period is twelve: $b_n = -1$ occurs exactly at multiples of twelve. Over one period the contributions are $3 \cdot 4 - 4 \cdot 3 = 0$. The finite sum counts multiples and gives (17.11). Its values for $N = 0, \dots, 11$ are $(0, 0, 0, 3, -1, -1, 2, 2, -2, 1, 1, 1)$ and repeat after adding twelve. $\square$

Theorem 17.4 (Logarithmic reconstruction of the constitutive reader). *For $|s| < 1$, the quotient f of (17.2) satisfies*

$$-\log f(s) = \sum_{n \geq 1} \frac{b_n}{n} s^n, \quad (17.12)$$

$$-\frac{sf'(s)}{f(s)} = \frac{3s^3}{1 - s^3} - \frac{4s^4}{1 - s^4}. \quad (17.13)$$

The holomorphic branch is fixed by $\log f(0) = 0$. For $0 < s < 1$, it agrees with the real logarithm.

Proof. The polynomials $1 - s^3$ and $1 - s^4$ do not vanish in the open disc. The expansion $-\log(1 - w) = \sum_{k \geq 1} w^k/k$, obtained by integrating the geometric series from zero, converges uniformly on compact discs. Substitution of $w = s^3, s^4$ and subtraction give $-\log f(s) = \sum_{k \geq 1} s^{3k}/k - \sum_{k \geq 1} s^{4k}/k$. The coefficient of s^n is b_n/n . Local uniform convergence permits differentiation and summation of the two geometric series, proving (17.13). The condition at zero fixes the integration constant and the branch. $\square$

Thus $b_0/12 = -1/12$ records the normalized dimension defect, whereas the full sequence b_n reconstructs a function. These are readings of different scope on the same pair of subspaces. Upon substituting $s = q_{\pm}^{30}$, the arguments remain outputs of the preceding angular construction; the expansion does not select them.

Theorem 17.5 (Depth moments and Mellin representation). *The series*

$$M_b(z) = \sum_{n \geq 1} \frac{b_n}{n^z} \quad (17.14)$$

converges locally uniformly for $\operatorname{Re} z > 0$. Writing $\sigma = \operatorname{Re} z > 0$, its remainder satisfies

$$\left| M_b(z) - \sum_{n=1}^N \frac{b_n}{n^z} \right| \leq 3 \left(1 + \frac{|z|}{\sigma} \right) (N+1)^{-\sigma}. \quad (17.15)$$

In the domain of absolute convergence $\operatorname{Re} z > 1$,

$$M_b(z) = (3^{1-z} - 4^{1-z})\zeta(z), \quad \zeta(z) = \sum_{n \geq 1} n^{-z}. \quad (17.16)$$

For $\operatorname{Re} z > 0$ one also has

$$\Gamma(z)M_b(z) = \int_0^\infty t^{z-1} \left(\frac{3}{e^{3t}-1} - \frac{4}{e^{4t}-1} \right) dt, \quad (17.17)$$

where $\Gamma(z) = \int_0^\infty u^{z-1} e^{-u} du$.

Proof. Summation by parts and $|B_n| \leq 3$ give, after taking the upper limit,

$$\sum_{n > N} b_n n^{-z} = -B_N(N+1)^{-z} + \sum_{n > N} B_n(n^{-z} - (n+1)^{-z}).$$

The modulus of the difference of powers is bounded by $|z| \int_n^{n+1} x^{-\sigma-1} dx$. This proves (17.15), convergence and uniformity on compact subsets. For $\sigma > 1$, absolute convergence permits multiples of three and four to be grouped separately, giving (17.16).

The geometric series of the transport, evaluated at $s = e^{-t}$, produces

$$\sum_{n \geq 1} b_n e^{-nt} = \frac{3}{e^{3t}-1} - \frac{4}{e^{4t}-1} =: K_b(t).$$

For $\sigma > 1$, termwise integration is justified by $\sum |b_n| n^{-\sigma} < \infty$. Substitution of $u = nt$ gives $\Gamma(z)n^{-z}$ and proves (17.17) on that half-plane. At the origin the $1/t$ terms cancel: $K_b(t) = 1/2 + O(t)$. At infinity, $K_b(t) = O(e^{-3t})$. The integral is therefore holomorphic for $\operatorname{Re} z > 0$ by local uniform convergence. The series is also holomorphic there by the preceding bound; the identity theorem extends equality to that domain. $\square$

For $0 < \operatorname{Re} z \leq 1$, increasing depth fixes the convergent ordering of the series. Its realization as the absolute trace of the diagonal operator applies for $\operatorname{Re} z > 1$.

Corollary 17.6 (Evaluations compatible with the constitutive endpoint). *One has*

$$M_b(2) = \frac{\zeta(2)}{12}, \quad M_b(1) = \log(4/3) = -\log f(1), \quad f(1) = \frac{3}{4}. \quad (17.18)$$

Proof. The first equality follows from $3^{-1} - 4^{-1} = 1/12$. For the second, the exact partial sums are $H_{[N/3]} - H_{[N/4]}$, with $H_j = \sum_{k=1}^j 1/k$ and $H_0 = 0$. Integral comparison for $1/x$ between these two indices gives the limit $\log(4/3)$. Cancelling $1 - s$ before evaluating the determinantal quotient gives $f(1) = 3/4$. $\square$

The affine register retains events and their interaction with previous memory; the trace sequence retains the incidence of the two subspaces at every power of transport; the moment applies a depth weighting to that already-produced sequence. Its spectral periodicity coexists with an event history that continues after phase return. Replacing C by its inverse preserves b_n , since $d \mid n$ is invariant under $n \mapsto -n$; the oriented resolvent coefficient in (17.5), with fixed marks, changes sign. The joint reading preserves a distinction not recorded by the scalar determinant alone. The quadratic form and coordinate sum of the register retain their mathematical types; this composition assigns them neither energy units nor electric-charge units.

17.6 Residual composition and products of moments

The subsequent composition of readers also retains the operation performed on depth. In the realization $\mathcal{H} = \ell^2(\mathbb{N}_{>0})$, set $N\delta_n = n\delta_n$. Let a, c be two already-produced periodic residual tables and let $Q_a\delta_n = a(n)\delta_n$, $Q_c\delta_n = c(n)\delta_n$. After extension to a common period, the two readers act on the same index. For $\text{Re } z > 1$, write $M_a(z) = \text{Tr}(N^{-z}Q_a)$ and similarly for M_c .

Proposition 17.7 (Diagonal composition and tensor composition). *The following identities hold:*

$$\text{Tr}(N^{-z}Q_aQ_c) = \sum_{n \geq 1} \frac{a(n)c(n)}{n^z}, \quad (17.19)$$

$$\begin{aligned} M_a(z)M_c(z) &= \text{Tr}_{\mathcal{H} \otimes \mathcal{H}}((N \otimes N)^{-z}(Q_a \otimes Q_c)) \\ &= \sum_{k \geq 1} \frac{(a * c)(k)}{k^z}, \end{aligned} \quad (17.20)$$

$$(a * c)(k) = \sum_{d \mid k} a(d)c(k/d), \quad \text{Re } z > 1. \quad (17.21)$$

Every operator appearing inside these traces is trace class.

Proof. The diagonal product acts on δ_n by $a(n)c(n)$, proving (17.19). On $\delta_m \otimes \delta_n$, the operator $N \otimes N$ has eigenvalue mn , and $Q_a \otimes Q_c$ acts by $a(m)c(n)$. If $\sigma = \text{Re } z > 1$, the sum of the moduli of these spectral coefficients is bounded by $\|a\|_\infty \|c\|_\infty (\sum n^{-\sigma})^2 < \infty$. This bound establishes the trace-class condition and permits the double sum to be factorized or grouped by $k = mn$. Pairs with product k are exactly $(d, k/d)$ with $d \mid k$. The analogous bound with a single sum proves the diagonal case. $\square$

For the readers in this section, taking $a(n) = b_n$ and $c(n) = \langle v, C^n u \rangle$ gives, at $n = 3$, $a(3)c(3) = -3$, whereas $(a * c)(3) = 3$. This coefficient difference exhibits the information retained by each operation. Diagonal composition retains a common depth; tensor composition retains two depths before grouping by their product. The tensor realization does not ascribe statistical independence to the HMT histories that produced the readers. Regularized derivatives and normalization changes of other moments are subsequent operations distinct from the two compositions proved here.

17.7 Factorization through the constitutive operator

Retain the lifted angular chart of Section 13. To make the angular conversion explicit, write

$$A_{\text{deg}} = \frac{180x}{\pi}, \quad C_{\text{deg}}^* = \frac{180y}{\pi}, \quad q_{\pm} = e^{-x \mp y}, \quad s_{\pm} = q_{\pm}^{30}. \quad (17.22)$$

The orientation considered in (15.3) corresponds to $0 < y < x$; sheet interchange is expressed by $y \mapsto -y$ in the larger domain $x > |y|$. Real exponentials act on these lifts, which preserve the values of $x \pm y$ during composition.

The oriented Barbero–Immirzi functional is

$$\gamma = \frac{\pi A_{\text{deg}} C_{\text{deg}}^*}{180} + 12[S_{90}(q_-) - S_{90}(q_+)] - [S_{120}(q_-) - S_{120}(q_+)]. \quad (17.23)$$

The first component is the angular evaluation; the second is the difference of the fundamental-chain readout on the conjugate channels. The arguments are received from the previously constructed angular transport. The formula determines a dimensionless scalar on that starting structure.

Let $\psi : (3/4, 1) \rightarrow (0, 1)$ be the inverse of f constructed through (15.7). Define

$$\mathcal{H}(s) = \frac{12s^3}{1-s^9} - \frac{s^4}{1-s^{12}} - \frac{(\log s)^2}{20\pi}, \quad 0 < s < 1. \quad (17.24)$$

This function is real analytic on its domain. The exponents 3, 4, 9, 12 result from expressing heights 90, 120 and returns 270, 360 in units of 30; the logarithmic coefficient is fixed when the angular contribution is transported.

Theorem 17.8 (Oriented constitutive expression of Barbero–Immirzi). *In chart* (17.22),

$$\gamma = \mathcal{H}(s_-) - \mathcal{H}(s_+) = \mathcal{H}(\psi(r_-)) - \mathcal{H}(\psi(r_+)). \quad (17.25)$$

On the two-dimensional channel representation,

$$\gamma = -\text{Tr}[\mathcal{R} \mathcal{H}(\psi(\mathcal{V}))], \quad (17.26)$$

where Tr is the ordinary trace, with $\text{Tr } I = 2$.

Proof. Substitution of $s = q^{30}$ into (14.2) gives

$$S_{90}(q) = \frac{s^3}{1-s^9}, \quad S_{120}(q) = \frac{s^4}{1-s^{12}}.$$

Moreover, (17.22) implies

$$\log s_{\pm} = -30(x \pm y) = -\frac{\pi}{6}(A_{\text{deg}} \pm C_{\text{deg}}^*).$$

By the difference of squares,

$$\frac{(\log s_+)^2 - (\log s_-)^2}{20\pi} = \frac{900[(x+y)^2 - (x-y)^2]}{20\pi} \quad (17.27)$$

$$= \frac{180xy}{\pi} = \frac{\pi A_{\text{deg}} C_{\text{deg}}^*}{180}. \quad (17.28)$$

Adding these identities proves the first equality in (17.25). The second uses the inversion $\psi(f(s_{\pm})) = s_{\pm}$ proved in Section 15.

For the operator expression, the sheet projectors have rank one and $\mathcal{R}P_{\pm} = \pm P_{\pm}$. Spectral calculus gives

$$\mathcal{H}(\psi(\mathcal{V})) = \mathcal{H}(s_+)P_+ + \mathcal{H}(s_-)P_-.$$

Multiplying by $\mathcal{R}$ and taking the ordinary trace gives $\mathcal{H}(s_+) - \mathcal{H}(s_-)$. Its negative is γ . $\square$

The orientation mark $\mathcal{R}$ belongs to the functional. If that mark is held fixed and the channels of $\mathcal{V}$ are interchanged, the sign of γ changes. Evaluation of the spectrum without its labels preserves the unordered pair, whereas (17.26) preserves the oriented contribution. If the normalized trace $\tau_2 = \text{Tr}/2$ is used, the same formula reads $\gamma = -2\tau_2[\mathcal{R}\mathcal{H}(\psi(\mathcal{V}))]$.

By contrast, a simultaneous change of representation $(\mathcal{R}, \mathcal{V}) \mapsto (U\mathcal{R}U^{-1}, U\mathcal{V}U^{-1})$ preserves the functional: spectral calculus is transported by conjugation and the trace is invariant. Interchanging the response relative to a fixed orientation and jointly transporting response and orientation are therefore different operations.

This normalization is compatible with the amplifications of (15.16). For $\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$ and $\mathcal{R}_m = \mathcal{R} \otimes I_{9^m}$,

$$\gamma = -\frac{1}{9^m} \text{Tr} [\mathcal{R}_m \mathcal{H}(\psi(\mathcal{V}_m))]. \quad (17.29)$$

Indeed, functional calculus preserves the tensor product and the trace of I_{9^m} is 9^m . Representation multiplicity is separated from the oriented evaluation.

17.8 Reconstruction from vacuum coordinates

Identities (15.6) express the same functional through normalized impedance and velocity:

$$\gamma = \mathcal{H}\left(\psi\left(\sqrt{\frac{\hat{Z}}{\hat{c}}}\right)\right) - \mathcal{H}\left(\psi\left(\frac{1}{\sqrt{\hat{Z}\hat{c}}}\right)\right). \quad (17.30)$$

The positive domain of this reconstruction is described by

$$1 < \hat{Z}\hat{c} < \frac{16}{9}, \quad \frac{9}{16} < \frac{\hat{Z}}{\hat{c}} < 1. \quad (17.31)$$

Indeed, these inequalities are equivalent to $3/4 < r_{\pm} < 1$ after substitution of the positive roots of (15.6). For orientation $y > 0$, one also retains $r_- < r_+$, equivalently $\hat{Z} < 1$. Domain points are used with the angular compatibility already established by the TPK; inversion is a subsequent reconstruction of its channels. The coordinates are the internal ones of (15.5): a change of units must be undone through (18.1) before ψ is applied.

Interchange $y \mapsto -y$ acts as $(\hat{Z}, \hat{c}) \mapsto (\hat{Z}^{-1}, \hat{c})$ and produces $\gamma \mapsto -\gamma$. In the balanced case $y = 0$, the channels coincide and (17.25) gives $\gamma = 0$. Normalized velocity preserves the product of the channels; its joint readout with impedance recovers both responses and the orientation required to evaluate the functional. In this balanced case $\mathcal{V} = rI$ has degenerate spectrum: the mark $\mathcal{R}$ is retained as representation data, although the scalar r no longer distinguishes its projectors. For simple spectrum, by contrast, $P_+ = (\mathcal{V} - r_- I)/(r_+ - r_-)$ and $P_- = I - P_+$.

18 Constitutive evaluation and dimensional realization

Evaluation now brings together the preceding constructions in their effective order. The regional expansions of π , φ , and e arise from nonadic refinement; the register K preserves its twelve positions. Its periodic readout determines the positional section of α . The action character and regional return act on it, followed by the angular pair and, finally, the constitutive response. This evaluation follows the selection of rules: none of its digits is used to choose a channel.

18.1 Evaluation of the internal chain

To reproduce the table, one uses the thousand-decimal regional expansions generated by the nonadic certificate and the integer N_K in (8.18). The readout used is

$$\alpha = \pi + e - \varphi - 4 - \frac{N_K}{1000^{12} - 1}.$$

The finite twelve-block section and the prolonged section have their definitions and exact difference in the register chapter. Here the latter is held fixed throughout evaluation. Readouts are not interchanged as precision increases, nor is the analytic operator of the second route replaced by a decimal coincidence with the first.

Object	Decimal evaluation (24 significant digits)
α	0.00729735256928380099728511 ...
H_5	1.05457181764615446962582 ...
η_{ret}	1.05457181691503906113369 ...
A_{deg}	7.29735256928380099728511 ...
C_{deg}^*	2.12373833896864572426195 ...
q_+	0.848377942576404374945737 ...
q_-	0.913660151150557285295519 ...
r_+	0.999999628548249363178695 ...
r_-	0.999724137189518364131517 ...
$\hat{\varepsilon}$	0.999999257096636702760442 ...
$\hat{\mu}$	0.999448350479326935089982 ...
$\hat{Z}$	0.999724508538937215455554 ...
$\hat{c}$	1.00027631048615752610639 ...
γ	0.274007672694459274747255 ...
G_{Cat}	0.915965594177219015054604 ...
ρ	0.267949192431122706472554 ...

The calculation uses 120 working digits; 24 significant digits are displayed to facilitate comparison between quantities of different orders. The 120 digits are an evaluation precision, not an experimental uncertainty or, by themselves, an interval error certificate. The package preserves the generated inputs, their hashes, the operations, and the residuals of the checked identities. The algebraic proofs in the main text establish these identities independently of this table.

The meaning of the final four constitutive coordinates is preserved under evaluation: $\hat{\varepsilon}$ and $\hat{\mu}$ are the quadratic responses per sheet, $\hat{Z}$ is their relative orientation, and $\hat{c}$ is the inverse of their product. The proximity of some digits to unity expresses the small defect of the 3–4 quotient on these channels. Its origin is seen exactly in

$$1 - f(s) = \frac{s^3}{1 + s + s^2 + s^3}.$$

The formula allows the response to be studied before a dimensional quantity is assigned to it: the sign of the defect, its cubic order, and its monotonicity are consequences of the same operator.

18.2 Unit charts and covariance

Let $\mathcal{L}_S$, $\mathcal{L}_Q$, and $\mathcal{L}_C$ be the action, charge, and velocity lines. A positive physical chart evaluates the sections u_S , u_Q , and u_C in action, charge, and velocity units. The realization (15.18) then produces an impedance, a velocity, a permeability, and a permittivity, with their respective dimensions. The coordinates in the preceding table remain dimensionless.

Proposition 18.1 (Covariance of the constitutive realization). *If the reference sections change by $u'_S = au_S$, $u'_Q = bu_Q$, $u'_C = du_C$, with $a, b, d > 0$, the coordinates representing the same quantities satisfy*

$$\hat{Z}' = a^{-1}b^2\hat{Z}, \quad \hat{c}' = d^{-1}\hat{c}, \quad \hat{\mu}' = a^{-1}db^2\hat{\mu}, \quad \hat{\varepsilon}' = adb^{-2}\hat{\varepsilon}.$$

The relations $\hat{\mu}'\hat{\varepsilon}' = (\hat{c}')^{-2}$ and $\hat{\mu}'/\hat{\varepsilon}' = (\hat{Z}')^2$ are preserved.

Proof. Equate each quantity in (15.18) to its expression in the new basis and solve for its coordinate. The factors in the product are d^2 , and those in the quotient are $a^{-2}b^4$, exactly those required by the transformed coordinates of c and Z . $\square$

This covariance distinguishes two acts: constructing a section and expressing that section in chosen units. The spectral theorem fixes the coordinates in the declared internal normalization. An SI

comparison must also exhibit the evaluation of the three reference sections. Fitting them using the quantities to be reproduced would calibrate the chart, not independently verify those quantities.

To reconstruct the channels from coordinates expressed in the primed chart, undo that change before using the cubic root:

$$\hat{Z} = ab^{-2}\hat{Z}', \quad \hat{c} = d\hat{c}', \quad \hat{\mu} = ad^{-1}b^{-2}\hat{\mu}', \quad \hat{\varepsilon} = a^{-1}d^{-1}b^2\hat{\varepsilon}'. \quad (18.1)$$

For example, the inputs to the spectral inverse are $r_+ = (\hat{Z}\hat{c})^{-1/2}$ and $r_- = (\hat{Z}/\hat{c})^{1/2}$, calculated after (18.1). Directly inserting the primed coordinates into those expressions would change the operator normalization and could take them outside the interval $(3/4, 1)$. Dimensional covariance preserves physical quantities; cubic inversion reconstructs the transport in the internal chart in which it was defined.

19 Generated coordinates, action, and electronic normalisation

The construction of the collections uses results from the preceding stages: the regional characters, the dodecaphase record, the action line, and the electronic section. We assemble here the operations connecting them and the proofs needed to compose them. Their order is causal: the regions and their records are produced on APP through TRIT and TPK; real evaluations are performed afterwards; the choice of units enters when dimensional magnitudes are realised.

Initial conditions and regional selection were developed in the section preceding the arithmetic of histories. The domain of the incidence ledger, its counts, and the reconstruction of the record are defined explicitly below. Selecting a specific collection of visits and traces is an additional operation beyond evaluating a ledger: it requires specifying its multiplicities, boundary transport, and coverage of the admitted traces. The reconstruction formulas proved here allow that record to be evaluated and checked; their invertibility does not by itself select the incidence collection.

19.1 Regional readers and compatibility of evaluation

Closure, propagation, and self-scaling are three regions of the same generation. The closure region has five emissions and multiplicity $432 + 4 \cdot 144 = 1008$. In the oriented gauge of the catalogue, its emissions are

(501, 614, 498, 169, 272, 272)	432
(810, 923, 870, 169, 272, 272)	144
(810, 983, 810, 169, 272, 272)	144
(870, 923, 810, 169, 272, 272)	144
(870, 923, 810, 418, 674, 521)	144.

The common ternary word is 010211; the representatives of propagation and self-scaling are 201101 and 121200. The census and orientation precede the evaluations that follow.

At the five positions of the closure reader, the uniform projection $P_5 = \mathbf{11}^T/5$, together with $P_5\lambda = \lambda$ and $\mathbf{1}^T\lambda = 1$, determines $\lambda = \mathbf{1}/5$. This normalisation of the reader positions differs from the seed distribution given in the table. In the elliptic regime $J^2 = -I$, multiplication gives

$$(I + aJ)(I + bJ) = (1 - ab)I + (a + b)J.$$

Whenever $1 - ab \neq 0$, the composite slope is $a \oplus b = (a + b)/(1 - ab)$. Two doublings of $1/5$ give $5/12$ and $120/119$. The positive compensator $1/q$, $q > 1$, that returns the slope to one satisfies

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1, \quad q = 239.$$

Thus the closure character of this reader is

$$\Pi_{cl} = 4 \left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \right). \quad (19.1)$$

The composition rule, its compensation, and the choice of angular branch fix this value; the conventional angular identity recognises it as π .

The propagation character is realised in a commutative formal collection P_t , normalised by $P_0 = 1$, $P'_0 = 1$, and $P_{t+s} = P_t P_s$. If $P_t = \sum_{n \geq 0} a_n t^n$, comparison of the coefficient of $s^n t$ gives

$$(n+1)a_{n+1} = a_n, \quad a_0 = 1.$$

Hence $a_n = 1/n!$; representation at one unit of the parameter is $\Pi_{\text{pr}} = \sum_{n \geq 0} 1/n!$, recognised as e . The unit and the sign of the generator belong to the reader: the semigroup law alone would admit other generators.

For self-scaling, TRIT selects the modes (Σ, Π, Π) . Their three cyclic transitions are $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Their incidence matrix, in areal–volumetric order, is

$$F = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (19.2)$$

The positive eigenvector normalised as $(1, x)^T$ satisfies $x = \lambda$, $1 + x = \lambda x$; hence $x^2 - x - 1 = 0$. Its positive root is the self-scaling coordinate φ . This equation recognises the character of the incidence already produced by the calendar.

Proposition 19.1 (Regional evaluation through compatible intervals). *The three preceding readers admit rational intervals of arbitrarily small width. Their finite decimal prefixes are determined by refinement and are compatible under truncation.*

Proof. For $0 < x < 1$, two consecutive alternating partial sums of $\sum_{j \geq 0} (-1)^j x^{2j+1}/(2j+1)$ enclose $\arctan x$. The first omitted term bounds the error. This provides rational intervals for (19.1). If $E_n = \sum_{j=0}^n 1/j!$, then, for $n \geq 1$,

$$E_n < \Pi_{\text{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

The bound follows by bounding successive ratios of the tail by $1/(n+2)$. Finally, F^n has entries F_{n-1}, F_n, F_{n+1} , where $F_0 = 0, F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$. The identity $F_{n+1}^2 - F_n F_{n+2} = (-1)^n$ shows that two consecutive ratios enclose φ with width $1/(F_n F_{n+1})$. Successive intervals may be intersected to preserve nestedness. After conventional recognition, the irrationality of π, e, φ ensures that no coordinate lies on a rational decimal boundary. At each finite depth, a sufficiently narrow interval lies in a unique decimal cell. Its index determines the prefix, and nestedness proves compatibility. $\square$

19.2 Signed dodecaphase chart of the full TPK state

The evolution producing the record is formulated on the complete state of the Topological Phase Kernel. Denote by TPKFullState the state space jointly preserving APP position, TRIT regime, route, sheet, orientation, carry, boundary, memory, incidence ledger, and terminal coordinates. In a canonical chart,

$$X = (X_{\text{APP}}, X_{\text{TRIT}}, X_{\text{ruta}}, X_{\text{mem}}, C_{\text{term}}, U_{12}^{\text{sgn}}, X_{\partial}, \dots).$$

The dynamics of one hundred and eight steps acts through the composition

$$X_{\text{term}} = (\mathcal{U}_{108} \circ \dots \circ \mathcal{U}_1)(X_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t, \quad (19.3)$$

where Sel_t applies the tritic phase selector, Tra_t transports sheet, route, and orientation, and Upd_t incorporates the carry and memory produced at instant t . Each factor therefore acts on the full state, and the composition preserves the genealogy of the twelve nonadic windows.

The incidence ledger of the following section is the coordinate $\mathcal{R}_{12}(X_{\text{term}})$. Its evaluation by the 90/120 channels defines

$$\varepsilon_{108}^{90,120} \circ \mathcal{R}_{12} : \text{TPKFullState} \longrightarrow \mathbb{Z}^{25}, \quad X \longmapsto C(X) = (b^{90}(X), b^{120}(X), Q(X)).$$

The signed dodecaphase projection is

$$\mathcal{S}_{12} := \text{pr}_{U_{12}^{\text{sgn}}} : \text{TPKFullState} \longrightarrow \mathbb{Z}^{12}.$$

On the compatible sublattice produced by the dynamics, the two charts satisfy the operator identity

$$\mathcal{S}_{12} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} = \Pi_H \circ \text{pr}_C. \quad (19.4)$$

Thus the dodecaphase signature is obtained by projection and evaluation of the same state that preserves the ledger; its domain includes the orientation and memory entering each count. The raw projection $\rho_{108} : \text{TPKFullState} \rightarrow \text{CT108RawProjection}$, by contrast, denotes the chart that forgets those coordinates after production. This separation of types fixes the causal order between generation, signed representation, and reduced reading.

19.3 From temporal incidences to the dodecaphase record

Definition 19.2 (Domain of the oriented incidence ledger). A ledger $\mathcal{L}$ contains a nonempty finite collection of routes, their identified visits, and a finite collection of traces. Each route preserves one visit per instant during the first 108 instants, and the later visits used by its traces. A visit preserves

$$v = (\gamma, t, i, j, s, \tau, w, \varepsilon_{\partial}),$$

where γ identifies the route, t its instant, $1 \leq i, j \leq 9$ its APP position, $s \in \{\Sigma, \Pi, 0\}$ its sheet, $\tau \in \{-1, 0, 1\}$ its tritic orientation, and w its positive integer multiplicity. The incidence chart evaluates

$$n_v = \begin{cases} i + j, & s = \Sigma, \\ ij, & s = \Pi, \\ 9, & s = 0. \end{cases} \quad r_v = 1 + ((n_v - 1) \bmod 9), \quad q_v = (n_v - r_v)/9.$$

The reading $n_v = 9$ fixes the threshold convention of this chart. For $r_v = 9$, the label ε_{∂} distinguishes corona from interior; it is not obtained from the remainder alone.

A trace $\theta = (v_1, \dots, v_N)$ traverses consecutive instants of the same route. It preserves a multiplicity $u(\theta) \in \mathbb{N}_{>0}$ and a reduced length unit $\lambda(\theta) \in \mathbb{N}_{>0}$, with

$$N\lambda(\theta) = 120(1 + 3j(\theta)), \quad j(\theta) \in \mathbb{N}_0.$$

Its channel is $\sigma(\theta) = \text{sgn}(\sum_{a=1}^N \tau(v_a))$. The collection counted in the 120 channels comprises traces of nonzero sign. Reduced length and number of instants are related through λ ; they are not identified with one another.

The definition specifies the evaluation domain, including the data that must be transmitted to the reader. Its multiplicities and trace collection belong to the dynamic selection of the ledger. An enumeration of the domain or a format check does not replace that selection.

Generation preserves a ledger $\mathcal{L}$ of visits and traces. The temporal chart divides it into windows $E_m = \{9m - 8, \dots, 9m\}$, $1 \leq m \leq 12$, equivalent to positions $m = \beta + 1$ of the record $\mathcal{R}_{12}$. Each window retains its subroute, orientation, boundary increment, and local incidences. Each visit records APP position, arithmetic sheet, instant, multiplicity, and boundary orientation. Each evaluation n_v preserves the division $n_v = r_v + 9q_v$, with $1 \leq r_v \leq 9$. In window m , the incidence reader expresses

$$\begin{aligned} A_m &= \sum_{v \in \mathcal{L}_m} w(v) \mathbf{1}_{\{1,3,5,7\}}(r_v), \\ C_m &= \sum_{j \geq 0} (\nu_{120,m}^-(j) - \nu_{120,m}^+(j)), \\ V_m &= \sum_{v \in \mathcal{L}_m} w(v) \mathbf{1}_{\{9\}}(r_v) \varepsilon_{\partial}(v). \end{aligned} \quad (19.5)$$

Here the traces in each window have finite support, $\varepsilon_\partial = +1$ on the corona and -1 in the interior, and the labels $\pm, 120$ belong to the preserved transport. The multiplicity w and the traces ν are those obtained from that transport; they are not reconstructed by selecting a value of α .

In terms of the identified traces, the second sum has the finite expression

$$C_m = - \sum_{\theta: t(v_1) \in E_m} u(\theta) \sigma(\theta).$$

Indeed, grouping by $j(\theta)$ and by sign produces $\nu_{120,m}^-(j) - \nu_{120,m}^+(j)$. This equality specifies the operation actually applied to each record; it keeps channel, multiplicity, and length separate.

Proposition 19.3 (Updating and concatenation of the ledger). *Once the collection of visits and traces admitted by transport is fixed, its counts are updated through integer increments. When a visit v in window m is added, the increment of (A_m, C_m, V_m) is*

$$(w(v) \mathbf{1}_{\{1,3,5,7\}}(r_v), 0, w(v) \mathbf{1}_{\{9\}}(r_v) \varepsilon_\partial(v)).$$

When a trace θ whose origin belongs to E_m is completed, the increment is $(0, -u(\theta) \sigma(\theta), 0)$. For two consecutive segments, the count of their concatenation is the sum of the counts of the visits and traces contained in each segment, plus that of the traces crossing their junction.

Proof. Each sum in (19.5) is indexed by visit or trace identifiers, so adding an identifier adds exactly the summand displayed. For concatenation, visits are distributed between the two segments. Traces are partitioned into those contained in the first, those contained in the second, and those crossing the junction. Additivity of integer sums proves the formula. Preserving the identifiers and pending prolongations prevents a trace from being counted twice or one that has not yet ended from being lost. Decimal reduction is applied after counting; the quotients of that reduction remain in memory. $\square$

Decimal reduction of these three counts produces

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}, \quad 0 \leq z_m < 1000. \quad (19.6)$$

Let $(D_3 z)_i = z_i - z_{i+3}$, $(D_4 z)_i = z_i - z_{i+4}$, with indices in $\mathbb{Z}/12\mathbb{Z}$, and $Q(z) = \sum_i z_i$. The reader preserves $\mathcal{D}(z) = (D_3 z, D_4 z, Q(z))$. Decimal reduction forgets multiples of ten in each count, whereas the original ledger preserves the visits and transports that generated them. The reading of even-code events on that same history, its decomposition $12 = 1 + 5 + 6$, and its integral inversion are developed in Proposition 22.1. That event vector does not replace the present incidence record: the domains and preserved coordinates remain distinct.

Proposition 19.4 (Incidence conjugation and decimal reduction). *Let $\varphi(m) = 13 - m$. Consider a conjugation of the ledger that maps the visits and traces of window m to those of $\varphi(m)$ through bijections preserving their multiplicities. Suppose it preserves the arithmetic class and boundary label of the visits, and interchanges the positive and negative channels of the traces. Then the counts in (19.5) satisfy*

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\varphi(m)}, -C_{\varphi(m)}, V_{\varphi(m)}). \quad (19.7)$$

Writing $u_m = [C_m]_{10}$, their decimal blocks satisfy

$$z_m^\dagger = z_{\varphi(m)} + 100 \mathbf{1}_{\{u_{\varphi(m)} \neq 0\}} - 20 u_{\varphi(m)}. \quad (19.8)$$

Proof. The bijections preserve every summand of A_m and V_m . Interchanging the channels permutes ν^- and ν^+ , thereby reversing the sign of C_m . This yields (19.7). If $C = 10q + u$, $0 \leq u < 10$, Euclidean division of $-C$ gives

$$[-C]_{10} = 10 \mathbf{1}_{\{u \neq 0\}} - u, \quad \left\lfloor \frac{-C}{10} \right\rfloor = -q - \mathbf{1}_{\{u \neq 0\}}.$$

In (19.6), only the middle digit changes. Its difference is $10([-C]_{10} - [C]_{10}) = 100 \mathbf{1}_{\{u \neq 0\}} - 20u$, which proves (19.8). The second identity simultaneously preserves the integer quotient of the reduction. $\square$

The proposition specifies the action on the ledger before transporting its decimal representation. Its application to route reversal also preserves the convention used to read the endpoints. Let $\gamma = (x_0, \dots, x_n)$ be an enriched route and let f be an integer observable evaluated before each advance. The reverse route $\gamma^{-1} = (x_n, \dots, x_0)$ satisfies

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0). \quad (19.9)$$

Indeed, the left-hand side sums indices $1, \dots, n$; the first term on the right sums $0, \dots, n-1$. Their difference is exactly the stated boundary term. The identity applies to each incidence observable with its preserved data. The endpoints are accounted for in the integer counts before reduction modulo ten; on a closed route the term vanishes, whereas in an open window it can change the block. Traces crossing a junction are preserved through the partition in Proposition 19.3.

This connects the counts to route conjugation in Section 21, with the specific action on each incidence kept explicit. Negating a channel and reversing a route are operations on the ledger; their decimal reduction is the representation just calculated.

Proposition 19.5 (Integral image and reconstruction of the record). *Let $b, c \in \mathbb{Z}^{12}$, $q \in \mathbb{Z}$, with indices from zero to eleven. Define*

$$w_i = c_i - b_{i+1}, \quad P_0 = 0, \quad P_i = \sum_{j=0}^{i-1} w_j, \quad T_w = \sum_{i=0}^{11} P_i.$$

There exists a unique $k \in \mathbb{Z}^{12}$ such that $D_3 k = b$, $D_4 k = c$, $Q(k) = q$, if and only if

$$b_i = w_i + w_{i+1} + w_{i+2}, \quad \sum_i w_i = 0, \quad q + T_w \equiv 0 \pmod{12}. \quad (19.10)$$

In that case,

$$k_i = \frac{q + T_w}{12} - P_i. \quad (19.11)$$

Proof. If k exists, then $c_i - b_{i+1} = k_i - k_{i+1}$. The cyclic sum of these differences is zero, their sum over length three is b_i , and $k_i = k_0 - P_i$. Summing the twelve coordinates gives $q = 12k_0 - T_w$, proving the necessary conditions and the formula. Conversely, the conditions make the formula integral and cyclic. Its differences over length three are b_i , and those over length four are $w_i + b_{i+1} = c_i$. Its sum is q . The difference between two solutions has all consecutive differences zero and sum zero; it is therefore zero. $\square$

Reconstruction is an operation on the typed image, not a retrospective selection of its arguments. In particular, $\mathcal{D}(z) = \mathcal{D}(z')$ implies $z = z'$. Since the three-digit expression of an integer in $[0, 999]$ is unique, two triples of counts express the same record if and only if they coincide componentwise modulo ten.

The signed change of representation uses the block

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I_4. \quad (19.12)$$

The operator H_{12} applies H_4 to the three ordered orbits $(1, 4, 7, 10)$, $(2, 5, 8, 11)$, $(3, 6, 9, 12)$. On the sublattice $\mathcal{L}_H = \{U \in \mathbb{Z}^{12} : H_{12}U \equiv 0 \pmod{4}\}$, the maps

$$U = H_{12}K, \quad K = \frac{1}{4}H_{12}U \quad (19.13)$$

are inverses. The congruence expresses exactly the integrality of the second map, and $H_{12}^2 = 4I$ proves both compositions.

The signed record is also calculated directly from the transverse coordinates before reconstructing K . With indices from zero to eleven, let, for $r = 0, 1, 2$,

$$S_r = \sum_{j=0}^3 c_{r+3j}, \quad a_0 = \frac{q + 2S_0 + S_1}{3}, \quad a_1 = a_0 - S_0, \quad a_2 = a_1 - S_1, \quad d_{r,j} = b_{r+3j}.$$

The harmonic map expresses by orbits

$$\Pi_H^{(r)}(b, c, q) = (a_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \quad (19.14)$$

To prove the formula on the integral image, let k be its preimage. If $s_r = \sum_j k_{r+3j}$, then $S_r = s_r - s_{r+1}$ and $q = s_0 + s_1 + s_2$, with orbit indices modulo three. Solving gives $s_r = a_r$. For the four coordinates (x_0, x_1, x_2, x_3) of an orbit, $d_j = x_j - x_{j+1}$. Their three combinations in (19.14) are respectively $x_0 + x_1 - x_2 - x_3$, $x_0 - x_1 + x_2 - x_3$, and $x_0 - x_1 - x_2 + x_3$. Together with $a_r = \sum_j x_j$, they are exactly $H_4(x_0, x_1, x_2, x_3)^T$. The effective composition is thus $\mathcal{L} \rightarrow (b, c, q) \rightarrow \Pi_H(b, c, q) = U \rightarrow H_{12}U/4 = K$. The preimage used in this proof verifies the identity; it is not an argument of the map Π_H .

For the terminal state of (19.3), the coordinate produced before signed representation is

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned} \quad (19.15)$$

Substituting these twenty-five fields into (19.14), the three signed orbits are

$$\begin{aligned} U_{\text{term}}^{(1)} &= (2378, -452, -668, -322), \\ U_{\text{term}}^{(2)} &= (1406, 998, -204, -28), \\ U_{\text{term}}^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns produces

$$U_{\text{term}} = \mathcal{S}_{12}(X_{\text{term}}) = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (19.16)$$

The orbital transform $H_{12}/4$ is defined on this sublattice by the exact divisibility of every block of four. Consequently, the chain

$$X_{\text{term}} \xrightarrow{\mathcal{R}_{12}} \mathcal{L}(X_{\text{term}}) \xrightarrow{\mathcal{E}_{108}^{90,120}} C_{\text{term}} \xrightarrow{\Pi_H} U_{\text{term}} \xrightarrow{H_{12}/4} K \quad (19.17)$$

is a typed composition from the full state to its periodic seal. Propositions 19.3 and 19.5 provide, respectively, the finite update of the ledger and the integral inverse of the last representation.

The generated record provides the ordered witness

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (19.18)$$

The leading zero in 058 forms part of the block reading. To check it through Proposition 19.5, its differences and sum are

$$\begin{aligned} b &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ c &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \\ q &= 6263. \end{aligned}$$

In the preceding three orbits, the signed record is

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Applying $H_4/4$ to these three vectors reproduces (19.18). These identities verify the representation of the incidences; their genesis remains that of the ledger (19.5).

19.4 Periodic representation of K and normalisation of α

Fix $B = 1000$, the phase origin, and the twelve positions. Write

$$N_K = \sum_{j=1}^{12} K_j B^{12-j}, \quad \kappa = \frac{N_K}{B^{12} - 1}.$$

This is the exact periodic evaluation of the record. Multiplication by $B^{12} - 1$ and successive Euclidean division recover its twelve blocks, including any leading zeros. For $\rho K = (K_2, \dots, K_{12}, K_1)$, the same concatenation gives $\kappa(\rho K) = B\kappa(K) - K_1$. Thus reversibility of the record preserves a specified phase origin.

Proposition 19.6 (Minimal period of the rational representation). *The periodic prolongation of the record (19.18) has minimal period 12 blocks in base 1000 and 36 digits in base 10.*

Proof. The twelve components of K are distinct. Any nonzero cyclic translation smaller than twelve would identify two components if it were a period. The minimal block period is therefore twelve.

Concatenating its three-digit blocks, including 058, produces a decimal period of length 36. This expansion is canonical: it is not eventually constant at the digit 9. Let d be its minimal decimal period. Then $d \mid 36$, because the shifts preserving a periodic word form a cyclic subgroup. Grouping the digits in threes, the minimal block period is the least $k > 0$ such that $d \mid 3k$, namely $d/\gcd(d, 3)$. No proper divisor of 36 satisfies $d/\gcd(d, 3) = 12$. It follows that $d = 36$. $\square$

This period belongs to the rational coordinate of the record, with the phase origin fixed. The following composition also preserves the regional coordinates and the normalisation quotients of its output.

Let B_j, E_j, Φ_j be the fractional blocks of the three regional characters already constructed, and extend K_j with period twelve. The composition of records is

$$Z_j = B_j + E_j - \Phi_j - K_j, \quad A_j = Z_j + c_{j+1} - Bc_j. \quad (19.19)$$

The integer quotient c_j transports the continuation towards the observed position. In a finite window, c_{N+1} is fixed first, and then

$$c_j = \left\lfloor \frac{Z_j + c_{j+1}}{B} \right\rfloor, \quad A_j = Z_j + c_{j+1} - Bc_j.$$

Euclidean division uniquely determines $0 \leq A_j < B$. Since $-1998 \leq Z_j \leq 1998$, the set $\{-2, -1, 0, 1, 2\}$ is stable for these quotients.

Theorem 19.7 (Exact composition at unlimited depth). *The complete evaluation of the normalisation, in the canonical decimal convention, is*

$$\alpha_A = \pi + e - \varphi - 4 - \kappa. \quad (19.20)$$

There is a unique canonical expansion (A_j) , with blocks between zero and $B - 1$, and a bounded sequence of integer quotients satisfying (19.19) and $c_1 = 0$.

Proof. The regional integer parts are 3, 2, 1. By absolute convergence,

$$y := \sum_{j \geq 1} Z_j B^{-j} = (\pi - 3) + (e - 2) - (\varphi - 1) - \kappa.$$

The first blocks are 141, 718, 618, 234, so that $Z_1 = 7$. The absolute value of the tail sum is at most $1998 \sum_{j \geq 2} B^{-j} = 0.002$; in particular, $0 < y < 1$. Take the canonical expansion $y = \sum A_j B^{-j}$, choosing the terminating expansion rather than the eventually constant $B - 1$ expansion if a boundary occurs. Set

$$c_j = \sum_{r \geq 0} (Z_{j+r} - A_{j+r}) B^{-r-1}.$$

Equality of the complete values shows that each c_j is the negative of a finite sum of integers weighted by powers of B ; it is therefore an integer, and $c_1 = 0$. Separating the first term gives $Bc_j = Z_j - A_j + c_{j+1}$. The tail bounds make the sequence bounded. More precisely, the tail of Z lies in $[-2, 2]$ and the canonical tail of A in $[0, 1)$, so $-3 < c_j \leq 2$. Being integral, c_j belongs to the stated set $\{-2, -1, 0, 1, 2\}$. Conversely, summing the recurrence through N gives

$$\sum_{j=1}^N A_j B^{-j} = \sum_{j=1}^N Z_j B^{-j} - c_1 + c_{N+1} B^{-N}.$$

Boundedness makes the final term vanish in the limit. Uniqueness of the canonical expansion of y determines A , and the tail formula then determines every c_j . $\square$

The finite evaluation $\kappa_{36} = N_K/B^{12}$ differs from κ by $B^{-12}\kappa$. Moreover, imposing $c_{13} = 0$ in a twelve-block window does not replace the quotient of the complete continuation. In the expressed record, the continuation gives $c_{13} = -1$: the twelfth block is 662, whereas the truncated window with zero terminal quotient gives 663. The complete evaluation begins

$$\alpha_A = 0.007297352569283800997285105472380662999 \dots$$

The period of K does not impose a period on (A_j) , since (19.19) also involves the three regional records and the integer quotients.

In the following formulas, α denotes this complete positional output α_A . The analytic vacancy route has its own operator; a polynomial truncated at order nine and the root of that polynomial are not identified here with the complete operator. The mass compositions that follow do not require that approximation to be promoted to an equality between the two infinite routes.

19.5 Three-dimensional construction of the electronic prefactor

Let $\mathcal{A}_3 = \{1, \dots, 9\}^3$, with uniform measure, and $\rho_9(n) = 1 + ((n-1) \bmod 9)$. On this domain act

$$\Sigma(x) = \rho_9(x_1 + x_2 + x_3), \quad \Pi(x) = \rho_9(x_1 x_2 x_3).$$

On $\ell^2(\mathcal{A}_3)$, let P and Q be the conditional averages over the fibres of Σ and Π , respectively. Each is idempotent and self-adjoint: within each fibre it replaces a vector by its constant component and preserves that component orthogonally. They are therefore orthogonal projectors.

The additive fibres have cardinality 81. The multiplicative fibres have cardinalities

$$m = (36, 36, 108, 36, 36, 108, 36, 36, 297).$$

The joint incidence $N_{ab} = \#\{x : \Sigma(x) = a, \Pi(x) = b\}$ is $9N_0$, where

$$N_0 = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 0 & 2 & 1 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 1 & 2 & 0 & 0 & 2 & 1 & 0 & 3 \end{pmatrix}. \quad (19.21)$$

This matrix is obtained by counting the 729 triples. It is also reproduced without rounding by $N_{ab} = \sum_{i,j,k=1}^9 \mathbf{1}_{\{a\}}(\rho_9(i+j+k)) \mathbf{1}_{\{b\}}(\rho_9(ijk))$. In the normalised bases of functions constant on each fibre, the intertwining has matrix $C_{ab} = N_{ab}/\sqrt{81m_b}$.

Proposition 19.8 (Norm of the noncommutativity of the two sheets). *The preceding projectors satisfy*

$$\|[P, Q]\| = \frac{\sqrt{3}}{4}. \quad (19.22)$$

Proof. On the space of the nine additive fibres, PQP has matrix $M = CC^\top$. Multiplication of the matrices in (19.21) gives eigenvalue one on the constant vector, $1/4$ on $(1, -1, 0, 1, -1, 0, 1, -1, 0)$, and $7/132$ on $(1, 1, -2, 1, 1, -2, 1, 1, -2)$. On coordinates $\{3, 6, 9\}$ with zero sum it acts by $1/18$, with multiplicity two. On each of the sets $\{1, 4, 7\}$ and $\{2, 5, 8\}$, the zero-sum vectors belong to the kernel; they give four dimensions. These orthogonal subspaces add up to the nine dimensions.

For a unit eigenvector $u \in \text{ran } P$ of PQP with eigenvalue $0 < \lambda < 1$, the vector $v = (Qu - \lambda u)/\sqrt{\lambda(1-\lambda)}$ is a unit vector orthogonal to $\text{ran } P$. In the basis (u, v) ,

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} \lambda & \sqrt{\lambda(1-\lambda)} \\ \sqrt{\lambda(1-\lambda)} & 1-\lambda \end{pmatrix}.$$

The commutator has norm $\sqrt{\lambda(1-\lambda)}$. The blocks corresponding to $\lambda = 0, 1$ commute. Among the calculated eigenvalues, the maximum is attained at $\lambda = 1/4$, since $\lambda(1-\lambda)$ is increasing on $[0, 1/2]$. Its square root is $\sqrt{3}/4$. $\square$

In the extremal block, $S = 2P - I$ and $J = 4[P, Q]/\sqrt{3}$ satisfy

$$S^2 = I, \quad J^2 = -I, \quad SJ = -JS.$$

The four matrices I, S, J, SJ are linearly independent, so they realise $\text{Cl}_{1,1} \simeq M_2(\mathbb{R})$. The elements $n_\pm = (S \pm J)/2$ satisfy $n_\pm^2 = 0$ and $n_+ n_- + n_- n_+ = I$, by direct expansion. The prefactor of the electronic equation thus arises from two evaluations of APP in dimension three and their commutator, before any reference mass.

19.6 Boundary reading and basal electronic composition

The incidence (19.2) has positive eigenvector $r = (1, \varphi)^\top$. The normalisation $p_{ij} = F_{ij}r_j/(\varphi r_i)$ produces

$$P_{\text{av}} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}, \quad \mu = \frac{(1, \varphi^2)}{1 + \varphi^2}.$$

The identity $\varphi^{-2} + \varphi^{-1} = 1$ proves that the rows sum to one. Since F is symmetric, weights proportional to r_i^2 satisfy detailed balance; hence $\mu P_{\text{av}} = \mu$. The incidence is irreducible and has a self-loop, so this stationary distribution is unique. The chronological frequency $(1/3, 2/3)$ counts the three instants of the calendar; μ weights its admissible trajectories with memory one.

The operator preserving the sheets, fixing the volumetric normalisation at one, and marking the areal sheet by the closure character is $D_\pi = \text{diag}(\pi, 1)$. By induction,

$$F^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

The ratio of marked diagonal returns is

$$\Theta_n = \pi \frac{F_{n-1}}{F_{n+1}} \longrightarrow \frac{\pi}{\varphi^2}.$$

Convergence follows from the dominant eigenvalue φ , or from the consecutive ratios already bounded. Under interchange of its arguments, the reader $\mathcal{R}(x, y) = x/y^2$ also expresses

$$\rho_A = \frac{\pi}{\varphi^2}, \quad \rho_E = \frac{\varphi}{\pi^2}, \quad \varphi^3 \rho_A^2 \rho_E = 1. \quad (19.23)$$

The last equality is checked by multiplying the fractions. It relates two orientations of the same reader and does not select the remaining coefficients of the electronic equation.

The typed central return has length 27, with support $\mathbb{Z}/9\mathbb{Z} \times \mathbb{F}_3$. In the fixed regional gauge, the five microfibre positions are inserted as $(4, 0), (5, 0), (6, 0), (6, 1), (6, 2)$. The complement has $27 - 5 = 22$ positions. Its incidence with the three tritic phase states has $5 \cdot 3 = 15$ pairs. This yields the integers -22 and 15 used by the electronic reader; orientation fixes the sign of the first term. The cubic composition uses three copies of the closure coordinate, so its evaluation is α^3 . The support, regional insertion, and this composition rule are distinct operational data; an isolated count does not replace their specification.

The global defect and its nonadic return are expressed through

$$d_4 = \frac{\pi - e \log \pi}{270}, \quad \Delta_4 = d_4 \left(1 + \frac{\pi}{729}\right). \quad (19.24)$$

The denominator 270 belongs to the corona without its closure unit; $729 = 9^3$ belongs to the cubic support. The function $f(x) = x - e \log x$ attains its minimum zero at $x = e$, since $f'(x) = 1 - e/x$ changes sign there. As $\pi \neq e$, we obtain $d_4 > 0$ and $\Delta_4 > 0$.

Definition 19.9 (Basal electronic coordinate). The composition of the preceding readers is

$$\varepsilon_e^{(0)} = \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3 \right] (1 + 15\Delta_4). \quad (19.25)$$

The exponential acts on orientation ρ_E of (19.23); the vacancy reading corrects that component, and the final factor preserves the return of the global defect. For $0 < \alpha < 0.01$, all factors are positive: $\exp(\varphi/\pi^2) > 1$, $22\alpha^3 < 22 \cdot 10^{-6} < 1$, and $\Delta_4 > 0$. This is a dimensionless coordinate. Its presence in a mass chart is defined after construction of the scale.

19.7 Determinantal scale and the reduced Planck constant

The local block (19.12) has determinant -16 . Its Smith normal form is $\text{diag}(1, 2, 2, 4)$. Indeed, the greatest common divisor of the entries is one; the minors of order two are even, and one has absolute value two. By the adjugate identity, the minors of order three have absolute value four, and the determinant has absolute value sixteen. The determinantal divisors are 1, 2, 4, 16, whose successive ratios give the four Smith factors. Consequently,

$$G_H = \text{coker } H_4 \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \quad |G_H| = 16. \quad (19.26)$$

The local normed reader multiplies the same coordinate α once for each class of this cokernel and applies the self-scaling coordinate once. Its definition produces

$$\Sigma_H = \varphi \prod_{g \in G_H} \alpha = \varphi \alpha^{16}. \quad (19.27)$$

The order of the cokernel determines the exponent of this reader. The Smith normal form and the choice of normed reader serve distinct functions: the former does not turn an arbitrary functional of the block into expression (19.27).

Proposition 19.10 (Decade of action). *The output (19.27) satisfies $10^{-34} < \Sigma_H < 10^{-33}$; its decimal valuation is -34 .*

Proof. The regional and positional evaluations give the intervals $1.618 < \varphi < 1.619$ and $0.007297 < \alpha < 0.007298$. By positivity,

$$\frac{1618 \, 7297^{16}}{10^{99}} < \Sigma_H < \frac{1619 \, 7298^{16}}{10^{99}}.$$

The integer inequalities $1618 \, 7297^{16} > 10^{65}$ and $1619 \, 7298^{16} < 10^{66}$ prove the result without using an external action constant. $\square$

The decade and the mantissa are representations of different stages. Before constructing the latter, the regional reader selects the persistent return of the closure microfibre. The projection $p_{\text{ret}}(u_1, \dots, u_6) = (u_4, u_5, u_6)$, applied to the five emissions of 19.1, produces the block $b_\pi = (169, 272, 272)$ four times and $b_{\bar{\pi}} = (418, 674, 521)$ once. These multiplicities count regional rows, not the seed multiplicities 432, 144.

Proposition 19.11 (Equivariant selector of the regional return). *Among rules preserving permutations of the five regions, selecting the terminal block of maximal multiplicity, equivariant under permutations of its three positions, and retaining a position fixed by the stabiliser of the block, the selected coordinate is unique and equals $\rho_\pi = 169$.*

Proof. Projection of the five rows shows that the block of maximal multiplicity is unique and is b_π . Its stabiliser in $\mathfrak{S}_3$ interchanges the two positions of value 272 and fixes only the position of value 169. Retaining that position is compatible with any simultaneous permutation of coordinates. Equivalently, the value is expressed without choosing an initial position as $\text{sing}(b_\pi) = \sum_j (b_\pi)_j - 2 \bmod a(b_\pi) = 169$. $\square$

Grading by length expresses a word of length n through the character $\chi_z([w]) = z^{|w|}$. Concatenation adds lengths, so $\chi_z([uv]) = \chi_z([u])\chi_z([v])$. Since the regional emissions belong to $\mathcal{U}_6$, the return impulse appears as $\rho_\pi z^6$. Degree six corresponds to the length of the emission, not the length of a closed walk traversing several anchors of the graph.

The prolongation weight is obtained on the determinant line of bilayer transport. For a real involution R in dimension two, with $\text{tr } R = 0$, let

$$x_A = \frac{50\pi}{9}\alpha, \quad \mathcal{T}(x_A, y) = \exp(-x_A I_2 - yR).$$

The oriented coordinate y is still formal. In an eigenbasis of R , the two entries are $e^{-x_A - y}$ and $e^{-x_A + y}$. Therefore,

$$\bigwedge^2 \mathcal{T} = \det \mathcal{T} = e^{-2x_A} = D_A.$$

Dependence on y cancels. Multiplicativity under concatenation determines the prolongation character $k \mapsto D_A^k$: any character with elementary value D_A must have that value at $k = 1 + \dots + 1$. This construction precedes evaluation of the conjugate angular coordinate.

The fifth-order regional reader uses eigenvalues 9, 5 of the central block $\begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$, the orbit of four positions, and the calendar of twelve windows. Its graded rule fixes

$$\begin{aligned} H_5 &= \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \\ D_A &= \exp\left(-\frac{100\pi\alpha}{9}\right), \\ \mathcal{C}_\pi &= 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \\ \eta_{\text{ret}} &= H_5 - \frac{90}{\pi}\mathcal{C}_\pi. \end{aligned} \tag{19.28}$$

In this rule, the ratios are $9/16 = 9/4^2$, $5/9$, $7/48 = (\text{tr } B_c/2)/(4 \cdot 12)$, and $1/54 = 1944/104976$. Their assignment to the degrees and signs forms part of the graded reader. The integer 169 is the representation of the selector proved in Proposition 19.11. Its regional datum precedes the evaluation of α ; the analytic reader subsequently composes it with the closure coordinate.

The tail of $\mathcal{C}_\pi$ has a renewal construction. The impulse $\rho_\pi z^6$ and the strict continuation belong to different summands. After recording the impulse once, the continuation is normalised by the unit of the determinant line; each new step multiplies it by D_A . The orientation of the regional gauge

places this return in the compensating sector subtracted from the fifth-degree character. With formal parameter z , the rule

$$R(z) = D_A z^7 + D_A z R(z) \quad (19.29)$$

uniquely determines $R(z) = D_A z^7 / (1 - D_A z)$. This can be proved by comparing coefficients or by inverting the formal series $1 - D_A z$. Normalisation of the first return, in addition to the recurrent ratio, is essential to this uniqueness. Since $0 < D_A < 1$ and $\alpha < 1$, evaluation at $z = \alpha$ converges absolutely.

Proposition 19.12 (Positivity and the action return ratio). *On the generated branch, $0 < \eta_{\text{ret}} < H_5$. The ratio*

$$R_{\text{act}} = \frac{H_5}{\eta_{\text{ret}}} \quad (19.30)$$

is positive, greater than one, and dimensionless.

Proof. The bounds $0.007 < \alpha < 0.008$ and $\varphi < 1.619$ give $\frac{1}{2}\sqrt{1000\alpha/\varphi} > 1$. The sum of the absolute values of the remaining negative terms of H_5 is less than 0.008001, so $H_5 > 0.99$. Since $D_A < 1$, $\pi > 3$, and $\alpha < 0.008$,

$$0 < \frac{90}{\pi} \mathcal{C}_\pi < 30 \left(169(0.008)^6 + \frac{(0.008)^7}{1 - 0.008} \right) < 10^{-6}.$$

It follows that $0 < \eta_{\text{ret}} < H_5$. The ratio of two action coordinates in the same basis is dimensionless and does not change when that basis is rescaled. $\square$

Let $\mathcal{U}_S$ be a basis of the action line. The two sections, before and after the return, are

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S. \quad (19.31)$$

Their physical reading is the proposed construction of the reduced Planck constant, with the return section made explicit. The unit $\mathcal{U}_S \mapsto \text{J s}$ is a subsequent metrological realisation. The coefficient of Σ_H within its decade is not H_5 : the determinant expresses the scale, and the regional reader expresses the coordinate occupying that scale. For each section, $h = 2\pi\hbar$ expresses the action of a complete circuit.

19.8 Angular coordinates and the clock of the absolute realisation

The angular pair uses the dimensionless coordinates already generated:

$$A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha), \quad x = \frac{\pi}{180} A_{\text{deg}}, \quad y = \frac{\pi}{180} C_{\text{deg}}^*. \quad (19.32)$$

The first operation shifts the base-thousand expansion by one block. The second is the bidirectional representation of the regional coordinate: equivalently, $y = \pi(H_5 + \alpha)/90 - \mathcal{C}_\pi$. This equality follows by substituting (19.28); it introduces no additional rule for selecting the counter-angle. The chart of one circuit has 360 angular units or 2π radians. Thus, for a lifted coordinate,

$$\hbar\theta_{\text{rad}} = h \frac{\theta_{\text{deg}}}{360}.$$

The factor 10^{-34} remains on the action line and is not inserted into the angular coordinates.

The positive channels are

$$q_+ = e^{-x-y}, \quad q_- = e^{-x+y}.$$

The pair is recovered through $x = -\frac{1}{2} \log(q_+ q_-)$, $y = \frac{1}{2} \log(q_- / q_+)$. Channel interchange preserves x and reverses the sign of y . The operator with oriented projectors $P_\pm$, $T = q_+ P_+ + q_- P_-$, preserves those labels when publishing its two eigenvalues.

For an internal time unit $t_0 > 0$, the closure of 108 steps determines

$$\omega_0 = \frac{2\pi}{108t_0}, \quad E_{\text{clk}} = \hbar_{\text{ret}} \omega_0, \quad m_{\text{clk}} = \frac{E_{\text{clk}}}{c_{\text{int}}^2}. \quad (19.33)$$

These formulas produce dimensions of frequency, energy, and mass. Velocity and length are obtained through the response of the same channels, as specified by the following identities. Expressing them in seconds, joules, or mass units requires the corresponding charts; those charts do not alter the channels or the electronic section normalised relative to the clock.

19.8.1 Identity between constitutive velocity and clock velocity

In the domain $x > |y|$, the two channels satisfy $0 < q_{\pm} < 1$. On the two-sheet realisation, with rank-one projectors P_+, P_- , the temporal response is defined by

$$V = (I - T^{90})(I - T^{120})^{-1} = r_+ P_+ + r_- P_-, \quad r_{\pm} = \frac{1 - q_{\pm}^{90}}{1 - q_{\pm}^{120}} > 0.$$

The inverse exists because both eigenvalues of T^{120} are less than one. The symmetric reading of the response is

$$\mathcal{E} = \log[(1 - q_+^{90})(1 - q_-^{90})] - \log[(1 - q_+^{120})(1 - q_-^{120})] = \log(r_+ r_-).$$

Let u_S, u_C, u_T be positive bases of action, velocity, and time, and $u_L = u_C u_T$ the length basis. The kinematic and constitutive readings are

$$c_{\text{int}} = e^{-\mathcal{E}} u_C, \quad c_{\text{vac}} = (r_+ r_-)^{-1} u_C.$$

Proposition 19.13 (Common realisation of propagation and absolute mass). *In the same dimensional chart,*

$$c_{\text{int}} = c_{\text{vac}} = \hat{c} u_C, \quad \hat{c} = (\det V)^{-1} = (r_+ r_-)^{-1}.$$

For $t_0 = \tau_0 u_T$, with $\tau_0 > 0$, the elementary length is

$$\ell_0 = c_{\text{int}} t_0 = \tau_0 \hat{c} u_L.$$

The chronological mass of (19.33) therefore retains the form

$$m_{\text{clk}} = \frac{2\pi \eta_{\text{ret}} 10^{-34}}{108 \tau_0 \hat{c}^2} \frac{\mathcal{U}_S}{u_T u_C^2}.$$

Proof. The identity $\mathcal{E} = \log(r_+ r_-)$ gives $e^{-\mathcal{E}} = (r_+ r_-)^{-1}$; the rank-one property of each sheet gives $\det V = r_+ r_-$. Sheet interchange permutes the factors and preserves their product. Length is obtained by multiplying the same velocity by t_0 . Substituting these expressions and (19.31) into (19.33) produces the mass formula. The factor $\mathcal{U}_S/(u_T u_C^2)$ has the dimension of mass; it does not enter the selection of the channels. $\square$

For a homogeneous route of $n > 0$ transitions, the additive readings $T(\gamma) = n t_0$ and $L(\gamma) = n \ell_0$ give $L(\gamma)/T(\gamma) = c_{\text{vac}}$. If the channel varies from edge to edge, their lengths are added; the ratio is then the mean velocity of those edges, not necessarily a common local velocity.

Equality of sections persists under changes of chart. If $u'_C = d u_C$, $u'_T = \eta u_T$, with $d, \eta > 0$, then

$$\hat{c}' = \hat{c}/d, \quad \tau'_0 = \tau_0/\eta, \quad u'_L = d \eta u_L, \quad \tau'_0 \hat{c}' u'_L = \ell_0.$$

In particular, choosing the adapted basis $u'_C = c_{\text{vac}}$ makes $\hat{c}' = 1$, but the reader coordinate in the internal chart remains $(r_+ r_-)^{-1}$.

19.9 Two readers of the electronic trajectory and beta evaluation

The central trajectory starts at $(r, c) = (5, 5)$, with orientation $(h, v) = (1, 1)$. TRIT repeats the cycle $(+1, -1, 0)$. In the first regime, $r + c$ is read and the advance $c \mapsto 1 + ((c - 1 + h) \bmod 9)$ follows; in the second, rc is read and the advance $r \mapsto 1 + ((r - 1 + v) \bmod 9)$ follows; in the zero regime, 9 is read without displacement. The class modulo three of the digital root of the reading is expressed. After steps 27, 54, 81, h, v are simultaneously reversed. The representation of 108 steps preserves the complete trajectory and gives

$$a = 100000220, \quad b = 120200000, \quad \gamma_e = (a^3 b^3)^2, \quad \tau_e = (+, +, +, -, -, -, +, +, +, -, -, -). \quad (19.34)$$

The identity is checked by traversing the nine instants of each window; at the third block, reversal changes a to b , and the second traversal of 54 steps repeats the representation. The state also preserves its memory coordinates.

The first reader measures cyclic disagreements of the ternary representation:

$$D_k = \#\{t \in \mathbb{Z}/108\mathbb{Z} : \gamma_e(t) \neq \gamma_e(t + k)\}.$$

Since the expressed word has period 54, every disagreement appears together with its translate by 54; all D_k are even. In the half $a^3 b^3$, the disagreement count for $k = 1, 3, 4, 27, 36$ gives respectively 27, 28, 34, 24, 16. Doubling it yields

$$(D_1, D_3, D_4, D_{27}, D_{36}) = (54, 56, 68, 48, 32).$$

Hence

$$K_D = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2} \right) = (80, 54, 6). \quad (19.35)$$

The second reader acts on the orientation of the twelve windows. Define, with cyclic indices,

$$C_k = \sum_{r=0}^{11} \tau_r \tau_{r+k}, \quad A_\ell = \sum_{r=0}^{11} \left(\sum_{j=0}^{\ell-1} \tau_{r+j} \right)^2, \quad P_6 = \sum_{r=0}^5 \tau_r \tau_{r+6}.$$

The oriented primitive integer combination cancelling the diagonal part of A_3, A_4 is $\Omega = 4A_3 - 3A_4$: its coefficients satisfy $3u + 4v = 0$, $\gcd(u, v) = 1$, and $u > 0$ is fixed. Expanding the square proves

$$A_\ell = \ell C_0 + 2 \sum_{k=1}^{\ell-1} (\ell - k) C_k, \quad \Omega = -2C_1 - 4C_2 - 6C_3.$$

For τ_e , we obtain $C_0 = 12, C_1 = 4, C_2 = -4, C_3 = -12, A_3 = 44, A_4 = 32, P_6 = 6$. Therefore,

$$K_\Omega = (\Omega, 54, P_6) = (80, 54, 6) = K_D. \quad (19.36)$$

The second component preserves the half-return of 108 steps; it is not introduced by diagonal cancellation. Equality of the two readers has been proved on the specified central trajectory; it is not extended by definition to all collections of the atlas.

Definition 19.14 (Electronic coordinate after the return). The complete electronic composition is

$$\varepsilon_e^\beta = \varepsilon_e^{(0)} R_{\text{act}}^{80-54\alpha+6\Delta_4}. \quad (19.37)$$

Positivity of (19.25) and (19.30) allows this power to be defined through the real logarithm. The integer triple arises equally from the two central readers just calculated. Relative evaluation preserves the reduced Planck constant through the ratio of its sections, and absolute realisation additionally uses (19.33) and the stated dimensional chart.

In the inherited tables, $\mathbf{e}_0$ denotes the basal coordinate $\varepsilon_e^{(0)}$ in the chart specified there. The section $\hbar_{\text{int}}$ used for a survival evaluation must be stated in the same chart: for the post-return realisation of

the present development, it is $\hbar_{\text{ret}}$. Historical rows retain the convention stated in their source table and are not recalculated through a silent change of this section.

If $\mathcal{U}_E$ is the energy basis of the electronic evaluation, $E_e = \varepsilon_e^\beta \mathcal{U}_E$. When the chronological quantum is used as the basis, the coefficient is

$$\hat{\varepsilon}_e = \varepsilon_e^\beta \frac{\mathcal{U}_E}{E_{\text{clk}}}, \quad E_e = \hat{\varepsilon}_e E_{\text{clk}}.$$

The equality preserves the object and transforms its coordinate. Hence a number presented in an MeV chart cannot be multiplied without conversion by an arbitrary different energy quantum. The electron retains its structural construction when expressed relative to a chronological scale: it is not replaced by the mass of that quantum.

Metrological comparison applies to the result of this composition, with the version of α , the action section, the Δ_4 reader, and the unit chart stated. In particular, d_4 and Δ_4 are not interchangeable: replacing them simultaneously in the basal factor and in the exponent changes the result by the positive ratio

$$\frac{1 + 15\Delta_4}{1 + 15d_4} R_{\text{act}}^{6(\Delta_4 - d_4)} > 1.$$

The preserved formulas allow that difference to be located without attributing it to an external mass or selecting the trajectory anew.

20 The particle as a class of persistent sections

The material construction begins with the state produced by APP, TRIT, and TPK. APP supplies the positions and the two arithmetic operations; TRIT distinguishes the regime and orientation of their transitions; TPK transports this information together with the quotients, boundary incidences, and memory. As the state is prolonged, a finite evaluation remains linked to the evaluations preceding it. The notion of a particle rests on this compatibility and on its material representation. Its mass will be an observable of the section thus constructed.

This sequence determines the meaning of the three objects to be used: the genealogical state preserves the complete extension; the route records its observable history; the material representation makes that history act on a state space and permits the evaluation of observables. The distinction is essential even when the three objects share the same electronic label or the same atlas coordinate.

20.1 Extension, observable history, and compatible section

Let $\mathcal{S}_m$ be the admissible states surviving through depth m , obtained by TPK prolongation, and let $\rho_{n,m} : \mathcal{S}_n \rightarrow \mathcal{S}_m$, for $n \geq m$, be their restriction maps. Compatibility of refinement means

$$\rho_{m,m} = I, \quad \rho_{n,m} \rho_{p,n} = \rho_{p,m} \quad (p \geq n \geq m). \quad (20.1)$$

A surviving extension is an element

$$x = (x_m)_{m \geq 0} \in \Omega_{\text{HMT}}^{\text{surv}} := \varprojlim_m \mathcal{S}_m, \quad \rho_{n,m}(x_n) = x_m. \quad (20.2)$$

The existence and admissibility of these states arise from the construction of the formal kernel. The inverse-limit expression describes the compatibilities preserved by a state already generated.

Let $\mathcal{R}_m$ be the space of observable histories at depth m . The route reader $\text{sh}_m : \mathcal{S}_m \rightarrow \mathcal{R}_m$ and the restriction $\text{tr}_{n,m}$ satisfy

$$\text{tr}_{n,m} \text{sh}_n = \text{sh}_m \rho_{n,m}. \quad (20.3)$$

Thus the reader transmits the restrictions of the state to its histories. The route of x is the collection $\gamma_x = (\gamma_x^{[m]})_m$, where $\gamma_x^{[m]} = \text{sh}_m(x_m)$.

Proposition 20.1 (Compatibility of the expressed history). *For every extension (20.2), its finite histories satisfy $\text{tr}_{n,m}(\gamma_x^{[n]}) = \gamma_x^{[m]}$.*

Proof. Applying (20.3) to the state x_n gives

$$\text{tr}_{n,m}(\gamma_x^{[n]}) = \text{sh}_m(\rho_{n,m}x_n) = \text{sh}_m(x_m) = \gamma_x^{[m]}.$$

The second equality uses precisely the inverse-limit condition. $\square$

The proposition establishes descent from the extension to the history. Its content does not include injectivity of the reader: two extensions can express the same finite history while preserving different memories. Hence a reused route retains its state of provenance, instead of reconstructing it from a numerical value or a visible word.

On each history $\gamma_x^{[m]}$, consider the discrete bundle $\mathcal{B}_{\gamma_x^{[m]}}$ of material states of the stated representation. Let $\mathcal{E}_m(x) = \Gamma(\mathcal{B}_{\gamma_x^{[m]}})$ be its spaces of sections, with restrictions $p_{n,m} : \mathcal{E}_n(x) \rightarrow \mathcal{E}_m(x)$ compatible with those of the state. A persistent section is a collection

$$s = (s_m)_{m \geq m_0}, \quad s_m \in \mathcal{E}_m(x), \quad p_{n,m}s_n = s_m \quad (n \geq m \geq m_0). \quad (20.4)$$

The initial depth m_0 allows a state to be described from any cut beyond which its representation is defined. Prior information entering its observables remains in the record transported by the extension.

20.2 Identity under refinement and equivalence of representations

Fix the internal symmetry groups G_m acting on the material fibres and preserving the stated observables. Two persistent sections s and s' represent the same material identity when, from some level m_1 onwards, there is a collection $g_m \in G_m$ such that

$$s'_m = g_m s_m, \quad p_{n,m}g_n = g_m p_{n,m}, \quad (20.5)$$

and the observable characters of the genealogical record coincide. The admissible symmetries are fixed before forming the quotient; their action transports sheet, orientation, memory, and representation according to the corresponding laws.

Proposition 20.2 (Compatible equivalence of sections). *Relation (20.5), on collections carrying the same types of representation and observables, is an equivalence relation. Its classes are unaffected by removing a finite number of initial levels, provided their transported genealogical information is preserved.*

Proof. The identity of each G_m gives reflexivity. The inverse of a compatible collection is again compatible: $p_{n,m}g_n = g_m p_{n,m}$ implies $p_{n,m}g_n^{-1} = g_m^{-1}p_{n,m}$. This gives symmetry. If $s' = gs$ and $s'' = hs'$, then from the larger of the two initial levels we have $s''_m = h_m g_m s_m$ and $p_{n,m}h_n g_n = h_m g_m p_{n,m}$. Composition also preserves the observables and proves transitivity. All these relations are formulated on a common tail; changing the initial point within that tail preserves the class. The last assertion concerns a change of presentation, not deletion of memory from the state. $\square$

Definition 20.3 (Persistent material particle). An HMT-MD particle is a class $[s]_{\text{pers}}$ of sections (20.4) under (20.5), accompanied by its extension and provenance record, admitting an irreducible material representation or a minimal invariant block and a stable spectral output under the specified coupling.

The definition gathers conditions at different levels. Prolongation and the record belong to the HMT state. Representation and coupling belong to its material realisation. Spectral stability is predicated of the operator and domain used by that coupling. This organisation permits identity, charge, spin, and mass to be studied together without turning one particular observable into the definition of all the others.

The nonadic return illustrates the distinction between identity and instantaneous state. For the phase reader g and memory record $\mathcal{B}$, the following can coexist:

$$g(k+9) = g(k), \quad p_{k+9,k}(s_{k+9}) = s_k, \quad \mathcal{B}_{k+9} \neq \mathcal{B}_k. \quad (20.6)$$

The first equality is periodic; the second expresses persistence; the third records advancement. The phase return belongs to the history of the same material identity, while the state preserves the fact that an additional circuit has been completed.

20.3 The species as a fibre: scope of the finite atlas

The atlas constructed by the APP-TRIT-TPK operations has support $\mathcal{C}_{81} = \{1, \dots, 9\}^2$. Its four orientations per cell form the domain of 324 oriented routes. The record map $\mathcal{R}$ distinguishes 56 route configurations, while the internal classifier

$$q : \mathcal{C}_{81} \longrightarrow \Sigma_{13} \quad (20.7)$$

gathers 13 types of collection and level. These censuses and their rules are developed in the part devoted to the atlas; here we specify which object they transmit to the definition of a particle.

For $y \in \Sigma_{13}$, the fibre $q^{-1}(y)$ preserves all its cells and records. Once the internal ordering of each fibre of $\mathcal{R}$ is fixed, let $k(c)$ be the rank of the cell within it. Its finite presentation is

$$\mathfrak{C}_y^{\text{atlas}} = \{(\mathcal{R}(c), k(c)) : q(c) = y\}. \quad (20.8)$$

Rank is added to separate cells possessing the same collection record; it does not select a mass. The pair $(\mathcal{R}(c), k(c))$ recovers the cell within the ordered chart because the second datum individualises its position in the fibre of the first.

If $\pi_{81} : \Omega_{\text{HMT}}^{\text{surv}} \rightarrow \mathcal{C}_{81}$ is the cellular reader, the lift of the species is

$$\tilde{\mathfrak{C}}_y = \{x \in \Omega_{\text{HMT}}^{\text{surv}} : q(\pi_{81}x) = y\}. \quad (20.9)$$

This preimage contains extensions, whereas (20.8) contains finite presentations. The construction and domain of the reader π_{81} form part of the passage between them. A particular particle is realised as a persistent section within the corresponding enriched multisection.

Theorem 20.4 (Central fixed point and equivariant selection). *Let $\rho_c(r, s) = (10 - r, 10 - s)$ be cellular inversion. In the ρ_c -stable fibres of the classifier (20.7), with trivial action on the label y , an equivariant point selection can take values only in cell (5, 5). The electronic fibre at level zero contains this unique fixed point; no other fibre admits such a selection.*

Proof. Solving $\rho_c(r, s) = (r, s)$ gives $r = s = 5$. If $a(y) \in q^{-1}(y)$ is an equivariant selection and y is fixed, then $a(y) = a(\rho_c y) = \rho_c a(y)$. Its value must be the fixed point (5, 5). The atlas places that cell in the electronic fibre at level zero, whose cardinality is one. For any other fibre, the selection condition would produce a fixed point not belonging to it. $\square$

The consequence is constructive: the other species are presented by orbits or multisections, and a realisation requiring a particular line must specify its representation and coupling. The electron provides the central rank-one case without turning all collections into copies of that case. Moreover, cell (5, 5), the raw central signature, the regional signature 555555, and the relative origin of an action line retain their own domains. Uniqueness of the point does not identify those records with one another.

20.4 From the material record to the mass observable

The chain used by the mass observable preserves the order

$$\text{extension} \longrightarrow \text{route} \longrightarrow \text{record} \longrightarrow \text{integer signature} \longrightarrow \text{character} \longrightarrow \text{towers} \longrightarrow \text{material operator}. \quad (20.10)$$

Each arrow retains the data required by the next. The signature is a reading of the record; the character evaluates that signature with the internal coefficients already constructed; the towers incorporate its refinement; finally, the operator acts in the representation of the fibre selected by the coupling. Its output can be a scalar, a multiplet, or a spectral distribution, according to that representation. Metrological comparison takes place after this construction.

A species is described by charge, spin representation, colour, flavour, generation, route configuration, and material operator. These data explain why two preparations with coinciding masses can preserve different information, and why the same identity can express different observables under different couplings. Identity resides in the persistent class and its admissible transports; measurement is a reading of that identity in the stated codomain.

The same hierarchy separates a persistent extension from a section with a finite horizon. If $\text{Ext}_L(s) \neq \emptyset$ and $\text{Ext}_{L+1}(s) = \emptyset$, the horizon L records an obstruction to prolongation. An intermediate participation of this kind in a composition does not by itself create a new asymptotic species. Similarly, a resonance requires the survival operator and its pole, as well as the identity of the section it transports. The distinction between persistence, finite prolongation, and spectral evaluation is thus preserved.

21 Conjugation, orientation, and return of material sections

The persistent identity of Section 20 admits inversion operations at different levels of its structure. Cellular inversion acts on the atlas; dodecaphase conjugation acts on the orientation record; lifting a loop acts on a fibre; a spinor rotation acts on a representation space. Specifying these domains allows the operations to be composed and explains what information is preserved in each return.

21.1 Dodecaphase involution and parity of observables

Let $\mathcal{F}_{12} = \{-1, 0, 1\}^{12}$ be the ambient space of tritic records with twelve positions. The admissible TPK domain is the subset satisfying the route compatibilities already constructed. The operation on the ambient space is

$$C_M : \mathcal{F}_{12} \longrightarrow \mathcal{F}_{12}, \quad (C_M \tau)_j = -\tau_{11-j}, \quad 0 \leq j < 12. \quad (21.1)$$

Restriction to admissible records uses stability of that domain under conjugation. Reversed order and sign reversal are simultaneous components of the operation.

Proposition 21.1 (Linear and quadratic parity of the record). *We have $C_M^2 = I$. If $a_j = a_{11-j}$ and $L_a(\tau) = \sum_j a_j \tau_j$, then $L_a(C_M \tau) = -L_a(\tau)$. If a quadratic form $Q_b(\tau) = \sum_{j,k} b_{jk} \tau_j \tau_k$ satisfies $b_{jk} = b_{11-j, 11-k}$, then $Q_b(C_M \tau) = Q_b(\tau)$.*

Proof. At each position, $[C_M(C_M \tau)]_j = -(-\tau_{11-(11-j)}) = \tau_j$. For the linear functional, substituting $k = 11 - j$ in the sum gives $L_a(C_M \tau) = -\sum_k a_{11-k} \tau_k = -L_a(\tau)$. In the quadratic expression the two sign reversals cancel, and the substitution $(r, s) = (11 - j, 11 - k)$ recovers Q_b through the symmetry of its coefficients. $\square$

Linear orientation and memory correlations therefore have different laws. Even a composition of these observables must preserve their parities: if L_a is odd, its exponential character satisfies

$$\exp(L_a(C_M \tau)) = \exp(L_a(\tau))^{-1}. \quad (21.2)$$

The equality expresses multiplicative inversion. Invariance of a complete material operator is studied on that operator, with its channels and representation, and is established by the corresponding covariance.

Cellular inversion in Theorem 20.4 is $\rho_c : \mathcal{C}_{81} \rightarrow \mathcal{C}_{81}$, whereas C_M has domain (21.1). Both are involutions, but comparing them requires the route record map and its transport law. This datum prevents conjugation of a history from being replaced by inversion of an isolated atlas coordinate.

21.2 Lifting to antisections and preservation of mass

Conjugation is applied to the particle through two compatible maps. The base map C_m^b transports state x_m to its conjugate state $\bar{x}_m$; the lift $\tilde{C}_m : \mathcal{E}_m(x) \rightarrow \mathcal{E}_m(\bar{x})$ transports its material sections. Let t_m be the reader of the dodecaphase record and $\bar{p}_{n,m}$ the restrictions in the conjugate collection. Their conditions are preservation of admissibility, involutivity, and compatibility:

$$\begin{aligned} \rho_{n,m} C_n^b &= C_m^b \rho_{n,m}, & t_m(C_m^b x_m) &= C_M t_m(x_m), \\ \bar{p}_{n,m} \tilde{C}_n &= \tilde{C}_m p_{n,m}, & \tilde{C}_{m,\bar{x}} \tilde{C}_{m,x} &= I. \end{aligned} \quad (21.3)$$

In the first row, the reader expresses the orientation record of the history. The remaining transport of the state accompanies that representation. The second row gives the laws of the lift; the subscripts $x, \bar{x}$ make its two source fibres explicit. If s is persistent, its antisection is $\bar{s} = (\tilde{C}_m s_m)_m$.

Proposition 21.2 (Persistence of the antisection). *Conditions (21.3) transport persistent sections to persistent sections and restore the section upon conjugating twice. If they also transport the internal symmetries by $g_m \mapsto \tilde{C}_m g_m \tilde{C}_m^{-1}$, they induce an involution on the material classes of Definition 20.3.*

Proof. Compatibility gives $\bar{p}_{n,m} \bar{s}_n = \tilde{C}_m p_{n,m} s_n = \tilde{C}_m s_m = \bar{s}_m$. Involutivity restores s . For $s'_m = g_m s_m$, their antisections are related through $\tilde{C}_m g_m \tilde{C}_m^{-1}$. The restriction law in (21.3) preserves compatibility of this collection, which belongs to the internal groups by the stated condition. Thus the construction descends to the classes. $\square$

The charge representation uses the odd part of the material record. If q is the stated charge character, its law is $q(\bar{s}) = -q(s)$. The elementary charge entering the expression of that coordinate on a dimensional line is the internal output already constructed; the representation is subsequently applied to the state without using a target charge to select its record.

For mass, covariance of its operator is required. Let $\hat{M}_s$ and $\hat{M}_{\bar{s}}$ be the self-adjoint operators of the two realisations, and let $\mathcal{C}$ be their unitary or antiunitary intertwiner. Its domain is transported by $\mathcal{C} \text{Dom}(\hat{M}_s) = \text{Dom}(\hat{M}_{\bar{s}})$. The operator condition is

$$\mathcal{C} \hat{M}_s \mathcal{C}^{-1} = \hat{M}_{\bar{s}}. \quad (21.4)$$

Proposition 21.3 (Spectral preservation under conjugation). *Under (21.4), transport of an eigensection with mass $m \in \mathbb{R}$ preserves that eigenvalue. The spectral measure is transported by $E_{\bar{s}}(B) = \mathcal{C} E_s(B) \mathcal{C}^{-1}$ for every real Borel set B . In particular, a scalar realisation preserves mass on passage to the antisection.*

Proof. If $\hat{M}_s \psi = m \psi$, covariance gives $\hat{M}_{\bar{s}} \mathcal{C} \psi = \mathcal{C}(m \psi) = m \mathcal{C} \psi$; the last equality also holds in the antiunitary case because m is real. The real functional calculus of self-adjoint operators transports their spectral projectors and proves the second assertion. $\square$

In a band record the same condition can be seen in elementary form. Given a real reading $E_0(\tau)$, let $E_{\pm} = (E_0 \pm E_0 \circ C_M)/2$ and let $\xi \in \{+1, -1\}$ be the sheet orientation. Then

$$E_{\text{band}}(\tau, \xi) = E_+(\tau) + \xi E_-(\tau), \quad E_{\text{band}}(C_M \tau, -\xi) = E_{\text{band}}(\tau, \xi). \quad (21.5)$$

The equality follows from the evenness of E_+ and oddness of E_- . It explains how band energy can be preserved by transporting record and sheet simultaneously; its identification with the mass of a particular realisation uses (21.4).

The electronic record $\tau_e = (+, +, +, -, -, -, +, +, +, -, -, -)$ satisfies $C_M \tau_e = \tau_e$. This equality concerns the twelve-position word. The pair (τ_e, ξ) preserves an additional orientation changed by conjugation. The distinction allows the unique electronic centre and the conjugate charge realisations to be retained together; a fixed word does not become a proof of neutrality.

21.3 Sign holonomy and the persistence band

Let ℓ be a return loop of the material reader of a surviving extension. Its circuit group is $\langle \ell \rangle \simeq \mathbb{Z}$. On this group, the sign character is specified by

$$\varepsilon : \langle \ell \rangle \longrightarrow \{+1, -1\}, \quad \varepsilon(\ell) = -1. \quad (21.6)$$

The circuit group records how many times the loop is traversed; its type differs from the finite phase C_9 . The interval bundle associated with the character is

$$\mathcal{M}_\ell = ([0, 1] \times [-1, 1]) / ((0, u) \sim (1, -u)). \quad (21.7)$$

Theorem 21.4 (Oriented return on the band). *The bundle (21.7) is a Möbius band. Its transport over ℓ reverses the sign of the fibre and over ℓ^2 restores it. Thus a nonzero vector of that line requires two circuits for its oriented return. This preserves the distinction between return of orientation and return of the complete state.*

Proof. The endpoint relation in (21.7) is the linear map $u \mapsto -u$, with negative determinant. It gives the nonorientable real line over the circle and its corresponding interval band. Composing two transports produces $u \mapsto u$. The memory of the extension is not part of this sign quotient; its updating continues to obey TPK transport and (20.6). $\square$

Parametrising one circuit of the base by 2π represents restoration of the cover by 4π . If the temporal reader assigns 108 steps to the relevant circuit, the oriented lift closes at 216 steps. These assignments require the stated loop and reader; they do not divide the history into two successive particles of 108 steps.

Proposition 21.5 (Decomposition of observables on the double cover). *Let $p : \tilde{X} \rightarrow X$ be a principal double cover, with involution ι . For $f \in C^0(\tilde{X}, \mathbb{R})$,*

$$f^+ = \frac{f + f \circ \iota}{2}, \quad f^- = \frac{f - f \circ \iota}{2}$$

give $f = f^+ + f^-$, with $f^+ \circ \iota = f^+$ and $f^- \circ \iota = -f^-$. The first part descends to X ; the second represents a section of the associated sign line. On the Möbius line, every continuous global section has a zero.

Proof. The identities follow by substituting $\iota^2 = I$. The invariant part is constant on each fibre and defines a function on the quotient. The anti-invariant part satisfies the transition law of the sign representation. On the circle, a section of this line is represented by a continuous real function u on $[0, 1]$ with $u(1) = -u(0)$. If an endpoint is zero, a zero already exists; otherwise the endpoints have opposite signs, and the intermediate value theorem provides an interior zero. $\square$

The decomposition distinguishes two classes of observables on the same cover. It represents odd orientation information and preserved even data. Applying it to a material record uses the lift (21.3); the sign character describes the transported line, not all the data of the particle.

21.4 Bidirectional lattice and channel conjugation

The integer angular record expresses two coordinates

$$W_+ = 6n_A + s_C, \quad W_- = 6n_A - s_C, \quad (n_A, s_C) \in \mathbb{Z}^2. \quad (21.8)$$

The coefficients n_A, s_C are internal route readings. The factor 6 belongs to the dodecaphase normalisation of that record.

Proposition 21.6 (Lattice index and interchange of readings). *The image of (21.8) has index 12 in $\mathbb{Z}^2$ and Smith normal form $\text{diag}(1, 12)$. It is characterised by $W_+ + W_- \equiv 0 \pmod{12}$. The operation $s_C \mapsto -s_C$ interchanges W_+ and W_- .*

Proof. The matrix of the map is $\begin{pmatrix} 6 & 1 \\ 6 & -1 \end{pmatrix}$. The greatest common divisor of its entries is 1, and its determinant is -12 , giving the stated invariant factors. The inverse coordinates are $n_A = (W_+ + W_-)/12$ and $s_C = (W_+ - W_-)/2$. If the sum is divisible by 12, the two integers have the same parity and both expressions are integral. The condition is also necessary by summing (21.8). The interchange assertion is checked by substituting $-s_C$. $\square$

In the real angular representation, the coordinates already generated, $a = A_{\text{rad}}$ and $b = C_{\text{rad}}^*$, produce the channels $q_{\pm} = \exp[-(a \pm b)]$. In the domain $a > |b|$, both belong to $(0, 1)$. Let $R^* = R$, $R^2 = I$ be the sheet grading and C a unitary interchange with $CRC^{-1} = -R$. For $T(b) = \exp[-(aI + bR)]$ we obtain

$$CT(b)C^{-1} = T(-b), \quad CP_{\pm}C^{-1} = P_{\mp}, \quad P_{\pm} = (I \pm R)/2. \quad (21.9)$$

Indeed, conjugation transforms the exponent and every term of its convergent series. This proof transmits the interchange to the channels. The integer lattice and the real operator have distinct domains and express the same interchange operation through their stated reading maps.

21.5 Spinor representation of the electronic block

The three-dimensional APP construction of the addition and multiplication projectors selects a principal electronic plane through the tritic mode $(1, -1, 0)^3$. Its derivation and the evaluation of the projectors are included among the antecedents of the present article. In an orthonormal basis of that plane, their restrictions are

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}, \quad S = 2P - I, \quad J = \frac{4}{\sqrt{3}}[P, Q]. \quad (21.10)$$

This representation receives the two constructed projectors and preserves their orientation. Multiplication gives $S^* = S$, $J^* = -J$, $S^2 = I$, $J^2 = -I$, and $SJ = -JS$. Specifically, $S = \text{diag}(1, -1)$ and $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Let $\mathcal{S}_e = E_e \otimes_{\mathbb{R}} \mathbb{C}$ be the complexification of the real plane. The scalar unit i belongs to the field of this representation; J remains the real endomorphism arising from the projectors.

Proposition 21.7 (Hermitian triple and representation of rotations). *In $\mathcal{S}_e$, define*

$$\sigma_1 = SJ, \quad \sigma_2 = -iJ, \quad \sigma_3 = S. \quad (21.11)$$

These operators are Hermitian and traceless, and satisfy

$$\sigma_a \sigma_b = \delta_{ab} I + i \sum_c \epsilon_{abc} \sigma_c, \quad L_a = \sigma_a / 2, \quad [L_a, L_b] = i \sum_c \epsilon_{abc} L_c, \quad \sum_a L_a^2 = \frac{3}{4} I. \quad (21.12)$$

They constitute the irreducible spin-1/2 representation on the electronic block.

Proof. The relations of S, J give $(SJ)^* = SJ$, $(-iJ)^* = -iJ$, and $\sigma_a^2 = I$. The cyclic products are $(SJ)(-iJ) = iS$, $(-iJ)S = iSJ$, and $S(SJ) = J = i(-iJ)$; reversing the order reverses the sign. This proves the product identity and the commutators. In the basis of (21.10), the three matrices are

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Their trace is zero and, together with I , they generate $M_2(\mathbb{C})$. A subspace invariant under all three is invariant under that entire algebra and can only be 0 or $\mathcal{S}_e$. Finally, summing $L_a^2 = I/4$ gives the value $3I/4$ of the quadratic operator. $\square$

Recognition of this triple as the Pauli matrices follows (21.10). The norm and product $\langle X, Y \rangle = \frac{1}{2} \text{tr}(XY)$ on the real space of traceless Hermitian matrices give its space of directions. For $n \in \mathbb{R}^3$, $|n| = 1$, let

$$H_e(n) = \sum_a n_a \sigma_a, \quad h_e(n) = \frac{1}{2} H_e(n), \quad \Pi_{\pm}(n) = \frac{1}{2} (I \pm H_e(n)). \quad (21.13)$$

Proposition 21.8 (Directional decomposition and double return). *The operators $\Pi_{\pm}(n)$ are rank-one orthogonal projectors and $h_e(n)\Pi_{\pm}(n) = \pm \frac{1}{2}\Pi_{\pm}(n)$. The rotation group around n is*

$$U_n(\theta) = e^{-i\theta H_e(n)/2} = \cos(\theta/2)I - i\sin(\theta/2)H_e(n), \quad U_n(2\pi) = -I, \quad U_n(4\pi) = I. \quad (21.14)$$

Proof. The antisymmetric part of (21.12) vanishes when contracted with $n_a n_b$, so $H_e(n)^2 = I$. Idempotence, orthogonality, and the sum $\Pi_+ + \Pi_- = I$ follow. They are Hermitian with trace one, hence their rank is one. Multiplication by $H_e(n)/2$ gives the eigenvalues. Separating even and odd powers of the exponential produces (21.14) and its group law. Angular normalisation is checked by

$$U_n(\theta)H_e(v)U_n(\theta)^* = H_e(v \cos \theta + (n \times v) \sin \theta + n(n \cdot v)(1 - \cos \theta)),$$

which follows from multiplication using the product identity of the triple. The induced action on directions is a rotation through angle θ ; the spinor lift carries its half-angle. $\square$

If n is realised as the direction of propagation, $h_e(n)$ is dimensionless helicity. Reversal of that direction satisfies $H_e(-n) = -H_e(n)$ and $\Pi_{\pm}(-n) = \Pi_{\mp}(n)$. This operation changes the directional labels; charge conjugation transports the material record through (21.3). Chirality is defined through the relativistic grading appropriate to the representation and its projectors, which differ from (21.13).

The scalar electronic evaluation acts as $\varepsilon_e^\beta I$ on S_e and commutes with the triple and with $\Pi_{\pm}(n)$. Rotational information preserves the structure of that evaluation without introducing another factor into mass. The Möbius band, in turn, transports a real line around a loop, and nonadic holonomy transports phase and memory. The three returns can enter the same realisation; their domains and intertwiners determine the precise composition.

22 From the trajectory record to the mass operator

The construction distinguishes two operations. First, the dynamics produces a trajectory record. Then a functional extracts integer coordinates and evaluates them through the internal constants already obtained. Preserving that order explains how mass can be a magnitude of persistence without exhausting its structure.

22.1 Chronological reading and oriented events

Let $d_1, \dots, d_{108}$ be the digital reading of an enriched route. In the stated chart, the action support is $\{1, 3, 5, 7\}$, and

$$n_A = \#\{t : d_t \in \{1, 3, 5, 7\}\}.$$

The 12 events of the 9-step blocks are also preserved:

$$e_m = \sigma_m \#\{t : 9m + 1 \leq t \leq 9m + 9, d_t \in \{2, 4, 6, 8\}\}, \quad 0 \leq m < 12,$$

where $\sigma = (+, +, +, -, -, -, +, +, +, -, -, -)$. The support of this extractor is an explicit part of the reading chart; its function must not be confused with the ternary phase selector of TPK.

The distribution of events matters even when their sums coincide. To see this, introduce

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}, \quad s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5).$$

Proposition 22.1 (Integral reconstruction of events). *The transformation*

$$(e_0, \dots, e_{11}) \mapsto (s_C, M_0, \dots, M_4, a_0, \dots, a_5)$$

is injective on $\mathbb{Z}^{12}$. Its image is characterised by

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6),$$

where $p_5 = (s_C - \sum M_i)/6$ and $p_i = p_5 + M_i$ for $i < 5$.

Proof. Adding the six expressions for p_i gives $s_C = 6p_5 + \sum_{i<5} M_i$. This recovers the p_i , and then

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}.$$

The congruences are necessary for these quantities to be integers and sufficient for the formulas to define an integral preimage. Substitution into the forward transformation recovers every coordinate. $\square$

The decomposition $12 = 1 + 5 + 6$ thus acquires an operational meaning: the total, differences, and orientations recover the events. Division by 6 belongs to this reconstruction. When the character uses $s_C/6$, s_C remains the integer sum; a coordinate already divided is not divided a second time.

22.2 Torsional memory and signature

For every occurrence of 9, the reading preserves its predecessor. Define, with cyclic indices,

$$j_2(d) = \begin{cases} 0, & d = 9, \\ 1, & d \in \{2, 4, 6, 8\}, \\ -1, & d \in \{1, 3, 5, 7\}, \end{cases} \quad z_t = \mathbf{1}_{\{d_t=9\}} j_2(d_{t-1}).$$

Then

$$T_z = \sum_{t=1}^{108} z_t, \quad C_z = \sum_{t \in \{27, 54, 81, 108\}} z_t, \quad k_{\Delta_4} = T_z - 2C_z.$$

The first sum preserves accumulated memory, and the second distinguishes corona closures. Removing the predecessors before evaluating these expressions changes the object being read.

With $\tau_m = \text{sgn}(e_m)$, the remaining coordinates are

$$b_{120} = \sum_{a=0}^3 \text{sgn}(e_a + e_{a+4} + e_{a+8}), \quad b_\zeta = \sum_{m=0}^{11} \tau_m \tau_{m+3},$$

where the second index is modulo 12. The signature obtained is

$$L_{\text{tick}}(\gamma) = \sigma(\gamma) = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta) \in \mathbb{Z}^5.$$

This map is deterministic in the stated chart. The signs and correlations explain why composition must take place on the record before extracting the 5 integers: adding signatures does not, in general, represent the composition of two histories.

22.3 Character and internal specialisation

Let $X = (X_A, X_C, X_4, X_{120}, X_{270})$. The formal character is

$$\chi(\sigma; X) = \exp(n_A X_A + s_C X_C + k_{\Delta_4} X_4 + b_{120} X_{120} + b_\zeta X_{270}).$$

For each X , it satisfies $\chi(\sigma + \sigma'; X) = \chi(\sigma; X) \chi(\sigma'; X)$ because the exponent is linear in the signature. This is a law on signatures; it does not assert that the extractor is additive on routes without preserving the interface.

Specialisation uses the internal coordinates constructed previously:

$$X_A = A_{\text{rad}}, \quad X_C = C_{\text{rad}}^*/6, \quad X_4 = \Delta_4, \quad X_{120} = s_{120}, \quad X_{270} = \zeta_{270} := 135\Delta_4^2.$$

The character is dimensionless and positive. Its effect is a homothety on the mass line. The exponential form determines how the coefficients act; their provenance is fixed by the preceding records and constructions.

22.4 Bilayer hierarchy and preservation of the two genealogies

Write $x = A_{\text{rad}}, y = C_{\text{rad}}^*$. In the chamber $x > y > 0$, the channels are

$$q_+ = e^{-x-y}, \quad q_- = e^{-x+y}, \quad 0 < q_+ < q_- < 1.$$

For an involution R , with $P_{\pm} = (I \pm R)/2$, the two-channel operator is

$$T = q_+ P_+ + q_- P_-.$$

Its scalar and oriented moments are expressed as

$$c_n = q_-^n + q_+^n, \quad s_n = q_-^n - q_+^n.$$

Multiplying these expressions proves

$$c_n^2 - s_n^2 = 4e^{-2nx}, \quad c_{m+n} = \frac{c_m c_n + s_m s_n}{2}, \quad s_{m+n} = \frac{s_m c_n + c_m s_n}{2}.$$

Both sequences satisfy

$$X_{n+2} = c_1 X_{n+1} - e^{-2x} X_n, \quad X \in \{c, s\}.$$

These relations show that the heights share the same operator. A different collection of moments is not defined for every species.

Proposition 22.2 (Semigroup of heights).

$$\langle 90, 120 \rangle = \{90, 120\} \cup \{30r : r \geq 6\}.$$

Proof. After division by 30, it suffices to study $\langle 3, 4 \rangle$. The integers 6, 7, 8 are $3 + 3, 3 + 4, 4 + 4$; adding 3 produces every subsequent integer. Below 6, the only positive elements are 3, 4. $\square$

The heights $360 = 4 \cdot 90 = 3 \cdot 120$ illustrate the distinction between equality of evaluation and memory of composition. The algebraic quotient identifies the two exponents; the record preserves whether transport used four stages of one type or three of the other.

The refined specialisation has exponent

$$\log \chi_{\text{ref}} = n_A x + \frac{s_C}{6} y + k_{\Delta_4} \Delta_4 + b_{\zeta} \zeta_{270} + \sum_{h \in \langle 90, 120 \rangle} N_h s_h, \quad N_h = \eta_h + b_{120} \mathbf{1}_{\{h=120\}}.$$

The pair (η_{120}, b_{120}) is preserved even though evaluation uses its sum. s_{270} and ζ_{270} also remain separate: the former is a bilayer moment, the latter a quadratic expression of refinement.

Proposition 22.3 (Convergence of the refined evaluation). *If $\sum_h |N_h s_h| < \infty$, the refined character defines a finite positive scalar, and its truncated evaluations converge to it.*

Proof. The assumption gives absolute convergence of the exponent. Continuity of the exponential permits passage to the limit; the exponential of a finite real number is strictly positive. $\square$

Convergence evaluates a sequence of coefficients already specified. Selection of those coefficients by the route is a distinct operation and must accompany every evaluation attributed to a species.

The notation χ_{full} retained in some sector evaluation formulas denotes this same character χ_{ref} , with the same arguments and coefficients. The change of symbol adds no further specialisation or tower.

22.5 Displacement and autocorrelation readers

The same route admits two readings of the return:

$$K_D = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2} \right), \quad K_\Omega = (\Omega, 54, P_6).$$

The first map preserves displacement defects; the second, return memory and autocorrelation. In the complete census of 324 orientations there are 14 profiles of the first, 9 of the second, and 40 joint pairs. The 18 coincidences have value (80, 54, 6). Equality of the two operators throughout the atlas is not inferred from this.

For the electronic route, the displacements are

$$(D_1, D_3, D_4, D_{27}, D_{36}) = (54, 56, 68, 48, 32).$$

Substitution produces $K_D = (80, 54, 6)$, coinciding with K_Ω . Its exponent is

$$\kappa_e = 80 - 54\alpha + 6\Delta_4.$$

The electronic trajectory is constructed in (19.34); its displacement and autocorrelation readers are evaluated, respectively, in (19.35) and (19.36). The coincidence $K_D = K_\Omega = (80, 54, 6)$ belongs to the beta-closure theorem in Subsection 19.9. Here that central exception is preserved, and the operation is extended to the atlas fibres.

22.6 Operator on a fibre and dimensional section

Let $\mathcal{H}_M$ be the finite Hilbert space with orthonormal basis (e_γ) labelled by the routes of a multisection. For $r \in \{D, \Omega\}$, define the return operator by $B_M^r e_\gamma = \kappa_\gamma^r e_\gamma$, with real exponents; it is therefore self-adjoint. Let $M_M^{(0)}$ be the positive basal operator, whose connection with the character is fixed below. The dimensionless ratio of the action sections is $R_{\text{act}} > 0$. Define

$$M_M^{\beta,r} = R_{\text{act}}^{B_M^r/2} M_M^{(0)} R_{\text{act}}^{B_M^r/2}.$$

Proposition 22.4 (Positivity and scalar restriction). *The operator $M_M^{\beta,r}$ is positive. If both operators are diagonal in the route basis, each basal value $m_\gamma^{(0)}$ is transformed into*

$$m_\gamma^\beta = m_\gamma^{(0)} R_{\text{act}}^{\kappa_\gamma}.$$

Proof. Let $S = R_{\text{act}}^{B_M^r/2}$, self-adjoint and invertible. For every v , $\langle v, SM_M^{(0)} Sv \rangle = \langle Sv, M_M^{(0)} Sv \rangle \geq 0$. In the diagonal basis, the two factors S multiply each value by $R_{\text{act}}^{\kappa_\gamma/2}$; their product gives the formula. $\square$

The absolute mass section is obtained from action and the clock:

$$\omega_0 = \frac{2\pi}{108t_0}, \quad m_{\text{clk}} = \hbar_{\text{ret}} \omega_0 c_{\text{int}}^{-2}.$$

In the chart normalised relative to the electron, the preceding basal operator is

$$M_M^{(0)} e_\gamma = m_e^{(0)} \chi_{\text{ref}}(\sigma_\gamma - \sigma_e, \eta_\gamma - \eta_e) e_\gamma.$$

Here $m_e^{(0)}$ is the realisation of the basal electronic coordinate, with its prefactor, refinement, and dimensional basis. If $E_e^{(0)} = \varepsilon_e^{(0)} \mathcal{U}_E$, then

$$m_e^{(0)} = \varepsilon_e^{(0)} \mathcal{U}_E c_{\text{int}}^{-2} = b_e m_{\text{clk}}, \quad b_e = \varepsilon_e^{(0)} \frac{\mathcal{U}_E}{E_{\text{clk}}}.$$

In the chart $\mathcal{U}_E = E_{\text{clk}}$, we obtain $b_e = \varepsilon_e^{(0)}$; in another chart, the stated ratio of bases is retained. If the tower coordinates are already relative to the electron, a second subtraction of η_e is omitted. This equality

links the operator to the signature of each route and explicitly preserves electronic normalisation. A different choice of reference section must also transport that normalisation. The scale of the reduced Planck constant belongs to this dimensional construction. The character and return factor subsequently act as scalars. The electron preserves its structure: it is not identified with the chronological quantum m_{clk} .

For two routes in the same absolute chart,

$$\frac{m_Y^\beta}{m_X^\beta} = \chi_{\text{ref}}(\sigma_Y - \sigma_X, \eta_Y - \eta_X) R_{\text{act}}^{\kappa_Y - \kappa_X}.$$

The common dimensional section cancels. The difference of the return exponents does not cancel unless it is zero. This identity preserves the distinction between the absolute scale and the relative structure of a collection.

22.7 The null propagation sector

The exponential takes strictly positive values. The null chart is constructed before that character, in the split realisation

$$\mathcal{A}_{\text{sp}} = \mathbb{R}[R]/(R^2 - 1), \quad z = r_+ P_+ + r_- P_-, \quad N(z) = r_+ r_-.$$

The directions $\mathbb{R}P_+$ and $\mathbb{R}P_-$ have zero norm. Propagation transported in one of them belongs to that sector; the bilayer material reading uses the coupling of both. In this way, the photon and the gluonic sector retain their place in the theory even though they do not form a row of the positive character. The realisation of their representations is presented alongside the corresponding sectors.

23 Action, gravity, and thermal information

23.1 Dodecaphasic period and circular realization of action

The calendar already constructed determines $T_{108} = 108t_0$. In its circular realization,

$$\omega_{108} = \frac{2\pi}{108t_0}, \quad R_{108} = \frac{c_{\text{int}}}{\omega_{108}} = \frac{54}{\pi} \ell_0, \quad \ell_0 = c_{\text{int}} t_0.$$

The unit ℓ_0 and the time and velocity charts remain explicit; the SI chart does not select the cycle. The longitudinal composition in Section 47.1 fixes $R_{108} = L_\alpha$ and $t_0 = \pi_{\text{HMT}} L_\alpha / (54c_{\text{int}})$ in this edition. The following circular argument is retained as a specialization of the unit character in Theorem 47.2. For each oriented action section $a \in \{\text{pre}, \text{ret}\}$,

$$E_{108,a} = \hbar_a \omega_{108}, \quad m_{108,a} = \frac{E_{108,a}}{c_{\text{int}}^2}.$$

The reduced and full lengths are, respectively,

$$\frac{\hbar_a}{m_{108,a} c_{\text{int}}} = R_{108}, \quad \frac{h_a}{m_{108,a} c_{\text{int}}} = 2\pi R_{108}.$$

The same action and duration enter both readings.

Proposition 23.1 (Radius-coincidence selector). *In the dimensional collection*

$$G_a(\vartheta) = \vartheta \frac{c_{\text{int}}^5 t_0^2}{\hbar_a}, \quad \vartheta > 0,$$

the circular-realization condition $G_a(\vartheta) m_{108,a} / c_{\text{int}}^2 = R_{108}$ has the unique solution

$$\vartheta_{108}^\circ = (54/\pi)^2.$$

Proof. The first radius is $r_g(\vartheta) = \vartheta(\pi/54)\ell_0$. Comparing it with $(54/\pi)\ell_0$, with $\ell_0 > 0$, reduces the equality to a strictly increasing linear equation. $\square$

The coincidence condition is an explicit part of this realization; the theorem solves for its coefficient without using a metrological value of G . The radius used here is Gm/c^2 , so the factor two from the convention $2Gm/c^2$ is not inadvertently transferred.

Proposition 23.2 (Reciprocal action and gravity, preserved area). *In the two preceding charts,*

$$\frac{G_{\text{pre}}}{G_{\text{ret}}} = R_{\text{act}}^{-1}, \quad \hbar_a G_a = \vartheta_{108}^5 c_{\text{int}}^5 t_0^2, \quad \ell_P^2 = \frac{\hbar_a G_a}{c_{\text{int}}^3} = \vartheta_{108}^2 \ell_0^2.$$

Proof. The definition of G_a contains $\hbar_a^{-1}$, and the remaining factors are common to both charts. Taking the quotient and product gives the three equalities. $\square$

23.2 Structural definition and function of the gravitational constant

Definition 23.3 (Gravitational section contragredient to action). For a positive section $\hbar_a$, a velocity $c_{\text{int}} > 0$, and the chronological radius R_{108} of section 23, the circular gravitational section is

$$G_a = \frac{c_{\text{int}}^3 R_{108}^2}{\hbar_a}, \quad R_{108} = \frac{54}{\pi} \ell_0, \quad \ell_0 = c_{\text{int}} t_0. \quad (23.1)$$

It is the action-reciprocal section realizing coincidence between the cycle's reduced gravitational radius and its chronological radius, in the convention $r_g = Gm/c_{\text{int}}^2$.

The radius-coincidence proposition of section 23 proves its uniqueness in the collection $G_a(\vartheta) = \vartheta c_{\text{int}}^5 t_0^2 / \hbar_a$. The circular condition fixes $\vartheta = (54/\pi)^2$ and leads to (23.1). That condition and the positive dimensional sections are part of the realization. Section 47 constructs its length and brings together the uniqueness proof for the general positive character, with $R_{108} = L_\alpha$. The circular selector and longitudinal readout operator therefore preserve the same normalization. Keeping them common to both orientations yields

$$\frac{G_{\text{pre}}}{G_{\text{ret}}} = R_{\text{act}}^{-1}, \quad \frac{G_a \hbar_a}{c_{\text{int}}^3} = R_{108}^2 = \ell_P^2. \quad (23.2)$$

The proof is direct substitution of (23.1); the action change is compensated by the reciprocal change in G_a .

24 Reduced information, energy, and thermal coordinates

24.1 Reader, memory, and state recovery

Fix a finite level of the development, and denote by Ω the set of admissible enriched states. The record preserves sheet, orientation, residue, quotient, carry, route, phase, and memory in the domains where each coordinate enters. Let

$$q : \Omega \longrightarrow Y, \quad M : \Omega \longrightarrow \mathcal{M}$$

be the visible representation and retained memory, respectively. For a distribution p on Ω , write $H(p) = -\sum_x p_x \log p_x$, with $0 \log 0 = 0$. The logarithm is applied to dimensionless probabilities.

Definition 24.1 (Genealogical recovery of a representation). The pair $\hat{q} = (q, M)$ is a recoverable reader on support Ω_p when there exists

$$d : \hat{q}(\Omega_p) \longrightarrow \Omega_p, \quad d \circ \hat{q} = \text{id}_{\Omega_p}.$$

The fiber information of q is $H_{\text{fib}}(q; p) = H(X \mid q(X))$, where X has law p .

Proposition 24.2 (Recovery identity and cost of reduction). *For a recoverable reader,*

$$H(X | q(X), M(X)) = 0, \quad (24.1)$$

$$H(X) = H(q(X)) + H(X | q(X)), \quad (24.2)$$

$$H(X | q(X)) = H(M(X) | q(X)). \quad (24.3)$$

Proof. The map d recovers X from the pair $(q(X), M(X))$, so the first conditional entropy is zero. Since q is a function of X , the chain rule gives the second equality. For the third, expand $H(X, M(X) | q(X))$ in two ways. One gives $H(X | q(X))$, because M is a function of X ; the other gives $H(M(X) | q(X)) + H(X | q(X), M(X))$. $\square$

Euclidean division of the APP sheets displays the elementary arithmetic mechanism. With $i, j \in \{1, \dots, 9\}$, one preserves

$$i + j = 9q_{ij}^+ + R_{ij}^+, \quad ij = 9q_{ij}^\times + R_{ij}^\times, \quad 1 \leq R_{ij}^\sigma \leq 9 \quad (\sigma \in \{+, \times\}).$$

The residue–quotient pair recovers the integer value of the operation. Summing the differences over the eighty-one positions gives

$$2025 - 810 = 54 + 9 \cdot 129.$$

Quotient memory restores the contribution not expressed by residual reduction. Recoverability of the complete state additionally requires preserving the relevant position and route coordinates; the residue–quotient pair here identifies the result of the operation.

24.2 Reversible update, tritic decision, and erasure

Let $U : \Omega \rightarrow \Omega$ be the TPK update on a domain where enriched transport has an inverse. The transported law is $p'_y = p_{U^{-1}y}$. The bijection preserves the multiset of probabilities, and hence

$$H(U(X)) = H(X), \quad H(X | U(X)) = 0. \quad (24.4)$$

If the observed output is $q(U(X))$, the unpublished information is quantified by $H(X | q(U(X)))$. The complete state, its representation, and the reset of a record are thus three distinct maps.

For the three oriented TRIT states, let $p = (p_{-1}, p_0, p_{+1})$ be a normalized distribution. Its information is

$$I_{\text{TRIT}}(p) = - \sum_{a=-1}^1 p_a \log p_a, \quad 0 \leq I_{\text{TRIT}}(p) \leq \log 3. \quad (24.5)$$

The upper bound corresponds to $p_a = 1/3$. Indeed,

$$D(p || (1/3, 1/3, 1/3)) = \sum_a p_a \log(3p_a) = \log 3 - I_{\text{TRIT}}(p) \geq 0;$$

the inequality follows from convexity of $t \log t$, and equality characterizes the uniform distribution. The lower bound corresponds to a distribution concentrated on one state.

The ideal Landauer realization uses a memory with energetically degenerate logical states, an isothermal environment at temperature Θ , and a balance that preserves the logical map's output and accounts for auxiliary records. In that model, the decrease in logical entropy is $k_B[H(X) - H(F(X))] = k_B H(X | F(X))$ for a deterministic map F . Entropy balance requires transferring at least that amount to the environment; its energy expression is $Q \geq k_B \Theta H(X | F(X))$. The distinction between reversible logical evolution and physical erasure corresponds to the Landauer–Bennett realization [15]. The invertible update (24.4) has zero logical contribution; resetting a uniform triple requires transferring $\log 3$ nats to the environment and satisfies $Q \geq k_B \Theta \log 3$. Control energy and dissipation in a particular physical realization belong to its additional thermodynamic balance.

24.3 Action, frequency, and decision energy

Let $\mathcal{L}_{\text{act}}$ and $\mathcal{L}_T$ be the dimensional lines of action and duration; the energy line is $\mathcal{L}_E = \mathcal{L}_{\text{act}} \otimes \mathcal{L}_T^*$. The preceding development supplies the positive sections $\hbar_a \in \mathcal{L}_{\text{act}}$, with $a \in \{\text{pre}, \text{ret}\}$, and the elementary duration $t_0 \in \mathcal{L}_T$. The complete calendar provides

$$T_{108} = 108t_0, \quad \omega_{108} = \frac{2\pi}{T_{108}}, \quad h_a = 2\pi\hbar_a.$$

The dimensional contraction between action and frequency defines

$$\epsilon_a := \hbar_a \omega_{108} = \frac{h_a}{108t_0} = \frac{\hbar_a}{t_P}, \quad t_P = \frac{54}{\pi}t_0. \quad (24.6)$$

The last member uses the same reduced circular period as the gravitational section. The three expressions correspond to one energy section, with an orientation a maintained in all factors. The normalized information of a uniform decision produces the energy-per-nat section

$$\epsilon_a := \frac{\epsilon_a}{\log 3} \in \mathcal{L}_E^+. \quad (24.7)$$

The previously constructed action determines the energy scale once duration is fixed; the cardinality of TRIT alternatives determines its information normalization. The value $\log 3$ enters through the count and measure declared in (24.5).

24.4 Dimensional definition of Boltzmann and coordinate selection

Let $\mathcal{L}_\Theta$ be the oriented temperature line. The dimensional entropy space is $\mathcal{L}_{\text{ent}} = \mathcal{L}_E \otimes \mathcal{L}_\Theta^*$. A positive linear map $k_B : \mathcal{L}_\Theta \rightarrow \mathcal{L}_E$ is, equivalently, a positive element of $\mathcal{L}_{\text{ent}}$. The same object can therefore act on a temperature or multiply dimensionless information.

Definition 24.3 (Thermal normalization of the tritic decision). A thermal pair compatible with oriented section a consists of a positive transducer k_B and a positive section $\Theta_{\text{clk},a} \in \mathcal{L}_\Theta$ such that

$$k_B(\Theta_{\text{clk},a}) = \epsilon_a, \quad k_B \Theta_{\text{clk},a} \log 3 = \epsilon_a. \quad (24.8)$$

The Boltzmann constant serves as the energy–temperature transducer of tritic information in this realization.

Theorem 24.4 (Uniqueness relative to the thermal section and dimensional covariance). *With a positive section $\Theta_{\text{clk},a}$ fixed, there is a unique positive linear map satisfying (24.8). For positive bases u_E, u_Θ , if*

$$\epsilon_a = \epsilon_a^{\text{num}} u_E, \quad \Theta_{\text{clk},a} = \theta_a u_\Theta, \quad k_B(u_\Theta) = b u_E,$$

its coordinate is

$$b = \frac{\epsilon_a^{\text{num}}}{\theta_a \log 3}. \quad (24.9)$$

Under $u'_E = s_E u_E$, $u'_\Theta = s_\Theta u_\Theta$, with $s_E, s_\Theta > 0$, one has

$$b' = \frac{s_\Theta}{s_E} b, \quad \theta'_a = \frac{\theta_a}{s_\Theta}, \quad (\epsilon_a^{\text{num}})' = \frac{\epsilon_a^{\text{num}}}{s_E}.$$

Proof. A nonzero section spans a line. Prescribing its image determines the unique linear map; positivity of both sections proves its positivity. Substituting the coordinates into (24.8) gives $b\theta_a \log 3 = \epsilon_a^{\text{num}}$. The three transformation laws follow by expressing the same sections and the same map in the new bases. $\square$

The same Boltzmann constant must act in both orientations. Since t_P is common, simultaneous compatibility is equivalent to

$$\frac{\Theta_{\text{clk,pre}}}{\Theta_{\text{clk,ret}}} = \frac{\epsilon_{\text{pre}}}{\epsilon_{\text{ret}}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}}. \quad (24.10)$$

Indeed, an isomorphism of lines preserves quotients of sections. Conversely, if the preceding equality holds, the isomorphism determined in one orientation also satisfies the normalization of the other. Thus return modifies the thermal section of the cycle in the same proportion as energy, while the transducer remains common to both readings.

The preceding equation makes explicit the choice required to express an individual value. If the thermal section has not yet been selected by an independent reader, then for every $\lambda > 0$ the pair

$$(k_B, \Theta_{\text{clk},a}) \longmapsto (\lambda k_B, \lambda^{-1} \Theta_{\text{clk},a})$$

preserves ϵ_a . Consequently, presenting an internally selected individual coordinate requires providing the reader of $\Theta_{\text{clk},a}$, the bases, and their dimensional transport. The action and information equations remain determined before that specification. Subsequent identification of the bases with joules and kelvins belongs to the metrological representation of the result.

24.5 Dimensional entropy and information of the reduced state

For a distribution p of admitted routes, or a reduced state with density matrix ρ , let

$$s(p) = - \sum_x p_x \log p_x, \quad s(\rho) = - \text{Tr}(\rho \log \rho).$$

Their dimensional representations are

$$S(p) = k_B s(p), \quad S(\rho) = k_B s(\rho) \quad \text{in } \mathcal{L}_{\text{ent}}. \quad (24.11)$$

Evaluation at Θ produces an energy $\Theta S = k_B \Theta s$. For the uniform triple at the cycle temperature, (24.8) gives exactly

$$S_{\text{TRIT}} = k_B \log 3, \quad \Theta_{\text{clk},a} S_{\text{TRIT}} = \epsilon_a.$$

25 Algebraic construction of quantum completions over APP

APP and TPK provide oriented states, transport, and sheet parity. Their linear realization enables construction of the symmetric and exterior algebras, with their creation, annihilation, and occupation operators. This section establishes these algebraic constructions, the cellular field sector, and reversible measurement coupling. Application of the occupation spectra to equilibrium is developed in Section 26. Each stage retains the conditions of the representation it uses.

25.1 Linear state space associated with a particle realization

Let S be a finite set of TPK route classes at the chosen depth, and let

$$\mathcal{H}_1 = \ell^2(S)$$

be the Hilbert space with orthonormal basis $(e_s)_{s \in S}$. A choice of positive weights E_s defines the diagonal operator

$$H e_s = E_s e_s.$$

The choice of S , its quotient by symmetries, and the weights are part of the model. Once these are fixed, the multiparticle completions are canonical.

25.2 Symmetric and exterior completions

Define

$$\mathcal{F}_+(\mathcal{H}_1) = \bigoplus_{n \geq 0} \text{Sym}^n \mathcal{H}_1, \quad \mathcal{F}_-(\mathcal{H}_1) = \bigoplus_{n \geq 0} \bigwedge^n \mathcal{H}_1.$$

The first is the bosonic completion; the second is the fermionic completion. In both, the degree-zero summand is the vacuum space. The exterior powers are equipped with the inner product for which ordered wedges of vectors from an orthonormal basis are orthonormal; the symmetric powers are used with the usual normalization of creation and annihilation operators.

25.3 Sheet grading and exchange algebra

Suppose that the oriented sheet defines an additive character

$$p : \text{Mor}(\text{P}_{\text{TPK}}) \longrightarrow \mathbb{Z}/2\mathbb{Z}, \quad p(\gamma\eta) = p(\gamma) + p(\eta).$$

On the corresponding graded linear space, introduce the symmetry

$$\tau(v \otimes w) = (-1)^{p(v)p(w)} w \otimes v.$$

Theorem 25.1 (Statistical completion determined by parity). *The quotient of the tensor algebra by the relations*

$$v \otimes w - (-1)^{p(v)p(w)} w \otimes v$$

is symmetric on the even sector and exterior on the odd sector. Two identical odd states have zero product; even states admit arbitrary occupation.

Proof. If $p(v) = p(w) = 0$, the relation is $vw = wv$. If $p(v) = p(w) = 1$, it is $vw = -wv$; setting $v = w$ and working in characteristic zero gives $v^2 = 0$. Mixed sectors commute without an additional sign. This is the universal property of the graded symmetric algebra. $\square$

25.4 Canonical operators and idempotence of occupation

The parity character determines the completion; the product of that completion then determines the operator algebra. In the exterior sector, creation $a_s^\dagger$ is exterior multiplication by e_s , and a_s is its adjoint.

Theorem 25.2 (Pauli exclusion principle in the exterior completion). *For every mode $s \in S$, the creation and annihilation operators of the exterior completion satisfy*

$$\{a_s, a_t^\dagger\} = \delta_{st} I, \quad \{a_s, a_t\} = \{a_s^\dagger, a_t^\dagger\} = 0. \quad (25.1)$$

In particular, the occupation operator $N_s = a_s^\dagger a_s$ is a projector:

$$N_s^2 = N_s, \quad \text{Spec}(N_s) \subseteq \{0, 1\}. \quad (25.2)$$

The eigenvalues 0 and 1 correspond, respectively, to the absence and presence of mode s .

Proof. On an ordered wedge, a_s removes the factor e_s , when present, with the sign determined by its position; when absent, it gives zero. Composing this operation with exterior multiplication proves relations (25.1): for $s = t$, the two compositions cover the complementary cases of absence and presence; for $s \neq t$, the exchange signs cancel. Antisymmetry implies $e_s \wedge e_s = 0$ and $a_s^2 = (a_s^\dagger)^2 = 0$. Therefore

$$N_s^2 = a_s^\dagger a_s a_s^\dagger a_s = a_s^\dagger (I - a_s^\dagger a_s) a_s = a_s^\dagger a_s = N_s. \quad (25.3)$$

The operator is self-adjoint, and every eigenvalue of an idempotent satisfies $\lambda^2 = \lambda$. Hence its spectrum is contained in $\{0, 1\}$. The vacuum and the state e_s realize both values [13, 10]. $\square$

In the symmetric sector, the bosonic operators $b_s^\dagger, b_s$ act on the normalized occupation basis with factors $\sqrt{m_s + 1}$ and $\sqrt{m_s}$, respectively. Their compositions give

$$[b_s, b_t^\dagger] = \delta_{st} I, \quad [b_s, b_t] = [b_s^\dagger, b_t^\dagger] = 0. \quad (25.4)$$

These identities are understood on the algebraic finite-particle domain, which is invariant under the operators considered.

Theorem 25.1 turns a compositional TPK parity into an exact occupation rule. To obtain the physical spin–statistics theorem, one must also construct a positive local Lorentzian theory and prove that sheet parity agrees with spin under rotations. This equality does not follow merely from both objects taking values modulo two.

26 Bose–Einstein and Fermi–Dirac distributions

The equilibrium formulation starts from the Hamiltonian and the total occupation number,

$$H = \sum_k E_k^{\text{occ}} n_k, \quad N = \sum_k n_k, \quad K_\mu := H - \mu N. \quad (26.1)$$

For $\beta > 0$, the grand canonical state is

$$\rho_{\beta, \mu} = \frac{e^{-\beta K_\mu}}{\mathcal{Z}_{\beta, \mu}}, \quad \mathcal{Z}_{\beta, \mu} = \text{Tr}(e^{-\beta K_\mu}). \quad (26.2)$$

Theorem 26.1 (Grand canonical factorization and occupation laws). *For energy levels E_k^{occ} with degeneracies g_k , the fermionic trace factorizes as*

$$\mathcal{Z}_{\text{FD}} = \prod_k (1 + e^{-\beta(E_k^{\text{occ}} - \mu)})^{g_k}, \quad \langle n_k \rangle_{\text{FD}} = \frac{g_k}{e^{\beta(E_k^{\text{occ}} - \mu)} + 1}. \quad (26.3)$$

In the bosonic sector, under the condition $\mu < \inf_k E_k^{\text{occ}}$,

$$\mathcal{Z}_{\text{BE}} = \prod_k (1 - e^{-\beta(E_k^{\text{occ}} - \mu)})^{-g_k}, \quad \langle n_k \rangle_{\text{BE}} = \frac{g_k}{e^{\beta(E_k^{\text{occ}} - \mu)} - 1}. \quad (26.4)$$

Proof. The operator K_μ is a sum of commuting occupation operators, so its trace factorizes over modes. In each fermionic mode, the occupations 0 and 1 are summed, and the corresponding factor is $1 + u_k$, where $u_k = e^{-\beta(E_k^{\text{occ}} - \mu)}$. In each bosonic mode, all nonnegative occupations are summed, giving the geometric series $(1 - u_k)^{-1}$; the condition on μ ensures $u_k < 1$. The powers g_k incorporate the degeneracy. Finally,

$$\langle n_k \rangle = -\frac{1}{\beta} \frac{\partial}{\partial E_k^{\text{occ}}} \log \mathcal{Z}$$

yields the two occupation laws [9, 11, 12, 10, 14]. $\square$

26.1 Constitutive speed and kinematic realization of the clock

Radiative propagation uses the speed section produced by the angular transport of the same HMT state. The identification is established before representation units are chosen. Let $x = A_{\text{rad}}$ and $y = C_{\text{rad}}^*$ be the angular coordinates generated in Section 11.4, with the channels (11.15), and let R be the self-adjoint involution of the two sheets, with rank-one projectors P_+ and P_- . On the domain $x > |y|$, the transport and its channels are

$$T = \exp(-xI - yR) = q_+ P_+ + q_- P_-, \quad q_+ = e^{-x-y}, \quad q_- = e^{-x+y}, \quad 0 < q_\pm < 1. \quad (26.5)$$

The response of the temporal blocks of lengths 90 and 120 is

$$V = (I - T^{90})(I - T^{120})^{-1} = r_+ P_+ + r_- P_-, \quad r_\pm = \frac{1 - q_\pm^{90}}{1 - q_\pm^{120}}. \quad (26.6)$$

The construction starts from the inherited channels; the symbols x, y and $q_{\pm}$ denote their coordinates, not parameters fitted to a target speed.

Proposition 26.2 (Identity of the constitutive and kinematic realizations). *In the same dimensional chart, the clock section c_{int} and the constitutive-response section c_{vac} coincide:*

$$c_{\text{int}} = c_{\text{vac}} = \hat{c} u_C, \quad \hat{c} = (r_+ r_-)^{-1} = \exp(-\mathcal{E}), \quad (26.7)$$

where

$$\mathcal{E} = \log[(1 - q_+^{90})(1 - q_-^{90})] - \log[(1 - q_+^{120})(1 - q_-^{120})]. \quad (26.8)$$

The equality preserves sheet exchange.

Proof. The factors appearing in the logarithms are positive. Therefore,

$$\mathcal{E} = \log(r_+ r_-) = \log \det V, \quad \exp(-\mathcal{E}) = (\det V)^{-1}.$$

The determinant is computed here in the two-sheet realization with rank-one projectors. The kinematic definition $c_{\text{int}} = \exp(-\mathcal{E})u_C$ and the constitutive definition $c_{\text{vac}} = (r_+ r_-)^{-1}u_C$ therefore give the same section. Exchanging $P_+ \leftrightarrow P_-$ interchanges r_+ and r_- and preserves their product. Finally, $\det T = e^{-2x}$ corresponds to the elementary transport, whereas $\det V = r_+ r_-$ corresponds to its temporal response; the two determinants have distinct roles in this composition. $\square$

26.1.1 Trajectory length and elementary duration

The positive dimensional frame (u_S, u_C, u_T, u_Q) induces the length basis $u_L = u_C u_T$. For an enriched trajectory γ of $n = |\gamma|$ transitions with a constant channel, the additive readers are

$$T_{\text{int}}(\gamma) = n u_T, \quad L_{\text{int}}(\gamma) = n \hat{c} u_L. \quad (26.9)$$

Their additivity concerns the count and realization of each transition: enriched routes additionally preserve orientation and memory.

Corollary 26.3 (Normalization of the elementary length). *For $n > 0$, $L_{\text{int}}(\gamma)/T_{\text{int}}(\gamma) = c_{\text{vac}}$. In the normalization $t_0 = u_T$,*

$$\ell_0 = c_{\text{int}} t_0 = \hat{c} u_L. \quad (26.10)$$

In a general temporal chart $t_0 = \tau_0 u_T$, with $\tau_0 > 0$, the corresponding expression is $\ell_0 = \tau_0 \hat{c} u_L$.

Proof. Dividing the expressions in (26.9) cancels the integer n and uses $u_L/u_T = u_C$. The elementary formula is obtained by multiplying the speed section by t_0 . $\square$

Generalization to an edge-dependent channel preserves the additive formula $L_{\text{int}}(\gamma) = \sum_{e \in \gamma} \hat{c}(e) u_L$. Its quotient by $n u_T$ is the mean of the speeds of those edges. The homogeneous radiative realization uses the constant channel specified in (26.9).

26.1.2 Change of chart and adapted normalization

Proposition 26.4 (Covariance of the identification). *Let $u'_C = d u_C$ and $u'_T = \eta u_T$, with $d, \eta > 0$. The same speed and the same elementary duration have coordinates*

$$\hat{c}' = d^{-1} \hat{c}, \quad \tau'_0 = \eta^{-1} \tau_0, \quad u'_L = d \eta u_L. \quad (26.11)$$

In particular, if $c_V = \hat{c}_V u_C^V$ and $u_C^V = d u_C^{II}$, equality of the sections is exactly equivalent to $d \hat{c}_V = \hat{c}_{III}$.

Proof. Express each section in the two bases. This gives $\hat{c}'u'_C = \hat{c}u_C$ and $\tau'_0 u'_T = \tau_0 u_T$. Their product satisfies

$$\tau'_0 \hat{c}' u'_L = \frac{\tau_0 \hat{c}}{\eta} d\eta u_L = \tau_0 \hat{c} u_L = \ell_0.$$

The final equivalence follows by substituting the relation between the bases. $\square$

The adapted choice $d = \hat{c}$ gives $u'_C = c_{\text{vac}}$ and $\hat{c}' = 1$. This choice expresses the already constructed section in a basis adapted to it. The spectral coefficient in the internal chart remains $(r_+ r_-)^{-1}$; spectral inversion uses that internal chart. For example, retaining the action and charge bases, the constitutive coordinates transform as

$$\hat{\mu}' = d\hat{\mu}, \quad \hat{\varepsilon}' = d\hat{\varepsilon}, \quad \hat{\mu}' \hat{\varepsilon}' = (\hat{c}')^{-2}.$$

In the adapted chart, $\hat{\mu}' = r_-/r_+$ and $\hat{\varepsilon}' = r_+/r_-$, since the internal coordinates are $\hat{\mu} = r_-^2$ and $\hat{\varepsilon} = r_+^2$.

27 Oriented radial recovery and symplectic transport

27.1 Recovery of Planck's action with orientation preserved

With the labelled semiaxes of the ellipse, define $R_\star = \sqrt{b_S/a_S}$. Then $q = R_\star^4 = e^{-2\eta_{\text{el}}} = (1 - \xi)/(1 + \xi)$, where $\xi = C^*/A$ and $\eta_{\text{el}} = \text{artanh } \xi$. This radial ratio retains the markings of both axes.

Theorem 27.1 (Radial recovery map for the return section). *In the positive chart, if $q = R_\star^4$, then*

$$\eta_{\text{ret}} = \alpha \frac{499 - 501q}{1 + q}, \quad (27.1)$$

$$\hbar_{\text{ret}} = \alpha \frac{499 - 501q}{1 + q} 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad (27.2)$$

$$R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5(1 + q)}{\alpha(499 - 501q)}. \quad (27.3)$$

The branch $\eta_{\text{ret}} > 0$ is equivalent to $q < 499/501$, within $q > 0$.

Proof. The angular identities give $\eta_{\text{ret}} = \alpha(500\xi - 1)$. Substitute $\xi = (1 - q)/(1 + q)$; the numerator is $499 - 501q$. Both the denominator and α are positive. The dimensional realization and the ratio are those of the action sections already constructed. The decade -34 and the unit $\mathcal{U}_S$ retain their preceding determinantal derivation. $\square$

This formula makes the Planck section within the counter-angle explicit. In terms of the action quantities, the angular relation is

$$C^* = 2 \left(\frac{\hbar_{\text{ret}}}{10^{-34} \mathcal{U}_S} + \alpha \right).$$

Division by the dimensional basis extracts the dimensionless coefficient; it does not add an action quantity to a dimensionless constant.

For the angular orientation $\varepsilon \in \{-1, +1\}$, define

$$C_\varepsilon^* = 2\varepsilon(\eta_{\text{ret}} + \alpha), \quad \xi_\varepsilon = C_\varepsilon^*/A, \quad q_\varepsilon = \frac{1 - \xi_\varepsilon}{1 + \xi_\varepsilon} > 0.$$

The recovery map on both charts is

$$\eta_{\text{ret}} = \alpha \left(500\varepsilon \frac{1 - q_\varepsilon}{1 + q_\varepsilon} - 1 \right). \quad (27.4)$$

Corollary 27.2 (Invariance of the conjugate recovery map). *The transformation $(\varepsilon, q) \mapsto (-\varepsilon, q^{-1})$ preserves η_{ret} , $\hbar_{\text{ret}}$, and R_{act} .*

Proof. The ratio $(1 - q)/(1 + q)$ changes sign when q is inverted. Its product with ε remains invariant. Apply (27.4), followed by the action representations. $\square$

If orientation is discarded and $\varepsilon = +1$ is incorrectly held fixed, the same expression gives $\eta_{\text{ret}}(q^{-1}) = -\eta_{\text{ret}}(q) - 2\alpha$. The register therefore distinguishes two algebraically different operations. Nor is the angular orientation ε automatically identified with the symplectic sign σ : exchanging the axes is antisymplectic, whereas the rotation that exchanges them preserves the area form.

27.2 Intertwining phase and anisotropy

Let

$$S_\eta = \text{diag}(e^{\eta/2}, e^{-\eta/2}), \quad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J_\eta = S_\eta J_0 S_\eta^{-1} = \begin{pmatrix} 0 & -e^\eta \\ e^{-\eta} & 0 \end{pmatrix}.$$

Then $\det S_\eta = 1$, $J_\eta^2 = -I$, and $C_\eta(\theta) = e^{\theta J_\eta} = S_\eta e^{\theta J_0} S_\eta^{-1}$. The law $S_{\eta+\delta} = S_\delta S_\eta$ proves the exact intertwining identity

$$C_{\eta+\delta}(\theta) S_\delta = S_\delta C_\eta(\theta). \quad (27.5)$$

The transformation preserves $dQ \wedge dP$ and carries the phase evolution to the new ellipse. With the normalization of (12.4), the swept action is $\hbar\theta$, or $\sigma\hbar\theta$ on the oriented sheet. The phase θ , the anisotropy η_{el} , and the parameter t of the flow of K thus retain distinct functions, related by explicit operators.

28 Positive domain of the elliptic chart

Proposition 28.1 (Membership of the representations in the elliptic chart). *The positive branch of (11.13), with the intervals (11.1), satisfies $A > 0$, $0 < C^* < A$, and $\hbar_s > 0$. Reversal of the odd coordinate preserves $|C^*/A| < 1$.*

Proof. Proposition 11.5 gives $0 < \eta_{\text{ret}} < H_5$ and positivity of both action sections. For an upper bound on H_5 , (11.1) implies $\alpha < 1/125$ and $\varphi > 8/5$. Retaining only the positive terms of (11.4) yields

$$H_5 < \frac{\sqrt{5}}{2} + \frac{9}{16 \cdot 125^2} + \frac{7}{48 \cdot 125^4}.$$

Since $5 < 81/16$, one has $\sqrt{5}/2 < 9/8$. Moreover,

$$\frac{9}{16 \cdot 125^2} + \frac{7}{48 \cdot 125^4} = \frac{421882}{11718750000} < \frac{1}{1000}.$$

Thus $H_5 < 9/8 + 1/1000 < 2$. Using $\alpha > 7/1000$ now gives

$$0 < C^* = 2(\eta_{\text{ret}} + \alpha) < 2\left(2 + \frac{1}{125}\right) = \frac{502}{125} < 7 < 1000\alpha = A.$$

The channel interchange in 11.10 replaces C^* by $-C^*$ and preserves A ; it therefore preserves the absolute inequality. $\square$

29 Primitive returns and lifted frequency

29.1 Local extension automaton

Let $\mathcal{X}_\Gamma$ be the finite set of local states compatible with the route. A state retains

$$x = (t, g, B, p, \varepsilon, c, w),$$

where t is discrete time, g the nonadic phase, B the block, p the prefix, ε the sheet, c the carry, and w the winding. An edge $x \rightarrow y$ exists when y is an admissible extension of x , respects the genealogical record, and survives G_0 pruning. The table of 36,864 carry compositions supplies the local addition law; the return tables determine the memory.

Let U_Γ be the normalized transfer operator of this graph and R_Γ the orientation involution. The required identities are

$$U_{C_M\Gamma} = JU_\Gamma J^{-1}, \quad R_{C_M\Gamma} = -JR_\Gamma J^{-1}.$$

29.2 Oriented primitive counts

For $n \geq 1$, the oriented trace

$$a_n(\Gamma) = \text{Tr}(R_\Gamma U_\Gamma^n)$$

counts signed closed returns. Möbius inversion separates primitive orbits:

$$p_n(\Gamma) = \frac{1}{n} \sum_{d|n} \mu(d) a_{n/d}(\Gamma).$$

The electron is taken as the affine origin of the relative corrections, and one defines

$$\boxed{\eta_h(\Gamma) = p_h(\Gamma) - p_h(\Gamma_e), \quad h \in \mathfrak{S}.}$$

This gives the map

$$\mathcal{F}_{\text{tower}} : \Gamma^{\text{enr}} \mapsto \eta(\Gamma) = (\eta_h(\Gamma))_{h \in \mathfrak{S}}.$$

There are no species-specific parameters. The same automaton, involution, and Möbius inversion act on all 324 routes.

29.3 Convergence of primitive refinement

Let $N_\Gamma = \dim \mathcal{X}_\Gamma$. If U_Γ is a contraction,

$$|a_n(\Gamma)| \leq N_\Gamma.$$

The number of divisors of n grows sublinearly, hence

$$|p_n(\Gamma)| \leq \frac{N_\Gamma \tau(n)}{n}.$$

Since $0 < q_+ < q_- < 1$,

$$|s_h| \leq q_-^h + q_+^h \leq 2q_-^h.$$

By comparison,

$$\sum_{h \in \mathcal{E}} |\eta_h(\Gamma) s_h| < \infty$$

uniformly on every finite collection of fibres. The refined character is therefore well defined.

29.4 Control identities

The construction must satisfy the following conditions simultaneously:

1. invariance under relabelling of states;
2. covariance under sheet conjugation;
3. compatibility with direct sums of components;
4. equality between recomputation from the genealogical record and from the certified graph;
5. preservation of the origin $\eta(\Gamma_e) = 0$;
6. complete insensitivity to permutations of the external mass columns;
7. determination of the tails by a bound established before comparison.

Exact representation of a given residual proves completeness of the codomain, but does not replace these seven tests.

30 Route clock and absolute normalization

30.1 Integer returns do not suffice to distinguish species

In the atlas of 324 routes, the first cell return is 27, the kinematic return within CT-108 is 54, and the extended kinematic return is 216. Being common to the routes, these integers fix the global temporal architecture, but cannot by themselves account for mass differences between species.

30.2 Temporal holonomy

The specific information resides in the holonomy of the extension operator. Let $\lambda_j(\Gamma)$ be a nontrivial eigenvalue of U_Γ on the persistent subspace. Write

$$\lambda_j(\Gamma) = r_j(\Gamma) e^{i\theta_j(\Gamma)}.$$

The associated internal frequency is

$$\omega_j(\Gamma) = \frac{\theta_j(\Gamma) + 2\pi w_j(\Gamma)}{108t_0},$$

where the integer w_j is fixed by the recorded winding, not by a subsequent branch choice. For a stable mode, $r_j = 1$; for a resonance, $r_j < 1$, and its attenuation supplies a width.

The ratio

$$\rho_{\omega,j}(\Gamma) = \frac{\omega_j(\Gamma)}{\omega_0}$$

is dimensionless and can modify mass ratios without changing the common decade -34 . The energy and mass of the mode are

$$E_j(\Gamma) = \hbar_{\text{ret}} \omega_j(\Gamma) \chi_{\text{full}}(\Gamma) R_{\text{act}}^{\kappa_j(\Gamma)},$$

$$m_j(\Gamma) = E_j(\Gamma) c_{\text{int}}^{-2}.$$

This equation brings together action, clock, character, and reader without identifying any of them with another. Its numerical application requires determining U_Γ , fixing the winding, and specifying the dimensional chart.

31 Antecedents of operator reconstruction and memory

These antecedents begin after the joint APP-TRIT-TPK production of the enriched state and the discrete structure of the continuum. We work with the dodecaphase record already generated and with declared realizations of its readout operators. The subsequent linear carrier and the candidates in a reconstruction do not replace the domain of generated histories. Nor is a subsequent reversible realization identified with nonadic holonomy Γ_9 merely through its notation.

In the chart and orientation of the record,

$$\begin{aligned} K &= (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601), \\ \mathbf{1}^*K &= 6263, \quad (Sx)_i = x_{i+1}, \quad i \pmod{12}. \end{aligned} \quad (31.1)$$

The record is used after its generation, not as a retrospective input for selecting an output. Here the operator S is the marked positional shift. Its domains will remain distinct from the canonical planes used later.

31.1 The full orbit as a reference for its carrier

In $V = \mathbb{R}^{12}$, with its canonical Euclidean inner product, consider

$$O_K = (K, SK, \dots, S^{11}K) : \mathbb{R}^{12} \longrightarrow V.$$

The shift S acts on positions in the record. Periodicity alone does not identify it with the full TPK evolution or with a subsequent exceptional symmetry.

Theorem 31.1 (Complete cyclic reference and exact Gram matrix). *The matrix O_K is invertible. Its Gram matrix has eigenvalues*

value	multiplicity
$39225169 = 6263^2$	1
$1248025 - 345420\sqrt{3}$	2
589609	2
1407529	2
1053157	2
$1248025 + 345420\sqrt{3}$	2
697225	1

and satisfies

$$589609\|c\|^2 \leq \|O_K c\|^2 \leq 39225169\|c\|^2, \quad \|O_K^{-1}\| = \frac{1}{\sqrt{589609}}, \quad \text{cond}(O_K) = \frac{6263}{\sqrt{589609}}. \quad (31.2)$$

The orbit of the centred record has rank eleven.

Proof. The Gram entries are the autocorrelations $\langle K, S^j K \rangle$. In order $j = 0, \dots, 11$, they are

$$\begin{aligned} &(4251257, 2999323, 3163385, 3287920, 3216553, 3344743, \\ &2950064, 3344743, 3216553, 3287920, 3163385, 2999323). \end{aligned}$$

Since S is orthogonal, the Gram matrix is circulant. Substituting the twelve characters $\exp(2\pi im/12)$ into that row gives the preceding real table. All values are positive. The smaller of the two values involving $\sqrt{3}$ exceeds 589609, since $658416 > 345420\sqrt{3}$, an inequality verified by squaring its two positive sides. The other values in the table are at least 589609. The uniform mode has value $(\sum K_i)^2 = 6263^2$; the triangle inequality for the Fourier transform, using $K_i \geq 0$, bounds all the others by that same value. The spectral theorem gives (31.2). Centring the record removes only the uniform mode; the remaining eleven retain the nonzero values in the table. $\square$

The determinant, with the column order and orientation of S fixed here, is

$$\det O_K = 5483119259214020183850397930008283625.$$

The absence of a proper period for the vector would not, by itself, prove invertibility: the decisive fact is that none of its Fourier modes vanishes. The condition number is approximately 8.15643. If K is normalized to unit norm, the extreme Gram eigenvalues are divided by $\|K\|^2 = 4251257$; the magnitude of its components must not be confused with an intrinsic improvement over every reference of equal norm.

Corollary 31.2 (Polar separation of form and orientation). *Let*

$$G_K = O_K^* O_K, \quad P_K = G_K^{1/2}, \quad F_K = O_K G_K^{-1/2}.$$

Then P_K is positive definite, F_K is orthogonal, and $O_K = F_K P_K$. An isometric realization $J : V \rightarrow \mathcal{Y}$ preserves G_K and P_K and transports the isometric factor to $J F_K$.

Proof. Positivity of the Gram matrix allows its square root and inverse to be defined by the spectral theorem. The identity $F_K^* F_K = G_K^{-1/2} G_K G_K^{-1/2} = I$ proves the claim. Moreover, $(J O_K)^* (J O_K) = O_K^* O_K$. $\square$

This polar decomposition is defined from the full Gram matrix: it cannot be obtained merely from the list of eigenvalues or the norm of K . The symbol P_K in this corollary is not the rank-ten dimensional projector of another incidence realization. This separation is a consequence of the cyclic reference, not an identification of all its readout operators.

31.2 Operator identification and relative holonomy

Theorem 31.3 (Twelve general responses or one cyclic response). *For any linear operator $H : V \rightarrow V$, its twelve responses $Y = (HK, HSK, \dots, HS^{11}K)$ determine*

$$H = Y O_K^{-1}.$$

If $HS = SH$, a single complete vector response $y = HK$ determines

$$H = O_y O_K^{-1}, \quad H = \sum_{j=0}^{11} h_j S^j, \quad h = O_K^{-1} y. \quad (31.3)$$

With errors E in the response matrix or δy in the vector, respectively,

$$\|\hat{H} - H\| \leq \frac{\|E\|}{\sqrt{589609}}, \quad (31.4)$$

$$\|\hat{h} - h\| \leq \frac{\|\delta y\|}{\sqrt{589609}}, \quad \|\hat{H} - H\| \leq \sqrt{\frac{12}{589609}} \|\delta y\|. \quad (31.5)$$

Proof. The first equality is $Y = H O_K$. If H commutes with S , then $HS^j K = S^j y$, hence $Y = O_y$. Commutation with the cycle determines every column of H from a single one; the twelve powers $I, S, \dots, S^{11}$ are independent and span that operator space. Thus $y = O_K h$, giving the second formula. The first two bounds use $\|O_K^{-1}\| = 1/\sqrt{589609}$. The last uses $\|O_{\delta y}\| \leq \|O_{\delta y}\|_F = \sqrt{12} \|\delta y\|$. $\square$

A response here means a twelve-component vector, not a scalar measurement. For example, if $H = 2I + 3S - S^5$, the response $y = HK$ returns exactly the coefficients 2, 3, -1 in their positions through (31.3); the inversion procedure requires y, K, S , not the filter coefficients. By contrast, the nonzero operator whose first row is $(543, -234, 0, \dots, 0)$ and whose remaining rows vanish annihilates K . A single response does not identify an arbitrary operator.

The reference role admits an exact generalization. On a cycle of n positions, $H \mapsto Hg$ identifies every operator commuting with the cycle if and only if $O_g = (g, Sg, \dots, S^{n-1}g)$ is invertible. Sufficiency is the preceding proof. If O_g is singular, a nonzero vector in its kernel gives a polynomial of degree less than n that annihilates g ; the operator is nonzero because the powers of the cycle are independent. The unit impulse also has an orthonormal orbit and serves as a reference. The specific consequence for K is that the same object already used in closure and incidence satisfies that condition, without being selected anew to optimize the test. No identification of nonlinear dynamics or arbitrary hidden states follows; an application to a linearization requires that it has been defined and justified.

Corollary 31.4 (Recovery of relative holonomy from the complement). *Let H_B be known and invertible, and let H_C be another transport of the same type. The complement*

$$D_\gamma = \frac{\sqrt{8}}{9}(H_C - H_B)$$

determines

$$H_B^{-1}H_C = I + \frac{9}{\sqrt{8}}H_B^{-1}D_\gamma.$$

If D_γ commutes with S , in particular if both transports do, $y = D_\gamma K$ suffices to recover

$$H_B^{-1}H_C = I + \frac{9}{\sqrt{8}}H_B^{-1}O_yO_K^{-1}. \quad (31.6)$$

The error in this reconstruction is bounded by

$$\frac{9}{\sqrt{8}}\|H_B^{-1}\|\sqrt{\frac{12}{589609}}\|\delta y\|.$$

Proof. Solve for H_C in the definition of the complement and apply theorem 31.3. The bound follows by submultiplicativity. $\square$

For $H_B = S^3$ and $H_C = S^4$, the rational datum $y/\sqrt{8} = (S^4 - S^3)K/9$ makes it possible to reconstruct the relative holonomy S without supplying H_C to the inversion procedure. If H_B is unitary, the error factor is approximately 0.01435510. Without commutation, the twelve general responses are retained, not a fictitious symmetry assumption. Selection of a physical loop and its representation still belong to its generating dynamics; this protocol does not replace them.

31.2.1 Consequence of the marked reference for two orientations

The consequence used in the notebook retains the preceding table as an antecedent. For a real operator H commuting with S , let $y_+ = HK$ and $y_- = H^\top K$. Their twelve Fourier intensities coincide, but $y_+ = y_-$ if and only if $H = H^\top$. Each complete vector response, together with K and the orientation of S , recovers its operator.

Proof. The operator H is circulant; its Fourier multipliers are λ_m , and those of $H^\top$ are their conjugates. Both responses therefore have intensities $|\lambda_m|^2|\hat{K}_m|^2$. If their vectors coincide, $(H - H^\top)K = 0$. Commutation implies that $H - H^\top$ also annihilates $S^j K$ for every j . These vectors form a basis, so $H - H^\top = 0$. Recovery is $H = O_{y_+}O_K^{-1}$, and analogously for $H^\top$. $\square$

In particular, the shifts S and S^{-1} give distinct responses with equal power spectra. The exact autocorrelations in the preceding table yield the identity already used in the notebook:

$$\|SK - S^{-1}K\|^2 = 2(\|K\|^2 - \langle K, S^2 K \rangle) = 2\,175\,744 > 0. \quad (31.7)$$

This number is a consequence of the autocorrelations, not an additional starting datum. The notebook's rational check recovers from the two responses the coefficients e_1 and e_{11} , respectively, by inverting O_K .

A spectral intensity or scalar balance does not replace the vector response. This corollary concerns the twelve-component cyclic carrier: it neither identifies S with the full enriched dynamics nor turns $S \leftrightarrow S^{-1}$ by itself into physical charge conjugation. For general H , transposition does not mean time reversal either; for S , it does coincide with S^{-1} .

31.3 Reversible coupling and nonstationary transport

31.3.1 Reversible extension of the balance between two forms

Let $B, C : V \rightarrow V'$ be isomorphisms preserving a nondegenerate form b between two realizations:

$$B^*b'B = C^*b'C = b.$$

The form may be a positive metric or an alternating form. In real spaces, $*$ denotes transposition with respect to the chosen coordinates; the equality expresses transport of the form between the specified realizations. In Hilbert spaces, the operators and their inverses are bounded.

With $s = \sqrt{8}$ and $Q_V = \frac{1}{3} \begin{pmatrix} sI & -I \\ I & sI \end{pmatrix}$, define

$$\mathcal{U}(B, C) = Q_V^* \text{diag}(B, C) Q_V = \frac{1}{9} \begin{pmatrix} 8B + C & \sqrt{8}(C - B) \\ \sqrt{8}(C - B) & B + 8C \end{pmatrix}. \quad (31.8)$$

Write $T = (8B + C)/9$, $D = \sqrt{8}(C - B)/9$, and $R = (B + 8C)/9$. The full update is

$$\begin{pmatrix} x' \\ m' \end{pmatrix} = \begin{pmatrix} T & D \\ D & R \end{pmatrix} \begin{pmatrix} x \\ m \end{pmatrix}. \quad (31.9)$$

Theorem 31.5. *The map (31.8) is reversible and preserves the form $b \oplus b$:*

$$\mathcal{U}(B, C)^*(b' \oplus b')\mathcal{U}(B, C) = b \oplus b, \quad \mathcal{U}(B, C)^{-1} = \mathcal{U}(B^{-1}, C^{-1}). \quad (31.10)$$

Proof. The equality $Q^*Q = I$ follows from $8 + 1 = 9$. Its scalar blocks make Q preserve $b \oplus b$ for any form b . The diagonal map preserves the forms by hypothesis. Composition proves (31.10), and inversion of the three factors gives the stated inverse. $\square$

For incoming memory $m = 0$, the balance $b = T^*b'T + D^*b'D$ is recovered. The second column determines the response to arbitrary incoming memory. Conservation of positive norms and conservation of oriented area are the respective specializations of (31.10).

Collective realization in the nine copies In V^9 , let $W_E = \text{diag}(I, \dots, I, E)$, with E in the ninth position. Let $A(a, b) = (a/\sqrt{8}, \dots, a/\sqrt{8}, b)$ and $L = AQ$. The uniform embedding is $Jx = (x, \dots, x)/3$. Then $L(x, 0) = Jx$ and

$$W_E L = L \mathcal{U}(I, E). \quad (31.11)$$

Proof. $W_E A = A \text{diag}(I, E)$; multiplication by Q and use of $AQ = L$ give (31.11). Orthogonality of the mean of the first eight positions to their seven differences also shows that the complement is fixed by W_E . $\square$

Let $P_{\text{cyc}}(v_0, \dots, v_8) = (v_8, v_0, \dots, v_7)$. The original cyclic step is $S_E = P_{\text{cyc}} W_E$. Thus $S_E L = (P_{\text{cyc}} L) \mathcal{U}(I, E)$, with different input and output charts. The collective reduction preserves the corresponding pattern: $S_E^9 = I_9 \otimes E$, whereas $W_E^9 = \text{diag}(I_8, E^9)$. When a readout operator uses the nine individual labels, they are retained in their original representation.

31.3.2 Exact composition and memory re-entry

For $B_1, C_1 : V_0 \rightarrow V_1$ and $B_2, C_2 : V_1 \rightarrow V_2$,

$$\boxed{\mathcal{U}(B_2, C_2)\mathcal{U}(B_1, C_1) = \mathcal{U}(B_2B_1, C_2C_1).} \quad (31.12)$$

Proof. In (31.8), the factors $Q_{V_1} Q_{V_1}^*$ cancel, and the two diagonal blocks multiply in the indicated order. Induction extends the formula to any finite number of transports, preserving their order. $\square$

The readout block contains the identity

$$\boxed{T_2T_1 + D_2D_1 = \frac{8B_2B_1 + C_2C_1}{9}.} \quad (31.13)$$

Starting from $(x, 0)$, the first stage produces (T_1x, D_1x) . The second readout is $T_2T_1x + D_2D_1x$. The second term is exactly the contribution of reintroduced memory. Storing D_1x outside the next coupling and continuing only with T_1x gives the separate-record procedure, whose readout is T_2T_1x . Equality (31.13) specifies the difference between the two procedures.

The realization U preserves the coupling state and the ordered products $B_N \cdots B_1, C_N \cdots C_1$. Preservation of all prefixes also includes the HMT emission record, which retains each factor together with its order.

Covariant increment equation For the separate record $x_{n+1} = T_nx_n, m_n = D_nx_n$, one has

$$x_{n+1} - B_nx_n = m_n/\sqrt{8}. \quad (31.14)$$

With $G_0 = I$ and $G_{n+1} = B_nG_n$, telescoping gives

$$x_0 = G_N^{-1}x_N - \frac{1}{\sqrt{8}} \sum_{n < N} G_{n+1}^{-1}m_n. \quad (31.15)$$

Memory is the increment relative to the transport between forms B_n , with the declared normalization. Equality (31.15) is a finite algebraic reconstruction; passing to an infinite series requires convergence. The following theorem constructs the stable inverse through the limit of positive operators.

31.3.3 Memory of a nonstationary sequence

In a Hilbert-space realization H , let C_n be unitary, with their order of composition declared. Write

$$T_n = (8I + C_n)/9, \quad D_n = \sqrt{8}(C_n - I)/9, \quad P_0 = I, \quad P_{n+1} = T_nP_n, \quad A_N = P_N^*P_N.$$

The index N counts compressions with separate storage, unlike full transport with re-entry. For variable metrics with isometric isomorphisms B_n , conjugation by G_n takes the transports to a fixed Hilbert space and yields this same case, with $\tilde{C}_n = G_{n+1}^{-1}C_nG_n$. The statement uses the fibre metrics and the declared isometries between them.

Theorem 31.6 (Positive terminal and nonstationary recovery). *There is a positive operator $0 \leq A_\infty \leq I$, the strong limit of A_N , and*

$$\boxed{I = A_\infty + \sum_{n \geq 0} P_n^* D_n^* D_n P_n.} \quad (31.16)$$

The map

$$\boxed{\mathcal{V}v = (A_\infty^{1/2}v, (D_n P_n v)_{n \geq 0})} \quad (31.17)$$

is isometric and has a stable left inverse

$$\mathcal{V}^*(a, (m_n)) = A_\infty^{1/2}a + \sum_{n \geq 0} P_n^* D_n^* m_n, \quad \|\mathcal{V}^*\| = 1. \quad (31.18)$$

Proof. The local identity $T_n^* T_n + D_n^* D_n = I$ gives $A_N - A_{N+1} = P_N^* D_N^* D_N P_N \geq 0$. The quadratic forms of the decreasing positive sequence converge to that of a positive operator A_∞ . For $B_N = A_N - A_\infty$, the inequality $0 \leq B_N^2 \leq B_N$ turns this convergence into strong convergence. Finite telescoping and passage to the limit prove (31.16). Evaluation on v proves (31.17). Its adjoint is (31.18); the series converges in norm because the truncations of a square-summable record converge and the adjoint is bounded. $\square$

Reconstruction with the limiting terminal and N records has exact error $(A_N - A_\infty)v$. The vector $A_\infty^{1/2}v$ is the canonical positive realization of the norm terminal. Its existence follows from the Gram limit, independently of vector convergence of $P_N v$.

31.3.4 Three different properties: exhaustion, distinguishability, and stability

Let $Mv = (D_n P_n v)_n$. By (31.16), $M^* M = I - A_\infty$. Hence:

1. The entire norm of v passes to the record exactly when $A_\infty v = 0$.
2. The record distinguishes every state exactly when $\ker(I - A_\infty) = \{0\}$.
3. Recovery from memory alone has a bounded inverse exactly when $I - A_\infty \geq \varepsilon I$ for some $\varepsilon > 0$.
In that case a left inverse is $(I - A_\infty)^{-1} M^*$.

Moreover,

$$\boxed{\ker M = \bigcap_{n \geq 0} \text{Fix}(C_n).} \quad (31.19)$$

Proof. If $Mv = 0$, then $D_0 v = 0$, whence $C_0 v = v$ and $P_1 v = v$. Inductively, $D_n P_n v = 0$ gives $C_n v = v$ and $P_{n+1} v = v$. The converse is immediate. The three equivalences follow from the Gram operator $M^* M$ and the characterization of operators bounded below. $\square$

Exact example separating the three properties In the algebraic example $C_0 = -I$ and $C_n = I$ for $n \geq 1$,

$$A_\infty = \frac{49}{81}I, \quad M^* M = \frac{32}{81}I, \quad m_0 = -\frac{2\sqrt{8}}{9}v, \quad v = -\frac{9}{2\sqrt{8}}m_0. \quad (31.20)$$

Memory contains 32/81 of the squared norm and permits recovery of the entire vector. The terminal retains 49/81 and is a positive operator distinct from a projector. Recoverability depends on the kernel and conditioning of the readout operator. Readout and memory are correlated components of an isometry.

This example separates the theorem's three logical consequences. An application to a physical trajectory also preserves the effective selection and provenance of its C_n through TPK.

The letter η in the following Gaussian realization denotes the deformation of the ellipse, namely the parameter ζ of the transport developments; it is not the action coefficient η_{ret} . A positive section $\hbar_a$ and the same unit for both coordinates of the angular ratio are maintained. The retained semiaxes are

$$a_S = \sqrt{2\hbar_a} e^{\eta/2}, \quad b_S = \sqrt{2\hbar_a} e^{-\eta/2}, \quad \eta = \text{artanh}(C^*/A).$$

31.4 Canonical covariance and reciprocity of the action ellipse

The preceding construction determines two complementary invariants: the product of the semiaxes fixes the action area, and their ratio retains the anisotropy of the angular pair. We now specify the canonical realization in which those semiaxes are twice the standard deviations. The starting structure is the ellipse already obtained from the angular and action outputs of APP–TRIT–TPK; the operator realization is established after that construction.

In this subsection, write $\eta = \eta_{\text{el}} = \text{artanh}(C^*/A)$, with $|C^*/A| < 1$ and $\hbar_a > 0$. The angular ratio uses the same unit for both components. In balanced coordinates (Q, P) , both having the dimensions of the square root of action, define

$$V_\eta := \frac{1}{4} \begin{pmatrix} a_S^2 & 0 \\ 0 & b_S^2 \end{pmatrix} = \frac{\hbar_a}{2} \begin{pmatrix} e^\eta & 0 \\ 0 & e^{-\eta} \end{pmatrix}. \quad (31.21)$$

This positive matrix has entries having the dimensions of action. Its interpretation as a quantum covariance is specified by the following realization.

Proposition 31.7 (Gaussian realization of the action covariance). *Consider the canonical representation on $L^2(\mathbb{R}, dQ)$, with $\hat{Q}f(Q) = Qf(Q)$ and $\hat{P}f(Q) = -i\hbar_a f'(Q)$, initially on the Schwartz space $\mathcal{S}(\mathbb{R})$. The normalized state*

$$\psi_\eta(Q) = (\pi\hbar_a e^\eta)^{-1/4} \exp\left(-\frac{Q^2}{2\hbar_a e^\eta}\right) \quad (31.22)$$

has zero means and covariance matrix V_η . In particular,

$$(\Delta Q)^2 = \frac{\hbar_a e^\eta}{2}, \quad (\Delta P)^2 = \frac{\hbar_a e^{-\eta}}{2}, \quad \text{Cov}(Q, P) = 0, \quad \det V_\eta = \frac{\hbar_a^2}{4}. \quad (31.23)$$

This realization saturates the Robertson–Schrödinger inequality.

Proof. The preceding operators are essentially self-adjoint on $\mathcal{S}(\mathbb{R})$, which is a common invariant core; on this space, $[\hat{Q}, \hat{P}] = i\hbar_a I$. For $\hat{Y} = (\hat{Q}, \hat{P})$, the symmetric covariance is

$$V_{jk} = \frac{1}{2} \langle (\hat{Y}_j - \langle \hat{Y}_j \rangle)(\hat{Y}_k - \langle \hat{Y}_k \rangle) + (\hat{Y}_k - \langle \hat{Y}_k \rangle)(\hat{Y}_j - \langle \hat{Y}_j \rangle) \rangle.$$

With $s = \hbar_a e^\eta > 0$, let $I_s = \int_{\mathbb{R}} e^{-Q^2/s} dQ$. The product of two equal integrals, evaluated in polar coordinates, gives $I_s^2 = 2\pi \int_0^\infty r e^{-r^2/s} dr = \pi s$. Moreover, the integral of $\frac{d}{dQ}(Qe^{-Q^2/s})$ is zero because of decay at both ends; hence $\int_{\mathbb{R}} Q^2 e^{-Q^2/s} dQ = sI_s/2$. These identities and parity prove normalization, the zero mean, and $\langle \hat{Q}^2 \rangle = s/2$. Moreover, $\hat{P}\psi_\eta = ie^{-\eta} Q\psi_\eta$. Thus $\langle \hat{P} \rangle = 0$, $\|\hat{P}\psi_\eta\|^2 = \hbar_a e^{-\eta}/2$, and $\text{Re}\langle \hat{Q}\psi_\eta, \hat{P}\psi_\eta \rangle = 0$.

For any normalized state in that domain, Cauchy–Schwarz applied to $(\hat{Q} - \langle \hat{Q} \rangle)\psi$ and $(\hat{P} - \langle \hat{P} \rangle)\psi$ gives

$$(\Delta Q)^2 (\Delta P)^2 \geq \text{Cov}(Q, P)^2 + \frac{1}{4} |[\hat{Q}, \hat{P}]|^2 = \text{Cov}(Q, P)^2 + \frac{\hbar_a^2}{4}.$$

The computed variances attain equality. Equivalently, the operator

$$a_\eta = \frac{e^{-\eta/2} \hat{Q} + ie^{\eta/2} \hat{P}}{\sqrt{2\hbar_a}}$$

satisfies $[a_\eta, a_\eta^\dagger] = I$ and $a_\eta \psi_\eta = 0$, by direct substitution. $\square$

31.5 Covariance, normal form, and Gaussian spectrum

The positive action section $\hbar_a$ and the preceding canonical realization determine the normalization used here. We prove the correspondence between covariance, spectrum, purity, and entropy for a finite number of modes; Fock space retains all its occupation numbers. Symplectic reduction and the Gaussian formulae are established results reproduced here to make their application to memory transport self-contained.

31.5.1 Canonical coordinates and characteristic function

On $L^2(\mathbb{R}^d)$, let $\hat{q}_j = \hat{Q}_j/\sqrt{\hbar_a}$, $\hat{p}_j = \hat{P}_j/\sqrt{\hbar_a}$, and $\hat{z} = (\hat{q}_1, \hat{p}_1, \dots, \hat{q}_d, \hat{p}_d)$. On Schwartz space,

$$[\hat{z}_j, \hat{z}_k] = iJ_{jk}I, \quad J = \bigoplus_{j=1}^d \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

This convention fixes the sign of the canonical form. The Weyl operators $\mathcal{W}(\xi) = \exp(i\xi^\top \hat{z})$ are defined through the self-adjoint closures of the indicated linear combinations. For a centred state ρ with finite second moments, the normalized covariance and its characteristic function are

$$W_{jk} = \text{Tr} \rho(\hat{z}_j \hat{z}_k + \hat{z}_k \hat{z}_j), \quad \chi_\rho(\xi) = \text{Tr}(\rho \mathcal{W}(\xi)), \quad V = \frac{\hbar_a}{2} W.$$

The state is centred Gaussian when

$$\chi_\rho(\xi) = \exp(-\xi^\top W \xi / 4). \quad (31.24)$$

The expectation of $(c^\top \hat{z})^\dagger (c^\top \hat{z})$ is $c^\dagger (W + iJ)c/2$; thus $W + iJ \geq 0$. Moreover, $W > 0$: if v is real and $v^\top W v = 0$, positivity implies $(W + iJ)v = 0$, hence $Jv = 0$ and $v = 0$.

Lemma 31.8 (Uniqueness of the characteristic function). *Two trace-class operators with the same characteristic function are equal.*

Proof. Regrouping positions and momenta as $\xi = (u, v)$,

$$(\mathcal{W}(u, v)f)(x) = e^{iu \cdot (x+v/2)} f(x+v).$$

For a finite-rank operator with Schwartz vectors and kernel K_A ,

$$\text{Tr}(A\mathcal{W}(u, v)) = \int K_A(x + v/2, x - v/2) e^{iu \cdot x} dx.$$

Plancherel's theorem and the change of variables with absolute Jacobian one give

$$\|A\|_{\text{HS}}^2 = (2\pi)^{-d} \int_{\mathbb{R}^{2d}} |\text{Tr}(A\mathcal{W}(\xi))|^2 d\xi.$$

These operators are dense in the trace class. Convergence in that norm implies Hilbert–Schmidt convergence and uniform convergence of the characteristic functions, since their difference is bounded by $\|A - B\|_1$. The equality extends in L^2 . Applying it to the difference of the operators proves the claim. $\square$

31.5.2 Symplectic reduction of the positive form

Theorem 31.9 (Williamson normal form). *For a real symmetric matrix $W > 0$ of order $2d$, there exist $S \in \text{Sp}(2d, \mathbb{R})$ and $\nu_j > 0$ such that*

$$W = SDS^\top, \quad D = \bigoplus_j \nu_j I_2.$$

The eigenvalues of JW are $\pm i\nu_j$. The condition $W + iJ \geq 0$ is equivalent to $\nu_j \geq 1$ for every j .

Proof. The matrix $B = W^{1/2}JW^{1/2}$ is skew-symmetric and invertible. If $-B^2u = \nu^2u$ and $\|u\| = 1$, the vector $v = -Bu/\nu$ has unit norm and is orthogonal to u ; it satisfies $Bu = -\nu v$, $Bv = \nu u$. The complement of this pair is invariant. Iteration produces an orthogonal matrix O with $O^TBO = JD$. Defining $S = W^{1/2}OD^{-1/2}$,

$$S^TJS = J, \quad SDS^T = W.$$

The similarity $W^{1/2}(JW)W^{-1/2} = B$ determines the values ν_j , with multiplicity. Congruence by S^{-1} transforms $W + iJ$ into $D + iJ$; each block has eigenvalues $\nu_j + 1$ and $\nu_j - 1$. $\square$

Lemma 31.10 (Unitary implementation in finitely many modes). *Every symplectic matrix S admits a unitary U_S satisfying $U_S^\dagger \mathcal{W}(\xi)U_S = \mathcal{W}(S^T\xi)$. The conjugation $\rho \mapsto U_S\rho U_S^\dagger$ transports covariance by $W \mapsto SWS^T$.*

Proof. In the polar decomposition $S = OP$, applying the identity $J^{-1}(S^T S)J = (S^T S)^{-1}$ to the positive square root through spectral calculus proves $PJP = J$. Then O is orthogonal symplectic. The eigenspaces of P are paired by J with eigenvalues λ, λ^{-1} ; the eigenspace for eigenvalue one is J -invariant. An orthogonal symplectic matrix K therefore reduces P to $\bigoplus_j \text{diag}(e^{r_j}, e^{-r_j})$.

Dilation is implemented by

$$(\mathcal{D}_r f)(x) = e^{-\frac{1}{2}\sum_j r_j} f(e^{-r_1}x_1, \dots, e^{-r_d}x_d).$$

A change of variables proves unitarity and the conjugation of $(\hat{q}_j, \hat{p}_j)$ to $(e^{r_j}\hat{q}_j, e^{-r_j}\hat{p}_j)$. An orthogonal symplectic transformation commutes with J and corresponds to a matrix $u \in U(d)$ acting on $a_j = (\hat{q}_j + i\hat{p}_j)/\sqrt{2}$. Its second quantization $\Gamma(u) = \bigoplus_{n \geq 0} u^{\otimes n}$ is unitary on $\mathcal{F}_s(\mathbb{C}^d)$, and the identity on finite symmetric products gives the transformation of the a_j .

The Hermite basis identifies this space with $L^2(\mathbb{R}^d)$. To justify completeness, if f is orthogonal to every polynomial multiplied by $e^{-|x|^2/2}$, the entire function $F(z) = \int f(x)e^{-|x|^2/2}e^{z \cdot x} dx$ has all its derivatives at zero equal to zero. Thus $F = 0$; restricting to $z = i\xi$, Fourier uniqueness gives $f = 0$. The normalizations and orthogonality follow from creation and annihilation. Composing the preceding implementations produces U_S . The Weyl equality implies $\chi_{U_S\rho U_S^\dagger}(\xi) = \chi_\rho(S^T\xi)$, which proves the covariance law. $\square$

31.5.3 Computing the spectrum of the isotropic density

In one mode, the normalized coherent states are

$$|z\rangle = e^{-|z|^2/2} \sum_{n \geq 0} \frac{z^n}{\sqrt{n!}} |n\rangle.$$

Their position wavefunction, $\pi^{-1/4} \exp[-(x - q_0)^2/2 + ip_0(x - q_0/2)]$, where $q_0 = \sqrt{2}\Re z$, $p_0 = \sqrt{2}\Im z$, gives by integration

$$\langle z | \mathcal{W}(u, v) | z \rangle = e^{-(u^2+v^2)/4} e^{i(uq_0+vp_0)}.$$

For $s > 0$, the integral

$$\rho^{(s)} = \frac{1}{\pi s} \int_{\mathbb{C}} e^{-|z|^2/s} |z\rangle \langle z| d^2z$$

converges in trace norm, is positive, and has trace one. Angular integration annihilates the off-diagonal elements in the occupation basis; radial integration gives

$$\langle n | \rho^{(s)} | n \rangle = \frac{s^n}{(1+s)^{n+1}}, \quad \chi_{\rho^{(s)}}(u, v) = e^{-(2s+1)(u^2+v^2)/4}.$$

With $\nu = 2s + 1$, the unique Gaussian state with covariance νI_2 is, by Lemma 31.8,

$$\rho_\nu = (1-t) \sum_{n \geq 0} t^n |n\rangle \langle n|, \quad t = \frac{\nu-1}{\nu+1}. \quad (31.25)$$

The case $\nu = 1$ is the vacuum projector.

Theorem 31.11 (Spectrum, purity, and entropy). *A centred Gaussian state with covariance W and symplectic eigenvalues $\nu_1, \dots, \nu_d$ is unitarily equivalent to $\bigotimes_j \rho_{\nu_j}$. Its eigenvalues are*

$$\lambda_{\mathbf{n}} = \prod_j (1 - t_j) t_j^{n_j}, \quad t_j = \frac{\nu_j - 1}{\nu_j + 1}, \quad \mathbf{n} \in \mathbb{N}^d.$$

With $0 \log 0 = 0$,

$$\text{Tr } \rho^2 = \prod_j \nu_j^{-1} = (\det W)^{-1/2}, \quad (31.26)$$

$$-\text{Tr}(\rho \log \rho) = \sum_j \left[\frac{\nu_j + 1}{2} \log \frac{\nu_j + 1}{2} - \frac{\nu_j - 1}{2} \log \frac{\nu_j - 1}{2} \right]. \quad (31.27)$$

Proof. Williamson's theorem gives $W = SDS^\top$. The product of the isotropic densities has characteristic function $e^{-\xi^\top D \xi/4}$. Unitary implementation and Weyl uniqueness prove the equivalence and the spectral formula. In one mode,

$$\sum_{n \geq 0} (1 - t)^2 t^{2n} = \frac{1 - t}{1 + t} = \nu^{-1},$$

and

$$-\sum_{n \geq 0} (1 - t) t^n \log((1 - t) t^n) = -\log(1 - t) - \frac{t}{1 - t} \log t.$$

Substitution of $t = (\nu - 1)/(\nu + 1)$ proves the formulae; the limit at $t = 0$ is zero. In the finite product, Tonelli's theorem justifies summing the nonnegative entropy terms, and $\det W = \prod_j \nu_j^2$ gives the determinantal expression. $\square$

If the state is the reduction of a pure bipartite vector, its eigenvalues are the squares of the Schmidt coefficients. The singular-value decomposition of the Hilbert-Schmidt operator defined by that vector proves equality of the nonzero spectra of both reductions. The entropy computed is then entanglement entropy.

31.5.4 Preparation and transport of the two covariances

The subsequent application uses two identical pure Gaussian inputs. In general, for $M \in \text{Sp}(2d, \mathbb{R})$, their covariance is $V_0 = (\hbar_a/2)MM^\top$, and the joint input has covariance $V_0 \oplus V_0$. The physical transport is obtained by conjugating U_H by $M \oplus M$. In the balanced chart, the normalized input covariance is I_{4d} ; with

$$T = \frac{8I + H}{9}, \quad D = \frac{\sqrt{8(H - I)}}{9},$$

the first reduction has

$$W = TT^\top + DD^\top = \frac{8I + HH^\top}{9}.$$

Its characteristic function is obtained by setting the arguments of the second group of modes to zero. It therefore remains Gaussian, and the preceding theorems compute its entire spectrum without truncating the occupation record.

31.6 Prolongation of memory transport: quantum realization

The quantum realization uses the reversible coupling of Section 31.3. We first consider a real canonical plane with alternating matrix $\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. The nonadic partition determines

$$Q = \frac{1}{3} \begin{pmatrix} \sqrt{8}I & -I \\ I & \sqrt{8}I \end{pmatrix}, \quad U_H = Q^\top \text{diag}(I, H)Q = \frac{1}{9} \begin{pmatrix} 8I + H & \sqrt{8}(H - I) \\ \sqrt{8}(H - I) & I + 8H \end{pmatrix}. \quad (31.28)$$

The operator $H \in \text{Sp}(2, \mathbb{R})$ transports that plane. The matrix Q preserves the Euclidean inner product and the form $\Omega \oplus \Omega$; hence U_H is symplectic. The first component defines the observed system and the second the memory record; the full transformation acts on both.

The ordered loop whose identities are reproduced in 31.6.1 is

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad H_n = C^{-1}P^{-n}CP^n = \begin{pmatrix} 1+n & n^2 \\ n & n^2 - n + 1 \end{pmatrix}. \quad (31.29)$$

For every $n \in \mathbb{Z}$, $\det H_n = 1$. In particular, for $n = 1$,

$$H_1 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = F_1^2, \quad F_1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = PF_{\text{av}}P^{-1}. \quad (31.30)$$

Its spectrum is $\{\varphi^2, \varphi^{-2}\}$, by subsequent recognition of the golden mode already generated. The integer n specifies the oriented number of iterations in the chosen loop. The collection of transports is algebraically determined; its realization as a physical protocol also retains the corresponding preparation and evolution data.

31.6.1 Algebraic identities of the loop and subsequent recognition

To preserve the proof of the identities used, their matrix composition is reproduced here. In the same chart,

$$N_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Let

$$P = I + N_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Since $N_0^2 = 0$, one has $P^n = I + nN_0$ for every $n \in \mathbb{Z}$. The ordered loop of four transports gives

$$H_n = C^{-1}P^{-n}CP^n = \begin{pmatrix} 1+n & n^2 \\ n & n^2 - n + 1 \end{pmatrix}. \quad (31.31)$$

For $n \neq 0$, its normalized difference is

$$F_n = \frac{H_n - I}{n} = \begin{pmatrix} 1 & n \\ 1 & n - 1 \end{pmatrix}.$$

Multiplying these matrices yields, without introducing spectral values,

$$\boxed{F_n^2 = H_n, \quad F_n^2 - nF_n - I = 0.} \quad (31.32)$$

With $H_B = I$, the memory is $D_{\gamma,n} = (\sqrt{8}/9)nF_n$. Consequently,

$$\boxed{H_n = \left(\frac{9}{n\sqrt{8}} D_{\gamma,n} \right)^2.} \quad (31.33)$$

The matrix memory response reconstructs the hyperbolic self-scale. For $n > 0$, subsequent spectral recognition identifies

$$\rho_{\text{ep},n} = \frac{n + \sqrt{n^2 + 4}}{2}, \quad \text{Spec } F_n = \{\rho_{\text{ep},n}, -\rho_{\text{ep},n}^{-1}\}, \quad \text{Spec } H_n = \{\rho_{\text{ep},n}^2, \rho_{\text{ep},n}^{-2}\}. \quad (31.34)$$

The case $n = 1$ produces the golden polynomial. The relation to the operator $F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ generated by the calendar preserves the change of chart:

$$F_1 = PF_{\text{av}}P^{-1}, \quad H_1 = PF_{\text{av}}^2P^{-1}. \quad (31.35)$$

The transport P , of determinant one, intertwines the two matrices and preserves Ω .

31.6.2 Initial state, transport, and readout

Proposition 31.7 constructs the Gaussian realization of the action ellipse. On the positive section $\hbar_a > 0$, with $A > 0$ and $|C^*/A| < 1$, its covariance is

$$V_\eta = \frac{\hbar_a}{2} M_\eta M_\eta^\top, \quad M_\eta = \text{diag}(e^{\eta/2}, e^{-\eta/2}), \quad \eta = \text{artanh}(C^*/A). \quad (31.36)$$

In the Schrödinger representation on $L^2(\mathbb{R}, dQ)$, with $\hat{Q}$ multiplication by Q and $\hat{P} = -i\hbar_a d/dQ$, the relation $[\hat{Q}, \hat{P}] = i\hbar_a I$ holds on Schwartz space. The normalized pure state is

$$\psi_\eta(Q) = (\pi\hbar_a e^\eta)^{-1/4} \exp[-Q^2/(2\hbar_a e^\eta)].$$

The preparation is the product $\psi_\eta \otimes \psi_\eta$. The transport in (31.28) is expressed in the same covariance chart by

$$\tilde{U} = (M_\eta \oplus M_\eta) U_H (M_\eta^{-1} \oplus M_\eta^{-1}). \quad (31.37)$$

Lemma 31.10 provides a unitary implementation of this symplectic matrix in two modes. Implementations differing by a global phase produce the same conjugation of the state. Covariance transforms by $V \mapsto \tilde{U}V\tilde{U}^\top$; the joint state remains pure, and its partial trace over the second mode determines the visible readout, in accordance with the construction of Section 31.5.

Theorem 31.12 (Mixedness of the reduced state from relative deformation). *The covariance of the visible readout is*

$$V_{\text{vis}} = \frac{\hbar_a}{2} M_\eta \frac{8I + HH^\top}{9} M_\eta^\top. \quad (31.38)$$

Its normalized symplectic eigenvalue is

$$\nu(H) := \frac{2\sqrt{\det V_{\text{vis}}}}{\hbar_a} = \sqrt{1 + \frac{8}{81} [\text{tr}(HH^\top) - 2]}. \quad (31.39)$$

Proof. Let $T = (8I + H)/9$ and $D = \sqrt{8}(H - I)/9$. The two initial modes are independent and have the same covariance. In the balanced chart of the initial state, the normalized visible covariance is $TT^\top + DD^\top$. Expansion gives

$$\begin{aligned} TT^\top + DD^\top &= \frac{64I + 8(H + H^\top) + HH^\top + 8[HH^\top - H - H^\top + I]}{81} \\ &= \frac{8I + HH^\top}{9}. \end{aligned}$$

This proves (31.38). Since $\det M_\eta = 1$ and $\det(HH^\top) = 1$, $\det(8I + HH^\top) = 65 + 8 \text{tr}(HH^\top)$. Division by 81 proves (31.39). The two positive eigenvalues of $HH^\top$ are reciprocal; their sum is at least two. Consequently, $\nu \geq 1$, with equality exactly when $HH^\top = I$. $\square$

The scale $\hbar_a$ and deformation η are retained until the determinant is taken. Their cancellation is covariant because state and operator are transported jointly. The expression $HH^\top$ uses the Euclidean inner product of the balanced chart of the initial state; a change of chart also transports that metric inner product.

An orthogonal symplectic transport H may have $D \neq 0$ while preserving the isotropic Gaussian product, with $\nu = 1$. The operator complement and state entanglement therefore have different criteria. In this preparation, nonisometric relative deformation characterizes exactly the case $\nu > 1$.

31.6.3 Exact evaluation of the golden loop

For (31.29),

$$\text{tr}(H_n H_n^\top) - 2 = n^2(2n^2 - 2n + 5), \quad \nu_n^2 = 1 + \frac{8}{81}n^2(2n^2 - 2n + 5). \quad (31.40)$$

The identity follows by summing the squares of the four entries. For $n = 1$, the sum is seven, and

$$\boxed{\nu_1 = \frac{11}{9}, \quad \text{Tr} \rho_{\text{vis}}^2 = \frac{9}{11}, \quad \bar{n}_W = \frac{1}{9}, \quad \frac{S_{\text{vis}}}{k_B} = \frac{10}{9} \log 10 - \log 9.} \quad (31.41)$$

Theorem 31.11 proves that the centred one-mode Gaussian state with symplectic eigenvalue ν is unitarily equivalent to the geometric density in (31.25), whose mean occupation is $\bar{n}_W = (\nu - 1)/2$. For $\nu = 11/9$, its eigenvalues are

$$\lambda_j = \frac{9}{10} 10^{-j}, \quad j = 0, 1, 2, \dots \quad (31.42)$$

The series sums to one. The sum of squares is $(9/10)^2/(1 - 10^{-2}) = 9/11$. Finally,

$$-\sum_j \lambda_j \log \lambda_j = -\log(9/10) + \log 10 \sum_j j \lambda_j = \frac{10}{9} \log 10 - \log 9.$$

Gaussian reduction and its spectrum are proved in Section 31.5; the preceding two series evaluate exactly the purity and entropy of this realization. Since the joint state is pure, S_{vis}/k_B is also its dimensionless entanglement entropy. The global transformation is unitary, whereas restriction to the first mode has positive entropy.

The result concerns the specified preparation and transport. Identifying it with a vacuum measurement requires the experimental realization of that protocol. The entropy of a reduction and transferred heat remain distinct quantities.

32 Operational memory and invariants of an evolving structure

32.1 Observable value, capacity to act, and compatibility

To know an object is to determine what its observable value preserves and which aspects of its capacity to act require an additional description. In the HMT construction, two present readouts may coincide while the orientation and memory retained produce different responses to the same continuation. Genealogy acquires physical and mathematical content when it determines these differences in response.

The origin of this investigation is the APP–TRIT–TPK composition, its enriched state, and the joint discrete structure. The coordinates π, φ, e, α retain their role as correlated outputs. The capacity, reciprocity, and action readout operators intervene afterwards, with their own domains. The following compositions examine those readout operators and retain the generator as their antecedent.

One conceptual characterization of constants is to study them as compatibility invariants of an evolving structure. For each constant, this characterization requires specifying the transformations under which it remains invariant, the information that may vary, and the readout operator through which it is identified. Radial reciprocity provides a proved case: the action section remains invariant when the radius is inverted and the corresponding change of orientation is retained. Extending this invariance to the full holonomy requires the explicit composition of the operations involved.

32.2 Retained memory and operationally visible memory

In the joint representation of readout and memory, the readout is updated by $x' = Tx + Dm$. For two states with the same present readout, one obtains

$$\Delta x' = D \Delta m. \quad (32.1)$$

The genealogical difference is visible at that step precisely when $D\Delta m \neq 0$. The identity identifies the continuation that transforms a difference in memory into a difference in readout.

Conversely, for a stationary transport C in this representation, one has

$$D = \frac{\sqrt{8}}{9}(C - I), \quad R = \frac{I + 8C}{9}. \quad (32.2)$$

If $C\Delta m = \Delta m$, then $D\Delta m = 0$ and $R\Delta m = \Delta m$. This difference in memory remains invisible under repetitions of the same protocol. A different operation may transfer it to the readout.

The sufficiency of a description therefore concerns the differences it must preserve in order to predict responses to the admissible operations. The full genealogy may contain more information than a particular observation requires. The observer's scope determines which reduction retains predictive capacity. This reduction is studied from the transports produced by HMT and preserves the provenance of its states and operators.

In the stationary case, the invisible memory differences are exactly $\text{Fix}(C)$. For a collection of available operations, the differences invisible under every continuation form the intersection of their fixed spaces. Necessity follows by taking each operation as the first step. Sufficiency is proved by induction: every operation fixes the difference in question and its transfer block annihilates that difference, so that it never appears in the readout. The result concerns this linear representation; the chronological counter of the holonomy retains its own map into it.

32.3 Regularity of operations and prolongation of the state

Vacancy cadences distinguish several levels of closure. A TRIT regime may return while the capacity phase remains displaced. A nonadic phase may return while the turn counter increases. Repetition of a type of operation is compatible with the production of a new history.

This distinction makes it possible to study the regularity of an unbounded prolongation: a finite rule may order the evolution without every new state reproducing an earlier one. Finiteness of the rule and finiteness of the set of reachable states are different properties. An aperiodic decimal readout may arise from dynamics subject to strict relations.

In the correlated regions, the object of study is the relation that jointly determines the next block of each region, the properties preserved by the mirror, and the memory that makes the prolongations compatible. The 21/22 and 6/7 cadences are capacity observations; their concrete maps relate them to the regional readout operators. The proof of arbitrary depth and the explanation of the shared structure are distinct and complementary results.

Composition of the cadences separates the levels exactly: completing the three changes of the selector leaves a capacity displacement of three or six classes modulo nine. The durations 437 and 459 quantify this distinction in the corresponding construction; their role here is to describe those cadences, without promoting them to additional fundamental constants.

32.4 Action character and oriented radial reconstruction

The character H_5 and the regional continuation determine an action section. The radial form makes it possible to reconstruct that section when orientation is retained. At fixed α , the coordinate $\chi_{\text{rad}} = \epsilon \log q_\epsilon$ classifies reciprocity and is monotonically related to the return section:

$$\frac{\eta_{\text{ret}}}{\alpha} = -500 \tanh\left(\frac{\chi_{\text{rad}}}{2}\right) - 1. \quad (32.3)$$

The oriented radial form and the action section are subject to a joint constraint. Action incorporates a condition of geometric compatibility; its coefficients, regional correction, and orientation must remain coordinated.

This relation distinguishes two levels of knowledge. Reparametrization proves equivalence of descriptions and makes it possible to recover either from the other. A comparison between readout operators provides additional information when both are evaluated from the preceding structure without introducing the value being checked as an input. Computing the radius through the inverse of the action formula and substituting it back verifies an algebraic identity. Producing the geometric readout through its antecedent operations and comparing it with the action character tests the causal composition. Each check has its own role.

The distinction identifies the developments that must be assembled and executed; it retains HMT genealogy as their common root. The comparison must be independent of the value being checked, not of the shared origin of its readout operators.

32.5 Action, information, and temperature

The state of the admissible continuations and their weights determines dimensionless information. The transducer k_B realizes its readout in entropy units. Action and period enter the relation

$$k_B \Theta_{\text{clk}} \log 3 = \frac{h}{T_{108}}. \quad (32.4)$$

The three equiprobable alternatives produce $\log 3$; a nonuniform distribution retains its weights in the evaluation of information.

The composition makes it possible to follow how the same evolution changes operational memory, compatible continuations, and the energy readout. The length of a history and its temperature correspond to different readouts. Thermal evaluation requires identifying the statistical state and the exchange operation that realize it. The precise problem is to determine which generated transformations preserve action, which redistribute accessible information, and which produce a calculable thermal difference.

When a path closes in a projection while its memory changes, evaluating its work requires declaring the space in which the cycle is considered complete. This is the problem of conservativity. Noncommutativity, by contrast, characterizes sensitivity to the order of operations. Preserving the distinction makes it possible to study the relation between the two properties.

32.6 Comparison programme and hierarchy of checks

Histories with a common readout and different future responses. The first comparison brings together two admissible routes produced by TPK that coincide under one readout operator, retains their memories, and applies the same continuation. The quantity examined is the exact separation of their subsequent readouts and the memory component that determines it. The difference between value and structure is thus expressed through a difference in predictive capacity. The examples already constructed retain their provenance and can be prolonged within this scheme.

Compatibility between regional action and geometric form. The second comparison follows the H_5 readout operator and its continuation, and the geometric readout operator with its own antecedents.

Their compatibility is examined before and after a return, distinguishing the limiting section from its finite approximations. The comparison identifies what remains constant while memory advances. The proved radial invariance provides the reference case and retains its specific scope.

Order of operations and memory re-entry. The third comparison uses two effective TPK operations and retains the intermediate charts. The 1 : 8 partition of the auxiliary realization corresponds to the regime of a fibre with identity reference. The application must verify that regime or transport the corresponding charts. The mathematical experiment compares the two orders with and without memory re-entry; the physical realization must specify the preparation, interaction, and observable that represent those operations.

The first two comparisons establish the conceptual priority: they determine which information has dynamical efficacy and which invariants support the compatibility of the architecture. The third provides a direct quantitative response within its proved regime. This hierarchy preserves the auxiliary role of the energy realization.

32.7 Scientific comparison and scope of interpretation

Memory in reduced dynamics has precedents in mathematical physics. The Mori–Zwanzig formalism makes it possible to study the evolution and memory terms of reduced observables; an example is the analysis by Hudson and Li.² The relevant comparison concerns the concrete composition examined here: APP–TRIT–TPK origin, correlated regions, action, orientation, and reconstruction. Mentioning memory alone does not determine priority for that composition.

The expository contribution consists in specifying consequences and comparisons within the construction. A historical assessment of novelty and physical corroboration retain their specific tests. The mathematical programme establishes objects and operations that can be subjected to scientific comparison without anticipating the outcome of those assessments.

Stored memory and operationally visible memory are characterized here through the kernel of D and the fixed space of C . This characterization follows from the declared update and retains the scope of its representation. The three comparisons in Section 32.6 constitute a work programme; their formulation distinguishes the definition of the mathematical experiment, the execution of TPK routes, and the realization of a physical experiment.

33 Chronological memory, balance, and vacancy response

The following construction receives the symbolic capacity factor and studies its subsequent readout operators. The internal ratio $3^6/10^3 = 729/1000$ and its calendar belong to the preceding construction (Sections 5.11 and 5.9.1). The APP–TRIT–TPK architecture, the regions, nonadic return, and the record K retain their prior causal position. The logarithms used in the capacity readout operator describe that factor and do not select the regions of π, φ, e, α anew. The propositions are focal consequences concerning observation and memory; their scope is distinguished from the full TPK state and from an attribution of historical priority for general results on mechanical sequences or information.

²Hudson and Li, *Coarse-graining of overdamped Langevin dynamics via the Mori–Zwanzig formalism*, arXiv:1810.08175.

33.1 Object, measure, and two readout operators

Let $\delta = \log_{10}(10/9)$ be the slope of the capacity calendar, and let θ be a phase uniformly distributed on $[0, 1)$. The measure represents uncertainty about the readout origin, while the mechanism remains determined. Define

$$\begin{aligned} z_j(\theta) &= \lfloor (j+1)\delta + \theta \rfloor - \lfloor j\delta + \theta \rfloor, \\ W_n(\theta) &= (z_0(\theta), \dots, z_{n-1}(\theta)), \\ B_n(\theta) &= \sum_{j=0}^{n-1} z_j(\theta). \end{aligned} \tag{33.1}$$

The block W_n retains chronology, and B_n retains balance. For a known exact phase and exact rule, both objects are deterministic; the entropies refer expressly to the set of possible phases.

Telescoping gives

$$B_n(\theta) = \lfloor n\delta + \theta \rfloor. \tag{33.2}$$

Thus B_n takes only the values $\lfloor n\delta \rfloor$ and $\lfloor n\delta \rfloor + 1$. The points $\{-j\delta\}$, for $j = 0, \dots, n$, divide the circle into $n+1$ phase intervals. Each interval carries a distinct sequence of length n . Indeed, the cumulative prefixes $\lfloor k\delta + \theta \rfloor$, for $k = 1, \dots, n$, are monotone in θ , and each boundary changes one of them; the sequence reconstructs all those prefixes.

The endpoints have measure zero. A half-open convention resolves their assignment and preserves the entropies.

33.2 Increasing information and zero entropy rate

Let $\ell_0, \dots, \ell_n$ be the lengths of the phase intervals, and let

$$H_n = - \sum_{j=0}^n \ell_j \log_2 \ell_j. \tag{33.3}$$

Proposition 33.1. *The following hold simultaneously:*

$$H_n \longrightarrow +\infty, \quad \frac{H_n}{n} \longrightarrow 0. \tag{33.4}$$

Proof. The upper bound is $H_n \leq \log_2(n+1)$. If $\ell_{\max}(n) = \max_j \ell_j$, then $H_n \geq -\log_2 \ell_{\max}(n)$. The slope δ is irrational: if $\delta = p/q$ with $q > 0$, then $(10/9)^q = 10^p$, contradicting the valuation at the prime 3. By irrationality, the orbit points are dense. The successive partitions have maximum mesh tending to zero: for every $\epsilon > 0$ there is a finite ϵ -dense segment, and later segments only subdivide its intervals. The lower bound therefore diverges. The upper bound divided by n tends to zero. $\square$

The interval lengths are not equal in general. The proof establishes the preceding bounds without asserting $H_n = \log_2(n+1)$ or an exact logarithmic asymptotic law that would require additional Diophantine conditions. The topological entropy of the factor is also zero because $\log(n+1)/n \rightarrow 0$. Block information and its rate are different quantities. Growth of the archive of a known trajectory does not, by itself, amount to production of independent statistical information.

33.3 Chronological information not determined by balance

Corollary 33.2. *For the same set of phases,*

$$H(W_n | B_n) = H_n - H(B_n) \geq H_n - 1 \longrightarrow +\infty. \tag{33.5}$$

Proof. B_n is a function of W_n , so the chain rule gives the equality. Its support has two elements, hence $H(B_n) \leq 1$ bit. Apply Proposition 33.1. $\square$

The conditional entropy that diverges is the average over balances; the same uncertainty is not assigned to every particular balance. Nor is that loss attributed to the full enriched state: when the retained memory recovers W_n , one has

$$H(W_n | B_n, M_n) = 0. \quad (33.6)$$

By cardinality, at least one of the two fibres of B_n contains $\lfloor (n+1)/2 \rfloor$ sequences or more. This combinatorial bound is distinct from the entropy bound. The result specifies the relation between value and genealogy for this readout operator: the two possible balances do not exhaust the compatible temporal histories.

33.4 Centred current: bounded balance and persistent activity

The centred readout operator is expressed by

$$J_j = z_j - \delta, \quad \sum_{j=0}^{n-1} J_j = B_n - n\delta. \quad (33.7)$$

Consequently, $\left| \sum_{j=0}^{n-1} J_j \right| < 1$ for every window. At the same time,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} J_j = 0, \quad \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} J_j^2 = \delta(1 - \delta) > 0. \quad (33.8)$$

Proof. One has $z_j^2 = z_j$ and $B_n/n \rightarrow \delta$ by the discrepancy bound. Expanding $(z_j - \delta)^2 = z_j(1 - 2\delta) + \delta^2$ and substituting the average of z_j gives the second identity. The argument holds for every phase and requires no random averaging. $\square$

Bounded cumulative balance, positive quadratic activity, and zero entropy rate are therefore compatible. The identities preserve the distinction between the centred readout operator and a representation of elementary charge; nor do they establish energy production without a balance.

If an already established dimensional realization takes $I_j = (u_Q/t_0)J_j$, one obtains

$$I_{\text{rms}}^2 = \left(\frac{u_Q}{t_0} \right)^2 \delta(1 - \delta). \quad (33.9)$$

If each I_j is maintained for an interval of duration t_0 , this discrete average coincides with the time average. As a subsequent physical comparison, an ideal resistive load $R_{\text{res}} > 0$ would have mean power

$$\bar{P}_{\text{res}} = R_{\text{res}} \left(\frac{u_Q}{t_0} \right)^2 \delta(1 - \delta) > 0. \quad (33.10)$$

This is a prediction of the model composed with that load. The resistance and the device retain their own specifications; they do not follow from the capacity sequence alone.

33.5 Current spectrum and orientation

The observation function is $f(\theta) = \mathbf{1}_{[1-\delta, 1)}(\theta) - \delta$, with $J_j = f(\theta + j\delta)$. Its Fourier coefficients, for $m \neq 0$, are

$$a_m = \frac{\exp(2\pi i m \delta) - 1}{2\pi i m}, \quad |a_m|^2 = \frac{\sin^2(\pi m \delta)}{\pi^2 m^2}, \quad a_0 = 0. \quad (33.11)$$

They are obtained by integrating the indicator function over its interval. Parseval's identity gives

$$\sum_{m \neq 0} |a_m|^2 = \delta(1 - \delta), \quad (33.12)$$

in agreement with quadratic activity. The temporal frequencies are $m\delta$ modulo 1. The stationary correlation, with uniform phase, is determined by these intensities; they do not contain the initial

phase. A translation multiplies a_m by a unit-modulus phase factor, and time reversal exchanges opposite frequencies.

Spectral power alone recovers neither orientation nor the temporal origin. A phase-sensitive readout or a marked reference supplies the complementary information. The deduction preserves the distinction between that general reference and the record K , as well as between symbolic rhythms and a particular experimental realization.

33.5.1 Cyclic reference and transports of equal spectral intensity

The cyclic property of the record K yields a finite consequence independent of the preceding spectrum. In $V_K = \mathbb{R}^{12}$, let S be the shift $(Sx)_i = x_{i+1}$ and let

$$O_K = (K, SK, \dots, S^{11}K). \quad (33.13)$$

We use the invertibility of O_K established by the cyclic reference of the record. For any real operator H commuting with S , put $y_+ = HK$ and $y_- = H^T K$.

Proposition 33.3. *The twelve Fourier intensities of y_+ and y_- coincide. However, $y_+ = y_-$ if and only if $H = H^T$. Each full vector response, together with K and the orientation of S , recovers its operator.*

Proof. H is circulant. Its Fourier multipliers are λ_m , and those of H^T are their conjugates, so the two responses have intensities $|\lambda_m|^2 |\hat{K}_m|^2$. If their vectors coincide, $(H - H^T)K = 0$. Commutation implies that $H - H^T$ also annihilates $S^j K$ for every j . These vectors form a basis; therefore $H - H^T = 0$. The converse is immediate. Recovery is expressed by

$$H = O_{y_+} O_K^{-1}, \quad H^T = O_{y_-} O_K^{-1}. \quad (33.14)$$

□

In particular, the shifts S and S^{-1} produce responses with the same power spectrum but different vectors. The exact autocorrelations of the record give

$$\|SK - S^{-1}K\|^2 = 2(\|K\|^2 - \langle K, S^2 K \rangle) = 2175744 > 0. \quad (33.15)$$

Rational inversion of O_K recovers from the two responses the coefficients e_1 and e_{11} , respectively. K is used after its generation as a marked reference. Spectral intensity or scalar balance does not replace the vector response. The corollary describes the twelve-component cyclic carrier; it preserves the difference between S and the full enriched dynamics, as well as the additional operators of a physical charge conjugation. For general H , transposition does not mean time reversal either; for S , it does coincide with S^{-1} .

33.6 Electronic operator link

Incidence between the APP sum and product fibres over D_9^3 defines C . Its tritic vector $m_{\text{ph}} = (1, -1, 0)^3$ satisfies

$$Cm_{\text{ph}} = C^T m_{\text{ph}} = -\frac{1}{2}m_{\text{ph}}. \quad (33.16)$$

The induced calendar traverses the regimes of the same tritic selector; the incidence selects the principal plane with singular value $1/2$. On that plane,

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}, \quad J = \frac{4}{\sqrt{3}}[P, Q], \quad J^2 = -I. \quad (33.17)$$

The incidence and the commutator norm are developed in Section 19.5. An effective relation between temporal organization and the electronic block is thereby recovered, preserving the selector, its labels, and the incidence. The charge representation and the dimensional realization retain their own operators: the link does not assign opposite elementary charges directly to $z_j - \delta$.

33.7 Retained information and energy composition

Fibre recovery establishes

$$H(X \mid q(X), M(X)) = 0 \quad (33.18)$$

when the enriched readout operator has an inverse; a bijective update preserves $H(X)$. These properties are assembled in (24.3) and (24.4). The coexistence of increasing H_n and zero rate is compatible with those properties and with Shannon's identities.

Erasure work depends on the accessible information retained. In the ideal isothermal regime of degenerate logical memories and in the appropriate asymptotic limit, the cost per copy with side information is

$$W_{\text{borrado,ideal}} = k_B T \ln(2) H(X \mid M). \quad (33.19)$$

If X can be recovered from M , the ideal logical contribution may be zero. Physical transport, readout, control, and the full cycle retain their own balances; the equality $h_{\text{top}} = 0$ does not remove them.

The relevant composition preserves the sequence

$$\begin{aligned} \text{chronology} &\longrightarrow \text{boundary record} \longrightarrow \text{current and state readout operator} \\ &\longrightarrow \text{retained memory} \longrightarrow \text{thermal balance.} \end{aligned}$$

The preceding development constructs the first two readout operators and their informational consequences. Its scope is distinguished from certification of a complete physical device and leaves intact the constructive genealogy of the constants.

34 Visible repetition and genealogical differentiation within a region

The biographies in (6.26) make it possible to specify the difference between a repetition of digits and a return of the state. For the channel recognized as e , the calendar (6.27) produces

$$201101 \mid 121221 \mid 102011 \mid 012222 \mid 102011.$$

Lifting precedes the decimal readout. The same matrices act on the other two channels, preserving their initial emissions and their distinct regional states. This common structure makes it possible to compare the biographies without identifying their coordinates.

Let w be a concatenation of L trits, and let N be its integer value in base three. The corresponding cylinder is

$$I_w = [N/3^L, (N+1)/3^L).$$

For d decimal places, the prefix is stable throughout the cylinder exactly when

$$\left\lfloor \frac{10^d N}{3^L} \right\rfloor = \left\lfloor \frac{10^d (N+1) - 1}{3^L} \right\rfloor.$$

This applies Lemma 7.12 to the decimal readout. The half-open right endpoint explains the term -1 in the integer division. Adding the integer part of the regional chart, the five successive readouts are

Tritic length	Stable prefix of the regional readout
6	2.71 ...
12	2.71828 ...
18	2.7182818 ...
24	2.7182818284 ...
30	2.71828182845904 ...

At length twenty-four, one has $N = 202864003874$ and $3^{24} = 282429536481$. At length thirty, one obtains $N = 147887858824447$ and $3^{30} = 205891132094649$. The preceding integer divisions certify the prefixes in the table. In particular, the first twelve fractional digits group as

$$718281828459 = 7 \underbrace{1828\ 1828}_{\text{positional repetition}} 459.$$

The twenty-four-trit cylinder already fixes both copies and the following digit. This localization distinguishes the decimal episode from the reappearance of the tritic block 102011, located in the third and fifth positions: their cuts and their readout bases differ.

Repetition of the visible block does not identify its prestates either. The prestates modulo twenty-seven retained in the development of this biography are

$$(25, 11, 7, 17, 8, 16), \quad (7, 8, 4, 23, 26, 7),$$

with respective subsequent quotients

$$(6, 13, 9, 2, 13, 1), \quad (15, 0, 3, 19, 23, 25).$$

Equality of the emission thus coexists with a differentiated genealogy. Its explanatory role is specific: observing a repeated block retains less information than the enriched continuation that produces it. To attribute a future difference exclusively to memory, the phases and comparison charts must also be retained.

This episode relates three levels: the transitions produce the tritic chain, the cylinders determine the stable digits, and the state retains the provenance of a repeated emission. Arbitrary depth belongs to the prolongation proved earlier; this finite readout specifies one episode of that biography. The regularity of interest lies in the compatibility between levels, not in prolonging a decimal fragment indefinitely: a third copy of 1828 would begin with 1, whereas the continuation already produced begins with 4.

35 Genealogical regularity, operator reconstruction, and the geometry of action

35.1 Production, compatibility, and reconstruction

The primary phenomenon is a dynamics that preserves provenance while producing new observations. Three operations with distinct mathematical roles are brought together:

1. *Production.* The APP-TRIT-TPK update prolongs the state and its correlated regions. The phase may return while memory advances.
2. *Compatibility.* The regional, angular, action, and incidence readouts obey common relations. Their variation must respect those relations in order to correspond to the same state.
3. *Reconstruction.* Retaining complementary information makes it possible to recover earlier values, products of operators, and differences between orders of composition.

A comparison between procedures for obtaining π must attend to these three levels. The specific HMT thesis lies in the joint construction and its relations, in addition to the production of numerical expansions. A decimal coincidence or the periodicity of a phase does not, by itself, describe the scope of the construction. The distinction identifies the mathematical object relevant to a historical comparison; the focal development does not, by itself, establish worldwide priority.

35.2 Vacancy regimes and return of the capacity phase

35.2.1 Capacity calendar and tritic selector

The capacity readout receives the internal compactification 729/1000 and distinguishes the refinement index from the capacity index. With the convention of Section 5.9.1,

$$\delta = \log_{10}(10/9), \quad p_j = \left\lfloor \frac{j}{\delta} \right\rfloor, \quad v_j = p_j - j, \quad j \geq 1. \quad (35.1)$$

The intervals between events are

$$h_j = p_{j+1} - p_j \in \{21, 22\}, \quad s_j = 22 - h_j. \quad (35.2)$$

The tritic class obeys

$$[v_{j+1}]_3 = [v_j - s_j]_3. \quad (35.3)$$

A short interval changes the class; a long one preserves it. The 6/7 spacings describe the plateaux of this induced regime. The construction retains its type as an observation factor of the TPK state; it neither replaces that state nor identifies those spacings with charge signs or with digits of π or e . The canonical selector is $M_{\text{ph}}(1 + [v_j]_9)$, whose pattern depends on the class modulo three.

35.2.2 Restoration of the local regime and displacement modulo nine

Let

$$\beta_{\text{cap}} = 22 - \delta^{-1}, \quad q_k = \left\lfloor \frac{k}{\beta_{\text{cap}}} \right\rfloor, \quad k \geq 1. \quad (35.4)$$

The positions q_k are the indices of the short intervals. Indeed,

$$p_j = 22j - [j\beta_{\text{cap}}], \quad s_j = [(j+1)\beta_{\text{cap}}] - [j\beta_{\text{cap}}]. \quad (35.5)$$

Irrationality of δ follows from the valuations at 2, 3, 5 in the equality that $(10/9)^q = 10^p$ would impose; it implies irrationality of β_{cap} . Consequently, positive indices have no endpoint ambiguity. The discontinuity corresponding to the integer k occurs at $j = [k/\beta_{\text{cap}}]$.

Proposition 35.1. *Between the events indexed by q_k and q_{k+3} , taking the intervals with indices in $[q_k, q_{k+3})$, the TRIT class returns while the capacity phase modulo nine changes.*

Proof. There are exactly three short intervals in this segment. The bounds $1/7 < \beta_{\text{cap}} < 3/20$ give $20 < 3/\beta_{\text{cap}} < 21$, hence

$$m_k = q_{k+3} - q_k \in \{20, 21\}. \quad (35.6)$$

The sum of lengths and the capacity gain are, respectively,

$$\Delta p = 22m_k - 3 \in \{437, 459\}, \quad \Delta v = 21m_k - 3 \in \{417, 438\}. \quad (35.7)$$

In both cases $\Delta v \equiv 0 \pmod{3}$. By contrast,

$$\Delta v \equiv 3 \text{ or } 6 \pmod{9}. \quad (35.8)$$

This proves the statement. The bounds on β_{cap} have integer certificates:

$$\begin{aligned} 10^{146} > 9^{153} &\implies \delta > \frac{7}{153}, \\ 10^{417} < 9^{437} &\implies \delta < \frac{20}{437}. \end{aligned} \quad (35.9)$$

Monotonicity of $22 - 1/x$ yields $1/7 < \beta_{\text{cap}} < 3/20$. Thus the bounds rest on integer inequalities, not on substitution of a rounded decimal. $\square$

The selector value $\tau_j = M_{\text{ph}}(1 + [v_j]_9)$ returns because it depends on the class modulo three. This return remains distinct from the full TRIT state and from a chronological turn of Γ_9 . Both durations occur infinitely often by density of the irrational rotation with slope $1/\beta_{\text{cap}}$.

Regularity has distinct levels: the local regime is restored while the capacity phase remains displaced. The nonadic return of the state also has its own memory increment. Each object therefore retains its own notion of return and its own duration. The numbers 437 and 459 are durations in this induced observation, without being assigned the status of additional universal constants. The result explicitly assembles the preceding formulae; presenting it as a focal consequence does not assert historical priority over developments that have not been exhaustively compared.

35.3 The contribution of memory to operator composition

35.3.1 Enlarged representation of readout and memory

The readout–memory representation is obtained by separating eight components from a ninth. In a single fibre, with identity reference transport, the internal transport C induces

$$T_C = \frac{8I + C}{9}, \quad D_C = \frac{\sqrt{8(C - I)}}{9}, \quad R_C = \frac{I + 8C}{9}, \quad (35.10)$$

$$u_C = \begin{pmatrix} T_C & D_C \\ D_C & R_C \end{pmatrix}. \quad (35.11)$$

The coupling arises from the change of basis in the nonadic decomposition. For transports preserving the declared form, it preserves the enlarged form. The composition identity is

$$u_{C_2} u_{C_1} = u_{C_2 C_1}, \quad T_2 T_1 + D_2 D_1 = \frac{8I + C_2 C_1}{9}, \quad (35.12)$$

with $T_i = T_{C_i}$ and $D_i = D_{C_i}$. The second term corresponds to memory produced in the first operation and reintroduced in the second. Setting it aside changes the procedure: it produces $T_2 T_1$ instead of the readout of the enlarged composition.

35.3.2 Partition of sensitivity to the order of composition

Proposition 35.2. *With the convention $[A, B] = AB - BA$,*

$$[T_2, T_1] = \frac{1}{81}[C_2, C_1], \quad [D_2, D_1] = \frac{8}{81}[C_2, C_1]. \quad (35.13)$$

Consequently, the readout difference between the two orders, with memory reintroduced, is

$$(T_2 T_1 + D_2 D_1) - (T_1 T_2 + D_1 D_2) = \frac{1}{9}[C_2, C_1]. \quad (35.14)$$

Proof. On expanding the products of T , the constant and linear terms cancel in the subtraction, leaving the commutator divided by 81. Expanding the products of D gives the same cancellation, with a factor of 8/81. Their sum is 1/9. $\square$

The re-entry mechanism belongs to the enlarged representation. Comparing the two orders makes one consequence explicit: in this representation, the memory contribution to the antisymmetric response is eight times that retained by composing isolated readouts. The full response is nine times the latter. These ratios concern response operators, not percentages of energy.

Reconstruction preserves noncommutativity and its readout intensity. In this case, isolated compression attenuates a nonzero commutator without annihilating it. An application to physical forces also retains the interaction operators and the corresponding work functional; the algebraic identity does not identify them by itself.

35.3.3 Operator comparison and nonadic normalization

For the general algebraic pattern $(n - 1) + 1$, the same calculations give the factors $1/n^2$, $(n - 1)/n^2$, and $1/n$. The nonadic architecture fixes $n = 9$. The composition law and the particular value of its normalization are thus distinguished.

When the effective operations of an application are represented by C_1, C_2 , procedures that set aside or reintroduce memory predict different order responses. The comparison concerns operational composition and the accessibility of memory, preserving the coefficients of the representation.

35.3.4 Readout and memory norms under central inversion

For $C = -I$, one obtains

$$T^*T = \frac{49}{81}I, \quad D^*D = \frac{32}{81}I. \quad (35.15)$$

The numerator 32 comes from $8|-1 - 1|^2$. In the nine memory components, the norm includes $8^2 + 8 = 72$, with denominator 27^2 . The square 64 enters that calculation together with the other eight contributions. Identification with a region of 64 cells requires the region's incidence map and is not obtained by doubling the numerator 32. A geometric readout of an 8×8 support against a 9×9 support therefore retains the problem of its connection to the particular application; an isolated numerical equality does not replace it.

35.4 Action character and complete reciprocity coordinate

35.4.1 Genealogical coefficients and regional continuation

The central matrix and its invariants are

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \lambda_+ = 9, \quad \lambda_- = 5. \quad (35.16)$$

The length-four orbit of the step-three shift, the length-twelve calendar, and the census quotient $104976/1944 = 54$ also enter. The coefficient relations are

$$\begin{aligned} \frac{9}{16} &= \frac{\lambda_+}{4^2}, & \frac{5}{9} &= \frac{\lambda_-}{\lambda_+}, \\ \frac{7}{48} &= \frac{\text{tr } B_c/2}{4 \cdot 12}, & \frac{1}{54} &= \frac{1944}{104976}. \end{aligned} \quad (35.17)$$

The assignment of degrees and signs belongs to the definition of the character; its isolated cardinalities do not determine it. The complete readout is

$$H_5 = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \quad (35.18)$$

$$D_A = \exp(-100\pi\alpha/9),$$

$$\mathcal{C}_\pi = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \quad (35.19)$$

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi}\mathcal{C}_\pi.$$

The fraction preserves the infinite continuation defined by

$$\mathcal{R}(z) = D_A z^7 + D_A z \mathcal{R}(z), \quad (35.20)$$

with unit initial weight. The continuation remains intact in the readout; a truncated polynomial does not replace that object. The construction of the coefficients and the renewal are assembled in Section 11.2.1.

35.4.2 Classification of oriented radial reciprocity

The relations $A_{\text{deg}} = 1000\alpha$, $C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha)$ and the radial recovery map of Theorem 27.1 are retained. To keep both orientations, let

$$\epsilon \in \{+1, -1\}, \quad q_{\text{rad}} = R_{\star}^4 > 0, \quad \Xi_{\text{rad}} = \epsilon \frac{1 - q_{\text{rad}}}{1 + q_{\text{rad}}}. \quad (35.21)$$

The recovery map (27.4) is written

$$\frac{\eta_{\text{ret}}}{\alpha} = 500\Xi_{\text{rad}} - 1. \quad (35.22)$$

The involution $(\epsilon, q_{\text{rad}}) \mapsto (-\epsilon, q_{\text{rad}}^{-1})$ preserves that action. The coordinate assembling this structure is

$$\chi_{\text{rad}} = \epsilon \log q_{\text{rad}}. \quad (35.23)$$

Proposition 35.3. χ_{rad} classifies exactly the orbits of this involution on $\{\pm 1\} \times \mathbb{R}_{>0}$.

Proof. Simultaneously changing the sign and inverting q_{rad} preserves χ_{rad} . If two pairs have the same value of χ_{rad} , equal signs imply equal q_{rad} ; with opposite signs, their fourth powers of the radii are reciprocal. These are exactly the two cases of belonging to the same orbit. The argument includes $q_{\text{rad}} = 1$, whose orbit retains both orientations. $\square$

The identity $(1 - q)/(1 + q) = -\tanh(\log q/2)$ gives

$$\Xi_{\text{rad}} = -\tanh(\chi_{\text{rad}}/2), \quad \eta_{\text{ret}} = \alpha[-500 \tanh(\chi_{\text{rad}}/2) - 1]. \quad (35.24)$$

Compatibility with the full character takes the form

$$\chi_{\text{rad}} = -2 \operatorname{artanh} \left(\frac{H_5 + \alpha - (90/\pi)\mathcal{C}_\pi}{500\alpha} \right). \quad (35.25)$$

The term $-\alpha$ in H_5 cancels the $+\alpha$ from the angular contrast before this readout. The denominator of the regional continuation remains complete. The equality receives the preceding HMT outputs and readout operators and assembles their two representations; it does not define α retroactively.

On the branch of positive action and finite radius,

$$0 < \eta_{\text{ret}} < 499\alpha, \quad \chi_{\text{rad}} < -2 \operatorname{artanh}(1/500). \quad (35.26)$$

At fixed α ,

$$\frac{\partial \eta_{\text{ret}}}{\partial \chi_{\text{rad}}} = -250\alpha \operatorname{sech}^2(\chi_{\text{rad}}/2) < 0. \quad (35.27)$$

Thus the dimensionless action coefficient uniquely determines the oriented radial class in this representation, and conversely. For joint variations,

$$d\eta_{\text{ret}} = \frac{\eta_{\text{ret}}}{\alpha} d\alpha - 250\alpha \operatorname{sech}^2(\chi_{\text{rad}}/2) d\chi_{\text{rad}}. \quad (35.28)$$

The character H_5 , the regional correction, and the oriented radial anisotropy thus satisfy a compatibility relation rather than constituting three independent parameters. The analytic character and the geometry of reciprocity represent the same return section. This structural level of the Planck equation remains explicit before action is composed with a frequency.

The action decade retains its determinantal genealogy, independently of this reparametrization:

$$\hbar_{\text{ret}} = 10^{-34} \mathcal{U}_S \eta_{\text{ret}}, \quad \hbar_{\text{pre}} = 10^{-34} \mathcal{U}_S H_5. \quad (35.29)$$

The action per turn $h = 2\pi\hbar$ and the thermal readout

$$k_B \Theta_{\text{clk}} \log 3 = \frac{h}{T_{108}} \quad (35.30)$$

are also retained within their scope. The physical clock of a realization retains its construction: the coordinate χ_{rad} classifies this reciprocity, not the totality of TPK histories. Classification by χ_{rad} and the differential relation assemble the existing representations of action; they retain their character as a focal formalization and do not constitute a second independent derivation of Planck's constant.

35.5 Phase circuits, memory, and conservativity

Nonadic holonomy restores the phase and increments memory, as described in Section 5.16. If a path closed in the phase projection increases the integer counter, that increment cannot be written as the difference of a potential depending only on phase.

The proof follows by telescoping: the sum

$$\sum_j [F(g_{j+1}) - F(g_j)] = 0 \quad (35.31)$$

around a phase circuit contrasts with the memory increment, equal to one per turn. On the enlarged state, the counter m records that increment. The path closed in the projection therefore remained open in the full state.

The study of conservativity begins by declaring the space in which the path closes and the coordinates participating in the dynamics. Noncommutativity and nonconservativity are different properties. Section 35.3 controls the order of operations; this section distinguishes a projected circuit from a full circuit. To study a particular force, the action or work functional must act on the same state space and the same transport.

Memory growth does not by itself assign thermodynamic production. Section 33.2 proves that block information can grow with zero entropy rate. Operator reconstruction adds a clarification: preserving information may mean preserving which transformations can be composed, in addition to retaining the stored symbols.

35.6 Dynamical compatibility between observations

The unit of study is dynamical compatibility between observations, with memory as part of the operation. The value of a constant constitutes a readout; its structure contains constraints on other readouts and on admissible transformations. Prolonging the analysis consists in obtaining relations that prevent value, orientation, memory, form, and action from being varied independently.

The three focal developments follow this same axis:

- A complete cycle of TRIT regimes still retains a nonadic capacity displacement: local regularity does not exhaust return.
- Memory re-entry changes exactly the response to the order of two operations: the history participates causally in the composition.
- The character H_5 and the oriented radial geometry mutually determine their return section: action admits a recoverable structural representation.

These consequences continue the genealogy of π, φ, e, α and make explicit what it means to preserve it when passing to operations and action. The subsequent composition assembles these levels through the common maps already constructed, maintaining the distinction between their indices and memories even when they share a name.

36 Operator antecedents of the criticality family

The criticality family is constructed from a readout of the TPK calendar. The two arithmetic sheets, tritic orientation and transport with memory first determine the states and their prolongations. The ordered reading of the continuum then allows analytic functions to be evaluated, parameters to be compared and roots to be selected. This development preserves the genealogical state and its two readouts: the Archimedean chart provides the evaluation space, while the incidence readout preserves boundary relations. The joint discrete structure of the continuum retains its five consubstantial constructions. The present exposition uses its ordered reader and preserves the common antecedent from which it originates.

36.1 Arithmetic cells, orientation and transport

We use the lifted cell (2.2), the tritic coordinates and their arithmetic law (2.6, 2.7), and the selector 2.3.2. To fix notation, let $D_9 = \{1, \dots, 9\}$, $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$ and, for every $n \in \mathbb{Z}$, $\rho_9(n) = 1 + ((n - 1) \bmod 9)$ and $q_9(n) = (n - \rho_9(n))/9$. The balanced representative is $r_3(n) \in \{-1, 0, +1\}$; its carry is $\chi(a, b) = (a + b - r_3(a + b))/3$. These formulae preserve the extension to integers of the kernel's positive evaluations.

Lifted transport and its reversibility are proved in 2.12; the observable configuration and its 108-step return, in 2.17 and 2.13. The update (2.12) preserves the full state. The holonomy in 2.22 and the returns in 4.22 distinguish phase restoration from memory advance. The concrete three-branch realization and its complete record are constructed next from these operations.

36.1.1 Edge-by-edge realization of the three nonadic extensions

This subsection specifies the autonomous model generated by the TPK extension relation entering the cardinal proof. The existence of an edge is not inferred from a coincidence of types: its initial and final states, APP relation, fiber transport, and unary update of the ledger are constructed. The construction uses exclusively the two APP sheets, the TRIT selector, and the TPK composition Sel–Tra–Upd.

Complete cells and three balanced reductions. In the chart $D_9 = \{1, \dots, 9\}$, we will always retain the complete APP cell

$$a_{p,q} := (p, q; R_{p,q}^\Sigma, q_{p,q}^\Sigma, R_{p,q}^\Pi, q_{p,q}^\Pi),$$

with

$$p + q = R_{p,q}^\Sigma + 9q_{p,q}^\Sigma, \quad pq = R_{p,q}^\Pi + 9q_{p,q}^\Pi.$$

The active sheet is recorded in a separate coordinate; $a_{p,q}$ is not split into a purported additive cell and a separate multiplicative cell. For $r \in D_9$, let

$$a(r) := \begin{cases} a_{1,9}, & r = 1, \\ a_{1,r-1}, & 2 \leq r \leq 9, \end{cases} \quad \tau(r) := M_{\text{ph}}(r), \quad m(r) := \tau(r) \in \{-1, 0, +1\}. \quad (36.1)$$

The additive residue of $a(r)$ supplies an APP representative of phase here; this does not make Σ the universally active sheet. The dynamic selector activates Σ for $r \in \{1, 4, 7\}$, activates Π for $r \in \{2, 5, 8\}$, and preserves both sheets without moving the cursor for $r \in \{3, 6, 9\}$. In all three cases, $R^\Sigma, q^\Sigma, R^\Pi, q^\Pi$ are transported jointly. We have

$$(m(1), \dots, m(9)) = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The three pairs ordered by the nonadic base are

$$\mathfrak{p}_{+1} := (1, 4), \quad \mathfrak{p}_{-1} := (2, 5), \quad \mathfrak{p}_0 := (3, 6). \quad (36.2)$$

These pairs form the normalized section that takes the first two occurrences of each TRIT type under the shift $r \mapsto r + 3$; the third phase block 7, 8, 9 preserves control of turn closure. Their two members belong to the same fiber of M_{ph} . The balanced reduction defined in these antecedents gives, without introducing a target coefficient,

ε	$m(\mathfrak{p}_\varepsilon^{(1)}) + m(\mathfrak{p}_\varepsilon^{(2)})$	$r_3(2\varepsilon)$	$\chi(\varepsilon, \varepsilon)$
+1	$1 + 1 = (-1) + 3 \cdot 1$	-1	+1
0	$0 + 0 = 0 + 3 \cdot 0$	0	0
-1	$(-1) + (-1) = 1 + 3 \cdot (-1)$	+1	-1

(36.3)

Therefore,

$$\chi(\varepsilon, \varepsilon) = \frac{2\varepsilon - r_3(2\varepsilon)}{3} = \varepsilon. \quad (36.4)$$

The branch sign is thus the output of the balanced cocycle applied to two TRIT emissions produced by APP. Each pair supplies one of the three carry outputs. For clarity, we subsequently index the pair by $\varepsilon(\mathfrak{p}) := \chi(m(\mathfrak{p}^{(1)}), m(\mathfrak{p}^{(2)}))$; the symbol ε does not supply a free carry to Upd^{cal} .

The phase architecture, balanced lift, and memory transport are realized here through twenty-seven local incidences and their extension graphs. The normalized section fixes representatives of the fibers of M_{ph} ; the covering to be proved uses the existence of these representatives and remains invariant when they are replaced by another compatible section.

Typed data of the nine edges. Let $K_{\text{Carr}} := \langle \kappa_{\text{Carr}} \rangle_{\text{Ab}}$ be the carry module. The actions are typed by

$$H_{\text{Carr}} : \mathcal{P}_{\text{TPK}}^{\text{op}} \longrightarrow \text{End}(K_{\text{Carr}}), \quad Y_{\text{Carr}} : \mathcal{P}_{\text{TPK}} \longrightarrow \text{End}(K_{\text{Carr}}).$$

In this generated model, take the trivial actions $H_{\text{Carr}}(e) = Y_{\text{Carr}}(e) = \text{id}_{K_{\text{Carr}}}$: the active sheet and the route are retained as coordinates of the ledger itself and do not act on the free carry summand. Also let (M_d, J_d, d) be the graded direct system of memory. To avoid confusing two representatives identified by the colimit, the state also retains the degree as a coordinate, and the labeled carrier

$$\tilde{M} := \bigsqcup_{d \geq 1} (\{d\} \times M_d), \quad \text{gr}_M(d, \mu) = d.$$

Publication in the limit memory is given by

$$\begin{aligned} M &:= \varinjlim_d M_d, & J_d &: M_d \longrightarrow M, \\ u &:= J_d(u_d), & \iota_M &: \mathbb{Z} \longrightarrow M, & n &\longmapsto nu. \end{aligned}$$

Here J_d is the canonical colimit map and the u_d form a compatible family. Its free component permits the reader $\deg_u(nu) = n$. The configuration, phase, memory, carry, route and survival projections will be typed on the calibrated carrier constructed recursively below; no ambient space will be imported for them. For each constructed edge e , the calibrated component $\text{Upd}_e^{\text{cal}} := \text{Mem}_e^{\text{cal}}$ will denote the third map in the factorization; its law is fixed field by field below and is not assumed as the restriction of an ambient relation.

Fix the initial depth $k_0 = 1$, a complete configuration $q_\star \in \mathcal{Q}_{\text{obs}}$ of phase one, and

$$q_{n,r} := F_{\text{obs}}^{\partial n + r - 1}(q_\star), \quad q_{n,10} := q_{n+1,1}.$$

Because the observable calendar is periodic, its value $q_{n,r}$ does not by itself distinguish occurrences separated by twelve turns. The calibrated model therefore uses the labeled lift

$$\tilde{q}_{n,r} := (q_{n,r}, n, r), \quad \tilde{q}_{n,10} := \tilde{q}_{n+1,1}, \quad \pi_{\text{obs}}(\tilde{q}_{n,r}) = q_{n,r}.$$

The lifted dynamics $\tilde{F}_{\text{obs}}(\tilde{q}_{n,r}) = \tilde{q}_{n,r+1}$ projects to F_{obs} ; the label does not alter the physical phase, but makes its levels of occurrence distinct. The copy of the cell at each level is labeled by

$$a_{n,r} := (n, r; a(r)), \quad (n, r)^+ := \begin{cases} (n, r+1), & 1 \leq r < 9, \\ (n+1, 1), & r = 9. \end{cases}$$

The label only prevents two occurrences of the same cell from being identified; the two Euclidean divisions and their four coordinates are those of $a(r)$. For turn $n \geq 0$ and phase $r \in D_9$, the symbol

$$e_{\varepsilon,n,r} : \tilde{q}_{n,r} \longrightarrow \tilde{q}_{n,r+1}, \quad r^+ := 1 + (r \bmod 9),$$

occupies level $d_{n,r} := k_n + r - 1$, where $k_n := k_0 + 9n$. Its local ledger contains the complete cell $a(r)$, the active sheet determined by $M_{\text{ph}}(r)$, the type $\tau(r)$, the membership mark in $\mathfrak{p}_\varepsilon$, and, upon reaching the second member of that pair, the certified identity (36.3). We define its operators solely from that ledger:

$$\mathcal{C}_M(e_{\varepsilon,n,r}) = J_{d_{n,r}+1, d_{n,r}}, \quad \kappa_M(e_{\varepsilon,n,r}) = \mathbf{1}_{\{r=9\}} u_{d_{n,r}+1}, \quad (36.5)$$

$$\text{Carr}(e_{\varepsilon,n,r}) = d_{\varepsilon,r} \kappa_{\text{Carr}}, \quad d_{\varepsilon,r} := \mathbf{1}_{\{r=\mathfrak{p}_\varepsilon^{(2)}\}} \chi(m(\mathfrak{p}_\varepsilon^{(1)}), m(\mathfrak{p}_\varepsilon^{(2)})). \quad (36.6)$$

For identities and every admissible composition, also define

$$C_M(\text{id}_{\tilde{q}}; d) := \text{id}_{M_d}, \quad \kappa_M(\text{id}_{\tilde{q}}; d) := 0, \quad (36.7)$$

$$C_M(e_t \cdots e_1) := C_M(e_t) \circ \cdots \circ C_M(e_1), \quad \kappa_M(e_2 e_1) := C_M(e_2)(\kappa_M(e_1)) + \kappa_M(e_2). \quad (36.8)$$

The identity carries degree d because the same observable configuration may recur at different depths. Iterated, the final equality defines $\kappa_M(e_t \cdots e_1)$ for every length and is the cocycle law of the direct system; the inclusions satisfy $J_{d+2,d+1} J_{d+1,d} = J_{d+2,d}$. In addition, set $s(e_{\varepsilon,n,r}) = \top$. The following twenty-seven entries certify the coordinates distinguishing the three extensions; the universal transport fields are specified immediately afterward. The symbols *I* and *II* designate, respectively, the first retained emission and the second emission performing the reduction; a dash denotes a zero carry increment, not the absence of an edge.

r	$M_{\text{ph}}(r)$	$\mathfrak{p}_{+1}$	$\mathfrak{p}_0$	$\mathfrak{p}_{-1}$	Δ	mem	
		mark	$d_{+1,r}$	mark	$d_{0,r}$	mark	$d_{-1,r}$
1	+1	I	0	–	0	–	0
2	–1	–	0	–	0	I	0
3	0	–	0	I	0	–	0
4	+1	II	+1	–	0	–	0
5	–1	–	0	–	0	II	–1
6	0	–	0	II	0	–	0
7	+1	–	0	–	0	–	0
8	–1	–	0	–	0	–	0
9	0	–	0	–	0	–	0
							u

Thus, $d_{\varepsilon,r}$ is computed from two outputs of the selector and is distinct from the cell quotients q^Σ, q^Π . The ledger also preserves the marked pair and the balanced residue; in particular, the zero branch is not confused with an absence of event.

The carry composition law used by the TPK category is

$$\text{Carr}(\text{id}_{\tilde{q}}) = 0, \quad \text{Carr}(e_2 e_1) = H_{\text{Carr}}(e_1)(\text{Carr}(e_2)) + Y_{\text{Carr}}(e_2)(\text{Carr}(e_1)). \quad (36.9)$$

On identities and paths, they are extended functorially:

$$\begin{aligned} H_{\text{Carr}}(\text{id}) &= Y_{\text{Carr}}(\text{id}) = \text{id}, \\ H_{\text{Carr}}(e_t \cdots e_1) &= H_{\text{Carr}}(e_1) \circ \cdots \circ H_{\text{Carr}}(e_t), \\ Y_{\text{Carr}}(e_t \cdots e_1) &= Y_{\text{Carr}}(e_t) \circ \cdots \circ Y_{\text{Carr}}(e_1). \end{aligned}$$

In this model all these actions remain the identity, but their typing makes explicit which endomorphism acts on each summand of (36.9). On calibrated edges, $H_{\text{Carr}} = Y_{\text{Carr}} = \text{id}$, so composition sums the increments already produced. Equation (36.9), rather than an identification between orientation and quotient, governs the accumulated register.

Independent construction of the transition tuples. A fiber state is written, in the order of its relevant fields, as

$$x = (a; q; \tau, o, h; b_{\text{ref}}; R^\Sigma, q^\Sigma, R^\Pi, q^\Pi; c; \text{Sig}; C, \partial C; p_{\text{APP}}, p_{\{-1,0,+1\}}; \lambda; \text{gr}_M; \text{mem}; s; \text{Led}).$$

The update notation $x[f \leftarrow v]$ changes only the indicated field. For $h \in \{(\Sigma, \Pi), (\Pi, \Sigma)\}$, define

$$h_r(h) = \begin{cases} (\Sigma, \Pi), & r \in \{1, 4, 7\}, \\ (\Pi, \Sigma), & r \in \{2, 5, 8\}, \\ h, & r \in \{3, 6, 9\}. \end{cases}$$

The complete sheet-and-arithmetic coordinate is abbreviated as

$$\hat{h}(x) := (h(x), R^\Sigma(x), q^\Sigma(x), R^\Pi(x), q^\Pi(x)).$$

Let $\hat{\mathcal{H}}_{\tilde{q},r}^{\text{cal}}$ be the set of quintuples

$$(h, R_{a_{n,r}}^{\Sigma}, q_{a_{n,r}}^{\Sigma}, R_{a_{n,r}}^{\Pi}, q_{a_{n,r}}^{\Pi}), \quad h \in \{(\Sigma, \Pi), (\Pi, \Sigma)\},$$

whose four arithmetic data are the two Euclidean divisions of the labeled APP cell $a_{n,r}$. This direct definition does not presuppose a state in a carrier not yet constructed. Let $\text{Hist}_{\tilde{q}}^{\text{cal}}$ be the set of admissible finite paths in the graph of labeled edges ending at $\tilde{q}$, including the identity, and let $\text{B}_{\tilde{q}}^{\text{cal}}$ be the set of their ordered endpoint sequences. Define

$$\text{S}_{\tilde{q}}^{\text{cal}} := \hat{\mathcal{H}}_{\tilde{q}}^{\text{cal}} \times \{o_{\rightarrow}, o_{\circ}, o_{\leftarrow}\} \times K_{\text{Carr}} \times \text{Hist}_{\tilde{q}}^{\text{cal}} \times \text{B}_{\tilde{q}}^{\text{cal}} \times \text{M}.$$

The calibrated signature is not postulated: it is the typed assembly map

$$\begin{aligned} \text{Sig}_{\tilde{q}}^{\text{cal}} : \hat{\mathcal{H}}_{\tilde{q}}^{\text{cal}} \times \{o_{\rightarrow}, o_{\circ}, o_{\leftarrow}\} \times K_{\text{Carr}} \times \text{Hist}_{\tilde{q}}^{\text{cal}} \times \text{B}_{\tilde{q}}^{\text{cal}} \times \text{M} &\longrightarrow \text{S}_{\tilde{q}}^{\text{cal}}, \\ (\hat{h}, o, c, \lambda, \partial C, \mu) &\longmapsto (\hat{h}, o, c, \lambda, \partial C, \mu). \end{aligned}$$

Likewise, $\pi_{\mathcal{F}}$ denotes the projection that removes only the first two coordinates $(a; q)$ of the state tuple; it preserves $\tau, o, h, b_{\text{ref}}$, the four Euclidean data, carry, signature, cylinder, boundary, prefixes, route, memory degree, memory, survival, and ledger. The corresponding orientation is fixed by $o(+1) = o_{\rightarrow}$, $o(0) = o_{\circ}$, and $o(-1) = o_{\leftarrow}$. The calibrated APP fiber over $\tilde{q}_{n,r}$ contains the already defined labeled copy $a_{n,r}$. The initial state is fixed without free coordinates:

$$\begin{aligned} \hat{h}_{\star} &:= ((\Sigma, \Pi), R_{1,9}^{\Sigma}, q_{1,9}^{\Sigma}, R_{1,9}^{\Pi}, q_{1,9}^{\Pi}). \\ x_{\star} &:= (a_{0,1}; \tilde{q}_{0,1}; +1, o_{\rightarrow}, (\Sigma, \Pi); \perp; R_{1,9}^{\Sigma}, q_{1,9}^{\Sigma}, R_{1,9}^{\Pi}, q_{1,9}^{\Pi}; 0; \\ &\quad \text{Sig}_{\tilde{q}_{0,1}}^{\text{cal}}(\hat{h}_{\star}, o_{\rightarrow}, 0, \text{id}_{\tilde{q}_{0,1}}, \emptyset, J_{k_0}(0_{\text{M}_{k_0}})); \emptyset, \emptyset; (a_{0,1}), (+1); \text{id}_{\tilde{q}_{0,1}}; k_0; 0_{\text{M}_{k_0}}; \top; \emptyset). \end{aligned} \quad (36.10)$$

The tuple contains no indeterminate coordinates and satisfies $s(x_{\star}) = \top$. Let $G_0 := \{x_{\star}\}$. Assuming G_n has been constructed, fix $x \in G_n$, $\varepsilon \in \{-1, 0, +1\}$, and $x_{\varepsilon,0}(x) := x$. For $r = 1, \dots, 9$, first construct the elementary ledger

$$\begin{aligned} \ell_{\varepsilon,n,r} &:= (e_{\varepsilon,n,r}, a_{n,r}, a_{(n,r)^+}, \tilde{q}_{n,r}, \tilde{q}_{n,r+1}, M_{\text{ph}}(r), \\ &\quad h_r(h(x_{\varepsilon,r-1}(x))), o(M_{\text{ph}}(r)), \mathfrak{p}_{\varepsilon}, d_{\varepsilon,r}, \\ &\quad C_{\text{M}}(e_{\varepsilon,n,r}), \kappa_{\text{M}}(e_{\varepsilon,n,r}), \text{Carr}(e_{\varepsilon,n,r})), \end{aligned} \quad (36.11)$$

and set $\text{Led}_{\varepsilon,n,r}^{\text{new}} := \text{Led}(x_{\varepsilon,r-1}(x)) \frown \ell_{\varepsilon,n,r}$. Then construct the tuple

$$y_{\varepsilon,r}(x) := x_{\varepsilon,r-1}(x)[\tau \leftarrow M_{\text{ph}}(r), h \leftarrow h_r(h(x_{\varepsilon,r-1}(x))), o \leftarrow o(M_{\text{ph}}(r)), b_{\text{ref}} \leftarrow \mathfrak{p}_{\varepsilon}], \quad (36.12)$$

where selection leaves the cell, quotients, carry, route, memory, cylinder, boundary, survival, and ledger unchanged; the coordinate b_{ref} makes the chosen balanced reduction explicit. Membership in an ambient relation is not presupposed here.

Next, define $z_{\varepsilon,r}(x)$ field by field: its configuration is $\tilde{q}_{n,r+1}$; its complete cell is $a_{(n,r)^+}$, with the four residues and quotients recomputed by the two Euclidean divisions; it preserves the selected sheet and orientation; and it performs the following six updates, in which only the common argument x is omitted:

$$\begin{aligned} C(z_{\varepsilon,r}) &:= C(y_{\varepsilon,r}) \frown e_{\varepsilon,n,r}, \\ \partial C(z_{\varepsilon,r}) &:= \partial C(y_{\varepsilon,r}) \frown (\tilde{q}_{n,r}, \tilde{q}_{n,r+1}), \\ p_{\text{APP}}(z_{\varepsilon,r}) &:= p_{\text{APP}}(y_{\varepsilon,r}) \frown a_{(n,r)^+}, \\ p_{\{-1,0,+1\}}(z_{\varepsilon,r}) &:= p_{\{-1,0,+1\}}(y_{\varepsilon,r}) \frown \tau(r), \\ \text{Led}(z_{\varepsilon,r}) &:= \text{Led}_{\varepsilon,n,r}^{\text{new}}, \\ \lambda(z_{\varepsilon,r}) &:= e_{\varepsilon,n,r} \lambda(y_{\varepsilon,r}), \end{aligned} \quad (36.13)$$

and still retains the fields $c, \text{gr}_{\text{M}}, \text{mem}, s$ of $y_{\varepsilon,r}(x)$. Its signature, by contrast, is retyped at the target configuration through the transported presignature

$$\begin{aligned} \text{Sig}(z_{\varepsilon,r}(x)) &:= \text{Sig}_{\tilde{q}_{n,r+1}}^{\text{cal}} \left(\hat{h}(z_{\varepsilon,r}(x)), o(z_{\varepsilon,r}(x)), c(y_{\varepsilon,r}(x)), \lambda(z_{\varepsilon,r}(x)), \right. \\ &\quad \left. \partial C(z_{\varepsilon,r}(x)), J_{d_{n,r}}(\text{mem}_{d_{n,r}}(y_{\varepsilon,r}(x))) \right). \end{aligned} \quad (36.14)$$

Thus, and only in Tra^{cal} , the edge, its endpoints, the mark $(p_\varepsilon, r, d_{\varepsilon,r})$, and its cocycles are added to the incidence ledger.

The initial state satisfies

$$c(x_\star) = 0 = \text{Carr}(\text{id}_{\tilde{q}_{0,1}}) = \text{Carr}(\lambda(x_\star)).$$

In the complete states $x_{\varepsilon,r-1}(x)$, and also after the selection $y_{\varepsilon,r}(x)$, the typed invariant $c(v) = \text{Carr}(\lambda(v))$ is preserved, since Sel^{cal} modifies neither field. Transport advances the route but leaves the carry pending; hence the pre-update tuple satisfies exactly

$$c(z_{\varepsilon,r}(x)) = c(y_{\varepsilon,r}(x)) = \text{Carr}(\lambda(y_{\varepsilon,r}(x))).$$

The map Upd^{cal} then restores $c(x_{\varepsilon,r}(x)) = \text{Carr}(\lambda(x_{\varepsilon,r}(x)))$ by means of the composition law, without attributing that invariant to the intermediate state z .

Finally, apply the unary function $\text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}} = \text{Mem}_{e_{\varepsilon,n,r}}^{\text{cal}}$ to that tuple. Its result $x_{\varepsilon,r}(x)$ is given by

$$\begin{aligned} c(x_{\varepsilon,r}(x)) &= \text{Carr}(\lambda(z_{\varepsilon,r}(x))) \\ &= H_{\text{Carr}}(\lambda(y_{\varepsilon,r}(x)))(\text{Carr}(e_{\varepsilon,n,r})) + Y_{\text{Carr}}(e_{\varepsilon,n,r})(c(y_{\varepsilon,r}(x))), \\ \lambda(x_{\varepsilon,r}(x)) &= \lambda(z_{\varepsilon,r}(x)), \\ \text{gr}_M(x_{\varepsilon,r}(x)) &= d_{n,r} + 1, \\ \text{mem}_{d_{n,r}+1}(x_{\varepsilon,r}(x)) &= C_M(e_{\varepsilon,n,r}) \text{mem}_{d_{n,r}}(z_{\varepsilon,r}(x)) + \kappa_M(e_{\varepsilon,n,r}), \\ s(x_{\varepsilon,r}(x)) &= s(e_{\varepsilon,n,r}) \wedge s(z_{\varepsilon,r}(x)), \\ \text{Sig}(x_{\varepsilon,r}(x)) &= \text{Sig}_{\tilde{q}_{n,r+1}}^{\text{cal}} \left(\hat{h}(z_{\varepsilon,r}(x)), o(z_{\varepsilon,r}(x)), c(x_{\varepsilon,r}(x)), \lambda(z_{\varepsilon,r}(x)), \partial C(z_{\varepsilon,r}(x)), \right. \\ &\quad \left. J_{d_{n,r}+1}(\text{mem}_{d_{n,r}+1}(x_{\varepsilon,r}(x))) \right). \end{aligned} \tag{36.15}$$

All fields not enumerated in (36.15) coincide with those of $z_{\varepsilon,r}(x)$. In particular, the signature is computed from the sheet, orientation, boundary, and route already fixed by Sel^{cal} and Tra^{cal} , and from the carry and memory computed in the preceding lines; it is not circularly defined from the final state. Neither ε , nor $d_{\varepsilon,r}$, nor u is an additional argument of Upd^{cal} : the function reads all three data from the already constructed edge.

Simultaneous induction of the levels. Once the preceding nine stages have been completed for an already constructed G_n , define immediately

$$\gamma_{\varepsilon,k_n}(x) := x_{\varepsilon,9}(x), \quad G_{n+1} := \{\gamma_{\varepsilon,k_n}(x) : x \in G_n, \varepsilon \in \{-1, 0, +1\}\}. \tag{36.16}$$

The base $G_0 = \{x_\star\}$, the intermediate tuples, and (36.16) constitute a single definition by recursion on n . Therefore, no subsequent carrier is used to assume the existence of G_n : the next level is introduced only after constructing all its edges from the preceding level.

TPK extension relation and effective membership. Let $\mathcal{E}_{\tilde{q}}^{\text{cal}}$ be the smallest labelled family containing $x_\star$ and, at each step of the preceding recurrence, the four tuples $x_{\varepsilon,r-1}(x), y_{\varepsilon,r}(x), z_{\varepsilon,r}(x), x_{\varepsilon,r}(x)$ in their source and target configurations. Also let $\mathcal{A}_{\tilde{q}_{n,r}}^{\text{cal}} := \{a_{n,r}\}$, and set

$$\tilde{\mathcal{Q}}_{\text{obs}} := \{\tilde{q}_{n,r} : n \geq 0, 1 \leq r \leq 9\}, \quad \mathcal{E}^{\text{cal}} := \bigsqcup_{\tilde{q}} \mathcal{E}_{\tilde{q}}^{\text{cal}}, \quad \mathcal{F}_{\tilde{q}}^{\text{cal}} := \pi_{\mathcal{F}}(\mathcal{E}_{\tilde{q}}^{\text{cal}}).$$

For each depth d , let $\mathcal{E}_d^{\text{cal}} := \text{gr}_M^{-1}(d) \subset \mathcal{E}^{\text{cal}}$. The explicit degree coordinate, rather than the mere membership of a representative in some image of the direct system, makes these carriers disjoint. On them, the projections are typed as

$$\begin{aligned} \text{cfg} : \mathcal{E}^{\text{cal}} &\longrightarrow \widetilde{\mathcal{Q}}_{\text{obs}}, \\ g(x) &:= \text{fase}(\pi_{\text{obs}}(\text{cfg}(x))) \in C_9, \\ c : \mathcal{E}^{\text{cal}} &\longrightarrow K_{\text{Carr}}, \\ s : \mathcal{E}^{\text{cal}} &\longrightarrow \{\perp, \top\}, \\ \lambda : \mathcal{E}_{\widetilde{q}}^{\text{cal}} &\longrightarrow \text{Hist}_{\widetilde{q}}^{\text{cal}}, \\ \text{gr}_M : \mathcal{E}^{\text{cal}} &\longrightarrow \mathbb{N}_{\geq 1}, \\ \text{mem}_d : \mathcal{E}_d^{\text{cal}} &\longrightarrow M_d, \\ \text{mem}(x) &:= J_d(\text{mem}_d(x)) \in M \quad (x \text{ of depth } d). \end{aligned}$$

Here $\text{cfg}(x)$ is the second coordinate of the tuple, and fase reads the phase of the observable calendar. For the three carrier classes, fix the diagonals

$$\begin{aligned} \Delta_{\text{APP}, \widetilde{q}} &:= \{(a, a) : a \in \mathcal{A}_{\widetilde{q}}^{\text{cal}}\}, \\ \Delta_{\text{fib}, \widetilde{q}} &:= \{(f, f) : f \in \mathcal{F}_{\widetilde{q}}^{\text{cal}}\}, \\ \Delta_{\text{enr}, \widetilde{q}} &:= \{(x, x) : x \in \mathcal{E}_{\widetilde{q}}^{\text{cal}}\}. \end{aligned}$$

In particular,

$$a_{0,1} \in \mathcal{A}_{\widetilde{q}_{0,1}}^{\text{cal}}, \quad x_{\star} \in \mathcal{E}_{\widetilde{q}_{0,1}}^{\text{cal}}.$$

On these carriers, the following relations are defined by their complete graphs, and not through necessary implications:

$$\text{Sel}_{e_{\varepsilon,n,r}}^{\text{cal}} := \{(x_{\varepsilon,r-1}(x), y_{\varepsilon,r}(x)) : x \in G_n\}, \quad (36.17)$$

$$\text{Tra}_{e_{\varepsilon,n,r}}^{\text{cal}} := \{(y_{\varepsilon,r}(x), z_{\varepsilon,r}(x)) : x \in G_n\}, \quad (36.18)$$

$$\text{Ext}_{e_{\varepsilon,n,r}}^{\text{APP}, \text{cal}} := \{(a_{n,r}, a_{(n,r)^+})\}, \quad (36.19)$$

$$\text{Ext}_{e_{\varepsilon,n,r}}^{\text{fib}, \text{cal}} := \{(\pi_{\mathcal{F}X_{\varepsilon,r-1}}(x), \pi_{\mathcal{F}X_{\varepsilon,r}}(x)) : x \in G_n\}. \quad (36.20)$$

The synchronized enriched extension is

$$\text{Ext}_{e_{\varepsilon,n,r}}^{\text{enr}, \text{cal}} := \{(x_{\varepsilon,r-1}(x), x_{\varepsilon,r}(x)) : x \in G_n\}. \quad (36.21)$$

The third map is defined by

$$\text{graph}(\text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}}) := \{(z_{\varepsilon,r}(x), x_{\varepsilon,r}(x)) : x \in G_n\}, \quad (36.22)$$

where the second component is exactly the one in (36.15). Consequently,

$$\mathcal{U}_{e_{\varepsilon,n,r}}^{(1), \text{cal}} := \text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}} \circ \text{Tra}_{e_{\varepsilon,n,r}}^{\text{cal}} \circ \text{Sel}_{e_{\varepsilon,n,r}}^{\text{cal}} = \{(x_{\varepsilon,r-1}(x), x_{\varepsilon,r}(x)) : x \in G_n\}. \quad (36.23)$$

Each operator is indexed by the edge $e_{\varepsilon,n,r}$ produced by the reduction $\mathfrak{p}_\varepsilon$, rather than by a trit supplied from outside. The coordinate b_{ref} also distinguishes their codomains. Therefore, $\text{Sel}_e^{\text{cal}}$, Tra^{cal} and Upd^{cal} are functions on their respective domains, whereas the union $\bigcup_{\varepsilon \in \{-1,0,+1\}} \mathcal{U}_{e_{\varepsilon,n,r}}^{(1), \text{cal}}$ deliberately preserves the three outputs of the trisection.

For $\bullet \in \{\text{APP}, \text{fib}, \text{enr}\}$, identities and compositions are not left implicit. Define

$$\text{Ext}_{\text{id}_{\widetilde{q}}}^{\bullet, \text{cal}} := \Delta_{\bullet, \widetilde{q}}, \quad (36.24)$$

$$\text{Ext}_{e_t \dots e_1}^{\bullet, \text{cal}} := \text{Ext}_{e_t}^{\bullet, \text{cal}} \circ \dots \circ \text{Ext}_{e_1}^{\bullet, \text{cal}} \quad (36.25)$$

for every admissible edge word. The two APP and fiber path closures are formed separately,

$$\text{Path}(\text{Ext}^{\text{APP},\text{cal}}), \quad \text{Path}(\text{Ext}^{\text{fib},\text{cal}}), \quad (36.26)$$

and Tra^{cal} synchronizes them edge by edge.

Finally, the levels of complete states and their truncations are fixed by the recursion itself:

$$H_{d_{n,r}}^{\text{cal}} := \{x_{\varepsilon,r-1}(x) : x \in G_n, \varepsilon \in \{-1, 0, +1\}\}, \quad (36.27)$$

$$H_{d_{n,r}+1}^{\text{cal}} := \{x_{\varepsilon,r}(x) : x \in G_n, \varepsilon \in \{-1, 0, +1\}\}, \quad (36.28)$$

$$\rho_{d_{n,r}+1,d_{n,r}}^{\text{cal}}(x_{\varepsilon,r}(x)) := x_{\varepsilon,r-1}(x). \quad (36.29)$$

The final ledger preserves $(n, r, \mathfrak{p}_\varepsilon)$, so the predecessor of the left-hand side is unique and ρ^{cal} is well-defined. For $b > a$, set $\rho_{b,a}^{\text{cal}} := \rho_{a+1,a}^{\text{cal}} \circ \dots \circ \rho_{b,b-1}^{\text{cal}}$. This map recovers the complete preceding state, including its signature; it does not purport to invert an overwritten coordinate without consulting the ledger.

Proposition 36.1 (Autonomous realization of the TPK through explicit transitions). *The family*

$$\begin{aligned} \mathfrak{T}_{\text{TPK}}^{\text{cal}} := & ((\tilde{q}_{n,r})_{n,r}, \pi_{\text{obs}}, \tilde{F}_{\text{obs}}, M_{\text{ph}}, \\ & (\mathcal{A}_{\tilde{q}}^{\text{cal}})_{\tilde{q}}, (\mathcal{E}_{\tilde{q}}^{\text{cal}})_{\tilde{q}}, (\mathcal{F}_{\tilde{q}}^{\text{cal}})_{\tilde{q}}, \text{Ext}^{\text{APP},\text{cal}}, \text{Ext}^{\text{fib},\text{cal}}, \text{Ext}^{\text{enr},\text{cal}}, \rho^{\text{cal}}, \\ & \text{Sel}^{\text{cal}}, \text{Tra}^{\text{cal}}, \text{Upd}^{\text{cal}}, C_M, \kappa_M, \text{gr}_M, H_{\text{Carr}}, Y_{\text{Carr}}, \\ & \text{Sig}^{\text{cal}}, \pi_{\mathcal{F}}). \end{aligned}$$

is an enriched model of the TPK. In particular, every pair on the right-hand side of (36.23) is an effective TPK edge, not a consequence inferred from a merely necessary condition.

Proof. The labels (n, r) make the carrier copies disjoint and unambiguously fix each source and target. The relation $\text{Ext}^{\text{APP},\text{cal}}$ transports the complete cell and preserves the two Euclidean identities; the relation $\text{Ext}^{\text{fib},\text{cal}}$ relates the fibers of the preceding and subsequent complete states: it transports sheet, orientation, prefix, cylinder, boundary, and ledger, and incorporates the updated carry, memory, and signature. Selection does not modify the ledger, and transport concatenates exactly one entry and produces the target presignature (36.14). The update recomputes the signature by means of the typed map $\text{Sig}_{\tilde{q}}^{\text{cal}}$ from the already determined sheet, orientation, carry, route, boundary, and memory. The map Upd^{cal} is unary because the edge contained in $z_{\varepsilon,r}(x)$ determines its cocycles.

The identities are the diagonals, and the word extensions are the compositions in (36.25). Concatenation of prefixes and ledgers is associative; memory satisfies the cocycle law, and carry satisfies (36.9). Therefore, the two parenthesizations of three edges produce the same state. Formula (36.29) defines truncation on complete states, and uniqueness of the final ledger proves that it recovers the source in every field. These are the identity, source, target, composition, memory, carry, and truncation laws of the enriched TPK. In particular, the extension relations are determined as graphs of effective maps rather than as consequences of merely necessary conditions. $\square$

Each γ_{ε,k_n} comprises exactly nine applications of $\text{Upd}_{e_{\varepsilon,n,r}}^{\text{cal}} \circ \text{Tra}_{e_{\varepsilon,n,r}}^{\text{cal}} \circ \text{Sel}_{e_{\varepsilon,n,r}}^{\text{cal}}$. Survival will be proved by means of a constructed tail and will not be used to define G_n .

Lemma 36.2 (Effective internal trisection of one turn). *For every $n \geq 0$, every $x \in G_n$, and every $\varepsilon \in \{-1, 0, +1\}$, the expression $\gamma_{\varepsilon,k_n}(x)$ is defined by nine TPK transitions, with its synchronized relations $\text{Ext}^{\text{APP},\text{cal}}$ and $\text{Ext}^{\text{fib},\text{cal}}$, and satisfies*

$$\rho_{k_n+9,k_n}^{\text{cal}} \gamma_{\varepsilon,k_n}(x) = x, \quad (36.30)$$

$$g(\gamma_{\varepsilon,k_n}(x)) = g(x), \quad J_{k_n+9}(\text{mem}_{k_n+9}(\gamma_{\varepsilon,k_n}(x))) = J_{k_n}(\text{mem}_{k_n}(x)) + u, \quad (36.31)$$

$$c(\gamma_{\varepsilon,k_n}(x)) = c(x) + \varepsilon \kappa_{\text{Carr}}. \quad (36.32)$$

The ordered ledger of the nine edges determines ε uniquely. The three images are therefore disjoint. Each endpoint belongs to G_{n+1} and again admits the three maps $\gamma_{-1,k_{n+1}}, \gamma_{0,k_{n+1}}, \gamma_{+1,k_{n+1}}$.

Proof. Formulas (36.1) traverse the nine phases once and return the copy $a_{n+1,1}$ of $a(1)$. At each step, the APP relation transports the complete sextuple; Sel^{cal} fixes the TRIT type from the phase mark r preserved in the state; and Tra^{cal} adds exactly one symbol to the prefix, one segment to the cylinder, and its ordered pair of endpoints to the boundary. By induction on r , the explicit tuples $y_{\varepsilon,r}(x)$ and $z_{\varepsilon,r}(x)$ satisfy the predicates of the indicated relations, and the unary update produces the state at the next level. The identity $\rho_{d_{n,r}+1,d_{n,r}}^{\text{cal}}(x_{\varepsilon,r}(x)) = x_{\varepsilon,r-1}(x)$ is the definition (36.29); its well-definedness follows from uniqueness of the final ledger. The composition of the nine truncation maps recovers x , proving (36.30) without separately inverting the overwritten fields.

The phase $9 \rightarrow 1$ occurs exactly once. By (36.5) and the cocycle law, the linear memory component transports the preceding memory, and its translational component adds exactly u in the direct limit. The compatibility $J_{d+1}J_{d+1,d} = J_d$ proves (36.31); the return concerns the phase, not the enriched state.

By (36.6), eight edges have zero increment, and the edge completing $\mathfrak{p}_\varepsilon$ has the computed increment $\chi(\varepsilon, \varepsilon)\kappa_{\text{Carr}}$. The laws (36.9) and (36.4) give (36.32). The ledger preserves $(\mathfrak{p}_\varepsilon, 2\varepsilon, r_3(2\varepsilon), \chi(\varepsilon, \varepsilon))$; the three values of this ledger are distinct even when the scalar increment is zero. Hence the images are not identified, and the block decoder recovers a unique ε .

The edge definitions depend only on the new phase and concatenate, rather than replace, the prefix, cylinder, boundary, and ledger. They are also invariant under $J_d \text{ mem}_d \mapsto J_d \text{ mem}_d + nu$ and under translation of the input carry. Consequently, the same construction applies at the endpoint of any turn. For every finite state, the explicit sequence

$$x, \quad \gamma_{0,k_n}(x), \quad \gamma_{0,k_{n+1}}\gamma_{0,k_n}(x), \quad \dots$$

is a compatible infinite tail. This proves the survival of all constructed states and the repeated availability of the three branches, without postulating either property. Consequently, in accordance with the preceding level definitions, we may finally identify

$$H_{k_n}^{\text{cal}} := G_n \tag{36.33}$$

for every $n \geq 0$. □

The preceding closure distinguishes the three clocks involved. Nine emissions return the phase in C_9 and advance memory. Twelve turns, that is, 108 emissions, also return the position of the observable calendar; even then, neither the ledger nor memory is erased. The operator $\mathcal{K}_{\text{ph}}$ acts only afterward, as a recognition of reduced memory from the already constructed turns; it never generates the edges of $\text{Ext}^{\text{enr,cal}}$.

36.2 Cylinder refinement and ordered readout

For $\mathbf{b} = (b_1, \dots, b_6) \in \mathbb{F}_3^6$, let $\nu : \mathbb{F}_3 \rightarrow \{-1, 0, +1\}$ be the balanced representative. The three prolongations constructed edge by edge define

$$\Gamma_{\mathbf{b},k}^{[54]} = \gamma_{\nu(b_6),k+45} \circ \dots \circ \gamma_{\nu(b_1),k}.$$

Starting from $Y_0 = \{x_\star\}$, put

$$Y_{N+1} = \{\Gamma_{\mathbf{b},1+54N}^{[54]}(x) : x \in Y_N, \mathbf{b} \in \mathbb{F}_3^6\}.$$

Truncation $\rho_{N+1,N}^{\text{blk}}$ removes the last fifty-four edges, recovering the previous state through the transition record.

Proposition 36.3 (Effective coding of prolongation blocks). *Reading the six tritic pairs of each block defines bijections*

$$\beta_N : Y_N \xrightarrow{\cong} (\mathbb{F}_3^6)^N$$

that commute with truncation. Each state of Y_N has exactly 729 block prolongations in Y_{N+1} .

Proof. On each turn, the record retains the tritic pair and its carry. The three pairs produce the three distinct outputs $\chi(\varepsilon, \varepsilon) = \varepsilon$, and the record allows recovery of the branch used, including the zero-carry branch. The six concatenations therefore produce all words of $\mathbb{F}_3^6$. Two different words are distinguished at the first recorded pair in which they differ. Recovery of the prefix proves compatibility with truncation. By induction on N , every word of length N has a unique antecedent state; the next word can be any of the $3^6 = 729$ possibilities. $\square$

Let $\Omega_{\Gamma}^{\text{cal}} = \varprojlim_N Y_N$. The preceding bijections induce

$$\Omega_{\Gamma}^{\text{cal}} \cong (\mathbb{F}_3^6)^{\mathbb{N}}.$$

The correspondence and its inverse are continuous: each finite coordinate depends on a finite prefix. The cylinders are open and closed. Each cylinder of depth N is partitioned into its 729 subcylinders of depth $N + 1$; each contains an infinite prolongation, obtained, for example, by continuing with the zero branch in all subsequent blocks.

36.3 Joint construction of the continuum through compatible histories and prolongations

The arithmetic expressed by a normal form preserves a precise readout of the state that produces it. To situate that readout within HMT, the histories, sheets and operations acting on them must also be retained. This chapter brings together that common substrate. Its different constructions arise from the same tree of APP–TRIT–TPK executions; the expository divisions allow their properties to be studied without turning them into independent domains chosen by a subsequent application.

The starting point is the formal kernel presented above. An admissible emission contains the APP operation, TRIT orientation, sheet change, remainder and quotient, together with the effective TPK memory update. When only a remainder is expressed, the execution history still belongs to the starting object. Two executions with identical final readout are not identified on that basis alone.

36.3.1 Survival by pruning prolongations

Retrospective memory distinguishes histories recorded by the APP–TRIT–TPK operations; prospective compatibility organizes their admissible prolongations. Let S be the space of complete states reachable through these operations. Each state retains sheets, residues, quotients, orientation, carry, memory, and boundary data; whenever a transition depends on depth, depth forms part of the state. Let $E \subseteq S \times S$ be the effective prolongation relation and $A \subseteq S$ the admissible set determined by the compatibility of these records. Admissibility is thus decided through the operations and boundary conditions that generate the history.

For $X \subseteq S$, define

$$\text{Pre}_E(X) = \{s \in S : \exists t \in X, (s, t) \in E\}, \quad F_{\text{surv}}(X) = A \cap \text{Pre}_E(X). \quad (36.34)$$

The pruning sequence is given by

$$V_0 = A, \quad V_{n+1} = F_{\text{surv}}(V_n), \quad V_{\infty} = \bigcap_{n \geq 0} V_n. \quad (36.35)$$

Proposition 36.4 (Survival and the greatest fixed point). *The set V_n consists exactly of the states in A admitting an n -step prolongation all of whose states belong to A . The sequence (V_n) is decreasing. If each state has finitely many admissible successors, then*

$$V_{\infty} = F_{\text{surv}}(V_{\infty}) = \nu F_{\text{surv}},$$

where νF_{surv} denotes the greatest fixed point under inclusion. Its elements are precisely the states admitting an infinite compatible prolongation.

Proof. For $n = 0$, a prolongation of length zero consists of the state itself in A . If the characterization holds for n , a state belongs to V_{n+1} exactly when it belongs to A and has a successor in V_n ; prefixing that edge produces an admissible prolongation of $n + 1$ steps. This equivalence proves the assertion by induction. Moreover, $V_1 \subseteq V_0$, and monotonicity of F_{surv} propagates $V_{n+1} \subseteq V_n$.

Let $s \in V_\infty$, and write $E(s) = \{t \in S : (s, t) \in E\}$. Since $s \in V_{n+1}$ for every n , the sets $E(s) \cap V_n$ are nonempty, finite, and decreasing. Their intersection contains an element t , which belongs to V_∞ and satisfies $(s, t) \in E$. Hence $V_\infty \subseteq F_{\text{surv}}(V_\infty)$. Conversely, monotonicity gives $F_{\text{surv}}(V_\infty) \subseteq V_{n+1}$ for every n ; the image also lies in $A = V_0$, and therefore belongs to V_∞ . This proves the fixed-point equality.

If $W = F_{\text{surv}}(W)$, then $W \subseteq A = V_0$. From $W \subseteq V_n$ it follows that $W = F_{\text{surv}}(W) \subseteq F_{\text{surv}}(V_n) = V_{n+1}$. Induction yields $W \subseteq V_\infty$, proving maximality. Finally, the prolongation tree of a state in V_∞ is finitely branching and has a nonempty level at every depth. König's lemma provides an infinite branch. Conversely, the prefixes of any infinite prolongation place its initial state in every V_n . $\square$

Branching and stabilization. The finiteness hypothesis has an explicit control. Consider a root with a distinct terminal branch of each positive integer length, all branches being admissible and disjoint after the root. The root admits prolongations of every finite length and belongs to all the V_n ; each of its successors belongs to a branch of bounded length. Consequently, the root has neither an infinite prolongation nor a successor in V_∞ . This tree has infinite branching at the root. In contrast, if A is finite, each strict inclusion $V_{n+1} \subsetneq V_n$ removes at least one state. Pruning reaches a fixed point after at most $|A|$ strict layer removals. This bound pertains to the finite domain A .

Uniqueness of continuation. For $s \in V_\infty$, the successors retaining survival form

$$E(s) \cap V_\infty.$$

The next state is uniquely selected exactly when

$$\#(E(s) \cap V_\infty) = 1. \quad (36.36)$$

An infinite prolongation is unique when this condition holds at each state of the path. Indeed, each surviving successor has an infinite prolongation by the preceding proposition; two distinct successors produce two distinct paths. When the successor is unique at each step, the prefixes are determined successively. Several viable successors express multiplicity of prolongations within the same survival domain.

Proposition 36.5 (Conjugation of the two directions of prolongation). *Let $C : S \rightarrow S$ be an involution such that $C(A) = A$ and*

$$(x, y) \in E \iff (Cy, Cx) \in E. \quad (36.37)$$

Let V_n^+ denote pruning for E , and V_n^- pruning for E^{-1} , both with initial set A . Then

$$C(V_n^+) = V_n^- \quad (n \geq 0), \quad C(V_\infty^+) = V_\infty^-.$$

Proof. If $s_0, \dots, s_n$ is an admissible path for E , the sequence $Cs_0, \dots, Cs_n$ is an admissible path for E^{-1} : for each j , the hypothesis converts $(s_j, s_{j+1}) \in E$ into $(Cs_{j+1}, Cs_j) \in E$. All its states belong to A . Applying the same operation again recovers the initial path because $C^2 = I$; it therefore establishes a bijection between finite prolongations in the two directions. The characterization of $V_n^\pm$ by paths gives $C(V_n^+) = V_n^-$. Since C is bijective, it commutes with the intersection of these sets, yielding the equality for $V_\infty^\pm$. $\square$

The involution transports the surviving set in one direction to the surviving set in the conjugate direction. Infinite prolongability in both directions corresponds to $V_\infty^+ \cap V_\infty^-$; uniqueness of a path retains the criterion in (36.36). Retaining a genealogy and restricting its continuation tree are two typed operations on the same records. Assigning an entropy to these readings additionally requires the corresponding measure on histories.

36.3.2 Admissible histories and memory by composition

Let $H_0 := \{*\}$. Level H_1 is the finite nonempty set of admissible APP-TRIT seeds, and, for $k \geq 1$, H_k is the set of enriched histories of length k that admit compatible TPK prolongations to every finite horizon. Its elements are constructed by successive emissions from the initial domain of the kernel. We write $\zeta\epsilon$ for the prolongation of ζ by an admissible emission ϵ , and $\rho_k(\zeta\epsilon) = \zeta$ for deletion of the last emission. The label produced by the emission is retained, even when different labels express the same scalar state. The space of complete histories is defined by

$$H_\infty^{\text{enr}} := \varprojlim_k (H_k, \rho_k).$$

The cofinal subsystem $X_N := H_{6N}$, with $X_0 = H_0$, agrees with the sextet clock used in the cardinality proof; cofinality gives a canonical identification $H_\infty^{\text{enr}} \cong \varprojlim_N (X_N, \rho_{6(N+1), 6N})$.

On the two arithmetic sheets, the local data satisfy the exact divisions

$$\delta_\diamond = r_\diamond + 9q_\diamond, \quad 0 \leq r_\diamond < 9, \quad \diamond \in \{\Sigma, \Pi\}.$$

The oriented TRIT coordinate likewise admits an expression $\Delta = o_r + 3\kappa$, with the oriented representative declared by the kernel. Preserving the quotient allows those representations to be composed without replacing an emission by its isolated remainder.

In a memory coordinatization, the update of an emission is written

$$U_\epsilon(m) = C_\epsilon m + \kappa_\epsilon.$$

For a trajectory $\alpha = (\epsilon_1, \dots, \epsilon_k)$, the effective operation is the ordered composition of these updates. Its linear part and its translation are

$$C_\alpha = C_{\epsilon_k} \cdots C_{\epsilon_1}, \quad \kappa_\alpha = \sum_{j=1}^k C_{\epsilon_k} \cdots C_{\epsilon_{j+1}} \kappa_{\epsilon_j}.$$

The empty product in the last sum is the identity.

Proposition 36.6 (Affine compatibility of concatenation). *If α precedes β , then*

$$C_{\beta\alpha} = C_\beta C_\alpha, \quad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta, \quad U_{\beta\alpha} = U_\beta U_\alpha.$$

Proof. Substituting $U_\alpha(m)$ into U_β produces $C_\beta C_\alpha m + C_\beta \kappa_\alpha + \kappa_\beta$. Decomposition into linear part and translation gives the three identities. Associativity of composition proves that the result does not depend on an auxiliary subdivision of the trajectory. $\square$

This cocycle identity is the precise content of accumulated memory. The quantity transported by a second trajectory depends on the coordinatization update it performs; in general, it is not the untransported sum of two records.

The nonadic return preserves a second distinction. In phase and turn coordinates, with $\bar{r} \in \{0, \dots, 8\}$, its representation is

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + 1 \pmod{9} \right), \quad \sigma_9^9(w, r) = (w + 1, r).$$

During nine steps, exactly one crossing from 8 to 0 occurs. The phase returns and the turn coordinate increases. The enriched holonomy also retains the memory updates accompanying that path.

36.3.3 Tree refinement and measure of histories

The initial measure is the canonical census measure: $\mu(*) = 1$ and $\mu(c) = 1/|H_1|$ for each seed $c \in H_1$. In this section each history has a finite positive number of prolongations. Deletion of emissions defines the refinement maps. Admissibility of the kernel supplies the prolongations; when a realization uses a neutral emission, this explicitly gives a section of ρ_k . Existence of that section refers to the tree containing that emission, not to every arbitrary restriction of routes.

Let $d(\zeta) \geq 1$ be the number of admissible emissions at ζ . The APP order and the census of prolongations determine the canonical law $p_{\text{can}}(\varepsilon \mid \zeta) = 1/d(\zeta)$. Without additional parameters, this law defines

$$\mu(\zeta\varepsilon) = \frac{\mu(\zeta)}{d(\zeta)}.$$

A nonuniform weighting may replace p_{can} only when it has already been produced and added to the enriched record. The correlative constructor of this chapter uses only p_{can} ; its measure is therefore a function of the APP-TRIT-TPK tree and not of an external choice.

Proposition 36.7 (Cylindrical compatibility and conditional expectation). *At each level we have*

$$\sum_{\rho_k(y)=x} \mu(y) = \mu(x).$$

In the spaces $L^2(H_k, \mu_k)$, the maps

$$(J_k f)(y) = f(\rho_k(y)), \quad (E_k g)(x) = \sum_{\rho_k(y)=x} \frac{\mu(y)}{\mu(x)} g(y)$$

satisfy $E_k = J_k^*$, $E_k J_k = I$ and $\|J_k f\| = \|f\|$.

Proof. The first identity is the normalization of local probabilities. Applying it to $|f(x)|^2$ gives the isometry. Grouping the sum by fibers of ρ_k gives the formula for E_k ; on a function constant on each fiber, that formula returns its value. $\square$

The compatible masses define a probability measure on the space of infinite histories: the cylinder of a finite history receives its mass $\mu(\zeta)$. The measure extends from the cylinder algebra, and its uniqueness follows because cylinders generate the σ -algebra. If the tree is finitely branching and every history has a prolongation, the inverse limit of its levels under emission deletion is nonempty. In the cylinder topology, every nonempty cylinder has positive measure.

This measure distinguishes histories that converge to the same representation. The uniform measure on terminal values would be a different readout, since it would remove the multiplicities that the construction has just preserved.

Enriched readers and fiber memory The difference between history and value admits an operational criterion. At a finite level H_k , let $q_k : H_k \rightarrow Z_k$ be a reader and let $M_k : H_k \rightarrow \mathcal{M}_k$ be the preserved register. The enriched reader is $\hat{q}_k = (q_k, M_k)$. Memory distinguishes states in the same reader fibre; its sufficiency is checked on the domain considered. This formulation incorporates into the article itself the complete development of the fibre information used below.

Proposition 36.8 (Recoverability at a finite level). *An inverse of $\hat{q}_k$ exists on its image if and only if $\hat{q}_k$ is injective. For a random variable Z taking values in H_k with distribution μ_k , we have*

$$H(Z) = H(q_k(Z)) + H(Z \mid q_k(Z)).$$

If $\hat{q}_k$ is injective on the support of Z , then

$$H(Z \mid q_k(Z), M_k(Z)) = 0.$$

Here $H(Z) = -\sum_z \Pr(Z = z) \log \Pr(Z = z)$, with $0 \log 0 = 0$, is finite entropy in nats; conditional entropy is the average entropy of the conditional distributions.

Proof. An inverse recovers a unique antecedent, which requires injectivity; conversely, injectivity allows each expressed pair to be assigned its unique antecedent. Since $q_k(Z)$ is a deterministic function of Z , grouping the sum defining $H(Z)$ by fibers of q_k gives the chain rule. If the pair $(q_k(Z), M_k(Z))$ distinguishes the support, each corresponding conditional distribution is concentrated on a single state and its entropy is zero. $\square$

Not expressing a coordinate and erasing the record that allows it to be recovered are distinct operations. This result does not identify entropy with the cardinality of the continuum or introduce a thermal interpretation: its domain, reader and distribution are those just fixed.

Equivariance of the measure under tree symmetries

Proposition 36.9 (Joint transport of the tree and its measure). *Let $g = (g_k)_{k \geq 0}$ be a collection of bijections $g_k : H_k \rightarrow H_k$ that commutes with truncations, fixes the root and bijectively transports admissible emissions, preserving their multiplicities. Then*

$$d(g_k \zeta) = d(\zeta), \quad (g_k)_* \mu_k = \mu_k, \quad (g_\infty)_* \mu = \mu.$$

More generally, if μ_ζ is the law of continuations of a history $\zeta \in H_k$, we have

$$(g_\infty)_* \mu_\zeta = \mu_{g_k \zeta}.$$

Proof. The emission bijection preserves their number with multiplicities. The uniform seed distribution is also invariant. Each path and its image therefore receive the same product of local weights $1/d(\zeta)$. Equality of measures follows at each level and on each cylinder. Coherence with truncations defines g_∞ , and uniqueness of the measure determined by cylinders proves equality in the limit. The same argument, starting at ζ , proves the assertion about continuations. $\square$

This gives naturality of the tree and the measure: a coherent change of representation transports the weights and does not require choosing them anew. Preservation of emissions and multiplicities is an essential hypothesis of the proposition, not a datum added after the construction.

36.3.4 Transport by sheets and exact accounting of outputs

First consider the space with an orthonormal basis formed by the histories at level k . The transport preserving the label of each prolongation is

$$\mathcal{U}_k e_\zeta = \sum_{\varepsilon \text{ admissible en } \zeta} \sqrt{p_{\text{can}}(\varepsilon \mid \zeta)} e_{\zeta \varepsilon}.$$

The successor fibers of distinct antecedents are disjoint. Normalization of p_{can} therefore proves $\mathcal{U}_k^* \mathcal{U}_k = I$. The representation by history masses is related to this one through the diagonal change of basis incorporating $\sqrt{\mu(\zeta)}$.

The sheet label is part of the complete register of a history. We write

$$\hat{\zeta} = ((\zeta_0, s_0), \dots, (\zeta_k, s_k)), \quad s_j \in \{\Sigma, \Pi\},$$

where s_j is the active sheet at step j . Prolongation adds the emission and the sheet s_{k+1} determined by the operational TPK selector, preserving the entire prefix, including the initial sheet. In the basis of these lifted histories,

$$\mathcal{U}_k e_{\hat{\zeta}} = \sum_{\varepsilon \text{ admissible en } \hat{\zeta}} \sqrt{p_{\text{can}}(\varepsilon \mid \hat{\zeta})} e_{\hat{\zeta} \varepsilon}, \quad R_k e_{\hat{\zeta}} = \mathbf{1}_{s_k = \Sigma} e_{\hat{\zeta}}, \quad Q_k = I - R_k.$$

Thus P_k reads the current sheet, while histories remain distinguished by their complete prefix. Two initial histories reaching the same active sheet retain distinct successor fibers. The isometry proof through disjoint collections of prolongations applies literally to this basis. Tritic orientation is another coordinate of the record: the value 0 preserves the threshold readout on each sheet and does not add a third sheet to this decomposition. We now separate sheet preservation and sheet exchange:

$$A_k = P_{k+1} \mathcal{U}_k P_k + Q_{k+1} \mathcal{U}_k Q_k, \quad \eta_k = Q_{k+1} \mathcal{U}_k P_k + P_{k+1} \mathcal{U}_k Q_k.$$

Proposition 36.10 (Gram identity by sheets). *For a linear transport U , whether isometric or not, the preceding decomposition satisfies*

$$A^* A + \eta^* \eta = P U^* U P + Q U^* U Q.$$

In particular, for $U = \mathcal{U}_k$,

$$A_k^* A_k + \eta_k^* \eta_k = I.$$

Proof. In $A^* A$ and $\eta^* \eta$, mixed terms containing $P_{k+1} Q_{k+1}$ vanish. Summing the remaining terms yields $P_k U^* (P_{k+1} + Q_{k+1}) U P_k$, and its analogue with Q_k . The first identity follows from complementarity of the projectors; the second also uses $U^* U = I$. $\square$

The first identity is the general formula. Replacing its right-hand side by $U^* U$ requires the cross blocks of that Gram matrix to vanish. Isometric preservation of history transport provides that condition here; it should not automatically be attributed to a subsequent compression.

Write $F_{0,0} = I$, $F_{0,n} = A_{n-1} \cdots A_0$ and $T_N = F_{0,N}^* F_{0,N}$.

Theorem 36.11 (Finite decomposition with terminal residual). *For each horizon N ,*

$$I = \sum_{n=0}^{N-1} F_{0,n}^* \eta_n^* \eta_n F_{0,n} + T_N.$$

The sequence T_N is positive and decreasing, and has a positive strong limit T_∞ . The sum of outputs converges strongly to $I - T_\infty$.

Proof. Proposition 36.10 gives $T_n - T_{n+1} = F_{0,n}^* \eta_n^* \eta_n F_{0,n}$. The telescoping sum proves the finite equality and monotonicity. For each vector, the quadratic forms decrease to a limit; polarization defines a bounded positive form and, by the representation theorem for bounded forms, an operator T_∞ . Convergence is strong because, for a positive operator $S \leq I$, $\|Sv\|^2 \leq \langle v, Sv \rangle$; this is applied to $S = T_N - T_\infty$. $\square$

The terminal residue is retained until its extinction in the domain under consideration is proved. This distinction prevents a phase return, by itself, from becoming an assumption of exhaustion of all histories.

36.3.5 Oriented balance and cancellation under refinement

A history carries its accumulated orientation $o(h)$ and the counts of its visits to the two sheets. The corresponding local balance is

$$b(h) = o(h)(N_\Sigma(h) - N_\Pi(h)), \quad b^+ = \frac{|b| + b}{2}, \quad b^- = \frac{|b| - b}{2}.$$

When a reader expresses the mean over prolongations, define $b_c = E b_f$. Orientation remains in the argument of b_f ; it is not removed before averaging.

Proposition 36.12 (Exact cancellation defect). *The quantity*

$$\kappa_{\text{can}} = \frac{E|b_f| - |Eb_f|}{2}$$

is nonnegative and satisfies

$$E(b_f^+) = (b_c)^+ + \kappa_{\text{can}}, \quad E(b_f^-) = (b_c)^- + \kappa_{\text{can}}.$$

Proof. The weighted triangle inequality gives $|Eb_f| \leq E|b_f|$. Writing the positive and negative parts through $(|b| \pm b)/2$ yields both equalities. $\square$

The information omitted when only the balance is expressed is the common cancellation of opposite contributions. This quantity is determined from the same prolongations used by the preceding transport. The factor $1/9$ appears only when the fiber contains nine prolongations with equal weights; conditional expectation is the rule that remains valid for general branching and weightings.

36.3.6 Ternary incidence of the sheets and rectangular complex

The two sheets determine the module $V = \mathbb{F}_3 e_\Sigma \oplus \mathbb{F}_3 e_\Pi$. Its four projective directions are the lines generated by $e_\Sigma, e_\Pi, e_\Sigma + e_\Pi, e_\Sigma - e_\Pi$. They give rise to six unordered pairs. The eight nonzero vectors are their oriented representatives. Thus, the counts 4, 6, 8 belong to distinct operations on the same sheet module.

On the six pair positions, we define

$$(Ba)_{\{L,L'\}} = a_L + a_{L'}, \quad s(q) = \sum_{\{L,L'\}} q_{\{L,L'\}}.$$

Each direction occurs in three pairs; hence $sB = 0$ in $\mathbb{F}_3$. The reader $W(q) = \mathbf{1}_{s(q)=0} - 1/3$ is invariant under $q \mapsto q + Ba$. On the uniform ambient space $\mathbb{F}_3^6$, its moments are

$$\mathbb{E}W = 0, \quad \mathbb{E}W^2 = \frac{2}{9}, \quad \mathbb{E}W^3 = \frac{2}{27}.$$

Indeed, s is surjective and each fiber contains 3^5 elements. The reader equals $2/3$ with probability $1/3$ and $-1/3$ with probability $2/3$. These means describe the uniform ambient space; the distribution induced by a subset of histories must be calculated with its own multiplicities, already preserved in H_k .

To fix the ports without losing multiplicities, at depth k retain the labeled readouts $\lambda_\Sigma(\zeta) = (\zeta, \Sigma)$ and $\lambda_\Pi(\zeta) = (\zeta, \Pi)$. Their sets are

$$\mathcal{P}_k^\Sigma = \{\lambda_\Sigma(\zeta) : \zeta \in H_k\}, \quad \mathcal{P}_k^\Pi = \{\lambda_\Pi(\zeta) : \zeta \in H_k\}.$$

They are finite and nonempty because H_k is. Values, quotients and remainders are expressed on those ports; equality of values does not identify their labels. The product $\mathcal{P}_k^\Sigma \times \mathcal{P}_k^\Pi$ is the domain of the rectangular readout, within which the support of admissible incidences is recorded.

With this level fixed, write $I = \mathcal{P}_k^\Sigma$, $J = \mathcal{P}_k^\Pi$ and $A_m = \mathbb{Z}/3^m\mathbb{Z}$. The APP lexicographic order determines the minimal surviving branch and, on it, the reference ports i_0, j_0 . No external choice is added: changing coordinates produces a canonically isomorphic complex. The maps

$$\kappa(u, v)_{ij} = u_i - v_j, \quad (\delta w)_{ij} = w_{ij} - w_{i j_0} - w_{i_0 j} + w_{i_0 j_0}$$

define the following sequence, valid over the finite ring A_m :

$$0 \longrightarrow A_m \longrightarrow A_m^I \oplus A_m^J \xrightarrow{\kappa} A_m^{I \times J} \xrightarrow{\delta} A_m^{(I \setminus \{i_0\}) \times (J \setminus \{j_0\})} \longrightarrow 0.$$

Proposition 36.13 (Exactness of rectangular incidence). *The preceding sequence is exact, with first arrow given by the constant diagonal.*

Proof. The equality $u_i - v_j = 0$ for each pair forces all coordinates of u and v to be the same constant. Direct substitution gives $\delta\kappa = 0$. If $\delta w = 0$, take $u_i = w_{i j_0}$ and $v_j = w_{i_0 j_0} - w_{i_0 j}$. Then $u_i - v_j = w_{i j}$. Finally, any matrix of the last term is extended with zero reference row and column; its image under δ is the initial matrix. The proof uses only addition and subtraction, so it does not require division by a unit of the ring. $\square$

Reducing from A_{m+1} to A_m while keeping $k, \mathcal{P}_k^\Sigma, \mathcal{P}_k^\Pi, i_0, j_0$ fixed commutes with both maps. The reconstruction formulas are the same at each modular depth. This is the arithmetic compatibility proved in this section. The prolongation of histories $k \rightarrow k+1$ preserves prefix labels; its maps between port sets are a datum distinct from the ring change $m \rightarrow m+1$.

These maps are obtained from emission removal:

$$\rho_k^\Sigma(\zeta\epsilon, \Sigma) = (\zeta, \Sigma), \quad \rho_k^\Pi(\zeta\epsilon, \Pi) = (\zeta, \Pi).$$

The minimal surviving branch of the APP order just constructed has compatible prolongations. Its two labels define compatible reference ports at every level. Coordinate transport is precomposition:

$$(L_k u)_{i'} = u_{\rho_k^\Sigma(i')}, \quad (L_k v)_{j'} = v_{\rho_k^\Pi(j')}, \quad (L_k w)_{i'j'} = w_{\rho_k^\Sigma(i'), \rho_k^\Pi(j')}.$$

For the last module of the sequence, first extend the matrix by zero on the reference row and column, apply this same formula and restrict to the new complement of the reference ports. This rule defines L_k even when a new port has the marked port as its antecedent.

Proposition 36.14 (Naturality of the rectangular complex under prolongation). *The preceding maps satisfy $L_k \kappa_k = \kappa_{k+1} L_k$ and $L_k \delta_k = \delta_{k+1} L_k$. They also commute with coefficient reduction $A_{m+1} \rightarrow A_m$ and compose according to prefix removal.*

Proof. The first identity follows from $u_{\rho_k^\Sigma(i')} - v_{\rho_k^\Pi(j')}$. In the second, each of the four terms defining δ is transported by the same precomposition. Compatibility of the marked ports identifies their rows and columns; when an antecedent is the marked port, the four terms cancel and agree with extension by zero. All formulas are sums and differences with integer coefficients, so they commute with modular reduction. Composition of precompositions is precomposition with the composite prefix, which proves the last assertion. $\square$

36.3.7 Memory complex and compatibility of incidences

The same accumulated memory coordinate $c_y \in \mathbb{Z}$, in a residence y , defines a free integer complex. Its reduction to A_m uses the same integer register before reducing it; successive depths are therefore compatible. Its generators are f_y in degree 2, e_y, b_y, r_y in degree 1, and g_y in degree 0:

$$\partial f_y = 3c_y e_y + b_y, \quad \partial e_y = g_y, \quad \partial b_y = -3c_y g_y, \quad \partial r_y = 0.$$

The identity $\partial^2 = 0$ is verified on f_y through $3c_y g_y - 3c_y g_y = 0$, and is immediate on the other generators.

If a trajectory $\alpha : y \rightarrow y'$ updates memory, its transport in the complex sends the correspondingly named generators to their new representatives, except

$$\Psi_\alpha(b_y) = b_{y'} + 3(c_{y'} - c_y)e_{y'}.$$

Proposition 36.15 (Naturality of the memory complex). *We have $\partial\Psi_\alpha = \Psi_\alpha\partial$ and $\Psi_{\beta\alpha} = \Psi_\beta\Psi_\alpha$. The maps extend linearly over the chosen coefficient ring.*

Proof. On f_y , the image of its boundary is $3c_y e_{y'} + b_{y'} + 3(c_{y'} - c_y) e_{y'} = 3c_{y'} e_{y'} + b_{y'}$. On b_y , the boundary of its image is $-3c_{y'} g_{y'} + 3(c_{y'} - c_y) g_{y'} = -3c_y g_{y'}$. The other two cases are immediate. In a concatenation, memory differences sum telescopically, proving the second identity. $\square$

The relation between two representatives of the same residence can also be written as an explicit boundary. For an integer coordinate v , reduced in the same ring where appropriate, let

$$z^- = r_y, \quad z^+ = r_y + 3c_y v e_y + v b_y, \quad h_* = -v f_y.$$

Then $\partial h_* = z^- - z^+$. Under a transport that also updates v , the correction $K_\alpha = (v_{y'} - v_y) f_{y'}$ satisfies $K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha$. The symbols K_α of this local homotopy do not denote the dodecaphasic seal K : they belong to another domain and their function is specified by the preceding identity. Their function can also be checked in the equality

$$z_{y'}^+ - \Psi_\alpha z_y^+ = \partial K_\alpha.$$

Both sides equal $(v_{y'} - v_y)(3c_{y'} e_{y'} + b_{y'})$.

Proposition 36.16 (Homology and memory of the local complex). *Over $\mathbb{Z}$, or after reduction to A_m , we have*

$$H_0(\Gamma) = H_2(\Gamma) = 0, \quad H_1(\Gamma) \cong A_m[r_y]$$

in the modular realization, and $H_1(\Gamma) \cong \mathbb{Z}[r_y]$ in the integer realization.

Proof. Define the homotopy of degree 1 by $h(g_y) = e_y$, $h(b_y) = f_y$, and zero on the other generators. Let p be the projection onto the summand generated by r_y and ι its inclusion. Checking on each generator gives

$$\partial h + h \partial = I - \iota p.$$

In particular, on b_y the terms $3c_y e_y$ and $-3c_y e_y$ cancel. The complex deformation retracts onto the module generated by r_y , concentrated in degree 1, which proves the assertion. $\square$

The transport Ψ_α is a shear of determinant 1. Thus, this complex explicitly preserves how c_y changes at the chain level, but its local homology does not distinguish the values of that coordinate. The full memory remains in the labels and transports of the source state; it is not recovered from H_1 alone. A memory-dependent numerical torsion requires the specified assembly and determinantal reading, in addition to this local complex.

36.3.8 Assembly with terminal paths

For a finite horizon, the residences and their prolongations form a path category. Each residence ν is assigned the free module $W(\nu) = A_m[\text{Path}(\nu, \text{Term})]$, which preserves each terminal continuation as a generator. An initial path acts on W by precomposition. The complex $\Gamma(\nu)$ of the preceding section acts covariantly through Ψ .

The assembly uses both transports in the same bigraded module:

$$\mathcal{B}_{q,r} = \bigoplus_{\nu_0 \rightarrow \dots \rightarrow \nu_q} W(\nu_q) \otimes_{A_m} \Gamma_r(\nu_0).$$

The normalized path complex is used: degenerate chains containing identities are omitted. The finite horizon then bounds the number of strict prolongations and the horizontal degree. The horizontal boundary removes an intermediate residence by composing the adjacent arrows. At the first endpoint

it transports Γ ; at the last it precomposes W . The alternating signs yield the path boundary d_{bar} . The total differential is

$$D = d_{\text{bar}} + (-1)^q \partial_\Gamma.$$

The face identities cancel the terms of d_{bar}^2 in pairs. The naturality of Proposition 36.15 implies that d_{bar} commutes with ∂_Γ ; the factor $(-1)^q$ cancels the cross terms of the square. Therefore, $D^2 = 0$. Compatibility is proved on path generators and is not introduced by a label of membership in a complex.

Terminal evaluation likewise does not require choosing a particular history. For each continuation h , the effective composition of its updates is applied. The result preserves the pair formed by h and its reading, with its multiplicity. If $k < \ell < N$, the continuations decompose as a disjoint union of prefixes up to ℓ and their continuations up to N . Proposition 36.6 proves that evaluating directly or evaluating these two parts produces the same reading. This yields the naturality square of the terminal.

The path modules, incidences, memory transport and their differentials arise from the same prolongations. Their assembly does not consist in adding to a state five coordinates that would be preserved by definition: their relations are constructed through sums, boundaries and effective compositions, and are verified on their generators.

36.3.9 Determinants, limit and arithmetic reading

In each realization, the local free complex Γ admits its graded determinant line

$$\text{Det } \Gamma = \bigotimes_r \left(\bigwedge^{\text{rk } \Gamma_r} \Gamma_r \right)^{(-1)^r}.$$

The total assembly of the preceding section, in turn, has the modules $\text{Tot}_d \mathcal{B} = \bigoplus_{q+r=d} \mathcal{B}_{q,r}$. At a finite horizon and over the same ring, they are free of finite rank and only finitely many of them are nonzero. Its line is

$$\text{Det}(\text{Tot } \mathcal{B}) = \bigotimes_{q,r} \left(\bigwedge^{\text{rk } \mathcal{B}_{q,r}} \mathcal{B}_{q,r} \right)^{(-1)^{q+r}}.$$

This formula follows from decomposition as a direct sum in each degree. It distinguishes the line of the local complex from that of the assembly with all paths; their bases are ordered through the preserved labels. Modular reduction commutes with these exterior powers because the modules and their bases come from the same free integer modules at a fixed horizon. The line is defined by the modules and their degrees, both over $\mathbb{Z}$ and after reduction to A_m ; the negative exponent denotes the dual module. A nonzero numerical torsion additionally requires specifying the complex over a field, the bases and the relevant homology identifications. To calculate Smith factors, the integer matrix produced before reduction is used, not an arbitrary lift of a modular matrix. A square integer block with nonzero determinant is invertible over $\mathbb{Q}$; its reduction is invertible over A_m only when that determinant is a unit, that is, when it is not divisible by 3. Refinement of memories and incidences allows the stability of the relevant invariant factors to be formulated and checked; these conditions do not follow from the existence of the determinant line.

Passage to the limit preserves the compatible projective systems of remainders, histories and their multiplicities. The squares proved in the preceding sections remain commutative because their maps are given by the same formulas at each level. For an Archimedean coordinate, the additional step consists in expressing nonempty cylinder intervals, compatible by inclusion and with diameters tending to zero. Their nested compact closures have a one-point intersection: existence follows by compactness and uniqueness by the diameter bound. That is the value of the limit readout. If the intervals are half-open, the point may belong only to their closures; the boundary representation rule then determines its positional expression. The value does not replace the history that produced it. The other representation preserves the incidences of the same state. Both destinations share their origin without thereby receiving the same information.

The coinductive regions, the dodecaphase record and exceptional incidence are expressed from that substrate through their respective maps. The Hilbert-space realization subsequently preserves phase labels and specifies its own unital inclusion; duality and screens act on the codomains constructed there. The joint operations brought together here transmit histories, sheets, orientation and memory to those representations. Identifying a particular functional with an external analytic form in turn requires writing its action on the generators and verifying its inner products: the conservation identities of this chapter do not replace that composition.

36.3.10 Mathematical scope of the joint construction

The state, tree and measure constructions; the correlative operations; and the terminal assembly with naturality and limit are developed in full in this article. In this chapter their balance, incidence and memory identities have been formulated and proved on the same generators. The five constructions thus remain operations of a single tree. Each specialization keeps its exact hypotheses visible: transport isometry, canonical weight law, treatment of the terminal remainder and nonsingularity of the determinantal block where applicable.

The preceding equations prove the memory cocycles, weighted cancellation, rectangular exactness and compatibility of the memory complex at every finite horizon, as well as their compatible passage to the limit. A subsequent analytic identification must compose its operator on these generators and verify the additional properties of its codomain; it neither alters nor replaces the joint construction proved here.

36.3.11 Joint emergence of the five constructions

For $0 \leq k \leq N$, let

$$\rho_{N,k} := \rho_k \circ \rho_{k+1} \circ \cdots \circ \rho_{N-1}, \quad \rho_{k,k} := \text{id}_{H_k},$$

and, for a surviving history $\zeta \in H_k$, define

$$\text{Desc}_{k,N}(\zeta) := \{x \in H_N : \rho_{N,k}(x) = \zeta\}.$$

The finite tree $\mathcal{T}_{k,N}(\zeta)$ has vertices

$$\prod_{j=k}^N \{y \in H_j : \rho_{j,k}(y) = \zeta\}$$

and has as edges only the TPK prolongations $a : y \rightarrow z$ with $z \in \rho_j^{-1}(y)$. Thus neither its vertices nor its prolongations come from an auxiliary tree.

For $x \in \text{Desc}_{k,N}(\zeta)$, the marked branch $\lambda_{\zeta,x}$ is the sequence

$$\zeta = x_k \longrightarrow x_{k+1} \longrightarrow \cdots \longrightarrow x_N = x$$

determined by the truncations of x . The common base register is

$$\begin{aligned} \mathcal{R}_{k,N}(\zeta; x) := & (\mathcal{T}_{k,N}(\zeta), \lambda_{\zeta,x}; \\ & (\varepsilon_a, s_a, o_a, (\delta_{\circ,a}, r_{\circ,a}, q_{\circ,a})_{\circ \in \{\Sigma, \Pi\}}, \text{Carr}_a, C_a, t_a, \text{Fr}_a, \text{Ruta}_a, \text{Mem}_a, \text{Surv}_a)_a; \\ & \text{src}, \text{tgt}, \circ, (\rho_j)_{k \leq j < N}), \end{aligned} \quad (\text{C1})$$

where

$$U_a(m_0) = C_a m_0 + t_a, \quad \delta_{\circ,a} = r_{\circ,a} + 9q_{\circ,a}.$$

The symbol m_0 in this formula is a memory coordinate; it is not the modular depth m used below. The true base space is

$$\mathfrak{R}_{k,N}(\zeta) := \{\mathcal{R}_{k,N}(\zeta; x) : x \in \text{Desc}_{k,N}(\zeta)\}.$$

The marked branch is not an added input: it is the sequence of actually produced edges ending at x . The rest of the register simultaneously preserves the lateral fibres of the tree; thus the mark does not replace branching with an isolated path.

Set $R_m := \mathbb{Z}/3^m\mathbb{Z}$, and let $V_j(\zeta) := \text{Desc}_{k,j}(\zeta)$. Transport, the sheet projectors and the terminal residue restrict to the subtree through

$$\begin{aligned} \mathcal{U}_j^\zeta &:= \mathcal{U}_j \upharpoonright_{\ell^2(V_j(\zeta), \mu_j)}, & P_j^\zeta &:= P_j \upharpoonright_{V_j(\zeta)}, & Q_j^\zeta &:= I - P_j^\zeta, \\ A_j^\zeta &:= P_{j+1}^\zeta \mathcal{U}_j^\zeta P_j^\zeta + Q_{j+1}^\zeta \mathcal{U}_j^\zeta Q_j^\zeta, & \eta_j^\zeta &:= Q_{j+1}^\zeta \mathcal{U}_j^\zeta P_j^\zeta + P_{j+1}^\zeta \mathcal{U}_j^\zeta Q_j^\zeta. \end{aligned}$$

For $k \leq n \leq N$, let

$$F_{k,k}^\zeta := I, \quad F_{k,n}^\zeta := A_{n-1}^\zeta \cdots A_k^\zeta, \quad T_{k,n}^\zeta := (F_{k,n}^\zeta)^* F_{k,n}^\zeta.$$

The ports belonging to the same subtree are

$$\mathcal{P}^\Sigma(\zeta) := \{(y, \Sigma) : y \in V_j(\zeta)\}, \quad \mathcal{P}^\Pi(\zeta) := \{(y, \Pi) : y \in V_j(\zeta)\};$$

we denote by $B_j^\zeta, \kappa_j^\zeta, \delta_j^\zeta$ the restrictions of the incidence and the rectangular complex to those ports, and by L_j^ζ the precomposition induced by removing prefixes.

To define the path reader without abbreviations, let

$$W_\ell^\zeta(y) := R_m[\text{Path}_{\mathcal{T}_{k,\ell}(\zeta)}(y, \text{Term}_\ell)]$$

and set

$$\mathcal{D}_{q,r;k,\ell}^{(m)}[\mathcal{R}] := \bigoplus_{y_0 \rightarrow \cdots \rightarrow y_q \text{ in } \mathcal{T}_{k,\ell}(\zeta)} W_\ell^\zeta(y_q) \otimes_{R_m} (\Gamma_r(y_0) \otimes_{\mathbb{Z}} R_m), \quad \text{Tot}_d \mathcal{D} := \bigoplus_{q+r=d} \mathcal{D}_{q,r;k,\ell}^{(m)}[\mathcal{R}].$$

The formulas already proved produce, on each $\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)$, the five objects

$$\begin{aligned} \mathcal{F}_{\text{sp}}[\mathcal{R}] &:= (\ell^2(V_j(\zeta), \mu_j), \mathcal{U}_j^\zeta, A_j^\zeta, \eta_j^\zeta, T_{k,j}^\zeta)_{k \leq j \leq N}, \\ \mathcal{F}_{\text{sol}}[\mathcal{R}] &:= (b, \kappa_{\text{can}}, C_\alpha, \kappa_\alpha)_{\alpha \in \mathcal{T}_{k,N}(\zeta)}, \\ \mathcal{F}_{\text{gau}}^{(m)}[\mathcal{R}] &:= (\mathcal{P}^\Sigma(\zeta), \mathcal{P}^\Pi(\zeta), B_j^\zeta, \kappa_j^\zeta, \delta_j^\zeta; R_m)_{k \leq j \leq N}, \\ \mathcal{F}_{\text{coh}}^{(m)}[\mathcal{R}] &:= (\Gamma_y \otimes R_m, \partial, \Psi_\alpha, H_\bullet)_{y, \alpha \in \mathcal{T}_{k,N}(\zeta)}, \\ \mathcal{F}_{\text{det}}^{(m)}[\mathcal{R}] &:= (\text{Det}(\text{Tot } \mathcal{D}_{\bullet, \bullet; k, \ell}^{(m)}[\mathcal{R}]))_{k \leq \ell \leq N}. \end{aligned} \tag{C2}$$

Here $\mathcal{D}_{\bullet, \bullet; k, \ell}^{(m)}[\mathcal{R}]$ is the normalized path-and-memory bicomplex of the preceding sections, restricted to the subtree of horizon ℓ and reduced over R_m . The last reader preserves the ordered collection of determinant lines up to horizon N ; in this way, its truncation is a canonical projection and does not presuppose a nonexistent determinantal factorization. The law μ is the canonical census law, and the ports are fixed through the lexicographically minimal branch of the APP order. Thus none of the five readers receives external weighting or an external port.

For $T \in \mathfrak{T} := \{\text{sp}, \text{sol}, \text{gau}, \text{coh}, \text{det}\}$, define the typed total space

$$\tilde{\mathcal{F}}_{T;k,N}^{(m)}(\zeta) := \prod_{\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)} \{\mathcal{F}_T^{(m)}[\mathcal{R}]\}, \quad p_T : \tilde{\mathcal{F}}_{T;k,N}^{(m)}(\zeta) \longrightarrow \mathfrak{R}_{k,N}(\zeta),$$

where $\mathcal{F}_{\text{sp}}^{(m)} := \mathcal{F}_{\text{sp}}$ and $\mathcal{F}_{\text{sol}}^{(m)} := \mathcal{F}_{\text{sol}}$; that is, the superscript m is inert for those two readers. The projection p_T forgets only the reader structure and preserves its source register. The correct fibre product is

$$\begin{aligned} \mathcal{J}_{k,N}^{(m)}(\zeta) &:= \tilde{\mathcal{F}}_{\text{sp};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{sol};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{gau};k,N}^{(m)}(\zeta) \\ &\quad \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{coh};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{det};k,N}^{(m)}(\zeta). \end{aligned} \tag{C3}$$

Equality of the five projections requires preserving the same tree, the same edges, the same produced branch and the same register. For each $\mathcal{R}$, the five constructions determine the element

$$s_{k,N}^{(m)}(\mathcal{R}) := (\mathcal{F}_T^{(m)}[\mathcal{R}])_{T \in \mathfrak{T}} \in \mathcal{J}_{k,N}^{(m)}(\zeta).$$

Existence of this element follows from the operations of (C2), not from the fibre-product notation.

Removal of the last layer induces

$$\text{res}_{N+1,N}^{\mathfrak{R}}(\mathcal{R}_{k,N+1}(\zeta; x)) := \mathcal{R}_{k,N}(\zeta; \rho_N(x)).$$

Restriction of each operator to the retained generators defines

$$r_{N+1,N}^m : \mathcal{J}_{k,N+1}^{(m)}(\zeta) \longrightarrow \mathcal{J}_{k,N}^{(m)}(\zeta).$$

In the determinantal reader, $r_{N+1,N}^m$ removes only the last coordinate of the collection of lines of (C2). The reduction $R_{m+1} \rightarrow R_m$ defines, by base change,

$$q_{m+1,m}^N : \mathcal{J}_{k,N}^{(m+1)}(\zeta) \longrightarrow \mathcal{J}_{k,N}^{(m)}(\zeta).$$

On each generator, we have

$$\begin{aligned} r_{N+1,N}^m q_{m+1,m}^{N+1} &= q_{m+1,m}^N r_{N+1,N}^{m+1}, \\ \partial \Psi_\alpha &= \Psi_\alpha \partial, \quad L_j^\zeta \kappa_j^\zeta = \kappa_{j+1}^\zeta L_j^\zeta, \quad L_j^\zeta \delta_j^\zeta = \delta_{j+1}^\zeta L_j^\zeta, \end{aligned} \tag{C4}$$

and compositions of the maps r and q satisfy the functorial identities. The first equality holds because deleting the last layer and reducing the coefficients act on distinct coordinates. The other three are the naturality identities proved earlier. A unique pro-system in (N, m) is obtained, not five juxtaposed limits.

At the root, define

$$\mathfrak{F}_{N,\bullet}^{(m)} := \mathcal{J}_{0,N}^{(m)}(*). \tag{C5}$$

The joint discrete structure of the continuum is the limit of those correlative objects:

$$\mathcal{C}_{\text{cont}}^{\text{disc}} := \lim_{\substack{\leftarrow \\ N \geq 1 \\ m \geq 1}} (\mathfrak{F}_{N,\bullet}^{(m)}, r_{N+1,N}^m, q_{m+1,m}^N). \tag{C6}$$

This is a definition through the pro-system generated by the complete tree; it is not the image or graph of a map retaining the input state as its first coordinate.

The registers have a produced terminal mark. Their compatible projections induce

$$\Pi : \mathcal{C}_{\text{cont}}^{\text{disc}} \longrightarrow H_\infty^{\text{enr}}.$$

Conversely, a complete history determines, through its already emitted prefixes, the compatible sections $s_{0,N}^{(m)}$. Thus, after having defined the continuum, one obtains

$$\text{Gen}_{\text{cont}} : H_\infty^{\text{enr}} \longrightarrow \mathcal{C}_{\text{cont}}^{\text{disc}}, \quad \Pi \circ \text{Gen}_{\text{cont}} = \text{id}_{H_\infty^{\text{enr}}}. \tag{C7}$$

This section proves recoverability; it is not used to define $\mathcal{C}_{\text{cont}}^{\text{disc}}$.

Theorem 36.17 (Correlative emergence of the five constructions). *For every surviving history $\zeta \in H_k$, every $N > k$, every $m \geq 1$ and every $\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)$, the five constructions of (C2) are defined on the same register, and (C3) contains $s_{k,N}^{(m)}(\mathcal{R})$. Their truncations, modular reductions and transports satisfy (C4). Furthermore, $\mathcal{C}_{\text{cont}}^{\text{disc}} \neq \emptyset$, the map Π is surjective and is split by Gen_{cont} . The five components are inseparable aspects of the same continuum, not five domains or five independent maps.*

Proof. The survival defining H_k and the TPK prolongation law give $\rho_k^{-1}(\zeta) \neq \emptyset$ for each vertex. Since the fibres are finite, Kőnig's lemma produces $H_\infty^{\text{enr}} \neq \emptyset$.

Given a register $\mathcal{R}$, the transport of histories and sheets produces $\mathcal{F}_{\text{sp}}[\mathcal{R}]$; affine composition and the exact cancellation defect produce $\mathcal{F}_{\text{sol}}[\mathcal{R}]$; incidences and the rectangular sequence produce $\mathcal{F}_{\text{gau}}^{(m)}[\mathcal{R}]$; the memory complex and its transport produce $\mathcal{F}_{\text{coh}}^{(m)}[\mathcal{R}]$; and the lines of the normalized path assembly

produce $\mathcal{F}_{\text{det}}^{(m)}[\mathcal{R}]$. Each construction acts on the same generated edges, not on an assumed subdomain or on the abstract ambient space of 729 words.

The affine, transport, incidence and memory compatibilities prove (C4). Truncating the determinantal collection of (C2) commutes by definition with r ; its base change commutes with q . The sections $s_{0,N}^{(m)}$ are compatible, so they produce Gen_{cont} . The identity (C7) simultaneously proves nonemptiness of the limit, surjectivity of Π and recoverability of each history without introducing it as a defining coordinate.

The fibre, relation, number, orientation, carry, memory, boundary and route remain in the common register; the multisection, terminals, and linear and probabilistic readers remain in the five correlative constructions. Passing to the limit therefore preserves dependency closure and does not separate the continuum. $\square$

Subsequent recognition of the uniquely constructed continuum through spectral, solenoidal, gauge, cohomological or determinantal language belongs to the later conventional comparison. It does not select its states or enter its APP–TRIT–TPK generation.

36.4 Extension of states and transfinite semantic extension

Let

$$\Sigma_{\text{HMT}} = (\dot{R}_{\text{APP}}, \dot{R}_{\text{TRIT}}, \dot{R}_{\text{TPK}}, \dot{R}_U, \dot{R}_{\Gamma_0}, \dot{R}_{\text{Ext}}, \dot{R}_\rho, \dot{R}_\pi, \dot{R}_\varphi, \dot{R}_e, \dot{R}_\alpha, \dot{R}_{\text{ar}}, \dot{R}_{\text{exc}}) \quad (33)$$

be the countable **relational** signature whose effective code is h . Each $\dot{R}_S$ is the graph of the operator S on the codes where it is defined; no external function is assumed to be total within each level. The graphs $\dot{R}_\pi, \dot{R}_\varphi, \dot{R}_e, \dot{R}_\alpha$ are the output readers constructed by coinductive generation and dodecaphase closure; they do not take the conventional values of the constants as arguments. Their presence in Σ_{HMT} ensures that the closure preserves the full state: it simultaneously integrates the routes that express value and the incidence that extends toward the exceptional realization.

At each horizon, the state code records the APP–TRIT–TPK coordinates, the route, sheet, carry, boundary, and memory; at its representation stages it also records P_N, Φ_N, E_N, A_N , where A_N is the prefix of α_{HMT} produced by the same state. These coordinates do not select the branch from outside: they are part of the history being encoded. Once a primitive-recursive numbering of the finite coordinates of a state is fixed, every inverse history $x_\infty = (x_N)_{N \geq 1} \in X_\infty^{\text{enr}}$ determines the code

$$\text{code}(x_\infty) := \{\langle N, j, c \rangle : (x_N)_j \text{ has code } c\} \subseteq \omega.$$

The coherence equations $\rho_{N+1,N}(x_{N+1}) = x_N$ make it possible to decode the history uniquely from that real. We write m_∞ for the code of the realized history and $m_\infty(n)$ for its characteristic function under the fixed numbering. The operator code h and this complete ledger are combined, by means of a primitive-recursive pairing function, into the real

$$u_{h,m} := \{\langle 0, \langle n, h(n) \rangle \rangle : n < \omega\} \cup \{\langle 1, \langle n, m_\infty(n) \rangle \rangle : n < \omega\}. \quad (33a)$$

The two primitive projections recover h and m_∞ from $u_{h,m}$.

The first extension is the metric and operator extension of the finite states. For $N \geq 1$, let $\hat{\mathcal{X}}_{6N}^{\text{enr,raw}}$ be the set of raw enriched states of length $6N$, and define the surviving subsystem set-theoretically by

$$\begin{aligned} X_N &:= \hat{\mathcal{X}}_{6N}^{\text{enr,surv}} \\ &:= \left\{ x \in \hat{\mathcal{X}}_{6N}^{\text{enr,raw}} : \right. \\ &\quad \forall M > N \exists z \in \hat{\mathcal{X}}_{6M}^{\text{enr,raw}} \text{ such that} \\ &\quad \left. \text{tr}_{6M,6N}(z) = x \wedge z \text{ satisfies every intermediate TPK transition} \right\}. \end{aligned}$$

and set

$$\rho_{N+1,N} := \text{tr}_{6N+6,6N} \upharpoonright X_{N+1},$$

and

$$\text{Ext}_N(x) := \{y \in X_{N+1} : (x, y) \in \text{Ext}_{6N+6,6N}\}.$$

The root level is included without any implicit exception:

$$X_0 := \{*\}, \quad \rho_{1,0} : X_1 \rightarrow X_0 \text{ is constant}, \quad \text{Ext}_0(*) := X_1.$$

The superscript *surv* denotes the pruned tree of states possessing compatible TPK descendants at every finite horizon. The TPK extension law established in the preceding sections proves that the emitted seeds belong to this tree; finite branching and Kőnig's lemma turn compatibility at every horizon into an infinite branch. In particular,

$$\forall N \geq 0 \forall x \in X_N \quad \emptyset \neq \text{Ext}_N(x) \subseteq \rho_{N+1,N}^{-1}(x). \quad (34)$$

By the same finitely branching tree lemma, every surviving seed has a compatible extension

$$x_\infty = (x_N)_{N \geq 1} \in X_\infty^{\text{enr}} := \varprojlim_N (X_N, \rho_{N+1,N}). \quad (35)$$

The second extension acts **afterward** on the complete code of that realized history. It is not inferred from the mere existence of the finite operator Ext_N : at the set-theoretic scale, it formalizes the HMT principle that only those objects possessing a genealogical derivation from the state and its rules belong to the full domain. The transfinite semantic extension is:

$$\begin{aligned} \mathfrak{G}_0(h, m_\infty) &:= \text{TC}(\{\omega, u_{h,m}, h, m_\infty\}), \\ \mathfrak{G}_{\beta+1}(h, m_\infty) &:= \text{Def}_{\Sigma_{\text{HMT}}}(\mathfrak{G}_\beta(h, m_\infty)), \\ \mathfrak{G}_\lambda(h, m_\infty) &:= \bigcup_{\beta < \lambda} \mathfrak{G}_\beta(h, m_\infty) \quad (\lambda \text{ limit}), \\ \mathfrak{G}_{\text{HMT}}(h, m_\infty) &:= \bigcup_{\beta \in \text{Ord}} \mathfrak{G}_\beta(h, m_\infty). \end{aligned} \quad (36)$$

The Continuum Hypothesis and a target bijection are not introduced among the conditions defining the successor operator; the operator is given prospectively by

$$\text{Def}_{\Sigma_{\text{HMT}}}(M) := M \cup \left\{ \{x \in M : \langle M, \in, (\dot{R}_S \cap M^{k_S})_{S \in \Sigma_{\text{HMT}}} \rangle \models \varphi(x, \bar{p})\} : \ulcorner \varphi \urcorner \in \omega, \bar{p} \in M^{<\omega} \right\}. \quad (37)$$

That is, it simultaneously expresses all subsets definable by a finite formula of the HMT relational language and parameters already produced by the genealogy. It consults neither a target real value, a conventional constant, a target cardinal, nor an external model. Formula (37) defines the transfinite closure over the previously constructed seeds, operators, enriched state, ledger, and double projection.

Lemma 36.18 (Joint coding of the HMT program and the realized ledger). *The pair formed by the operator code h and the ledger m_∞ is encoded by a single real $u_{h,m} \subseteq \omega$, and every formula of Σ_{HMT} is uniformly translated into a formula of the language $\{\in, u_{h,m}\}$.*

Proof. Both h and m_∞ are sequences of natural numbers: h encodes the signature, the finite tables, and the operator programs; m_∞ encodes one realized history, not the complete history space. Formula (33a) and its two decoders prove the first assertion. Each symbol in (33) denotes its **graph relation encoded in h** , never the external graph of all its possible evaluations. APP, TRIT, U_t , Γ_9 , the restrictions ρ , and the finite extensions have primitive-recursive graphs on integer codes and, therefore, Δ_0 formulas absolute between transitive levels. The two representations are expressed as graph relations through the formulas

$$\begin{aligned}\dot{R}_{\text{ar}}(x, y) &\iff \forall k \in \omega \exists N \forall n \geq N \quad |\mu_n(x) - y| < 2^{-k}, \\ \dot{R}_{\text{exc}}(x, c, z) &\iff \forall N \in \omega \quad z \upharpoonright N = E_N(x \upharpoonright N, c \upharpoonright N),\end{aligned}\tag{37c_0}$$

where μ_n and E_N are the rational/finite functions whose codes occur in h . Every atomic formula containing an HMT symbol is inductively replaced by its graph formula with parameter $u_{h,m}$. Connectives and quantifiers are preserved. This induction on syntactic complexity produces a translation $\varphi \mapsto \varphi^*$ such that, for every transitive level M of (36) containing the parameters,

$$\langle M, \in, (\dot{R}_S \cap M^{ks})_S \rangle \models \varphi(\bar{a}) \iff M \models \varphi^*(\bar{a}; u_{h,m}).\tag{37c}$$

Thus, (37) cannot consult an oracle, a target decimal table, or an uncoded relation that introduces uncountably many objects in a single step. $\square$

HMT semantics is now defined **independently of identifying it with $\mathfrak{G}$** . Let $\mathcal{T}_0^{\text{HMT}}$ be the set of canonical names of the elements of $\mathfrak{G}_0(h, m_\infty) = \text{TC}(\{\omega, u_{h,m}, h, m_\infty\})$. In particular, the initial level contains the realized ledger m_∞ , not every history in X_∞^{enf} as primitive parameters. The admissible terms are the well-founded trees obtained by

$$\begin{aligned}\mathcal{T}_{\beta+1}^{\text{HMT}} &:= \mathcal{T}_\beta^{\text{HMT}} \cup \{\langle \beta, \ulcorner \varphi \urcorner, t_0, \dots, t_{k-1} \rangle : t_i \in \mathcal{T}_\beta^{\text{HMT}}\}, \\ \mathcal{T}_\lambda^{\text{HMT}} &:= \bigcup_{\beta < \lambda} \mathcal{T}_\beta^{\text{HMT}}, \quad \mathcal{T}^{\text{HMT}} := \bigcup_{\beta \in \text{Ord}} \mathcal{T}_\beta^{\text{HMT}}.\end{aligned}\tag{37a}$$

The value of a successor term is the subset of the preceding level defined by φ in the induced relational structure, with the values of $t_0, \dots, t_{k-1}$ as parameters. Before any set-theoretic comparison, define

$$\text{Ob}_{\text{sem}}^{\text{HMT}}(h, m_\infty) := \{\llbracket t \rrbracket : t \in \mathcal{T}^{\text{HMT}}\}.\tag{37b}$$

This is the rule of **generative exhaustiveness** adopted by HMT: every object in the entire domain must have one of those derivation trees and every admissible tree expresses an object. It is not a consequence of finite branching. The continuum hypothesis is not introduced into this rule; it is obtained through the proof developed in the article. Neither an enumeration of the reals nor a target bijection is introduced as data.

Theorem 36.19 (Equivalence between term semantics and the genealogical hierarchy). *The evaluation of the preceding terms satisfies*

$$\text{Ob}_{\text{sem}}^{\text{HMT}}(h, m_\infty) = \mathfrak{G}_{\text{HMT}}(h, m_\infty).\tag{37d}$$

Proof. By transfinite induction. At the initial level, the two classes coincide. If every value of a term in $\mathcal{T}_\beta^{\text{HMT}}$ belongs to $\mathfrak{G}_\beta$, the semantics of a successor term is one of the subsets in (37). Conversely, every subset in (37) is given by a formula code and a finite tuple of parameters; by the induction hypothesis those parameters possess terms, and (37a) forms the term that denotes it. At a limit, both constructions take the same union. $\square$

The state encoder is compatible with truncations and sends every TPK extension to a successor term. Let $\mathcal{T}_N^{\text{st}} \subset \mathcal{T}^{\text{HMT}}$ be the subclass of terms encoding complete states of depth N , and let $\tau_{N+1,N} : \mathcal{T}_{N+1}^{\text{st}} \rightarrow \mathcal{T}_N^{\text{st}}$ delete the last block together with the corresponding truncation of the genealogical ledger, carry, and memory. If $J_N : X_N \rightarrow \mathcal{T}_N^{\text{st}}$ is the complete encoder, compatibility with Ext_N proves the well-typed identity

$$\forall x \in X_N \forall y \in \text{Ext}_N(x) \quad \tau_{N+1,N}(J_{N+1}(y)) = J_N(x). \quad (37e_0)$$

Define within the semantics

$$\text{Succ}_{N+1}^{\text{TPK}}(J_N(x)) := \{t \in \mathcal{T}_{N+1}^{\text{st}} : \exists y \in X_{N+1} (y \in \text{Ext}_N(x) \wedge t = J_{N+1}(y))\}.$$

Choose β such that $\mathfrak{G}_\beta$ contains x, X_N, X_{N+1} , the finite graphs of $\text{Ext}_N, J_N, J_{N+1}$, and their codes in h . Then the primitive-recursive formula for the graph of Ext_N gives

$$J_{N+1}[\text{Ext}_N(x)] = \text{Succ}_{N+1}^{\text{TPK}}(J_N(x)), \quad J_{N+1}[\text{Ext}_N(x)] \in \mathfrak{G}_{\beta+1}, \quad (37e)$$

and both sides are formed by the graph formula of the same TPK extension. Equations (37e₀)–(37e) are the explicit semantic link: the subclass of successor terms is exactly the image of the admissible extensions, and its truncation commutes. The complete fiber of ρ is not identified with Ext_N , nor is Def identified with TPK merely by nomenclature.

Lemma 36.20 (Transitivity, axioms, and well-ordering of the genealogical model). *$\mathfrak{G}_{\text{HMT}}(h, m_\infty)$ is a transitive class, contains all ordinals, satisfies Zermelo–Fraenkel set theory (ZF), and possesses a definable global well-order; it therefore satisfies Choice and all axioms used in the subsequent cardinal comparison.*

Proof. Abbreviate $B = \mathfrak{G}_0$ and $u = u_{h,m}$. Induction on (36) proves simultaneously that every level is transitive and that $\mathfrak{G}_\alpha \in \mathfrak{G}_{\alpha+1}$; if all ordinals smaller than α are present, their set is definable at the corresponding level and expresses α . Hence the union contains all ordinals. Extensionality and Foundation are inherited from the ambient universe; Empty Set and Infinity belong to B . Once a level containing the parameters is chosen, the formulas defining $\{a, b\}$, $\bigcup a$, and subsets by Separation express them at a successor level.

For Collection–Replacement, let F be a functional relation definable in $\mathfrak{G}$ on the set a . The reflection scheme relative to the definable hierarchy $(\mathfrak{G}_\xi, \in, u_{h,m})_{\xi \in \text{Ord}}$, applied in the ambient metatheory to the finite list of formulas defining F , produces a level $\mathfrak{G}_\theta$ containing a and the parameters and correct for those formulas. The ranks of the witnesses in $F[a]$ are then bounded by θ ; Separation over $\mathfrak{G}_\theta$ forms the image in $\mathfrak{G}_{\theta+1}$. For Power Set, the internal set to be constructed is not presupposed. Given $a \in \mathfrak{G}_\gamma$, consider in the ambient metatheory the already existing set $\mathcal{P}^V(a)$. For every $b \in \mathcal{P}^V(a) \cap \mathfrak{G}$, let $r(b)$ be its first genealogical stage. Since $\mathfrak{G}$ is a definable class, Separation in the ambient metatheory first forms the set $\mathcal{P}^V(a) \cap \mathfrak{G}$. Replacement **in the ambient metatheory**, applied to that set, bounds the ordinals $r(b)$ by some θ . Consequently,

$$\mathcal{P}^\mathfrak{G}(a) = \{b \in \mathfrak{G}_\theta : b \subseteq a\},$$

and the right-hand side is expressed by Separation in $\mathfrak{G}_{\theta+1}$. It does not introduce into $\mathfrak{G}$ the subsets external to the semantics (37b). This is the noncircular rank-bounding argument.

Finally, each object is assigned its first level of birth and its least **derivation name**: a well-founded, finitely branching derivation tree (and, by (37a), finite for each individual name), labeled by a level ordinal, a formula code, and the tuple of names of its parameters. These names are ordered by transfinite recursion: first by the ordinal label, then by the formula code, and then lexicographically by the already ordered parameter names. The least name of each value defines a class relation $<_\mathfrak{G}$ that compares all objects. It is a definable global well-order and, by taking the least element of every nonempty set, proves Choice. $\square$

Reflection by witness closure and construction of the image. The Collection–Replacement step of Lemma 36.20 admits the following explicit proof. We retain the hierarchy already generated from the operator code and the realized record, and abbreviate $\mathfrak{G} = \mathfrak{G}_{\text{HMT}}(h, m_\infty)$. The argument uses Separation and Replacement in the ambient metatheory, without presupposing these schemes in $\mathfrak{G}$.

Let Φ be a finite collection of formulas. First translate their universal quantifiers using negation and existential quantification, and close the resulting collection under subformulas. For every initial level there is a later limit ordinal θ such that

$$\mathfrak{G}_\theta \models \varphi(\bar{a}) \iff \mathfrak{G} \models \varphi(\bar{a}) \quad (\varphi \in \Phi, \bar{a} \in \mathfrak{G}_\theta^{<\omega}).$$

For each fixed formula, satisfaction in the class is interpreted by relativizing its quantifiers to the definable class $\mathfrak{G}$; no uniform truth predicate is introduced.

Proof. Fix α . For each existential subformula $\exists y \psi(y, \bar{x})$ of Φ , the tuples $\bar{a} \in \mathfrak{G}_\alpha^{<\omega}$ of the corresponding arity that have a witness in $\mathfrak{G}$ form a set by Separation in the metatheory. Assign to each tuple the least ordinal β for which some $y \in \mathfrak{G}_\beta$ satisfies $\mathfrak{G} \models \psi(y, \bar{a})$. This ordinal exists because $\mathfrak{G}$ is the union of its levels. Replacement in the metatheory bounds these first stages of witnesses. Since Φ is finite, a single bound $f(\alpha) > \alpha$, valid for all its existential subformulas, can be chosen definably. Thus every tuple under consideration has *at least one witness* in $\mathfrak{G}_{f(\alpha)}$.

Take α_0 large enough to contain the initial parameters and set

$$\alpha_{n+1} = f(\alpha_n), \quad \theta = \sup_{n < \omega} \alpha_n.$$

By continuity of the hierarchy, every finite tuple from $\mathfrak{G}_\theta$ belongs to some $\mathfrak{G}_{\alpha_n}$. If an existential formula of Φ with that tuple is true in $\mathfrak{G}$, it has a witness in $\mathfrak{G}_{\alpha_{n+1}} \subseteq \mathfrak{G}_\theta$. Induction on formulas now proves the equivalence: atomic formulas are preserved in the induced structure; connectives preserve the equivalence; the existential step uses that witness in one direction and the same witness, viewed in $\mathfrak{G}$, in the other. The prior translation of universal quantifiers also includes the existential formulas expressing their possible counterexamples. This proves the required finite reflection. $\square$

Now let $F(x, y, \bar{p})$ be functional on a in $\mathfrak{G}$. Apply the construction to the formulas expressing F , the existence of its witnesses, and its image, with $a, \bar{p} \in \mathfrak{G}_\theta$. Transitivity gives $a \subseteq \mathfrak{G}_\theta$. Every $x \in a$ has its value within that level, and reflection identifies the exact image as

$$b = \{y \in \mathfrak{G}_\theta : \mathfrak{G}_\theta \models \exists x \in a F(x, y, \bar{p})\}.$$

This set is definable over $\mathfrak{G}_\theta$, with parameters from the same level. By the successor operation of the hierarchy, $b \in \text{Def}(\mathfrak{G}_\theta) \subseteq \mathfrak{G}_{\theta+1}$. This establishes Replacement in $\mathfrak{G}$. If the existence of witnesses is retained and uniqueness is omitted, the same construction provides a set of witnesses for Collection.

Arity of names and internal selection. In the well-order of Lemma 36.20, the enumeration of the base assigns to each element its least natural-number code. At each successor stage, formula codes are ordered first; once a code is fixed, the arity of its parameter tuple is fixed. Lexicographic order is then applied to a *finite product of fixed arity* of the previously constructed well-orders. The least name represents each new value, and limit levels retain the order of first appearance. This specifies the ordering of tuples without combining different arities into a single lexicographic order. Selecting the least element of every nonempty set, together with Replacement already established, forms internally the choice functions used subsequently. This construction of names precedes the cardinal injections and receives no bijection of the continuum as data.

36.5 Ordered readout of block cylinders

For the representative $[b]_3 \in \{0, 1, 2\}$, fix

$$\nu(\mathbf{b}) = \sum_{i=1}^6 [b_i]_3 3^{6-i} \in \{0, \dots, 728\}.$$

In the HMT ordered completion, a prefix $s = (m; \mathbf{b}_0, \dots, \mathbf{b}_{N-1})$, $m \in \mathbb{Z}$, determines

$$a_N(s) = m + \sum_{j=0}^{N-1} \nu(\mathbf{b}_j) 729^{-j-1}, \quad I_N(s) = [a_N(s), a_N(s) + 729^{-N}]_{\text{HMT}}.$$

The carry preserves the joining of the two representatives of a boundary. If $C_N(s) = \overline{I_N(s)}$, the preceding relations give

$$I_N(s) = \bigsqcup_{\mathbf{b} \in \mathbb{F}_3^6} I_{N+1}(s; \mathbf{b}), \quad \text{diam } C_N(s) = 729^{-N} \longrightarrow 0.$$

Proposition 36.21 (Coverage of the Archimedean readout). *The map*

$$P_{\text{ar}} : \mathbb{Z} \times \Omega_{\Gamma}^{\text{cal}} \longrightarrow \mathbb{R}_{\text{HMT}}, \quad \{P_{\text{ar}}(m, \xi)\} = \bigcap_N C_N(m; \beta_N(\rho_{\infty, N} \xi))$$

is well defined, is surjective and determines

$$(\mathbb{Z} \times \Omega_{\Gamma}^{\text{cal}}) / \sim_{P_{\text{ar}}} \cong \mathbb{R}_{\text{HMT}}.$$

Proof. The left endpoints form a Cauchy sequence in the HMT completion, since the intervals are nested and their diameters tend to zero. Their limit belongs to all the closed intervals and is unique. For any element of the completion, first choose the integer interval containing it and then the unique half-open subinterval of each partition containing it. The preceding proposition realizes each of those choices through an effective TPK prolongation. The compatible prefixes define ξ , whose readout is the chosen element. At boundaries, the carry identifies equivalent representations after their construction. By definition and surjectivity, the quotient by equality of readout is in bijection with the completion. $\square$

The construction is also carried out in the internal genealogical universe $\mathfrak{G} = \mathfrak{G}_{\text{HMT}}(h, m_{\infty})$. Here h encodes the operator rules and m_{∞} the compatible history; the universe is obtained through initial transitive closure, definability at each successor and union at limit stages:

$$G_0 = \text{TC}\{\omega, u_{h, m}, h, m_{\infty}\}, \quad G_{\beta+1} = \text{Def}_{\Sigma_{\text{HMT}}}(G_{\beta}), \quad G_{\lambda} = \bigcup_{\beta < \lambda} G_{\beta}.$$

The relations of Σ_{HMT} are those of the operations already constructed and their codes. The internal domain is therefore retained in all coverage claims. Denote its completion by $\mathbb{R}_{\text{HMT}}^{\mathfrak{G}}$ and the ordered realization within the same universe by $\mathbb{R}^{\mathfrak{G}}$.

Proposition 36.22 (Internal ordered chart). *There exists a unique isomorphism of complete Archimedean ordered fields*

$$\iota : \mathbb{R}_{\text{HMT}}^{\mathfrak{G}} \xrightarrow{\cong} \mathbb{R}^{\mathfrak{G}}$$

that fixes the unit. It preserves rational endpoints, positive square roots and limits of convergent sequences.

Proof. The unit fixes the integers and rationals. Each element of the HMT completion is associated with the cut of rationals less than it; completeness of the ordered realization provides the unique element with that cut. Rational density proves injectivity and strict order preservation. The same procedure in the reverse direction proves surjectivity. Rational approximations from above and below preserve addition and multiplication and thereby the field structure. The positive element whose square is $a > 0$ is transported to the positive element whose square is $\iota(a)$. If a sequence converges, for every positive rational tolerance its final terms lie in the rational neighborhood of the limit; order preservation transports that condition. All these operations are performed within $\mathfrak{G}$. Uniqueness follows from preservation of rational cuts. $\square$

36.6 Descent along fibers and preservation of the selector

Proposition 36.23 (Descent criterion for an operation). *Let $q : \Omega \rightarrow J$ and $\star : D \subseteq \Omega^2 \rightarrow \Omega$, with D saturated by the fibers of $q \times q$. There exists a unique operation $\bar{\star}$ on $(q \times q)(D)$ such that*

$$q(x \star y) = q(x) \bar{\star} q(y)$$

if and only if

$$q(x) = q(x'), \quad q(y) = q(y') \implies q(x \star y) = q(x' \star y')$$

for pairs in the domain.

Proof. If $\bar{\star}$ exists, both results are the evaluation of the same pair of arguments. Conversely, choose antecedents of the visible arguments and define the result as the readout of their operation. Saturation fixes the domain; the condition on fibers makes the result independent of the chosen antecedents. Surjectivity proves uniqueness. $\square$

In an application through P_{ar} , this criterion identifies the precise condition for transporting an operation from complete states to their readout. In the chart ι , the field structure and limits already satisfy that condition. Next, the analytic profile is constructed from the calendar and this compatibility is used to transport its iteration and roots.

36.7 Dodecaphase readout and quadratic normalization

The oriented criticality section on the twelve calendar windows is

$$\mathcal{P}_{\text{crit}}(m) = \varphi_m = (30m - 10) \frac{\pi_{\text{HMT}}}{180} = \pi_{\text{HMT}} \frac{3m - 1}{18}, \quad m = 1, \dots, 12.$$

This section fixes the origin and orientation of its angular representatives. The uniform reader of the twelve windows is

$$\eta_0(F) = \frac{1}{12} \sum_{m=1}^{12} F(\varphi_m).$$

The function η_0 , which assigns weights to these windows, is distinguished from the ordered chart ι . Uniform normalization and the angular section are part of the operator data of this profile.

Setting $a_m = (3m - 1)/18$, one obtains

$$D_0 = \frac{1}{12} \sum_{m=1}^{12} a_m^2 = \frac{899}{648}, \quad s_0 = \frac{36}{\pi_{\text{HMT}} \sqrt{899}}, \quad k_m = s_0 \varphi_m = \frac{2(3m - 1)}{\sqrt{899}}.$$

Indeed, $\sum_{m=1}^{12} (3m - 1)^2 = 5394 = 6 \cdot 899$, so that $s_0^2 \pi_{\text{HMT}}^2 D_0 = 2$. The profile is

$$u_0(x) = \frac{1}{12} \sum_{m=1}^{12} [1 - \cos(k_m x)], \quad f_\lambda(x) = 1 - \lambda u_0(x). \quad (36.38)$$

The constant π_{HMT} enters as an angular readout of the HMT kernel and cancels algebraically in the normalization of the profile. The scale λ is the family parameter, whose critical value will be selected through the dynamics.

Proposition 36.24 (Normalization and closed form of the profile). *The profile u_0 is even and analytic, satisfies $u_0(0) = u'_0(0) = 0$ and $u''_0(0) = 2$, and, for $y = x/\sqrt{899}$, admits the expression*

$$u_0(x) = 1 - \frac{\cos(37y) \sin(36y)}{12 \sin(3y)} = 1 - \frac{\sin(73y) - \sin(y)}{24 \sin(3y)}.$$

The apparent singularities are interpreted by continuity. Its initial expansion is

$$u_0(x) = x^2 - \frac{715769}{2424603} x^4 + O(x^6).$$

Proof. The finite cosine sum preserves evenness and analyticity. Its second derivative at zero is $12^{-1} \sum_m k_m^2 = 2$. The sum of the trigonometric progression $\sum_{m=1}^{12} \cos((6m-2)y)$ is $\cos(37y) \sin(36y) / \sin(3y)$; the product-sum identity provides the second expression. The cosine series and $\sum_m (3m-1)^4 = 4294614$ give the stated fourth-degree coefficient. The expression as a finite sum fixes the extension at the zeros of the closed form's denominator. $\square$

36.8 Transport of iterates, roots and minima

Within $\mathbb{R}_{\text{HMT}}^{\mathbb{G}}$, the convergent series

$$\cos_{\text{HMT}}(t) = \sum_{j=0}^{\infty} \frac{(-1)^j t^{2j}}{(2j)!}$$

defines the corresponding analytic realization. Let

$$u_{\text{HMT}}(x) = \frac{1}{12} \sum_{m=1}^{12} \left[1 - \cos_{\text{HMT}} \left(\frac{2(3m-1)x}{\sqrt{899}_{\text{HMT}}} \right) \right], \quad F_a(x) = 1 - a u_{\text{HMT}}(x),$$

$$S_n^{\text{HMT}}(a) = F_a^{2^n}(0), \quad S_n(\lambda) = f_{\lambda}^{2^n}(0).$$

Proposition 36.25 (Ordered conjugacy of the family and the root selector). *For the internal arguments of the preceding expressions,*

$$\iota \circ F_a = f_{\iota(a)} \circ \iota, \quad \iota(S_n^{\text{HMT}}(a)) = S_n(\iota(a)).$$

The chart transports exactly the conditions of return, exact period and parameter order. If a set of admissible roots has a minimum, the image of that minimum is the minimum of the image set.

Proof. The chart fixes the rationals, transports the positive square root and preserves limits of the cosine partial sums. Hence, $\iota(u_{\text{HMT}}(x)) = u_0(\iota(x))$. Compatibility with addition and multiplication proves the first equality; induction on the number of iterations proves the second. A root is characterized by equality to zero, which is preserved in both directions. The exact period 2^n adds the inequalities $F_a^{2^j}(0) \neq 0$, $0 \leq j < n$, also preserved by the injectivity of ι . If the parameter is required to exceed a preceding center, that inequality is transported by order preservation. Finally, $a \leq b$ for every b in the set is equivalent to $\iota(a) \leq \iota(b)$ for every element of its image. Surjectivity onto the sets thus defined completes the assertion about the minimum. $\square$

The domain of this transport is the indicated internal continuum. The existence of each minimum and its identification with a specific dynamical branch are results of the criticality construction that follows; the chart preserves those properties once they have been established.

36.9 Finite spectral representation of the uniform reader

The squared frequencies of the profile define the positive measure

$$s_m = k_m^2 = \frac{4(3m-1)^2}{899}, \quad \nu_0 = \frac{1}{12} \sum_{m=1}^{12} \delta_{s_m}, \quad M_r = \int s^r d\nu_0(s).$$

Its twelve nodes are distinct and positive. For $1 \leq r \leq 12$, the Hankel matrix $H_r = (M_{i+j})_{0 \leq i, j < r}$ is positive definite. Indeed, for p of degree less than r ,

$$\mathbf{p}^T H_r \mathbf{p} = \frac{1}{12} \sum_{m=1}^{12} p(s_m)^2 > 0$$

unless $p = 0$, because a polynomial of degree at most eleven that vanishes at twelve distinct nodes is zero. At degree twelve appears the polynomial $\prod_m (s - s_m)$, with zero norm in this measure. The corresponding spectral representation has dimension twelve.

Let $p_0, \dots, p_{11}$ be the monic orthogonal polynomials, $h_j = \int p_j^2 d\nu_0 > 0$ and $\psi_j = p_j/\sqrt{h_j}$. Since ν_0 is a probability measure, $\psi_0 = 1$. Multiplication by s in $L^2(\nu_0)$ is self-adjoint and has a tridiagonal matrix J_{12} in this basis.

Proposition 36.26 (Exact recurrence and spectral intertwining). *With*

$$\alpha_j = \frac{\int s p_j(s)^2 d\nu_0(s)}{h_j}, \quad \beta_j = \frac{h_j}{h_{j-1}} > 0 \quad (1 \leq j \leq 11),$$

the matrix J_{12} has diagonal α_j and subdiagonal $\sqrt{\beta_j}$. If

$$Q_{mj} = \frac{\psi_j(s_m)}{\sqrt{12}} \quad (1 \leq m \leq 12, 0 \leq j \leq 11), \quad D = \text{diag}(s_1, \dots, s_{12}),$$

then

$$Q^T Q = I, \quad DQ = QJ_{12}, \quad J_{12} = Q^T DQ,$$

and the profile satisfies

$$u_0(x) = e_0^T [I - \cos(x\sqrt{J_{12}})] e_0. \quad (36.39)$$

Proof. Positivity of the Hankel systems uniquely provides the monic orthogonal polynomials. On expanding sp_j in the orthogonal basis, the coefficients of p_i , $i \leq j-2$, vanish because $\langle sp_j, p_i \rangle = \langle p_j, sp_i \rangle = 0$. The coefficient of p_{j+1} is one by monicity, that of p_j is α_j and that of p_{j-1} is h_j/h_{j-1} . At the last degree, the same equality is understood in $L^2(\nu_0)$, where the monic polynomial of degree twelve vanishing at the nodes represents the zero vector. Normalization gives the stated positive subdiagonal.

Orthonormality is equivalent to $Q^T Q = I$. The recurrence equality at each node gives $DQ = QJ_{12}$. Since Q is square, we also have $QQ^T = I$, and obtain the diagonalization of J_{12} . Its positive eigenvalues allow its positive square root to be defined. For every function defined at those twelve nodes,

$$e_0^T \Phi(J_{12}) e_0 = \frac{1}{12} \sum_{m=1}^{12} \Phi(s_m),$$

since $Qe_0 = (1, \dots, 1)^T/\sqrt{12}$. The choice $\Phi(s) = 1 - \cos(x\sqrt{s})$ proves (36.39). $\square$

The representation preserves exactly the uniform functional, its moments and its profile. Its first coefficients are

$$\alpha_0 = M_1 = 2, \quad \beta_1 = M_2 - M_1^2 = \frac{2493348}{808201}.$$

The complete genealogy remains in the TPK state from which the reader originates; J_{12} represents its specific spectral readout. The Gram–Schmidt construction and the Jacobi recurrence are applied to this atomic measure, terminating in dimension twelve.

The operator chain used in the following sections is thus fixed: APP cells and TRIT orientation feed the TPK transition; its prolongations produce cylinders and their ordered readout; the twelve oriented windows and the uniform functional produce u_0 ; the family f_λ determines returns and roots; the internal chart preserves the order of their selectors. The spectral realization describes the same profile through a finite self-adjoint matrix and a cyclic vector.

37 Dodecaphase profile and quadratic return

37.1 Genealogy and object of the proof

The initial object is the critical readout of the oriented dodecaphase wheel already constructed in the corpus. APP evaluates addition and multiplication on the two sheets of the digit lattice, preserving quotient and remainder; TRIT determines regime and orientation; TPK transport selects, transports and updates the state with its carry, boundary and memory records. The nonadic connection prolongs that state, with phase return and memory advance. Its readers act on the same discrete structure of the continuum.

The focal reader of the source is

$$\mathcal{P}_{\text{crit}} : U_{12}^{\text{sign}} \longrightarrow \mathbb{R}/2\pi\mathbb{Z}, \quad m \longmapsto \phi_m = \frac{(3m-1)\pi}{18}, \quad 1 \leq m \leq 12.$$

To evaluate the angular moment, the oriented representative of that chart is retained: replacing it with another representative modulo 2π would change the moment. The uniform trace fixes $w_m = 1/12$. The focal chain is

$$\begin{aligned} \text{APP-TRIT-TPK} &\longrightarrow \text{discrete structure of the continuum} \\ &\longrightarrow U_{12}^{\text{sign}} \xrightarrow{\mathcal{P}_{\text{crit}}} (\phi_m, w_m) \\ &\longrightarrow u_0 \longrightarrow f_\lambda \longrightarrow \text{returns and renormalization.} \end{aligned}$$

The abbreviated notation retains the complete state and its prolongations as antecedent; the scalar family is a readout of that antecedent. The already fixed reader, with its order, orientation and uniform sector, is taken as an explicit starting structure. The conclusions concern that particular reader; uniqueness of that reader among all possible TPK readouts lies outside these proofs.

The coordinate π_{HMT} used by the chart belongs to the internal outputs of the corpus. Its factor cancels exactly when normalizing the second moment. The Feigenbaum values enter exclusively in subsequent recognition, never in the definition of the family or the certification of roots.

37.2 Exact profile and quadratic criticality

Let $s_m = 3m - 1$. The integer sum

$$\sum_{m=1}^{12} s_m^2 = 5394 = 6 \cdot 899$$

determines

$$\begin{aligned} \kappa_0 &= \frac{36}{\pi\sqrt{899}}, \quad k_m = \kappa_0 \phi_m = \frac{2s_m}{\sqrt{899}}, \\ u_0(x) &= \frac{1}{12} \sum_{m=1}^{12} (1 - \cos(k_m x)), \quad f_{\lambda,0}(x) = 1 - \lambda u_0(x). \end{aligned}$$

By the second-moment identity,

$$u_0(0) = u'_0(0) = 0, \quad u''_0(0) = 2, \quad f''_{\lambda,0}(0) = -2\lambda.$$

The analytic representation preserves rational coefficients:

$$u_0(z) = \sum_{j \geq 1} (-1)^{j+1} \frac{4^j \sum_m s_m^{2j}}{12 \cdot 899^j (2j)!} z^{2j} = z^2 - \frac{715769}{2424603} z^4 + \frac{1353832978}{32695771455} z^6 + \dots$$

Thus the finite record of twelve sectors produces an entire profile, with quadratic criticality and scale fixed before iteration.

37.3 Structural robustness with uniform bounds

The deformation already present in the originating code is

$$w_m(\eta) = \frac{1 + \eta p_m}{12}, \quad p_m = 12 \cos(3\phi_m) - \cos(4\phi_m).$$

We define

$$M_{2,\eta} = \sum_m w_m(\eta) \phi_m^2, \quad \kappa_\eta = \sqrt{2/M_{2,\eta}}, \quad k_{m,\eta} = \kappa_\eta \phi_m,$$

$$u_\eta(x) = \sum_m w_m(\eta) (1 - \cos(k_{m,\eta} x)), \quad f_{\lambda,\eta} = 1 - \lambda u_\eta.$$

Structural robustness theorem. For $|\eta| \leq 1/40$, the weights are positive and sum to one. The profile is even and entire, satisfies $u_\eta''(0) = 2$ and is strictly increasing on $(0, 1]$. For $0 < \lambda \leq 2/u_\eta(1)$, the map sends $[-1, 1]$ into itself, has a unique critical point in that interval and satisfies

$$Sf_{\lambda,\eta}(x) \leq -\frac{640}{47647} < 0, \quad 0 < |x| \leq 1.$$

Proof. The sums of cosines $3\phi_m$ and $4\phi_m$ vanish through cycles of lengths four and three. Since $|p_m| \leq 13$,

$$\frac{9}{160} \leq w_m \leq \frac{53}{480}, \quad \frac{27}{40} M_{2,0} \leq M_{2,\eta} \leq \frac{53}{40} M_{2,0}.$$

It follows that

$$k_{\max,\eta}^2 \leq \frac{196000}{24273} < 9 < \pi^2, \quad k_{\min,\eta}^2 \geq \frac{640}{47647}.$$

For $0 < x \leq 1$, all sines $\sin(k_{m,\eta} x)$ are positive, whence

$$u_\eta'(x) = \sum_m w_m k_m \sin(k_m x) > 0, \quad u_\eta'''(x) = -\sum_m w_m k_m^3 \sin(k_m x) < 0.$$

The quotient u_η'''/u_η' is the negative of a positive mean of the k_m^2 . Invariance of the Schwarzian under affine changes of the value gives

$$Sf_{\lambda,\eta} = \frac{u_\eta'''}{u_\eta'} - \frac{3}{2} \left(\frac{u_\eta''}{u_\eta'} \right)^2 \leq -k_{\min,\eta}^2.$$

Evenness covers $x < 0$, and the image of the interval is $[1 - \lambda u_\eta(1), 1]$. $\square$

The width $1/40$ is a choice made to obtain uniform bounds, distinct from a fundamental constant or an optimal threshold. The original sector remains $\eta = 0$.

37.4 Global quadratic holomorphic coordinate on a strip

Holomorphic factorization theorem. For $|\eta| \leq 1/40$, in

$$\mathcal{S} = \{z \in \mathbb{C} : |\operatorname{Re} z| < \pi/3\}$$

there exists a holomorphic, odd and univalent function b_η , with $b_\eta'(0) = 1$, such that

$$\boxed{u_\eta(z) = b_\eta(z)^2, \quad f_{\lambda,\eta}(z) = 1 - \lambda b_\eta(z)^2.}$$

Proof. For $z = x + iy$ in the right half-strip,

$$\operatorname{Re} u_\eta'(z) = \sum_m w_m k_m \sin(k_m x) \cosh(k_m y) > 0.$$

The integral of u_η' over the segment between two points, divided by the difference of the endpoints, has positive real part. Convexity of the half-strip proves injectivity there. Evenness gives the corresponding result on the left.

Moreover,

$$\operatorname{Im} u_\eta(x + iy) = \sum_m w_m \sin(k_m x) \sinh(k_m y)$$

has the sign of y for $x > 0$. On the positive real axis $u_\eta > 0$; on the imaginary axis,

$$u_\eta(iy) = \sum_m w_m (1 - \cosh(k_m y))$$

is strictly negative away from zero and decreases strictly with $|y|$. These separations, injectivity on each half-strip and evenness prove that the fibers of u_η in $\mathcal{S}$ are exactly $\{z, -z\}$, except for the fiber of the origin.

The function $u_\eta(z)/z^2$, extended with value one at zero, is holomorphic and has no zeros in the simply connected strip. It has a holomorphic logarithm L_η with $L_\eta(0) = 0$, which is even by uniqueness of the normalization. Define

$$b_\eta(z) = z \exp\left(\frac{1}{2}L_\eta(z)\right).$$

Then $b_\eta^2 = u_\eta$, b_η is odd and $b'_\eta(0) = 1$. The equality $b_\eta(z) = b_\eta(w)$ forces $w = z$ or $w = -z$. The second case gives equality only at the origin. This proves its univalence. $\square$

The twelve phases thus determine an explicit conformal deformation of the quadratic fold. The complete dynamical conjugacy preserves that deformation:

$$b_\eta \circ f_{\lambda, \eta} \circ b_\eta^{-1}(w) = b_\eta(1 - \lambda w^2),$$

on their composition domains. The factorization proved and a conjugacy to the entire quadratic family are distinct assertions.

37.5 Return, rescaling and memory

Let $a = -f(1) > 0$, $H_a(x) = -ax$, $J = H_a([-1, 1])$. When $J \subset [-1, 1]$ is admissible for the return, $f^2(J) \subset J$ and the compositions are defined,

$$\mathcal{R}f = H_a^{-1}f^2H_a, \quad \mathcal{R}f(x) = -a^{-1}f(f(-ax)).$$

If in addition $f(J) \subset (0, 1]$, a unique central quadratic maximum is preserved:

$$(\mathcal{R}f)(0) = 1, \quad (\mathcal{R}f)''(0) = -af'(1)f''(0) < 0.$$

Composition of Schwarzians gives, at regular points,

$$S(\mathcal{R}f)(x) = a^2 [(Sf)(f(-ax))f'(-ax)^2 + (Sf)(-ax)] < 0.$$

Domain inclusions are part of the hypotheses and are checked at every scale.

Let $v_\eta = (u_\eta|_{[0,1]})^{-1}$. The inverse branches are

$$\beta_\sigma(y) = \sigma v_\eta((1 - y)/\lambda), \quad \sigma \in \{-1, +1\},$$

with the critical record $\beta_0(1) = 0$. The step

$$(x, w) \mapsto (f(x), w \parallel \sigma(x))$$

is inverted on its image by reading the last sheet and applying β_σ . The word preserves branch decisions; the previous value is reconstructed through the map equation.

The return groups two transitions. From the endpoint x' and its sheets (σ_0, σ_1) one recovers

$$x = H_a^{-1}\beta_{\sigma_0}(\beta_{\sigma_1}(H_a x')).$$

The intermediate points are also reconstructed. The APP-TRIT-TPK records of sheet, carry, route, boundary and memory are concatenated while preserving their identity. The sign of the analytic branch is an additional record: it does not replace those data with a binary label.

If C_N groups histories of length $2N$ into pairs, the restrictions satisfy

$$\pi_{N,M}^{\text{ret}} C_N = C_M \pi_{2N,2M}^f, \quad M \leq N.$$

Both sides remove the same terminal pairs and preserve the same inverse branches. Compatible transport of finite histories is thus proved.

37.6 Accumulated scales and derivative of the operator

Let $c_n = f^{2^n}(0)$, $c_0 = 1$ and $H_n(x) = c_n x$. As long as the compositions are admissible and $c_n \neq 0$,

$$\boxed{\mathcal{R}^n f = H_n^{-1} f^{2^n} H_n}, \quad (\mathcal{R}^n f)(1) = \frac{c_{n+1}}{c_n} = -a_n.$$

Induction follows from $H_n H_{a_n} = H_{n+1}$. The convention $a_n > 0$ requires sign alternation of c_n . At a superstable root of period 2^N , $c_N = 0$: that normalization becomes singular and the preceding levels or a controlled limit must be used.

For $z = -ax$, $w = f(z)$ and a perturbation h , the complete derivative is

$$\boxed{D\mathcal{R}_f[h](x) = -\frac{h(w) + f'(w)h(z)}{a} + \frac{h(1)}{a} [\mathcal{R}f(x) - xf'(w)f'(z)]}.$$

It is differentiated using $\dot{a} = -h(1)$, $\dot{z} = xh(1)$, $\dot{w} = h(z) + xf'(z)h(1)$ and the variation of the denominator. If $h(0) = 0$, this gives $D\mathcal{R}_f[h](0) = 0$. The initial HMT parameter direction is $h = -u_\eta$.

This formula includes scale variation; freezing a would give a different linearized operator.

37.7 Certified roots and itineraries through period 256

Let $F_n(\lambda) = f_{\lambda,0}^{2^n}(0)$. Compute

$$x_0 = p_0 = 0, \quad x_{j+1} = 1 - \lambda u_0(x_j), \quad p_{j+1} = -u_0(x_j) - \lambda u'_0(x_j)p_j.$$

Then $F_n = x_{2^n}$ and $F'_n = p_{2^n}$.

The attached program uses rational intervals with outward integer rounding, square roots enclosed by integer square roots, and sine/cosine with an explicit Taylor remainder. The exact bound $|u''_0| \leq 2$ allows centered evaluation with a quadratic remainder.

The historical approximations only propose boxes of radius 10^{-42} . Opposite signs at their endpoints, separation of F'_n from zero throughout the box, and exclusion of returns with periods that properly divide 2^n are certified. The intermediate value theorem and monotonicity prove local existence and uniqueness; exclusion of divisors proves the least period.

Level	Least period	Box center, rounded
1	2	1.345913990471832792210949
2	4	1.763379787846910127290794
3	8	1.850413796950136464530567
4	16	1.868979756606798125150574
5	32	1.872955034435659740004205
6	64	1.873806367467030119412262
7	128	1.873988695221365362307169
8	256	1.874027744154450672897008

The first root has the exact expression $\lambda_1 = 1/u_0(1)$. The complete boxes and their checks are in the JSON certificate.

The 502 preclosure states of these eight levels, all separated from zero, and their complete sign words have been enclosed. In particular,

$$\text{sign } f_{\lambda_n,0}^{2^j}(0) = (-1)^j, \quad 0 \leq j < n.$$

The finite orientations of the rescalings preceding closure are established. Inclusions of restrictive intervals at every scale constitute an additional check.

For $\delta_{F,n} = (\lambda_{n-1} - \lambda_{n-2})/(\lambda_n - \lambda_{n-1})$, an abbreviated outer decimal interval is

$$\begin{aligned} 4.66921218915020415455609621588966392804 &< \delta_{F,8}, \\ \delta_{F,8} &< 4.66921218915020415455609621588966392863. \end{aligned}$$

The width of the complete interval is less than $5.808 \cdot 10^{-37}$. It measures the numerical error of the finite ratio $\delta_{F,8}$, distinct from the error in approximating the universal limit.

37.8 Analytic persistence of the deformed roots

For each certified level, $F'_n(\lambda_n) \neq 0$. The implicit function theorem produces a local analytic branch $\lambda_n(\eta)$ with

$$f_{\lambda_n(\eta),\eta}^{2^n}(0) = 0.$$

By continuity, the proper returns remain separated from zero for η sufficiently small; the least period and itinerary are preserved. The existence of those local neighborhoods is proved. Their common width is not yet identified with the entire structural strip $1/40$.

With $M = M_{2,\eta}$, the profile response is

$$v_\eta(x) = \partial_\eta u_\eta(x) = \sum_m \frac{p_m}{12} (1 - \cos(k_{m,\eta}x)) - \frac{M'}{2M} x u'_\eta(x).$$

The second term preserves normalization of the second moment. If $q_j = \partial_\eta x_j$ at fixed λ ,

$$q_0 = 0, \quad q_{j+1} = -\lambda u_\eta(x_j) - \lambda u'_\eta(x_j) q_j, \quad \boxed{\lambda'_n(\eta) = -q_{2^n}/p_{2^n}}.$$

The weights–scale–response–superstable-parameter chain is thus explicit. This finite nondegeneracy is distinct from asymptotic transversality to the stable leaf of the universal fixed point.

37.9 Analytic tails and error propagation

For $\eta = 0$, let

$$\mu_{2j} = \frac{1}{12} \sum_m k_m^{2j}, \quad p_N(z) = \sum_{j=1}^N (-1)^{j+1} \frac{\mu_{2j}}{(2j)!} z^{2j}, \quad \Omega = \frac{70}{\sqrt{899}}.$$

If $q_N = \Omega^2 r^2 / [(2N+3)(2N+4)] < 1$,

$$\|u_0 - p_N\|_r \leq \frac{\mu_{2N+2} r^{2N+2}}{(2N+2)!(1 - q_N)}.$$

The inequality $\mu_{2j+2} \leq \Omega^2 \mu_{2j}$ bounds the tail by a geometric series. For $0 \leq s \leq 2N+2$,

$$\|(u_0 - p_N)^{(s)}\|_r \leq \frac{\mu_{2N+2} r^{2N+2-s}}{(2N+2-s)!} \left(1 - \frac{\Omega^2 r^2}{(2N+3-s)(2N+4-s)} \right)^{-1},$$

when the denominator is positive.

To transport errors through a return, its domains are also controlled: if $\|f - g\| \leq \varepsilon$, $a_f, a_g \geq a_* > 0$, the compositions lie in a common convex domain and $\|f'\| \leq L$, $\|g\| \leq M$ there, then

$$\|\mathcal{R}f - \mathcal{R}g\|_r \leq \varepsilon \left[\frac{1 + L + L^2 r}{a_*} + \frac{M}{a_*^2} \right].$$

The proof adds the outer perturbation, the inner perturbation, the change of argument due to $|a_f - a_g| \leq \varepsilon$ and the variation of the denominator. These bounds prevent reuse of the initial cosine sum as if all renormalized profiles retained that same form.

38 Nonadic depth and persistence of the itinerary

38.1 Outer growth and inner resolution

The refinement law of article XI, with integer base $B \geq 2$, is

$$L_c(x, y) = (Bx - c, By + c), \quad 0 \leq c < B.$$

For a word $c_1 \dots c_m$, its record is

$$C_m = \sum_{j=1}^m c_j B^{m-j}, \quad (x_m, y_m) = (B^m x_0 - C_m, B^m y_0 + C_m).$$

The outer sum is $B^m M_0$, with $M_0 = x_0 + y_0$. The normalized reading of the record belongs to the cylinder

$$I_m(C_m) = \left[\frac{C_m}{B^m}, \frac{C_m + 1}{B^m} \right], \quad \text{diam } I_m = B^{-m}.$$

The integer pair and the scale recover the quotients, remainders and leading zeros of the word. End-points with two expansions retain their boundary mark. Support growth and readout refinement are thus coordinated without identifying distinct histories solely by their limit image.

The composition is exact:

$$L_D^{[b]} \circ L_C^{[a]} = L_{B^b C + D}^{[a+b]}.$$

By associativity, grouping a history of eighteen events into nine blocks of two or two blocks of nine produces the same record. At return level 2^N , the common horizon is $9 \cdot 2^N$. The proof is an equality of composition on the same ordered history; it retains its carries and its recoverable subblocks.

The restriction of states and the prolongation of their readers are expressed through the following limits:

$$X_\infty = \varprojlim X_m, \quad r^* a = a \circ r, \quad r^* e_x = \sum_{r(y)=x} e_y, \quad \mathfrak{H}_{\text{cyl}} = \varinjlim \mathfrak{H}_m.$$

States restrict; readers extend. Pullback preserves products, the unit and idempotents. An exactly compatible family induces a reader on the direct limit. Approximants that are compatible only up to an error additionally require a convergence estimate in their completion. The next section provides that estimate for the critical return.

38.1.1 Capacity and vacancies of the record

The capacity reader of XI compares blocks of 729 and 1000 possibilities. With $\rho = \log_{1000} 729 = \log_{10} 9$, it retains

$$\mathcal{C}_k = \lfloor (k+1)\rho \rfloor, \quad \mathcal{C}_k - \mathcal{C}_{k-1} = 1 - z_k, \\ \mathcal{C}_b - \mathcal{C}_a + \sum_{k=a+1}^b z_k = b - a, \quad T_{\text{cap}}(a) = \lceil a/\rho \rceil - 1 \quad (a \geq 1).$$

Each vacancy records a transition during which capacity is maintained; chronological length is conserved in the balance. Capacity phase and chronological phase are distinct readers.

To store a prefix of m base-nine digits, a sufficient decimal capacity is

$$a(m) = \left\lceil \frac{m\rho}{3} \right\rceil.$$

The quotient and remainder allow the integer C_m to be converted between bases, keeping the depth m separate. The following hold

$$9^m \leq 1000^{a(m)} \leq 729^{T_{\text{cap}}(a(m))+1}.$$

These inequalities are also decided by integer powers, without rounding logarithms. The readout budget of the next section is thus translated into record capacity while preserving units. This count sizes the storage; the survival rules additionally determine which native histories are admissible.

38.2 Explicit depth budget for each return

Theorem. With $|\eta| \leq 1/40$ fixed, normalize

$$t = \lambda u_\eta(1)/2 \in [0, 1], \quad z = (x + 1)/2,$$

$$F_{t,\eta}(z) = 1 - t \frac{u_\eta(2z - 1)}{u_\eta(1)}, \quad z \in [0, 1].$$

For every integer $q \geq 1$ and two parameters t, s ,

$$|F_{t,\eta}^q(1/2) - F_{s,\eta}^q(1/2)| \leq S_q |t - s|, \quad S_q = \sum_{j=0}^{q-1} (6\sqrt{2})^j < 9^q.$$

Consequently, a base-nine parameter cylinder of depth $m = 2^N + r$ encloses the readout of return $q = 2^N$ in an interval of diameter less than 9^{-r} .

Proof. The function $(1 - \cos v)/v^2$ decreases for $0 < v < \pi$: the sign of its derivative is that of $v \sin v - 2(1 - \cos v)$, which is negative because $\tan(v/2) > v/2$. By the second moment,

$$u_\eta(1) \geq \frac{1 - \cos 3}{9} \sum_j w_j k_j^2 = \frac{2(1 - \cos 3)}{9} > \frac{1}{3}.$$

The last inequality is verified, for example, by bounding $\cos 3$ by its alternating Taylor sum through degree eight, which is less than $-1/2$. Moreover, Cauchy–Schwarz gives $|u'_\eta(x)| \leq \sqrt{2}$ on the real line. It follows that

$$\text{Lip}_z F_{t,\eta} \leq 6\sqrt{2} < 9, \quad \text{Lip}_t F_{t,\eta} \leq 1.$$

If d_j is the difference between the two orbits, then $d_0 = 0$ and $d_{j+1} \leq 6\sqrt{2} d_j + |t - s|$. Induction proves the geometric sum. For $|t - s| \leq 9^{-m}$, the diameter is less than 9^{q-m} . Substituting $q = 2^N$ and $m = q + r$ completes the proof. $\square$

The bound controls parameter uncertainty with η fixed. An implementation separately adds and propagates cosine-evaluation and rounding errors; the preceding rational certificate provides these operations. The bound is sufficient and conservative, with a cost that grows with the horizon.

Limit reader. Realize t from $(x_0, y_0) = (1, 0)$, $B = 9$, so that

$$(x_m, y_m) = (9^m - C_m, C_m), \quad t_m = C_m/9^m.$$

The readers $E_{q,m} = F_{t_m,\eta}^q(1/2)$, brought to a common level, satisfy

$$\|E_{q,m+a} - r^* E_{q,m}\|_\infty \leq S_q 9^{-m}.$$

They converge uniformly to a continuous reader E_q in the completion of the cylinder algebra. The set-theoretic version

$$J_{q,m}(C) = \{F_{t,\eta}^q(1/2) : t \in I_m(C)\}$$

is exactly nested and its diameters tend to zero. This result gives the passage to the limit of the **fixed-horizon readout**, with depth control at any requested horizon.

38.3 Doubling word and nonadic reader of chronology

The eight itineraries certified in the preceding work obey

$$W_1 = +, \quad W_{n+1} = W_n (-1)^n W_n.$$

Its unique prefix continuation is

$$\tau_j = (-1)^{v_2(j)}, \quad j \geq 1; \quad \tau_{2j} = -\tau_j, \quad \tau_{2j+1} = +.$$

The proof uses $v_2(2^n + r) = v_2(r)$ for $0 < r < 2^n$. It is a recurrence at every depth, and agreement with the eight computed returns constitutes its finite check in the family.

In the parity-lexicographic order of a map with a critical maximum, the orientation factor preceding position k is $\prod_{j < k} (-\tau_j)$. The word τ is strictly greater than any of its positive shifts. Indeed, the first disagreement with an odd shift occurs at $k = 1$ or 2 ; an even shift halves the comparison and doubles k . The first disagreement is therefore at $k = 2^r$. There

$$\prod_{j < 2^r} (-\tau_j) = (-1)^r = \tau_{2^r},$$

and the oriented difference with the opposite sign is positive. This argument also proves aperiodicity.

The chronological record retains

$$K = \rho_9(K) + 9q_9(K).$$

Its dyadic reader is

$$\ell_n(B_K) = [\rho_9(K) + 9q_9(K)]_{2^n} = [K]_{2^n}.$$

On the chart of histories in which $\Gamma_9 B_K = B_{K+9}$,

$$\ell_n(\Gamma_9 B_K) = \ell_n(B_K) + 9 \pmod{2^n}, \quad \pi_{n+1,n} \ell_{n+1} = \ell_n.$$

Since nine is odd, advancement by nine visits the 2^n positions. Its conjugacy with unit advancement is $j \mapsto 9j$. For $0 < j < 2^n$,

$$v_2(9j \bmod 2^n) = v_2(j),$$

so that nonadic resampling also preserves the oriented word in each certified cycle.

This reader uses remainder **and** quotient. The phases of $K = 1$ and $K = 10$ agree modulo nine, while their dyadic signs are opposite. Memory distinguishes the two histories. The arithmetic property of resampling holds for every odd multiplier; the HMT calendar fixes the choice of nine here.

38.4 Compatible bilateral memory at infinite depth

Let $\mathcal{H}_n = \ell^2(\mathbb{Z}/2^n\mathbb{Z})$,

$$C_n e_j = e_{j+1}, \quad J_n e_j = (e_j + e_{j+2^n})/\sqrt{2}.$$

A check on the basis gives $J_n^* J_n = I$ and $C_{n+1} J_n = J_n C_n$, including the boundary terms.

Article XI provides the projectors

$$P_0 = \frac{1}{9} \begin{pmatrix} 8 & -\sqrt{8} \\ -\sqrt{8} & 1 \end{pmatrix}, \quad P_1 = \frac{1}{9} \begin{pmatrix} 1 & \sqrt{8} \\ \sqrt{8} & 8 \end{pmatrix}.$$

They are orthogonal and complementary. Therefore

$$\mathbb{U}_n = P_0 \otimes I + P_1 \otimes C_n = \begin{pmatrix} T_n & D_n \\ D_n & R_n \end{pmatrix}$$

is unitary, with

$$T_n = (8I + C_n)/9, \quad D_n = \sqrt{8}(C_n - I)/9, \quad R_n = (I + 8C_n)/9.$$

The naturality already proved implies

$$\mathbb{U}_{n+1}(I_2 \otimes J_n) = (I_2 \otimes J_n)\mathbb{U}_n, \quad \mathbb{U}_n^a = P_0 \otimes I + P_1 \otimes C_n^a \quad (a \in \mathbb{Z}).$$

The Hilbert limit is identified with $L^2(\mathbb{Z}_2, \mu_{\text{Haar}})$, taking $e_j^{(n)}$ to $2^{n/2} \mathbf{1}_{j+2^n\mathbb{Z}_2}$. The identities extend by density and boundedness. For every vector v and every integer time a ,

$$\|T_{\infty,a} v\|^2 + \|D_{\infty,a} v\|^2 = \|v\|^2,$$

$$v = T_{\infty,a}^*(T_{\infty,a}v) + D_{\infty,a}^*(D_{\infty,a}v).$$

Here $T_{\infty,a} = (8I + C_{\infty}^a)/9$. Bilateral composition preserves the archive between steps; composing only T_n describes a different operation.

The permutation $S_n e_j = e_{9j \bmod 2^n}$ satisfies

$$S_{n+1}J_n = J_n S_n, \quad S_n C_n S_n^{-1} = C_n^9.$$

Thus one obtains in the limit

$$S_{\infty} C_{\infty} S_{\infty}^{-1} = C_{\infty}^9, \quad (I_2 \otimes S_{\infty}) \mathbb{U}_{\infty} (I_2 \otimes S_{\infty}^{-1}) = \mathbb{U}_{\infty}^9.$$

The temporal representation is faithful: $C_{\infty}^a = C_{\infty}^b$ forces $2^n \mid a - b$ for every n , whence $a = b$. Particular preparations may preserve temporal symmetries. The operator identity preserves this distinction.

The projective point $([K]_{2^n})_n$ and a vector of L^2 have different types: a point history induces a Dirac measure, whereas normalizable amplitudes belong to the Hilbert space. The result constructs an equivariant reader of chronology; the remaining native coordinates of the TPK accompany the complete record.

38.5 Existence and separation of asymptotic obligations

38.5.1 Realization of the infinite word throughout the strip

Existence theorem. For each $|\eta| \leq 1/40$ there exists at least one

$$\lambda_{\infty}(\eta) \in \left[\frac{1}{2u_{\eta}(1)}, \frac{2}{u_{\eta}(1)} \right]$$

such that

$$\text{sign } f_{\lambda_{\infty}(\eta), \eta}^j(0) = (-1)^{v_2(j)} \quad \text{for every } j \geq 1.$$

The conclusion realizes the infinite doubling itinerary within the fixed family. The existence quantifier remains separate from a unique selection and from the law of distances between superstable parameters.

Proof. We fix η . We write K_{λ} for the sign word of the critical orbit; when it first returns to zero, the word ends at that zero. We use the unimodal order with a maximum, whose factor before comparing symbol j is the product of the signs $-K_i$, $i < j$.

At the left endpoint the image of the entire interval is in $[1/2, 1]$; hence $K_- = + + + \dots$. At the right endpoint the orbit is $0 \mapsto 1 \mapsto -1 \mapsto -1$, whence $K_+ = + - - \dots$. Comparing the second and third symbols, respectively, gives

$$K_- < \tau < K_+.$$

If K_{λ_0} is infinite and distinct from τ , the first difference occurs at a finite position. Continuity of the corresponding iterates makes the side of the comparison locally constant. It remains to examine a superstable parameter of first period p , with word $W0$, where $|W| = p - 1$.

For λ near λ_0 , let $g_{\lambda} = f_{\lambda, \eta}^p$ and $a = g_{\lambda}(0)$. We have $g'_{\lambda}(0) = 0$, and a uniform local bound on the second derivative gives

$$g_{\lambda}(x) = a + O(x^2).$$

For $|a|$ sufficiently small and nonzero, g_{λ} maps $[-2|a|, 2|a|]$ into $[a - |a|/2, a + |a|/2]$. The returns retain the sign of a , and the $p - 1$ intermediate positions retain W . The neighboring words are therefore $(W+)^{\infty}$ or $(W-)^{\infty}$; parameters with $a = 0$ retain $W0$.

If τ differs from W before position p , that difference fixes the same side for all three words. If it shares W , put

$$s = \tau_p, \quad e = \prod_{j < p} (-\tau_j), \quad A = Ws.$$

The sign of the comparison of τ with $W0$ and with $(W, -s)^\infty$ is es . To compare it with A^∞ , let q be the first position at which $\tau_{p+q} \neq \tau_q$. It exists by aperiodicity. Up to position $p + q - 1$, both words agree, and the orientation sign factors as $O_p O_{q-1}$, with $O_p = -es$. The strict maximality already proved gives

$$O_{q-1}(\tau_q - \tau_{p+q}) > 0.$$

Consequently, the comparison of τ with A^∞ has sign $-O_p = es$. The three local possibilities lie on the same side of τ .

If no parameter realized τ , the sets $\{\lambda : K_\lambda < \tau\}$ and $\{\lambda : K_\lambda > \tau\}$ would be open, disjoint, nonempty and would cover the parameter interval. Connectedness of the interval excludes that partition. The stated parameter therefore exists. $\square$

This itinerary-continuity argument applies after the HMT generator and its analytic family have been fixed; its hypotheses are verified directly here. The construction preserves the infinite word without introducing a parameter of the quadratic family or the universal value as an input.

38.5.2 Margins preventing loss of the itinerary when taking limits

Write $c_p = f_{\lambda, \eta}^p(0)$. A sufficient uniform rational bound is

$$|f'|, |f''| \leq 16, \quad B_p := 16^p \frac{16^p - 1}{15} \geq \|(f^p)''\|_{[-1,1]}.$$

Indeed, $1 - \cos v \geq v^2/2 - v^4/24 \geq v^2/8$ for $|v| \leq 3$, so that $u_\eta(1) \geq 1/4$, $\lambda \leq 8$ and both derivatives are bounded by 16, using the moments. The chain rule proves the bound B_p by induction.

The word τ satisfies $\tau_{2p} = -\tau_p$ for every p . For any of its realizations, c_{2p} and c_p have opposite signs. Since $(f^p)'(0) = 0$, Taylor gives

$$|c_{2p} - c_p| \leq \frac{1}{2} B_p |c_p|^2.$$

The left-hand side is greater than $|c_p|$, and therefore

$$|c_p| > 2/B_p > 0.$$

For each fixed horizon there is thus a uniform margin preventing a sequence of realizations from losing its sign upon convergence. In the compact coordinates (η, b) , $b = \lambda u_\eta(1) \in [1/2, 2]$, the set of realizations of τ is closed and compact, and its projection onto the entire η strip is surjective.

The same argument applies to a prefix tree: if all prefixes of τ are realized, a subsequence in the compact set preserves every sign, because sufficiently long prefixes simultaneously include positions p and $2p$. The existence theorem now provides the nonemptiness that this argument requires.

38.5.3 Invariant return intervals at every level

Nested returns theorem. For any of the realizations of the existence theorem in section 38.5.1, let $c_j = f^j(0)$ and

$$J_n = \text{conv}\{c_{2^n}, c_{2^{n+1}}\}, \quad n \geq 1.$$

Then $0 \in \text{int } J_n$, $J_{n+1} \subset J_n$,

$$f^{2^n}(J_n) = J_n,$$

and the 2^n images $J_n, f(J_n), \dots, f^{2^n-1}(J_n)$ are disjoint. The normalized returns are unimodal with a maximum, with critical value one and the same word τ . These intervals are invariant return cores; a normalization requiring periodic endpoints may enlarge the core to the corresponding maximal restrictive interval and must specify that step.

Proof of the first return. Let F be a unimodal map with a maximum, with $F(0) = 1$ and word τ , and $d_j = F^j(0)$. Its first signs are $+, -, +, +, +, -, +$. The interval $I = [d_2, 1]$ is invariant: $F(d_2) = d_3 > 0$,

$F(1) = d_2 < 0$, and the maximum is one. Put $J = [d_2, d_4]$. Since F decreases on positive arguments and

$$F(d_3) = d_4 > 0 > d_6 = F(d_5),$$

we have $d_3 < d_5$, and therefore $F(J) = [d_3, 1]$.

Moreover, $d_3 > d_4$. To check this, F^2 is strictly increasing between the two positive points d_3, d_4 , because their images d_4, d_5 are positive. Its values are $F^2(d_3) = d_5 > 0 > d_6 = F^2(d_4)$, which forces that order. Thus J and $F(J)$ are separated by an open interval, and

$$F^2(J) = F([d_3, 1]) = [d_2, d_4] = J.$$

The map $F^2|_J$ has a unique critical minimum: the image $F(J)$ remains positive. The conjugacy $h(x) = d_2x$ reverses orientation and normalizes the return as a maximum with value one. Its itinerary is

$$\text{sign } \frac{d_{2j}}{d_2} = -\tau_{2j} = \tau_j.$$

The same argument can be repeated at any depth.

Inductive disjointness. Suppose the $p = 2^n$ images of the parent J_n are disjoint, and let $g = f^p$. The first return applied to the normalized map produces $B = J_{n+1}$, $A = g(B)$, with $A \cap B = \emptyset$, $g(B) = A$, $g(A) = B$. For $0 \leq j < p$, an intersection of $f^j(A)$ and $f^j(B)$ would, upon applying f^{p-j} , produce an intersection of B and A . Those two images are therefore disjoint. Images corresponding to distinct j belong to disjoint images of the parent. This proves the $2p$ disjoint images and the return $f^{2p}(B) = B$. The endpoint formula results from composing the normalizations $x \mapsto c_{2^n}x$. $\square$

The compact sets

$$\Lambda_n = \bigcup_{0 \leq j < 2^n} f^j(J_n)$$

are nonempty and nested. Their intersection Λ_∞ has a continuous reader onto the dyadic odometer: each point is assigned the label of the unique component it occupies at each level. Every compatible sequence of labels has a preimage by compactness, and advancement by f adds one to the labels. This map is a surjective semiconjugacy. Promoting it to a pointwise conjugacy requires proving that the component diameters tend to zero; that metric promotion remains separate.

39 Compatibility of readers and recovery of the parameter

39.1 Compatibility of readers, minima and recovery of history

The prolongation operations and ordered readers preserve the information required for root selection. The following identities specify their domains and composition.

39.1.1 Recovered forward rule

The deterministic update starts from the R36 boundary state with character χ and internal remainder. The update is

$$u_K^\chi = \text{Upd}_K^\chi \circ \text{Tra}_K^\chi \circ \text{Sel}_K^\chi, \quad N_{K+1} = 729N_K + d_K^\chi. \quad (39.1)$$

The digit and the update depend on the preceding state. Two forward sections with the same complete R36 state coincide by induction. The subrelation $\text{Ext}^{\chi, \rightarrow}$ is the graph of that update; the total relation Ext admits multiple continuations depending on the data and characters.

Compatible prolongations transport the regional readers and the joint signature over the same prefixes.

These rules transport orientation, remainder, quotient, carry, boundary and memory. Their application to the cascade must also preserve ordered selection of the critical reader's parameter.

39.1.2 Independence of the antecedent constants from the free parameter

Let B be the HMT state providing the wheel and its antecedent constants. Write $\mathcal{C}(B)$ for the set of those readouts and u_B for the critical profile. In the domain of the critical reader, the subsequent construction is

$$(B, \lambda) \mapsto f_{B, \lambda}(x) = 1 - \lambda u_B(x), \quad 0 < \lambda \leq 2/u_B(1). \quad (39.2)$$

For the same B , the reading of the preceding constants factors through projection onto the first component:

$$\tilde{\mathcal{C}}(B, \lambda) = \mathcal{C} \circ \text{pr}_B(B, \lambda) = \mathcal{C}(B). \quad (39.3)$$

Consequently, two distinct values of λ preserve the same antecedent readouts. The equality is a property of this causal construction: the profile is fixed before λ is varied, and the factor π_{HMT} cancels in its quadratic normalization.

Lemma. Suppose a condition A is evaluated solely on those antecedent readouts. For every fixed B ,

$$\{\lambda \in \Lambda : A(\tilde{\mathcal{C}}(B, \lambda))\} = \begin{cases} \Lambda, & A(\mathcal{C}(B)) \text{ holds,} \\ \emptyset, & A(\mathcal{C}(B)) \text{ fails.} \end{cases} \quad (39.4)$$

Proof. Substituting (39.3) makes the predicate independent of λ . Therefore all parameters in the domain satisfy it, or none do. This includes comparison of the antecedent constants with metrological intervals.

Uniqueness of the section B and the subsequent restriction $f_{B, \lambda}^{2^n}(0) = 0$ produce a solution fiber in λ . The first-root rule acts on this fiber through the return and its itinerary. Selection and its prolongation are proved in the following results.

This result maintains the HMT order: metrology compares previously constructed outputs, and root selection is proved within the reader that produces it.

39.1.3 Lemma on ordered transport of the selector

Let P_n be an ordered set of genealogical candidates with minimum p_n^* . Let Z_n be the corpus's original set of candidate roots, with the real order, and let $L_n : P_n \rightarrow Z_n$ be a monotone reader. Its image is said to be **coinitial** when

$$\forall z \in Z_n \quad \exists p \in P_n : L_n(p) \leq z. \quad (39.5)$$

Then

$$L_n(p_n^*) = \min Z_n \iff L_n(P_n) \text{ is coinitial in } Z_n. \quad (39.6)$$

Proof. If the image of the minimum is the global minimum, take $p = p_n^*$ in (39.5). Conversely, given $z \in Z_n$, coinitiality provides p with $L_n(p) \leq z$. Monotonicity gives $L_n(p_n^*) \leq L_n(p) \leq z$. Since $L_n(p_n^*) \in Z_n$, it is the global minimum. Surjectivity of L_n is a sufficient condition, stronger than (39.5).

The ordered isomorphism of the continuum chart satisfies reader surjectivity. The concrete application to the return and its complete set of zeros is proved in the ordered-transport section; classification and isolation then determine its dynamical continuation.

39.1.4 Minimal selection through survival at arbitrary depth

Composition with survival also produces a precise global selector. The uniform family $\eta = 0$, constructed in section 42.1, is retained, and the theorems on realization of the infinite itinerary and preservation of its signs are used. Let

$$I = \left[\frac{1}{2u_0(1)}, \frac{2}{u_0(1)} \right], \quad c_j(\lambda) = f_\lambda^j(0), \quad \tau_j = (-1)^{v_2(j)}.$$

The set

$$\Lambda_\tau = \{\lambda \in I : \text{sign } c_j(\lambda) = \tau_j \text{ for every } j \geq 1\}$$

is compact and nonempty. The derivative control in that source provides

$$B_j = \frac{16^j(16^j - 1)}{15}, \quad |c_j(\lambda)| > \frac{2}{B_j} \quad (\lambda \in \Lambda_\tau). \quad (39.7)$$

Define

$$\varepsilon_j = \frac{1}{B_j}, \quad A_N = \{\lambda \in I : \tau_j c_j(\lambda) \geq \varepsilon_j, 1 \leq j \leq N\}. \quad (39.8)$$

Each A_N is compact by continuity of the iterates and nonempty because it contains Λ_τ . Moreover,

$$A_{N+1} \subseteq A_N, \quad \bigcap_{N \geq 1} A_N = \Lambda_\tau. \quad (39.9)$$

The right-to-left inclusion follows from (39.7). The reverse inclusion follows from the strictly positive margins in (39.8), which fix each sign.

Surviving-minimum theorem. The minima $a_N = \min A_N$ satisfy

$$a_N \uparrow a_* = \min \Lambda_\tau. \quad (39.10)$$

Proof. The inclusion (39.9) makes the sequence of minima increasing and bounded within I ; it therefore has a limit a . For each r , all terms a_N , $N \geq r$, belong to the closed set A_r ; thus $a \in A_r$. By (39.9), $a \in \Lambda_\tau$. For any $\lambda \in \Lambda_\tau$, we have $a_N \leq \lambda$ at every level, and on passing to the limit, $a \leq \lambda$. This proves (39.10).

In the HMT cylindrical reader, the object a_* has compatible prefixes at every depth, with orientation, remainder and memory preserved. Its choice uses the order of the parameter reading and complete survival. Compactness proves the existence and convergence of these minima.

The parameters a_N solve sign constraints with a margin. Their boundaries may satisfy $\tau_j c_j = \varepsilon_j$, whereas the superstable roots satisfy $c_{2^n} = 0$. Therefore (39.10) establishes a global selector for infinite realization and preserves, as a distinct object, the corpus's first-root selector.

39.1.5 Recovery of the parameter from the complete critical history

The second term of the critical history provides an exact inverse reader:

$$c_1(\lambda) = 1, \quad c_2(\lambda) = 1 - \lambda u_0(1), \quad \lambda = \frac{1 - c_2(\lambda)}{u_0(1)}. \quad (39.11)$$

The positivity $u_0(1) > 0$, proved for the profile, makes division valid. Consequently, two identical complete critical histories have the same parameter. The sign word τ is a reduced readout of that history; it preserves its combinatorics and omits the magnitude of c_2 .

Formulas (39.10)–(39.11) unite survival-based selection and recovery of the parameter from the critical history. Equality of its limit with the stable crossing is proved by first exit, and identification of the centers by analytic transport of their hybrid classes.

40 Ordered transport and Jacobi representation

40.1 Complete transport from the continuum and spectral realization of the profile

The ordered transport of the continuum and the spectral representation of the reader are composed before parameter selection.

40.1.1 Coverage of all roots and transport of their minimum

The ordered construction of the continuum developed in the preceding works produces the surjective Archimedean readout from compatible cylinders and their ordered quotient. Within its declared universe $\mathfrak{G}$, it proves the isomorphism of complete ordered fields

$$\iota : \mathbb{R}_{\text{HMT}}^{\mathfrak{G}} \xrightarrow{\cong} \mathbb{R}^{\mathfrak{G}}. \quad (40.1)$$

The name η used in the source work is changed here to ι , to distinguish it from the weighting parameter, fixed at zero. The second projection preserves incidence, boundary and memory over the same antecedent; the evaluation quotient and the complete history retain their respective domains.

Let us construct in the HMT field, before selecting a root,

$$\begin{aligned} \cos_{\text{H}}(t) &= \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k)!}, & u_{\text{H}}(x) &= \frac{1}{12} \sum_{m=1}^{12} \left[1 - \cos_{\text{H}} \left(\frac{2(3m-1)x}{\sqrt[4]{899}} \right) \right], \\ F_a(x) &= 1 - a u_{\text{H}}(x), & S_n^{\text{H}}(a) &= F_a^{2^n}(0). \end{aligned} \quad (40.2)$$

The integers and the second moment are those already derived from the dodecaphase reader. The square root is the positive one in the ordered field. The convergent series analytically realizes that reader; it preserves the profile and the normalizing coefficient of section 42.1.

Return transport theorem. For every parameter in the internal domain of (40.2),

$$\iota(F_a(x)) = f_{\iota(a)}(\iota(x)), \quad \iota(S_n^{\text{H}}(a)) = S_n(\iota(a)). \quad (40.3)$$

Proof. The ordered isomorphism fixes the rationals and preserves the positive square root. It preserves convergent limits: every neighborhood of positive rational radius is transported to another of the same radius. Applied to the partial sums of (40.2), it transports the cosine and u_{H} . The first equality follows from preservation of addition and multiplication; the second follows by induction on the number of iterations.

With a previous center a_{n-1} fixed, define Z_n^{H} by zero return, exact critical period 2^n and $a > a_{n-1}$, within the declared parameter domain. Let Z_n be the set defined by the same conditions for f_{λ} , to the right of $\iota(a_{n-1})$. Then

$$\boxed{\iota[Z_n^{\text{H}}] = Z_n, \quad \iota(\min Z_n^{\text{H}}) = \min Z_n.} \quad (40.4)$$

The second equality is asserted when the minimum exists; the first also preserves the nonexistence of candidates.

Proof. (40.3) preserves the zero return and the earlier returns determining the exact period. Order preserves position relative to the previous center. Surjectivity of (40.1), applied to any root parameter, proves reverse coverage. If a_* is the minimum, every $z \in Z_n$ has the form $\iota(a)$, with $a \in Z_n^{\text{H}}$, whence $\iota(a_*) \leq z$. The converse assertion uses ι^{-1} .

Surjectivity covers every parameter in the chart, including those whose roots have not been isolated by a finite computation. The isomorphism preserves the declared universe of interpretation and the selection order.

40.1.2 Jacobi representation of the complete dodecaphase profile

Define the atomic measure of the reader,

$$s_m = \frac{4(3m-1)^2}{899}, \quad \nu_0 = \frac{1}{12} \sum_{m=1}^{12} \delta_{s_m}, \quad M_r = \int s^r d\nu_0(s). \quad (40.5)$$

The twelve nodes are distinct and positive. For a polynomial $p \neq 0$ of degree less than $r \leq 12$,

$$\sum_{i,j=0}^{r-1} p_i M_{i+j} p_j = \frac{1}{12} \sum_{m=1}^{12} p(s_m)^2 > 0. \quad (40.6)$$

A polynomial of degree less than twelve does not vanish at all twelve nodes. At degree twelve, $\prod_m (s - s_m)$ appears, with zero norm.

Orthogonalization with respect to this measure produces orthonormal polynomials $\psi_0, \dots, \psi_{11}$, with $\psi_0 = 1$. Let

$$Q_{mj} = \psi_j(s_m)/\sqrt{12}, \quad D = \text{diag}(s_m), \quad J_{12} = Q^T D Q.$$

The following hold

$$Q^T Q = I, \quad DQ = QJ_{12}, \quad \boxed{u_0(x) = e_0^T [I - \cos(x\sqrt{J_{12}})] e_0.} \quad (40.7)$$

Proof. Orthogonality follows from (40.6). Multiplication by s produces a three-term recurrence: entries separated by more than one position vanish by orthogonality and degree. Choosing positive leading coefficients makes the subdiagonals positive. By functional calculus, $\cos(x\sqrt{J_{12}}) = Q^T \cos(x\sqrt{D})Q$. The vector Qe_0 is constant, with value $1/\sqrt{12}$; substituting it recovers the sum in section 42.1.

The article on Grassmannians develops this mechanism for marked measures and refinements. Here it is applied to (40.5); rank twelve belongs to this reader. The genealogical marks are retained alongside the reader, in accordance with the marked faithfulness of the source work. J_{12} represents the analytic profile; the complete state additionally preserves routes, orientation and memory.

The verifier for this extension checks, with exact fractions, the twelve degrees of positive norm, vanishing at degree twelve, the intertwiner $VT = DV$ in the monic basis and weighted self-adjointness $HT = T^T H$. It checks moments of degree zero through thirty; the identity for every degree follows from the intertwiner. Its first coefficients are

$$a_0 = 2, \quad \beta_1 = \frac{2493348}{808201}. \quad (40.8)$$

40.1.3 Oriented sensitivity of the return

For an exact return of period $p = 2^n$, let

$$c_j = f_\lambda^j(0), \quad q_j = \partial_\lambda c_j, \quad D_j = \prod_{k=1}^j f'_\lambda(c_k), \quad D_0 = 1.$$

Before the critical return, the factors of D_{p-1} are nonzero. Differentiation and telescoping give

$$q_{j+1} = -u_0(c_j) + f'_\lambda(c_j)q_j, \quad \boxed{\frac{q_p}{D_{p-1}} = -\sum_{j=1}^{p-1} \frac{u_0(c_j)}{D_j}.} \quad (40.9)$$

In the canonical word, $\text{sign } c_j = (-1)^{v_2(j)}$. The sign of $f'_\lambda(c_j)$ is opposite; therefore,

$$\text{sign } D_j = (-1)^{j + \sum_{k=1}^j v_2(k)} = (-1)^{s_2(j)}, \quad (40.10)$$

using $\sum_{k=1}^j v_2(k) = j - s_2(j)$, where $s_2(j)$ counts the binary ones. The normalized sensitivity is

$$\mathcal{T}_n = -\sum_{j=1}^{2^n-1} (-1)^{s_2(j)} t_j, \quad t_j = \frac{u_0(c_j)}{|D_j|} > 0. \quad (40.11)$$

Nodal positivity of (40.6) determines u_0 ; (40.11) additionally retains the orbital products and their orientations.

The additional moment representation would require a positive measure ρ on $[0, 1]$ with $t_j = \int z^{j-1} d\rho$. Under that condition,

$$\mathcal{T}_n = \int_0^1 \frac{1 - \prod_{r=0}^{n-1} (1 - z^{2^r})}{z} d\rho(z) > 0 \quad (40.12)$$

for $\rho \neq 0$, extending the integrand by its value 1 at zero. The identity is obtained by expanding the binary product.

The rational check at the previously certified period-four root obtains

$$t_1 \approx 0.40425224267600139730, \quad t_2 \approx 0.04925249167432646403, \quad t_3 \approx 0.15740870098294833737$$

and the rigorous outer interval

$$\begin{aligned} 0.296096033367379523967764444238832345360107 &< \mathcal{J}_2, \\ \mathcal{J}_2 &< 0.296096033367379523967764444238832345360112. \end{aligned} \tag{40.13}$$

It also certifies $t_3 > t_2$. A positive measure on $[0, 1]$ would impose $t_3 \leq t_2$; that specific lifting of orbital weights to positive moments is ruled out. The positive sign of (40.13) remains certified. This check distinguishes the failure of a proposed proof from the failure of the property it sought to prove.

41 Classification and continuation of the first-root selector

41.1 Object and provenance

This appendix preserves the family rendered by the HMT dodecaphase reader:

$$u_0(x) = \frac{1}{12} \sum_{m=1}^{12} \left(1 - \cos \frac{2(3m-1)x}{\sqrt{899}} \right), \quad f_\lambda(x) = 1 - \lambda u_0(x).$$

The family originates in the operator and analytic antecedents presented in this development. The existence and compactness of its realizations of the infinite itinerary were proved in the subsections on realization of the word and preservation of its signs.

The following argument is original and uses only those properties and the return operations of the fixed family. It does not use a parameter from another family or the value of a universal constant as an input. The sheets of the itineraries used here are the two monotone branches of the critical reader; the remaining coordinates and HMT memory retain their residence in the state from which the reader originates.

We write

$$c_j(\lambda) = f_\lambda^j(0), \quad \tau_j = (-1)^{v_2(j)}, \quad W_n = \tau_1 \cdots \tau_{2n-1}.$$

The itinerary of a critical orbit that returns to the origin for the first time ends in the zero symbol. The itinerary order is the unimodal order with a maximum: before comparing the symbol of index j , the orientation factor is

$$O_{j-1}(K) = \prod_{i=1}^{j-1} (-K_i).$$

The ordinary order of the symbols is $-1 < 0 < 1$.

41.2 Classification before the infinite word

Classification theorem. Let $F : I \rightarrow I$, with $I = [\ell, 1]$, $\ell < 0$, be a continuous map, strictly increasing to the left of zero and strictly decreasing to its right, whose maximum satisfies $F(0) = 1$. Suppose zero has exact period 2^n , with $n \geq 1$, and its critical itinerary K satisfies $K < \tau$. Then

$$\boxed{K = W_n 0.}$$

The conclusion concerns the itinerary and allows distinct parameters in a family to have that same itinerary.

Proof. Write $d_j = F^j(0)$. For period two the itinerary is $+0 = W_1 0$. Consider an exact period $p = 2^n \geq 4$.

We have $d_2 < 0$. Indeed, $d_2 = 0$ would give period two. If $d_2 > 0$, monotonicity of the positive branch gives

$$F([d_2, 1]) = [d_2, d_3] \subset [d_2, 1],$$

and the critical orbit remains in a strictly positive interval, incompatible with a return to zero.

The initial signs are then $+, -$. The orientation factor for comparing the third sign is -1 ; hence $K < \tau$ forces $d_3 > 0$. A zero at that position would produce period three.

Nor can $d_4 < 0$ occur. Under this hypothesis,

$$d_2 < d_4 < 0, \quad F([d_2, d_4]) = [d_3, d_5] \subset (0, 1],$$

since $d_4 = F(d_3) > F(1) = d_2$ and $d_5 = F(d_4) > F(d_2) = d_3$. The positive branch is decreasing, so that

$$F^2([d_2, d_4]) = [d_6, d_4] \subset [d_2, d_4].$$

Here $d_6 \geq d_2$, because $d_5 \leq 1$, and $d_6 < d_4$, because $d_5 > d_3$. This strictly negative interval traps the even returns of the critical orbit and excludes a return to zero. Therefore $d_4 \geq 0$. If $p = 4$, one obtains precisely $+ - + 0 = W_2 0$.

Now suppose $p \geq 8$. Then $d_4 > 0$. The orientation factors for the fifth and sixth signs are, respectively, -1 and $+1$. Since $\tau_5 = +1$ and $\tau_6 = -1$, the inequality $K < \tau$ forces

$$\text{sign}(d_1, \dots, d_6) = (+, -, +, +, +, -). \quad (41.1)$$

Zero signs at those positions are excluded by the exact period.

We construct the return interval

$$J = [d_2, d_4].$$

The inequality $F(d_3) = d_4 > 0 > d_6 = F(d_5)$, together with $d_3, d_5 > 0$, implies $d_3 < d_5$; therefore

$$F(J) = [d_3, 1].$$

Moreover, $d_4 < d_3$. To check this, the image under F of the interval with endpoints d_3, d_4 is strictly positive, since its endpoint images are $d_4, d_5 > 0$. On that interval F^2 is strictly increasing. Its endpoint values satisfy

$$F^2(d_3) = d_5 > 0 > d_6 = F^2(d_4),$$

which forces $d_3 > d_4$. Consequently,

$$J \cap F(J) = \emptyset, \quad F^2(J) = F([d_3, 1]) = [d_2, d_4] = J. \quad (41.2)$$

The map $F^2|_J$ has a unique minimum at zero. Conjugacy by $h(x) = d_2 x$, which reverses orientation, defines

$$G = h^{-1} \circ F^2 \circ h : \left[\frac{d_4}{d_2}, 1 \right] \longrightarrow \left[\frac{d_4}{d_2}, 1 \right]. \quad (41.3)$$

This map satisfies the same maximum hypotheses as F , with $G(0) = 1$, and its critical point has exact period $p/2$. Its itinerary is

$$K'_j = -K_{2j}. \quad (41.4)$$

All odd-indexed symbols of K are positive, because the even positions belong to J and the odd positions to $F(J) \subset (0, 1]$.

The transformation (41.4) preserves the comparison order with τ . Indeed, $\tau_{2j-1} = +1$ and $\tau_{2j} = -\tau_j$. If q is the first position at which K' and τ differ, the first position of difference between K and τ is $2q$, and

$$O_{2q-1}(K) = -O_{q-1}(K'), \quad K_{2q} - \tau_{2q} = -(K'_q - \tau_q). \quad (41.5)$$

The product determining the order is identical. The identities remain valid when the first distinct symbol is the final zero. Thus $K' < \tau$.

Induction applied to G gives $K' = W_{n-1}0$. The positive odd positions and (41.4) reconstruct exactly $K = W_n0$. The theorem is proved. $\square$

The theorem also proves that each classified map has the restrictive doubling returns preceding its superstable return, with the domains of (41.2)–(41.3). It does not presuppose monotonicity of a family parameter.

Symmetric normalization of the fixed family. When F is even and defined on $[-1, 1]$, the preceding inequalities additionally give

$$F(d_4) = d_5 > d_3 = F(d_2) = F(-d_2).$$

Since both arguments $d_4, -d_2$ are positive and F decreases on that branch,

$$0 < d_4 < -d_2 < 1. \quad (41.6)$$

Therefore $F^2([d_2, -d_2]) = [d_2, d_4]$, and the same conjugacy $h(x) = d_2x$ defines on all of $[-1, 1]$ an even map with a maximum whose image is $[d_4/d_2, 1] \subset [-1, 1]$. It agrees exactly with the operator $RF = -F(F(-ax))/a$, $a = -F(1)$, used in the corpus. This symmetric extension preserves the critical return, its period and its itinerary.

41.3 First parameter with infinite itinerary

We work on the compact interval already used in the existence proof,

$$L = \left[\frac{1}{2u_0(1)}, \frac{2}{u_0(1)} \right].$$

Let

$$\Lambda_\tau = \{\lambda \in L : \text{sign } c_j(\lambda) = \tau_j \text{ for every } j \geq 1\}.$$

By the existence and sign-separation proofs of the prolongation, this set is nonempty and compact. Therefore there exists

$$a = \min \Lambda_\tau. \quad (41.7)$$

Comparison lemma. For every $\lambda \in [1/(2u_0(1)), a)$, we have $K_\lambda < \tau$.

Proof. At the left endpoint, the itinerary is $+\infty < \tau$. The separation argument for realization of the infinite word proves that the strict comparison sets with τ are open: in an infinite itinerary distinct from τ , the first finite difference persists by continuity; at a superstable parameter, the two neighboring periodic words and the word with a final zero lie on the same side of τ , by its strict maximality. By definition, the interval preceding a contains no parameters with itinerary τ . Its connectedness maintains the initial comparison throughout that interval. $\square$

41.4 Existence of each first new root

We retain the original selector. The first root is

$$\lambda_1 = \frac{1}{u_0(1)},$$

and, for $n \geq 2$, λ_n is the first root of c_{2^n} to the right of λ_{n-1} satisfying

$$c_{2^j}(\lambda_n) \neq 0 \quad (1 \leq j < n). \quad (41.8)$$

Since every first period dividing 2^n is a power of two, (41.8) is equivalent to requiring exact critical period 2^n .

Existence lemma. All roots of this selector exist and satisfy

$$\lambda_1 < \lambda_2 < \dots < \lambda_n < a. \quad (41.9)$$

Proof. The sign $c_2(a) < 0$ implies $\lambda_1 < a$. Suppose $\lambda_{n-1} < a$ has been constructed, and put $p = 2^{n-1}$. The analytic function $c_p(\lambda)$ is not identically zero, since $c_p(a) \neq 0$. Its zero set in $[\lambda_{n-1}, a]$ is finite, nonempty and does not contain a . Let $z < a$ be its last zero.

On $(z, a]$, the sign of c_p is constant and equal to $s = \tau_p$. For $g_\lambda = f_\lambda^p$, we have

$$g_\lambda(0) = c_p(\lambda), \quad g'_\lambda(0) = 0.$$

A uniform local bound on the second derivative gives

$$c_{2p}(\lambda) = g_\lambda(c_p(\lambda)) = c_p(\lambda) + O(c_p(\lambda)^2). \quad (41.10)$$

As λ approaches z from the right, the first-order term dominates; hence $c_{2p}(\lambda)$ has sign s . At a , its sign is

$$\tau_{2p} = -\tau_p = -s.$$

The intermediate value theorem provides a zero $\beta \in (z, a)$ of c_{2p} . Since $c_p \neq 0$ on that interval, the critical period at β is exactly $2p$.

Analyticity and $c_{2p}(a) \neq 0$ imply that the zeros of c_{2p} in $[\lambda_{n-1}, a]$ are finite. Among its zeros of exact period $2p$ to the right of λ_{n-1} , there is therefore a first one. It is the original selector λ_n , and it satisfies $\lambda_n \leq \beta < a$. $\square$

Use of the last auxiliary zero z belongs solely to the existence proof. The selected parameter remains the **first new root** λ_n .

41.5 Convergence of the global selector to the first infinite itinerary

Global continuation theorem. For the fixed HMT family and the original selector, all selected itineraries are canonical and

$$K_{\lambda_n} = W_n 0, \quad \lambda_n \nearrow \min \Lambda_\tau. \quad (41.11)$$

Proof. The comparison and existence lemmas place each selected parameter below a , with an itinerary less than τ . The classification theorem identifies that itinerary with $W_n 0$. The sequence is increasing and bounded by a ; let $b \leq a$ be its limit.

Fix $p \geq 1$. For n sufficiently large, $2p < 2^n$. Then the signs of $c_p(\lambda_n)$ and $c_{2p}(\lambda_n)$ agree with τ_p and $\tau_{2p} = -\tau_p$, respectively. The uniform bound on second derivatives,

$$\|(f_\lambda^p)''\| \leq B_p, \quad B_p = 16^p \frac{16^p - 1}{15},$$

and the identity $(f_\lambda^p)'(0) = 0$ imply

$$|c_{2p}(\lambda_n) - c_p(\lambda_n)| \leq \frac{1}{2} B_p |c_p(\lambda_n)|^2.$$

Since the two terms on the left-hand side have opposite signs,

$$|c_p(\lambda_n)| > 2/B_p. \quad (41.12)$$

By continuity, $c_p(b)$ retains the sign τ_p and satisfies $|c_p(b)| \geq 2/B_p > 0$. This holds for each p , so that $b \in \Lambda_\tau$. Minimality of a gives $b \geq a$, and therefore $b = a$. $\square$

The complete proof preserves the global selector and produces its itineraries and its limit. The information from the first eight levels is compatible with the result, but the proof of (41.11) does not extrapolate those eight levels: it uses induction, invariant intervals and uniform separation of every sign.

41.6 Logical checks and falsifiers

1. The classification theorem requires a strict unimodal maximum and an invariant interval. If either branch loses monotonicity, the intervals (41.2) may cease to be restrictive and the classification must be checked again.
2. The condition $K < \tau$ is essential. Beyond the limit word, centers of period 2^n with other combinatorics may exist; the theorem does not identify them with $W_n 0$.
3. Analyticity in the parameter provides isolation and finiteness of the zeros on the compact set. For a merely continuous family, the existence of a strictly subsequent first root requires an additional argument.
4. A limit of critical roots may, in general, lose signs upon reaching a return of lower period. Inequality (41.12) excludes precisely that phenomenon for this sequence and each fixed horizon.
5. Convergence of the first roots to the first parameter with itinerary τ is not equivalent to uniqueness of all parameters with that itinerary and does not by itself determine a metric rate. Transverse crossing and branch identification provide those subsequent conclusions.

42 Transverse entry and local scaling

42.1 Provenance and result

APP evaluates addition and multiplication on the two sheets of the lattice, preserving quotient and remainder. TRIT determines regime and orientation. TPK transport selects, transports and updates the state with its carry, boundary and memory records. Its readers act on the same discrete structure of the continuum. The prolongation $w6 \rightarrow w12 \rightarrow w18 \rightarrow w24 \rightarrow w30 \rightarrow R36 \rightarrow G9$ preserves this provenance and distinguishes phase return from memory increment.

The critical reader of the dodecaphase wheel, constructed in the operator antecedents and transported by compatible refinement, produces

$$\phi_j = \frac{(3j-1)\pi_{\text{HMT}}}{18}, \quad w_j = \frac{1}{12}, \quad k_j = \sqrt{\frac{2}{\sum_{\ell=1}^{12} w_\ell \phi_\ell^2}} \phi_j = \frac{2(3j-1)}{\sqrt{899}}.$$

The coordinate π_{HMT} retains its APP-TRIT-TPK origin; its common factor cancels in this second moment. The resulting reader is

$$u_0(x) = \frac{1}{12} \sum_{j=1}^{12} (1 - \cos k_j x), \quad f_\lambda(x) = 1 - \lambda u_0(x). \quad (42.1)$$

The focal sector is the uniform one, $\eta = 0$. The preceding extension separately retains the weighted family $|\eta| \leq 1/40$. The new numerical bounds correspond to (42.1).

The family is even and entire, with $f_\lambda(0) = 1$, a quadratic critical point and second moment $\sum_j w_j k_j^2 = 2$. The parameter λ ranges over a family downstream of the HMT generator. The target Feigenbaum constant remains outside the inputs of (42.1).

Main result. There exists a unique

$$\lambda_* \in [1.874038, 1.874039] \quad (42.2)$$

for which the fourth normalized return of (42.1) belongs to the local stable leaf of the real doubling fixed point. The crossing of the family with that leaf is transverse, with quantified separation. For every $n \geq 0$,

$$\|R^{4+n} f_{\lambda_*} - g\|_{\mathcal{A}} < 0.001666 (0.83996)^n. \quad (42.3)$$

The local superstable cascade associated with this crossing has parameters μ_j and constants $C \neq 0$, $0 < \theta < 1$, such that

$$\mu_j - \lambda_* = C \delta_F^{-j} [1 + O(\theta^j)], \quad \frac{\mu_{j-1} - \mu_{j-2}}{\mu_j - \mu_{j-1}} \longrightarrow \delta_F. \quad (42.4)$$

Here $\delta_F > 1$ is the unstable eigenvalue of the doubling operator. Selection of μ_j is performed through the local returns of section 42.8. Identification with the global selector is proved in sections 43.1–43.3: all global terms retain the combinatorics and the centers agree at every sufficiently high level when their periods are matched.

42.2 Analytic chart and effective return

The chart of even functions is used

$$s = x^2, \quad z = \frac{s-1}{5/2}, \quad f(x) = 1 - x^2 V(z), \quad V(z) = \sum_{j \geq 0} v_j z^j.$$

The coordinates of the Banach space are

$$u = 10v_0, \quad v = (v_1, v_2, \dots), \quad \|(\Delta u, \Delta v)\|_{\mathcal{A}} = |\Delta u| + \sum_{j \geq 1} |\Delta v_j|. \quad (42.5)$$

In particular, the constant coefficient of V has weight ten. This normalization agrees with the Hertling–Spandl chart $u/10$, section 2 of the source.

Let $a = -f(1) = v_0 - 1$. For admissible real returns,

$$(Rf)(x) = -\frac{f(f(-ax))}{a}. \quad (42.6)$$

The program evaluates an exact factorization of (42.6). Defining

$$S = \frac{a^2 s - 1}{5/2}, \quad b = V(S), \quad h = 1 - a^2 s b, \quad t = \frac{h^2 - 1}{5/2},$$

$$\text{shift } V(z) = \sum_{j \geq 0} v_{j+1} z^j,$$

one obtains

$$V_R = a b (h + 1) \left[V(t) + \frac{2}{5} \text{shift } V(t) \right]. \quad (42.7)$$

To verify it, one uses $v_0 = 1 + a$, $h^2 - 1 = -a^2 s b (h + 1)$ and $V(t) - v_0 = t \text{shift } V(t)$. Thus, $a + 1 - h^2 V(t) = a^2 s b (h + 1) [V(t) + (2/5) \text{shift } V(t)]$, which is precisely $a V_R$.

The initial moments are rational:

$$\frac{1}{12} \sum_j k_j^{2m} = \frac{1}{12} \left(\frac{4}{899} \right)^m \sum_{j=1}^{12} (3j-1)^{2m}. \quad (42.8)$$

Therefore both (42.7) and the initial construction use rational intervals; the initial trigonometric functions are represented by their series with an outer remainder bound.

42.3 External bounds used as subsequent recognition

Once (42.1) has been constructed, Lanford’s hyperbolicity theorem is used as an analytic certificate for the operator (42.6). Let g_0 be the published central polynomial and g the fixed point. The estimates used are

$$\|g - g_0\|_{\mathcal{A}} < \varepsilon_0 = 4 \cdot 10^{-5},$$

and, in the ball $\|f - g_0\|_{\mathcal{A}} < 0.01$, the blocks of DR , written $R = (U, V)$, satisfy

$$U_u \geq a_0 = 4.521, \quad \|U_v\| \leq b_0 = 0.560, \quad \|V_u\| \leq c_0 = 0.756, \quad \|V_v\| \leq d_0 = 0.719. \quad (42.9)$$

The fixed point has a unique unstable direction, with positive real eigenvalue $\delta_F > 1$; the rest of the spectrum lies inside the unit disk. The figures in (42.9) come from the published estimates, not from a new reproduction of Lanford’s proof in this dossier. The coefficients of g_0 are explicitly incorporated into the program’s central reference polynomial, in accordance with table 2 of Hertling–Spandl.

Sources: Lanford, hyperbolicity estimates; Hertling–Spandl, analytic chart and table 2.

42.4 Rational entry and direction certificate

The renormalization entry certificate divides (42.2) into sixteen equal rational intervals. It propagates four returns of (42.7) and the exact derivative with respect to λ , using differentiation of the composition and outward rounding to one hundred decimal places.

Each series consists of a polynomial of degree 32 with interval coefficients and an arbitrary analytic error E , $\|E\|_1 \leq \epsilon$. This error may also modify low-order coefficients. For $\|q\|_1 < 1$,

$$\|E \circ q\|_1 \leq \epsilon, \quad \|E' \circ q\|_1 \leq \frac{\epsilon}{(1 - \|q\|_1)^2}, \quad \|\text{shift } E\|_1 \leq \epsilon. \quad (42.10)$$

The products include polynomial–error and error–error terms. The initial tail is bounded by its first absolute term and a geometric ratio, using $\|1 + (5/2)z\|_1 = 7/2$. Each composed argument remains strictly inside the unit ball of this algebra.

For $\Gamma(\lambda) = R^4 f_\lambda = (u_\lambda, v_\lambda)$, the certificate gives the following conservative bounds, uniform in (42.2):

Quantity	Certified outer bound
$\ \Gamma(\lambda) - g_0\ _{\mathcal{A}}$	< 0.004993128
$\ v_\lambda - v_{g_0}\ _1$	< 0.001396077
u'_λ	> 3821.289115
$\ v'_\lambda\ _1$	< 551.757000
$\ v'_\lambda\ _1 / u'_\lambda$	$< 0.144386 < 1/4$
Analytic error of the function	$< 8.482 \cdot 10^{-19}$
Analytic error of the tangent	$< 4.569 \cdot 10^{-15}$

The complete decimals, the sixteen subintervals and every intermediate return are retained in the rational entry certificate.

The cone $\|\dot{v}\|_1 \leq |\dot{u}|/4$ is invariant under (42.9):

$$\begin{aligned} |\dot{u}_{\text{nuevo}}| &\geq (4.521 - 0.560/4)|\dot{u}| = 4.381|\dot{u}|, \\ \|\dot{v}_{\text{nuevo}}\|_1 &\leq (0.756 + 0.719/4)|\dot{u}| = 0.93575|\dot{u}| < \frac{1}{4}(4.381)|\dot{u}|. \end{aligned} \quad (42.11)$$

The certified parameter direction therefore enters a uniformly expanding cone of the operator.

42.5 Quantitative construction of the stable leaf

Translate g to the origin and write the coordinates as $(x, y) = (u - u_g, v - v_g)$. Fix

$$L = 0.16, \quad r = 0.004, \quad A = a_0 - Lc_0 = 4.40004, \quad q = d_0 + c_0L = 0.83996. \quad (42.12)$$

The cylinder $|x| \leq Lr$, $\|y\|_1 \leq r$ lies inside the ball of (42.9), since $(1 + L)r + \epsilon_0 = 0.00468 < 0.01$.

Consider the complete space of functions $\sigma : B_r \rightarrow \mathbb{R}$ with $\sigma(0) = 0$ and Lipschitz constant at most L , equipped with the uniform distance. For each y , define $(\mathcal{G}\sigma)(y)$ as the root in $[-L\|y\|_1, L\|y\|_1]$ of

$$F_\sigma(x, y) = U(x, y) - \sigma(V(x, y)). \quad (42.13)$$

On that interval, $\|V(x, y)\|_1 \leq q\|y\|_1 < r$. As x increases, (42.9) shows that F_σ grows with slope at least A . At its endpoints,

$$F_\sigma(L\|y\|_1, y) \geq [AL - b_0 - Ld_0]\|y\|_1,$$

and the opposite inequality holds at the negative endpoint. The margin is

$$AL - b_0 - Ld_0 = 0.0289664 > 0. \quad (42.14)$$

A unique root is obtained. Comparing (42.13) for two values of y ,

$$\text{Lip}(\mathcal{G}\sigma) \leq \frac{b_0 + Ld_0}{A} = \frac{0.67504}{4.40004} < 0.16.$$

Comparing two graphs,

$$\|\mathcal{G}\sigma - \mathcal{G}\tilde{\sigma}\|_{\infty} \leq A^{-1}\|\sigma - \tilde{\sigma}\|_{\infty}, \quad A^{-1} < 0.228. \quad (42.15)$$

The contraction theorem produces a unique fixed graph $x = \sigma_*(y)$. On it,

$$\|y_n\|_1 \leq q^n \|y_0\|_1, \quad \|(x_n, y_n)\|_{\mathcal{A}} \leq (1 + L)q^n \|y_0\|_1. \quad (42.16)$$

Moreover, the vertical distance $d(x, y) = x - \sigma_*(y)$ satisfies

$$|d(R(x, y))| \geq A|d(x, y)|$$

as long as the orbit remains in the cylinder. Every orbit that remains there indefinitely belongs to the graph. This graph identifies exactly the local stable leaf and provides estimate (42.16). Its local regularity agrees with the analytic stable manifold of the hyperbolic fixed point.

42.6 Existence, uniqueness and transversality of the parameter

For λ in (42.2), define

$$H(\lambda) = u_{\lambda} - u_g - \sigma_*(v_{\lambda} - v_g).$$

The certificate ensures $\|v_{\lambda} - v_g\|_1 < 0.001436077 < r$. For $\lambda_2 > \lambda_1$,

$$H(\lambda_2) - H(\lambda_1) \geq \int_{\lambda_1}^{\lambda_2} [u'_t - L\|v'_t\|_1] dt > 3733.007995 (\lambda_2 - \lambda_1). \quad (42.17)$$

To determine the endpoint signs, $\|g - g_0\|_{\mathcal{A}} < \varepsilon_0$ and $|\sigma_*(y)| \leq L\|y\|_1$ are used. Rational point evaluation produces

$$H(1.874038) < -0.0011856110945, \quad H(1.874039) > 0.0021866053399. \quad (42.18)$$

By continuity, a root exists; (42.17) proves its uniqueness and the transversality of the crossing. At that root, (42.16) and $(1 + L)(0.001396077 + 0.00004) < 0.001666$ prove (42.3).

The bound is uniform in the number of returns: it converts increasing depth into an explicit geometric operator error.

42.7 Real admissibility of all returns

The four initial returns are accompanied, on each rational subinterval, by the inequalities

$$0 < a = -f(1) < 1, \quad b = f(a) > a, \quad f(b) < a. \quad (42.19)$$

The program stores $a, b, f(b)$ before each application of (42.7). Unimodality, evenness and the quadratic critical point are preserved under these normalized returns.

For subsequent iterations, a uniform check in the ball of (42.9) suffices. If $E = V - V_{g_0}$, the norm (42.5) gives, for $-0.4 \leq z \leq 0$,

$$|E(z)| \leq 0.004, \quad |E'(z)| \leq 0.01.$$

The second inequality uses $\sup_{j \geq 1} j(0.4)^{j-1} = 1$. Rational evaluation of the central coefficients provides

$$\frac{6}{5} < V(z) < \frac{8}{5}, \quad |V'(z)| < \frac{1}{2}, \quad 0.39 < a < 0.41. \quad (42.20)$$

Consequently,

$$f'(x) = -2x \left[V(z) + \frac{2}{5}x^2 V'(z) \right],$$

and the bracketed expression exceeds one for $0 \leq x \leq 1$. The map is unimodal and preserves $[-1, 1]$. Moreover,

$$\begin{aligned} b = f(a) &> 1 - (0.41)^2 \frac{8}{5} = \frac{4569}{6250} > a, \\ f(b) &< 1 - \left(\frac{4569}{6250}\right)^2 \frac{6}{5} = \frac{35028967}{97656250} < 0.39 < a. \end{aligned} \quad (42.21)$$

The orbits of the stable leaf satisfy (42.19) at every depth. Thus the analytic iterations of the operator correspond to real doubling returns, with their domains and orientations preserved.

42.8 Scaling of the local superstable cascade

The metric conclusion requires two crossings: that of the family with the stable leaf, proved in section 42.6, and that of the unstable manifold with the superstable target. The second is proved by Eckmann–Wittwer for $\Sigma_2 = \{\psi : \psi(1) = 0\}$, corresponding to the period-two superstable cycle. Normalization (42.6) agrees with that of the even real operator used there.

The general theorem of Collet–Eckmann–Lanford, section 6 of the cited source, proposition 6.1 and theorems 6.2–6.3, applies to a Banach-space map C^2 , a fixed point with a unique unstable eigenvalue $\delta_F > 1$, a curve transverse to its stable leaf and a target transverse to its unstable manifold. The local preimages of the target intersect the curve at points satisfying

$$\delta_F^j(\mu_j - \lambda_*) \longrightarrow C \neq 0. \quad (42.22)$$

The index j counts local returns. Since the initial curve here is $R^4 f_\lambda$, the physical period is 2^{j+5} when the target is taken in Σ_2 ; any fixed number of steps used to bring the target into the local neighborhood is incorporated into a fixed index shift and into C .

The real conditions of section 42.7 and the target’s real doubling branch fix the exact period and itinerary. The sequence is selected by this oriented local continuation. Taking differences in (42.22) gives the ratio limit of (42.4).

Sources of the two results used: Eckmann–Wittwer, *A complete proof of the Feigenbaum conjectures*; Collet–Eckmann–Lanford, section 6.

42.8.1 Power-law asymptotic remainder

The analyticity of our family allows (42.22) to be made more precise. In the normal-coordinate construction of Collet–Eckmann–Lanford, only the stable manifold, the unstable manifold and the target hypersurface are flattened. These three submanifolds admit local C^2 changes of coordinates with bounded derivatives. The smooth cutoff used is applied in the scalar unstable coordinate; the complete foliation is then obtained through the constructed change of coordinates.

In those coordinates, the correction is written as a series of increments w_j whose norm C^1 satisfies

$$\|w_j\|_{C^1} \leq C_1 \rho^j, \quad 0 < \rho < 1.$$

The chain rule for the cut-off map, with uniformly bounded first- and second-order derivatives, provides

$$\|w_j\|_{C^2} \leq C_2 B^j$$

for some $B \geq 1$. The weighted norm of the construction controls the ordinary C^1 norm on the domain considered. By interpolation,

$$[Dw_j]_\beta \leq C_3 (\rho^{1-\beta} B^\beta)^j.$$

Choose

$$0 < \beta < \min \left\{ 1, \frac{-\log \rho}{\log B - \log \rho} \right\}. \quad (42.23)$$

The series converges in $C^{1,\beta}$. Let z be the resulting normal coordinate, normalized so that the preimages of the target satisfy $z = \delta_F^{-j}$, absorbing the fixed index shift. Since our curve is analytic and transverse,

$$z(\Gamma(\lambda)) = c(\lambda - \lambda_*) + O(|\lambda - \lambda_*|^{1+\beta}), \quad c \neq 0.$$

Local inversion gives

$$\mu_j - \lambda_* = C\delta_F^{-j} + O\left(\delta_F^{-(1+\beta)j}\right). \quad (42.24)$$

This proves the remainder in (42.4), with $\theta = \delta_F^{-\beta}$. The statement asserts the existence of the constants in the parameter remainder. The explicit operator rate 0.83996 of (42.3), for its part, controls convergence of the renormalized maps.

42.8.2 Conversion of the remainder into ratio error

If $\mu_j - \lambda_* = C\delta_F^{-j}(1 + e_j)$, with $|e_j| \leq M\theta^j$, write

$$\mu_{j-1} - \mu_j = C(\delta_F - 1)\delta_F^{-j}(1 + r_j), \quad r_j = \frac{\delta_F e_{j-1} - e_j}{\delta_F - 1}.$$

With $K = M(\delta_F + \theta)/(\delta_F - 1)$, we have $|r_j| \leq K\theta^{j-1}$. Therefore,

$$\left| \frac{\mu_{j-2} - \mu_{j-1}}{\mu_{j-1} - \mu_j} - \delta_F \right| \leq \frac{\delta_F K(1 + \theta)\theta^{j-2}}{1 - K\theta^{j-1}} \quad (42.25)$$

when $K\theta^{j-1} < 1$. This formula separates the evaluation precision of each root from the truncation error of the cascade.

42.9 Certified continuation of the eight initial roots

The finite continuation certificate preserves the family (42.1) and the previous eight-root certificate. For $2 \leq n \leq 8$, it studies

$$S_n(\lambda) = f_\lambda^{2^n}(0)$$

from the certified box of λ_{n-1} to 1.874039. The complete rational covering contains exactly the inherited root λ_{n-1} and the new root λ_n . In particular, λ_n is the unique new root in $(\lambda_{n-1}, 1.874039]$.

Boxes are excluded by the sign of the return or by a local derivative of certified sign. Anchor boxes retain existence and uniqueness through endpoint signs and local monotonicity. Their endpoints cover each interval exactly; adjacencies are checked. Evaluation uses the moments (42.8), order-32 Horner evaluation, outer geometric tails and $|u_0''| \leq 2$. The uniform tail of the potential is less than $1.294 \cdot 10^{-58}$ on $|x| \leq 3/2$.

The final certificate contains 650 processed boxes and 332 covering leaves. The first level has the exact expression $\lambda_1 = 1/u_0(1)$. The following approximations are shown solely for guidance; their complete boxes are retained in the certificates:

Level	Superstable parameter
1	1.345913990471832792210949095...
2	1.763379787846910127290794030...
3	1.850413796950136464530566778...
4	1.868979756606798125150574309...
5	1.872955034435659740004205128...
6	1.873806367467030119412262311...
7	1.873988695221365362307168780...
8	1.874027744154450672897007573...

43 Isolation of the limit and scaling of the global selector

43.1 Indefinite continuation of the first root and exclusion of the intermediate interval

The complete classification proved above preserves the family and the first-new-root selector. Ordered transport preserves the zeros and their minimum; the return word preserves the nonadic chronological reading.

43.1.1 Classification and existence at every depth

Write $W_n = \tau_1 \cdots \tau_{2^n-1}$, with $\tau_j = (-1)^{v_2(j)}$, and $a_* = \min \Lambda_\tau$. The existence and compactness of Λ_τ are proved in the prolongation development, 38.5.1–38.5.2; its minimum exists by compactness and order.

Classification. Every strictly unimodal map with a maximum, on an invariant interval, whose critical point has exact period 2^n and whose itinerary K satisfies $K < \tau$, has itinerary $W_n 0$.

The induction proceeds through the effective return. For period at least eight, comparison with τ , together with the intervals that would trap orbits with incompatible signs, forces the prefix $(+, -, +, +, +, -, -)$. This yields

$$J_1 = [d_2, d_4], \quad F(J_1) = [d_3, 1], \quad d_4 < d_3, \quad F^2(J_1) = J_1.$$

The return normalized by $h(x) = d_2 x$ has maximum one, half the period, and itinerary $K'_j = -K_{2j} < \tau$. Order is preserved because

$$O_{2q-1}(K) = -O_{q-1}(K'), \quad K_{2q} - \tau_{2q} = -(K'_q - \tau_q).$$

The initial cases are $+0$ and $+ - +0$; induction reconstructs $W_n 0$. For even maps, also $0 < d_4 < -d_2 < 1$, so the return extends to $[-1, 1]$ and coincides with (42.6).

Before a_* , all itineraries are smaller than τ : the strict-comparison sets are open by the separation proved in 38.5.1; the interval preceding a_* is connected and begins with $+\infty < \tau$.

Existence of each first root. Suppose $\lambda_{n-1} < a_*$ has been constructed, and let $p = 2^{n-1}$. The zeros of c_p in $[\lambda_{n-1}, a_*]$ form a nonempty finite set: the function is analytic and $c_p(a_*) \neq 0$. Let $z < a_*$ be its last zero. In $(z, a_*]$, c_p has sign τ_p . For $g_\lambda = f_\lambda^p$,

$$g'_\lambda(0) = 0, \quad c_{2p}(\lambda) = c_p(\lambda) + O(c_p(\lambda)^2).$$

Near z , on the right, c_{2p} has sign τ_p ; at a_* , it has the opposite sign. There is a zero of c_{2p} in (z, a_*) , and its period is exactly $2p$, since there $c_p \neq 0$. Finiteness of the zeros provides the first new root in (λ_{n-1}, a_*) . The last auxiliary zero proves existence; the effective choice remains the first root. The base case is $\lambda_1 = 1/u_0(1) < a_*$.

43.1.2 Limit of the original sequence

Theorem. The original selector exists at every level and satisfies

$$K_{\lambda_n} = W_n 0, \quad \lambda_n \nearrow a_* = \min \Lambda_\tau. \quad (43.1)$$

Proof. The preceding existence result gives an increasing sequence bounded by a_* . The classification fixes all its itineraries. Let $b \leq a_*$ be its limit. For each p , sufficiently late terms preserve the opposite signs τ_p, τ_{2p} . Taylor's formula and the uniform bound $B_p = 16^p(16^p - 1)/15$ give

$$|c_{2p}(\lambda_n) - c_p(\lambda_n)| \leq \frac{1}{2} B_p |c_p(\lambda_n)|^2, \quad |c_p(\lambda_n)| > 2/B_p.$$

Passing to the limit preserves sign $c_p(b) = \tau_p$, with absolute value at least $2/B_p > 0$. For every p , this implies $b \in \Lambda_\tau$; minimality gives $b = a_*$.

The result establishes indefinite existence, preservation of the symbolic cascade, and selection of its first limiting parameter for the original global rule. The HMT completion transports this sequence and its limit while preserving order and reading.

43.1.3 Certificate excluding the intermediate interval

The bridge program certifies

$$\Lambda_\tau \cap [\lambda_{8,\text{inf}}, 1.874038] = \emptyset, \quad (43.2)$$

where

$$\lambda_{8,\text{inf}} = 1.874027744154450672897007573595092408435133414523352839687903472138975$$

is the rational lower endpoint of the certified interval for λ_8 .

The cover contains 374 exactly adjacent intervals and no unresolved region: 319 with $c_{512} > 0$, 10 with $c_{1024} < 0$, two with $c_{2048} > 0$, all opposite to the sign of τ , and 43 near centers of periods 256, 512 or 1024, excluded by their second return.

In the latter case, $g = f_\lambda^p$, $d = g(0)$ and a uniform bound $|g''| \leq M$ between zero and d satisfy $M|d| < 2$. Then

$$|g(d) - d| \leq \frac{1}{2}M|d|^2 < |d| \quad (d \neq 0).$$

Both returns have the same sign; for $d = 0$, both are zero. Both situations exclude $\tau_{2p} = -\tau_p$ with strict signs.

The evaluation uses rational moments, geometric tails of the potential and its second derivative, and $|u_0''| < 6$. Parameter centering intersects the natural enclosure with

$$c_j(\lambda_{\text{medio}}) + \partial_\lambda c_j(I)(I - \lambda_{\text{medio}});$$

the derivative is propagated over the full box. The self-map property of $[-1, 1]$ is proved before restricting the enclosures to it.

The complete certificate retains the cover and checks its exact tiling, together with every sign or Taylor inequality.

43.1.4 Joint localization

The local crossing λ_* realizes τ through its real returns at every depth, so that $a_* \leq \lambda_*$. By (43.1), $\lambda_8 < a_*$; together with (43.2),

$$1.874038 < a_* \leq \lambda_* < 1.874039. \quad (43.3)$$

The localization preserves the canonical itinerary and places its first limiting parameter in the same interval as the stable crossing.

43.2 First-exit isolation and coincidence of the limits

This section completes the isolation identified in section 43.1.4. It preserves the HMT family (42.1), the minimal selection (43.1), the chart (42.5) and the stable sheet of sections 42.5–42.7. The proof uses the order $a_* \leq \lambda_*$: it suffices to exclude a separation toward the negative side of the expanding cone.

43.2.1 Upper and lower bounds for transverse transport

Besides (42.9), Lemma 2.1 of Hertling–Spandl, section 2 provides $\|D\Phi\| < 0.9$ in the same ball of radius 0.01, where

$$\Phi_u(u, v) = u - \frac{U(u, v) - u}{3.669}.$$

Therefore,

$$\left|1 - \frac{U_u - 1}{3.669}\right| < 0.9, \quad U_u < 1 + 1.9(3.669) = 7.9711. \quad (43.4)$$

The number 3.669 belongs to the published preconditioner of this analytic certificate; the generated family (42.1) retains its prior coefficients.

Take $\kappa = 0.21$. For two points in the ball joined by a secant with $\Delta u < 0$, $\|\Delta v\|_1 \leq \kappa|\Delta u|$, integration of the blocks of DR along the segment gives

$$4.4034|\Delta u| \leq |\Delta u'| \leq 8.0887|\Delta u|, \quad \Delta u' < 0, \quad (43.5)$$

$$\|\Delta v'\|_1 \leq (0.756 + 0.719\kappa)|\Delta u| = 0.90699|\Delta u| < \kappa(4.4034)|\Delta u|. \quad (43.6)$$

Here $4.4034 = 4.521 - 0.560\kappa$ and $8.0887 = 7.9711 + 0.560\kappa$. Thus, the secant cone preserves orientation and expands while the segment remains in the ball.

43.2.2 Reduction of an infinite separation to a finite region

Suppose $a_* < \lambda_*$, and set

$$f_n = R^{4+n} f_{a_*}, \quad s_n = R^{4+n} f_{\lambda_*}, \quad (\Delta u_n, \Delta v_n) = f_n - s_n.$$

The positive bound for u' and $\|v'\|_1/u' < 0.144386 < \kappa$ on $\Gamma(J)$ place the initial secant in the negative cone. The stable sheet is written relative to g , with

$$s_n = g + e_n, \quad \|e_n\|_{\mathcal{A}} < 0.001666(0.83996)^n, \quad |(e_n)_u| \leq 0.16\|(e_n)_v\|_1. \quad (43.7)$$

The entry checks imply

$$|\Delta u_0| < 0.004993128 + 0.00004 + 0.16(0.001396077 + 0.00004) = 0.00526290032 < h, \quad h = 0.006. \quad (43.8)$$

While $|\Delta u_n| < h$, (43.6)–(43.7) give

$$\|f_n - g_0\|_{\mathcal{A}} < (1 + \kappa)h + 0.001666 + 0.00004 = 0.008966 < 0.01. \quad (43.9)$$

The point s_n and the segment between the two also remain in that convex ball. The lower expansion (43.5) forces a finite first exit N , and the upper bound applied to the preceding step provides

$$0.006 \leq -\Delta u_N < 0.0485322, \quad \|\Delta v_N\|_1 \leq 0.21|\Delta u_N|. \quad (43.10)$$

The inequality $8.0887h = 0.0485322$ is used; the full jump, not only the first-exit surface, is included.

Consequently, f_N belongs to the following region:

$$\begin{aligned} \mathcal{C}_- = \{g + (d_u, d_v) + e : & -0.0485322 \leq d_u \leq -0.006, \\ & \|d_v\|_1 \leq 0.21|d_u|, \quad |e_u| + \|e_v\|_1 \leq 0.001666, \\ & |e_u| \leq 0.16\|e_v\|_1 \}. \end{aligned} \quad (43.11)$$

Since a_* realizes τ , all its normalized returns are unimodal self-maps of $[-1, 1]$ with the same itinerary. This property follows from return induction, independently of the distance from f_N to g_0 . This is why the following certificate treats $\mathcal{C}_-$ intersected with that class of self-maps.

43.2.3 Certified exclusion of the first-exit region

The rational first-exit exclusion certificate, reproduced in the computational supplement, covers the interval of (43.10) exactly with 64 adjacent rational bands. All are excluded; no additional subdivision or unresolved region enters the result.

Of the 64 bands, 54 display an opposite sign and ten a contracting return of period 16 or 32. The largest Lipschitz bound for those returns is less than 0.475146491; the maximum exclusion horizon is iterate 64.

The evaluation retains the first forty perturbation coefficients as shared variables, together with an explicit bound on their infinite tail. For each band, the center is g_0 shifted in coordinate u . If $a, b \geq 0$

bound the dual sensitivities in u, v , the support of the uncertainty in the fixed point and the stable residual is

$$0.00004 \max(a, b) + 0.001666 \max\left(b, \frac{0.16a + b}{1.16}\right). \quad (43.12)$$

The second expression is obtained by maximizing $a|e_u| + b\|e_v\|_1$ under the two constraints of (43.7). This uses the orientation of the stable sheet proved previously, as well as its norm.

Let x_j be the center's orbit and L_j its linear sensitivity to all shared coefficients. The Taylor model has the form

$$c_j = x_j + L_j(\Delta v) + r_j, \quad |r_j| \leq E_j.$$

If ρ_j bounds $|L_j(\Delta v)| + E_j$, the outward recurrence used is

$$E_{j+1} \leq |f'_0(x_j)|E_j + \frac{1}{2} \sup |f''_0| \rho_j^2 + \sup |\Delta f'| \rho_j. \quad (43.13)$$

The derivatives are bounded along the segment between the two states. The central orbit is verified within $[-1, 1]$; the other orbits belong to that interval by the self-map property included in the certified domain. The calculations use outward rational rounding, including for the geometric tails.

Each band is excluded by one of two inequalities:

1. A critical value has a sign strictly opposite to τ_j .
2. For $p = 2^k$, the return $H = f^p$ has Lipschitz constant $L_p < 1$ on the segment between 0 and $H(0)$. Then

$$|H(H(0)) - H(0)| \leq L_p |H(0)|.$$

If $H(0) \neq 0$, the two values have the same sign; if $H(0) = 0$, both vanish. In both cases the strict itinerary $\tau_{2p} = -\tau_p$ is excluded.

The constant L_p is obtained by propagating the initial segment $[-r, r]$, $r \geq |H(0)|$, alongside the critical-orbit boxes and multiplying the outward derivative bounds. The maps of the full band are bounded, retaining (43.12).

43.2.4 Coincidence theorem for the accumulation parameter

Theorem. For the uniform HMT family (42.1), the limit of the original first-root selection coincides with the certified stable crossing:

$$\lambda_n \nearrow a_* = \lambda_*, \quad 1.874038 < \lambda_* < 1.874039. \quad (43.14)$$

Proof. (43.1)–(43.3) give $\lambda_n \uparrow a_* \leq \lambda_*$, with both parameters in J . If the inequality were strict, (43.5)–(43.10) would produce a return $f_N \in \mathcal{C}_-$ realizing τ . The complete cover of section 43.2.3 excludes precisely that possibility. Hence $a_* = \lambda_*$.

The proof controls every depth through a finite first exit and a uniform cover of its region. The identity (43.14) links the global selector's limit to the transverse crossing.

43.3 Identification of the original centers and global scaling

The final step preserves the selection of (43.1). The local continuation and the original parameters are compared through their exact period and their hybrid class; coincidence is obtained through uniqueness in a fixed parameter neighborhood.

43.3.1 Degree-two analytic realization and transversality

The fixed point g of section 42.3 is even, analytic, with a quadratic critical point and stationary doubling combinatorics. Theorem 2.1 of de Faria–de Melo–Pinto, applied to that single combinatorics, gives a one-element limit set, with a proper holomorphic extension of degree two between nested topological disks. Its attraction on the interval identifies that element with g , since $Rg = g$.

Lemma 3.2 of the same source extends a fixed number of returns to a full analytic neighborhood of g , with degree-two images and positive modulus. The norm (42.5) controls the uniform norm on a sufficiently small complex neighborhood of $[-1, 1]$, because $|(x^2 - 1)/2.5| < 1$ there. The convergence (42.3) enters that neighborhood; analytic dependence of the returns provides, for some fixed integer d , a family

$$\mathcal{F}_\lambda = R^d f_\lambda \quad (43.15)$$

of degree-two maps, analytic in a complex neighborhood of λ_* . The disks and the number of returns are fixed before varying λ in that neighborhood.

The parameter tangent belongs to the expanding cone, whereas the stable sheet satisfies the slope bound of section 42.5. The returns preserve that separation. The degree-two realization locally identifies the stable sheet with the hybrid class of the infinitely renormalizable map; hence (43.15) is transverse to that class.

The change of analytic space is justified in detail in 44.1.2, devoted to the transport of transversality. A direction tangent to the hybrid class would produce decreasing renormalized tangents, by uniform convergence on a hybrid disk and Cauchy's estimate. The certified direction, by contrast, satisfies $|\dot{f}(1)| = |\dot{u}|/10$ and grows under the returns by the expanding cone. This common evaluation of both representatives proves transversality without assuming equivalence of their norms.

43.3.2 Uniqueness in a fixed neighborhood

For $n > d$, the classification of section 43.1 gives

$$K(\mathcal{F}_{\lambda_n}) = W_{n-d}0. \quad (43.16)$$

Straightening a degree-two map preserves the critical orbit; (43.16) identifies its hybrid class with that of the canonical quadratic center of period 2^{n-d} . The a priori bounds for the real doubling sequence are those used in Lyubich's universality theorem.

Lyubich's Theorem 7.4, pp. 387–388 gives a single intersection with each of those classes, at sufficiently high levels, on the analytic transversal and within a fixed neighborhood of the accumulation parameter.

By (43.14), λ_n eventually enters that neighborhood. The local continuation of section 42.8, indexed by the same period, belongs to the same hybrid class. Uniqueness implies

$$\lambda_n = \hat{\mu}_n \quad (n \geq n_0), \quad (43.17)$$

where $\hat{\mu}_n$ denotes the local center of physical period 2^n . This is a reindexing of μ_j by a fixed shift determined by the returns and the target of section 42.8. The original selection of λ_n remains intact.

The integer n_0 is existential in this proof. Continuation at every level and the word $W_n 0$ are already proved in (43.1); (43.17) provides the metric identity needed for the limit.

43.3.3 Scaling theorem for the first HMT roots

Theorem. For the uniform family (42.1), let λ_n be the successive first new roots of exact critical period 2^n . There exist $C > 0$, $0 < \theta < 1$ and $M < \infty$ such that, for every sufficiently large n ,

$$\lambda_* - \lambda_n = C\delta_F^{-n}(1 + e_n), \quad |e_n| \leq M\theta^n, \quad (43.18)$$

and

$$\lim_{n \rightarrow \infty} \frac{\lambda_{n-1} - \lambda_{n-2}}{\lambda_n - \lambda_{n-1}} = \delta_F. \quad (43.19)$$

Proof. (43.17) transports the local scaling (42.24) to the original roots; the finite index shift is absorbed into C, M . Its sign is positive because $\lambda_n < \lambda_*$.

The power-law remainder also admits a direct justification through the holonomy of hybrid classes. Lyubich's Lemma 7.3 gives $C^{1+\beta}$ regularity, with nonzero derivative, on the transversal at the accumulation point. The one-dimensional unstable dynamics is analytic and has multiplier $\delta_F > 1$; its linearizing coordinate places successive centers at distances proportional to δ_F^{-n} . Composing with the inverse holonomy gives a nonzero linear term and a remainder $O(\delta_F^{-(1+\beta)n})$. Thus one may take $\theta = \delta_F^{-\beta}$.

Writing

$$r_n = \frac{\delta_F e_{n-1} - e_n}{\delta_F - 1}, \quad K = \frac{M(\delta_F + \theta)}{\delta_F - 1},$$

one obtains

$$\lambda_n - \lambda_{n-1} = C(\delta_F - 1)\delta_F^{-n}(1 + r_n), \quad |r_n| \leq K\theta^{n-1}.$$

Therefore,

$$\left| \frac{\lambda_{n-1} - \lambda_{n-2}}{\lambda_n - \lambda_{n-1}} - \delta_F \right| \leq \frac{\delta_F K(1 + \theta)\theta^{n-2}}{1 - K\theta^{n-1}}, \quad K\theta^{n-1} < 1. \quad (43.20)$$

The right-hand side tends to zero, proving (43.19).

43.3.4 Chain of results and scope of the closure

The effective composition is: APP-TRIT-TPK dodecaphase reader; profile (42.1); ordered transport of returns (40.2)–(40.4); classification and existence of the first roots (43.1); localization (43.3); first-exit isolation (43.14); eventual identity of centers (43.17); scaling (43.18)–(43.20). The Jacobi representation (40.7) preserves this reader's full profile and its marks remain in the antecedent.

The spatial expansion of the HMT reader and nonadic refinement provide the genealogy and evaluation at arbitrary depth. Scaling between parameters is proved through the return dynamics, transversality and isolation that we have just composed. The cited analytic proofs act on the family already constructed; their comparison values do not select its phases or coefficients.

The closure concerns the original global selector of the uniform sector $\eta = 0$, with the listed theorem dependencies. The asymptotic law retains its existential quantifiers. The width of a finite-root box measures evaluation precision; the estimate (43.20) expresses convergence toward the limit.

The programs and certificates in the reproducibility supplement make it possible to reproduce the finite parts of the proof. Continuation at every depth rests on the inductions, the uniform first-exit argument and the identification theorems, with their explicit domains.

44 Analytic identification of the original centers

44.1 Degree-two realization and metric identification of the centers

The equality between the first parameter of the infinite itinerary and the stable crossing, proved by first exit, provides

$$\boxed{a_* := \min \Lambda_\tau = \lambda_*}. \quad (44.1)$$

The family, dodecaphase profile, returns and selector remain exactly as in sections 41.1–41.5. The external results used below recognize the dynamics of that already constructed object. Their universal values do not enter the selection of coefficients or parameters of the HMT family.

Metric transfer theorem. The identification (44.1), together with the previously proved entry, admissibility and transversality bounds, implies that the original sequence of first new roots satisfies, for certain constants $C > 0$ and $0 < \theta < 1$,

$$\boxed{\lambda_* - \lambda_n = C\delta_F^{-n}(1 + O(\theta^n))}, \quad (44.2)$$

where $\delta_F > 1$ is the unstable eigenvalue of the doubling operator at the recognized fixed point. In particular,

$$\frac{\lambda_{n-1} - \lambda_{n-2}}{\lambda_n - \lambda_{n-1}} = \delta_F + O(\theta^n). \quad (44.3)$$

The sequence eventually coincides with the centers of the local continuation, once both indices refer to the same physical period. The proof retains the first-root selector of section 41.4.

44.1.1 Obtaining a common quadratic-like domain

Write g for the renormalization fixed point and $\Gamma(\lambda) = R^4 f_\lambda$. The analytic chart used in the certificate is

$$f(x) = 1 - x^2 V\left(\frac{x^2 - 1}{2.5}\right), \quad V(z) = \frac{u}{10} + \sum_{k \geq 1} v_k z^k,$$

with difference norm $|\Delta u| + \|\Delta v\|_1$. The admissibility inequalities prove that g is even, analytic, unimodal, quadratic and of doubling type; moreover, $g(x) = \varphi(x^2)$ with $\varphi' < 0$ in $[0, 1]$.

Theorem 2.1 of de Faria–de Melo–Pinto, applied to the combinatorial set consisting solely of doubling, provides a one-point limit set with a quadratic-like extension. Since $Rg = g$, the real convergence in that theorem identifies this point with g . Lemma 3.2 provides, in an analytic neighborhood of g , a finite renormalization iterate with a quadratic-like domain and uniformly positive modulus. The locations are sections 2.1–2.3, pp. 736–738, and section 3.1, Lemma 3.2, p. 745, of de Faria–de Melo–Pinto, *Global hyperbolicity of renormalization for C^r unimodal mappings*.

To check membership in the neighborhood of that lemma, let Ω_ϵ be a sufficiently small complex neighborhood of $[-1, 1]$, with

$$\rho_\epsilon := \sup_{x \in \Omega_\epsilon} \left| \frac{x^2 - 1}{2.5} \right| < 1, \quad M_\epsilon := \sup_{x \in \Omega_\epsilon} |x|^2 < \infty.$$

The restriction of the certified chart satisfies

$$\|\Delta f\|_{\Omega_\epsilon} \leq M_\epsilon \left(\frac{|\Delta u|}{10} + \rho_\epsilon \|\Delta v\|_1 \right) \leq C_\epsilon \|\Delta f\|_{\mathcal{A}}. \quad (44.4)$$

The certificate gives

$$\|R^m \Gamma(\lambda_*) - g\|_{\mathcal{A}} < 0.001666(0.83996)^m.$$

A fixed integer m is therefore chosen to place this image in the neighborhood of Lemma 3.2. Continuity of the finite composition provides an interval I_* around λ_* where the same operations are defined. If N is the iterate supplied by that lemma, we set

$$d = 4 + m + N, \quad \mathcal{F}_\lambda = R^d f_\lambda, \quad \lambda \in I_*. \quad (44.5)$$

The family $\mathcal{F}$ has quadratic-like representatives of degree two, on a common domain and with uniformly positive modulus. Its real-analytic dependence admits a local complexification of the parameter. The number d is fixed and counts the doublings performed before applying the parameter theorem.

This construction fixes a common domain and a finite number of returns d , sufficient to apply the parameter theorem.

44.1.2 Transport of certified transversality

The direction of Γ satisfies the cone condition

$$\|\dot{v}\|_1 \leq \frac{1}{4} |\dot{u}|, \quad \dot{u} \neq 0.$$

The certified differential bounds imply invariance of this cone and

$$|\dot{u}_{k+1}| \geq 4.381 |\dot{u}_k| \quad (44.6)$$

along the orbit of $\Gamma(\lambda_*)$. All its points remain in the certified domain. In particular, the direction transported through the finite steps of (44.5) retains a nonzero expanding component.

The hybrid class of the fixed point coincides with its stable manifold in the space of quadratic-like germs: Lyubich, *Feigenbaum–Coullet–Tresser universality and Milnor’s hairiness conjecture*, Theorem 6.1, pp. 371–372. Transversality remains valid on passing to that space, as can be checked without identifying different Banach norms. Indeed, if $\partial_\lambda \mathcal{F}_{\lambda_*}$ were tangent to the hybrid class, it would be the tangent of an analytic disk contained in it. Uniform exponential convergence of renormalization on a compact subdisk, followed by Cauchy’s estimate in its parameter, would make the evaluation at $x = 1$ of its renormalized tangents converge to zero. However, in the preceding chart,

$$\dot{f}(1) = -\dot{u}/10,$$

and (44.6) makes its absolute value grow. Both derivatives represent the same germ and have the same real evaluation. The contradiction proves

$$\partial_\lambda \mathcal{F}_{\lambda_*} \notin T_{\mathcal{F}_{\lambda_*}} \mathcal{H}_{c_\infty}. \quad (44.7)$$

Thus, $S = \{\mathcal{F}_\lambda\}$ is a complex-analytic transversal to the hybrid class of the doubling limit.

44.1.3 Eventual identification of the first roots by their period

Denote by c_r the canonical quadratic center of period 2^r , obtained through r successive doublings of the period-one center. This symbol denotes the parameter of the quadratic recognition family; it is distinct from the orbital functions $c_j(\lambda) = f_\lambda^j(0)$ of the preceding sections. We write c_r^{quad} to preserve the distinction.

The global continuation theorem and (44.1) give $\lambda_n \rightarrow \lambda_*$, hence $\lambda_n \in I_*$ for sufficiently large n . The classification and restrictive intervals of section 41.2 show that, when $n > d$,

$$\mathcal{F}_{\lambda_n} = R^d f_{\lambda_n} \quad \text{has exact critical period } 2^{n-d} \quad \text{and itinerary } W_{n-d}0. \quad (44.8)$$

Its critical orbit remains in the invariant real interval, contained in the quadratic-like domain; its filled Julia set is therefore connected. Straightening preserves the postcritical orbit and the real branches. Following its restrictive returns gives $n - d$ canonical doublings and, finally, a fixed critical point. The latter straightens to the center 0; the successive inverses of the straightening maps of the doubling copies give the unique center c_{n-d}^{quad} . This is the concrete application of the combinatorial parameterization and the straightening homeomorphism of the copies described in Lyubich, sections 5.1–5.3, pp. 362–364. We conclude

$$\mathcal{F}_{\lambda_n} \in \mathcal{H}_{c_{n-d}^{\text{quad}}}. \quad (44.9)$$

Lyubich’s Theorem 7.4, pp. 387–388, gives a unique local intersection of S with each class $\mathcal{H}_{c_r^{\text{quad}}}$ for sufficiently large r . Its assumptions of a priori complex bounds hold for the real quadratic doubling cascade, according to Theorem 5.6, p. 367, of the same work. Write ζ_r for the parameter of that intersection. Equations (44.8)–(44.9), convergence to the crossing and local uniqueness imply

$$\boxed{\lambda_n = \zeta_{n-d} \quad \text{for every } n \text{ sufficiently large.}} \quad (44.10)$$

The uniqueness used here concerns **each canonical hybrid class** in a fixed neighborhood of the crossing. It is not inferred merely from the existence of a local cascade or from uniqueness of the crossing with the stable sheet. Equation (44.10) identifies the original selector, already defined globally, with the local intersections; it does not introduce a replacement selector.

In particular, let μ_j be the local cascade of section 42.8, and let s be the fixed integer determined by its exact physical period:

$$\text{per}_{\text{crit}}(f_{\mu_j}) = 2^{j+s}.$$

In the presentation with curve $R^4 f_\lambda$ and period-two target, $s = 5$; the additional fixed steps used to localize the target modify s . By (44.9) and the same uniqueness,

$$\mu_j = \zeta_{j+s-d}, \quad \boxed{\lambda_n = \mu_{n-s} \quad (n \text{ sufficiently large})}. \quad (44.11)$$

The shift d in the quadratic-like preparation and the shift s in the target periods are distinct objects. Identification is made through the physical period, not by equating those two integers.

44.1.4 Power-law remainder through transverse holonomy

Lyubich's Lemma 7.3, p. 384, with proof on pp. 384–387, provides $C^{1+\beta}$ -conformal regularity, for some $\beta > 0$, of the holonomy between transversals to the hybrid class of the Feigenbaum point. Its definition on p. 384 concerns the connectedness loci on the transversals; these contain, in particular, the centers considered here. We may reduce the exponent so that $0 < \beta \leq 1$.

Let h be the holonomy from S to the unstable manifold, and let z be an analytic coordinate linearizing the one-dimensional restriction of renormalization:

$$z(g) = 0, \quad z(Rq) = \delta_F z(q). \quad (44.12)$$

The existence of this coordinate follows by applying the linearization of a repelling holomorphic germ, with multiplier $\delta_F > 1$, to the analytic unstable manifold. The expanding direction of (44.6), cone invariance and the stationary return identify that multiplier with the positive unstable eigenvalue used in the certificate.

If q_r is the center of the class $\mathcal{H}_{c_F^{\text{quad}}}$ lying on the unstable manifold, compatibility of straightening and renormalization gives $Rq_r = q_{r-1}$. This identity is the one used in the proof of Theorem 7.4, p. 388. By (44.12), there exists $b \neq 0$, absorbing any fixed initial index, such that

$$z(q_r) = b\delta_F^{-r} \quad (44.13)$$

for every sufficiently large r .

The regularity of h and transversality (44.7) give, on the connectedness locus near λ_* ,

$$H(\lambda) := z(h(\mathcal{F}_\lambda)) = A(\lambda - \lambda_*) + O(|\lambda - \lambda_*|^{1+\beta}), \quad A \neq 0. \quad (44.14)$$

Since $H(\zeta_r) = b\delta_F^{-r}$, the lower estimate $|H(\lambda)| \geq |A| |\lambda - \lambda_*|/2$ in a sufficiently small neighborhood first implies $|\zeta_r - \lambda_*| = O(\delta_F^{-r})$. Substituting this bound into the remainder of (44.14) yields

$$\zeta_r - \lambda_* = \frac{b}{A} \delta_F^{-r} + O(\delta_F^{-(1+\beta)r}). \quad (44.15)$$

We apply (44.10) with $r = n - d$. The finite shift modifies the leading coefficient and the remainder constant. Since $\lambda_n < \lambda_*$, we obtain (44.2), with

$$C = -\frac{b}{A} \delta_F^d > 0, \quad \theta = \delta_F^{-\beta} \in (0, 1).$$

Finally,

$$\lambda_n - \lambda_{n-1} = C(\delta_F - 1)\delta_F^{-n}(1 + O(\theta^n)),$$

which proves (44.3). $\square$

44.1.5 Quantitative scope of the theorem

The identification of limits proved by first exit activates metric transfer for the global first roots of the uniform sector. The operator rate 0.83996 controls the renormalized maps on the stable sheet. The parameter rate $\theta = \delta_F^{-\beta}$ follows from the regularity of transverse holonomy. Both estimates retain their variable and domain.

The theorem establishes the existence of β, θ, C , the neighborhood I_* , the finite number of returns d and an index n_0 beyond which the centers coincide. Their individual numerical values are not assigned in this asymptotic statement. This formulation completes the limit and its power-law remainder; numerical evaluation of those witnesses is a distinct quantitative specialization.

45 Return, memory and universality of the dodecaphase cascade

45.1 The generating structure and the critical reading

The Feigenbaum construction in Triadic Modular Holography links an oriented arithmetic organization to an asymptotic scaling law. Its starting point is the effective composition of APP, TRIT and TPK. APP evaluates sum and product on the two sheets of the digit lattice and preserves the quotients and remainders of those operations. TRIT determines the local regime and the orientation of its reading. TPK selects and transports the states, updates their records and prolongs their routes with carry, boundary and memory. The joint discrete structure of the continuum provides the common residence of those states and of the readers acting on them.

The prolongation

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

organizes this persistence. The initial seed is lifted while preserving its prefixes; successive lifts change the regime without losing orientation or memory; the coinductive boundary allows the construction to continue, and the nine charts bring together its local readings. Nonadic holonomy expresses phase return accompanied by an increase in memory. Thus, a visible turn preserves its position within the complete history. From the outset, the return operation acquires two inseparable components: reiteration of a rule and preservation of its provenance.

The critical reader fixed on the oriented dodecaphase wheel turns this structure into twelve phases with a uniform trace. Their representatives are

$$\phi_m = \frac{(3m-1)\pi_{\text{HMT}}}{18}, \quad w_m = \frac{1}{12}, \quad 1 \leq m \leq 12.$$

The choice of oriented representative matters because the second moment uses its magnitude. The angular quotient records the phase, while the representative retains the information needed to normalize it. Coordinate π_{HMT} comes from the coinductive readers of the same APP–TRIT–TPK core. Within its correlated image, coordinates π, φ, e and the dodecaphase closure of α_{HMT} retain their constructive order and their own domains. The critical reader considered here uses the angular coordinate already generated.

Normalization is determined through an integer sum:

$$\sum_{m=1}^{12} (3m-1)^2 = 5394 = 6 \cdot 899, \quad k_m = \frac{2(3m-1)}{\sqrt{899}}.$$

The common factor π_{HMT} cancels when the second moment is fixed. This cancellation preserves the provenance of the phases and produces a particularly precise representation of their normalized reading. We obtain

$$u_0(x) = \frac{1}{12} \sum_{m=1}^{12} (1 - \cos(k_m x)), \quad f_\lambda(x) = 1 - \lambda u_0(x), \quad \sum_m w_m k_m^2 = 2.$$

The profile is even and entire; it satisfies $u_0(0) = u'_0(0) = 0$ and $u''_0(0) = 2$. For $0 < \lambda \leq 2/u_0(1)$, the family maps $[-1, 1]$ into itself and has a quadratic critical maximum with $f''_\lambda(0) = -2\lambda$. Parameter λ ranges over this family after the reader, orientation, weights and scale have been fixed. The universal number recognized at the end is thus separated from the selection of the coefficients.

45.2 Twelve phases, moments and spectral representation

The profile admits a second expression bringing its phases together in a finite operator. The nodes

$$s_m = k_m^2 = \frac{4(3m-1)^2}{899}$$

determine the positive measure $\mu_0 = 12^{-1} \sum_m \delta_{s_m}$. Its moments $M_r = \int s^r d\mu_0(s)$ define strictly positive Hankel forms up to rank twelve. This positivity follows from evaluating a polynomial at twelve distinct nodes: a polynomial of degree less than twelve retains at least one nonzero evaluation. At degree twelve, the product of the twelve nodal factors annihilates the norm and determines the reader's rank exactly.

Orthogonalization of these moments produces a Jacobi matrix J_{12} , real symmetric, with positive subdiagonal entries. Its functional calculus recovers the complete profile:

$$u_0(x) = e_0^T [I - \cos(x\sqrt{J_{12}})] e_0.$$

The cosine sum and the Jacobi operator are two realizations of the same reader. The moments retain all frequencies and their weights; the orientation, route and memory marks remain in the state that generates them. Rank twelve identifies the spectral dimension of this specific reading. The depth of its prolongations belongs to another dimension of the construction: the length of compatible histories and the resolution at which they are evaluated.

The analytic geometry is also made explicit. For the weighted deformation u_η , in the strip $|\eta| \leq 1/40$, the weights remain positive and the second moment remains normalized. In $|\operatorname{Re} z| < \pi/3$ there exists an odd, univalent holomorphic function b_η , normalized by $b'_\eta(0) = 1$, such that $u_\eta = b_\eta^2$. The wheel therefore produces a conformal deformation of the quadratic fold. Its dynamical conjugacy preserves that deformation through $b_\eta(1 - \lambda w^2)$. The parameter-scaling theorem developed here concerns the uniform member $\eta = 0$; the structural properties of the weighted strip and the local persistence of its roots retain their specific quantifiers.

45.3 Reading depth and recovery of histories

Nonadic depth refines the reading of a parameter, whereas dynamical doubling changes the return horizon. Refinement with carry expresses the first operation through

$$L_c(x, y) = (Bx - c, By + c), \quad C_m = \sum_{j=1}^m c_j B^{m-j},$$

$$(x_m, y_m) = (B^m x_0 - C_m, B^m y_0 + C_m).$$

The outer support grows as B^m , and the normalized record cylinder has diameter B^{-m} . The integer pair, depth and boundary mark preserve reconstruction of quotients, remainders and prefixes. Outer growth and inner resolution thus describe compatible operations on the same history.

The exact composition $L_D^{[b]} L_C^{[a]} = L_{B^b C + D}^{[a+b]}$ allows a history to be grouped into blocks without altering its data. A doubling return combines two transitions; the nonadic calendar combines nine. Both groupings are compared over a common horizon of $9 \cdot 2^N$ events, preserving the order of the subblocks and their carries.

The contribution of memory can be observed directly in the profile's inverse branches. If $v = (u_0|_{[0,1]})^{-1}$, the two branches are

$$\beta_\sigma(y) = \sigma v\left(\frac{1-y}{\lambda}\right), \quad \sigma \in \{-1, +1\}.$$

The final value, together with the last sheet, reconstructs the preceding value. A sequence of sheets allows all intermediate steps to be recovered. The branch word accompanies the APP-TRIT-TPK

records of route, boundary, carry and memory; each record retains its function. This concrete recovery explains what information is required to invert the critical reading.

The resolution of each return also has an explicit bound. With $t = \lambda u_\eta(1)/2$, $z = (x + 1)/2$ and

$$F_{t,\eta}(z) = 1 - t \frac{u_\eta(2z - 1)}{u_\eta(1)},$$

one obtains

$$|F_{t,\eta}^q(1/2) - F_{s,\eta}^q(1/2)| \leq S_q |t - s|, \quad S_q = \sum_{j=0}^{q-1} (6\sqrt{2})^j < 9^q.$$

A base-nine cylinder of depth $m = 2^N + r$ therefore allows the return of horizon 2^N to be read with diameter less than 9^{-r} . This estimate gives quantitative content to arbitrary depth: for every requested horizon and precision, it determines a sufficient depth. Evaluation of the analytic functions and rounding have their own remainders, which propagate together with the parameter uncertainty.

Vacancies enter the capacity balance of the record. Comparison between blocks of 729 and 1000 possibilities distinguishes accumulated capacity from chronological length, preserving the steps at which the former remains constant. This accounting determines the space required to preserve a reading and its depth. The admissibility of histories follows, in turn, from the survival rules. Both operations contribute to the construction and retain distinct mathematical objects.

45.4 Nonadic chronology and doubling combinatorics

The critical doubling word has an expression at every depth:

$$\tau_j = (-1)^{v_2(j)}, \quad \tau_{2j} = -\tau_j, \quad \tau_{2j+1} = +1.$$

Its prefixes satisfy $W_{n+1} = W_n(-1)^n W_n$. The rule determines how the two copies of the return are oriented and where their separation appears. The nonadic reading of chronology preserves the complete integer $K = \rho_9(K) + 9q_9(K)$. Reducing it modulo 2^n , the holonomy that advances nine events induces translation by nine on that cycle. Since nine is odd, this translation visits all its positions. The permutation $j \mapsto 9j \pmod{2^n}$, which conjugates it to unit advance, preserves the dyadic valuation of each nonzero position. The chronological remainder and quotient jointly support that compatibility.

Bilateral memory also has an operator realization compatible with refinement. On cycles of length 2^n , the shift operator C_n is incorporated into

$$\mathbb{U}_n = P_0 \otimes I + P_1 \otimes C_n,$$

with the complementary projectors of the nonadic reader. In the limit, its visible and complementary components satisfy

$$\|T_{\infty,a}v\|^2 + \|D_{\infty,a}v\|^2 = \|v\|^2, \quad v = T_{\infty,a}^* T_{\infty,a}v + D_{\infty,a}^* D_{\infty,a}v.$$

These identities describe preservation and recovery within the declared space and representation. Contraction of the renormalized maps, which will appear later, measures another operation: the approach of their profiles to a stable form. The history preserved by the lift and the convergence of a functional reading are articulated at different levels.

45.5 The ordered chart and global root selection

The Archimedean readout of the HMT continuum transports the operations through an isomorphism of complete ordered fields, within the declared universe $\mathfrak{G}$:

$$\iota : \mathbb{R}_{\text{HMT}}^{\mathfrak{G}} \longrightarrow \mathbb{R}^{\mathfrak{G}}.$$

The internal construction of the cosine through its series, the positive square root of 899, the profile and its iterates precedes zero selection. Preservation of sum, product, order and convergent limits gives

$$\iota(S_n^H(a)) = S_n(\iota(a)), \quad S_n(\lambda) = f_\lambda^{2^n}(0).$$

Surjectivity covers all roots in the domain of that chart; order preserves preceding returns, exact period and the condition of lying to the right of the preceding root. In this way, the minimum of each candidate set is also transported. The genealogy reaches the effective parameter selection, with the second projection and its marks preserved on the same antecedent.

The selector is fixed by $\lambda_1 = 1/u_0(1)$ and by the first new root of exact critical period 2^n lying to the right of λ_{n-1} . The itinerary classification proves that each selected root has word $W_n 0$. Restrictive returns halve its period while preserving unimodal order, and induction restores the complete word. Analyticity of returns that are not identically zero isolates their zeros in each compact set; comparison of signs before the first infinite-itinerary parameter guarantees a new root at every level.

Let Λ_τ be the nonempty compact set of parameters realizing the infinite word, and let a_* be its minimum. It is proved that

$$\lambda_n \nearrow a_*.$$

Passage to the limit preserves each sign through a uniform bound at a fixed horizon. If $c_p = f_\lambda^p(0)$, the opposite signs of c_p and c_{2p} , together with $(f_\lambda^p)'(0) = 0$, imply

$$|c_p| > \frac{2}{B_p}, \quad B_p = 16^p \frac{16^p - 1}{15}.$$

Thus, the prefixes of increasing periods preserve the infinite word as they accumulate. The proof uses the original sequence of first roots from its global definition.

45.6 The stable sheet and identification of the selected limit

The normalized return

$$(Rf)(x) = -\frac{f(f(-ax))}{a}, \quad a = -f(1) > 0,$$

combines two transitions, changes the scale and preserves the orientation needed to restore a maximum of value one. The restrictive domains establish its interpretation as a real return. The derivative of this operator includes the variation of scale a ; that contribution determines the parameter sensitivity of the complete transformation.

In the chart $f(x) = 1 - x^2 V((x^2 - 1)/(5/2))$, with coordinates $u = 10v_0$ and $v = (v_1, v_2, \dots)$, the fourth return of the family enters a quantified neighborhood of the fixed point g . The curve crosses its stable sheet transversely at a unique local parameter

$$1.874038 < \lambda_* < 1.874039,$$

and satisfies

$$\|R^{4+n} f_{\lambda_*} - g\|_A < 0.001666(0.83996)^n.$$

The stable sheet is a graph of slope at most 0.16. The parameter secants belong to a transverse expanding cone. Both properties distinguish the decrease in stable deformations from the growth of separation between parameters.

Identifying a_* with λ_* brings this local geometry together with the global minimum. First, the interval between the eighth root and the lower endpoint of the local box is excluded. Then the order $a_* \leq \lambda_*$ allows a hypothetical separation to be studied through the negative cone. While its transverse coordinate has absolute value less than 0.006, the secant remains in the analytic domain and grows by a factor between 4.4034 and 8.0887. A persistent separation would therefore produce a first exit with amplitude between 0.006 and 0.0485322.

The stable-orbit residual retains information more precise than its norm: its components satisfy $|e_u| \leq 0.16\|e_v\|_1$. This relation allows their effects on the critical orbit to be bounded jointly. The cover

of the entire first-exit region shows in each case a sign incompatible with τ , or a contracting return forcing two successive dyadic critical values to have the same sign. The infinite word requires opposite signs. The contradiction identifies the parameters:

$$\lambda_n \nearrow a_* = \lambda_*.$$

The proof combines an induction valid at every depth with a uniform finite exclusion of the region that any separation would reach. Its infinite scope rests on that composition.

45.7 Universal scaling and the meaning of precision

The degree-two holomorphic realization of the returns and the established transversality allow their hybrid classes to be compared. For each sufficiently high period, the canonical class has a unique local intersection with the family. The global first roots already converge to the crossing and preserve the doubling combinatorics; hence they eventually coincide with the local centers when their physical periods are matched. The proof transports the original selection to the universal metric law.

There exist $C > 0$, $M < \infty$ and $0 < \theta < 1$ such that

$$\lambda_* - \lambda_n = C\delta_F^{-n}(1 + e_n), \quad |e_n| \leq M\theta^n,$$

and, consequently,

$$\lim_{n \rightarrow \infty} \frac{\lambda_{n-1} - \lambda_{n-2}}{\lambda_n - \lambda_{n-1}} = \delta_F.$$

The factor δ_F is the expanding eigenvalue of the doubling operator recognized at the end of the construction. The specific family preserves its twelve phases, moments and provenance history; the limit of its parameter ratios belongs to the same universality class of quadratic criticality. This relation jointly explains the specificity of the generator and the universality of the output.

Precision has three residences here. The root intervals control their finite evaluation. The estimate $0.001666(0.83996)^n$ controls operator convergence on the stable sheet. The term $M\theta^n$ controls the transfer of that geometry to the parameter scale through transverse holonomy. Each estimate corresponds to its variable and operation. The eighth quotient is enclosed around $4.669212189150204154556096215889663928$, with width less than $5.808 \cdot 10^{-37}$; that width measures the evaluation of $\delta_{F,8}$. The distance from $\delta_{F,8}$ to the limit requires quantifying the constants of the parameter remainder, whose existence is already proved.

The notation keeps separate δ_F , the Feigenbaum parameter-scaling factor; α_F , the spatial constant associated with its renormalization problem; and α_{HMT} , the corpus's fine-structure constant. The present result identifies δ_F through the first roots of the uniform sector $\eta = 0$. The weighted strip retains its structural and existence theorems, and its quantified metric extension constitutes a distinct specific statement.

The contribution of the chain lies in continuity between generation, reading, memory, selection and limit. Oriented arithmetic determines the profile; the spectral representation preserves its twelve components; the cylinders allow it to be resolved with arbitrary precision at each horizon; history transport restores its paths; the ordered chart preserves all roots and their minima; classification prolongs the selector; and transverse expansion, together with stable contraction, determines its asymptotic law. Universality emerges as a proved property of this complete construction and its analytic realization.

46 Articulation of the critical, spectral and constitutive readers

The APP–TRIT–TPK composition precedes the readers in this section. APP preserves the additive and multiplicative evaluations in each cell, with their remainders and quotients; tritic reduction determines the regime, orientation and carry of the transition. The effective composition of selection, transport and memory update produces the enriched state and its prolongations, developed in 36.1 and 36.1.1. Nonadic holonomy composes those transitions: phase return advances memory and preserves history. The readers act on the same joint discrete structure of the continuum, with its five consubstantial constructions and with the two readouts of the same antecedent: ordered evaluation and boundary incidence.

The oriented wheel and angular channels are outputs of this composition. Their already generated HMT coordinates are used in subsequent operations, with their marks and domains; they are not metrological numbers introduced to select an answer. The ordered readout allows the following formulas to be evaluated, while the provenance state preserves sheet, route, remainder, quotient, carry, boundary and memory. The recoveries that are proved recover the object of their declared domain; the evaluated representation remains accompanied by those records.

46.1 Critical separation as an intrinsic asymptotic reading

In the uniform sector of the dodecaphase wheel, the oriented representatives and their weights are

$$\phi_m = \frac{(3m-1)\pi_{\text{HMT}}}{18}, \quad w_m = \frac{1}{12}, \quad 1 \leq m \leq 12.$$

The second moment is normalized through $\sum_{m=1}^{12} (3m-1)^2 = 5394 = 6 \cdot 899$. Thus the common angular factor cancels, leaving

$$k_m = \frac{2(3m-1)}{\sqrt{899}}, \quad u_0(z) = \frac{1}{12} \sum_{m=1}^{12} (1 - \cos(k_m z)), \quad f_\lambda(z) = 1 - \lambda u_0(z). \quad (46.1)$$

Variable λ ranges over the family after the reader, its weights and normalization have been fixed. In $0 < \lambda \leq 2/u_0(1)$, the dynamical domain is $[-1, 1]$. The first root is $\lambda_1 = 1/u_0(1)$; for $n \geq 2$, λ_n is the smallest root greater than λ_{n-1} whose critical point has exact period 2^n . The classification, existence of each minimum and metric identification of these centers have been proved in 43.1–43.3.

Proposition 46.1 (Structural reading of the scaling constant). *For the uniform reader (46.1) and its first-root selector, the critical-separation constant is*

$$\delta_{\text{HMT,crit}} := \lim_{n \rightarrow \infty} \frac{\lambda_n - \lambda_{n-1}}{\lambda_{n+1} - \lambda_n} = \delta_F. \quad (46.2)$$

It is the asymptotic parameter-separation invariant of the critical returns of that reader, with its selection order preserved.

Proof. The scaling theorem (43.18) provides $\lambda_* - \lambda_n = C\delta_F^{-n}(1+e_n)$, with $C > 0$ and $e_n \rightarrow 0$. Subtracting two consecutive terms,

$$\lambda_n - \lambda_{n-1} = C\delta_F^{-n}[\delta_F - 1 + \delta_F e_{n-1} - e_n].$$

The quotient of this equality and the one at index $n+1$ converges to δ_F . The eventual identity of the local centers with the first roots, proved in (43.17), allows the scaling to be applied precisely to the selected sequence and not to another cascade chosen afterward. $\square$

This formulation preserves more information than the terminal numeral: it identifies the wheel, its reading, the family and the selector whose limit is evaluated. Its interpretation is intrinsic to that HMT composition; recognition of the universal constant subsequently uses the analytic theorem applied to the already constructed reader.

46.2 The finite measure and spectral recovery of the profile

The squared frequencies produce the atomic probability measure

$$s_m = k_m^2 = \frac{4(3m-1)^2}{899}, \quad \nu_0 = \frac{1}{12} \sum_{m=1}^{12} \delta_{s_m}, \quad M_j = \int s^j d\nu_0(s). \quad (46.3)$$

Here δ_{s_m} is the Dirac mass at a node, distinct from the constant δ_F . The measure preserves the twelve frequencies and their weights. For a nonzero polynomial of degree less than $r \leq 12$,

$$\int p(s)^2 d\nu_0(s) = \frac{1}{12} \sum_{m=1}^{12} p(s_m)^2 > 0,$$

since a polynomial of degree less than twelve cannot vanish at the twelve distinct nodes. Thus the Hankel matrices $(M_{i+j})_{0 \leq i, j < r}$ are positive definite. The polynomial $\prod_m (s - s_m)$ has zero norm; the space $L^2(\nu_0)$ has dimension twelve.

Proposition 46.2 (Exact representation of the critical reader). *Let $\psi_0, \dots, \psi_{11}$ be the orthonormal polynomials with positive leading coefficient for ν_0 , with $\psi_0 = 1$. Multiplication by s has Jacobi matrix J_{12} , real symmetric with positive subdiagonal entries. With $e_0 = (1, 0, \dots, 0)^T$,*

$$u_0(z) = \int (1 - \cos(z\sqrt{s})) d\nu_0(s) = e_0^T [I - \cos(z\sqrt{J_{12}})] e_0. \quad (46.4)$$

Proof. The preceding positivity allows orthogonalization through degree eleven. If $i < j - 1$, self-adjointness of multiplication gives $\langle s\psi_j, \psi_i \rangle = \langle \psi_j, s\psi_i \rangle = 0$. By symmetry, only the diagonal and its two neighbors remain; the positive ratio of the leading coefficients establishes positivity of the subdiagonal. Define

$$Q_{mj} = \frac{\psi_j(s_m)}{\sqrt{12}}, \quad D = \text{diag}(s_1, \dots, s_{12}).$$

Orthonormality gives $Q^T Q = I$, and evaluating the recurrence at the nodes gives $DQ = QJ_{12}$. Since Q is square, $J_{12} = Q^T DQ$. Its eigenvalues are positive, so the positive square root and cosine are calculated in this diagonalization. Finally, $Qe_0 = (1, \dots, 1)^T / \sqrt{12}$, which proves (46.4). $\square$

The realization through (J_{12}, e_0) recovers the same profile; therefore, preserving the family and selector also recovers its returns and first roots. Spectral positivity justifies this representation. Parameter transversality also uses the derivative of the return and its orbital products, developed in 40.1.3 and 42.6. These are properties of different operations: the scaling proof preserves both and does not replace tangent control with positivity of the moments.

46.3 Constitutive channels and their labeled inversion

The constitutive reader acts on another typed object on the same support: two-sheet angular transport. Its already generated coordinates are $x = A_{\text{rad}}$ and $y = C_{\text{rad}}^*$, in the contracting chamber $x > |y|$. The sheet involution $R = R^* = R^{-1}$ determines the marked projectors $P_{\pm} = (I \pm R)/2$. The transport representation is

$$q_+ = \exp[-(x + y)], \quad q_- = \exp[-(x - y)], \quad T = q_+ P_+ + q_- P_-, \quad 0 < q_{\pm} < 1. \quad (46.5)$$

Exponential evaluation realizes those channels after their angular construction. The transport incidences 90 and 120 fix the response

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1}.$$

The inverse exists because $1 - q_{\pm}^{120} > 0$. Writing $s = q_{\pm}^{30}$, with $30 = \gcd(90, 120)$, gives

$$f(s) = \frac{1 - s^3}{1 - s^4} = \frac{1 + s + s^2}{1 + s + s^2 + s^3}, \quad r_{\pm} = f(q_{\pm}^{30}), \quad \mathcal{V} = r_+ P_+ + r_- P_-. \quad (46.6)$$

In this subsection f denotes the constitutive response; f_λ still denotes the dynamical map of (46.1). Their arguments and domains are distinct.

Proposition 46.3 (Recovery of the channels and orientation). *The constitutive function $f : (0, 1) \rightarrow (3/4, 1)$ is a decreasing bijection. Its inverse ψ assigns to r the unique root $s \in (0, 1)$ of*

$$rs^3 + (r - 1)(1 + s + s^2) = 0. \quad (46.7)$$

The normalized constitutive readings, with their labeled sheets,

$$\hat{\varepsilon} = r_+^2, \quad \hat{\mu} = r_-^2, \quad \hat{Z} = \frac{r_-}{r_+}, \quad \hat{c} = \frac{1}{r_+ r_-},$$

with $\hat{\varepsilon}, \hat{\mu} \in (9/16, 1)$, allow recovery of

$$\begin{aligned} r_+ &= \sqrt{\hat{\varepsilon}}, & r_- &= \sqrt{\hat{\mu}}, \\ s_\pm &= \psi(r_\pm), & q_\pm &= s_\pm^{1/30}. \end{aligned} \quad (46.8)$$

$$x = -\frac{1}{2} \log(q_+ q_-), \quad y = \frac{1}{2} \log \frac{q_-}{q_+}. \quad (46.9)$$

Proof. Differentiating the rational quotient gives

$$f'(s) = -\frac{s^2(3 + 2s + s^2)}{(1 + s + s^2 + s^3)^2} < 0 \quad (0 < s < 1). \quad (46.10)$$

The limits $f(0^+) = 1$ and $f(1^-) = 3/4$, together with continuity, prove bijectivity. Multiplying $f(s) = r$ by the positive denominator produces (46.7). Positivity of $r_\pm$ determines the square roots and that of $s_\pm$ determines the thirtieth roots. By (46.5), $\log q_+ = -x - y$ and $\log q_- = -x + y$; adding and subtracting recovers x, y . Conversely, the recovered roots belong to $(0, 1)$, so that $x \pm y = -\log q_\pm > 0$; the resulting pair remains in the chamber $x > |y|$. $\square$

In particular, $\hat{\mu}\hat{\varepsilon} = \hat{c}^{-2}$ and $\hat{\mu}/\hat{\varepsilon} = \hat{Z}^2$. Exchanging the labels permutes q_+ and q_- , preserves x and changes the sign of y . Recovery requires that orientation: the isolated product $r_+ r_-$ preserves common propagation, but does not determine the two labeled responses. The positive roots and cubic inversion recover the channels within the fixed reader; the complete record of their generation remains in the enriched state.

46.4 The Catalan moment and functional recovery

The same incidences determine a relation between functions, before either channel is evaluated. For $0 < s < 1$, let $\mathcal{D} = s \, d/ds$, and define the oriented moment

$$\mathcal{C}_2(s) = \int_0^s \log \frac{s}{t} \frac{1+t}{1+t+t^2} f(t) \, dt = \int_0^s \frac{\log(s/t)}{1+t^2} \, dt. \quad (46.11)$$

The second equality follows from the factorization $1 + t + t^2 + t^3 = (1+t)(1+t^2)$. The continuous positive weight $dt/(1+t^2)$ is thus produced by the rational composition of the constitutive reader, not by an identification with ν_0 .

Proposition 46.4 (Differential inversion, uniqueness and endpoint reading). *The moment $\mathcal{C}_2$ is the unique function $F \in C^2((0, 1))$ satisfying*

$$\mathcal{D}^2 F(s) = \frac{s(1+s)}{1+s+s^2} f(s), \quad \lim_{s \downarrow 0} F(s) = 0. \quad (46.12)$$

The constitutive response is recovered through

$$\mathcal{D}^2 \mathcal{C}_2(s) = \frac{s}{1+s^2}, \quad f(s) = \frac{1+s+s^2}{s(1+s)} \mathcal{D}^2 \mathcal{C}_2(s). \quad (46.13)$$

Its continuous extension to the closed interval has the series

$$\mathcal{C}_2(s) = \sum_{j=0}^{\infty} \frac{(-1)^j s^{2j+1}}{(2j+1)^2}, \quad 0 \leq s \leq 1, \quad \mathcal{C}_2(1) = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^2} = G_{\text{Cat}}. \quad (46.14)$$

Proof. The integrand of (46.11) is nonnegative and

$$0 \leq \mathcal{C}_2(s) \leq \int_0^s \log(s/t) dt = s.$$

This proves integrability at zero and the required limit. The identity $\log(s/t) = \int_t^s du/u$, followed by interchanging nonnegative integrals, provides

$$\mathcal{C}_2(s) = \int_0^s \frac{1}{u} \left(\int_0^u \frac{dt}{1+t^2} \right) du.$$

On every compact interval of the open domain, the fundamental theorem of calculus applies. Therefore,

$$\mathcal{D}\mathcal{C}_2(s) = \int_0^s \frac{dt}{1+t^2}, \quad \mathcal{D}^2\mathcal{C}_2(s) = \frac{s}{1+s^2}.$$

The rational factorization used in (46.11) gives (46.12); solving for f gives (46.13), whose denominators are positive in $(0, 1)$.

If two solutions have the required zero limit, their difference H satisfies $\mathcal{D}^2H = 0$. The substitution $z = \log s$ transforms this equation into $d^2H(e^z)/dz^2 = 0$, whence $H(s) = a \log s + b$. Existence of the zero limit forces $a = b = 0$, proving uniqueness.

For $s < 1$, the geometric expansion of $(1+t^2)^{-1}$ can be integrated term by term with the logarithmic weight, since

$$\int_0^s t^{2j} \log(s/t) dt = \frac{s^{2j+1}}{(2j+1)^2}, \quad \sum_{j \geq 0} \frac{s^{2j+1}}{(2j+1)^2} < \infty.$$

This proves the series in the interior. The bound $1/(2j+1)^2$ gives absolute and uniform convergence of that series on $[0, 1]$. In the integral, the integrand extended by zero for $t > s$ is dominated by $\log(1/t)$ on $(0, 1)$. Dominated convergence allows us to reach $s = 1$ and agrees with the extension of the series. Its endpoint value is precisely the stated Catalan series. $\square$

Differential inversion uses the family $\mathcal{C}_2(s)$, not only its endpoint value G_{Cat} . In particular, $\mathcal{D}^2\mathcal{C}_2(s) \rightarrow 1/2$ when $s \uparrow 1$. The series differentiated twice with $\mathcal{D}$ has a radial Abel limit there; the series of the moment itself is absolutely convergent at the boundary. This distinction preserves the precise conditions of each passage to the limit.

46.5 Composition of the recoveries and joint scope

The preceding interrelations have explicit operations and domains. The moments of the atomic measure ν_0 produce J_{12} , and its functional calculus recovers the profile u_0 . The profile and first-root selector determine the sequence whose scaling yields the readout δ_F . On the other hand, the constitutive function f produces $\mathcal{C}_2$ through the integral moment, and (46.13) recovers f from that family. Its labeled evaluations at $s_{\pm} = q_{\pm}^{30}$ determine the constitutive responses; cubic inversion recovers both channels and their angular coordinates. The endpoint reading of the moment gives G_{Cat} .

The measure ν_0 is positive, atomic and of rank twelve. The Catalan moment integrates a continuous positive weight and has an alternating expansion. The signs of that expansion are coefficients of the geometric series; they are not negative weights of ν_0 , nor do they turn the positive integral into the same spectral representation. Each reader retains its support, orientation, normalization and evaluation operation.

Unity lies in the APP–TRIT–TPK structure that generates and transports their antecedents; the functional relations are established through the proved compositions. Equality of origin does not identify operators with different domains. In particular, this articulation does not postulate a scalar formula determining δ_F from Catalan’s constant and other physical constants. It places critical scaling alongside effective recoveries between readers on the same support, preserving the information required by each direction.

47 Joint normalization of action, length, and gravitational coupling

The determination of a quantity is preserved when the readouts of the same state are composed. Here the inscribed return, the action section, the constitutive speed, and the positive route character are brought together. Their common antecedent is APP-TRIT-TPK generation, with its sheets, orientation, carry, memory, and prolongations; the regional coordinates and α enter as outputs of that generation. The following construction uses those outputs before performing the dimensional recognition of G .

The normalization problem comprises three distinct operations: producing a length from the recorded return, transporting the character without losing its complement, and comparing the resulting energy and length readouts. The identities in this section jointly fix those operations. Their proofs concern finite fibres or bounded, uniformly positive operators; the null sector retains its separate treatment.

47.1 Inscribed return and longitudinal section

The marked carrier of the transport retains the support

$$\Omega = \{(j, k, q, u, v) : j \in \mathbb{Z}/12\mathbb{Z}, k - j \in \{0, 3, 6, 9\}, 1 \leq q \leq 8, u, v \in \{1, 2, 4, 5, 7, 8\}, u < v\}.$$

The index j retains the dodecaphase phase; the four displacements retain the quarter-turn, and the remaining indices retain the marked incidence. Thus $N = |\Omega| = 12 \cdot 4 \cdot 8 \binom{6}{2} = 5760$ counts incidence positions. The space of this register is transported together with genealogical memory; its finite dimension belongs to this particular carrier.

Let $B : V_U \rightarrow V_K$ be the transported Hadamard inverter, $B^*B = BB^* = I/4$, $u_K = N^{-1/2}(1, \dots, 1)$, $P_K = u_K u_K^*$ and $Q_K = I - P_K$. Define

$$C_0 = Q_K B, \quad M_0 = P_K B, \quad U = 2B.$$

Orthogonality of the two projectors proves

$$C_0^* C_0 + M_0^* M_0 = I/4, \quad C_0^* M_0 = 0, \quad U^* U = U U^* = I.$$

Thus $x \mapsto (2C_0 x, 2M_0 x)$ preserves the complete archive of this transport. U is its unitary normalization; the enriched evolution U_t retains its own domain and update law.

Let $E_i = e_i e_i^*$ be the resolution of the register and $\Delta_\Omega(A) = \sum_i E_i A E_i$ its diagonal expectation.

Lemma 47.1 (Inscribed longitudinal return). *The return and its inscribed readout satisfy*

$$C_0 C_0^* = Q_K/4, \quad \Delta_\Omega(C_0 C_0^*) = r_N I, \quad r_N = \frac{N-1}{4N} = \frac{5759}{23040}.$$

With the local scalar norm $n_H = \alpha^{16}$ of the cokernel of H_4 , the longitudinal composition on a length cochain β_* is

$$\mathcal{T}_L(\beta_*) = n_H(\Delta_\Omega(C_0 C_0^*) \otimes I) \beta_* = q_\alpha \beta_*, \quad q_\alpha = \alpha^{16} r_N.$$

Proof. The equality $BB^* = I/4$ gives $C_0 C_0^* = Q_K/4$. Every diagonal entry of P_K equals $1/N$, proving the expectation. Multiplying a cochain by that scalar preserves its longitudinal type and gives the last equality. If $\beta_*(\bar{a}) = -U_a \beta_*(a)$, the image satisfies the same reversal. Likewise, $\beta_m(a) = \sum_j U_j^{-1} \beta_{m+1}(a_j)$ yields the refined identity by multiplying both sides by q_α . $\square$

The expectation has an explicit inscribed realization: $W e_i = e_i \otimes e_i$ and $W^*(A \otimes I)W = \Delta_\Omega(A)$. For $R_K = Q_K/4$ and $R_W = WW^*$, the complement $Z_\perp = (I - R_W)(R_K \otimes I)W$ satisfies

$$(R_K \otimes I)W = W r_N + Z_\perp, \quad W^* Z_\perp = 0, \quad Z_\perp^* Z_\perp = \frac{N-1}{16N^2} I.$$

Indeed, $R_K^2 = R_K/4$, so the squared full norm is $r_N/4$, whereas the squared norm of the inscribed component is r_N^2 . Their difference is precisely the stated complement. Preservation of that complement accompanies the longitudinal readout; adding it to another readout would change the observable.

The expectation is unique among bimodular maps into the diagonal algebra that fix its projectors: on a matrix unit E_{ij} , bimodularity requires $\mathcal{E}(E_{ij}) = E_i \mathcal{E}(E_{ij}) E_j$, which is zero if $i \neq j$, and $\mathcal{E}(E_{ii}) = E_{ii}$. This uniqueness preserves the marked resolution.

With the reference longitudinal section L_* already specified, one obtains

$$L_\alpha = q_\alpha L_*, \quad D = L_\alpha U. \quad (47.1)$$

The readout operator uses the return as a linear operator on a length. A quadratic readout of the source would produce a different type of quantity. The section L_* , the inscribed resolution, and the local norm are explicit antecedents of this composition.

47.2 Polar transport and reciprocity of the same character

Let X be the operator character of the non-null positive sector, retaining the route and its normalizations. In finite dimension $X > 0$ is required; in infinite dimension X must be bounded, self-adjoint, and satisfy $X \geq \epsilon I$ for some $\epsilon > 0$. Write $Y = UXU^*$. The polar radial representation is

$$R = (DX^2D^*)^{1/2}.$$

The positive square root is unique, and substituting $D = L_\alpha U$ yields $R = L_\alpha Y$. Equivalently, $\|Rv\|^2 = \|XD^*v\|^2$ for every v : polarization recovers $R^2 = DX^2D^*$, and positivity determines R . This condition specifies the induced metric that is preserved.

The action $\hbar$ and the constitutive speed c belong to the same section. The compatible circular realization fixes

$$\omega_0 = \frac{c}{L_\alpha}, \quad t_0 = \frac{\pi_{\text{HMT}} L_\alpha}{54c}, \quad R_{108} = \frac{54}{\pi_{\text{HMT}}} c t_0 = L_\alpha. \quad (47.2)$$

The length has been constructed before the circular clock is expressed. This equality allows the notation R_{108} to be retained in subsequent developments and fixes its normalization. The dimensional bases are transported jointly; L_* and the edge basis u_L have different roles.

Theorem 47.2 (Joint determination of radial coupling). *The readout operators on the same character,*

$$H = \frac{\hbar c}{L_\alpha} Y, \quad \Lambda = L_\alpha Y^{-1}, \quad R = L_\alpha Y, \quad M = H/c^2,$$

satisfy

$$H\Lambda = \hbar c I, \quad R\Lambda = L_\alpha^2 I, \quad R = \frac{L_\alpha^2}{\hbar c} H. \quad (47.3)$$

In the dimensional identification of R as the reduced gravitational radius, $R = GM/c^2$, the common coefficient is unique:

$$G = \frac{c^3 L_\alpha^2}{\hbar}. \quad (47.4)$$

Proof. Multiplication of the readout operators gives the first two identities. Multiplying them on the right by Λ^{-1} proves the third. Substituting $M = H/c^2$ identifies the coefficient. If G' preserves the same readouts, $(G' - G)M/c^2 = 0$; in the non-null fibre, invertibility of M implies $G' = G$. All products are defined on the complete fibre under the hypotheses of the statement. $\square$

The proof applies to every positive mode of this realization, without choosing the electronic mass to normalize G . The identification of the reduced radius is explicit; the radius $2GM/c^2$ corresponds to another convention for reading the same constant. The null sector remains outside the inverse. Extension to unbounded operators requires a development of domains and limits distinct from the present theorem.

The square of L_α comes from the product of conjugate lengths. Action enters inversely when Λ is eliminated between energy and length. The passage from energy to mass and from mass to radius contributes c^4 , which, combined with the c in $H\Lambda = \hbar c I$, produces c^3 . The two normalizations fix the dimensionless factor at one. Dimensional analysis verifies the exponents; composition determines the coefficient.

47.3 Planck scales and specialization of the unit character

The constructed length, the action section, and the speed determine

$$t_P = \frac{L_\alpha}{c}, \quad m_{P,a} = \frac{\hbar_a}{cL_\alpha}, \quad E_{P,a} = \frac{\hbar_a c}{L_\alpha}.$$

Substituting (47.4) into $\sqrt{\hbar_a G_a / c^5}$, $\sqrt{\hbar_a c / G_a}$ and $\sqrt{\hbar_a c^5 / G_a}$ proves these identities, respectively; positivity fixes each square root. The length precedes the conventional recognition of these scales. For a mode of character $y > 0$,

$$m(y) = m_{P,a} y, \quad r_g(y) = L_\alpha y, \quad \bar{\lambda}(y) = L_\alpha / y.$$

Equality of the two lengths is equivalent to $y = y^{-1}$ and hence to $y = 1$. The areal product is preserved for every positive mode; coincidence of radii characterizes the unit mode.

The chronological step satisfies $t_0 = (\pi_{\text{HMT}}/54)t_P$; the phase period is $108t_0 = 2\pi_{\text{HMT}}t_P$ and $\omega_{108} = 1/t_P$. A horizon $R_h = \zeta_h L_\alpha$ retains its own geometric condition. For the Schwarzschild specialization of the same mass, $R_h = 2r_g = 2L_\alpha y$. Comparing sections with fixed L_α, c and $\rho_a = \hbar_b / \hbar_a$ yields $G_b = G_a / \rho_a$, $m_{P,b} = \rho_a m_{P,a}$ and $E_{P,b} = \rho_a E_{P,a}$. This comparison of sections preserves L_α and t_P .

47.4 Area and reconstruction of the memory sectors

Boundary incidence retains the grouping of six trits. If $x_j = r_{2j-1} + 3r_{2j}$, $r_i \in \{0, 1, 2\}$, the map

$$(x_1, x_2, x_3) \mapsto (r_1 + 3r_2 + 9r_3, r_4 + 3r_5 + 9r_6)$$

is a bijection between $(\mathbb{Z}/9\mathbb{Z})^3$ and $(\mathbb{Z}/27\mathbb{Z})^2$. Its inverse extracts the six digits in order, preserving the position at which carry is recorded. On the boundary torus, the block $A = \{0, \dots, 8\}^2$ has two collections of links:

$$\begin{aligned} \partial A_{90} &= \{((-1, j), (0, j)), ((8, j), (9, j)) : 0 \leq j \leq 8\}, \\ \partial A_{120} &= \{((i, -1), (i, 0)), ((i, 8), (i, 9)) : 0 \leq i \leq 8\}. \end{aligned}$$

Each collection contains eighteen links. Their labels retain channel provenance; each link carries a number operator with spectrum $\{0, 1, 2\}$. Let N_{90}, N_{120} be the corresponding sums.

The angular areal scale retains the counterangle already constructed:

$$\theta_{\text{clk}} = \frac{\pi C_{\text{deg}}^*}{1080}, \quad a_0 = \frac{1 + 2 \cos(\pi/18)}{3} \theta_{\text{clk}}.$$

The prefactor is the normalized real trace of $\text{diag}(1, e^{i\pi/18}, e^{-i\pi/18})$. Thus a_0 retains its meaning as a dimensionless area normalization. The area readout is taken on the positive branch $\theta_{\text{clk}} > 0$ and $\gamma > 0$. With the Barbero–Immirzi functional γ and the length already determined,

$$\mathcal{A} = \gamma a_0 L_\alpha^2 (N_{90} + N_{120}).$$

The thirty-six local operators act on distinct tensor factors. The spectrum of $N = N_{90} + N_{120}$ is therefore $\{0, \dots, 72\}$, and the multiplicity of n is the coefficient of z^n in $(1 + z + z^2)^{36}$. The proof consists in multiplying the generating functions of the factors. The new length transports the areal unit and preserves this combinatorics.

Proposition 47.3 (Joint readout of area and sectoral provenance). *The pair $N = N_{90} + N_{120}$ and $\mathcal{D}_\partial = 90N_{90} + 120N_{120}$ recovers the two sectors:*

$$N_{90} = 4N - \frac{\mathcal{D}_\partial}{30}, \quad N_{120} = \frac{\mathcal{D}_\partial}{30} - 3N.$$

Proof. The transformation matrix is $\begin{pmatrix} 1 & 1 \\ 90 & 120 \end{pmatrix}$, with determinant 30. Its inverse gives the formulas. Commutativity of the two number operators implies that of the two readout operators. $\square$

In the modular realization with $\rho_A = Z_A^{-1} \exp(-\Delta_{\text{mod}} \mathcal{D}_\partial)$, $\Delta_{\text{mod}} > 0$, functional calculus gives $-\log \rho_A = (\log Z_A)I + \Delta_{\text{mod}} \mathcal{D}_\partial$. With the normalizations retained, the pair consisting of area and the modular Hamiltonian recovers the two occupations. The reconstruction proof holds for each constructed value of Δ_{mod} ; the new longitudinal readout operator does not redefine that coefficient or the modular state. The sum of occupations and sectoral provenance express distinct observables. Likewise, the thirty-six response links and the eighty-one unit capacities of the cubic cut retain their roles.

47.5 Joint variation and compatible memory reduction

For norm-differentiable collections that are uniformly positive in a neighbourhood, let $\kappa = L_\alpha^2/(\hbar c)$. Then

$$\delta R = \delta \kappa H + \kappa \delta H, \quad \frac{\delta G}{G} = 3 \frac{\delta c}{c} + 2 \frac{\delta L_\alpha}{L_\alpha} - \frac{\delta \hbar}{\hbar}.$$

The first identity follows by differentiating $R = L_\alpha Y$ and $H = (\hbar c/L_\alpha)Y$; it does not require $[Y, \delta Y] = 0$. An admissible connection variation acting on Y and preserving the three sections maintains $\delta G = 0$. The statement covers variations of this radial realization; equivalence between complete field functionals has its own domains and boundary conditions.

Now let $Y = \begin{pmatrix} A & C \\ C^* & D_i \end{pmatrix} > 0$ under the separation of readout and memory, with $S = A - CD_i^{-1}C^* > 0$. The minimizing extension of a vector b is $(b, -D_i^{-1}C^*b)$, as completing the square shows:

$$\left\langle \begin{pmatrix} b \\ m \end{pmatrix}, Y \begin{pmatrix} b \\ m \end{pmatrix} \right\rangle = \langle b, Sb \rangle + \langle m + D_i^{-1}C^*b, D_i(m + D_i^{-1}C^*b) \rangle.$$

The coherent reduction therefore gives

$$H_{\text{eff}} = \frac{\hbar c}{L_\alpha} S, \quad R_{\text{eff}} = L_\alpha S, \quad \Lambda_b = L_\alpha S^{-1}.$$

The compression of Y^{-1} onto the readout block is S^{-1} ; the direct compression of Y is A . Memory distinguishes these operations.

If dynamical elimination produces a bounded temporal metric $Z \geq z_0 I$, $z_0 > 0$, take $T = Z^{1/2}$, bounded and invertible. Canonical normalization requires the dual transports

$$H_c = T^{-*} H_{\text{eff}} T^{-1}, \quad R_c = T^{-*} R_{\text{eff}} T^{-1}, \quad \Lambda_c = T \Lambda_b T^*.$$

Multiplying them yields $H_c \Lambda_c = \hbar c I$ and $R_c \Lambda_c = L_\alpha^2 I$, without a commutativity hypothesis. In the resolvent expansion of an energy with blocks H_{bb}, V, H_{ii} , with H_{ii} positive and invertible and $|\hbar \omega| < \|H_{ii}^{-1}\|^{-1}$,

$$K_b(\omega) = H_{bb} - \hbar \omega I - V(H_{ii} - \hbar \omega I)^{-1} V^* = H_{\text{stat}} - \hbar \omega Z + O(\omega^2), \quad Z = I + V H_{ii}^{-2} V^*.$$

The last expression is local around zero and retains the order of approximation; the preceding dual products are exact identities of the normalized readout operators.

Proportionality is also preserved under full dynamical elimination. For z such that $H_{ii} - zI$ is invertible, define

$$\mathcal{K}_H(z) = H_{bb} - zI - V(H_{ii} - zI)^{-1} V^*, \quad \kappa = \frac{L_\alpha^2}{\hbar c}.$$

The same readout-memory decomposition applied to $R = \kappa H$ produces

$$\mathcal{K}_R(\kappa z) = \kappa \mathcal{K}_H(z). \quad (47.5)$$

Indeed, its blocks are $R_{bb} = \kappa H_{bb}$, $R_{bi} = \kappa V$ and $R_{ii} = \kappa H_{ii}$, and

$$(R_{ii} - \kappa z I)^{-1} = \kappa^{-1} (H_{ii} - z I)^{-1}.$$

Substitution into the Schur complement proves (47.5), preserving the order of the factors. When both complements are invertible, their inverses satisfy $\mathcal{K}_R(\kappa z)^{-1} = \kappa^{-1} \mathcal{K}_H(z)^{-1}$. The energy parameter is $z = \hbar\omega$, and its image in the radial readout operator is κz : both retain their respective dimensions. The identity holds throughout this resolvent domain and therefore jointly transports all orders of dynamical memory. The preceding local expansion describes its first coefficients; this equality explains the exact preservation of radial proportionality under reduction.

47.6 Temporal rigidity and boundary memory

Preserving only area allows $R \mapsto \zeta R$, $\Lambda \mapsto \Lambda/\zeta$. Preserving $H, \hbar, c$ and $H\Lambda = \hbar c I$ as well forces $\zeta = 1$. If $H \mapsto \zeta H$ is also applied, the quotient R/H and the coefficient G remain unchanged. Comparisons must preserve the same joint observations.

The lifted phase supplies another effective constraint. In a bounded realization $H = H_{\text{clk}} + D_0^* W D_0$ with $W > 0$, the evolution $P(t) = \exp(-itH/\hbar)$ determines $H = i\hbar \dot{P}(0)$. With $H_{\text{clk}}, D_0, \hbar$ and the clock fixed, equality of the evolutions implies $D_0^*(W' - W)D_0 = 0$. For a route with fixed initial endpoint, D_0 is triangular and invertible, so $W' = W$ follows. For a general graph, the form is determined on $\text{Ran } D_0$; subsequently extending the residue domain would change the elimination problem. A single return $P(\tau)$ admits spectral aliases, whereas the lifted phase or compatible times tending to zero distinguish them.

In the nonadic capacity chart, let $\rho = \log_{10} 9$, $\delta = 1 - \rho$, $p_j = \lfloor j/\delta \rfloor$ and $v_j = p_j - j$. The capacity of a window is

$$C(j, n) = p_{j+n} - p_j - n.$$

For $n = 35$ its values are 729 or 730, depending on the origin; capacity return modulo nine distinguishes the former within this class. The window $j = 20 \rightarrow 55$ satisfies $p_j : 437 \rightarrow 1201$ and $v_j : 417 \rightarrow 1146$. Its capacity phase returns to 3, and the quotient increases from 46 to 127. The six capacities of 20 and the twenty-nine of 21 sum to 729; accumulated memory remains part of the state.

For the first arrival $T(k) = \lfloor k/\rho \rfloor - 1$, the count is $N(k, C) = T(k + C) - T(k) - C$. Writing $\epsilon(k) = \lfloor k/\rho \rfloor - k/\rho$ gives exactly

$$N(k, C) = (\rho^{-1} - 1)C + \epsilon(k + C) - \epsilon(k).$$

Capacity-729 windows with counts 34 and 35 retain different boundary terms. Even $k = 21$ and $k = 2289$ share $(k \bmod 108, T(k) \bmod 108) = (21, 22)$ and accumulate 1 and 109 vacancies, respectively. Coincidence of finite calendars retains less information than the complete state. These calculations interpret the return; the longitudinal readout operator in (47.1) has its own construction map and does not select its scale through these windows.

47.7 Geometric normalization and evaluation

The central APP block $B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$ satisfies $B_c^\top g_c B_c = 45 g_c$, $g_c = \text{diag}(1, -1)$. Its positive unimodular normalization is $B_c/\sqrt{45}$: an additional positive factor a would give determinant a^2 . The areal-volumetric incidence $T_{\text{av}} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ preserves $G_{\text{av}} = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$ by direct multiplication. In a Minkowski realization, the causal form and dimensional soldering have distinct roles. The cochain $\beta_m = \ell_m \sigma_m W_\square e_\lambda$, $\ell_m = \ell_0/9^m$, and the relation $\ell_0 = ct_0$ retain the scale together with the cone. Altering the soldering while preserving only the signature changes that realization.

The expanding Pell unit with norm one and trace four satisfies $x + x^{-1} = 4$, $x > 1$, and hence $x = 2 + \sqrt{3}$. The orbit $r_n = r_0 x^n$ preserves $r_n r_{-n} = r_0^2$. When $r_0 = L_\alpha$, the product preserves the same areal section. The norm, marked form, return, and longitudinal section constrain different objects; their roles are retained when they are composed.

In the declared SI section, $L_* = 1 \text{ m}$, $S_* = 10^{-34} \text{ J s}$ and $c = 299792458 \text{ m s}^{-1}$, structural substitution produces

$$L_\alpha = 1.616254982625126330 \dots 10^{-35} \text{ m}, \quad \hbar_{\text{ret}} = 1.054571816915039061 \dots 10^{-34} \text{ J s},$$

$$G_{\text{ret}} = 6.674299659414022706 \dots 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}. \quad (47.6)$$

The evaluator substitutes the regional coordinates and register K , reconstructs α , the return of H_5 , and L_α ; metrological comparison follows. The CODATA 2022 reference is $6.67430(15) 10^{-11}$ in those units [4]. The difference is $-3.40586 10^{-18}$, equivalent to -0.00227 standard uncertainties. The calculated digits describe the mathematical evaluation in this section, not an experimental uncertainty. The pre-return and post-return action sections retain their labels and $\hbar_a G_a = c^3 L_\alpha^2$.

The structural meaning of G is thereby determined: it is the coefficient that converts the energy readout into conjugate length in the common radial realization while simultaneously preserving action and area. Memory, sheet markers, and bases belong to the construction that fixes this coefficient. Their joint preservation explains the proved uniqueness.

48 Geometric and material background of the joint action

The seeds and operations of APP, the orientation and regime of TRIT, and the transport of TPK produce the enriched routes on which the readers of the common core act. Sheet, orientation, carry, boundary and memory are jointly preserved; the phase return does not reset that state. The constructions of the continuum belong to the same object and are not turned here into independent sectors. The realization that follows transports this already constructed structure to geometry and matter. Its units and couplings are earlier HMT outputs: no numerical comparison acts as a generator.

To make the action and its reduction readable within this manuscript, we bring together the screen operations, the effective variation of transport, and the spinorial current they use. The Cartan–Holst elimination is proved in 50.4.1; the total action and its canonical reduction, in 51. The formulas in this section specify the data and calculations entering both.

48.1 Incidence, screen and soldering

The cubic representation of a route boundary has the six faces $F = \{(i, \epsilon) : 1 \leq i \leq 3, \epsilon = \pm 1\}$. On $H_F = \mathbb{R}^F$, opposition is $Re_{i,\epsilon} = e_{i,-\epsilon}$. With

$$u = \sum_{i,\epsilon} e_{i,\epsilon}, \quad P_u = \frac{uu^\top}{6}, \quad P_- = \frac{I - R}{2}, \quad P_0 = \frac{I + R}{2} - P_u,$$

the three projectors are orthogonal, sum to I and have ranks 1, 3, 2. The incidence between faces and vertices $v \in \{-1, 1\}^3$ is $M_{(i,\epsilon),v} = \mathbf{1}_{\{v_i=\epsilon\}}$.

Proposition 48.1 (Screen of oriented incidence). *We have*

$$MM^\top = 12P_u + 4P_-, \quad \ker M^\top = P_0 H_F.$$

The quotient $Q = H_F / P_0 H_F$, identified with $(P_u + P_-)H_F$, is four-dimensional. The form $B = P_- - P_u$ on Q has signature $(- + ++)$.

Proof. Each face contains four vertices; two opposite faces share none, and two faces on different axes share two. Therefore,

$$MM^\top = 4I + 2(uu^\top - I - R) = 12P_u + 4P_-.$$

The identity $\|M^\top x\|^2 = \langle x, MM^\top x \rangle$ determines the kernel. In the orthonormal basis

$$t = u/\sqrt{6}, \quad d_i = (e_{i,+} - e_{i,-})/\sqrt{2}$$

of Q , the matrix of B is $\text{diag}(-1, 1, 1, 1)$. □

Let $W_{\square} = \sqrt{6}P_u + \sqrt{2}P_v$. Then

$$W_{\square}e_{i,\epsilon} = t + \epsilon d_i =: n_{i,\epsilon}, \quad B(n_{i,\epsilon}, n_{i,\epsilon}) = 0, \quad MM^T|_Q = 2W_{\square}^*W_{\square}|_Q.$$

These equalities are obtained by applying the projectors to the face basis. The proper rotations of the cube preserve the projectors, the time orientation t and the form B .

The enriched route provides each edge a with its face, its sign $\sigma_m(a)$ and the transport $U_m(a) : Q_{s(a)} \rightarrow Q_{t(a)}$. Its cellular soldering is

$$\beta_m(a) = \ell_m \sigma_m(a) n_{\lambda(\underline{a})}^{s(a)}, \quad \ell_m = \ell_0 9^{-m}, \quad \beta_m(\bar{a}) = -U_m(a) \beta_m(a). \quad (48.1)$$

The inversion $U_m(\bar{a}) = U_m(a)^{-1}$ implies that inverting twice recovers the soldering. The ordered carry record is also preserved: it is not replaced by the arrival vector.

Proposition 48.2 (Naturality of soldering). *Let $a = a_1 \cdots a_9$ be a compatible refinement: the child incidences transported to the origin of the predecessor preserve face and sign, and $\ell_{m+1} = \ell_m/9$. If $\mathcal{U}_j : Q_{m,s(a)} \rightarrow Q_{m+1,s(a_j)}$ includes the identification by refinement and the transport to $s(a_j)$, then*

$$\beta_m(a) = \sum_{j=1}^9 \mathcal{U}_j^{-1} \beta_{m+1}(a_j).$$

Proof. Each summand in the source fibre is $(\ell_m/9)\sigma_m(a)n_{\lambda(\underline{a})}$. Their sum gives (48.1). Reversing the route reverses the order of the transports and each transport; the soldering inversion formula gives the global sign. Concatenation preserves the memory of the nine incidences in the same order. The equality iterates by composition of compatible refinements. $\square$

48.2 Transport representation and geometric realization

The record of a route γ counts inversions and effective changes of sheet through the cocycle $c(\gamma) = (c_g(\gamma), c_s(\gamma)) \in \mathbb{Z}^2$. The count is cumulative on forward routes; its extension to the groupoid is oriented and assigns opposite increments to the inverse route. For composable routes, $c(\gamma_2\gamma_1) = c(\gamma_2) + c(\gamma_1)$, and $c(\bar{\gamma}) = -c(\gamma)$. Let R_4 be an element of order four in $\text{Spin}^+(3, 1)$ whose vector representation $\Lambda(R_4)$ fixes the unit time vector u_0 . The received affine representation is

$$\rho_C^{\text{disc}}(\gamma) = (R_4^{c_g(\gamma)}, \ell_0 c_s(\gamma) u_0).$$

With the product $(R, a)(S, b) = (RS, a + \Lambda(R)b)$, the fixing of u_0 and the additivity of c directly prove that ρ_C^{disc} preserves identity, concatenation and inversion. When the rotational component returns to the identity, the memory translation remains determined by $c_s(\gamma)$, which is not reset.

Normalization

$$N_{\ell_0}(R, a) = (\Lambda(R), a/\ell_0), \quad \rho_C = N_{\ell_0} \circ \rho_C^{\text{disc}}$$

is also a homomorphism, since $(a + \Lambda(R)b)/\ell_0 = a/\ell_0 + \Lambda(R)(b/\ell_0)$. The lift $R_4^{c_g(\gamma)}$ is retained when spinors act; it cannot be recovered from $\Lambda(R_4^{c_g(\gamma)})$ alone, whose representation has kernel $\{1, -1\}$.

In a screen realization $J_{\text{scr},m}$ that transports the form and the soldering, $e_m = J_{\text{scr},m} \beta_m$. The smooth realization considered consists of an oriented four-dimensional region $\mathcal{U}$, a nondegenerate cotetrad e , a Lorentz connection ω and a route realization $\mathfrak{R}_C$ with the compatibility condition

$$\text{Hol}_{\mathcal{A}_{\ell_0}}(\mathfrak{R}_C(\gamma)) = \rho_C(\text{Hol}_{\text{TPK}}(\gamma)), \quad \mathcal{A}_{\ell_0} = \begin{pmatrix} \omega & \ell_0^{-1}e \\ 0 & 0 \end{pmatrix}. \quad (48.2)$$

The argument of ρ_C preserves the cocycle of the enriched holonomy; it is not replaced by its reduced phase alone. The scale $\ell_0 > 0$ is constant in the coordinatization. These are the data and the condition of the smooth realization used; preserving path composition neither identifies distinct paths with the same endpoints nor removes their memory. The metric is $g = \eta_{IJ} e^I \otimes e^J$, $\eta = \text{diag}(-1, 1, 1, 1)$.

Proposition 48.3 (Curvature, covariance and Bianchi). *With $F_\omega = d\omega + \omega \wedge \omega$ and $T = de + \omega \wedge e$,*

$$\mathcal{F}_{\ell_0} = \begin{pmatrix} F_\omega & \ell_0^{-1}T \\ 0 & 0 \end{pmatrix}, \quad D_\omega T = F_\omega \wedge e, \quad D_\omega F_\omega = 0.$$

A change of frame $g_L : \mathcal{U} \rightarrow SO^+(3,1)$ acts by $e' = g_L e$, $\omega' = g_L \omega g_L^{-1} - dg_L g_L^{-1}$; then $T' = g_L T$, $F_{\omega'} = g_L F_\omega g_L^{-1}$.

Proof. Multiplying matrices of forms gives the diagonal block $d\omega + \omega \wedge \omega$ and the translational block $\ell_0^{-1}(de + \omega \wedge e)$. On substituting the transformed fields, the terms involving dg_L cancel. Expanding $D_\omega T$ leaves $(d\omega + \omega \wedge \omega) \wedge e$. In $D_\omega F_\omega$, the derivatives of ω and the cubic products cancel in pairs. $\square$

The tritic sheet $\tau \in \{+1, 0, -1\}$ preserves its regime in the Cartan realization with generators $J_{ab}, P_a^{(\tau)}$, the Lorentz vector action and $[P_a^{(\tau)}, P_b^{(\tau)}] = -\tau J_{ab}$. For $\mathcal{A}_\tau = \frac{1}{2}\omega^{ab}J_{ab} + \ell^{-1}e^a P_a^{(\tau)}$,

$$\mathcal{F}_\tau = \frac{1}{2}(F_\omega^{ab} - \tau \ell^{-2}e^a \wedge e^b)J_{ab} + \ell^{-1}T^a P_a^{(\tau)}.$$

Indeed, the Lorentz bracket gives F_ω , the mixed bracket gives $D_\omega e$ and the translational bracket gives the term $-\tau e \wedge e/(2\ell^2)$. The condition (48.2) corresponds to the central sheet; the extreme sheets use their own representation $\rho_{C,\tau}$ and holonomy condition.

48.3 Transport differential and material current

In a finite cellular realization, let $U_a : \mathcal{H}_{s(a)} \rightarrow \mathcal{H}_{t(a)}$ be an invertible transport between Hermitian fibres. It is not assumed that every Lorentz transport is unitary for the energy inner product. One orientation per edge is chosen. For

$$(Df)_a = f_{t(a)} - U_a f_{s(a)}, \quad r = Df, \quad q = Wr, \quad v_a = U_a f_{s(a)}, \quad E = \langle r, Wr \rangle,$$

the weight $W = W^* > 0$ may couple edges. This weight is not $W_\square$. The variations correspond to the connection representation and the chosen boundary problem.

Proposition 48.4 (Full variation). *For a differentiable curve of data, with $A_a = \dot{U}_a U_a^{-1}$, we have*

$$\dot{E} = 2 \operatorname{Re} \sum_a \langle q_a, \dot{f}_{t(a)} - U_a \dot{f}_{s(a)} - A_a v_a \rangle + \langle r, \dot{W}r \rangle. \quad (48.3)$$

With the field and weight fixed, the full transport gradient is $G_a = -2q_a v_a^$:*

$$\delta_U E = \sum_a \operatorname{Re} \operatorname{Tr}(G_a^* A_a).$$

In the unitary subclass $A_a^ = -A_a$, its anti-Hermitian part $J_a = v_a q_a^* - q_a v_a^*$ suffices.*

Proof. The product rule gives $\dot{E} = 2 \operatorname{Re} \langle Wr, \dot{r} \rangle + \langle r, \dot{W}r \rangle$. Differentiating each residual yields (48.3). Since $\operatorname{Tr}(v_a q_a^* A_a) = q_a^* A_a v_a$, we obtain G_a . The Hermitian part of G_a is orthogonal, with respect to $\operatorname{Re} \operatorname{Tr}(X^* Y)$, to the anti-Hermitian variations. This reduction applies only in that subclass. $\square$

For a route with $U_\gamma = U_N \cdots U_1$, let $B_k = U_N \cdots U_{k+1}$. The product rule proves

$$\dot{U}_\gamma U_\gamma^{-1} = \sum_k B_k A_k B_k^{-1}. \quad (48.4)$$

Consequently, a gradient G_γ is transported to the edge k as $G_k = B_k^* G_\gamma (B_k^{-1})^*$. Cyclicity of the trace proves this formula even if B_k is not unitary.

If the real coordinates of connection variation are θ_b , its effective differential is written as $A_a(\theta) = \sum_b B_{ab} \theta_b$. For $S_{\text{mem}} = aE$, with the field, weight and action section a fixed,

$$\delta S_{\text{mem}} = -\frac{1}{2} \sum_b \theta_b s_b, \quad s_b = 4a \operatorname{Re} \sum_a \langle q_a, B_{ab} v_a \rangle. \quad (48.5)$$

Substituting $A_a(\theta)$ into the variation proves the equality and fixes the factor 4. If the oriented cellular pairing is an invertible matrix M , defined by $\delta S_{\text{mem}} = -\frac{1}{2}\theta^T M \sigma$, the current is $\sigma = M^{-1}s$. This matrix preserves volumes and orientation; it cannot be replaced by scalar division without fixing a basis in which that is appropriate. If f, W, a also vary, all the terms of (48.3) and $E\delta a$ are retained.

For a weight per edge, inversion preserves the full response by jointly transforming

$$U_{\bar{a}} = U_a^{-1}, \quad r_{\bar{a}} = -U_a^{-1}r_a, \quad W_{\bar{a}} = U_a^*W_aU_a.$$

Substitution gives $r_{\bar{a}}^*W_{\bar{a}}r_{\bar{a}} = r_a^*W_ar_a$. Its derivative necessarily retains $\dot{W}_{\bar{a}} = U_a^*(A_a^*W_a + \dot{W}_a + W_aA_a)U_a$: removing this term would alter the current. If the weight couples several edges, the same congruence is applied to the full matrix with block-diagonal transport; its cross-blocks are also retained.

Proposition 48.5 (Variational compatibility under refinement). *For fixed maps J_m, K_m , suppose that, throughout the variation considered,*

$$D_{m+1}J_m = K_mD_m, \quad K_m^*W_{m+1}K_m = W_m.$$

Then $E_{m+1}(J_m f) = E_m(f)$ and their differentials coincide. If the actions also coincide and P_m transports the connection variations to the refinement, their currents satisfy

$$s_m = P_m^T s_{m+1}, \quad M_m \sigma_m = P_m^T M_{m+1} \sigma_{m+1}.$$

Proof. Substituting $D_{m+1}J_m = K_mD_m$ into the energy and then the weight congruence proves the first equality along the entire curve. Differentiating it and using $-\frac{1}{2}\theta^T s_m = -\frac{1}{2}(P_m \theta)^T s_{m+1}$ proves the others. If J_m varies, its differential contributes $2 \operatorname{Re}\langle D_{m+1}J_m f, W_{m+1}D_{m+1}\dot{J}_m f \rangle$; it is not discarded. $\square$

The memory current thus obtained comes from its action. The following spinorial current comes from the affine action of the fields. They combine when they are terms of a single action; they are not identified merely because they share transport. In particular, eliminating a contorsion on which the prepared field, the incidence or its minimum also depend requires retaining those differentials. The linear Cartan formula used below corresponds to the expressly stated affine material problem.

48.4 Spinorial representation and affine current

The spin structure of the geometric realization is fixed. The inherited matrices, after complexifying the real module, are

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (48.6)$$

$$g^0 = J \otimes I_2, \quad g^1 = R \otimes I_2, \quad g^2 = Z \otimes R, \quad g^3 = Z \otimes Z.$$

Their squares are $-I_4, I_4, I_4, I_4$, and distinct matrices anticommute. It follows that $\{g^I, g^J\} = 2\eta^{IJ}I_4$. We define

$$g_I = \eta_{IJ}g^J, \quad \mathbb{B}_{IJ} = \frac{1}{4}[g_I, g_J], \quad A_D = -ig^0, \quad \bar{\psi} = \psi^\dagger A_D, \quad \Omega = \frac{1}{2}\omega^{IJ}\mathbb{B}_{IJ}.$$

Expanding the three-matrix commutator gives $[\mathbb{B}_{IJ}, g^K] = \delta_J^K g_I - \delta_I^K g_J$. Thus Ω represents the same metric connection, with $D\psi = d\psi + \Omega\psi$ and $D\bar{\psi} = d\bar{\psi} - \bar{\psi}\Omega$. The matrix A_D is Hermitian and $A_D g^I$ is anti-Hermitian; these fix the adjoint and the action factor in signature $(-+++)$.

Let $\eta_I = \iota_{E_I} \operatorname{vol}_e$, so that $e^J \wedge \eta_I = \delta_I^J \operatorname{vol}_e$. The affine action is

$$S_D = \int_M \left\{ \frac{\hbar}{2} [\bar{\psi} g^I D\psi - (D\bar{\psi}) g^I \psi] \wedge \eta_I - mc \bar{\psi} \psi \operatorname{vol}_e \right\}. \quad (48.7)$$

The mass and the positive sections $c, \hbar$ are the received readings, and $[\psi] = L^{-3/2}$. The alternative notation $\Gamma^I = -ig^I$, with principal term $i\hbar\Gamma^I D_I$, switches to the opposite Clifford signature; the two conventions are not mixed.

Proposition 48.6 (Normalized spinorial current). *With $B^{IJK} = \bar{\psi} g^{[I} g^J g^{K]} \psi$, where antisymmetrization is normalized, the convention $\delta_\omega S_D = -\frac{1}{2} \int \delta \omega^{JK} \wedge \sigma_{JK}$ gives*

$$\sigma_{JK} = -\frac{\hbar}{2} \bar{\psi} \{g^I, \mathbb{B}_{JK}\} \psi \eta_I = -\frac{\hbar}{2} B^I_{JK} \eta_I. \quad (48.8)$$

The current is independent of the contorsion when e, ψ is fixed.

Proof. We have $\delta \Omega = \frac{1}{2} \delta \omega^{JK} \mathbb{B}_{JK}$. The two kinetic terms of (48.7) give

$$\delta_\omega S_D = \frac{\hbar}{4} \int \delta \omega^{JK} \wedge \eta_I \bar{\psi} \{g^I, \mathbb{B}_{JK}\} \psi.$$

Comparison with the variational convention determines the sign and factor. Expanding the anticommutator using Clifford relations cancels the degree-one contractions and leaves $\{g^I, \mathbb{B}_{JK}\} = g^{[I} g_J g_K]$. The action is linear in Ω , which proves independence. $\square$

If internal connections are included, they act on the material factor of the spinorial module, whereas $\mathbb{B}_{IJ}$ acts on the spin factor. Thus, $[\mathbb{B}_{IJ} \otimes I, I \otimes \rho(A)] = 0$. Varying ω produces the same formula, contracted over the internal components and summed over the multiplets present. This preserves their chiral, colour and family representations; it adds no multiplets to those already constructed.

48.5 Torsional contraction and cross-terms between species

The real inversion of Cartan–Holst and of the map $T^I \mapsto T^I \wedge e^J - e^I \wedge T^J$, proved in 50.4.1, unambiguously fixes the contorsion $\mathfrak{k}_*$ for the affine current (48.8). These formulas retain $\gamma \in \mathbb{R} \setminus \{0\}$, a nondegenerate cotetrad, and the sections $G, c, \hbar, \gamma$ fixed during variation. Elimination uses the linear interaction only once and gives $-\frac{1}{4} \mathfrak{k}_*^{IJ} \wedge \sigma_{IJ}$.

Theorem 48.7 (Coefficient of the spinorial contraction). *Let $\kappa = 8\pi G/c^4$. The effective interaction is*

$$\mathcal{L}_{\text{spin}}^{\text{ef}} = C_\gamma q(B) \text{vol}_e, \quad C_\gamma = \frac{3\kappa c \hbar^2}{16} \frac{\gamma^2}{1 + \gamma^2},$$

where

$$q(B) = \frac{1}{3!} B^{IJK} B_{IJK} = -(B^{012})^2 - (B^{013})^2 - (B^{023})^2 + (B^{123})^2.$$

Proof. We first calculate

$$y^{IJ} = \kappa c \frac{\gamma^2}{1 + \gamma^2} (-\star + \gamma^{-1} I) \sigma^{IJ}, \quad \mathcal{B}^I = \sum_J \iota_{E_J} y^{IJ}, \quad T^I = \mathcal{B}^I - \frac{1}{4} e^I \wedge \sum_L \iota_{E_L} \mathcal{B}^L,$$

and then $\mathfrak{k}_{IJK} = (T_{KIJ} - T_{IJK} - T_{JKI})/2$. These are linear compositions of the already proved inverses. To make all the contraction coefficients explicit, the trivector is ordered as (012, 013, 023, 123) and the factor $\kappa c \hbar^2 \gamma^2 / (1 + \gamma^2)$ is extracted. Substituting $\sigma_{JK}/\hbar = -\frac{1}{2} B^I_{JK} \eta_I$ into these compositions and into $-\frac{1}{4} \mathfrak{k}_*^{IJ} \wedge \sigma_{IJ}$ gives

operator on σ	012	013	023	123	six polarization coefficients
$-\star$	-3/16	-3/16	-3/16	3/16	0
I	0	0	0	0	0

Each entry uses $e^I \wedge \eta_J = \delta^I_J \text{vol}_e$ and the contractions written above. The four diagonal entries and the six polarizations enumerate the full quadratic form. The part $\gamma^{-1} I$ vanishes and the remainder is $(3/16)q(B)$, which proves the formula. $\square$

For several species, variation is performed on their common action: $\sigma = \sum_s \sigma_s$. Since the response $\mathfrak{f}_* = C_e(\sigma)$ is linear,

$$-\frac{1}{4}C_e\left(\sum_s \sigma_s\right) \wedge \sum_t \sigma_t = -\frac{1}{4} \sum_{s,t} C_e(\sigma_s) \wedge \sigma_t. \quad (48.9)$$

This expansion proves that the terms $s \neq t$ remain. Calculating separate responses and retaining only $s = t$ would change the total action.

48.6 Normal ordering of the total current

The finite fermionic realization of a cell uses $\mathcal{F} = \bigoplus_n \Lambda^n V$, with the relations $\{a_r, a_s^\dagger\} = \delta_{rs}$, $\{a_r, a_s\} = 0$ and the occupation vacuum as the normal-ordering reference. We write $d\Gamma(M) = \sum_{r,s} a_r^\dagger M_{rs} a_s$; this Γ denotes second quantization, not the nonadic holonomy. For the four-component spinorial module, the matrices are

$$\begin{array}{c|cccc} abc & 012 & 013 & 023 & 123 \\ \hline M_{abc} = A_D g^a g^b g^c & iJ \otimes R & iJ \otimes Z & iI_2 \otimes J & iZ \otimes J \end{array}.$$

Multiplying (48.6) gives these matrices; they are Hermitian, traceless and square to I_4 .

Proposition 48.8 (Normal contraction and preservation of cross-terms). *For every matrix M ,*

$$: d\Gamma(M)^2 := d\Gamma(M)^2 - d\Gamma(M^2).$$

In the four-component module,

$$\widehat{\mathcal{Q}}_{\text{sp}} = \sum_{a=1}^4 s_a d\Gamma(M_a)^2 + 2\widehat{N}, \quad s = (-1, -1, -1, +1), \quad \widehat{N} = d\Gamma(I_4).$$

The operator is self-adjoint and preserves occupation. In a sum of multiplets, the same first identity is used with the total matrices $\widehat{M}_a$; the contractions between species are retained.

Proof. In $d\Gamma(M)^2$, the relation $a_s a_t^\dagger = \delta_{st} - a_t^\dagger a_s$ separates the contraction $d\Gamma(M^2)$ from the term with two creation and two annihilation operators. The latter is, by definition, the normal product. Since $M_a^2 = I_4$ and $\sum_a s_a = -2$, the contraction subtraction equals $+2\widehat{N}$. Hermiticity of the matrices makes the summands self-adjoint, and each monomial preserves occupation.

If $\widetilde{M} = \bigoplus_s M_s$, then $d\Gamma(\widetilde{M}) = \sum_s d\Gamma(M_s)$ and $d\Gamma(\widetilde{M}^2) = \sum_s d\Gamma(M_s^2)$. Bilinears of distinct multiplets commute; squaring the first sum leaves $2d\Gamma(M_s)d\Gamma(M_t)$ for $s < t$. Subtracting an occupation does not remove these cross-terms. This is the operator version of (48.9). $\square$

Invariance under internal unitary changes follows because they act on the material indices and intertwine the current matrices: second quantization transports the full formula by conjugation. The density and its evaluation on a state also use the declared cell volume and material state. The mean current and its second moment are not identified. These data complete the background used by the joint action, without introducing a second torsion elimination or modifying its boundary conditions.

49 Chromatic residue and internal representation of the joint action

The APP-TRIT-TPK chain preserves the enriched state, its routes, orientations, sheets and memory within the joint discrete structure of the continuum. The residue read below is a finite readout of that state; it neither replaces it nor resets its history. The phase coordinates, including the complex unit and the regional coordinate π , are HMT outputs received before this algebraic realization. The representation is constructed on the residual support with the declared chart, orientation and phase character. A gauge group is not inferred from the number of labels alone.

49.1 Residual support and preserved orientation

In the positional copy of APP, let

$$I_9 = \{1, \dots, 9\}, \quad \mathcal{C} = I_9 \times I_9, \quad \mathcal{C}_q = \{(r, c) \in \mathcal{C} : r \text{ is even}\}.$$

The residual observable and the cellular inversion are

$$\kappa_{\text{col}}(r, c) = r - c \pmod{3}, \quad \rho_c(r, c) = (10 - r, 10 - c).$$

The symbol κ_{col} does not denote the gravitational coefficient of the action. This observable is lifted to the persistent sections through projection to the atlas and preserves the remaining records; it does not select a cell by its evaluated mass.

Proposition 49.1 (Ternary fibres and chromatic balance). *The functional $\kappa_3 : \mathbb{F}_3^2 \rightarrow \mathbb{F}_3$, $\kappa_3(r, c) = r - c$, has three affine fibres of cardinality three. On $\mathcal{C}_q$, the three residues have cardinalities 12, 12, 12. Cellular inversion negates the residue.*

Proof. For each $j \in \mathbb{F}_3$, the fibre is $\{(c + j, c) : c \in \mathbb{F}_3\}$, a translate of the kernel $\{(c, c)\}$. In $\mathcal{C}_q$ there are four even values of r ; for each one, $c = 1, \dots, 9$ runs three times through each class modulo three. Each residue has $4 \cdot 3 = 12$ representatives. Finally,

$$\kappa_{\text{col}}(\rho_c(r, c)) = (10 - r) - (10 - c) = -(r - c) \pmod{3}.$$

The zero class is fixed, and classes one and two are exchanged. □

This balance does not identify flavour, depth and colour. Nor does it assert that the three lines to be constructed are three complete states: the routes and memory sharing a residual reading remain in the enriched fibre.

49.2 Qutrit chart, Weyl pair and colour algebra

We fix the chart

$$V_{\text{col}} = \mathbb{C}[\mathbb{F}_3] \simeq \mathbb{C}^3, \quad (e_0, e_1, e_2), \quad \omega_3 = e^{2\pi i/3},$$

with orthonormal delta basis, orientation $j \mapsto j + 1$ and character $j \mapsto \omega_3^j$. The operators are

$$Xe_j = e_{j+1 \bmod 3}, \quad Ze_j = \omega_3^j e_j, \quad X^3 = Z^3 = I, \quad ZX = \omega_3 XZ.$$

The last equality is checked by applying both sides to each e_j . The map from the atlas to this chart is

$$(r, c) \longmapsto \mathbb{C}e_{\kappa_{\text{col}}(r, c)}.$$

Thus, X permutes the lines and Z records their phases. Changing the affine chart and jointly transporting the basis, orientation and character conjugates the operators by the corresponding unitary change-of-basis matrix. It is not equivalent to changing just one label while keeping the other data fixed.

Theorem 49.2 (Matrix algebra and compact real form). *The nine operators $W_{ab} = X^a Z^b$, $(a, b) \in \mathbb{F}_3^2$, form an orthogonal basis of $\text{End}_{\mathbb{C}}(V_{\text{col}})$. The eight nonzero indices linearly span $\mathfrak{sl}_3(\mathbb{C})$, whose real form of anti-Hermitian matrices is $\mathfrak{su}(3)$.*

Proof. If $a \neq 0$, the matrix $X^a Z^b$ has no diagonal entries. If $a = 0$, its trace is $1 + \omega_3^b + \omega_3^{2b}$, which is zero for $b \neq 0$ and three for $b = 0$. The Weyl relation then gives

$$\text{tr}(W_{ab}^\dagger W_{cd}) = 3\delta_{ac}\delta_{bd}.$$

Independence and dimension nine prove the first assertion; the remaining eight form a basis of the traceless subspace. Moreover,

$$W_{ab}^\dagger = \omega_3^{ab} W_{-a, -b}.$$

The eight nonzero indices form four adjoint pairs. Taking $\mathcal{R} = \{(1, 0), (0, 1), (1, 1), (1, 2)\}$, the matrices

$$A_u = W_u - W_u^\dagger, \quad B_u = i(W_u + W_u^\dagger), \quad u \in \mathcal{R},$$

are anti-Hermitian and traceless. Their real independence follows from the complex independence of the eight W_u, W_{-u} : in each pair, the coefficients of a vanishing real combination force both real coefficients to zero. They form an eight-dimensional real basis. Therefore,

$$\{A \in \mathfrak{sl}_3(\mathbb{C}) : -A^\dagger = A\} = \mathfrak{su}(3).$$

□

The fixed Hermitian form and complex orientation determine the realization group

$$SU(V_{\text{col}}) = \{U : U^\dagger U = I, \det U = 1\}.$$

Its tangent algebra is exactly the preceding one: differentiating both equations at the identity gives $A^\dagger = -A$ and $\text{tr} A = 0$; conversely, e^{tA} satisfies both. Every group element can be written as the exponential of a matrix in that algebra. Indeed, unitarily diagonalize U with eigenvalues $e^{i\theta_j}$; since the determinant is one, $\sum_j \theta_j = 2\pi n$. Subtracting $2\pi n$ from one of the angles does not alter U and produces a traceless anti-Hermitian logarithm. This continuous realization is not identified with the finite group generated solely by X, Z . It preserves the compact form and introduces no value for the strong coupling.

49.3 Chromatic transport and independent internal factors

In a finite oriented cellularization compatible with the routes, we assign $U_e \in SU(V_{\text{col}})$ and $U_{\bar{e}} = U_e^{-1}$. For $g_v \in SU(V_{\text{col}})$,

$$U_e^g = g_{t(e)} U_e g_{s(e)}^{-1}, \quad \psi_v^g = g_v \psi_v.$$

By substitution,

$$U_e^g \psi_{s(e)}^g - \psi_{t(e)}^g = g_{t(e)} (U_e \psi_{s(e)} - \psi_{t(e)}).$$

The opposite sign, used in D_T , has the same covariance. An oriented product around a face transforms by conjugation with the gauge at the base vertex: the factors at intermediate vertices cancel consecutively. In particular, for $\beta \geq 0$,

$$S_{\text{col}}^{(\mathcal{K}, \beta)}[U] = \frac{\beta}{3} \sum_f (3 - \text{Re tr } U_f)$$

is invariant and nonnegative. Invariance uses the cyclic trace; positivity follows from $\text{Re tr } U_f \leq 3$. The parameter and cellularization retain their status as data of this realization. This proof does not determine them by comparison with chromodynamic values.

The matter of the joint action uses this factor, not another partition of the atlas. For left-handed quarks, its internal fibre has factors $V_{\text{col}} \otimes \mathbb{C}^2 \otimes F$, where F preserves families; for each right-handed type, $V_{\text{col}} \otimes F$ is used. The colour factor is trivial in leptons. The spinorial factor is retained separately. For colour matrices A and weak matrices B , we have

$$[A \otimes I, I \otimes B] = 0,$$

since both products are $A \otimes B$. The hypercharge characters act as scalars on each multiplet and also commute. This identity allows the internal connections and the spin connection to be composed in the same derivative without confusing colour, chirality and flavour.

The Yukawa maps of the subsequent action are the identity on the chromatic factor and maps on the Higgs-family factors. Therefore $Y_q(H)(g_{\text{col}} \otimes I) = (g_{\text{col}} \otimes I)Y_q(H)$. Weak and hypercharge compatibility is proved there with their explicit columns and weights; it does not follow from the census $12 + 12 + 12$. This preserves the residual-quitrit-algebra-transport chain required by the common action, without asserting a Yang-Mills quantum limit in this background or replacing its source documents.

50 Gravity, energy and self-inertia of the joint system

The joint normalization allows an operator-level question to be formulated: how geometry, matter and memory enter a single realization. Three links are brought together. The positive character simultaneously preserves its energy, mass and radial readings; variation of the joint transport produces geometric and internal currents of the same energy; and the self-inertial response is expressed through the gravitational radius of that character. Preservation of these links under memory reduction and propagation of the constraints of the coupled action during its regular evolution are proved.

The starting point is the common core of the article. APP produces both sheets with their residues, quotients and incidences; TRIT preserves regime and orientation: +1 corresponds to return and memory, 0 to the present threshold and -1 to opening. TPK selects, transports, folds, returns and preserves the enriched state and its history. The prolongation $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ retains prefix, carry, survival, route and boundary. The dynamics, its nonadic holonomy and the reduced readout follow the order $U_t \rightarrow \Gamma_9 \rightarrow K_{ph}$: phase returns, memory advances and the state is not reset. The five consubstantial constructions of the discrete structure of the continuum are not separated when their readers are realized.

The constants and scales used are outputs of this genealogy. The quadrirelation $(\pi, \varphi, e, \alpha)$ preserves its common origin and its internal constructive order; action, constitutive speed and Barbero–Immirzi are received with their readers. Operator and variational language composes and recognizes these outputs without introducing metrological values to select states or coefficients.

50.1 Energy, mass and radius of the same character

The marked realization of the preceding section is retained:

$$\Omega = \{(j, k, q, u, v) : j, k \in \mathbb{Z}/12\mathbb{Z}, k - j \in \{0, 3, 6, 9\}, 1 \leq q \leq 8, u < v, u, v \in \{1, 2, 4, 5, 7, 8\}\}.$$

Its cardinality $N_\Omega = 12 \cdot 4 \cdot 8 \binom{6}{2} = 5760$ counts incidences, not matter states or lapses. For the transported inverter $B : V_{in} \rightarrow V_{out}$, with $B^*B = BB^* = I/4$, the projector $P = uu^*$ of the common mode has $|u_i|^2 = 1/N_\Omega$. The archive retains

$$C_0 = (I - P)B, \quad M_0 = PB, \quad C_0 + M_0 = B, \quad U = 2B.$$

The images of C_0, M_0 are orthogonal and U is unitary. The marked expectation $\Delta_\Omega(A) = \sum_i R_i A R_i^*$, $R_i = e_i e_i^*$, is unique among bimodular expectations onto the diagonal algebra that fix the R_i : on a matrix unit, bimodularity requires $\mathcal{E}(E_{ij}) = R_i \mathcal{E}(E_{ij}) R_i^*$, which is zero if $i \neq j$ and equals E_{ii} if $i = j$.

From $C_0 C_0^* = (I - P)/4$ and $R_i P R_i^* = R_i/N_\Omega$ follows

$$\Delta_\Omega(C_0 C_0^*) = r_\Omega I, \quad \Delta_\Omega(M_0 M_0^*) = \frac{I}{4N_\Omega}, \quad r_\Omega = \frac{5759}{23040}. \quad (50.1)$$

Both readings sum to $I/4$. The longitudinal constitutive rule applies the first coefficient linearly to a length section:

$$L = \alpha^{16} r_\Omega L_*, \quad D = LU. \quad (50.2)$$

The local norm α^{16} and the section L_* are explicit antecedents; taking an intensity root would be another reader. The radial metric rule is $\|R_X v\|^2 = \|XD^* v\|^2$. For $X = X^* > 0$ in the finite fibre, polarization and the unique positive root give $R_X^2 = DX^2 D^* = L^2 UX^2 U^*$ and $R_X = LY$, with $Y = UXU^* > 0$. The spectrum is not selected from a desired gravitational response.

The same section $\hbar > 0$ and the conjugate clock $\omega L = c$ determine $E_L = \hbar c/L$ and the readers

$$H_X = E_L Y, \quad M_X = H_X/c^2, \quad R_X = LY, \quad \Lambda_X = LY^{-1}. \quad (50.3)$$

The notation H_X is reserved for this reader of the character. The joint Hamiltonian $\mathbb{H}$ introduced later has another domain and is not identified with H_X merely because it shares the name energy.

Proposition 50.1 (Common radial coefficient). *In the explicit identification of R_X as a reduced gravitational radius, $R_X = GM_X/c^2$, there is a unique compatible coefficient:*

$$G = \frac{c^3 L^2}{\hbar}, \quad R_X = \frac{G}{c^4} H_X = \frac{G}{c^2} M_X, \quad R_X \Lambda_X = L^2 I, \quad H_X \Lambda_X = \hbar c I. \quad (50.4)$$

Proof. Multiplying the readers proves both products. The radial identification is equivalent to $LY = G\hbar Y/(c^3 L)$. Cancelling the invertible character determines G ; back-substitution verifies the equality for every positive character. If two coefficients satisfy it, their difference multiplied by Y is zero and they coincide. $\square$

The longitudinal and metric rules fix the realization; the reduced-radius convention fixes the physical meaning of G . The Schwarzschild radius of the same mass is $2R_X$. The equality preserves superpositions and off-diagonal elements in any transported basis, not only mean values. For a differentiable family and a common frame connection $\mathcal{A}_t$, with G, c fixed,

$$\mathcal{D}_t R_X = \frac{G}{c^4} \mathcal{D}_t H_X, \quad \mathcal{D}_t M_X = c^{-2} \mathcal{D}_t H_X, \quad \mathcal{D}_t O = \dot{O} + [\mathcal{A}_t, O]. \quad (50.5)$$

This follows by differentiating the full identity, including the frame. It does not imply that the state is motionless.

For positive modes, or for operators on distinct tensor factors,

$$m(y) = \frac{\hbar}{cL} y, \quad \frac{Gm_1 m_2}{\hbar c} = y_1 y_2, \quad \frac{R(y)}{\bar{\lambda}(y)} = y^2, \quad \bar{\lambda}(y) = L/y. \quad (50.6)$$

Substitution cancels the dimensional sections; operators on distinct factors commute. The dimensionless size depends on the character, without any other free parameter per species. This does not imply a spatial force law or a numerical evaluation without evaluating the character.

The calendar preserves $t_P = L/c$, $t_0 = \pi L/(54c)$ and $108t_0 = 2\pi t_P$. Therefore $L = (54/\pi)\ell_0$, with $\ell_0 = ct_0$; the two lengths are not identified. Between sections $\hbar_b = \rho \hbar_a$, $\rho > 0$, with L, c, y_i fixed, we have $G_b = G_a/\rho$, $m_{i,b} = \rho m_{i,a}$ and the quantity $Gm_1 m_2/(\hbar c)$ remains invariant. Joint sections are compared, not a temporal variation with the other dimensional coordinates artificially held fixed.

50.2 Static and dynamic memory, and spectral domain

In finite dimension, or with bounded coercive blocks, write

$$Y = \begin{pmatrix} A & F \\ F^* & C \end{pmatrix} > 0, \quad Y_{\text{ef}} = A - FC^{-1}F^*, \quad \eta = y + C^{-1}F^*b.$$

Completing the square proves

$$\left\langle \begin{pmatrix} b \\ y \end{pmatrix}, Y \begin{pmatrix} b \\ y \end{pmatrix} \right\rangle = \langle b, Y_{\text{ef}} b \rangle + \langle \eta, C \eta \rangle. \quad (50.7)$$

Hence $Y_{\text{ef}} > 0$, the minimizing extension $y = -C^{-1}F^*b$ and the reconstruction $y = \eta - C^{-1}F^*b$. The residual η is retained. Since $\text{Schur}(sY) = sY_{\text{ef}}$ for $s > 0$, we obtain $H_{X,\text{ef}} = E_L Y_{\text{ef}}$ and $R_{X,\text{ef}} = L Y_{\text{ef}}$. Solving $Y(x, y) = (f, 0)$ shows that the boundary block of Y^{-1} is Y_{ef}^{-1} , hence $\Lambda_b = L Y_{\text{ef}}^{-1}$. Direct compression of Y would give A : it is not the same operation.

Let $\chi = L/E_L = G/c^4 > 0$ and $R_X = \chi H_X$. In a decomposition compatible with the domains, we define

$$\mathcal{K}_{H_X}(z) = H_{bb} - zI - V(H_{ii} - zI)^{-1}V^*.$$

Proposition 50.2 (Full dynamic memory). *In the resolvent domain of the interior block,*

$$\mathcal{K}_{R_X}(\chi z) = \chi \mathcal{K}_{H_X}(z). \quad (50.8)$$

If both complements are invertible, their inverses carry the factor χ^{-1} .

Proof. Each radial block is χ times the energy block and $(\chi H_{ii} - \chi z I)^{-1} = \chi^{-1}(H_{ii} - z I)^{-1}$. Substitution leaves a factor χ in each term, preserving order. Taking inverses proves the last assertion. $\square$

The energy parameter z is transported to the length χz ; the identity preserves every order of spectral memory. For $|z| < \|H_{ii}^{-1}\|^{-1}$, the resolvent series gives

$$\mathcal{K}_{H_X}(z) = H_{\text{est}} - zZ + O(z^2), \quad H_{\text{est}} = H_{bb} - V H_{ii}^{-1} V^*, \quad Z = I + V H_{ii}^{-2} V^* \geq I.$$

Positivity follows from $Z - I = (H_{ii}^{-1} V^*)^* (H_{ii}^{-1} V^*)$. The static extension $(b, -H_{ii}^{-1} V^* b)$ has squared norm $\langle b, Zb \rangle$: energy and temporal norm come from the same extension.

If $T = Z^{1/2}$ is bounded and invertible, transport is dual:

$$H_{X,c} = T^{-*} H_{X,\text{ef}} T^{-1}, \quad R_{X,c} = T^{-*} R_{X,\text{ef}} T^{-1}, \quad \Lambda_c = T \Lambda_b T^*. \quad (50.9)$$

Multiplication proves $R_{X,c} = \chi H_{X,c}$, $R_{X,c} \Lambda_c = L^2 I$ and $H_{X,c} \Lambda_c = \hbar c I$, without commutativity of T with the character. If $\psi = T(t)b$, $\dot{\psi} = \dot{T}b + T\dot{b}$ is preserved. The temporal connection term of the generator is not retrospectively incorporated into the radial character without declaring its map.

Proposition 50.3 (Spectral extension and compatible refinement). *Let X be self-adjoint, positive and injective, and U unitary. The readers of (50.3), defined by spectral calculus of $Y = UXU^*$, satisfy $R_X = \chi H_X$ on $\text{Dom}(Y)$. Their products with Λ_X hold on $\text{Dom}(Y^{-1})$ and have the bounded extensions of (50.4). If isometric embeddings satisfy $X'J = JX$, $U'J = KU$ and preserve L, E_L , then $R'_X K = KR_X$ and $H'_X K = KH_X$ on the transported domains.*

Proof. On $P_{\varepsilon,M} = \mathbf{1}_{[\varepsilon,M]}(Y)$, readers and inverses are bounded and the identities are multiplications of functions of Y . For $\psi \in \text{Dom}(Y)$,

$$\|Y(P_{\varepsilon,M} - I)\psi\|^2 = \int_{(0,\infty) \setminus [\varepsilon,M]} \lambda^2 d\langle \psi, E_Y(\lambda)\psi \rangle \longrightarrow 0.$$

Injectivity removes spectral mass at zero and also $P_{\varepsilon,M}\psi \rightarrow \psi$. This is convergence in the graph norm. If $\psi \in \text{Dom}(Y^{-1})$, then $Y^{-1}\psi \in \text{Dom}(Y)$ and $YY^{-1}\psi = \psi$, which proves the products and their extensions. Substituting the intertwining relations into $R'_X = LU'X'U'^*$ proves $R'_X K = KR_X$; the energy calculation is the same. $\square$

The null sector remains separate. This specifies the unbounded extension of the preceding normalization. An arbitrary orthogonal projection is not thereby entitled to the block formulas: unbounded operators require compatible domains or closed forms with controlled blocks.

50.3 Backreaction in the same transport

The geometric record x , the internal links u and the fields f come from the same enriched state. In an oriented cellularization,

$$E(f, x, u) = \langle r, W(x, u)r \rangle, \quad r = D_T f, \quad (D_T f)_a = f_{t(a)} - T_a f_{s(a)}, \quad T_a = U_a^{\text{sp}}(x) \otimes R_{\text{int}}(u_a). \quad (50.10)$$

$W = W^* > 0$ preserves the blocks between incidences. Spin and internal representation have distinct factors; the chiral realization preserves its blocks and the equivariant Higgs–Yukawa map.

Proposition 50.4 (Differential of a single energy). *With fixed fields and $q = Wr$,*

$$\begin{aligned} \delta E &= -2 \operatorname{Re} \sum_a \langle q_a, \delta T_a f_{s(a)} \rangle + \langle r, \delta W r \rangle, \\ \delta T_a &= \delta U_a^{\text{sp}} \otimes R_{\text{int}}(u_a) + U_a^{\text{sp}} \otimes \delta R_{\text{int}}(u_a). \end{aligned} \quad (50.11)$$

Proof. Differentiating the quadratic form gives two terms conjugate through $W = W^*$ and the term $\langle r, \delta W r \rangle$. Using $\delta r_a = -\delta T_a f_{s(a)}$ proves the first formula; the product rule proves the second. Both variations act on the same r, q , with no independently chosen currents. $\square$

50.3.1 One quantum space and one joint operator

We fix a finite cellularization, a finite set of geometric records $\mathcal{X}$ and a finite fermionic Fock space $\mathcal{F}$. The links range over the received compact product $\mathcal{G}^E$; the doublets $\Phi_v \in \mathbb{C}^2$ have no amplitude cutoff. The invariant measures $d\mu_x = Z_x^{-1} e^{-S_x} d\mu_0$ have bounded positive densities with bounded inverse and do not depend on Φ in this chart. The space is

$$\mathcal{H} = \bigoplus_{x \in \mathcal{X}} L^2(\mathcal{G}^E \times \mathbb{C}^{2|V|}, \mu_x \otimes d\Phi; \mathcal{F}).$$

The identifications $A_x f = \sqrt{Z_x} e^{S_x/2} f$ are unitary from the common measure and transport the clock to $H_{\text{clk}}^\mu = A(H_{\text{clk}} \otimes I)A^*$. The clock mixes geometric records of the same space. The operator is realized through the form of

$$\mathbb{H} = H_{\text{clk}}^\mu + \hbar c K_{\text{gauge}} + d\Gamma(D_T^* W D_T) + H_{\Phi, \text{cin}} + V_\Phi + \hat{H}_Y + \hat{H}_{\text{tor}}. \quad (50.12)$$

The Yukawa map is linear in $\Phi, \bar{\Phi}$ and satisfies $Y(g\Phi)\rho_R(g) = \rho_L(g)Y(\Phi)$. The eliminated contorsion is not simultaneously retained as an independent variable in the base terms of this effective chart.

The scalar-momentum kinetic term has the form $\int \langle \nabla_\Phi \Psi, A(x, u) \nabla_\Phi \Psi \rangle$, with A bounded, uniformly positive and independent of Φ . We require $A(x, u^g) = O_g A(x, u) O_g^\top$ for the real action O_g of the doublets; positivity alone does not prove that covariance. The scalar spatial energy is a positive quadratic potential bounded by $C_{\text{grad}} r^2$, $r^2 = \sum_v |\Phi_v|^2$, not a bounded operator on all of L^2 . The volumes satisfy $0 < w_{\min} \leq w_v(x) \leq w_{\max}$ and $\lambda_\Phi > 0$. The clock, gauge block, transports, W and torsional coefficients are bounded in the chart. The potential retains its absolute value:

$$V_\Phi = \sum_v w_v \lambda_\Phi (|\Phi_v|^2 - v_0^2/2)^2 - \sum_v w_v \lambda_\Phi v_0^4/4. \quad (50.13)$$

The last contribution depends on volume and is not subtracted.

Theorem 50.5 (Closed form and joint evolution). *The form of (50.12) is closed and bounded below on $\mathcal{Q} = \{\Psi \in \mathcal{H} : \nabla_\Phi \Psi \in L^2, r^2 \Psi \in L^2\}$. It determines a unique associated self-adjoint operator $\mathbb{H}$ and the unitary evolution $e^{-it\mathbb{H}/\hbar}$.*

Proof. Cauchy–Schwarz gives $\sum_v w_v \lambda_\Phi |\Phi_v|^4 \geq ar^4$, $a = \lambda_\Phi w_{\min}/|V| > 0$. Finite Fock space and linear Yukawa give $\|\hat{H}_Y(\Phi)\| \leq C_Y r$; torsion is bounded. For $\varepsilon > 0$,

$$C_Y r \leq \varepsilon r^4 + \frac{3C_Y^{4/3}}{4^{4/3}\varepsilon^{1/3}}, \quad br^2 \leq \varepsilon r^4 + \frac{b^2}{4\varepsilon}.$$

The first bound maximizes $C_Y r - \varepsilon r^4$; the second completes the square in r^2 . Applying them with $\varepsilon = a/4$ to Yukawa and to the negative quadratic term gives

$$q(\Psi) \geq c_1 \|\nabla_\Phi \Psi\|^2 + (a/2) \|r^2 \Psi\|^2 - C \|\Psi\|^2, \quad c_1 > 0.$$

The norm $\|\Psi\|^2 + \|\nabla_\Phi \Psi\|^2 + \|r^2 \Psi\|^2$ is complete by closedness of the weak derivative and multiplication. These bounds and the upper bounds make the form norm, after adding a constant, equivalent to that complete norm. The representation theorem for closed forms produces the associated operator, and its spectral calculus gives the evolution. Uniqueness refers to this form and domain, not to every alternative formal domain. $\square$

Covariance of D_T, W , invariance of measures, the Yukawa identity and internal torsional contraction preserve q and $\mathcal{Q}$. Uniqueness of the associated operator implies that the compact representation commutes with $\mathbb{H}$; its Haar average reduces the operator and its evolution. It is not a subsequent corrective average.

The computational cutoff is removed with complete Peter–Weyl blocks and complete isotropic Hermite shells. Regularization and smooth truncation approximate derivatives and the weight r^2 ;

bounded densities preserve equivalent norms. Their union is dense in $\mathcal{Q}$. For the orthogonal projections P_n in the measures μ_x and the Galerkin operators $\mathbb{H}_n$,

$$(\mathbb{H}_n + \lambda)^{-1} P_n f \longrightarrow (\mathbb{H} + \lambda)^{-1} f \quad (\lambda > C).$$

The coercive form equation and its Galerkin version give error orthogonality; continuity and coercivity bound it by the best approximation error, which tends to zero. The scalar cutoff and the link-function cutoff of this chart are removed, not the spatial, geometric or fermionic cutoffs. Self-adjointness does not imply that $e^{-t\mathbb{H}}$ is trace class either.

50.3.2 Exchange and conservation

For $V = \sum_x P_x \otimes B_x$ and an element h_{xy} of the clock, block multiplication gives

$$[H_{\text{clk}} \otimes I, V]_{xy} = h_{xy}(B_y - B_x). \quad (50.14)$$

If $h_{xy} \neq 0$ and $B_y \neq B_x$, there is geometric–material exchange. If $\mathbb{H} = A + B + V$, for bounded operators or a common core justifying the derivatives,

$$\frac{d}{dt} \langle A \rangle = \frac{i}{\hbar} \langle [\mathbb{H}, A] \rangle, \quad \frac{d}{dt} \langle B \rangle = \frac{i}{\hbar} \langle [\mathbb{H}, B] \rangle, \quad \frac{d}{dt} \langle V \rangle = \frac{i}{\hbar} \langle [\mathbb{H}, V] \rangle.$$

The sum is zero by $[\mathbb{H}, \mathbb{H}] = 0$. Energy, including interaction, is conserved, not each sector separately. Geometry belongs to the quantum space; it is not added as an external background. The extended histories retain memory and return; their refinement is not by definition equivalent to spatial refinement.

50.4 The same action and its currents

The geometric realization uses a nondegenerate cotetrad e , a Lorentz connection ω and internal connections of the same matter. Let $\Sigma^{IJ} = e^I \wedge e^J$, $P_\gamma = \star + \gamma^{-1}I$, $\gamma \in \mathbb{R} \setminus \{0\}$ and $\star^2 = -I$ on bivectors. The couplings remain fixed during variation. With $\kappa = 8\pi G/c^4$, the same action is

$$S_{\text{tot}} = \frac{\hbar}{16\pi L^2} \int \Sigma_{IJ} \wedge (P_\gamma F_\omega)^{IJ} - \frac{\Lambda \hbar}{8\pi L^2} \int \text{vol}_e + S_m, \quad (50.15)$$

where $S_m = S_{\text{YM}} + S_\Phi + S_{\text{Weyl}} + S_Y$ contracts its indices with the same e , which is varied. The coefficients follow from $(2\kappa c)^{-1} = \hbar/(16\pi L^2)$.

With $\delta_\omega S_m = -\frac{1}{2} \int \delta\omega^{IJ} \wedge \sigma_{IJ}$, $\delta F = D_\omega \delta\omega$ and integration by parts,

$$D_\omega(P_\gamma \Sigma) = \kappa c \sigma. \quad (50.16)$$

Interior variations or the compatible boundary completion are used. Spin current and metric source are derivatives of the same matter, not two independent inputs.

50.4.1 Torsional elimination and cross-terms

The real inverse is $P_\gamma^{-1} = \gamma^2(-\star + \gamma^{-1}I)/(1 + \gamma^2)$: the product of the two polynomials in $\star$ before normalization equals $(1 + \gamma^{-2})I$. Denote by $y^{IJ} = \kappa c(P_\gamma^{-1}\sigma)^{IJ}$ the degree-three bivector-valued form, distinct from the character Y . For $T^I = D_\omega e^I$, $D_\omega \Sigma^{IJ} = T^I \wedge e^J - e^I \wedge T^J$. With the dual frame E_I , its inverse is

$$\mathcal{B}^I = \sum_J \iota_{E_J} y^{IJ}, \quad \vartheta = \frac{1}{4} \sum_I \iota_{E_I} \mathcal{B}^I, \quad T^I = \mathcal{B}^I - e^I \wedge \vartheta.$$

Indeed, if $t = \sum_J \iota_{E_J} T^J$, contractions give $\mathcal{B}^I = T^I + e^I \wedge t$ and then $4t$. The map has zero kernel between spaces of dimension twenty-four, hence is an isomorphism. For $\omega = \hat{\omega}(e) + \mathfrak{k}$,

$$T_{IJK} = \mathfrak{k}_{IKJ} - \mathfrak{k}_{IJK}, \quad \mathfrak{k}_{IJK} = \frac{1}{2}(T_{KIJ} - T_{IJK} - T_{JKI}).$$

The cyclic combination and antisymmetry in the first two indices prove these formulas. For matter affine in $\mathfrak{k}$, with current independent of it, there is a unique $\mathfrak{k}_*[e, \sigma]$ linear in the total current $\sigma = \sum_s \sigma_s$.

The curvature expansion preserves the boundary

$$S_\partial[e, \mathfrak{k}] = \frac{1}{2\kappa c} \int_{\partial M} \Sigma_{IJ} \wedge (P_\gamma \mathfrak{k})^{IJ}.$$

The contorsion-dependent interior density is $\mathcal{Q}_{e,\gamma}(\mathfrak{k}) - \frac{1}{2} \mathfrak{k}^{IJ} \wedge \sigma_{IJ}$, where $\mathcal{Q}$ is homogeneous of degree two. Its stationarity evaluated at $\delta \mathfrak{k} = \mathfrak{k}_*$ gives $2\mathcal{Q}(\mathfrak{k}_*) = \frac{1}{2} \mathfrak{k}_*^{IJ} \wedge \sigma_{IJ}$. Therefore,

$$\mathcal{L}_{\text{spin}}^{\text{ef}} = -\frac{1}{4} \mathfrak{k}_*^{IJ} \wedge \sigma_{IJ} = -\mathcal{Q}_{e,\gamma}(\mathfrak{k}_*). \quad (50.17)$$

Linearity preserves cross-terms between species. Contorsion is eliminated once; its eliminated linear interaction and the effective quartic term are not added simultaneously.

In the spinorial specialization, the received coefficient $C_\gamma = 3\kappa c \hbar^2 \gamma^2 / [16(1 + \gamma^2)]$ gives

$$\frac{C_\gamma}{\hbar} = \frac{3\pi}{2} L^2 \frac{\gamma^2}{1 + \gamma^2}. \quad (50.18)$$

It suffices to substitute $\kappa = 8\pi G/c^4$ and $\hbar G = c^3 L^2$, without changing the branch of γ . In the Fock space of the chart, the total current preserves the contraction

$$\hat{Q}_{\text{tot},v} = \sum_{a=1}^4 s_a \{d\Gamma(\tilde{M}_{a,v})^2 - d\Gamma(\tilde{M}_{a,v}^2)\}, \quad s = (-1, -1, -1, +1),$$

$$\hat{H}_{\text{tor}} = -\sum_v \frac{N_v c C_\gamma}{\mathcal{V}_v} \hat{Q}_{\text{tot},v}.$$

The matrices $\tilde{M}_{a,v}$ represent the current components. N_v is the lapse, distinct from N_Ω and from an occupation. The normal-ordering identity removes the one-particle contraction, not the cross-terms between species. The signature remains in s_a .

50.4.2 Metric source and Legendre transform of the same action

Stationary elimination preserves variation of the other variables: the chain rule contains a term in $\delta \mathfrak{k}_*$ multiplied by the connection equation, which is zero. The reduced action is

$$S_{\text{red}} = \frac{1}{2\kappa c} \int (R[g] - 2\Lambda) \text{vol}_e + S_m^{\text{LC}} + S_{\text{spin}}^{\text{ef}} + S_\partial.$$

We define the action source by $\delta_g(S_m^{\text{LC}} + S_{\text{spin}}^{\text{ef}}) = -\frac{1}{2} \int \mathcal{T}_{\mu\nu}^S \delta g^{\mu\nu} \text{vol}_e$. The interior gravitational variation produces

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa c \mathcal{T}_{\mu\nu}^S = \kappa (T_{\mu\nu}^{\text{LC}} + U_{\mu\nu}^{\text{spin}}), \quad T^{\text{LC}} + U^{\text{spin}} = c \mathcal{T}^S. \quad (50.19)$$

The volume variation of the constant term of the potential is included. The factor c corresponds to the cotetrad on the length line; action and physical sources are not mixed.

With $a = (2\kappa c)^{-1}$ and $\mathcal{B}_{IJ} = (P_\gamma \Sigma)_{IJ}$, the geometric potential is $\theta_{\text{geom}} = a \mathcal{B}_{IJ} \wedge \delta \omega^{IJ}$. Decomposing $\omega = \omega_0 dt + \underline{\omega}$ and $F_{0a} = \dot{\omega}_a - D_a \omega_0$ identifies the kinetic term; integration by parts preserves Gauss and the boundary. With $e_0 = Nn + N^a e_a$ and the Legendre transform of the same matter,

$$S_{\text{tot}} = \int dt [\Theta_{\text{red}}(\dot{z}) - \mathcal{C}[N] - \mathcal{V}[N^a] - \mathcal{G}_L[\omega_0] - \mathcal{G}_{\text{int}}[A_0]] + S_\partial. \quad (50.20)$$

Lapse, shift and gauge retain their degenerate directions. Second-class constraints are reduced before using Θ_{red} or are expressly retained. First-order spinors are not assigned a fictitious velocity inverse.

For contorsion and its momentum, the constraints are $\pi_{\mathfrak{k}} = 0$ and $\partial\mathcal{Q}/\partial\mathfrak{k} - \sigma/2 = 0$. Their matrix has blocks $\begin{pmatrix} 0 & -M \\ M^\top & B \end{pmatrix}$, with M invertible by the preceding reconstruction; it is therefore invertible. The Dirac bracket performs this elimination and separately preserves the graded fermionic constraints. In a regular bosonic direction,

$$H_\Phi = N \left(\frac{p^2}{2w} + wV + H_{\text{grad}} \right), \quad p = \frac{w}{N} D_t \phi, \quad L_\Phi = \frac{w}{2N} |D_t \phi|^2 - NwV - NH_{\text{grad}}.$$

The volume $w = \sqrt{\det q}$ is not frozen. For $K_{ij} = (\dot{q}_{ij} - \mathcal{L}_{N^a} q_{ij})/(2N)$,

$$\Pi^{ij} = \frac{\sqrt{q}}{2\kappa c} (K^{ij} - q^{ij} K), \quad K_{ij} = \frac{2\kappa c}{\sqrt{q}} (\Pi_{ij} - \frac{1}{2} q_{ij} \Pi).$$

The trace and the traceless part prove the inverse. If the total momentum contains a material frame contribution, $\Pi = p - p_m$ is used, preserving p_m . The same action and its source are thereby recovered. The Legendre transform does not by itself prove that an arbitrary quantization of (50.20) coincides with (50.12).

50.5 Constraint propagation with the total source

Let $E_{\mu\nu} = G_{\mu\nu} + \Lambda g_{\mu\nu} - \kappa \mathcal{T}_{\mu\nu}^S$. On the matter equations, diffeomorphism invariance of the reduced functional implies $\nabla^\mu \mathcal{T}_{\mu\nu}^S = 0$: an interior variation generated by ξ leaves the integral of $\mathcal{T}_{\mu\nu}^S \nabla^\mu \xi^\nu$, while the matter equations cancel the other terms. Integrating by parts and varying ξ proves zero divergence, before imposing $E = 0$. Bianchi and the fixed couplings give $\nabla_\mu E^{\mu\nu} = 0$.

Proposition 50.6 (Preservation of constraint data). *A regular solution of the matter and metric evolution equations preserves the initially satisfied constraints throughout its interval of regular existence.*

Proof. In Gaussian coordinates $g = -d\tau^2 + q_{ij}(\tau, x) dx^i dx^j$, we impose $E_{ij} = 0$ and write $C = E_{00}$, $C_i = E_{0i}$, $K_{ij} = \dot{q}_{ij}/2$ and $\mathcal{K} = q^{ij} K_{ij}$. We have $E^{00} = C$, $E^{0i} = -C^i$, $E^{ij} = 0$, $\Gamma^i_{0j} = K^i_j$ and $\Gamma^\mu_{\mu 0} = \mathcal{K}$. The temporal and spatial components of the divergence give

$$\partial_\tau C - D_i C^i + \mathcal{K} C = 0, \quad \partial_\tau C^i + \mathcal{K} C^i + 2K^i_j C^j = 0.$$

On lowering the index, $\dot{q}_{ij} = 2K_{ij}$ cancels the last spatial term. Exactly the following remain

$$\partial_\tau C - D_i C^i + \mathcal{K} C = 0, \quad \partial_\tau C_i + \mathcal{K} C_i = 0. \quad (50.21)$$

The second homogeneous equation maintains $C_i = 0$ and the first becomes $\dot{C} + \mathcal{K} C = 0$, preserving $C = 0$. q and $\mathcal{K}$ are those of the coupled solution, not a prescribed background. $\square$

The Gaussian chart locally proves a tensorial assertion, without freezing q^{-1} or replacing the source. Global singularity-free existence is not asserted for every datum. Nor is variational propagation by itself a quantum commutator identity for arbitrary normal deformations.

50.6 Boundary and self-inertial operator

Let $b : \mathcal{X}_{\infty}^{\text{enr}} \rightarrow B_{\text{ef}}$, $B_{\text{ef}} = \text{im}(b)$. A response I_v factors uniquely as $I_v = \bar{I}_v \circ b$ if and only if it is constant on the fibres of b . Necessity is immediate; for sufficiency, $\bar{I}_v(\beta) = I_v(x)$ with $b(x) = \beta$ is independent of the representative. The effective image gives uniqueness. Strong Machian dependence adds states with the same local restriction and different responses.

The hinge of the same state produces $r_{\text{av}} = \pi/\varphi^2$ and $r_{\text{av}}^J = \varphi/\pi^2$. In the common-amplitude sector, $\mathcal{L}_{\text{ar}} = \pi\mu_{\text{ar}}\mathcal{M}(x)$ and $\mathcal{L}_{\text{vol}} = \mu_{\text{vol}}\mathcal{M}(x)$, $\mathcal{M}(x) > 0$, with $\mu_{\text{ar}}/\mu_{\text{vol}} = \varphi^{-2}$. Dividing by $\mathcal{L}_{\text{vol}}$ gives readers of the same type $A_\partial = r_{\text{av}}$ and $V_\partial = 1$. The quotient cancels the amplitude, not the history preserved in the archive.

For an orientation a , $q_a(x) = h_a(x)/(108t_0(x)) = \hbar_a(x)/t_P(x)$ descends to $Q : B_{\text{ef}} \rightarrow \mathcal{L}_E^+$ if and only if $b(x) = b(x')$ implies $h_a(x)/t_0(x) = h_a(x')/t_0(x')$. This is the factorization criterion applied to the clock. It is satisfied by fixed charts and by variable charts whose quotient descends. In the radial chart, $Q = \hbar_a c/L$. With energies on that same line, $E_1 E_2 / Q^2$ is dimensionless and basis-independent.

For complementary orthogonal projections,

$$w_A = \frac{A_\partial r_{\text{av}}}{A_\partial r_{\text{av}} + V_\partial r_{\text{av}}^J} = \frac{\pi^4}{\pi^4 + \varphi^5}, \quad w_V = 1 - w_A, \quad (50.22)$$

$$W_{\text{av}} = w_A P_A + w_V P_V, \quad \bar{I}_v = \frac{E_1 E_2}{Q^2} W_{\text{av}}. \quad (50.23)$$

Multiplying the numerator and denominator of $r_{\text{av}}^2/(r_{\text{av}}^2 + r_{\text{av}}^J)$ by $\varphi^4 \pi^2$ proves the evaluation. The distinction between areal reader and weighting is preserved: the regional mark is generated once. $0 < w_A, w_V < 1$ makes W_{av} positive and invertible, and

$$\langle z, \bar{I}_v z \rangle = \frac{E_1 E_2}{Q^2} (w_A \|P_A z\|^2 + w_V \|P_V z\|^2) > 0 \quad (z \neq 0).$$

Fully exchanging $(A_\partial, r_{\text{av}}, P_A)$ and $(V_\partial, r_{\text{av}}^J, P_V)$ permutes the summands. Unitary transport conjugates the response and preserves its spectrum; implementing the exchange within one space requires equal sector dimensions. A common rescaling of A_∂, V_∂ does not change the weights and does not prove global dependence. With fixed energies and weights, $Q \mapsto qQ$ changes the response by q^{-2} ; the strong witness also preserves the same local restriction.

Theorem 50.7 (Self-inertia and radius of the same character). *With the same clock and orientation,*

$$\bar{I}_v = \frac{G_a E_1 E_2}{\hbar_a c^5} W_{\text{av}} = \frac{G_a m_1 m_2}{\hbar_a c} W_{\text{av}}. \quad (50.24)$$

For a probe $y_p > 0$, $E_p = Q y_p$ and $\bar{\lambda}_p = L/y_p$, the spectral extension satisfies

$$\hat{I}_{p,Y} = y_p Y \otimes W_{\text{av}} = \frac{E_p}{Q^2} H_X \otimes W_{\text{av}} = \frac{R_X}{\bar{\lambda}_p} \otimes W_{\text{av}} = \frac{G_a m_p}{\hbar_a c} M_X \otimes W_{\text{av}}. \quad (50.25)$$

Proof. $Q^2 = \hbar_a^2 c^2 / L^2$ and $\hbar_a G_a = c^3 L^2$ give $Q^{-2} = G_a / (\hbar_a c^5)$; $E_i = m_i c^2$ is then used. For $Y = \sum_j y_j P_j$, applying the reader to each energy $Q y_j$ produces $\sum_j y_p y_j P_j \otimes W_{\text{av}}$, independently of the eigenbasis. Spectral calculus extends the linear reader to the general positive self-adjoint character. Decomposition by P_A, P_V gives the operators $y_p w_A Y$ and $y_p w_V Y$, with their multiplicities: they are positive self-adjoint operators on the domain of $Y \otimes I$. Substituting the radial readers proves the other expressions. $\square$

If an additional realization effectively identifies its full operator with $H_X = QY$, it preserves that identification and its domains. For $H_X = \sum_r H_r$ on a common core, $\hat{I}_{p,Y} = (E_p/Q^2) \sum_r H_r \otimes W_{\text{av}}$. Positivity of each summand is not required; positivity of the character of the identified operator is. No terms are suppressed or energy origin changed to impose it. The theorem does not automatically assign this reader to $\mathbb{H}$.

In the blocks of (50.7), the interior inverse of $y_p Y \otimes W_{\text{av}}$ is $y_p^{-1} C^{-1} \otimes W_{\text{av}}^{-1}$. Multiplication gives

$$\text{Schur}(\hat{I}_{p,Y}) = y_p Y_{\text{ef}} \otimes W_{\text{av}} = \frac{R_{X,\text{ef}}}{\bar{\lambda}_p} \otimes W_{\text{av}}. \quad (50.26)$$

The cross-blocks are not deleted. For $a_s = y_p w_s > 0$, $s = A, V$, the same calculation gives $\mathcal{K}_{a_s Y}(a_s z) = a_s \mathcal{K}_Y(z)$ in the interior resolvent domain, preserving full memory. If $Y'K = KY$ and $W_{\text{av}}'J = JW_{\text{av}}$, with y_p and the intensive section fixed,

$$\hat{I}_{p,Y'}(K \otimes J) = (K \otimes J) \hat{I}_{p,Y}. \quad (50.27)$$

This is proved by substitution. The spectral cutoffs of Proposition 50.3 converge in graph norm; the two fixed weights transmit convergence to the response. Naturality is that of these effective intertwiners, not that of a different family of spatial maps identified by name alone.

50.7 Self-inertia of the state and acceleration of its projection

In a differentiable realization, the joint state $\Gamma = (x, \mathcal{O}, m)$ has connection ∇^{tot} on $\mathcal{N}$. A smooth projection $\pi_S : \mathcal{N} \rightarrow \mathcal{N}_S$, with connection ∇^S , preserves the transports and readers it realizes. The observed frame is $\mathfrak{F}_{\mathcal{O}}(\Gamma) = \text{Res}_{\mathcal{O}}(\Gamma, \nabla^{\text{tot}})$. For a twice differentiable curve and $\gamma_S = \pi_S \circ \Gamma$, self-inertia means $\nabla_{\dot{\Gamma}}^{\text{tot}} \dot{\Gamma} = 0$. The defect is

$$\mathcal{K}_{\pi_S}(\dot{\Gamma}, \dot{\Gamma}) = \nabla_{\dot{\gamma}_S}^S \dot{\gamma}_S - d\pi_S(\nabla_{\dot{\Gamma}}^{\text{tot}} \dot{\Gamma}).$$

Proposition 50.8 (Readout acceleration). *For a self-inertial trajectory,*

$$a_S = \mathcal{K}_{\pi_S}(\dot{\Gamma}, \dot{\Gamma}), \quad (\mathcal{K}_{\pi_S})^a_{BC} = \partial_B \partial_C \pi_S^a + (\Gamma^S)^a_{bc} \partial_B \pi_S^b \partial_C \pi_S^c - \partial_D \pi_S^a (\Gamma^{\text{tot}})^D_{BC}. \quad (50.28)$$

Proof. The chain rule gives $\ddot{\gamma}_S^a = \partial_B \pi_S^a \ddot{\Gamma}^B + \partial_B \partial_C \pi_S^a \dot{\Gamma}^B \dot{\Gamma}^C$. Adding the subsystem connection and subtracting the image of the total acceleration cancels the terms in $\ddot{\Gamma}$. The remaining terms are the displayed tensor contracted with the velocities. Self-inertia sets the total acceleration to zero and leaves the equality. $\square$

The geometric response factors through b when the connection, projection and kinematic datum descend to the boundary, or the latter is fixed in the comparison. If velocity changes, it is retained as an argument. The self-inertial whole can thus yield accelerated subsystems as readouts without an external absolute frame.

Self-inertia does not by itself cancel curvature or holonomy. A datum transported around a loop returns if and only if it lies in the fixed subspace of its holonomy. Return of the full state additionally requires an injective realization in the sector being compared: visible return and return of history remain distinct.

50.8 Result and scope checks

The same character fixes mass, energy and radius; their proportionality preserves static and spectral memory and dual normalization. Geometric and internal transports enter the differential of a single energy. The joint action produces the geometric and torsional sources and preserves the constraints on its regular solution. The self-inertial operator is the radius of the character divided by the conjugate length of the probe, while acceleration is the projection defect of the state.

The last two objects have distinct types: $\hat{I}_{p,Y}$ is identified neither with $\mathcal{K}_{\pi_S}$ nor with the Einstein tensor. Nor is the proved incorporation replaced by the different requirement of cancelling a total quantum curvature for arbitrary normal deformations. That statement has its own generators and domains; it is neither asserted here nor declared absent from the corpus because it is not this conclusion.

The falsifiers are concrete. Changing only G or $\hbar$ with L, Y fixed breaks (50.4); replacing L with ℓ_0 without $54/\pi$ alters the clock; deleting F modifies the reduction; jointly rescaling the areal and volumetric readers does not prove Machian separation; a map that does not intertwine Y does not satisfy (50.27). The rational tests are reproducible finite checks, not target values of the generator or substitutes for the domain and limit proofs.

51 Joint action, backreaction and canonical closure of the constraints

The causal sequence of this development is

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \text{enriched state} \longrightarrow \text{joint discrete structure of the continuum.}$$

The sheets, orientation, residues, carry, routes, boundary and memory of this construction are preserved. The action scale and the couplings are HMT outputs received before using variational and symplectic language. They are not selected by target values or by the curvature conclusion to be proved. Nor does phase return reset the enriched state.

The previously constructed geometric realization provides a nondegenerate cotetrad e , its metric connection and the spinorial lift of transport. Here that geometry is varied together with the matter fields. We shall bring together in a single proof the common action, its torsional reduction, the Legendre transformation, the first-class property, constraint propagation and the characteristic flatness of the canonical connection resulting from the same action. This last realization is not identified by name with another interacting Hamiltonian.

51.1 Starting structure and action with the full source

Let M be a smooth oriented four-dimensional region with spin structure, signature $(-+++)$ and a regular spatial foliation. The cotetrad takes values in the length line of the received realization. We write

$$\Sigma^{IJ} = e^I \wedge e^J, \quad F_\omega = d\omega + \omega \wedge \omega, \quad P_\gamma = \star + \gamma^{-1}I, \quad \star^2 = -I.$$

The star in this formula acts on internal bivectors; it is not exterior duality of forms. During variation, the received sections $c > 0, \hbar > 0, G > 0, \gamma \in \mathbb{R} \setminus \{0\}$ and Λ remain fixed. With $\kappa = 8\pi G/c^4$, the total action is

$$\begin{aligned} S_{\text{tot}} = & \frac{1}{2\kappa c} \int_M \Sigma_{IJ} \wedge (P_\gamma F_\omega)^{IJ} - \frac{\Lambda}{\kappa c} \int_M \text{vol}_e \\ & + S_{\text{YM}}[e, A] + S_H[e, A, H] + S_W[e, \omega, A, \Psi] + S_Y[e, H, \Psi] + S_\partial. \end{aligned} \quad (51.1)$$

Definition 51.1 (Variational and canonical unification). HMT variational and canonical unification is the composition of the geometric, chromatic and electroweak realizations of the same enriched state in the common action (51.1), whose variations produce joint sources and whose reduction preserves a common constraint algebra and its propagation. The electromagnetic component belongs to the electroweak realization. Theorem 51.2 establishes this composition in the regular domain and under the stated boundary conditions.

The electroweak couplings g, g' and the self-coupling λ_H are inherited from the construction in 16 and Theorem 16.2; the present variation holds these sections fixed. Comparison between sections is a distinct operation, preserved in the compatibility-horizon development.

The symbol H without a subscript denotes the Higgs field here, not a Hamiltonian. All spacetime contractions of matter use the same e . The boundary problem is that received from the Cartan–Holst variation: interior variations, compatible fixed boundary data, or a completion cancelling the same boundary term.

To fix the content of (51.1), the internal connections act on the chiral bundle

$$(S_L \otimes E_L) \oplus (S_R \otimes E_R).$$

That bundle is not replaced by a vector coupling. In the integer normalization $y = 6Y$, the received multiplets have weights $1, 4, -2, -3, -6$, respectively for $q_L, u_R, d_R, \ell_L, e_R$; the doublet H has weight 3. Colour acts on the factor $\mathbb{C}^3$ of quarks and trivially on leptons. The family factor and its maps Y_u, Y_d, Y_e are preserved. This block does not add a right-handed partner to a multiplet that has none.

With $\tilde{H} = i\sigma_2 \bar{H}$, the material map is

$$\begin{aligned} Y_q(H)(u, d) &= \tilde{H} \otimes Y_u u + H \otimes Y_d d, \\ Y_\ell(H)e &= H \otimes Y_e e. \end{aligned} \quad (51.2)$$

The identity on the colour factor is understood in the first line. The action term is $-(\bar{\Psi}_L Y(H) \Psi_R + \text{h.c.})$, with the received dimensional sections. Equivariance is not merely nominal: for $U \in SU(2)$, $i\sigma_2 \bar{U} = U i\sigma_2$, so that $\tilde{H} \mapsto U \tilde{H}$. The weights of the three columns are $-3 + 4 = 1$, $3 - 2 = 1$ and $3 - 6 = -3$. Colour is intertwined by the identity, and the family maps commute with the internal action. Therefore, for each internal transformation u ,

$$Y(uH)\rho_R(u) = \rho_L(u)Y(H). \quad (51.3)$$

If the preceding material realization also contains other multiplets, their maps are incorporated with their own equivariance identity; this proof invents no new states.

In a units chart $\hbar = c = 1$, the material density summarizing these actions has the form

$$\begin{aligned} \mathcal{L}_m = & -\frac{1}{4}F_{B,\mu\nu}F_B^{\mu\nu} - \frac{1}{4}\sum_a F_{W,\mu\nu}^a F_W^{a,\mu\nu} - \frac{1}{4}\sum_a F_{C,\mu\nu}^a F_C^{a,\mu\nu} \\ & - (\nabla_\mu H)^\dagger \nabla^\mu H - V(H) + \mathcal{L}_W(\Psi_L, \nabla_L) + \mathcal{L}_W(\Psi_R, \nabla_R) \\ & - (\bar{\Psi}_L \Upsilon(H) \Psi_R + \text{h.c.}). \end{aligned} \quad (51.4)$$

The curvatures include their couplings; these and the component normalizations are those of their source documents. Writing in that chart neither removes the action section nor recalibrates the coefficients. In particular, no new numerical value for the strong coupling is introduced here. The spin connection of ∇_L, ∇_R is that of e, ω ; the connections B, W, C act in the representations just described. The constant of V is retained, since its volume variation belongs to the gravitational source.

The spinorial part preserves the explicit conventions of the preceding development: $\{g^I, g^J\} = 2\eta^{IJ}I$, $\mathbb{B}_{IJ} = \frac{1}{4}[g_I, g_J]$, $A_D = -ig^0$, $\bar{\psi} = \psi^\dagger A_D$ and $D\psi = d\psi + \frac{1}{2}\omega^{IJ}\mathbb{B}_{IJ}\psi + A\psi$. Its symmetric kinetic term is

$$\frac{\hbar}{2}[\bar{\psi}g^I D\psi - (D\bar{\psi})g^I \psi] \wedge \eta_I, \quad \eta_I = \iota_{E_I} \text{vol}_e.$$

The sum runs over the fields actually present, with their chiral projections, not over a vector duplication of the catalogue. The Yukawa terms are algebraic and do not depend on ω . The total current is fixed by

$$\delta_\omega S_m = -\frac{1}{2} \int_M \delta\omega^{IJ} \wedge \sigma_{IJ}, \quad \sigma_{JK} = \sum_s -\frac{\hbar}{2} \bar{\psi}_s \{g^I, \mathbb{B}_{JK}\} \psi_s \eta_I. \quad (51.5)$$

51.2 A proof of compatibility, propagation and closure

Theorem 51.2 (Canonical compatibility of the joint action). *Consider (51.1) on the preceding nondegenerate domain, with matter affine in the contorsion, and in a regular chart of the restricted Legendre transformation. Retain the first-order fermionic constraints and eliminate the second-class constraints through their Dirac bracket. For parameters with interior support, or with the compatible boundary without improper charges, the following hold jointly: the metric equation has the total material source of that same action; its diffeomorphism and gauge generators are first class; the constraints propagate on every regular coupled solution; and the total canonical potential induces a prequantum connection flat in the characteristic directions of the regular constraint surface.*

The assertion concerns this action and this canonical realization. It does not automatically identify its connection with a different interacting Hamiltonian or prove its spatial–chiral limit.

Proof. 1. Torsional elimination without duplication. Write $\omega = \hat{\omega}(e) + \mathfrak{f}$. The expansion $F_\omega = F_{\hat{\omega}} + D_{\hat{\omega}}\mathfrak{f} + \mathfrak{f} \wedge \mathfrak{f}$ and $D_{\hat{\omega}}\Sigma = 0$ separate the boundary contribution

$$S_{\partial, \mathfrak{f}} = \frac{1}{2\kappa c} \int_{\partial M} \Sigma_{IJ} \wedge (P_\gamma \mathfrak{f})^{IJ}$$

from the algebraic density

$$\mathcal{Q}_{e,\gamma}(\mathfrak{f}) - \frac{1}{2}\mathfrak{f}^{IJ} \wedge \sigma_{IJ}, \quad \mathcal{Q}_{e,\gamma}(\mathfrak{f}) = \frac{1}{2\kappa c} \Sigma_{IJ} \wedge P_\gamma(\mathfrak{f} \wedge \mathfrak{f})^{IJ}. \quad (51.6)$$

The original variation gives $D_\omega(P_\gamma \Sigma) = \kappa c \sigma$. The Holst inverse and the torsion–contorsion inverse from the preceding development are invertible for the declared coframe and γ . They thus determine a unique $\mathfrak{f}_*[e, \sigma]$, linear in σ . The radial variation of (51.6) at its stationary point gives $2\mathcal{Q}(\mathfrak{f}_*) = \frac{1}{2}\mathfrak{f}_*^{IJ} \wedge \sigma_{IJ}$. Consequently,

$$S_{\text{red}} = \frac{1}{2\kappa c} \int_M (R - 2\Lambda) \text{vol}_e + S_m^{\text{LC}} - \frac{1}{4} \int_M \mathfrak{f}_*^{IJ} \wedge \sigma_{IJ} + S_{\partial, \text{red}}. \quad (51.7)$$

The first Bianchi identity cancels the Levi–Civita Holst term in this expression. The chain rule preserves the equations of e, A, H, Ψ , since the coefficient of $\delta \mathfrak{f}_*$ vanishes by stationarity. Since $\sigma = \sum_s \sigma_s$, the effective contribution includes $\mathfrak{f}_*[\sigma_s] \wedge \sigma_t$ for $s \neq t$. It is not replaced by squares of isolated species, nor is the same $\mathfrak{f}$ simultaneously retained as an independent variable.

2. Full source and backreaction. Define the action tensor by varying all the material terms of (51.7):

$$\delta_g S_{m,\text{red}} = -\frac{1}{2} \int_M \mathcal{T}_{\mu\nu}^S \delta g^{\mu\nu} \text{vol}_e, \quad E_{\mu\nu} := G_{\mu\nu} + \Lambda g_{\mu\nu} - \kappa c \mathcal{T}_{\mu\nu}^S. \quad (51.8)$$

The metric equation is $E_{\mu\nu} = 0$. The factor κc corresponds to the length cotetrad; if $T^{\text{phys}} = c\mathcal{T}^S$, we write κT^{phys} . The matter fields and internal connections are also varied. Thus the fields determine the source, and the geometry solving the equation in turn determines their connections, volume and evolution. This is backreaction, not evolution on a fixed background. The variation includes the volume of the Higgs potential and the metric dependence of the torsional current, as well as its cross-terms.

3. Restricted Legendre transform and canonical reduction. The total canonical potential is obtained from $\delta L = \mathcal{E}\delta z + d\theta(\delta z)$ by integrating θ over the spatial leaf and performing the second-class reduction. Its geometric part before that reduction is

$$\theta_{\text{geom}}(\delta) = \frac{1}{2\kappa c} (P_\gamma \Sigma)_{IJ} \wedge \delta \omega^{IJ}.$$

The decomposition $F_{0a} = \dot{\omega}_a - D_a \omega_0$, together with the same decomposition of the material actions, produces

$$S_{\text{can}} = \int d\tau [\Theta_{\text{red}}(\dot{z}) - \mathcal{C}[N] - \mathcal{D}[\xi] - \mathcal{G}_L[\lambda] - \mathcal{G}_{\text{int}}[\chi]] + S_{\partial,\text{red}}. \quad (51.9)$$

The absent velocities of lapse, shift or temporal connections are not artificially inverted.

The algebraic contorsion has constraints $\pi_{\mathfrak{f}} = 0$ and $\partial Q / \partial \mathfrak{f} - \sigma/2 = 0$. Their bracket matrix has the block

$$\begin{pmatrix} 0 & -M \\ M^T & B \end{pmatrix}, \quad M = \frac{\partial^2 Q}{\partial \mathfrak{f}^2}.$$

The torsional inverse makes M invertible, and hence this matrix. These are second-class constraints. We use

$$\{F, G\}_D = \{F, G\} - \{F, \chi_a\} (C^{-1})^{ab} \{\chi_b, G\}. \quad (51.10)$$

For the variables that do not depend on the contorsion–momentum pair, the lower-right block of C^{-1} is zero. This elimination produces no additional bracket between them. The first-order spinorial constraints are retained separately; in the graded phase space, the graded bracket and even generators are used. Θ_{red} denotes the full result, not only its geometric part.

Legendre inversion in regular directions recovers the same action. For example, $N[p^2/(2w) + wV + H_{\text{grad}}]$, with $w = \sqrt{\det q}$, gives $p = (w/N)D_\tau \phi$, and by substitution

$$L_H = \frac{w}{2N} |D_\tau \phi|^2 - NwV - NH_{\text{grad}}.$$

D_τ preserves the shift and temporal connection; w is not frozen. For $K_{ij} = (\dot{q}_{ij} - \mathcal{L}_\xi q_{ij})/(2N)$, the gravitational contribution to momentum and its inverse are

$$\Pi^{ij} = \frac{\sqrt{q}}{2\kappa c} (K^{ij} - q^{ij}K), \quad K_{ij} = \frac{2\kappa c}{\sqrt{q}} (\Pi_{ij} - \frac{1}{2} q_{ij} \Pi). \quad (51.11)$$

They yield

$$\mathcal{C}_{\text{grav}} = \frac{2\kappa c}{\sqrt{q}} (\Pi^{ij} \Pi_{ij} - \frac{1}{2} \Pi^2) - \frac{\sqrt{q}}{2\kappa c} (R^{(3)} - 2\Lambda).$$

If the spinor chart contributes a material shift of the total momentum p , we use $\Pi = p - p_m$; p_m is not discarded. The indefinite character of the DeWitt form does not prevent this algebraic inverse. No velocity inverse is required of the first-order fermionic kinetic term.

4. Noether–Legendre identity and first-class property. The coincidence of the generators with the constraints of the same action can be seen before imposing its equations. Set $a = (2\kappa c)^{-1}$, $B = P_\gamma \Sigma$, $u^I = \iota_\zeta e^I$ and $\lambda^{IJ} = \iota_\zeta \omega^{IJ}$. From $\mathcal{L}_\zeta \omega = \iota_\zeta F + D_\omega \lambda$ we obtain

$$\begin{aligned} j_\zeta^{\text{geom}} &= d(aB_{IJ}\lambda^{IJ}) - a(D_\omega B_{IJ})\lambda^{IJ} \\ &\quad - 2au^I e^J \wedge (P_\gamma F)_{IJ} + 2a\Lambda u^I \eta_I. \end{aligned} \quad (51.12)$$

The material part adds exactly its contributions to the equations of e, ω, A , its vertical generators and its boundary. Decomposing the same expression on a spatial leaf gives (51.9). Stationary reduction and Dirac reduction transport these transformations instead of replacing them by momenta of an external action.

With the usual canonical convention for the constraints, the total brackets are, modulo the preserved Gauss generators,

$$\begin{aligned} \{\mathcal{D}[\xi], \mathcal{D}[\eta]\}_D &\simeq \mathcal{D}[[\xi, \eta]], \\ \{\mathcal{D}[\xi], \mathcal{C}[N]\}_D &\simeq \mathcal{C}[\mathcal{L}_\xi N], \\ \{\mathcal{C}[N], \mathcal{C}[M]\}_D &\simeq \mathcal{D}[q^{ij}(N\partial_j M - M\partial_j N)\partial_i]. \end{aligned} \quad (51.13)$$

Indeed, tangential transformations have the Lie bracket and transport the lapse as a scalar. For normal transformations, differentiating their orthogonality to the tangent vectors and $g(n, n) = -1$ determines the tangential variation of n ; with K_{ij} from (51.11) we obtain $\delta_N n = \text{grad}_q N$. The deformation commutator, converted to the canonical bracket with these conventions, gives the last line of (51.13). Covariantized transport additionally preserves frame and gauge rotations. The transformations are tangent to the constraint surface of the invariant action, yielding the first-class property. The functions q^{ij} depend on the state and are not frozen. With interior support, no central boundary charge is added; an improper generator with nonzero charge is not declared a zero constraint.

5. Propagation in the geometry that responds to matter. Diffeomorphism invariance of $S_{m,\text{red}}$, evaluated on its matter equations, gives by integration by parts $\mathring{\nabla}^\mu \mathcal{T}_{\mu\nu}^S = 0$. The identity uses the full tensor of (51.8); it does not postulate separate conservation of each contribution. Bianchi then implies $\mathring{\nabla}_\mu E^{\mu\nu} = 0$, even before imposing all the metric equations.

In a local Gaussian chart, $ds^2 = -d\tau^2 + q_{ij}dx^i dx^j$, impose the evolution equations $E_{ij} = 0$ and set $C = E_{00}$, $C_i = E_{0i}$, $K_{ij} = \dot{q}_{ij}/2$, $K = q^{ij}K_{ij}$. Here K is only the trace of the second fundamental form. The two components of the divergence identity are

$$\partial_\tau C - D_i C^i + KC = 0, \quad \partial_\tau C_i + KC_i = 0. \quad (51.14)$$

To check the signs, we use $E^{00} = C$, $E^{0i} = -C^i$, $E^{ij} = 0$, $\Gamma^i_{0j} = K^i_j$ and $\Gamma^\mu_{\mu 0} = K$. On lowering the index of the spatial equation, $\dot{q}_{ij} = 2K_{ij}$ cancels the two connection terms. If initially $C_i = 0$, its homogeneous equation maintains $C_i = 0$; the first becomes $\dot{C} + KC = 0$ and preserves $C = 0$. q and K are those of the coupled solution. The Gaussian choice proves the local assertion; its tensorial form allows lapse and shift to be changed. Likewise, the Noether gauge identity relates the covariant divergence of the gauge equation to the matter equations and propagates Gauss when the spatial equations hold. Neither global existence nor absence of singularities for every solution is inferred.

6. The same action induces characteristic closure. Let $\mathcal{P}_{\text{red}}$ be the regular canonical chart obtained and $\mathcal{Z} = \{C_A = 0\}$ its regular surface of total constraints, including Gauss. We use a symbol distinct from Σ^{IJ} for this surface. The potential $\Theta = \Theta_{\text{red}}$ of (51.9) is retained, including the material and mixed terms. We fix

$$\Omega = -d_{\mathcal{P}}\Theta, \quad \iota_{X_f}\Omega = d_{\mathcal{P}}f, \quad \{f, g\}_D = \Omega(X_f, X_g). \quad (51.15)$$

In a canonical pair, $\Theta = p dq$ gives $X_f = f_p \partial_q - f_q \partial_p$, $\{q, p\}_D = 1$, $X_f(g) = -\{f, g\}_D$ and $[X_f, X_g] = -X_{\{f, g\}_D}$. In the graded phase space, this expression applies to the relevant even functions and constraints.

On the phase line trivialized over this chart, we construct

$$\nabla_X = X - \frac{i}{\hbar} \Theta(X) = \delta_X + \frac{i}{\hbar} H_X^{\text{can}}, \quad H_X^{\text{can}} = -\Theta(X). \quad (51.16)$$

No arbitrary conjugation is chosen to make it flat: its coefficient comes from the action potential. For any fields X, Y of the chart, calculation gives

$$\begin{aligned} \mathcal{F}_{X,Y}^{\text{can}} &= XH_Y^{\text{can}} - YH_X^{\text{can}} - H_{[X,Y]}^{\text{can}} + \frac{i}{\hbar} [H_X^{\text{can}}, H_Y^{\text{can}}] \\ &= -X\Theta(Y) + Y\Theta(X) + \Theta([X, Y]) \\ &= -d_{\mathcal{P}}\Theta(X, Y) = \Omega(X, Y). \end{aligned} \quad (51.17)$$

The H_X^{can} are multipliers on the line, so their commutator is zero; derivatives of the coefficients and fields are not suppressed.

The already proved first-class property means $\{C_A, C_B\}_D = f_{AB}{}^D C_D$. For V tangent to $\mathcal{Z}$, $\Omega(X_{C_A}, V) = d_{\mathcal{P}}C_A(V) = 0$. The constraint fields are therefore characteristic. Under the declared regularity they generate the characteristic distribution of the coisotropic surface. Tensoriality of Ω extends the calculation to all their smooth combinations and proves

$$\mathcal{F}_{X,Y}^{\text{can}}|_{\mathcal{Z}} = 0 \quad \text{for } X, Y \text{ characteristic.} \quad (51.18)$$

For two lapses, $[X, Y]$ includes the tangential deformation and the Gauss actions of (51.13). On passing to the gauge quotient, the equality is expressed modulo these vertical actions. It is not asserted that Ω is zero on the whole phase space or that every global holonomy of a leaf is trivial.

7. Prequantum expression of the same closure. The differential realization associated with (51.16), on smooth sections where the products are defined, is

$$Q_{\text{pre}}(f) = -i\hbar \nabla_{X_f} + f = -i\hbar X_f + f - \Theta(X_f). \quad (51.19)$$

With $R^\nabla = (i/\hbar)\Omega$, expansion of the commutator and $X_f(g) = -\{f, g\}_D$ give

$$[Q_{\text{pre}}(f), Q_{\text{pre}}(g)] = i\hbar Q_{\text{pre}}(\{f, g\}_D). \quad (51.20)$$

The product rule preserving the structure functions is

$$Q_{\text{pre}}(aC) = aQ_{\text{pre}}(C) + C(Q_{\text{pre}}(a) - a). \quad (51.21)$$

It follows from $X_{aC} = aX_C + CX_a$ and (51.19). The second term is not removed outside $\mathcal{Z}$. On $\mathcal{Z}$, $Q_{\text{pre}}(C_A) = -i\hbar \nabla_{X_{C_A}}$; thus (51.20) and (51.18) are the operator and geometric expressions of the same canonical closure, not two independent incorporations of gravity. $\square$

51.3 Domain of the result and comparison of realizations

The proof brings together an action with dynamical gravity and a joint source, its Legendre transform and its characteristic connection. The qualifier *arbitrary* for deformations refers to admissible lapses and shifts generating characteristic directions in that regular region; not to any phase-space vector, any boundary or any singular solution. Local flatness preserves possible global monodromy and does not identify two TPK histories merely because they have the same endpoints.

The coefficient H_X^{can} of a line connection is not by definition the clock-and-response operator $H_{\text{clk}} + D^*WD$, or its interacting extension. $Q_{\text{pre}}(C)$ additionally includes a derivation on phase space; it is

not an alternative name for those operators either. Identifying a concrete realization through a map I requires the typed equality

$$\nabla_N^{\text{op}} I = I \nabla_N^{\text{can}}$$

on their effective domains. If I depends on geometry, its derivative is part of this equality.

The distinction has an exact check: in the chart $\Theta = p \, dq$,

$$Q_{\text{pre}}(p^2/2) = -i\hbar p \partial_q - p^2/2. \quad (51.22)$$

Acting on the function 1, it produces $-p^2/2$. It therefore does not preserve by mere restriction the polarization of functions independent of p , and it is not $-\hbar^2 \partial_q^2/2$ there. The check does not refute the received Hamiltonian: it prevents replacing one representation by another without its map.

No infinite-dimensional Liouville measure has been assumed, nor is self-adjointness deduced from the preceding differential identities. If refinement inclusions intertwine the connections and preserve a common core, their curvatures intertwine by composition; a limit additionally requires its own domain and convergence controls. That condition is not presented here as verified for the interacting operator and the joint spatial–chiral limit. The earlier quantum constructions are preserved; the present result specifies exactly the canonical closure just proved.

52 Structural compatibility of action, mass and operational information

Composing the equations identifies a relation more precise than the equivalence between mass and energy or the proportionality between energy and frequency: the same route structure determines a proper frequency, a relation between lengths and a thermal weighting. The angular pair participates in these three readouts because it contains the action coefficient and determines the transport channels. The following exposition preserves together the construction of the objects, their readout operations and the conditions of their physical realizations.

This composition also characterizes the imprint left by the order of a sequence of operations. This imprint can be expressed through the angular pair and admits an inverse map that recovers the action coefficient. In the constitutive setting, the vacuum response and the Catalan spectral moment are brought together through the same functional collection. The explanatory content of these relations lies in the constraints they jointly impose, not in the mere accumulation of identities.

The APP–TRIT–TPK genealogy provides the objects being composed. The subsequent readouts preserve different aspects of the same structure: scale, orientation, phase, memory, incidence and response. Their joint analysis establishes which relations remain determined under a change of representation, which information a projection removes and which complementary observations allow its recovery. The proofs of the compositions and of their domains are developed in the proof sections.

52.1 Action in the angular composition and the electronic structure

The Planck section enters before the final energy evaluation. Its coefficient participates in the angular geometry subsequently used by the mass law. The link, with units preserved, is

$$\hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S, \quad A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha). \quad (52.1)$$

The coefficient η_{ret} is dimensionless and can therefore be composed with α in the angular expression. The dimensionful constant $\hbar_{\text{ret}}$ retains its unit of action. This distinction allows the same dependence to be traced without adding quantities of different dimensions.

Action thus has two related roles. On the one hand, it fixes a dimensional scale; on the other, it participates in the structure determining the distribution of transport between the two orientations. After the single conversion from degrees to radians, the angle and counter-angle produce the channels q_+ and q_- . These channels enter the hexad–octad incidences, the Barbero–Immirzi functional, the constitutive response and the terms of the mass law.

Replacing a constant by its structural expression reveals dependencies that remain implicit when the equation is written in terms of isolated constants. The expanded expression transmits the operation that determines the value, the coordinates on which it depends and the conditions under which it can be reused. Its explanatory role derives from this information transmitted to the next stage.

For the electron, the complete expression preserves the exponent φ/π^2 , the corrections constructed with α and Δ_4 , and the return factor

$$\left(\frac{H_5}{\eta_{\text{ret}}}\right)^{80-54\alpha+6\Delta_4}. \quad (52.2)$$

The same η_{ret} that enters the counter-angle participates in the complete electronic readout. The angular geometry and the electron's action correction therefore share an effective dependence. Studying both equations together determines which relations are imposed by the involvement of the same structural coordinate in different operations.

52.2 Proper frequency and structural normalization of mass

The general mass law preserves its electronic prefactor. Let Ξ_Γ denote the complete structural factor of a route Γ : it combines the electronic normalization, the route character with its signature and towers, and the corresponding power of the action ratio H_5/η_{ret} . This notation abbreviates a composition of operations already defined; its content remains tied to those operations.

The common clock has angular frequency $\omega_0 = 2\pi/(108t_0)$, with $t_0 > 0$. Composing the mass law with the action section gives

$$m_\Gamma = \frac{\hbar_{\text{ret}}\omega_0}{c^2} \Xi_\Gamma, \quad E_\Gamma = \hbar_{\text{ret}}\omega_0 \Xi_\Gamma, \quad \Omega_\Gamma = \frac{E_\Gamma}{\hbar_{\text{ret}}} = \omega_0 \Xi_\Gamma. \quad (52.3)$$

Here E_Γ is the rest energy and Ω_Γ its associated energy frequency.

This readout separates the common rhythm from the contribution specific to each structure. Two sections can share the elementary period and have different masses and proper frequencies: their signatures, returns and memories produce different factors Ξ_Γ . The diversity of the specific readouts is thus referred to the structure of the section, while the common clock establishes their temporal normalization.

In the quotient $E_\Gamma/\hbar_{\text{ret}}$, the absolute action factor appearing explicitly in (52.3) cancels. Its structural participation in the channels and in the return ratio contained in Ξ_Γ nevertheless remains. The cancellation of a factor in a subsequent readout must be distinguished from removing the relations through which that factor was constructed and came into play again.

Mass ratios coincide with the ratios between these proper frequencies. A common change in the units of action, time or speed leaves these ratios unchanged. The relative structure therefore has its own requirement for testing: its relations between sections must remain coherent, independently of the common normalization chosen.

The frequency associated with a lifted phase also preserves the number of turns in the history. Two phases equal after reduction modulo 2π can correspond to different lifts. Phase return preserves a regularity, while the history distinguishes the path that produced that return. Identifying the complete state with the reduced phase would remove precisely this distinction.

52.3 Reciprocal dilation of the spatial readouts of mass

In the circular gravitational realization, action and gravitation satisfy

$$G = \frac{c^3 L_\alpha^2}{\hbar_{\text{ret}}}, \quad R_{108} = L_\alpha = \frac{c}{\omega_0}. \quad (52.4)$$

Composition with the mass in (52.3) determines

$$r_{g,\Gamma} = R_{108} \Xi_\Gamma, \quad \bar{\lambda}_\Gamma = \frac{R_{108}}{\Xi_\Gamma}, \quad r_{g,\Gamma} \bar{\lambda}_\Gamma = R_{108}^2. \quad (52.5)$$

The reduced gravitational radius convention of the derivation is used; the construction of L_α and joint normalization are proved in Section 47. The clock and radius are two compatible expressions of that same length. $\bar{\lambda}_\Gamma$ denotes the reduced Compton wavelength. These reciprocal identities are formulated for $\Xi_\Gamma > 0$; the readout R_{108}/Ξ_Γ requires this positive-mass domain.

The same factor that increases the proper frequency dilates the gravitational readout and contracts the length associated with action and mass. The two transformations preserve a common area. In this realization, the mass factor acts as a reciprocal dilation parameter: one length increases with it and the other decreases inversely.

This relation gives precise mathematical content to the connection between mass and spatial configuration. The hierarchy of the composition is essential: the route and its structural factor are determined first; their temporal, energy and spatial readouts are then obtained. The structure is transmitted to the readouts through specified operations, rather than being reconstructed retrospectively from a measured mass.

The circular condition belongs to the result (52.5). The two lengths are readouts defined by that realization; interpreting them as material radii of a particle would require the corresponding physical identification. The equality here establishes an exact reciprocity between the defined quantities, with the action chart, clock and speed preserved.

Incorporating momentum retains this structural mass in the propagation equation. In the corresponding free realization, the proper frequency determines the mass term in the dispersion relation, while momentum describes spatial variation. The persistent structure of the section and its state of motion are thus distinguished.

52.4 Channel orientation and the constitutive response of the vacuum

Composing permittivity, permeability and the Barbero–Immirzi functional distinguishes the value of a response from the structure that determines it. The constitutive readout operator uses two labelled channels, with responses r_+ and r_- . The normalized speed depends on their product; the impedance depends on their quotient. Speed preserves a common relation between the channels, while impedance distinguishes the distribution of the response between their orientations.

Within the normalized collection, speed and the Barbero–Immirzi functional together allow the two labelled responses to be recovered. The functional provides oriented information that speed alone does not determine. Reconstruction rests on the complete composition of their readout operators and preserves the distinction between each channel and their aggregate.

Let x and y be the coordinates in radians of the angle and counter-angle. For $x > 0$ and $|y| < x$, the response satisfies

$$\hat{c}(x, y) \geq \hat{c}(x, 0), \quad (52.6)$$

with equality only when $y = 0$. More precisely, with x fixed, $\log \hat{c}$ is an even and strictly convex function of y . The proof of this property belongs to the subsequent constitutive derivation.

Speed loses the sign of the orientation but retains an effect of the separation between the channels. Near the symmetric configuration, that effect appears at second order; impedance distinguishes the orientation already at first order. Replacing both channels by their mean before evaluating the response therefore changes the result. A scalar readout can conceal an orientation while at the same time retaining a consequence of it.

This behaviour specifies which information each operation transmits. Loss of sign, preservation of a separation and recovery of the distribution are different properties. The structure retains observable consequences even when one of its readouts is expressed by a single number.

The inequality (52.6) describes the mathematical response collection within its domain. A physical evolution must lie within the states and trajectories actually produced by TPK. The analytic parametrization allows the readout operator's sensitivity to be studied; identifying a parametric variation with a physical trajectory retains its realization requirement.

52.5 Boltzmann, structural frequency and ternary thermal weighting

The thermal section relates the energy of the cycle to the Boltzmann constant and to its corresponding temperature:

$$k_B \Theta_{108} \log 3 = E_{\text{clk}}. \quad (52.7)$$

Since the energy of a section is $E_\Gamma = E_{\text{clk}} \Xi_\Gamma$, the thermal weight evaluated at that temperature is

$$\exp\left(-\frac{E_\Gamma}{k_B \Theta_{108}}\right) = 3^{-\Xi_\Gamma}. \quad (52.8)$$

The ternary base appears by composing the informational normalization with action and the clock. The route structure determining mass also determines the exponent of the thermal weight.

Weight and probability have different roles. Probabilities are obtained by normalizing the weights and preserving the multiplicities of the states. Likewise, several equivalent histories require an identification of states before receiving a physical multiplicity: the number of histories and the degeneracy of the state refer to different objects.

The composition brings the readouts together in a common identity:

$$\Xi_\Gamma = \frac{\Omega_\Gamma}{\omega_0} = \frac{r_{g,\Gamma}}{R_{108}} = \frac{R_{108}}{\bar{\lambda}_\Gamma} = \frac{E_\Gamma}{k_B \Theta_{108} \log 3}. \quad (52.9)$$

This gives four coordinated ways of reading the same structural component: temporal, spatial, gravitational and thermal. Each equality preserves the conditions of its realization, especially the circular gravitational condition and the cycle temperature.

The identity (52.9) specifies the relation shared by these quantities and provides a criterion for testing their compatibility. Relative mass can be characterized through relative frequency; the same coordinate determines a reciprocal dilation of lengths and a thermal weighting. The readout functions are different, but their dependence is linked by the common genealogy.

The thermal relation identifies the state and normalization in which Θ_{108} enters. This identification keeps the section's energy separate from its possible transfer as heat; it also keeps the meaning of the cycle temperature localized, without extending it to the universe as a single temperature. The explanatory capacity derives from the explicit conditions of the composition.

52.6 The action ellipse and transport of geometric shape

The action ellipse has semiaxes whose product is $2\hbar$. The oriented area accumulated during one turn is $h = 2\pi\hbar$. An elliptic deformation redistributes the semiaxes while preserving their product. In the electric and magnetic realization, two contributions proportional to $\cos^2 \theta$ and $\sin^2 \theta$ are obtained, whose sum is $\hbar\omega$. The structure therefore determines the distribution of energy during the cycle, in addition to its total value.

Transporting an ellipse whose shape changes requires preserving the relation between deformation and action. Let ζ be the deformation coordinate, determined by the ratio of the counter-angle to the angle. For a differentiable deformation, exact transport introduces

$$H_{\text{tot}} = \frac{\omega}{2} (e^{-\zeta} Q^2 + e^\zeta P^2) + \frac{\dot{\zeta}}{2} QP. \quad (52.10)$$

The first term describes evolution on an ellipse of fixed shape. The second arises from transporting a variable shape. The proof of the transport identity establishes that this term exactly compensates the geometric variation and preserves the corresponding quadratic action. Omitting it or reversing its sign destroys that compensation.

The interpretation requires two situations to be distinguished. When only the coordinates change, the additional term is a connection associated with the representation. A physical exchange, by contrast,

requires a relative variation between the system and its reference. Re-expressing a Hamiltonian preserves the content of the representation; attributing an interaction requires determining the relative operations acting on the system.

The governing angular pair preserves another decisive dependence: fixing α and the action section also fixes ζ . An evolution of shape must therefore be extracted from the effective states and operations. The parametric deformation alone describes a collection of representations; identifying it as an HMT evolution transmits the relations producing the angle and counter-angle.

Joint transport of the shape and action section preserves the normalized action, while the unnormalized area changes according to the quotient of the two sections. This construction distinguishes three operations: redistributing a conserved action, transporting an action section and changing the representation or units. The distinction makes it possible to examine interactions while retaining a balance that includes the source of the variation, instead of attributing energy generation to a mere coordinate deformation.

52.7 Joint compatibility and relational testing

The research focus is to determine the compatibility conditions that the same structure imposes on its different physical manifestations. Two lines of composition complement one another.

The first brings together static relations: complementary observables recover the channels, and the recovered structure determines action, the constitutive response and the mass factors. The relevant test is a joint one. A modification preserving one numerical value must also preserve the other relations used by the same construction. Speed, Barbero–Immirzi and impedance provide one system of checks; mass and frequency ratios provide another.

The second line studies effective histories and compares their results after coherently transporting the system and its reference. Differences in relative phase, relative constitutive response or transfer between modes have a content distinct from differences due to coordinate names. Memory, orientation and returns enter through operations that a scalar readout may remove.

A structural relation acquires explanatory and testing capacity when several independent readouts must agree because they derive from the same structure. Its value lies in the rigidity of that agreement: action, rhythm, spatial response and informational weighting can no longer be chosen separately within the declared realization. The composition brings these constraints together and allows their consequences to be proved without replacing their genealogy by individual adjustments of the results.

52.8 Phase closure and relative holonomy of the state

The dodecaphase cycle contains the plane in which the internal root of -1 is realized. The inclusion of that plane in the twelve components is explicit; the quarter-turn used below comes from that construction. The shape of the action ellipse determined by the angular pair acts on this plane, with coordinate

$$\xi = \frac{C^*}{A}, \quad \zeta = \operatorname{artanh} \xi. \quad (52.11)$$

The quotient uses both angles expressed in the same unit and has the same value in degrees and radians. The real shape coordinate is defined for $|\xi| < 1$. The conjugate shape corresponds to reversing the sign of ζ .

The quarter-turn of one shape is composed with that of its conjugate and with their inverses. If U_+ and U_- denote the two operations, the exact composition is

$$H_{\text{lazo}} = U_+ U_- U_+^{-1} U_-^{-1} = \begin{pmatrix} \rho_{\text{lazo}} & 0 \\ 0 & \rho_{\text{lazo}}^{-1} \end{pmatrix}, \quad \rho_{\text{lazo}} = \left(\frac{1 + \xi}{1 - \xi} \right)^2. \quad (52.12)$$

The composition dilates one coordinate and contracts its conjugate by the inverse factor: it preserves area and changes the state. For $\xi \neq 0$, its result differs from the identity.

The four operations include their inverses, and their declared phase increments sum to zero. The composition order nevertheless retains an effect. A description retaining only the increments that are added and cancelled removes this relative effect. Closure of phase bookkeeping and return of the complete state are therefore distinct properties of the sequence.

The equality admits a lift to the original twelve-component space. The operator acts by reciprocal dilation on the plane under consideration and as the identity on its complement. The link to the dodecaphase cycle is realized through an explicit transformation; the plane thus preserves its provenance within phase space.

The notation H_{lazo} denotes this subsequent composition and keeps it distinct from the nonadic holonomy Γ_9 . The realization concretely illustrates a structural distinction: closure of a readout of the path can coexist with a persistent transformation of the state. The provenance of the operators and the scope of their lift remain set out in the proof development.

52.9 Reconstruction of the action coefficient from relative transport

The angular pair in (52.1) can be expressed with $s_0 = 10^{-34}\mathcal{U}_S$ and $\hbar_{\text{ret}} = s_0\eta_{\text{ret}}$. The coefficient η_{ret} remains dimensionless, while s_0 retains the dimensional scale. Incorporating the complete counter-angle into (52.12) gives

$$\rho_{\text{lazo}} = \left(\frac{501\alpha + \eta_{\text{ret}}}{499\alpha - \eta_{\text{ret}}} \right)^2. \quad (52.13)$$

This expression uses the positive branch $0 < \eta_{\text{ret}} < 499\alpha$. The numbers 501 and 499 arise from 500 ± 1 , and 500 arises from $1000/2$: they are consequences of the normalizations already entering the angular pair.

The relation between α and the action coefficient determines the separation between the conjugate responses after the composite path. Action participates both in the dimensional scale and in the shape governing transport. The formula makes this dual involvement explicit without identifying the two roles.

The relation admits the inversion

$$\eta_{\text{ret}} = \alpha \left[500 \frac{\sqrt{\rho_{\text{lazo}} - 1}}{\sqrt{\rho_{\text{lazo}} + 1}} - 1 \right]. \quad (52.14)$$

With α and the oriented axes preserved, the loop response allows the action coefficient to be recovered. The readout is strictly increasing with respect to ρ_{lazo} on the declared branch.

This yields an operational characterization: the action coefficient can be recovered through the relative dilation produced by the conjugate shapes of the angular pair. Its generating definition remains in the preceding construction; the operational characterization establishes how to recover it in a different representation.

The scale s_0 is still necessary to recover the dimensionful constant $\hbar_{\text{ret}}$. The shape of the transport preserves a dimensionless relation; the action scale determines the complete physical normalization. This separation identifies precisely which datum the inversion returns and which additional structure the dimensional readout transmits.

52.10 Holonomy iteration and operational information of the history

Repeating the loop preserves the same programme of operations and produces an explicit accumulation law:

$$H_{\text{lazo}}^n = \begin{pmatrix} \rho_{\text{lazo}}^n & 0 \\ 0 & \rho_{\text{lazo}}^{-n} \end{pmatrix}. \quad (52.15)$$

A preparation (Q_0, P_0) transforms into

$$Q_n = \rho_{\text{lazo}}^n Q_0, \quad P_n = \rho_{\text{lazo}}^{-n} P_0. \quad (52.16)$$

The product $Q_n P_n$ remains constant and the transport preserves symplectic area, while the proportion between the two coordinates changes.

A repeated finite rule can leave an accumulating imprint that distinguishes its iterations. A known preparation and a preserved reference allow n to be recovered through the logarithmic increment of a nonzero coordinate, for $\rho_{\text{lazo}} \neq 1$. When only the final state is preserved and the preparation is unknown, the same reconstruction remains undetermined.

The accessible information depends on the retained readout: balance, amplitude, oriented proportion, preparation or complete operator. Each choice can identify histories that another readout distinguishes. Comparing

$$U_+ U_- U_+^{-1} U_-^{-1} \quad \text{and} \quad U_+ U_+^{-1} U_- U_-^{-1} \quad (52.17)$$

provides a direct test. Both products contain the same operations with the same multiplicities. The second gives the identity; the first gives H_{lazo} . Statistics of occurrence frequencies are insufficient to predict this response: the order must also be preserved.

This result specifies which statistical description is insufficient for the object studied and which observations recover the difference. Shannon information theory admits distributions over ordered sequences and can represent that order. The operational distinction concerns the content retained by the chosen observable.

The operational information of a history is characterized through its capacity to produce distinguishable responses for specified preparations and detectors. This definition makes it possible to specify how much a readout preserves. Its scope corresponds to the representation used: different histories may induce the same reduced operator while retaining differences in other components of the TPK genealogy.

The trace of H_{lazo} equals that of H_{lazo}^{-1} . It detects the intensity of the effect but removes its orientation. A readout preserving the axes and comparing their responses distinguishes the two directions. The choice of detector therefore forms part of the reconstruction of the path.

The finiteness of the programme, the growth of an observable memory and an entropy rate describe different properties. A zero rate of combinatorial growth can coexist with distinguishable states. The entropy of the complete dynamics requires its state space and, where appropriate, its measure to be specified. The composition proves accumulation and its reconstruction in the declared representation; the evaluation of a thermodynamic entropy retains its own variables and physical conditions.

52.11 Spectral potential and differential reciprocity of the responses

The constitutive channels preserve the coordinates $x = A_{\text{rad}}$, $y = C_{\text{rad}}^*$ and the incidences 90/120. Their composition determines the identity

$$\boxed{\frac{\partial \log \hat{c}}{\partial y} = \frac{\partial \log \hat{Z}}{\partial x}}. \quad (52.18)$$

The quantities $\hat{c}$ and $\hat{Z}$ denote the readout operator's normalized speed and impedance.

The sensitivity of propagation to the oriented separation is linked to the sensitivity of impedance to the common coordinate. The two responses share a spectral function, and their variations must maintain that relation to continue belonging to the same construction.

There is a function $\mathcal{F}(x, y)$ whose derivatives are

$$\partial_x \mathcal{F} = \log \hat{c}, \quad \partial_y \mathcal{F} = \log \hat{Z}. \quad (52.19)$$

This is a spectral potential defined from the channels 90/120, with a convergent series and explicit normalization. Its Hessian is negative definite in $x > |y|$, a property allowing the uniqueness and sensitivity of reconstruction to be proved.

Equality of the cross derivatives imposes a constraint on the joint variation of the responses within this realization collection. A modification that destroys the identity (52.18) changes the constitutive

readout operator. Testing thus concerns the relations between variations, as well as the evaluated values.

The potential is single-valued: the integral of its differential over a closed loop in the two coordinates vanishes. The ordered composition of transports, by contrast, can retain a nontrivial holonomy. The constitutive projection and operator transport retain different aspects of the history; comparing them requires preserving their domains and readout operations.

52.12 The Catalan moment and contractive response of the incidences

The connection between the oriented Catalan moment and the constitutive potential is expressed through the depth-two moment

$$D_2(z) = \sum_{n \geq 1} \frac{z^n}{n^2}. \quad (52.20)$$

The Catalan moment is obtained as the imaginary part of $D_2(is)$; at $s = 1$ it recovers Catalan's constant. The constitutive potential uses the same function on the channel contractions, at $q_{\pm}^{90}$ and $q_{\pm}^{120}$, with the corresponding normalizations.

The oriented quarter-turn and the contractive response 90/120 are brought together through the same spectral moment, evaluated on different supports. Catalan acquires meaning here through the spectral operation it represents and the relation of that operation to the constitutive response. The functional collection preserves information exceeding that of its endpoint value.

The composition also maintains the differences between its objects. The isolated value of Catalan does not determine all the channels; the relation uses the complete spectral moment, its arguments and its orientation. Convergence of the series, the identities of its derivatives and its compatibility with amplification are part of the proof development of this connection.

52.13 Exact reconstruction and stability under finite precision

The pair $(\hat{c}, \hat{Z})$ uniquely recovers the two parameters of the readout operator in the declared domain. The sensitivity bounds also relate the response error to the error in the recovered parameters. Exact uniqueness and stability of the inversion are distinct and complementary results.

This distinction separates information removed by the projection from information preserved but difficult to resolve at finite precision. In regions of strong contractions, appreciable parameter variations can produce very small differences in the responses. The inversion remains unique for exact data, while finite uncertainty can prevent those variations from being distinguished.

Genealogical reconstruction requires specifying both which structure can be recovered and the stability with which it can be recovered. The reconstruction margin denotes the bound quantifying that stability on a domain. The choice of observable must attend to the difference one seeks to resolve: a readout can be correct yet have little sensitivity to it.

The loop trace provides an example of this selection. Its readout removes the direction of the path; the response along oriented axes preserves it. Determining the appropriate detector belongs to the proof of recoverability and to the design of the test.

52.14 Synthesis: coordinates of the section and memory of the operations

The identity (52.9) retains as its axis the relation between a section's structural factor, its proper frequency, the two reciprocal lengths and its thermal readout. The conditions remain explicit: the same action chart, the circular gravitational realization and the cycle temperature.

The operator composition adds a central distinction. The factor of the section determines its co-ordinated readouts; transport determines the result of composing operations on it and the imprint retaining their order. The mass factor and the dilation ρ_{lazo} are different objects. Their relation must

pass through the corresponding readout operators to distinguish a transformation of the state from a mass identification.

Planck enters the normalization of action and the angular shape of transport. Boltzmann enters the relation between energy and thermal weighting. Gravitational reciprocity preserves the product of the lengths in its declared realization. Speed and impedance provide complementary readouts of the channels. This architecture jointly admits a description of compatibilities and a description of operational memory.

The explanatory capacity of the structure lies in determining several responses, imposing relations between them and specifying which history can be recovered from its observations. Coincidences of values are accompanied by operations, compatibility conditions and reconstruction criteria.

The composition distinguishes the passive change of coordinates, which telescopes when chart and phase return, from relative transport, whose final operator is $H_{\text{lazo}} \neq I$. A physical exchange requires the operations, their inverses and the reference to be realized within the same system, including the controller in the balance. This criterion preserves the mathematical content of an energy variation without attributing energy production to a mere change of readout.

The proofs of composition, iteration, reconstruction of the action coefficient, constitutive reciprocity and sensitivity support these characterizations in their respective domains. Narrative, definitions and proofs thus form a continuous exposition: the narrative identifies the meaning of the relations, the definitions fix the objects, and the proofs establish the operations allowing their composition and recovery.

53 Composition and reconstruction of the angular, electronic and constitutive readouts

53.1 Action coefficient and construction of the counterangle

The functional abbreviation $B(\alpha)$ collects an evaluation at α ; its argument belongs to the function and does not denote the product of an independent constant and α . To keep the operations involved in this abbreviation explicit, write

$$B(\alpha) = \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5 - \frac{90}{\pi}C_\pi(\alpha), \quad (53.1)$$

$$C_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \quad D_A = \exp\left(-\frac{100\pi\alpha}{9}\right). \quad (53.2)$$

The complete identity is

$$C_{\text{deg}}^* = \sqrt{\frac{1000\alpha}{\varphi}} + 2B(\alpha). \quad (53.3)$$

In particular, α belongs to the radicand, and the radical term is retained together with $2B(\alpha)$. This identity rewrites the map constructed earlier while preserving its object and genealogy.

The original definition keeps the dimensionless coefficient separate from the action section:

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi}C_\pi, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34}U_S. \quad (53.4)$$

Hence,

$$C_{\text{deg}}^* = 2\left(\frac{\hbar_{\text{ret}}}{10^{-34}U_S} + \alpha\right) = 2(\eta_{\text{ret}} + \alpha). \quad (53.5)$$

The expression uses the dimensionless coefficient of the reduced Planck section. The constant h is subsequently obtained through $h = 2\pi\hbar$. The composition extracts the action coefficient, combines it with α and applies bidirectional closure. This preserves the dimensional type of each operation.

The mass construction expresses particle and antiparticle as two oriented sections of the same genealogical band. The dodecaphasic operation is

$$C_M(\tau) = -\text{rev}(\tau), \quad \tau_e = +++--+++-, \quad C_M(\tau_e) = \tau_e, \quad (53.6)$$

extended to the register and the charge representation. The central route is fixed under C_M , whereas the charge representation distinguishes the electron from the positron. This composition gives precise content to the bidirectional readout. The angular factor of two retains its closure function; physical conjugation additionally retains the stated operators. The return at 216 restores the oriented lift: its structure does not consist in assigning 108 updates to the particle and another 108 to the antiparticle.

53.1.1 Radial reconstruction of the action coefficient

Write $q_{\text{rad}} = q_\varepsilon$ and $\varepsilon = \varepsilon$ for the radial ratio and orientation in (27.4). The oriented radial invariant $\chi_{\text{rad}} = \varepsilon \log q_{\text{rad}}$ allows the previously constructed reconstruction to be written as

$$\frac{\eta_{\text{ret}}}{\alpha} = -500 \tanh\left(\frac{\chi_{\text{rad}}}{2}\right) - 1. \quad (53.7)$$

With $\xi = C_{\text{deg}}^*/A_{\text{deg}}$ and $A_{\text{deg}} = 1000\alpha$, one also obtains

$$\eta_{\text{ret}} = \alpha(500\xi - 1). \quad (53.8)$$

These are downstream reconstruction maps for the same section; its construction remains in H_5 and in the regional return.

53.2 Angular charts and conservation of action

The degree chart has a full turn of 360 units and preserves the partitions $12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120 = 360$. The constructed angular pair is

$$A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha). \quad (53.9)$$

Conversion is performed once:

$$x = \frac{\pi}{180}A_{\text{deg}}, \quad y = \frac{\pi}{180}C_{\text{deg}}^*, \quad q_{\pm} = \exp(-x \mp y). \quad (53.10)$$

The exponents use lifted real representatives and retain the number of turns. The action satisfies

$$h\theta_{\text{rad}} = \frac{h\theta_{\text{deg}}}{360}. \quad (53.11)$$

For the bilinear Barbero functional, chart transport gives

$$\frac{\pi A_{\text{deg}} C_{\text{deg}}^*}{180} = \frac{180xy}{\pi}. \quad (53.12)$$

The bilinear transformation is part of the same conversion applied to the exponential arguments; omitting it would change the functional.

53.3 Complete electronic readout and action return

The two stages of the same electronic coordinate are retained:

$$\varepsilon_e^{(0)} = \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3 \right] (1 + 15\Delta_4), \quad \Delta_4 = \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729}\right), \quad (53.13)$$

$$\boxed{\varepsilon_e^\beta = \varepsilon_e^{(0)} \left(\frac{H_5}{\eta_{\text{ret}}}\right)^{80-54\alpha+6\Delta_4}}. \quad (53.14)$$

The centre (5, 5), its inversion, the deformations (8, 2)/(2, 8) and the unit quotient defect precede the value. The norm $\sqrt{3}/4$ comes from the commutator between additive and multiplicative projections; the exponential character uses the orientation φ/π^2 ; $22 = 27 - 5$ retains a regional complement and $15 = 3 \cdot 5$ the corresponding incidences. The route readout maps produce (80, 54, 6). The ratio H_5/η_{ret} incorporates the action return into the same central section.

The role of the counterangle is made explicit by substituting its proven inverse:

$$\varepsilon_e^\beta = \varepsilon_e^{(0)} \left(\frac{H_5}{C_{\text{deg}}^*/2 - \alpha} \right)^{80-54\alpha+6\Delta_4}. \quad (53.15)$$

This expression preserves the preceding electronic generation. The same action section that participates in the counterangle determines the factor taking the basal readout to the complete one. The ratio cancels the common dimensional basis; the absolute mass realization retains its temporal readout operator and dimensional chart. The substitution composes constructed dependencies without introducing an external angle as an electronic generator.

53.4 Barbero functional and the hexad–octad transition

The incidence preserves the same marked pair as the support is enlarged:

$$\mathcal{F}_6 = H \times \binom{H}{2} \hookrightarrow \mathcal{F}_8 = O \times \binom{H}{2}, \quad |\mathcal{F}_6| = 90, \quad |\mathcal{F}_8| = 120. \quad (53.16)$$

The cardinalities are $6 \cdot 15$ and $8 \cdot 15$; the shared reference allows the two incidences to be compared. The normalized difference is $-1/12$. The dodecaphasic chain retains twelve hexadic contributions and one octadic boundary contribution with the opposite sign. The readout operator

$$S_m(q) = \frac{q^m}{1 - q^{3m}} = \sum_{j \geq 0} q^{m+3mj} \quad (53.17)$$

retains height and returns. This yields $12S_{90} - S_{120}$ and its difference between the two orientations. The complete functional is

$$\gamma = \frac{\pi A_{\text{deg}} C_{\text{deg}}^*}{180} + 12[S_{90}(q_-) - S_{90}(q_+)] - [S_{120}(q_-) - S_{120}(q_+)]. \quad (53.18)$$

In this realization, Barbero is the joint oriented evaluation of the angular pair and the returns of the incidence transition. Its participation in the area scale is subsequently composed with $\ell_P^2 = G\hbar/c^3$ and with the occupation operator. Entanglement entropy additionally retains the reduced state and its probabilities; its reconstruction requires those data and is not determined by scalar area alone.

53.4.1 Elimination of the common electron–Barbero coordinate

Define $\kappa_e = 80 - 54\alpha + 6\Delta_4$. On the positive branch, with $\kappa_e \neq 0$, the electronic equation implies exactly

$$\eta_{\text{ret}} = H_5 \left(\frac{\varepsilon_e^{(0)}}{\varepsilon_e^\beta} \right)^{1/\kappa_e}. \quad (53.19)$$

To express the composition while keeping its operations explicit, put $x = 50\pi\alpha/9$, $y = \pi(\eta_{\text{ret}} + \alpha)/90$ and $\mathcal{R}(q) = 12S_{90}(q) - S_{120}(q)$. Then

$$\gamma = \frac{100\pi}{9}\alpha(\eta_{\text{ret}} + \alpha) + \mathcal{R}(e^{-x+y}) - \mathcal{R}(e^{-x-y}). \quad (53.20)$$

Substitution of (53.19) into this identity yields an explicit relation between the complete electronic readout and Barbero, retaining α , H_5 , Δ_4 and the incidences. It is a compatibility consequence of the

construction. The angular physical branch used here satisfies $0 < y < x$; the operation introduces no metrological mass as generating data.

In the extension with orientation label $\epsilon = \pm 1$, $C_{\text{deg},\epsilon}^* = 2\epsilon(\eta_{\text{ret}} + \alpha)$, the exchange $\epsilon \mapsto -\epsilon$ fixes η_{ret} , preserves the electronic expression and interchanges q_+ and q_- . The Barbero functional changes sign. This covariance distinguishes a scalar action readout from an oriented evaluation. Identification with charge conjugation additionally retains C_M and the electromagnetic sheet representation; the angular sign alone does not replace these operators.

53.5 Oriented Catalan moment and constitutive response

The dodecaphasic cycle realizes an oriented quarter-turn plane. Its resolvent coefficient is $k_4(s) = s/(1 + s^2)$. The depth-two moment is fixed by

$$\mathcal{C}_2(s) = \sum_{k \geq 0} \frac{(-1)^k s^{2k+1}}{(2k+1)^2}, \quad \mathcal{D} = s \frac{d}{ds}, \quad \mathcal{D}^2 \mathcal{C}_2(s) = k_4(s), \quad \mathcal{C}_2(1) = G_{\text{Cat}}. \quad (53.21)$$

The differential identity is used for $0 < s < 1$; the endpoint value is understood with the previously established convergence and moment reconstruction conditions. The intrinsic definition of Catalan is the endpoint value of this oriented depth moment. Its collection retains more information than the endpoint value. The constitutive response uses that collection:

$$f(s) = \frac{1 + s + s^2}{s(1 + s)} \mathcal{D}^2 \mathcal{C}_2(s) = \frac{1 - s^3}{1 - s^4}, \quad r_{\pm} = f(s_{\pm}), \quad s_{\pm} = q_{\pm}^{30}. \quad (53.22)$$

The powers 3 and 4 correspond to $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$.

Proposition 53.1 (Functional reconstruction with a boundary condition). *The constitutive collection f determines the oriented moment through*

$$\mathcal{C}_2(s) = \int_0^s \log \frac{s}{t} \frac{1+t}{1+t+t^2} f(t) dt, \quad 0 < s < 1.$$

It is the unique solution of $\mathcal{D}^2 \mathcal{C}_2(s) = s(1+s)f(s)/(1+s+s^2)$ that tends to zero at the origin.

Proof. For the constructed collection, $(1+t)f(t)/(1+t+t^2) = 1/(1+t^2)$. The integral converges, and its logarithmic derivative is $\mathcal{D} \mathcal{C}_2(s) = \int_0^s (1+t^2)^{-1} dt$. Applying $\mathcal{D}$ again yields $s/(1+s^2)$. The integral is $O(s)$ at the origin. Two solutions differ by $a + b \log s$; the zero limit forces $a = b = 0$. For $s < 1$, integrating the geometric series on compact subsets and using $\int_0^s t^{2k} \log(s/t) dt = s^{2k+1}/(2k+1)^2$ recovers the series in (53.21). Its dominated absolute convergence allows $s \uparrow 1$ and recovers Catalan's constant. $\square$

The boundary condition fixes the two integration constants. The reconstruction uses the constitutive function together with its domain; the endpoint value of Catalan's constant and the two channel values are readouts of that already constructed function. At the endpoint, the twice-differentiated identity retains its interpretation through an Abel limit.

The constitutive operator $V = r_+ P_+ + r_- P_-$ determines

$$\hat{\epsilon} = r_+^2, \quad \hat{\mu} = r_-^2, \quad \hat{Z} = \frac{r_-}{r_+}, \quad \hat{c} = (r_+ r_-)^{-1}. \quad (53.23)$$

With $\psi = f^{-1}$ and

$$\mathcal{H}_B(s) = \frac{12s^3}{1-s^9} - \frac{s^4}{1-s^{12}} - \frac{(\log s)^2}{20\pi}, \quad (53.24)$$

the constructed factorization takes the form

$$\gamma = \mathcal{H}_B(\psi(r_-)) - \mathcal{H}_B(\psi(r_+)). \quad (53.25)$$

The logarithmic part reconstructs the angular contribution to Barbero exactly, retained in degrees or transported in full to radians.

53.6 Reconstruction from normalized speed and Barbero

Proposition. In the normalized constitutive collection of two labelled channels, the map

$$(r_+, r_-) \mapsto (\hat{c}, \gamma), \quad (3/4, 1)^2 \longrightarrow (1, 16/9) \times \mathbb{R} \quad (53.26)$$

is a diffeomorphism. Its inverse recovers the two labelled eigenvalues. The sheet projectors belong to the initial representation: two scalars do not recover an arbitrary spatial basis.

Proof. The regular expression $f(s) = (1 + s + s^2)/(1 + s + s^2 + s^3)$ gives

$$f'(s) = -\frac{s^2(s^2 + 2s + 3)}{(1 + s + s^2 + s^3)^2} < 0. \quad (53.27)$$

Thus f is analytic and strictly decreasing, with $f(0^+) = 1$ and $f(1^-) = 3/4$. For $0 < s < 1$,

$$\mathcal{H}'_B(s) = \frac{36s^2(1 + 2s^9)}{(1 - s^9)^2} - \frac{4s^3(1 + 2s^{12})}{(1 - s^{12})^2} - \frac{\log s}{10\pi s} > 0. \quad (53.28)$$

Indeed, the ratio of the first positive term to the second positive term before its subtraction is

$$\frac{9}{s} \frac{1 + 2s^9}{1 + 2s^{12}} \left(\frac{1 - s^{12}}{1 - s^9} \right)^2 > 9. \quad (53.29)$$

The last summand of (53.28) is also positive. Hence $F_B = \mathcal{H}_B \circ \psi$ has strictly negative derivative, $F_B(3/4^+) = +\infty$ and $F_B(1^-) = -\infty$.

Fix $c = \hat{c}$, write $z = \log \hat{Z}$ and $r_{\pm} = c^{-1/2} \exp(\mp z/2)$. The admissible interval is

$$|z| < a(c) := \min \left\{ \log c, \log \frac{16}{9c} \right\}. \quad (53.30)$$

The function $\Gamma_c(z) = F_B(r_-) - F_B(r_+)$ satisfies

$$\Gamma'_c(z) = \frac{1}{2} [r_- F'_B(r_-) + r_+ F'_B(r_+)] < 0. \quad (53.31)$$

As z increases to $a(c)$, either r_- reaches 1, or r_+ reaches $3/4$, or both occur; Γ_c tends to $-\infty$. At the opposite endpoint it tends to $+\infty$. There is therefore a unique solution for every real γ . The nonzero derivative and the implicit function theorem provide a smooth global inverse. This proves the statement. In the original orientation $y > 0$, one has $z < 0$ and $\gamma > 0$.

Normalized speed retains the product of the responses; Barbero monotonically distinguishes their oriented distribution. Together they reconstruct the spectral information that neither readout determines alone.

The statement concerns the collection of constitutive readout operators. The image actually produced by the full HMT generator is a subset of that collection: restriction to that image preserves injectivity and causal order. Normalization and sheet markings are necessary data for reconstruction. The proof neither introduces CODATA values to select channels nor asserts that every synthetic point in the collection is a generated state.

53.7 Reconstruction on the generated image

Once $r_{\pm}$ have been recovered from $(\hat{c}, \gamma)$, one obtains $s_{\pm} = \psi(r_{\pm})$. Then

$$A_{\text{deg}} = -\frac{3}{\pi} \log(s_+ s_-), \quad C_{\text{deg}}^* = \frac{3}{\pi} \log \frac{s_-}{s_+}. \quad (53.32)$$

On the generated image and in its positive orientation,

$$\alpha = \frac{A_{\text{deg}}}{1000}, \quad \eta_{\text{ret}} = \frac{C_{\text{deg}}^*}{2} - \alpha > 0. \quad (53.33)$$

With the regional coordinates π, φ, e and the dimensional sections already constructed, this reconstruction recovers H_5 , the factor $R_{\text{act}} = H_5/\eta_{\text{ret}}$ and the complete electronic readout of Section 53.3. It also recovers the Planck section. If the twelve-block positional base, origin and carry are retained as well, the identity

$$\kappa_{\text{ag}} = \pi + e - \varphi - 4 - \alpha \quad (53.34)$$

allows reconstruction of the register K described by its readout operator. This is a reconstruction path; its reverse direction does not constitute a new generating arrow from external physical values.

Gravitation and thermal conversion retain the sections of the same state. Theorem 47.2 and (47.2) first determine L_α and its circular representation $R_{108} = L_\alpha$:

$$G_a = \frac{c_{\text{int}}^3 R_{108}^2}{\hbar_a}, \quad \frac{G_a \hbar_a}{c_{\text{int}}^3} = R_{108}^2, \quad k_B \Theta_{\text{clk},a} \log 3 = \frac{\hbar_a}{108 t_0}. \quad (53.35)$$

The chronological radius, physical speed and thermal basis retain their construction maps. In particular, the clock of the circular representation is tied to L_α and c_{int} . The first composition makes action–gravitation reciprocity explicit; the second combines action, duration and tritic information, with temperature and k_B individually specified in their declared bases.

53.8 Structural definitions of the composite readouts

α . Scalar coordinate of dodecaphasic closure compatible with the regions and the arithmetic–geometric register K ; it transmits that compatibility to the angular and action sectors.

Counterangle. Angular coordinate of bidirectional closure combining the coefficient of the return action section with α in the 360-unit chart.

Reduced Planck constant. Action section whose regional coefficient and determinantal scale are constructed before the counterangle; in the ellipse, oriented area measure per unit of parametric phase.

Electron. Persistent central section; its scalar readout composes additive–multiplicative incompatibility, regional transfer, closure defect and action return. The complete equation retains the section in addition to its mass value.

Barbero–Immirzi. Oriented functional of angular transport and of returns in the hexad–octad incidence; in the constitutive collection, the coordinate that, together with normalized speed, reconstructs the labelled spectrum.

Catalan. Endpoint value of the oriented depth-two moment of the dodecaphasic quarter-turn. Its collection is linked differentially to the vacuum response.

G . Common coefficient of the radial realization relating energy and conjugate length while simultaneously preserving $H\Lambda = \hbar c I$ and $R\Lambda = L_\alpha^2 I$. Longitudinal and polar composition determines its normalization; its dimensional expression is $c^3 L_\alpha^2 / \hbar$.

Boltzmann. Transducer between the thermal line and decision energy, with declared tritic information normalization and thermal basis.

These intrinsic formulations rest on the constructed objects and equations. Their scope is that of this realization; the physical identifications retain their corresponding proofs and do not follow from structural terminology alone.

53.9 Transport of the nilpotent electronic block

Proposition. Let

$$S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad n_\pm = \frac{S \pm J}{2}, \quad D = \text{diag}(r, r^{-1}), \quad r > 0. \quad (53.36)$$

Define $n_{\pm,D} = D n_\pm D^{-1}$ and $g = D^{-T} D^{-1}$. Then

$$n_{\pm,D}^2 = 0, \quad \{n_{+,D}, n_{-,D}\} = I, \quad (n_{+,D})^\dagger g = n_{-,D}, \quad (53.37)$$

where $A^\dagger g = g^{-1} A^T g$. The ellipse transported from $Y^T Y = 2\hbar$ by $X = DY$ satisfies $X^T g X = 2\hbar$.

Proof. Direct matrix calculation gives $S^2 = I$, $J^2 = -I$ and $SJ + JS = 0$. Consequently,

$$n_{\pm}^2 = \frac{1}{4}(S^2 + J^2 \pm SJ \pm JS) = 0, \quad n_+ n_- + n_- n_+ = \frac{1}{2}(S^2 - J^2) = I.$$

Moreover, $S^T = S$ and $J^T = -J$ imply $n_+^T = n_-$. Conjugation by D preserves products, squares and the identity. For the metric adjoint,

$$g^{-1}(Dn_+D^{-1})^T g = DD^T D^{-T} n_- D^T D^{-T} D^{-1} = Dn_- D^{-1} = n_{-,D}.$$

Finally, $D^T g D = I$, whence $X^T g X = Y^T D^T g D Y = Y^T Y = 2\hbar$. This proves the claims.

The exact algebraic verification takes place in $\mathbb{Q}[r, r^{-1}]$; the restriction $r > 0$ fixes the oriented elliptic realization used here. This composition preserves the specific roles of the different factors $1/2$ in the development: a numerical coincidence between them does not identify their operators.

54 Action, mass and propagation: structural compatibility and transport

54.1 Shared action section and mass law

Let

$$s_0 = 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} s_0, \quad R_{\text{act}} = \frac{H_5}{\eta_{\text{ret}}}. \quad (54.1)$$

The coordinates constructed in degrees are retained:

$$A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha). \quad (54.2)$$

Only afterwards are they transported to radians through $x = \pi A_{\text{deg}}/180$ and $y = \pi C_{\text{deg}}^*/180$. The channels are $q_{\pm} = \exp(-x \mp y)$, with $x > |y|$. The elliptic shape is expressed by

$$\zeta = \text{artanh}(y/x). \quad (54.3)$$

The composition first uses the dimensionless coefficient η_{ret} , thereby preserving the types of action and angle. In this notation, η_{ret} is the action coefficient, $\eta_h(\Gamma)$ belongs to a route tower, and ζ expresses the deformation of the ellipse; their roles remain distinct.

The working domain has $t_0, c_{\text{int}}, \eta_{\text{ret}} > 0$, a convergent character and a factor $\Xi_{\Gamma} > 0$. The constructed mass readout operator determines

$$\omega_0 = \frac{2\pi}{108t_0}, \quad E_{\text{clk}} = \hbar_{\text{ret}} \omega_0, \quad m_{\text{clk}} = \frac{E_{\text{clk}}}{c_{\text{int}}^2}. \quad (54.4)$$

For a route with a scalar realization,

$$m_{\Gamma}^{\beta} = m_{\text{clk}} \Xi_{\Gamma}, \quad \Xi_{\Gamma} = b_e \chi_{\text{ref}}(\sigma_{\Gamma} - \sigma_e, \eta_{\Gamma} - \eta_e) R_{\text{act}}^{\kappa_{\Gamma}}. \quad (54.5)$$

Ξ_{Γ} abbreviates the structural route factor. The following normalization is explicitly retained:

$$b_e = \frac{\varepsilon_e^{(0)} \mathcal{U}_E}{E_{\text{clk}}}. \quad (54.6)$$

The central state retains its own normalization relative to the chronological quantum:

$$\varepsilon_e^{(0)} = \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3 \right] (1 + 15\Delta_4), \quad \varepsilon_e^{\beta} = \varepsilon_e^{(0)} R_{\text{act}}^{80-54\alpha+6\Delta_4}. \quad (54.7)$$

The refined character includes the terms

$$n_A x + \frac{s_C}{6} y + k_{\Delta_4} \Delta_4 + b_{\zeta} \zeta_{270} + \sum_h N_h (q_-^h - q_+^h), \quad N_h = \eta_h + b_{120} \mathbf{1}_{\{h=120\}}, \quad (54.8)$$

with h in the semigroup generated by 90 and 120. The height-120 term is included once. The primitive return coefficients are constructed through

$$\begin{aligned} a_n(\Gamma) &= \text{Tr}(R_\Gamma U_\Gamma^n), \\ p_n(\Gamma) &= \frac{1}{n} \sum_{d|n} \mu_{\text{Mob}}(d) a_{n/d}(\Gamma), \\ \eta_h(\Gamma) &= p_h(\Gamma) - p_h(\Gamma_e), \end{aligned} \quad (54.9)$$

where μ_{Mob} is the arithmetic Möbius function. Action participates in the absolute scale, in the ratio R_{act} and in y , which determines the channels. Consequently, cancellation of a dimensional scale retains the remaining structural dependencies. When the clock varies, $b_e = \varepsilon_e^{(0)} \mathcal{U}_E/E_{\text{clk}}$ varies with it: preserving the electronic readout requires transporting the complete character, rather than fixing that quotient separately. The integrated longitudinal realization uses $t_0 = \pi_{\text{HMT}} L_\alpha / (54 c_{\text{int}})$ from (47.2).

54.1.1 Frequency and length of each massive section

Define $E_\Gamma = m_\Gamma c_{\text{int}}^2$ as rest energy, $\Omega_\Gamma = E_\Gamma / \hbar_{\text{ret}}$, $\nu_\Gamma = \Omega_\Gamma / (2\pi)$ and $R_{108} = c_{\text{int}} / \omega_0$. Then

$$\begin{aligned} E_\Gamma &= \hbar_{\text{ret}} \omega_0 \Xi_\Gamma, \quad \Omega_\Gamma = \omega_0 \Xi_\Gamma, \\ \nu_\Gamma &= \frac{\Xi_\Gamma}{108 t_0}, \quad \bar{\lambda}_\Gamma = \frac{\hbar_{\text{ret}}}{m_\Gamma c_{\text{int}}} = \frac{R_{108}}{\Xi_\Gamma}. \end{aligned} \quad (54.10)$$

Proof. Substitute $m_\Gamma = m_{\text{clk}} \Xi_\Gamma$ and perform the cancellations in the same dimensional chart. For positive routes X, Y , the character property and common normalization give

$$\frac{\Omega_Y}{\Omega_X} = \frac{m_Y}{m_X} = \chi_{\text{ref}}(\sigma_Y - \sigma_X, \eta_Y - \eta_X) R_{\text{act}}^{\kappa_Y - \kappa_X}. \quad (54.11)$$

□

The cancellation is dimensional; the channels and relative memory remain. The zero sector lies outside the domain of inverse mass. In a nonscalar fibre, the operator or its spectral measure is retained, rather than selecting an arbitrary point representative.

54.1.2 Lifted frequency and compatibility of normalizations

The chronological construction also produces the lifted holonomy frequency

$$\omega_j = \frac{\theta_j + 2\pi w_j}{108 t_0}, \quad \lambda_j = r_j \exp(i\theta_j), \quad (54.12)$$

where w_j retains the winding of the history. Its energy readout is

$$E_j = \hbar_{\text{ret}} \omega_j \chi_{\text{full}} R_{\text{act}}^{\kappa_j}. \quad (54.13)$$

This frequency, ω_0 and Ω_Γ correspond to distinct readout operators. If both energy expressions represent the same mode and the same exponent, their exact compatibility requires

$$\frac{\omega_j}{\omega_0} \chi_{\text{full}} = b_e \chi_{\text{ref,rel}}. \quad (54.14)$$

The two normalizations are identified through this equality and are not multiplied together. The reduced phase θ_j alone does not recover w_j . A phase return therefore remains distinct from the complete chronological history. Winding retains its construction map; this distinction does not authorize choosing it freely or producing energy without a dynamics.

54.2 Spatial reciprocity and gravitational composition

The circular gravitational section expresses the result of Theorem 47.2: $G_a = c_{\text{int}}^3 L_\alpha^2 / \hbar_a$ and $R_{108} = L_\alpha$. The length arises from the inscribed return and precedes this circular identification. Retaining $\hbar_a = \hbar_{\text{ret}}$ and the reduced convention $r_g = Gm/c_{\text{int}}^2$, the preceding mass determines

$$r_{g,\Gamma} = R_{108} \Xi_\Gamma, \quad \bar{\lambda}_\Gamma = \frac{R_{108}}{\Xi_\Gamma}, \quad r_{g,\Gamma} \bar{\lambda}_\Gamma = R_{108}^2. \quad (54.15)$$

Proof. Substitution of G_a and m_Γ gives

$$\frac{G_a m_\Gamma}{c_{\text{int}}^2} = \frac{c_{\text{int}}^3 R_{108}^2}{\hbar_{\text{ret}}} \frac{\hbar_{\text{ret}} \omega_0 \Xi_\Gamma}{c_{\text{int}}^4} = R_{108} \Xi_\Gamma.$$

The second equality follows from (54.10); their product proves the third. $\square$

The same route coordinate enlarges one radial readout and contracts the other. In logarithmic form,

$$\log \frac{r_{g,\Gamma}}{R_{108}} = \log \Xi_\Gamma, \quad \log \frac{\bar{\lambda}_\Gamma}{R_{108}} = -\log \Xi_\Gamma. \quad (54.16)$$

The geometric centre of the two readouts is fixed by the area R_{108}^2 .

In the bounded operator version, let X be bounded and self-adjoint, with $X \geq \delta I$ for some $\delta > 0$, on a massive fibre. The operators

$$M = m_{\text{clk}} X, \quad \Lambda = R_{108} X^{-1}, \quad R_g = R_{108} X \quad (54.17)$$

satisfy $R_g \Lambda = \Lambda R_g = R_{108}^2 I$ throughout the space. The claim uses functional calculus for the same operator. The nullspace is excluded from the inverse. The present statement retains the bounded, uniformly positive regime; extension to unbounded operators requires separate proofs for the domains, compatible truncations, and passage to the limit. Section 47.5 also proves preservation of both identities under noncommutative variations and compatible memory reduction.

The proposed interpretation characterizes the relative mass factor as the reciprocal-dilation parameter of these two readouts. The circular condition belongs to the result. This equality alone does not identify the related lengths with measured material radii or with an electronic horizon. The radial identity in isolation is used for its role in the composition, without attributing independent historical priority to it.

54.3 Constitutive response, orientation and magnitude

With $T = \exp(-xI - yR)$, $R = P_+ - P_-$ and the sheet projectors retained, the temporal readout operator is

$$V = (I - T^{90})(I - T^{120})^{-1} = r_+ P_+ + r_- P_-, \quad (54.18)$$

$$r_\pm = f(s_\pm), \quad s_\pm = \exp[-30(x \pm y)], \quad f(s) = \frac{1 - s^3}{1 - s^4}.$$

This recovers $\hat{\varepsilon} = r_+^2$, $\hat{\mu} = r_-^2$, $\hat{Z} = r_-/r_+$ and $\hat{c} = (r_+ r_-)^{-1}$. The constitutive speed relation establishes $c_{\text{int}} = c_{\text{vac}} = \hat{c} u_C$ in the same chart. The oriented Catalan collection participates in f through its resolvent moment, retaining the structure that precedes the endpoint value.

The exact decomposition

$$V = \hat{c}^{-1/2} \exp\left[-\frac{1}{2}(\log \hat{Z})R\right] \quad (54.19)$$

distinguishes common dilation from distribution between sheets. Sheet exchange preserves $\hat{c}$ and inverts $\hat{Z}$; the Barbero functional changes sign. Section 53.6 proves that $(\hat{c}, \gamma)$ recovers r_+, r_- in the normalized collection. Restriction to the generated image preserves this recoverability; the inversion statement does not declare every synthetic pair to be generated.

54.3.1 Convexity and evaluation of the separated channels

Proposition 54.1. *For fixed $x > 0$ and $|y| < x$, the function $y \mapsto \log \hat{c}(x, y)$ is even and strictly convex. Consequently,*

$$\hat{c}(x, y) \geq \hat{c}(x, 0) = \frac{1}{f(e^{-30x})^2}, \quad (54.20)$$

with equality precisely when $y = 0$.

Proof. Let $g(u) = \log f(e^{-u})$ for $u > 0$. Then

$$g'(u) = \frac{3}{e^{3u} - 1} - \frac{4}{e^{4u} - 1} > 0, \quad (54.21)$$

$$g''(u) = -\frac{9}{4 \sinh^2(3u/2)} + \frac{16}{4 \sinh^2(2u)} < 0. \quad (54.22)$$

The first inequality follows because $a/(\exp(au) - 1)$ decreases in $a > 0$. For the second, the function $a/\sinh(au/2)$ is strictly decreasing: its derivative has the sign of $\sinh v - v \cosh v$, which is negative for $v > 0$. Since

$$\log \hat{c}(x, y) = -g(30(x + y)) - g(30(x - y)),$$

its second derivative with respect to y is

$$-900[g''(30(x + y)) + g''(30(x - y))] > 0. \quad (54.23)$$

Evenness fixes the unique minimum at zero. $\square$

One also obtains

$$\begin{aligned} \log \hat{c}(x, y) &= -2g(30x) - 900g''(30x)y^2 + O(y^4), \\ \log \hat{Z}(x, y) &= -60g'(30x)y + O(y^3). \end{aligned} \quad (54.24)$$

Speed loses the orientation sign but retains a quadratic response to channel separation. Impedance retains oriented information from first order. Averaging $x + y$ and $x - y$ before applying the readout operator removes that contribution. The claim is a proven consequence of the constitutive response, distinct from a new measurement.

On the positive action branch, $\eta_{\text{ret}} = (90/\pi)(y - x/500)$; hence,

$$\hbar_{\text{ret}} = s_0 \frac{90}{\pi}(y - x/500), \quad E_k = s_0 \frac{90u_c |k|}{\pi}(y - x/500) \hat{c}(x, y). \quad (54.25)$$

For fixed x , $|k| > 0$ and $x/500 < y < x$, the latter function is strictly increasing in y . This is a property of the homogeneous radiative realization collection. Its application to a generated trajectory retains the construction map for y ; variation of a parameter of this collection is not in itself an experimental variation of a universal constant. Under orientation reversal, the action label is transported as well, retaining the corresponding positive branch.

54.4 Action, clock and thermal information

The thermal section retains $k_B \Theta_{108} \log 3 = E_{\text{clk}}$, with $\Theta_{108} = \Theta_{\text{clk,ret}}$ in the return section. Define $\beta_{108} = (k_B \Theta_{108})^{-1}$. For $E_\Gamma = E_{\text{clk}} \Xi_\Gamma$, it follows exactly that

$$\beta_{108} E_\Gamma = (\log 3) \Xi_\Gamma, \quad \boxed{\exp(-\beta_{108} E_\Gamma) = 3^{-\Xi_\Gamma}}. \quad (54.26)$$

The expression is an unnormalized Gibbs weight. For a finite collection of states, or a collection with a convergent partition sum, and multiplicities g_Γ , one obtains

$$Z_{\text{part}} = \sum_{\Gamma} g_\Gamma 3^{-\Xi_\Gamma}, \quad p_\Gamma = \frac{g_\Gamma 3^{-\Xi_\Gamma}}{Z_{\text{part}}}, \quad \frac{p_Y/g_Y}{p_X/g_X} = 3^{-(\Xi_Y - \Xi_X)}. \quad (54.27)$$

A canonical realization at Θ_{108} and a common energy origin are assumed. When a chemical potential is involved, it is incorporated into the corresponding energy. The index Γ enumerates distinct physical states or groups; multiple equivalent histories do not automatically produce a multiplicity. The dimensionless entropies are

$$\begin{aligned} H_{\text{micro}} &= \log Z_{\text{part}} + (\log 3) \sum_{\Gamma} p_{\Gamma} \Xi_{\Gamma}, \\ H_{\text{grupos}} &= H_{\text{micro}} - \sum_{\Gamma} p_{\Gamma} \log g_{\Gamma}. \end{aligned} \quad (54.28)$$

Finiteness of entropy additionally requires

$$\sum_{\Gamma} g_{\Gamma} \Xi_{\Gamma} 3^{-\Xi_{\Gamma}} < \infty. \quad (54.29)$$

Finiteness of Z_{part} alone ensures normalization. Both conditions hold for finite collections.

The formula links mass structure with proper frequency and thermal weighting in the same framework. It preserves the distinction between energy and heat, between the real coordinate Ξ_{Γ} and an integer trit count, and between a realization at Θ_{108} and a claim about the temperature of the universe. For free bosonic or fermionic occupations per mode, with zero chemical potential, the same substitution gives

$$\bar{n}_{\Gamma} = \frac{1}{3^{\Xi_{\Gamma}} \mp 1}. \quad (54.30)$$

The minus sign corresponds to Bose and the plus sign to Fermi. The total for the degenerate group is $g_{\Gamma} \bar{n}_{\Gamma}$. For Bose, $\Xi_{\Gamma} > 0$ is required and the zero mode is treated separately. These are downstream statistical realizations applied to an already constructed spectrum.

54.4.1 Joint compatibility of four readouts

Composing the preceding sections under the same hypotheses gives

$$\Xi_{\Gamma} = \frac{\Omega_{\Gamma}}{\omega_0} = \frac{r_{g,\Gamma}}{R_{108}} = \frac{R_{108}}{\bar{\lambda}_{\Gamma}} = \frac{E_{\Gamma}}{k_B \Theta_{108} \log 3}. \quad (54.31)$$

In the circular collection, $\Xi = 1$ identifies the section with frequency ω_0 , coincident radii R_{108} and energy E_{clk} . Its unnormalized thermal weight is $1/3$. This is a compatibility point of the readouts. The existence of an atlas species with $\Xi = 1$, or of a second generated species for every reciprocal section $1/\Xi$, requires its generated membership in the atlas; it does not follow from this equality.

54.5 Energy identity of the ellipse and exact transport

For the action ellipse,

$$a_S = \sqrt{2\hbar} e^{\zeta/2}, \quad b_S = \sqrt{2\hbar} e^{-\zeta/2}, \quad Q = a_S \cos \theta, \quad P = b_S \sin \theta, \quad (54.32)$$

with $\zeta = \text{artanh}(C_{\text{deg}}^*/A_{\text{deg}})$, one has $a_S b_S = 2\hbar$, oriented area $\mathcal{A}(\theta) = \hbar \theta$ and $\mathcal{A}(2\pi) = h$. The parametric phase θ and the polar angle of a point on the ellipse retain their respective geometric meanings.

The declared LC realization uses

$$L_{\zeta} = \frac{Z_*}{\omega} e^{-\zeta}, \quad C_{\zeta} = \frac{1}{Z_* \omega} e^{\zeta}. \quad (54.33)$$

In action coordinates $Q = \sqrt{Z_*} q$ and $P = \Phi/\sqrt{Z_*}$,

$$H_0 = \frac{\omega}{2} (e^{-\zeta} Q^2 + e^{\zeta} P^2), \quad U_E = \hbar \omega \cos^2 \theta, \quad U_B = \hbar \omega \sin^2 \theta. \quad (54.34)$$

Composition of the two contributions yields $H_0 = \hbar \omega$. The clock and transducer belong to the declared realization; the section $\hbar$ retains its construction prior to the circuit. The impedance $Z_{\text{LC}}/Z_* = e^{-\zeta}$ and the constitutive vacuum impedance $f(s_-)/f(s_+)$ are distinct readouts. Their common angular antecedent does not establish equality between them.

54.5.1 Discrete transport of the quadratic form

Proposition 54.2. *Let $S_\zeta = \text{diag}(e^{\zeta/2}, e^{-\zeta/2})$, $g_\zeta = S_\zeta^{-T} S_\zeta^{-1}$ and $R(-\theta)$ be a rotation. For shape values ζ_n and phases θ_n , the transport*

$$X_{n+1} = S_{\zeta_{n+1}} R(-\theta_n) S_{\zeta_n}^{-1} X_n = T_n X_n \quad (54.35)$$

preserves $I_n = X_n^T g_{\zeta_n} X_n / 2$ exactly and has determinant one.

Proof. Substitution and the identity $R^T R = I$ give

$$T_n^T g_{\zeta_{n+1}} T_n = g_{\zeta_n}. \quad (54.36)$$

Evaluating both sides on X_n shows preservation of the quadratic form. The factors of T_n have determinant one, whence $\det T_n = 1$. $\square$

This composes the previously constructed radial transport. If ζ_n, θ_n are extracted from a TPK route, the identity applies to that sequence; the algebraic proposition keeps route selection distinct.

54.5.2 Differentiable transport and connection term

If $\zeta(t)$ is of class C^1 and $\omega(t) > 0$ is continuous, the relation $X = S_\zeta Y$ and the evolution $\dot{Y} = -\omega J_0 Y$, with

$$J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad (54.37)$$

give

$$\dot{Q} = \frac{\dot{\zeta}}{2} Q + \omega e^\zeta P, \quad \dot{P} = -\omega e^{-\zeta} Q - \frac{\dot{\zeta}}{2} P. \quad (54.38)$$

With the conventions $\dot{Q} = \partial_P H$ and $\dot{P} = -\partial_Q H$, the Hamiltonian is, up to a summand depending only on time,

$$H_{\text{tot}} = \frac{\omega}{2} (e^{-\zeta} Q^2 + e^\zeta P^2) + \frac{\dot{\zeta}}{2} QP. \quad (54.39)$$

The invariant

$$I = \frac{1}{2} (e^{-\zeta} Q^2 + e^\zeta P^2) \quad (54.40)$$

is preserved exactly, without requiring an adiabatic approximation.

Proof. Differentiation gives

$$\dot{I} = \frac{\dot{\zeta}}{2} (e^\zeta P^2 - e^{-\zeta} Q^2) + e^{-\zeta} Q \dot{Q} + e^\zeta P \dot{P}.$$

Substitution of (54.38) cancels the terms proportional to $\dot{\zeta}$ and the two ωQP terms. Thus $\dot{I} = 0$. If the mixed term in (54.39) is omitted, the remainder is

$$\dot{I} = \frac{\dot{\zeta}}{2} (e^\zeta P^2 - e^{-\zeta} Q^2). \quad (54.41)$$

Reversing the sign of the mixed term doubles this defect. $\square$

On the ellipse, $QP = I \sin(2\theta)$ and $\dot{\theta} = -\omega$, so

$$H_{\text{tot}} = I\omega + \frac{I\dot{\zeta}}{2} \sin(2\theta). \quad (54.42)$$

The matrix of the quadratic form has determinant $\omega^2 - \dot{\zeta}^2/4$. For $\omega > 0$, it is positive definite precisely when $|\dot{\zeta}| < 2\omega$, and positive semidefinite at equality. This threshold concerns positivity: existence of the transport and physical stability retain their respective conditions. Conservation of I does not imply conservation of H_{tot} , which depends explicitly on time.

The quantum version requires symmetrization of the mixed term, $\dot{\zeta}(QP + PQ)/4$. This expression specifies the required operation without presupposing an additional quantum representation.

54.5.3 Compatibility with the action section generating the counterangle

In the governing construction,

$$\zeta = \operatorname{artanh} \left[\frac{2(\eta_{\text{ret}} + \alpha)}{1000\alpha} \right]. \quad (54.43)$$

Fixing α and η_{ret} also fixes ζ . The result with variable $\zeta(t)$ is therefore an extension of transport between shapes. To represent a physical trajectory of the governing pair, the variation must arise from the corresponding genealogy, rather than from an independently chosen deformation.

Transport can also be composed exactly when the positive action section changes between two states while the same dimensional chart is retained. Let

$$\hbar_n > 0, \quad \rho_{\text{act},n} = \frac{\hbar_{n+1}}{\hbar_n}, \quad \tilde{T}_n = \sqrt{\rho_{\text{act},n}} S_{\zeta_{n+1}} R(-\theta_n) S_{\zeta_n}^{-1}. \quad (54.44)$$

Then

$$\det \tilde{T}_n = \rho_{\text{act},n}, \quad \tilde{T}_n^T g_{\zeta_{n+1}} \tilde{T}_n = \rho_{\text{act},n} g_{\zeta_n}, \quad \frac{I_{n+1}}{\hbar_{n+1}} = \frac{I_n}{\hbar_n}. \quad (54.45)$$

The equalities follow from (54.36) by multiplication by the scalar factor. This is conformally symplectic transport: it preserves normalized action, and it preserves unnormalized action area precisely when $\rho_{\text{act},n} = 1$. Its differentiable limit has trace $\dot{\hbar}/\hbar$. A linear Hamiltonian generator on a plane with a fixed symplectic form has trace zero; a material variation of $\hbar$ therefore requires specification of the transported form or the degrees of freedom with which action is exchanged. The mixed deformation term by itself retains trace zero.

The result is a compatibility criterion for the realizations. To interpret numerical variations due to a change of units, the bases are transported first: the chart change preserves the physical section. The composition does not assert experimental variability of the Planck constant.

54.5.4 Representation transformation and material deformation

If S_ζ is a passive coordinate change, the mixed term is a representation connection. The observables are transported as well:

$$L_X(S_\zeta Y) = L_Y(Y). \quad (54.46)$$

This equality preserves the physical readout as the Hamiltonian expression changes, without introducing an additional interaction or physical work.

A material deformation requires deriving ζ from the TPK update and showing a relative change between system and reference, both transported consistently. One can test a relative phase, a response $Z_{\text{sistema}}/Z_{\text{referencia}}$ or a transfer between modes defined by the detector. Two TPK histories inequivalent under a representation change must be distinguished, and their readout conditions retained. When an energy exchange is attributed, the balance includes field, memory and apparatus. The transport theorem delineates this distinction while keeping separate the dynamical law that realizes it physically.

54.6 Momentum, tests and scope of the composition

The Clifford–Dirac realization receives the route mass. Within the same action section, its coefficient is written

$$\frac{m_{\Gamma} c}{\hbar} = \frac{\omega_0 \Xi_{\Gamma}}{c}. \quad (54.47)$$

In a free homogeneous realization with the declared relativistic dispersion relation, the latter is re-expressed as

$$\Omega^2 = c^2 |k|^2 + \omega_0^2 \Xi_{\Gamma}^2. \quad (54.48)$$

The rewriting preserves the hypotheses of that realization and its antecedents in the space–time connection. The return moments $q_-^h - q_+^h$ are tower evaluations and are distinguished from mechanical momenta.

The factor Ξ_Γ reappears as mass weight, relative frequency, radial dilation and thermal exponent. The constitutive response determines the common spatial conversion. Barbero retains oriented information complementary to speed. Action fixes the areal measure; the clock introduces the temporal rate; shape and its transport specify the distribution and its evolution.

The corresponding tests comprise mass and frequency ratios; compatibility of speed, impedance and Barbero; preservation of the quadratic form under transport; and thermal weights once temperature and multiplicities are fixed. These relations preserve more data than an isolated decimal coincidence. Their proofs retain the energy balance and the conditions of the entropies considered; they do not imply zero thermodynamic entropy.

55 Marked Reading and the Thermometric Conjugacy of Action

Oriented action, the calendar, the capacity readers and the memory archive arise from the same enriched APP-TRIT-TPK dynamics. The two sheets preserve remainder and quotient; the tritic regime preserves orientation; transport U_t and its holonomy Γ_9 precede the reduced reading K_{ph} . Phase return preserves a history that continues to grow. The present construction composes these results in a reference-and-test register, within the previously constructed joint discrete structure of the continuum. Its thermometric rule is specified by two operations on that register.

55.1 Capacity, Spectral Reference and Retained Origin

The capacity of a prefix indexed by $t \geq 0$ and its remainder are

$$C_t = \lfloor \log_{10}(9^{t+1}) \rfloor, \quad 9^{t+1} = 10^{C_t} R_t, \quad 1 \leq R_t < 10. \quad (55.1)$$

The count is evaluated through integer powers. The inequalities $10^{20} \leq 9^{21} < 10^{21}$ and $10^{20} \leq 9^{22} < 10^{21}$, together with $10^{j-1} \leq 9^j < 10^j$ for $1 \leq j \leq 21$, prove that the first positive vacancy has index 21 and capacity 20. In the tritic reading of capacity, residues 0, 1, 2 correspond to +1, -1, 0, respectively. The capacities at $t = 21, 22, 23, 24$ are 20, 21, 22, 23; the first subsequent return of the regime is therefore $t = 24$. The prefix contains 25 positions and yields capacity 23. The distinct roles of (20, 24, 25, 23) are thereby retained. The return of the regime retains a different nonadic phase; the phase returns at $t = 30$, with capacity 29.

The carrier of the reader is $\mathcal{D} = \mathcal{D}_0 \oplus \mathcal{D}_1 \oplus \mathcal{D}_2$, with ranks (1, 380, 649) and grading $N = 0, 1, 2$. The construction preserves two gradings: N distinguishes the prefix and L the occupation of the exterior or symmetric space. For $m = 20, n = 24$, let $\mathcal{F}_2^J$ be the degree-two exterior fixed space of two exchanged sheets and $\mathcal{B}_{\leq 2}^J$ the symmetric fixed space through degree two. The carrier is $\mathbb{C}\Omega_{\text{ext}} \oplus \mathcal{F}_2^J \oplus \mathcal{B}_{\leq 2}^J$. The involution J exchanges the two sheets through the graded interchange; in the exterior sector, each of its eigenspaces has dimension m . The characters of the fixed subspaces are

$$\begin{aligned} \chi_F(z) &= \frac{(1+z)^{2m} + (1-z^2)^m}{2}, \\ \chi_B(z) &= \frac{P_n(z)^2 + P_n(z^2)}{2}, \\ P_n(z) &= \prod_{j \geq 1} (1 - z^j)^{-n}. \end{aligned} \quad (55.2)$$

Indeed, the projector $(I + J)/2$ averages the total character and the trace of interchange. In the exterior space, the eigenvalues +1, -1 of J produce $(1 + z)^m(1 - z^2)^m$; in the symmetric product, only identical pairs contribute to the trace of interchange, producing $P_n(z^2)$. For coefficients through degree two, it suffices to use $P_n(z) = 1 + nz + n(n+3)z^2/2 + O(z^3)$; the formal products are finite at each degree. Coefficient extraction proves

$$\begin{aligned} \dim \mathcal{F}_2^J &= m(m-1) = 380, \\ (\dim \mathcal{B}_0^J, \dim \mathcal{B}_1^J, \dim \mathcal{B}_2^J) &= (1, n, n(n+2)) = (1, 24, 624). \end{aligned} \quad (55.3)$$

Thus, the simultaneous decomposition $(N, L; \text{multiplicity})$ is

$$(0, 0; 1), (1, 2; 380), (2, 0; 1), \\ (2, 1; 24), (2, 2; 624).$$

The rank $649 = 1 + 24 + 624 = 25^2 + 24$ retains three occupations; the external vacuum and the vacuum of the symmetric summand remain orthogonal. The joint character is $1 + 380xy^2 + x^2(1 + 24y + 624y^2)$. Assigning $N = 0, 1, 2$ to the cuts 24, 27, 30, with capacities 23, 26, 29, specifies the reader used in this realization, while retaining L as a concomitant observable. In particular, $\text{Tr } q^L = 2 + 24q + 1004q^2$ and $\text{Tr } q^N = 1 + 380q + 649q^2$ preserve distinct operations.

The spectral reference originates in (31.28)–(31.42). The nonadic 8 : 1 coupling and the ordered loop $H_1 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, with the preparation of two identical pure modes and the covariance chart transported jointly, yield $\nu^2 = 1 + (8/81)[\text{tr}(H_1 H_1^\top) - 2] = 121/81$. The sum of the squared entries of H_1 is seven, so that $\nu = 11/9$. Theorem 31.11 then gives $r = (\nu - 1)/(\nu + 1) = 1/10$ and eigenvalues $(1 - r)r^j$. This established preparation is used, including its memory reduction. The division $j = 3n + b$, $0 \leq b < 3$, preserves quotient and residue and satisfies

$$(1 - r)r^{3n+b} = (1 - r^3)(r^3)^n \frac{r^b}{1 + r + r^2}. \quad (55.4)$$

Consequently, the quotient has ratio $q = r^3 = 1/1000$, while the residue retains the distribution $\text{diag}(100, 10, 1)/111$. Increasing the quotient after completing the three residues preserves the carry. If the elementary energy is ϵ , the indexed energy is $\epsilon(3n + b + 1/2)$: the interblock spacing is 3ϵ at the same reference temperature.

Define the positive reference operator and its intensity by

$$\eta = \bigoplus_{n=0}^2 r^{23+3n} I_{d_n}, \quad (d_0, d_1, d_2) = (1, 380, 649), \quad (55.5) \\ b_* := \text{Tr } \eta = \frac{1380649}{10^{29}}.$$

In particular, $b_* = 1.380649 \times 10^{-23}$ and $0 < b_* < 1$. The origin 23 is retained in η ; normalizing its state alone removes that factor from the relative probabilities.

The action section of (54.4) supplies $\epsilon_a = h_a/(108t_0)$ and $\beta_{\text{clk}} = \ln 3/\epsilon_a$. For the additive energy lift of this capacity reader,

$$H_{\text{tot}}(t) = 2(t + 1)\epsilon_a, \\ H_{\text{mem}}(t) = \frac{\epsilon_a}{\ln 3} \ln R_t, \quad (55.6) \\ H_{\text{cap}}(t) = H_{\text{tot}}(t) - H_{\text{mem}}(t) = \frac{\epsilon_a \ln 10}{\ln 3} C_t.$$

Taking logarithms in (55.1) proves the identity. Substitution yields $e^{-\beta_{\text{clk}} H_{\text{cap}}(t)} = 10^{-C_t}$. On $\mathcal{D}$, this gives $H_{\text{cap}} = (\epsilon_a \ln 10 / \ln 3)(23I + 3N)$ and $\text{Tr } e^{-\beta_{\text{clk}} H_{\text{cap}}} = b_*$. The remainder remains recorded as an energy contribution; compensation is a reading distinct from eliminating a factor by partial trace.

55.2 Tensor Identity and Specification of the Readers

Lemma 55.1 (Incremental information of the product register). *For every positive matrix η and density matrix ρ , with $\mathcal{H}(X) = -\text{Tr}(X \ln X)$ and $0 \ln 0 = 0$,*

$$\mathcal{H}(\eta \otimes \rho) - \mathcal{H}(\eta) = \text{Tr}(\eta)s(\rho), \\ s(\rho) = -\text{Tr}(\rho \ln \rho). \quad (55.7)$$

Proof. On the supports, the logarithm of the tensor product is $\ln \eta \otimes I + I \otimes \ln \rho$. Trace factorization and $\text{Tr } \rho = 1$ give $\mathcal{H}(\eta \otimes \rho) = \mathcal{H}(\eta) + \text{Tr}(\eta)s(\rho)$. Continuity of $x \ln x$ at zero extends the identity to nontrivial kernels. $\square$

For (55.5), form a normalized register and its marker:

$$\hat{\eta} = \eta \oplus (1 - b_*)|f\rangle\langle f|, \quad \Omega_\rho = \hat{\eta} \otimes \rho, \quad P = I_{\mathcal{D}} \oplus 0, \quad (55.8)$$

where P acts as the identity on the test factor. The flag f completes the probability of this realization. Any additional microscopic archive is transported as such; the flag preserves only this selection distinction. For a fixed Hamiltonian H , specify

$$\begin{aligned} \mathcal{S}_P(\rho) &= -\text{Tr}[P\Omega_\rho(I \otimes \ln \rho)] = b_* s(\rho), \\ \mathcal{E}_P(\rho) &= \frac{\text{Tr}[P\Omega_\rho(I \otimes H)]}{\text{Tr}(P\Omega_\rho)} = \text{Tr}(\rho H). \end{aligned} \quad (55.9)$$

The first operation reads information before conditioning; the second reads energy conditioned on the marker. Both act on the same register, and the probability of the marker remains b_* . The reference remains fixed as the test state varies.

Theorem 55.2 (Thermometric conjugacy of the marked reader). *Let H be a fixed self-adjoint Hamiltonian on a finite-dimensional space and $\rho_\beta = e^{-\beta H}/Z(\beta)$, with $\beta > 0$ and $\text{Var}_{\rho_\beta}(H) > 0$. For the pair (55.9), the conjugate temperature satisfies*

$$\Theta_P := \left(\frac{d\mathcal{S}_P}{d\mathcal{E}_P} \right)^{-1}, \quad b_* \Theta_P = \beta^{-1}. \quad (55.10)$$

Proof. Write $E(\beta) = \text{Tr}(\rho_\beta H)$. Differentiation of the finite spectral sum gives $Z'/Z = -E$ and $E' = -\text{Var}_{\rho_\beta}(H)$. Since $s(\rho_\beta) = \beta E + \ln Z$, one has $s' = \beta E'$. The fixed reference permits differentiation of $\mathcal{S}_P = b_* s$ without an additional term, and the positive variance permits division by E' . Hence $d\mathcal{S}_P/d\mathcal{E}_P = b_* \beta$, proving the claim. $\square$

Normalization is part of the operational definition. If information and energy are both read conditionally, their values are (s, E) and their differential ratio is β . If both are read before conditioning, their values are $(b_* s, b_* E)$ and the same ratio is β . The pair (55.9) yields $b_* \beta$ because it retains intensity in the first operation and uses energy per selected test in the second. Preserving that pair fixes its coefficient: replacing it by cb_* , with $c \neq 1$, changes the equation. If the reference varied, $d(b_* s) = b_* ds + s db_*$; that protocol would have a different response and would require retaining the second term.

55.3 Variational Selection and Information Balance

Theorem 55.3 (Variational principle in the marked realization). *For $\Theta > 0$, the functional*

$$\Phi_\Theta(\rho) = \mathcal{E}_P(\rho) - \Theta \mathcal{S}_P(\rho) = \text{Tr}(\rho H) - b_* \Theta s(\rho) \quad (55.11)$$

has a unique minimum over density matrices, attained at

$$\rho_\Theta^* = \frac{e^{-H/(b_* \Theta)}}{\text{Tr} e^{-H/(b_* \Theta)}}. \quad (55.12)$$

Moreover,

$$\Phi_\Theta(\rho) - \Phi_\Theta(\rho_\Theta^*) = b_* \Theta D(\rho \| \rho_\Theta^*) \geq 0. \quad (55.13)$$

Proof. The state (55.12) is faithful in finite dimension. Substituting $\ln \rho_\Theta^* = -H/(b_* \Theta) - \ln Z_\Theta$ in $D(\rho \| \rho_\Theta^*) = \text{Tr}(\rho(\ln \rho - \ln \rho_\Theta^*))$ gives the identity. To verify positivity, diagonalize ρ and the faithful state $\sigma = \rho_\Theta^*$, with eigenvalues r_i, s_j and the bistochastic overlap matrix p_{ij} . Concavity of $\ln$ gives $D(\rho \| \sigma) \geq \sum_i r_i \ln(r_i/t_i)$, where $t_i = \sum_j p_{ij} s_j > 0$ and $\sum_i t_i = 1$. The inequality $-\ln u \geq 1 - u$ shows that this sum is nonnegative. Equality requires $r_i = t_i$ at every index and each eigenvector of ρ to lie in an eigenspace of σ with eigenvalue t_i ; thus it requires $\rho = \sigma$. This proves the minimum and its uniqueness. The convention on the kernel includes singular states; faithfulness of σ prevents such a state from attaining equality. $\square$

In this realization, the same coefficient enters both differential conjugacy and variational equilibrium. Generalization to infinite systems separately retains the conditions for trace, differentiability, and finite energy and entropy; the preceding theorems have finite-dimensional domains.

55.4 Composition with Action, Area and Gravitational Reciprocity

Use the previously constructed oriented sections in a common chart:

$$\begin{aligned}\epsilon_a &= \frac{h_a}{108t_0} = \frac{\hbar_a c_{\text{int}}}{L_\alpha}, & q_A &= \gamma a_0 L_\alpha^2, \\ \mathcal{F}_{I/A} &= \frac{\epsilon_a}{q_A}, & G_a &= \frac{c_{\text{int}}^3 L_\alpha^2}{\hbar_a}.\end{aligned}\tag{55.14}$$

The longitudinal identity retains the clock of (47.2); $q_A > 0$ fixes the areal orientation used. The functional γ is that of (53.18) and retains the hexad–octad incidences and the 90/120 channels. Its occurrence in the area preserves that complete dependency. Section 47.4 constructs a_0 as a normalized tritic angular character and proves $\mathcal{A} = q_A(N_{90} + N_{120})$ on thirty-six labelled links. Thus q_A is the area increment per unit occupation in that section, rather than a factor inferred from a predetermined entropy.

Corollary 55.4 (Compatibility of thermal, areal and gravitational readers). *With the tritic reference $\beta_{\text{clk}} = \ln 3/\epsilon_a$ and the same pair (55.9),*

$$\begin{aligned}b_* \Theta_{\text{clk},P} &= \frac{\epsilon_a}{\ln 3} = \frac{\mathcal{F}_{I/A} \gamma a_0 L_\alpha^2}{\ln 3}, \\ G_a b_* \Theta_{\text{clk},P} \ln 3 &= c_{\text{int}}^4 L_\alpha.\end{aligned}\tag{55.15}$$

For the horizon section $\tau_R = \hbar_a c_{\text{int}}/(2\pi R)$,

$$b_* T_{R,P} = \tau_R, \quad \frac{T_{R,P}}{\Theta_{\text{clk},P}} = \frac{\ln 3}{2\pi} \frac{L_\alpha}{R}.\tag{55.16}$$

Proof. Applying (55.10) to the clock gives the first identity. Substituting q_A in $\mathcal{F}_{I/A} q_A = \epsilon_a$ gives the second. Multiplication by G_a and (55.14) yield $G_a \epsilon_a = c_{\text{int}}^4 L_\alpha$. For the horizon, apply the same theorem with $\beta_R = 1/\tau_R$ and divide by the clock identity. $\square$

Conversions between sections i, j preserve $\Theta_{i,P}/\Theta_{j,P} = \tau_i/\tau_j$. Every conversion cycle closes by telescoping cancellation. Horizon and clock temperatures agree precisely when their thermal energy sections agree.

In an internal chart, b_* is the coefficient of the transducer defined by (55.9). The thermal unit is specified through the conjugacy between those readings and the chosen energy section. Comparison with the application called Boltzmann in another protocol also transports its readers, dimensional lines and bases; the numeral alone retains its scalar scope. In particular, this extension introduces an explicit thermometric realization and its proofs. Its rule is incorporated as such, without retrospectively attributing it to every earlier thermal operator or identifying its internal unit with an external metrological unit merely by name.

56 Horizon composition and thermal-geometric recovery

The composition of the constants distinguishes what each physical reading retains of the same generated state. We use the action and length sections, the angular pair, and the constitutive velocity already constructed from APP–TRIT–TPK. Abbreviate $\eta = \eta_{\text{ret}}$ and denote by s_0 the action section whose SI expression is 10^{-34} J s. Thus,

$$\hbar = \eta s_0, \quad L_\alpha = r_N \alpha^{16} L_*, \quad r_N = \frac{5759}{23040}, \quad G = \frac{c^3 L_\alpha^2}{\eta s_0}.$$

The inscribed return determines r_N ; the local norm construction determines the power 16; the section L_* retains the longitudinal coordinatization. The function $\eta = H_5 - 90C_\pi/\pi$, with its coefficients and regional dependencies, is the one obtained in (53.4). The following substitutions use these outputs and retain the units and domains of their readers.

56.1 Fixed area, fixed mass, and preservation of the thermal product

For comparison with an Einstein-gravity horizon, the Bekenstein–Hawking expression is $S_{\text{BH}} = k_B c^3 \mathcal{A} / (4G\hbar)$. The coefficient 1/4 belongs here to this comparison law. For a stationary Schwarzschild horizon without charge or rotation, $R_S = 2GM/c^2$, $\mathcal{A} = 4\pi R_S^2$, and $k_B T_H = \hbar c / (4\pi R_S)$.

Proposition 56.1 (Two constraints on horizon composition). *In the specified realization, the structural outputs give*

$$\begin{aligned} S_{\mathcal{A}} &= \frac{k_B \mathcal{A}}{4L_\alpha^2}, & T_{\mathcal{A}} &= \frac{\eta s_0 c}{2k_B \sqrt{\pi \mathcal{A}}}, \\ S_M &= \frac{4\pi k_B c^2 L_\alpha^2 M^2}{\eta^2 s_0^2}, & T_M &= \frac{\eta^2 s_0^2}{8\pi k_B L_\alpha^2 M}, & S_M T_M &= \frac{Mc^2}{2}. \end{aligned}$$

In a comparison between sections that retains c, L_α, k_B , the fixed-area reading preserves $S_{\mathcal{A}}$ and makes $T_{\mathcal{A}}$ proportional to η ; the fixed-mass reading gives $S_M \propto \eta^{-2}$ and $T_M \propto \eta^2$.

Proof. The identity $G\hbar = c^3 L_\alpha^2$ immediately gives the first entropy. Substituting $R_S = \sqrt{\mathcal{A}/(4\pi)}$ into the temperature gives the first temperature. At fixed mass, $R_S = 2cL_\alpha^2 M / (\eta s_0)$; its area and reciprocal produce the other two formulas. Multiplication cancels the common factors, leaving $Mc^2/2$. $\square$

The cancellation of action in the fixed-area entropy expresses a compatibility between its gravitational and longitudinal readings. At fixed mass, the same action section enters the horizon geometry, and its dependence remains in the entropy. The balance must retain the chosen constraint throughout the comparison. The radial parameter R in the earlier realization $\hbar c / (2\pi R)$ equals $2R_S$ in this comparison: both expressions then represent the same temperature. This identification retains the factor of two between geometric radius and acceleration scale.

56.2 Angular separation and recovery of the action section

The modular boundary realization retains eighteen tritic occupations in each incidence sector. Write $X = N_{90}$, $Y = N_{120}$, $N = X + Y$, and $M = X - Y$. Its specified separation is

$$\Delta_{\text{mod}} = \frac{(A_{\text{deg}} - C_{\text{deg}}^*)\pi/180}{9 \cdot 90} \left(1 - \frac{7}{12} \frac{\pi}{729}\right), \quad d = 30\Delta_{\text{mod}}, \quad f_\pi = 1 - \frac{7\pi}{8748}.$$

The coordinate d retains both the angular conversion $\pi/180$ and the incidence difference $120 - 90 = 30$. In this realization, the distribution over microscopic configurations is proportional to $\exp[-d(3X + 4Y)]$. We use $A_{\text{deg}} = 1000\alpha$, $C_{\text{deg}}^* = 2(\eta + \alpha)$, and set $\xi = C_{\text{deg}}^* / A_{\text{deg}} = \tanh s$, on the positive-action branch $1/500 < \xi < 1$. The shape rapidity s is dimensionless; both coordinates of the angular ratio are expressed in degrees.

Proposition 56.2 (Compatibility curve and recovery reader). *With $d_* = 50\pi f_\pi \alpha / 243$,*

$$d = d_*(1 - \xi) = d_*(1 - \tanh s), \quad 0 < d < \frac{499}{500}d_*.$$

For $m = 1, \dots, 36$, the preceding distribution satisfies

$$\frac{\Pr(M = m)}{\Pr(M = -m)} = e^{dm}.$$

Consequently, each probability ratio determines the same d and recovers the coordinates

$$\xi = 1 - \frac{d}{d_*}, \quad \eta = 499\alpha - \frac{2430d}{\pi f_\pi}, \quad C_{\text{deg}}^* = 1000\alpha \left(1 - \frac{d}{d_*}\right).$$

Proof. Substitution of the angular pair gives $d = \pi f_\pi (499\alpha - \eta) / 2430$. Since $\eta = 500\alpha\xi - \alpha$, this gives the first equality and, from the specified branch, the interval. For the second identity, each occupation configuration (X, Y) has a counterpart (Y, X) with the same multiplicity. The ratio of their weights is $e^{-d(3X+4Y)+d(3Y+4X)} = e^{d(X-Y)}$. Summing over $X - Y = m$ retains that factor. Taking the logarithm and solving for ξ and η gives the three recovery formulas. $\square$

This composition establishes a joint check: the thirty-six logarithmic ratios, divided by their respective m , must give the same coordinate. Recovery acts on already generated outputs and retains their causal order. Its result connects the occupation bias to the elliptical shape and action section, without fixing an additional radius or energy unit.

The channels of the Barbero functional are then recovered as

$$q_- = e^{-27d/f_\pi}, \quad q_+ = D_A/q_-, \quad D_A = e^{-100\pi\alpha/9}.$$

Indeed, $27d/f_\pi = \pi(A_{\text{deg}} - C_{\text{deg}}^*)/180$, whereas $q_+q_- = D_A$. Substitution of these channels and $C_{\text{deg}}^*(d)$ into (53.18) determines $\gamma(d)$. The area coefficient then retains its composition $\gamma(d)a_0(d)L_\alpha^2$. For an area eigenvalue $\mathcal{A}_n = \gamma a_0 L_\alpha^2 n$, the horizon comparison gives $S_{\text{BH}}(\mathcal{A}_n)/k_B = \gamma a_0 n/4$. The entropy of the reduced state additionally retains its probabilities and bipartition. Both evaluations remain available with their respective objects: horizon geometry and entanglement distribution.

57 Electronic History and Mass Conservation under Transport

Equation (53.14) evaluates a persistent central section and preserves its action return. The trajectory, its memory and its mass are distinct readings of the same APP-TRIT-TPK state. Transport of a route may modify events and memory while preserving the mass reader's value. This is decided using the complete character of (54.8), together with the return exponent.

57.1 Separation and Reconstruction of the Two Central Histories

The expressed event registers of the two orientations are

$$e_{++} = (2, 2, 5, -4, -3, -2)^2, \quad e_{--} = (4, 3, 2, -2, -2, -5)^2, \quad (57.1)$$

where the exponent 2 denotes concatenation of two blocks. They have zero total and the same absolute histogram. For a register $e \in \mathbb{Z}^{12}$, let

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6},$$

$$s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5). \quad (57.2)$$

Proposition 57.1 (Integral reconstruction of the register). *The data $(s_C, M_0, \dots, M_4, \alpha_0, \dots, \alpha_5)$ determine the register through*

$$\begin{aligned} p_5 &= \frac{s_C - \sum_{i=0}^4 M_i}{6}, & p_i &= p_5 + M_i, \\ e_i &= \frac{p_i + \alpha_i}{2}, & e_{i+6} &= \frac{p_i - \alpha_i}{2}. \end{aligned} \quad (57.3)$$

Its integral image is characterized by divisibility of the first numerator by six and the congruence $p_i \equiv \alpha_i \pmod{2}$ for each i .

Proof. Summing $p_i = p_5 + M_i$ over the first five indices and adding p_5 gives $s_C = 6p_5 + \sum M_i$. Adding and subtracting the definitions of p_i, α_i gives the remaining formulas. The stated conditions are necessary for integer values and sufficient to reconstruct them; substitution recovers all the original data. $\square$

For (57.1), $\alpha = 0$ and $M_{++} = (8, 8, 14, -4, -2)$, whereas $M_{--} = (18, 16, 14, 6, 6)$. Reconstruction distinguishes both histories even when particular scalar readers share their value. The symbols α_i in this decomposition are event differences and are distinguished from the areal character α_0 by their domain.

57.2 Spectral Form of the Return Reader

For $\tau \in \mathbb{R}^{12}$, with cyclic indices, define

$$C_s = \sum_{j=0}^{11} \tau_j \tau_{j+s}, \quad T_k = \sum_{j=0}^{11} \tau_j e^{-2\pi i k j / 12}, \quad \theta_k = 2\pi k / 12. \quad (57.4)$$

The character representation is applied after the register has been produced. The return reader retains $\Omega = -2C_1 - 4C_2 - 6C_3$ and $P_6 = C_6/2$.

Proposition 57.2 (Exact harmonic decomposition).

$$\begin{aligned} C_s &= \frac{1}{12} \sum_{k=0}^{11} |T_k|^2 \cos(s\theta_k), \\ \Omega &= \frac{1}{12} \sum_{k=0}^{11} |T_k|^2 [-2 \cos \theta_k - 4 \cos 2\theta_k - 6 \cos 3\theta_k], \\ P_6 &= \frac{1}{24} \sum_{k=0}^{11} |T_k|^2 (-1)^k. \end{aligned} \quad (57.5)$$

Proof. Expanding $|T_k|^2$ gives $\sum_{j,l} \tau_j \tau_l e^{-i\theta_k(j-l)}$. The sum $\sum_{k=0}^{11} e^{i\theta_k m}$ is twelve when 12 divides m and zero otherwise. Applying this identity to the shift s recovers C_s . Since τ is real, imaginary parts cancel in pairs $k, 12 - k$, leaving the cosine. The other two formulas follow by linear combination and by $\cos(6\theta_k) = (-1)^k$. $\square$

For $\tau_e = (+++---)^2$, the only nonzero powers are $|T_2|^2 = 64, |T_6|^2 = 16, |T_{10}|^2 = 64$. Their weights in Ω are 7, 4, 7, so that $\Omega = 80$ and $P_6 = 6$. This recovers the harmonic part of $\kappa_e = 80 - 54\alpha + 6\Delta_4$ in the complete electronic equation. The alternating register $(+-)^6$, with the same histogram, has $|T_6|^2 = 144$ and $\Omega = 48$: order is an effective dependency of the reader. Cyclic translations preserve $|T_k|^2$ while changing the phase of T_k ; this spectral reading preserves correlations, rather than the entire history.

57.3 Criterion for Transported Mass and Its Associated Readings

Let $\mathcal{H}_{\mathcal{F}}$ be the fibre of a finite route domain, with basis $\{|\Gamma\rangle\}$. Fix one return reader, $r = D$ or $r = \Omega$, and a common chart with $R_{\text{act}} > 0$. The mass law of (54.5) determines

$$\frac{m_{\Gamma}^{\beta}}{m_e^{\beta}} = \chi_{\text{ref}}(\sigma_{\Gamma} - \sigma_e, \eta_{\Gamma} - \eta_e) R_{\text{act}}^{\kappa_{\Gamma} - \kappa_e}, \quad (57.6)$$

$$\Lambda_{\Gamma} = \ln(m_{\Gamma}^{\beta}/m_e^{\beta}).$$

The logarithm acts on a dimensionless ratio. On a nonscalar fibre, the operator and its declared diagonal realization are retained; the finite domain may be an orbit or multisection, without choosing an arbitrary individual cell.

Theorem 57.3 (Conservation of the mass reader on a stable domain). *Let F be a TPK transport acting bijectively on the fixed domain, and let $V|\Gamma\rangle = |F\Gamma\rangle$. For $M|\Gamma\rangle = m_e^{\beta} e^{\Lambda_{\Gamma}} |\Gamma\rangle$,*

$$[M, V]|\Gamma\rangle = m_e^{\beta} (e^{\Lambda_{F\Gamma}} - e^{\Lambda_{\Gamma}}) |F\Gamma\rangle. \quad (57.7)$$

Thus $[M, V] = 0$ is equivalent to $\Lambda_{F\Gamma} = \Lambda_{\Gamma}$ for every route. With chart parameters fixed, the condition is

$$\delta n_A x + \frac{\delta s_C}{6} y + \delta k_{\Delta_4} \Delta_4 + \delta b_{\zeta} \zeta_{270} + \sum_h \delta N_h (q_-^h - q_+^h) + \delta \kappa \ln R_{\text{act}} = 0, \quad (57.8)$$

where δ compares the transported route with the initial one.

Proof. Applying MV and VM to every basis vector gives (57.7). Positivity of m_e^{β} and injectivity of the real exponential give the equivalence. Taking the logarithm of (57.6) and subtracting the two route characters gives (57.8). For infinite towers, require for each route $\sum_h |N_h(q_-^h - q_+^h)| < \infty$; absolute convergence legitimizes the subtraction. Finiteness of the route domain preserves the bounded operators used in this proof. $\square$

Transport is first produced by TPK; the identity subsequently determines what it preserves. Individual events and memory may change and compensate in (57.8). With the action section, clock and circular realization of (54.15) fixed, commutation with M also preserves the proper-frequency, gravitational-radius, reduced-Compton-wavelength and thermal-weight readings that are functions of the same positive operator. Inversion of mass is restricted to the complement of the null sector. In the marked realization of (55.15), the fourth reading is

$$\Xi_{\Gamma} = \frac{\Omega_{\Gamma}}{\omega_0} = \frac{r_{g,\Gamma}}{L_{\alpha}} = \frac{L_{\alpha}}{\lambda_{\Gamma}} = \frac{E_{\Gamma}}{b_* \Theta_{\text{clk},P} \ln 3}. \quad (57.9)$$

The proof substitutes the identities already established in a common chart. This criterion describes conservation under a declared transport; the physical selectors of species, representation and interaction retain their own conditions. The trajectory and its harmonic reading add information to the scalar mass without replacing those conditions.

57.4 Transport of the electronic section to the electroweak sector

The central section in (54.7) jointly retains the prefactor, the correction and the return memory. Its dependencies arise from the APP sheets, the TRIT orientation and the route recorded by TPK; the angular character and the towers subsequently act on multisections of the same discrete structure of the continuum. Write

$$r_e = R_{\text{act}}^{\kappa_e}, \quad E_e^{\beta} = \mathcal{U}_E \varepsilon_e^{(0)} r_e, \quad \kappa_e = 80 - 54\alpha + 6\Delta_4. \quad (57.10)$$

Here $\mathcal{U}_E$ is the section of the energy line and r_e denotes the electronic return factor. The action decade remains in its own line; the realised energy coordinate is transported exactly once.

The angular signatures in (16.4) produce, in the same chart, $E_s^\angle = E_e^\beta \exp(n_s x + s_s y/6)$. In particular,

$$E_W^\angle = \mathcal{U}_E \varepsilon_e^{(0)} r_e e^{94x-y/6}, \quad D = e^{-2x}. \quad (57.11)$$

This is the angular realisation of W. Refinements in another tower retain their own factors when a composition uses them.

Proposition 57.4 (Electronic scale and tree-level Fermi coupling). *In the tree-level electroweak realisation, with the energy coordinates expressed in one unit and $g^2 = 4\pi\alpha/(1-D)$, the relations $v = 2E_W^\angle/g$ and $G_F^{(0)} = 1/(\sqrt{2}v^2)$ give*

$$G_F^{(0)} = \frac{\pi\alpha e^{-188x+y/3}}{\sqrt{2}(1-D)\mathcal{U}_E^2(\varepsilon_e^{(0)})^2 r_e^2}. \quad (57.12)$$

With the signatures, channels and energy chart fixed, the passage from the basal to the returned stage satisfies

$$\frac{(G_F^{(0)})_\beta}{(G_F^{(0)})_0} = r_e^{-2}, \quad \frac{v_\beta}{v_0} = r_e. \quad (57.13)$$

The ratios $E_W^\angle/E_Z^\angle$, $E_H^\angle/E_W^\angle$ and $\lambda_H = (g^2/8)(E_H^\angle/E_W^\angle)^2$ are independent of this common factor r_e .

Proof. Substitution of v gives $G_F^{(0)} = g^2/[4\sqrt{2}(E_W^\angle)^2]$. Using the expression for g^2 and squaring (57.11) yields (57.12): the coefficients 188 and $1/3$ arise respectively from $2 \cdot 94$ and $2/6$. Replacing E_e^0 by $E_e^\beta = r_e E_e^0$ multiplies all angular energies by r_e . The definition of v retains its first power; that of $G_F^{(0)}$ retains its inverse square. Each energy ratio cancels the same positive factor, and the formula for λ_H preserves this cancellation. The couplings g, g' depend on α, x and remain fixed in this comparison. $\square$

The quantity $G_F^{(0)}$ has inverse-energy-squared dimension. The declared tree-level realisation specifies the physical scope of this composition; radiative corrections have their own subsequent operators and order. Comparing the two stages of the reader retains their charts and signatures, rather than postulating independent temporal variation of the correlated constants.

57.5 Preservation of route refinements

Composition with the complete law preserves the character and tower of each route. For two comparable positive eigensections in the same chart and with the same reader $r \in \{D, \Omega\}$, let

$$\rho_{W/e}^r = \frac{\chi_{\text{full}}^r(\sigma_W, \eta_W)}{\chi_{\text{full}}^r(\sigma_e, \eta_e)} R_{\text{act}}^{\kappa_r(\Gamma_W) - \kappa_e}. \quad (57.14)$$

The corresponding energy is $E_W = E_e^\beta \rho_{W/e}^r$. In the Fermi realisation compatible with that section, one obtains

$$G_F^{(0)} = \frac{\pi\alpha}{\sqrt{2}(1-D)(E_e^\beta)^2(\rho_{W/e}^r)^2}. \quad (57.15)$$

The proof substitutes the complete ratio into the preceding quadratic identity. In the explicit angular chart, that ratio takes the form in (57.11); in a refined realisation it also retains the tower and reader differences in (54.11). If $E_s = E_e^\beta \chi_s f_s$, then $E_s/E_t = (\chi_s/\chi_t)(f_s/f_t)$. This identity shows exactly which factor cancels and which species-specific information remains.

57.6 Mass congruence and spectral inversion without commutation

Let $\hat{M}^0 > 0$ be the basal operator on a finite positive fibre, including its character and towers, and let $\hat{B}^r$ be the self-adjoint return reader. Action transport produces $C = R_{\text{act}}^{\hat{B}^r/2} > 0$ and the law $\hat{M}^\beta = C\hat{M}^0C$.

Proposition 57.5 (Operator reciprocity of the action length). *For $\hbar_a, c > 0$ in the same dimensional chart,*

$$\begin{aligned} (\hat{M}^\beta)^{-1} &= C^{-1}(\hat{M}^0)^{-1}C^{-1}, \\ \hat{L}_C^\beta &:= \frac{\hbar_a}{c}(\hat{M}^\beta)^{-1} = C^{-1}\hat{L}_C^0C^{-1}, \quad \hat{L}_C^0 = \frac{\hbar_a}{c}(\hat{M}^0)^{-1}. \end{aligned} \quad (57.16)$$

These identities allow $[C, \hat{M}^0] \neq 0$.

Proof. Positivity of C implies invertibility. For $u \neq 0$, $\langle u, C\hat{M}^0Cu \rangle = \langle Cu, \hat{M}^0Cu \rangle > 0$, so the final operator is also invertible. Direct multiplication in both orders gives

$$(C\hat{M}^0C)(C^{-1}(\hat{M}^0)^{-1}C^{-1}) = I, \quad (C^{-1}(\hat{M}^0)^{-1}C^{-1})(C\hat{M}^0C) = I.$$

Multiplication by $\hbar_a/c$ proves the second identity. The spectral decomposition of the final operator, $\hat{M}^\beta = \sum_j m_j P_j$ with $m_j > 0$, yields $\hat{L}_C^\beta = \sum_j \hbar_a/(cm_j) P_j$. The reciprocal function is injective on the positive eigenvalues, preserving their final projectors and multiplicities. $\square$

Congruence may change the basal operator's projectors; spectral inversion preserves those of the final operator. The null sector is outside the domain of this inversion. A preparation occupying several eigenspaces retains its spectral measure and multiplicities. In rank one, the identity recovers the electronic length $\tilde{\lambda}_{C,e} = \hbar_a c / E_e^\beta$ and then $a_{B,e} = \tilde{\lambda}_{C,e} / \alpha$. The result connects the electronic scale, the electroweak response and the action length through explicit operations on their shared structures.

58 Constitutive compatibility, relative holonomy and operational information

58.1 Differential reciprocity of speed and impedance

The constitutive antecedent in Section 15 constructs, on the retained sheets of its representation,

$$\begin{aligned} T &= e^{-xI-y\mathcal{R}}, \quad r_\pm = f(e^{-30(x\pm y)}), \\ V &= (I - T^{90})(I - T^{120})^{-1} = r_+P_+ + r_-P_-. \end{aligned} \quad (58.1)$$

where

$$f(s) = \frac{1-s^3}{1-s^4}, \quad x > |y|.$$

The sheet projectors remain fixed. The marked physical chamber $x > y > 0$ and its conjugate lie within this analytic collection. Define the normalized logarithmic coordinates

$$c_\ell = \log \hat{c} = -\log r_+ - \log r_-, \quad z_\ell = \log \hat{Z} = \log r_- - \log r_+. \quad (58.2)$$

These letters denote neither new constants nor the counterangle C^* . Let

$$u = 30(x+y), \quad v = 30(x-y), \quad g(w) = \log f(e^{-w}), \quad a = g'(u), \quad b = g'(v).$$

Proposition 58.1 (Differential constitutive reciprocity). *In the domain $x > |y|$, one has*

$$g'(w) = \frac{3}{e^{3w}-1} - \frac{4}{e^{4w}-1} > 0, \quad D(c_\ell, z_\ell) = -30 \begin{pmatrix} a+b & a-b \\ a-b & a+b \end{pmatrix}. \quad (58.3)$$

The eigenvalues of the Jacobian are $-60a$ and $-60b$. It is therefore symmetric and negative definite, its determinant is $3600ab > 0$, and

$$\boxed{\partial_y \log \hat{c} = \partial_x \log \hat{Z}}. \quad (58.4)$$

Proof. Differentiate the identities $c_\ell = -g(u) - g(v)$ and $z_\ell = g(v) - g(u)$. Positivity of g' follows from the strict decrease, in $k > 0$, of $k/(e^{kw} - 1)$. It also follows from the positive rational expression in Section 58.3. The resulting matrix has the stated eigenvalues and determinant; equality of its off-diagonal entries gives (58.4). $\square$

The equality relates the response of speed to oriented separation with the response of impedance to the common coordinate. Both sensitivities arise from the same spectral function. This is a reciprocity of the present constitutive collection; it is not identified without proof with a law of thermodynamics or macroscopic electromagnetism.

58.2 Spectral potential and the depth-two connection with Catalan

58.2.1 Construction of the potential

From the already constructed channels, define

$$\mathcal{F}(x, y) = \sum_{\sigma=\pm 1} \sum_{n \geq 1} \left[\frac{e^{-120n(x+\sigma y)}}{120n^2} - \frac{e^{-90n(x+\sigma y)}}{90n^2} \right]. \quad (58.5)$$

Proposition 58.2 (Potential of the two logarithmic responses). *The series (58.5) defines on $x > |y|$ a potential satisfying*

$$\partial_x \mathcal{F} = c_\ell, \quad \partial_y \mathcal{F} = z_\ell. \quad (58.6)$$

Its Hessian is the Jacobian in (58.3); consequently, $\mathcal{F}$ is strictly concave. The additive constant is fixed by $\mathcal{F} \rightarrow 0$ as $x - |y| \rightarrow \infty$.

Proof. On every compact subset of the domain $x > |y|$, the quantities $x \pm y$ have a positive minimum. A power of n multiplied by a decreasing exponential is summable; hence the series and its derivatives of every fixed order converge uniformly on that compact set. Termwise differentiation, using $-\log(1-z) = \sum_{n \geq 1} z^n/n$, yields the two identities in (58.6). The Hessian then coincides with (58.3), which is negative definite. The normalization at infinity is that of the displayed series. $\square$

$\mathcal{F}$ is a mathematical response potential. No energy units are assigned to it, nor is it declared to be a thermodynamic potential.

58.2.2 Realization through a common spectral moment

Let

$$D_2(z) = \sum_{n \geq 1} \frac{z^n}{n^2}.$$

Its recognition as the dilogarithm follows this series definition. For $|z| < 1$, it satisfies $(z\partial_z)^2 D_2(z) = z/(1-z)$. In this notation,

$$\mathcal{F} = \sum_{\sigma \in \{+, -\}} \left[\frac{D_2(q_\sigma^{120})}{120} - \frac{D_2(q_\sigma^{90})}{90} \right]. \quad (58.7)$$

The oriented Catalan moment constructed in Section 17 satisfies

$$\mathcal{C}_2(s) = \text{Im } D_2(is) = \sum_{k \geq 0} \frac{(-1)^k s^{2k+1}}{(2k+1)^2}, \quad \mathcal{C}_2(1) = G_{\text{Cat}}. \quad (58.8)$$

Proof. Even powers of i are real, and odd powers have imaginary part $(-1)^k$. This gives the series in (58.8), which converges absolutely even at $s = 1$. For $0 \leq s < 1$, applying $s\partial_s$ twice termwise returns $s/(1+s^2)$, the oriented coefficient of the quarter-turn resolvent already present in the corpus. If used at $s = 1$, the differentiated identity is interpreted through analytic continuation or an Abel limit: the twice-differentiated series does not converge in the ordinary sense at that endpoint. The identification $\mathcal{C}_2(1) = G_{\text{Cat}}$ follows directly from the original absolutely convergent series. $\square$

Thus the Catalan moment and the constitutive potential are brought together through a common depth-two spectral moment, evaluated on different supports: the oriented quarter-turn and the contractions of the 90/120 sectors. This composition neither identifies Catalan with $\mathcal{F}$ nor asserts that its single endpoint value determines all channels. The analytic collection and its arguments retain information that an isolated value does not contain.

58.2.3 Compatibility with amplification

For $T_m = T \otimes I_{9^m}$ and $\tau_m = \text{Tr} / (2 \cdot 9^m)$, one has exactly

$$\mathcal{F}_m = 2\tau_m \left[\frac{D_2(T_m^{120})}{120} - \frac{D_2(T_m^{90})}{90} \right] = \mathcal{F}. \quad (58.9)$$

The multiplicity 9^m cancels the normalization. The same argument preserves the derivatives. This establishes amplification of the representation, not every conceivable refinement of the TPK. The temporal operator is not replaced by a list of eigenvalues lacking sheet markings.

For every smooth closed loop in this analytic collection,

$$\oint (c_\ell dx + z_\ell dy) = 0. \quad (58.10)$$

This equality concerns the two-coordinate constitutive projection. It does not imply that the enriched history or its operator holonomy is trivial.

58.3 Exact reconstruction and finite sensitivity

Proposition 58.3 (Analytic reconstruction of the angular coordinates). *With $L = \log(16/9)$, the map $(x, y) \mapsto (c_\ell, z_\ell)$ is a real-analytic diffeomorphism from the domain $x > |y|$ onto*

$$0 < c_\ell + z_\ell < L, \quad 0 < c_\ell - z_\ell < L. \quad (58.11)$$

Proof. The function g is strictly increasing from $\log(3/4)$ to 0. The combinations

$$c_\ell + z_\ell = -2g(u), \quad c_\ell - z_\ell = -2g(v)$$

recover u, v through g^{-1} . Then $x = (u + v)/60$ and $y = (u - v)/60$. Injectivity is preserved under restriction to the image actually produced by HMT; no assertion is made that HMT produces the entire cone. $\square$

With $s = e^{-w}$, the derivative has the rational form

$$a(w) = g'(w) = \frac{s^3(s^2 + 2s + 3)}{(1 + s + s^2)(1 + s + s^2 + s^3)}, \quad 0 < a(w) < \frac{1}{2}. \quad (58.12)$$

The function is strictly decreasing, with $a(0+) = 1/2$, and $a(w) \sim 3e^{-3w}$ at infinity. To prove the decrease, use

$$a'(w) = -\frac{9}{4 \sinh^2(3w/2)} + \frac{16}{4 \sinh^2(2w)} < 0, \quad (58.13)$$

since $k/\sinh(kw/2)$ is strictly decreasing in $k > 0$. Moreover,

$$\frac{1}{2}e^{-3w} < a(w) < 3e^{-3w}. \quad (58.14)$$

The lower bound follows by multiplying the denominators and using

$$\begin{aligned} 2(s^2 + 2s + 3) - (1 + s + s^2)(1 + s + s^2 + s^3) \\ = (1 - s)(5 + 7s + 6s^2 + 3s^3 + s^4) > 0. \end{aligned} \quad (58.15)$$

The upper bound follows from

$$3(1 + s + s^2)(1 + s + s^2 + s^3) - (s^2 + 2s + 3) > 0.$$

If $m = 30(x - |y|)$ and $M = 30(x + |y|)$, the norm of the inverse Jacobian is $1/[60a(M)]$. To distinguish the vector-valued map from the scalar potential, write $\mathbf{F} = (c_\ell, z_\ell)$.

Proposition 58.4 (Finite reconstruction bounds). *On a convex region in which both arguments u, v belong to $[m_0, M_0] \subset (0, \infty)$, the following bounds hold:*

$$60a(M_0)\|p - q\| \leq \|\mathbf{F}(p) - \mathbf{F}(q)\| \leq 60a(m_0)\|p - q\|. \quad (58.16)$$

Proof. Integrate the Hessian of $\mathcal{F}$, namely the Jacobian of $\mathbf{F}$, along the segment. For the lower bound, use

$$-\langle \mathbf{F}(p) - \mathbf{F}(q), p - q \rangle \geq 60a(M_0)\|p - q\|^2$$

and the Cauchy–Schwarz inequality. The upper bound uses the maximum norm of the Jacobian. These are finite-error bounds on the specified domain. $\square$

At $y = 0$, as $x \rightarrow \infty$, both directions contract equally: the matrix condition number is one, but the norm of the inverse grows as $e^{90x}/180$. Exact invertibility and recovery robust to absolute error are therefore different properties. As $|y| \rightarrow x$ with fixed $x > 0$, there is no differential singularity: one derivative tends to $1/2$, and the other to $a(60x) > 0$.

58.4 The quarter-turn arises from the dodecaphasic cycle

The construction of (17.4) is recovered. In $\mathbb{R}^{12}$, let

$$C_{12}e_j = e_{j+1 \bmod 12}, \quad \Pi_4 = P_4^{\text{cyc}} \frac{I - C_{12}^6}{2},$$

and

$$u_0 = (1, 0, -1, 0, 1, 0, -1, 0, 1, 0, -1, 0)^\top, \quad v_0 = C_{12}u_0, \quad (58.17)$$

$$W = (u_0, v_0)/\sqrt{6}.$$

The symbol Π_4 distinguishes the projector from the action coordinates Q, P . The following identities hold:

$$W^\top W = I, \quad WW^\top = \Pi_4, \quad C_{12}W = WJ_0, \quad (58.18)$$

$$J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J_0^2 = -I.$$

In particular, $C_{12}^3W = -WJ_0$. The plane thus has an explicit intertwiner with the cycle; no external rotation has been introduced to fit a response.

The radial antecedent of (27.5) realizes

$$S_\zeta = \text{diag}(e^{\zeta/2}, e^{-\zeta/2}), \quad J_\zeta = S_\zeta J_0 S_\zeta^{-1}.$$

On this plane, define

$$U_\zeta(\theta) = S_\zeta e^{-\theta J_0} S_\zeta^{-1} = \begin{pmatrix} \cos \theta & e^\zeta \sin \theta \\ -e^{-\zeta} \sin \theta & \cos \theta \end{pmatrix}. \quad (58.19)$$

The sign of θ fixes the chosen orientation. The generated ratio $\xi = C^*/A$, with $|\xi| < 1$ and both angles expressed in the same unit, satisfies $\zeta = \text{artanh } \xi$. In the conjugate orientation, $(\varepsilon, \xi) \mapsto (-\varepsilon, -\xi)$ preserves the positive action section through the oriented reconstruction map of the corpus; ζ changes sign.

58.5 Relative loop of two transports

Let

$$U = U_{\zeta_s}(\theta), \quad V = U_{\zeta_r}(\vartheta), \quad H_{\text{lazo}} = UVU^{-1}V^{-1}. \quad (58.20)$$

Products act from right to left. Here ϑ denotes the second phase, to distinguish it from the regional coordinate φ . Put

$$\delta = \zeta_s - \zeta_r, \quad \chi = \sinh \delta \sin \theta \sin \vartheta.$$

Proposition 58.5 (Trace and spectrum of the relative loop). *The composition (58.20) satisfies*

$$\boxed{\begin{aligned} \det H_{\text{lazo}} &= 1, & \operatorname{tr} H_{\text{lazo}} &= 2 + 4\chi^2, \\ H_{\text{lazo}} &= I \iff \chi &= 0. \end{aligned}} \quad (58.21)$$

Its eigenvalues are

$$\lambda_{\pm} = (\sqrt{1 + \chi^2} \pm |\chi|)^2 = \exp(\pm 2 \operatorname{arsinh} |\chi|). \quad (58.22)$$

Proof. Under simultaneous conjugation by $S_{\zeta_r}^{-1}$, the operator V becomes $R(-\vartheta)$, and U has terms $e^{\pm \delta} \sin \theta$. Direct multiplication gives, in that reference,

$$UV - VU = -2\chi \operatorname{diag}(1, -1).$$

Since

$$H_{\text{lazo}} - I = (UV - VU)U^{-1}V^{-1},$$

the equivalence with $\chi = 0$ follows. Multiplication, or the trace identity for unimodular matrices, gives $\operatorname{tr} H_{\text{lazo}} = 2 + 4\chi^2$. The characteristic polynomial is $\lambda^2 - (2 + 4\chi^2)\lambda + 1$, whose roots are those in (58.22). $\square$

This is a relative holonomy of the declared composition. It is not identified with Γ_0 or with a complete TPK update on the basis of similar terminology.

58.5.1 Specialization fully determined by the conjugate pair

Let $U_+ = U_{\zeta}(\pi/2)$ and $U_- = U_{-\zeta}(\pi/2)$. Both satisfy $U_+^2 = U_-^2 = -I$, but

$$\begin{aligned} U_+ U_- &= -\operatorname{diag}(e^{2\zeta}, e^{-2\zeta}), \\ H_{\text{lazo}} &= (U_+ U_-)^2 = \operatorname{diag}(e^{4\zeta}, e^{-4\zeta}). \end{aligned} \quad (58.23)$$

Writing $\xi = \tanh \zeta = C^*/A$, with both angles expressed in the same unit, gives

$$\begin{aligned} H_{\text{lazo}} &= \operatorname{diag}\left(\left[\frac{1+\xi}{1-\xi}\right]^2, \left[\frac{1-\xi}{1+\xi}\right]^2\right), \\ \operatorname{tr} H_{\text{lazo}} - 2 &= \frac{16\xi^2}{(1-\xi^2)^2}. \end{aligned} \quad (58.24)$$

The formula for H_{lazo} composes the quarter-turn and the conjugate shapes of the same pair. Its expression is exact in the generated angular ratio ξ .

On the positive action branch of the corpus, $A_{\text{deg}} = 1000\alpha$ and $C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha)$, with $\hbar_{\text{ret}} = s_0\eta_{\text{ret}}$. Hence,

$$\begin{aligned} \xi &= \frac{\eta_{\text{ret}} + \alpha}{500\alpha}, \\ \rho_{\text{lazo}} = e^{4\zeta} &= \left(\frac{501\alpha + \eta_{\text{ret}}}{499\alpha - \eta_{\text{ret}}}\right)^2, \quad 0 < \eta_{\text{ret}} < 499\alpha. \end{aligned} \quad (58.25)$$

The coefficients 501 and 499 are 500 ± 1 ; 500 comes from the factor $1000/2$ in the angular pair and is not a new parameter. With the oriented axes retained, reconstruction is

$$\eta_{\text{ret}} = \alpha \left[500 \frac{\sqrt{\rho_{\text{lazo}}} - 1}{\sqrt{\rho_{\text{lazo}}} + 1} - 1 \right]. \quad (58.26)$$

The derivative

$$\frac{d\eta_{\text{ret}}}{d\rho_{\text{lazo}}} = \frac{500\alpha}{\sqrt{\rho_{\text{lazo}}}(\sqrt{\rho_{\text{lazo}}} + 1)^2}$$

is positive. The chamber $\eta_{\text{ret}} > 0$ corresponds to $\rho_{\text{lazo}} > (501/499)^2$; the boundary $\eta_{\text{ret}} \rightarrow 499\alpha$ corresponds to $\rho_{\text{lazo}} \rightarrow \infty$. This is an inverse readout of the dimensionless action coefficient from relative transport and the already constructed coordinate α . It retains the scale s_0 as an antecedent: a unimodular shape transformation does not by itself determine that dimensional scale. The conjugate orientation inverts ρ_{lazo} and transports the action label; it does not turn η_{ret} into negative action.

58.5.2 Loop iteration and oriented accumulation

Let

$$\rho_{\text{lazo}} = e^{4\zeta} = \left(\frac{1 + \xi}{1 - \xi} \right)^2.$$

For every integer n ,

$$\begin{aligned} H_{\text{lazo}}^n &= \text{diag}(\rho_{\text{lazo}}^n, \rho_{\text{lazo}}^{-n}), \\ Q_n &= \rho_{\text{lazo}}^n Q_0, \quad P_n = \rho_{\text{lazo}}^{-n} P_0. \end{aligned} \quad (58.27)$$

The proof is multiplication of diagonal matrices, extended to $n < 0$ through the inverse. The product $Q_n P_n = Q_0 P_0$ and symplectic area are preserved, whereas the state changes for $\zeta \neq 0$ and $X_0 \neq 0$. In the four-operation protocol, the declared phase increments sum to zero at each repetition; that phase accounting does not determine the total state transport.

For $\zeta \neq 0$ and $Q_0 \neq 0$, one recovers exactly

$$n = \frac{\log |Q_n/Q_0|}{4\zeta}.$$

If $Q_0 = 0$, the condition $P_0 \neq 0$ allows use of

$$n = -\frac{\log |P_n/P_0|}{4\zeta}.$$

The readout requires knowledge of the preparation and preservation of a reference. The multiplicative extension $H_{\text{lazo}}^{n+m} = H_{\text{lazo}}^n H_{\text{lazo}}^m$ corresponds to an additive logarithmic increment $4(n+m)\zeta$. Reversing the loop changes the sign of that increment.

Thus a fixed repeated protocol admits a response that records its iteration. This is a mathematical realization of operational accumulation in the composition studied. The integer n is not identified with all carry or memory of Γ_9 , nor is zero entropy for the complete dynamics inferred from periodicity of its phase protocol. If the initial amplitude is unknown and only the final state is available, n is not determined by that isolated readout.

58.5.3 Explicit lift to the twelve phases

Let

$$\tilde{S}_\zeta = I - \Pi_4 + W S_\zeta W^\top, \quad L_\pm = \tilde{S}_{\pm\zeta} C_{12}^3 \tilde{S}_{\pm\zeta}^{-1}.$$

On $\text{Ran } \Pi_4$, these act as U_+ , U_- , respectively; on the complement, both act as C_{12}^3 . Hence,

$$L_+ L_- L_+^{-1} L_-^{-1} = I - \Pi_4 + W H_{\text{lazo}} W^\top. \quad (58.28)$$

The complementary action cancels. The trace of the twelve-phase operator is $10 + \text{tr } H_{\text{lazo}}$. The intertwining and composition are exact and have been checked rationally using the integer vectors u_0, v_0 . This lift brings together the dodecaphasic antecedent and elliptic transport; it does not assert that its sequence is in itself the original autonomous TPK update.

58.5.4 Refinement and limit

The amplification $H_{\text{lazo},m} = H_{\text{lazo}} \otimes I_{9^m}$ preserves

$$\frac{1}{9^m} \text{Tr } H_{\text{lazo},m} = \text{Tr } H_{\text{lazo}},$$

as well as the norm and spectrum, apart from multiplicities. With the inclusions $v \mapsto v \otimes e_0$, one has

$$H_{\text{lazo},m+1} \iota_m = \iota_m H_{\text{lazo},m}.$$

The uniform operator defines a bounded extension on the Hilbert-space inductive limit, with the same norm. This is the limit of this amplification; it does not replace any other refinement system in the corpus.

58.6 Distinguishing a chart change from a relative effect

The coordinate transport

$$T_n = S_{\zeta_{n+1}} R(-\theta_n) S_{\zeta_n}^{-1}$$

telescopes:

$$T_{N-1} \cdots T_0 = S_{\zeta_N} R\left(-\sum_{n=0}^{N-1} \theta_n\right) S_{\zeta_0}^{-1}. \quad (58.29)$$

If chart and phase both return, the result is I . The commutator in Section 58.5 composes evolutions of different shapes in a common reference; it is not that passive change.

Under a common representation change P , the operators U, V, H_{lazo} are conjugated by P . The trace and spectrum of H_{lazo} remain invariant. If $X \mapsto PX$ and $g \mapsto P^{-\top} g P^{-1}$ as well, then $X^\top g X$ and every correctly transported energy readout are preserved. Thus $H_{\text{lazo}} \neq I$ cannot be removed by renaming coordinates.

This constitutes a closed relational mathematical protocol. Its experimental implementation requires the evolutions and their inverses to be available physical operations and the system, reference and controller to have a common realization. No new force or energy source is asserted here.

58.6.1 Area conservation and conservation of a common energy

Proposition 58.6 (Absence of a common positive metric). *Each quarter-turn preserves its own metric: $g_\zeta = \text{diag}(e^{-\zeta}, e^\zeta)$ for U_+ , and $g_{-\zeta}$ for U_- . When $\zeta \neq 0$, no common positive metric exists.*

Proof. The conditions

$$G = \begin{pmatrix} a & b \\ b & d \end{pmatrix} > 0, \quad U_+^\top G U_+ = G$$

force $b = 0$ and $d = ae^{2\zeta}$. The condition $U_-^\top G U_- = G$ requires $d = ae^{-2\zeta}$. The two are compatible only for $\zeta = 0$. $\square$

Preserving symplectic area is therefore not equivalent to preserving the same energy form throughout every composition of these transports. In the reference $g_r = S_r^{-\top} S_r^{-1}$, with $\omega_r > 0$, $X \neq 0$ and $E_r(X) = \omega_r X^\top g_r X / 2$, one has

$$\frac{E_r(H_{\text{lazo}} X)}{E_r(X)} = \frac{x^\top \bar{H}_{\text{lazo}}^\top \bar{H}_{\text{lazo}} x}{x^\top x}, \quad (58.30)$$

$$\bar{H}_{\text{lazo}} = S_r^{-1} H_{\text{lazo}} S_r, \quad x = S_r^{-1} X.$$

The extrema are the squared singular values. In the diagonal conjugate case, they are $e^{\pm 8|\zeta|}$, whereas the state dilations are $e^{\pm 4\zeta}$. This difference of exponents is necessary. Physically switching operations requires accounting for the controller; gain in a readout does not establish energy creation.

58.6.2 Order requires an oriented readout

Interchanging U_+ and U_- inverts H_{lazo} . The trace and the set of extremal gains are the same for H_{lazo} and H_{lazo}^{-1} . These scalars detect the loop effect but not its orientation. Two independent inputs with their complete vector responses reconstruct H_{lazo} ; a scalar measurement is generally insufficient.

In the conjugate case, with marked reference axes, let

$$G_Q = \frac{E_r(H_{\text{lazo}}e_Q)}{E_r(e_Q)}, \quad G_P = \frac{E_r(H_{\text{lazo}}e_P)}{E_r(e_P)}.$$

The readout

$$\mathcal{O} = \frac{1}{4} \log(G_Q/G_P) = 4\zeta \quad (58.31)$$

recovers orientation and changes sign under loop reversal. Axes and metric must be transported together under a representation change. Orientation is thereby distinguished without attributing it to a trace that removes it.

58.7 History information versus occupation statistics

The products

$$U_+ U_- U_+^{-1} U_-^{-1} \quad \text{and} \quad U_+ U_+^{-1} U_- U_-^{-1}$$

contain the same four operations with the same multiplicities. The second equals I ; the first is $H_{\text{lazo}} \neq I$ for $\zeta \neq 0$. Any statistic depending only on multiplicities therefore assigns the same result to two protocols with different actions on the state.

The exact conclusion is that such a statistic does not constitute a sufficient description for predicting the response of the composition. This does not refute Shannon: a distribution on ordered sequences can represent order. What is distinguished is the information retained in the chosen observable.

Definition 58.7 (Operational information of a history). In this representation, the operational information of a history is proposed to mean the class of its ordered transport distinguishable by the admitted readouts. Let $\mathcal{D}$ be a collection of detectors defined on operators. It may include

$$D_{X,L}(H_{\text{lazo}}) = L(H_{\text{lazo}}X), \quad D_{\text{tr}}(H_{\text{lazo}}) = \text{tr } H_{\text{lazo}},$$

with preparation X and linear readout operator L . Two histories are equivalent if

$$D(H_{\text{lazo},1}) = D(H_{\text{lazo},2}) \quad \text{for every } D \in \mathcal{D}.$$

Enlarging the collection may separate previously indistinguishable classes. Complete vector responses on a basis reconstruct every linear operator; with trace alone, $H_{\text{lazo}} \sim H_{\text{lazo}}^{-1}$ may persist. This distinguishes the detector of a state response from the spectral detector of the operator.

The definition does not identify transport with the entire TPK genealogy: different histories may produce the same operator in a representation. Information latent in other sheets, incidences and extensions is retained by its own readout operators.

The statistical and operational dimensions are compatible. The combinatorial growth rate, or topological entropy, measures asymptotic growth of admissible configurations; a Shannon entropy rate additionally requires a distribution. An observation map determines which transformations can be reconstructed. None of the preceding proofs asserts universally zero thermodynamic entropy. The contribution is to specify which description suffices for a prediction or reconstruction task.

58.8 Consequences and proposed new definitions

The preceding results bring together five notions, retaining their domains and readout operators:

1. **Constitutive spectral potential.** $\mathcal{F}$, defined by the 90/120 moments, has the logarithms of normalized speed and normalized impedance as its two derivatives. This is a mathematical definition downstream of the channels, with no added energy interpretation.
2. **Differential constitutive reciprocity.** The equality of cross-sensitivities proved in Section 58.1 provides a verifiable joint constraint in addition to the values.
3. **Relative closure defect.** $\text{tr } H_{\text{lazo}} - 2$ detects the nontrivial loop. It is accompanied by an oriented readout, such as $\mathcal{O}$, when order must be distinguished. Trace alone does not define all memory.
4. **Operational reconstruction.** This is recovery of the action of a history from a specified collection of observables. It is distinguished from two-channel reconstruction and from reconstruction of the full state.
5. **Reconstruction margin.** The bound $60a(M_0)$ in Section 58.3 quantifies absolute stability on a domain, whereas injectivity asserts only exact uniqueness.

The mass, frequency, radius and temperature coordinates of the preceding development remain unchanged. This development adds two levels: differential compatibility between scalar readout operators and order memory in operator compositions. The dilation factor of H_{lazo} is not identified with a new mass without passing through the corresponding mass readout operator.

59 Conclusions

The unifying result of this work is the joint determination of structures that scalar evaluations present separately. Explicit compatibility and reconstruction maps have been constructed, the invariants of their transformations have been identified, and the observations that recover the retained information have been established. The composition relates the constants through their shared operations. Action participates in a dimensional scale, in the counterangle, and in the electronic return quotient. The angular pair determines oriented channels; hexad–octad incidences produce constitutive readout operators and the Barbero–Immirzi functional. The composition preserves this genealogy and establishes joint properties beyond the individual evaluation of each readout operator. The starting point remains the APP–TRIT–TPK composition: the inverse relations act on its constructed outputs and preserve the orientations, domains, and dimensional data that they use.

Chronological memory and operator response. The chronological analysis specifies the information retained by an extension. In the capacity factor, with the stated phase measure, block entropy grows without bound while its rate per position tends to zero. The balance, which takes two values, loses an increasing amount of chronological information. The quadratic activity of the current readout operator remains positive even when its mean vanishes. These properties concern different observations of the same process and can coexist without identifying the symbolic rate with a temperature.

Operator reconstruction adds a dynamical distinction. In the nonadic representation of readout and memory, order sensitivity is distributed between the contributions $1/81$ and $8/81$ of the internal commutator. Reintroducing memory combines them and produces $1/9$. The retained history therefore participates in composing the response: knowledge of the present balance and knowledge of the capacity to respond to a continuation constitute different information.

Spectral determination and reconstruction stability. Spectral reconstruction establishes a joint determination result. In the normalized constitutive collection, velocity fixes the product of the two responses, and the Barbero functional distinguishes their oriented distribution monotonically. Together, these data recover the labelled eigenvalues. On the generated image, inversion composes with angular reconstruction and reconstruction of the action coefficient. Sheet markers and dimensional bases are retained: they are representation data used in the reconstruction.

The differential characterization makes this reconstruction precise. The logarithms of velocity and impedance are derivatives of a single spectral potential, defined through the incidence channels. Its negative definite Hessian yields an exact reciprocity of sensitivities and bounds the inverse on controlled domains. Uniqueness for exact data and sensitivity to finite errors are thus characterized by distinct results. This distinction is particularly relevant when strong contraction preserves injectivity while reducing the observable separation between states.

Constitutive composition and electroweak reconstruction. The cubic inverse transports multiplication of contractions to an associative law on responses. Its logarithmic coordinate is additive and retains the direction of composition on each sheet. Electroweak reconstruction specifies another complementarity: the ratio of gauge couplings recovers the common contraction, whereas Higgs determines the oriented coordinate within the specified domain. Marked projectors turn that scalar recovery into reconstruction of the constitutive operator. This preserves a proved connection between distinct readers of the same channels.

Catalan’s constant and the constitutive potential. The depth-two moment connects two evaluations of the same functional structure. On the oriented quarter-turn, its imaginary part produces the collection whose extremal value is Catalan’s constant; on the contractive channels, the same moment enters the constitutive potential. The arguments, orientation, and normalization of each evaluation remain specified. The functional structure carries more information than any one of its extremal values considered in isolation. The integral formula in Proposition 53.1 also recovers the moment from the constitutive response. The zero-limit condition at the origin uniquely fixes the integration constants. Production and reconstruction are thus linked on a single functional collection, together with its domain and endpoint value.

Reconstruction of action and memory of order. Relative composition shows that order leaves a recoverable operator signature. The quarter-turns of conjugate forms and their inverses produce a unimodular loop with factor

$$\rho_{\text{lazo}} = \left(\frac{501\alpha + \eta_{\text{ret}}}{499\alpha - \eta_{\text{ret}}} \right)^2, \quad 0 < \eta_{\text{ret}} < 499\alpha.$$

The stated branch and references allow η_{ret} to be recovered. Its dimensional scale remains the necessary antecedent for reconstructing the Planck section. The lift to the twelve-phase cycle establishes where this transformation acts, and iteration expresses its accumulation through reciprocal powers. Nonadic holonomy retains its own domain and update law; the composition studied here constitutes a well-defined subsequent transport.

Operational information thereby acquires a precise definition: two histories are distinguishable with respect to the admissible preparations and detectors when their transports produce different responses. Operation multiplicities do not generally determine this response; order can be indispensable. The trace and vector responses likewise retain different data. A sufficient collection of readouts reconstructs the operator, whereas other readouts assign the same value to oppositely oriented transformations. This definition makes the relation between observation, memory, and reconstruction explicit, while distinguishing a representation from the complete genealogy of the state.

Mass reciprocity and the interrelation of physical quantities. Joint normalization has a prior operator proof: the inscribed return produces L_α , and polar transport of the same character determines the readout operators R and Λ . The identities $HA = \hbar cI$ and $RA = L_\alpha^2 I$ fix $G = c^3 L_\alpha^2 / \hbar$ as the unique coefficient of the radial realization. The proof retains noncommutative variations and compatible elimination of memory, including dual transport of the action length. Its returned evaluation in the SI section is $6.674299659414022706 \dots 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$; the CODATA comparison follows that evaluation.

The composition brings together four readouts of a single positive structural mass coordinate. Under the stated circular and thermal conditions, the proper frequency, reduced gravitational radius, reduced Compton wavelength, and energy normalized by the tritic decision satisfy

$$\Xi_\Gamma = \frac{\Omega_\Gamma}{\omega_0} = \frac{r_{g,\Gamma}}{R_{108}} = \frac{R_{108}}{\bar{\lambda}_\Gamma} = \frac{E_\Gamma}{k_B \Theta_{108} \log 3}.$$

The electronic prefactor, character, and towers belong to the construction of Ξ_Γ . The identity preserves their roles and normalizations. In this realization, the mass factor is characterized as a reciprocal dilation parameter for the two lengths; their product preserves the area R_{108}^2 . The thermal weight of the cycle is expressed by $3^{-\Xi_\Gamma}$, subject to the corresponding normalization and convergence conditions. Here $R_{108} = L_\alpha$; the circular form preserves the length already constructed. Varying the clock also transports the electronic quotient b_e , preserving the same mass readout. Chronological windows, reduced phases, and terminal costs describe aspects of the state; preserving them in isolation allows changes that the full composition distinguishes.

Electronic scale, Fermi and action length. Substitution of the complete electronic section retains its prefactor and return within the electroweak realisation. The common scale enters tree-level Fermi with an inverse-square power and the Higgs vacuum expectation value with a direct power; it cancels in angular ratios and in the self-coupling. Species-specific factors remain in their ratios. Positive congruence also transports the Compton length by inverse congruence, without requiring commutation between basal mass and return. These proofs distinguish the absolute scale, angular information and spectral structure preserved by each operation.

Electronic reconstruction and preservation under transport. The electronic readout retains a verifiable dependence on order. The sum and differences between opposite positions reconstruct the record, including its integrality conditions; autocorrelations and the spectrum distinguish records with equal multiplicities. The coefficient 80 evaluates the specified central record and changes when its order is altered. For a finite collection of multisections, the mass operator is positive, and its commutator with a transport determines exactly when that transport preserves mass. This criterion transmits the same preservation to the temporal, spatial, gravitational, and thermal readouts, with their references fixed.

Marked trace and thermometric response. The composition of capacities and occupations produces

$$b_* = 10^{-23} + 380 \cdot 10^{-26} + 649 \cdot 10^{-29} = 1.380649 \times 10^{-23}.$$

The grading of the 649 states retains the sectors 1, 24, and 624 and their respective memory indices. The spectrum with ratio 1/10 follows from the memory transport already proved. An explicit thermometric realization is constructed on that reference: marked information is $b_*s(\rho)$, and energy is the readout conditioned on the marked sector. For Gibbs variations with fixed Hamiltonian and reference, the temperature satisfies $b_*\Theta = \beta^{-1}$. The coefficient is thereby characterized through a concrete thermal operation. Conditioning both quantities in the same way gives a different normalization; varying the reference adds the corresponding differential term. The proof identifies these dependencies and retains the dimensional chart required for the physical Boltzmann readout. Composition with the tritic clock, entanglement area, and gravitational reciprocity yields the joint relation among decision energy, action, and area.

Physical content of the transformation. The analysis constructively distinguishes passive transport from relative response. The former telescopes upon return of the chart and phase. The latter retains a final operator different from the identity when the commutator is nontrivial. Symplectic area preservation coexists with the absence of a common positive energy metric for the two conjugate forms. Interpreting an energy gain requires the realization of the controller and the reference to be included. The mathematical result determines what must be tested and which data must be retained to attribute a response to the physical system.

Structural definition and unity of the result. The developments refine the definitions used throughout the work. The action coefficient characterizes a regional section that participates both in angular construction and in its reconstruction through transport. The Barbero–Immirzi functional distinguishes the oriented distribution of constitutive channels; together with normalized velocity, it determines their spectrum. Catalan’s constant appears as an extremal evaluation of the oriented moment whose collection also enters the response potential. In the specified realization, the mass coordinate parametrizes a reciprocal dilation preserving the product of the lengths. Boltzmann’s constant relates the temperature and decision-energy lines through the stated positive transducer and tritic normalization. Each definition supplements the value with an operation, a domain, and a role within the composition.

Critical separation and reconstruction of readers. Parametric scaling is established for the sequence selected by the construction itself: the first return roots of the uniform dodecaphasic reader. APP–TRIT–TPK, the joint continuum, orientation and memory remain its antecedent. Identification of centres and remainder control in 43.1–43.3 culminate in (46.2): the Feigenbaum constant is the asymptotic separation invariant of those parameters. The same development recovers the profile from its twelve-node measure through Jacobi and, through constitutive inversion, recovers the oriented channels and the moment whose endpoint is Catalan’s constant. The result thus brings generation, selection and reconstruction together, preserving the domain and information that distinguish each reader.

Integration of the four interactions and total canonical closure. Gravitation and the strong, weak and electromagnetic interactions act on the same material realization of the APP–TRIT–TPK structure. Transport preserves the sheets, orientation, memory and incidence of the enriched state; its spin and gauge realizations compose on the same material fibres. The electromagnetic component belongs to the electroweak realization: it is not introduced as an additional interaction disconnected from it. The total action uses the coefficients and readers already constructed, including the Planck section, radial length and Barbero–Immirzi functional.

The content of this integration is not limited to collecting terms in an expression. Variation of the same action produces the total spin current and the joint metric source. Eliminating contorsion preserves the cross-terms between species; the Legendre transformation preserves material momenta and produces the total constraints. The proofs in sections 50 and 51 establish backreaction, the first-class property and propagation of those constraints in the specified regular realization.

In particular, let Θ_{tot} be the total canonical potential obtained in that transformation, $\mathcal{Z} = \{C_A = 0\}$ the regular constraint surface, and $X_N = N^A X_{C_A}$, $X_M = M^B X_{C_B}$ two admissible characteristic deformations.

The connection induced by the action has $H_N^{\text{tot,can}} = -\Theta_{\text{tot}}(X_N)$. Its curvature, calculated without omitting derivatives of the deformation coefficients, is

$$\begin{aligned}\mathcal{F}_{N,M}^{\text{tot,can}} &= \delta_N H_M^{\text{tot,can}} - \delta_M H_N^{\text{tot,can}} - H_{[X_N, X_M]}^{\text{tot,can}} + \frac{i}{\hbar} [H_N^{\text{tot,can}}, H_M^{\text{tot,can}}] \\ &= -d\Theta_{\text{tot}}(X_N, X_M) = N^A M^B f_{AB}{}^D C_D.\end{aligned}\tag{59.1}$$

Therefore,

$$\mathcal{F}_{N,M}^{\text{tot,can}}|_{\mathcal{Z}} = 0.$$

The formula includes the tangential deformation and the gauge actions of the total bracket; on the quotient, it expresses the same equality modulo those vertical directions. The prequantum representation preserves this closure through the commutator identity proved in (51.20).

Thus, gravitational incorporation is part of the same action, its sources and its constraint algebra, not an equation juxtaposed after constructing the other interactions. The proved vanishing concerns the canonical characteristic curvature; it does not require spacetime curvature or gauge-field strengths to vanish. Its domain preserves the nondegenerate coframe, the regular chart and the boundary conditions used in the proof. Nor does it suppress the memory or global monodromy of the histories that gave rise to the realization.

Horizon composition distinguishes the action dependence at fixed area from that at fixed mass. Section 56 also proves a compatibility curve connecting thermal bias, elliptical shape, and the Barbero channels. Probability ratios recover the same action section and provide a joint check of these readings.

Production, compatibility, and reconstruction therefore constitute three complementary levels of a single description. Genealogy determines the objects, joint relations constrain their readouts, and detectors determine the recoverable structure. The new characterizations rest on the operations presented here and retain their conditions, allowing them to be incorporated into specialized developments while preserving the unity of the argument.

A Compensated transport, geometric memory and energy tests

A.1 Operator origin and domain of the realization

The construction retains APP–TRIT–TPK and the enriched state. APP brings together the arithmetic sheets with their residues and quotients; TRIT retains regime and orientation; TPK composes selection, transport and memory updating. The matrix representations used below are downstream realizations. The nonadic holonomy Γ_9 and its memory retain their own operators.

The antecedents used are the elliptic, hyperbolic and parabolic quadratic representations of TRIT, the differential of invertible transports, the energy functional $E = \langle Df, WDf \rangle$ with $W > 0$, return with translational memory, the action ellipse with semiaxis product $2\hbar$ and its LC realization, vacancy polarization and reversible two-mode coupling. Translational return and the commutator below are distinct objects. The elliptic–parabolic loop and its oriented-area balance will subsequently be used as a test collection.

The compensated composition and test protocol bring together these realizations. The algebraic theory of commutator traces in $SL(2)$ is an established mathematical antecedent.³ The subject of this appendix is its specific composition with the declared readout operators. The parameters a, b are realization parameters, not fitted physical constants.

The scope ends at this local realization. Identification of TPK routes admitting a material device with those parameters requires composition of their effective readout operators. An arbitrary invertible matrix does not by itself establish a TPK route, and identification of the commutation defect with Cartan torsion requires its connection and soldering.

A.2 Compensation of operations and area conservation

The regime matrices are

$$J_e = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad J_h = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (\text{A.1})$$

They satisfy $J_e^2 = -I$, $J_h^2 = I$ and $[J_e, J_h] = 2 \operatorname{diag}(1, -1)$. Define

$$R(a) = e^{aJ_e}, \quad B(b) = e^{bJ_h}, \quad C_{\text{aux}}(a, b) = R(a)B(b)R(a)^{-1}B(b)^{-1}. \quad (\text{A.2})$$

The action on columns is read from right to left. The protocol contains each transformation and its inverse, but the sequence is not a full retracing of the path. A full retracing would, for example, have the form $RBB^{-1}R^{-1} = I$. The symbol C_{aux} distinguishes this elliptic–hyperbolic commutator from the counterangle C^* and from the conjugate-shape loop H_{lazo} studied in the main text.

Proposition A.1. For $a, b \in \mathbb{R}$,

$$\det C_{\text{aux}} = 1, \quad \operatorname{tr} C_{\text{aux}} = 2 + 4 \sin^2 a \sinh^2 b. \quad (\text{A.3})$$

Proof. Put $c_a = \cos a$, $s_a = \sin a$, $u_b = \cosh b$ and $v_b = \sinh b$. Then

$$R = \begin{pmatrix} c_a & s_a \\ -s_a & c_a \end{pmatrix}, \quad B = \begin{pmatrix} u_b & v_b \\ v_b & u_b \end{pmatrix},$$

with $c_a^2 + s_a^2 = 1$ and $u_b^2 - v_b^2 = 1$. All the matrices have determinant one. Multiplication gives

$$\begin{aligned} (C_{\text{aux}})_{11} &= u_b^2 + 2c_a s_a u_b v_b - (c_a^2 - s_a^2) v_b^2, \\ (C_{\text{aux}})_{12} &= -2s_a^2 u_b v_b - 2c_a s_a v_b^2, \\ (C_{\text{aux}})_{21} &= -2s_a^2 u_b v_b + 2c_a s_a v_b^2, \\ (C_{\text{aux}})_{22} &= u_b^2 - 2c_a s_a u_b v_b - (c_a^2 - s_a^2) v_b^2. \end{aligned} \quad (\text{A.4})$$

The diagonal sum is $2u_b^2 - 2(c_a^2 - s_a^2)v_b^2 = 2 + 4s_a^2 v_b^2$, as claimed. $\square$

³Goldman, *Trace Coordinates on Fricke spaces*, arXiv:0901.1404.

When $s_a v_b \neq 0$, the trace exceeds two and the commutator is hyperbolic, with reciprocal positive eigenvalues

$$e^\lambda, e^{-\lambda}, \quad \lambda = 2 \operatorname{arsinh} |\sin a \sinh b| > 0. \quad (\text{A.5})$$

This follows from $\cosh \lambda = \operatorname{tr}(C_{\text{aux}})/2$ and $\cosh(2 \operatorname{arsinh} x) = 1 + 2x^2$. If $s_a v_b = 0$, then $b = 0$ or $R = \pm I$, and $C_{\text{aux}} = I$. The local expansion is

$$C_{\text{aux}} = I + 2ab \operatorname{diag}(1, -1) + O(|a|^2|b| + |a||b|^2). \quad (\text{A.6})$$

The residual begins with the product of the two operations. Applying either operation followed by its own inverse yields the identity.

A two-dimensional real matrix of determinant one preserves the area form. Thus C_{aux} transforms the action ellipse into another ellipse of area h . If $\Sigma > 0$ is a covariance matrix,

$$\Sigma' = C_{\text{aux}} \Sigma C_{\text{aux}}^\top, \quad \det \Sigma' = \det \Sigma. \quad (\text{A.7})$$

The distribution and orientation of the semiaxes may change. The eigenvalues $e^{\pm\lambda}$ describe eigendirections and iterations; singular values determine a different readout and do not, in general, coincide with them. The variation of Euclidean semiaxes in a single application requires the corresponding metric.

Conservation of action area is compatible with accumulation of oriented deformation. The geometric observable λ depends on a, b and does not constitute an additional universal constant. Conservation of determinant or area alone does not establish conservation of physical energy.

A.3 Positive readout of the residual and path reversal

On a return fibre, with declared field f and Hermitian weight $W > 0$, the energy readout operator takes the form

$$E_C(f) = \langle (I - C_{\text{aux}})f, W(I - C_{\text{aux}})f \rangle. \quad (\text{A.8})$$

For $s_a v_b \neq 0$, the commutator has no eigenvalue one and $E_C(f) > 0$ for every $f \neq 0$. Full retracing yields zero residual. Dimensional normalization belongs to the initial functional: its numerical coordinates require their realization before being interpreted as joules, heat, mass or conduction current.

Complete path reversal inverts C_{aux} . The trace and λ assign the same readout to the operator and its inverse. Reconstruction of direction requires an additional readout: the residual vector, transport components or a phase-resolved probe. This comparison places a concrete limit on the sufficiency of an exclusively scalar readout.

A rational example, independent of physical data, is

$$R = \begin{pmatrix} 3/5 & 4/5 \\ -4/5 & 3/5 \end{pmatrix}, \quad B = \begin{pmatrix} 5/4 & 3/4 \\ 3/4 & 5/4 \end{pmatrix}. \quad (\text{A.9})$$

It yields $\operatorname{tr} C_{\text{aux}} = 86/25$, $\det C_{\text{aux}} = 1$ and eigenvalues $(43 \pm 6\sqrt{34})/25$. The rational checks in the supplement include this example and 169 pairs, covering zero cases, full retracing and preservation of the covariance determinant. These are finite checks of the proven identities.

A.4 Reversible coupling and reduced entanglement readout

For $H \in \operatorname{Sp}(2, \mathbb{R})$, the two-mode realization uses the coupling

$$U_H = \frac{1}{9} \begin{pmatrix} 8I + H & \sqrt{8}(H - I) \\ \sqrt{8}(H - I) & I + 8H \end{pmatrix}. \quad (\text{A.10})$$

With two identical pure Gaussian states prepared from the action ellipse and joint transport of the initial chart, the reduced readout satisfies

$$\nu(H)^2 = 1 + \frac{8}{81} \{\operatorname{tr}(HH^\top) - 2\}, \quad \operatorname{Tr} \rho_{\text{vis}}^2 = \nu(H)^{-1}. \quad (\text{A.11})$$

Preparation and readout operators remain fixed. The joint transformation is reversible and preserves global purity.

Corollary A.2. For $C_{\text{aux}}(a, b)$,

$$\text{tr}(C_{\text{aux}} C_{\text{aux}}^T) - 2 = 16 \sin^2 a \sinh^2 b \cosh^2 b, \quad (\text{A.12})$$

and

$$\boxed{\nu_C^2 = 1 + \frac{32}{81} \sin^2 a \sinh^2(2b)}. \quad (\text{A.13})$$

Proof. The four entries of (A.4) can be written as $d + k, l - m, l + m, d - k$, where

$$d = 1 + 2s_a^2 v_b^2, \quad k = 2c_a s_a u_b v_b, \quad l = -2s_a^2 u_b v_b, \quad m = 2c_a s_a v_b^2.$$

Their sum of squares is $2(d^2 + k^2 + l^2 + m^2) = 2 + 16s_a^2 u_b^2 v_b^2$. Substitution in (A.11) and $\sinh(2b) = 2 \sinh b \cosh b$ prove the result. $\square$

Example (A.9) gives $\nu_C^2 = 17/9$ and purity $3/\sqrt{17}$.

Cancellation of $\hbar$ and of the initial deformation uses the declared preparation and joint chart transport. A protocol with $s_a v_b \neq 0$ produces reduced mixedness; complete inversion of the coupling can undo it. Reduced mixedness, stored energy and heat retain their roles as distinct readouts.

A.4.1 Signed test in an elliptic-parabolic collection

The constructed loop collection is

$$H_n = \begin{pmatrix} 1+n & n^2 \\ n & n^2 - n + 1 \end{pmatrix}, \quad n \in \mathbb{Z}. \quad (\text{A.14})$$

Its alternating memory balance depends on n^2 , whereas the covariance-determinant ratio is

$$\mathcal{R}_n = \frac{\det V_{\text{vis},n}}{\det V_0} = 1 + \frac{8}{81} n^2 (2n^2 - 2n + 5). \quad (\text{A.15})$$

Subtracting the two orientations cancels the even terms:

$$\boxed{\mathcal{R}_{-n} - \mathcal{R}_n = \frac{32n^3}{81}}. \quad (\text{A.16})$$

For $n = 1$ and $n = -1$, the responses are $121/81$ and $153/81$, with purities $9/11$ and $3/\sqrt{17}$, respectively. This collection satisfies $H_{-n} \neq H_n^{-1}$ in general; it is not the iteration H_1^n either. Changing the oriented parameter must be distinguished from fully retracing the operations. Its alternating area readout retains an identification distinct from that of the Barbero functional.

The mixedness criterion uses the commutator $[D, J]$, with $D = \sqrt{8}(H - I)/9$. If D commutes with the complex structure compatible with the initial state, it may transport memory without making that Gaussian product mixed. If it does not commute, the readout operator determines the mixedness. Absence of mixedness for that state is a property distinct from absence of material dissipation.

The difference (A.16) is a corollary of the collection and its readout operator, not a universal constant. The formula for C_{aux} composes the same readout operator with another collection admitted by the theorem. A material test must determine transports and couplings before measurement; arbitrarily programming these matrices establishes their emulation and retains a different scope.

A.5 Second moment of the vacancy current

The capacity readout operator fixes

$$\delta = \log_{10}(10/9), \quad z_n \in \{0, 1\}, \quad R_N = \sum_{n < N} (z_n - \delta), \quad |R_b - P_a| < 1.$$

On equal intervals t_0 , the centred current is $I_n = A_I(z_n - \delta)$, with $A_I = u_Q/t_0$. The symbol A_I distinguishes this amplitude from the principal angle. For $L = b - a$,

$$\frac{1}{A_I^2} \sum_{n=a}^{b-1} I_n^2 = L\delta(1 - \delta) + (1 - 2\delta)(R_b - P_a). \quad (\text{A.17})$$

The proof expands $(z_n - \delta)^2$, uses $z_n^2 = z_n$ and sums. The signed imbalance remains bounded, whereas $I_{\text{rms}}/|A_I| \rightarrow \sqrt{\delta(1 - \delta)}$. The amplitude A_I is the jump between the two current levels. The test retains the temporal realization and must account for any coupling that filters the sequence.

A zero signed balance can coexist with a positive quadratic readout. If an ideal resistance $R_E > 0$ is added and that current is maintained over the intervals, the downstream test model gives

$$Q_{R_E} = R_E t_0 \sum_n I_n^2. \quad (\text{A.18})$$

The resistance belongs to this downstream realization. Its role is to test a possible heat reduction against an explicit balance.

A.6 Conditions for a material test

The protocol separates two questions. For order memory, compensated paths, order exchanges and full retracings are compared through a remaining vector or phase-resolved response. An isolated intensity or net charge may lose the distinction sought. For energy recovery, input, useful or returned energy, storage variation and heat are measured, including preparation and restoration of the apparatus. Phase return remains distinct from variation in stored energy.

In a scalar one-port test, the relation $v_2(t) = v_1(T - t)$ preserves the amplitude distribution and power spectrum. In a periodic regime, the same linear time-invariant system predicts equal mean power. An instantaneous law $i = g(v)$ also preserves the integral of vi . A reproducible difference would probe history dependence, but attributing it requires comparison with hysteresis, heating and alternative dynamical models. In multiport systems, cross-spectra must also be retained.

A reduction in dissipation is tested at equal useful transfer, fidelity and rate. Attenuation must distinguish energy, signal amplitude, carrier population and conserved charge before it is attributed to a gravitational origin. The distinction between storage, recovery and losses has experimental precedents in electrocaloric recovery.⁴

The appendix yields a general identity within the specified matrix collection, a positive readout operator for the residual, an inversion check and an exact law for the second moment of the current. Its scope distinguishes these results from a new universal constant, a prototype without heating or a proof of gravitational electron decay. The prospective link with a device must be fixed before obtaining the test data.

⁴Defay et al., *Enhanced electrocaloric efficiency via energy recovery* (2018), doi:10.1038/s41467-018-04027-9. This antecedent does not constitute experimental evidence for the HMT transport composed here.

B Action, reduced memory and recoverable work

B.1 Composition of action with an energy realization

The composition receives action outputs and memory transports from APP-TRIT-TPK, retaining the enriched state as an antecedent. Its domain is the declared Gaussian and energy realization. Passivity and ergotropy are established concepts; they are used to study the specific coupling and the readout operators already constructed.

The action equation keeps its coefficients explicit:

$$H_5 = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \quad (\text{B.1})$$

$$D_A = e^{-100\pi\alpha/9}, \quad \mathcal{C}_\pi = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \quad (\text{B.2})$$

$$\begin{aligned} \hbar_{\text{pre}} &= s_0 H_5, \\ \hbar_{\text{ret}} &= s_0 \left(H_5 - \frac{90}{\pi} \mathcal{C}_\pi \right), \quad h_s = 2\pi \hbar_s, \quad s_0 = 10^{-34} \mathcal{U}_S. \end{aligned} \quad (\text{B.3})$$

The decimal order and dimensional chart retain distinct antecedents. The formula keeps the coordinates π, φ , the orientation and scale of the readout operator, in addition to α . The dimensional equality retains its unit of action.

Fix a positive section and write $\hbar = \hbar_s$. The ellipse has semiaxes $\sqrt{2\hbar e^{\zeta/2}}$ and $\sqrt{2\hbar e^{-\zeta/2}}$, where $\zeta = \text{artanh}(C^*/A)$. Its oriented area satisfies

$$\mathcal{A}(\theta) = \hbar\theta, \quad \mathcal{A}(2\pi) = h. \quad (\text{B.4})$$

In the LC realization, with declared clock ω and reference impedance,

$$\mathcal{H}_\zeta(Q, P) = \frac{\omega}{2} (e^{-\zeta} Q^2 + e^{\zeta} P^2). \quad (\text{B.5})$$

On the orbit, $\mathcal{H}_\zeta = \hbar\omega = hf$, with $f = \omega/(2\pi)$. The Hamiltonian convention gives $\dot{\theta} = -\omega$, and hence $d\mathcal{A}/dt = -\hbar\omega$. Positive energy coincides with the magnitude of that areal velocity in the realization and fixed section considered. Accumulated area and the Lagrangian action of an arbitrary trajectory are distinct objects.

The internal equation determines the action scale per unit phase; the clock fixes its rate. Ellipse deformation and transport order retain information beyond hf . The Gaussian preparation used subsequently differs from a point on the classical orbit: its initial energy is $hf/2$, rather than hf .

B.2 Preparation and readout operators for the energy result

Two pure, centred, identical Gaussian modes are prepared with covariance

$$V_0 = \frac{\hbar}{2} M M^\top, \quad M = \text{diag}(e^{\zeta/2}, e^{-\zeta/2}). \quad (\text{B.6})$$

The operator U_H of (A.10) acts, transported to the initial chart by $M \oplus M$. The joint state remains pure. The final energy readout operators have the same frequency ω , retain the initial metric $G_\zeta = M^{-\top} M^{-1}$ and include no residual interaction energy between the modes. Extraction operations begin and end with these same readout operators. Physical accessibility of the admitted operations is a hypothesis of the work bound, distinct from construction of a device realizing them.

Define

$$\begin{aligned} \tau &= \text{tr}(H H^\top) - 2 \geq 0, \\ W_v &= \frac{8I + H H^\top}{9}, \quad W_m = \frac{I + 8H H^\top}{9}. \end{aligned} \quad (\text{B.7})$$

The covariances are $V_j = (\hbar/2) M W_j M^\top$. Visible purity is denoted by $\mu = \text{Tr} \rho_v^2$ and its normalized symplectic eigenvalue by $\nu = \mu^{-1}$. In this appendix, ν has this role; ordinary frequency is written f and angular frequency ω .

B.3 Identity between visible energy and purity

Proposition B.1. *With the preceding preparation and readout operators, the visible energy above the initial state satisfies*

$$\frac{\Delta E_v}{hf} = \frac{\tau}{36}, \quad \nu^2 = 1 + \frac{8\tau}{81}, \quad \boxed{\mu^{-2} - 1 = \frac{32}{9} \frac{\Delta E_v}{hf}}. \quad (\text{B.8})$$

Proof. The state is centred and the quadratic readout operator remains fixed. Hence,

$$E_v = \frac{\omega}{2} \text{tr}(G_\zeta V_v) = \frac{\hbar\omega}{4} \text{tr} W_v.$$

Since $\text{tr} W_v = 2 + \tau/9$, subtracting $\hbar\omega/2$ gives the first equality. The reduced covariance satisfies $\det W_v = 1 + 8\tau/81$ and $\mu = (\det W_v)^{-1/2}$. Eliminating τ proves the boxed identity. The proof requires neither symmetry of H nor coincidence between its eigenvalues and singular values. $\square$

The law eliminates the particular path parameters. The coefficient $32/9$ comes from the coupling split, and the scale h is retained until normalization. To check the specificity of that split, consider $W_p = pI + (1-p)HH^\top$ with $0 < p < 1$. One obtains

$$\mu^{-2} - 1 = 4p \frac{\Delta E}{hf}. \quad (\text{B.9})$$

The declared choice is $p = 8/9$. When $\Delta E > 0$, a test can estimate

$$p_{\text{obs}} = \frac{\mu^{-2} - 1}{4\Delta E/(hf)} \quad (\text{B.10})$$

and compare it with $8/9$ without fitting p to those same data. Programming a mixer with that split verifies an emulation; attribution of the split to a natural mechanism retains its independent proof.

The identity uses centred Gaussian states and the specified preparation. A coherent displacement adds energy without changing purity. Initial thermal noise and metric changes alter the relation. These variations delimit the statement and provide falsifiers for an extension to arbitrary states.

The convex combination W_v combines covariance matrices produced by reducing a joint Gaussian state. A probabilistic mixture of two pure states is a different object, generally non-Gaussian; its purity is not given by the determinant of that covariance.

B.4 Recoverable work under local operations

Ergotropy is the maximum energy decrease under cyclic unitary operations with a fixed energy readout operator. Its definition does not automatically include the net cost of constructing, preparing or controlling the apparatus.

The reduced spectrum has populations

$$\lambda_j = (1-r)r^j, \quad r = \frac{\nu-1}{\nu+1}. \quad (\text{B.11})$$

They decrease with oscillator energy. The passive state therefore has energy $E_{v,\text{pas}} = hf\nu/2$. It is reached by a Gaussian operation:

$$S = \sqrt{\nu} W_v^{-1/2}, \quad \det S = 1, \quad S W_v S^\top = \nu I. \quad (\text{B.12})$$

Population ordering also proves minimality among all unitaries, not only Gaussian ones.

Consequently,

$$\boxed{\frac{W_v}{hf} = \frac{(\nu-1)(9\nu-7)}{32}, \quad \frac{E_{v,\text{pas}} - hf/2}{hf} = \frac{\nu-1}{2}}. \quad (\text{B.13})$$

The two quantities sum to $\Delta E_v/(hf) = 9(\nu^2 - 1)/32$. The passive term is energy inaccessible under the declared local operations; its presence does not amount to an actual heat flow.

B.4.1 Joint access to correlations

The second mode satisfies $\det W_m = \det W_v = \nu^2$ and $\Delta E_m = 8\Delta E_v$. Thus,

$$\Delta E_{\text{tot}} = 9\Delta E_v = hf \frac{\tau}{4}. \quad (\text{B.14})$$

Joint inversion of the coupling restores the initial product. The total state is pure, and that product is the ground state of the joint readout operator; joint access ideally allows recovery of ΔE_{tot} . Independent local operations leave two passive states, each with energy $hf\nu/2$. It follows that

$$\boxed{\mathcal{W}_{\text{conjunta}} - \mathcal{W}_{\text{local}} = hf(\nu - 1) = hf(\mu^{-1} - 1)}. \quad (\text{B.15})$$

The identity quantifies the additional work accessible through correlations. The initial preparation required energy, and the balance of a complete cycle includes that contribution. The relation compares energy readouts and operations on the state, without identifying information and energy as the same physical quantity.

Here “local” means product unitaries $U_v \otimes U_m$; measurements, classical communication and feedback lie outside that class. The gain follows from the operational restriction on the correlated state and from its marginal spectra, not from residual interaction energy. The 1 : 8 split concerns energies above the vacuum, not total energies or ergotropies.

B.5 Exact cases and transport orientation

For H_n in (A.14),

$$\tau_n = n^2(2n^2 - 2n + 5). \quad (\text{B.16})$$

At $n = 1$, one obtains

$$\begin{aligned} \nu &= 11/9, & \mu &= 9/11, & \Delta E_v &= \frac{5}{36}hf, \\ \mathcal{W}_v &= \frac{1}{36}hf, & E_{v,\text{pas}} - hf/2 &= \frac{1}{9}hf. \end{aligned} \quad (\text{B.17})$$

The joint system has $\Delta E_{\text{tot}} = 5hf/4$, local work $37hf/36$ and joint advantage $2hf/9$. The case $n = -1$ shares the eigenvalues of H_1 but has $\tau = 9$ and

$$\nu = \frac{\sqrt{17}}{3}, \quad \Delta E_v = \frac{hf}{4}, \quad \mathcal{W}_v = \frac{9 - 2\sqrt{17}}{12}hf. \quad (\text{B.18})$$

Added visible energy differs by a ratio 9/5 at equal frequency and action section. The change $n \mapsto -n$ is not full retracing: $H_{-n} \neq H_n^{-1}$ in general. True matrix inversion preserves τ in dimension two. The parametric asymmetry of this collection thus remains distinct from complete reversal.

For C_{aux} , $\tau_C = 4 \sin^2 a \sinh^2(2b)$. The energy composition gives

$$\boxed{\Delta E_v(C_{\text{aux}}) = \frac{hf}{9} \sin^2 a \sinh^2(2b)}, \quad (\text{B.19})$$

$$\mathcal{W}_{\text{conjunta}} - \mathcal{W}_{\text{local}} = hf \left(\sqrt{1 + \frac{32}{81} \sin^2 a \sinh^2(2b)} - 1 \right). \quad (\text{B.20})$$

The connection with Planck remains explicit: in the prior section, $hf = 2\pi s_0 H_5(\alpha, \varphi)f$; in the return section, H_5 is replaced by η_{ret} . The composition fixes the section used; it assumes neither an experimental variation of α nor free alternation between dimensional sections.

B.6 Boltzmann and the equivalent temperature of the passive state

For $\nu > 1$, the passive state has equivalent temperature

$$k_B T_{\text{pas}} = \frac{hf}{\log[(\nu + 1)/(\nu - 1)]}. \quad (\text{B.21})$$

When the transducer clock is used, $hf = k_B \Theta_{\text{clk}} \log 3$, giving

$$\frac{T_{\text{pas}}}{\Theta_{\text{clk}}} = \frac{\log 3}{\log[(\nu + 1)/(\nu - 1)]}. \quad (\text{B.22})$$

For $n = 1$, this recovers $\log 3 / \log 10$. The entropy is that of the Gaussian spectrum, with occupation $(\nu - 1)/2$. The energy composition retains this previously constructed thermal readout. The equivalent temperature of the passive state does not imply that the preceding reduced state was in equilibrium or that heat has been transferred.

B.7 Specificity and testing conditions

The general relation between coherence, correlations and work extraction belongs to the existing literature. The composition examined here specifies a joint law: coefficient 32/9, energy split 1 : 8, oriented collection H_n , action scale and the difference between local and joint access.

Two experimental levels are distinguished. Emulation realizes U_H and checks its identities; it verifies that realization of the model. Physical prediction requires prospective determination, from the construction, of the device, degrees of freedom, frequency, preparation and coupling realizing its objects, without fitting 8/9 to the test data. Energy, covariance, purity, control costs and restoration energy are measured separately.

An independent test obtains temperature and purity through readout operators independent of the law under examination. Recovered work is compared with a balance that includes preparation, controls and memory restoration.

External comparisons include studies of Gaussian ergotropy and passivation,⁵ coherent ergotropy extraction with explicit energy costs,⁶ and coherent control between quantum-machine cycles.⁷

The result is an algebraic composition with explicit initial structure. Its proofs establish identities and operational restrictions. An observed experimental advantage, a new universal constant, a violation of thermodynamics or a material mechanism for heat-free transport would require their respective demonstrations and tests; they are not conclusions of the auxiliary identities established here.

C Exhaustive enumeration and finite determination of transport operators

This appendix develops two finite operations with different starting data. The first enumerates the initial conditions of the two cursors and constructs their emissions through the declared arithmetic recurrence. The second determines transport operators from the regional transition register. Separating their domains specifies what each proof produces: enumeration constructs the emission catalogue; matrix reconstruction determines the operators compatible with the register. The reconstruction theorem is not used as a proof of the prior production of that register.

The primitives of the first operation are the support I_9^2 , the four cardinal orientations, the tritic selector, the pre-displacement readout, orientation reversals and the emission rule. Their definitions are brought together below. Those of the second operation are the four transition matrices in (C.7). Their unique reconstruction is proved in Theorem C.2; the resulting calendar is given in (C.9). These determinations preserve the starting data specified for each operation.

⁵Kua, Serafini and Genoni, *Daemonic ergotropy of Gaussian quantum states and the role of measurement-induced purification via general-dyne detection*, arXiv:2506.22288. Its role is comparison with general formulas for energy, purity and passivity; it does not validate the HMT coupling or its constants.

⁶Experimental Extraction of Coherent Ergotropy and Its Energetic Cost in a Superconducting Qubit, arXiv:2506.16881.

⁷Hou et al., *Combining energy efficiency and quantum advantage in cyclic machines*, doi:10.1038/s41467-025-60179-5. The cited experiment does not measure the nonadic transport composed in this appendix.

C.1 Complete domain of initial conditions

Let $I_9 = \{0, \dots, 8\}$ and let

$$\mathcal{D} = \{N, E, S, O\}, \quad v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

An initial cursor condition is $s = (i, j, d) \in \mathcal{S} = I_9^2 \times \mathcal{D}$. The two cursors are independent in the initial domain

$$\Omega = \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}| = 9^2 \cdot 4 = 324, \quad |\Omega| = 324^2 = 104\,976. \quad (\text{C.1})$$

Index i corresponds to the positive APP label $i + 1$. For positive integers, positive reduction is $\rho_9(n) = 1 + [(n - 1)]_9$, where $[a]_b$ denotes the representative in $\{0, \dots, b - 1\}$. The residual evaluations of the two sheets are

$$\sigma(i, j) = \rho_9(i + j + 2), \quad \varpi(i, j) = \rho_9((i + 1)(j + 1)).$$

The finite multiplicative readout uses

$$\begin{array}{c|cccccccc} a & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline \ell_9(a) & 0 & 1 & 0 & 2 & 5 & 0 & 4 & 3 & 0 \end{array}. \quad (\text{C.2})$$

On units, ℓ_9 is the discrete exponent in base 2. On 3, 6, 9, the value 0 constitutes the declared extension of the emission observable; membership in that sector remains in the trajectory register.

For $1 \leq t \leq 54$, define

$$r_t = 1 + [(t - 1)]_9, \quad \tau_t = \begin{cases} +1, & r_t \in \{1, 4, 7\}, \\ -1, & r_t \in \{2, 5, 8\}, \\ 0, & r_t \in \{3, 6, 9\}. \end{cases}$$

Each cursor preserves its position z_t immediately before step t and its orientation vector v_t . The additive cursor is active when $\tau_t = +1$; the multiplicative cursor, when $\tau_t = -1$. For each of them, with activity indicator a_t , the update is

$$z_{t+1} = [z_t + a_t v_t]_9, \quad v_{t+1} = \begin{cases} -v_t, & t \in \{27, 54\}, \\ v_t, & t \notin \{27, 54\}. \end{cases} \quad (\text{C.3})$$

The readout at step t is taken at z_t , before displacement. The neutral event preserves both positions. This convention fixes unambiguously the window endpoints and the effect of cardinal reversals.

In the windows $W_m = \{9m - 8, \dots, 9m\}$, $1 \leq m \leq 6$, let

$$\begin{aligned} S_m(s_+) &= \sum_{\substack{t \in W_m \\ \tau_t = +1}} \sigma(z_t^+), & E_m(s_\times) &= \sum_{\substack{t \in W_m \\ \tau_t = -1}} \ell_9(\varpi(z_t^\times)), \\ s_m &= [S_m]_{10}, & e_m &= [E_m]_{10}. \end{aligned} \quad (\text{C.4})$$

Each window contains three events of each type. The signatures s and e are the six components of these residues, not the integers S_m, E_m before reduction.

C.2 Census of signatures and emissions

The emission of a pair of initial conditions is determined by

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}, \quad 1 \leq m \leq 6. \quad (\text{C.5})$$

The final term 1 is $[7 \cdot 3]_{10}$, corresponding to the three neutral events. This rule's coefficients belong to the declared recurrence. The real constants studied subsequently do not enter enumeration of its domain or its evaluation.

Proposition C.1 (Injective factorization of the finite catalogue). *Emission distinguishes every signature pair (s, e) . There are 18 additive and 26 multiplicative signatures; their pairs produce 468 emissions. The multiplicities of these emissions are 144, 432 and 1944, with 432, 18 and 18 emissions, respectively.*

Proof. The hundreds digit of U_m recovers s_m . If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit verifies the third congruence of (C.5). Thus, the coupling is injective.

The census is obtained by traversing, in the full Cartesian order, $i = 0, \dots, 8$, $j = 0, \dots, 8$ and $d = N, E, S, O$, and applying the 54 steps of (C.3). The 324 additive conditions are distributed over the following 18 signatures, each with 18 preimages. In the tables, a sequence of six digits represents the six ordered components of a signature.

122564	122898	212645
212988	221456	221889
456221	465898	546988
564122	645212	654889
889221	889654	898122
898465	988212	988546

For the multiplicative cursor, the signatures and their multiplicities are

Signature	Multiplicity	Signature	Multiplicity
000000	108	555555	24
177393	8	555717	8
177555	8	555771	8
177771	8	555933	8
339555	8	717555	8
339771	8	717717	8
339933	8	717933	8
393177	8	771177	8
393393	8	771339	8
393555	8	771555	8
555177	8	933339	8
555339	8	933555	8
555393	8	933717	8

These tables result from the finite sums (C.4), with domain, readout order and reversals already specified. Their sums of multiplicities are $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$. The Cartesian domain (C.1) permits every signature pair to be realized. Injectivity gives $18 \cdot 26 = 468$ emissions and the multiplicities

$$(18 \cdot 24, 18 \cdot 8) = (432, 144), \quad (18, 18 \cdot 24) = (18, 432), \quad (18, 18 \cdot 108) = (18, 1944),$$

where each pair indicates the number of emissions and multiplicity of their fibres. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

Enumeration thus exhausts the complete domain of initial conditions. $\square$

The oriented ternary reduction is

$$\Psi(U) = ([-U_1]_3, \dots, [-U_6]_3).$$

Applied to the $18 \cdot 26$ emissions of the preceding tables, it produces the following fibre census. Here r counts distinct emissions, not initial conditions.

$$\begin{array}{c|cccccc} r & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline \#\{w : \#\Psi^{-1}(w) = r\} & 132 & 66 & 18 & 3 & 6 & 18 \end{array} \quad (\text{C.6})$$

The sum of the second row is 243, while its weighted sum with weights r is 468. The image of 243 words is therefore distinguished from the ambient space $\mathbb{F}_3^6$, containing 729 words. The accompanying certificate preserves all signatures with their preimages, the 468 emissions and their ternary reduction; the preceding formulas provide a complete procedure for reconstructing them.

C.3 Transition register and reconstruction of the lifts

This subsection fixes the regional register of the cited source. Let $b_0^\chi, \dots, b_4^\chi \in \mathbb{F}_3^6$, with $\chi \in \{C, P, A\}$, be its three sequences. The register organizes transitions $b_0 \rightarrow b_1 \rightarrow b_2$ in X_0, Y_0 and transitions $b_2 \rightarrow b_3 \rightarrow b_4$ in X_1, Y_1 , using row vectors and the same regional order:

$$\begin{aligned} X_0 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, & Y_0 &= \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}. \end{aligned} \tag{C.7}$$

The preceding production is expressed in its source through the composition of selection, transport and update:

$$x_{j+1}^\chi = \mathcal{U}_{9j+9} \circ \dots \circ \mathcal{U}_{9j+1}(x_j^\chi), \quad b_j^\chi = \Pi_6(x_j^\chi), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

The following result starts from register (C.7). Its proof reconstructs the operators and verifies transition compatibility; it does not attribute coordinatewise production of states x_j^χ to matrix inversion.

Proposition C.2 (Uniqueness on the regional register). *The systems $X_a L_a = Y_a$, $a = 0, 1$, have a unique solution over $\mathbb{F}_3$. The two solutions are*

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}. \tag{C.8}$$

Proof. Elimination in $\mathbb{F}_3$ gives $\det X_0 = 1$ and $\det X_1 = 2 = -1$. Each map $L \mapsto X_a L$ is therefore an automorphism of the 36-dimensional matrix space; under columnwise vectorization it is represented by $I_6 \otimes X_a$. One obtains $L_a = X_a^{-1} Y_a$. Multiplication of the matrices in (C.8) by the respective matrices X_a reproduces the two matrices Y_a of (C.7); existence and uniqueness are established on that register. $\square$

The three resulting sequences, explicitly separated into blocks, are

$$\begin{aligned} B^C &= 010211 \mid 012222 \mid 010211 \mid 002111 \mid 110221, \\ B^P &= 201101 \mid 121221 \mid 102011 \mid 012222 \mid 102011, \\ B^A &= 121200 \mid 112202 \mid 121020 \mid 010210 \mid 010200. \end{aligned}$$

Each block preserves its six positions. Concatenation is not identified with a base-ten integer.

C.4 Exhaustiveness of the calendar and stability of reconstruction

A binary calendar is $c = (c_0, c_1, c_2, c_3) \in \{0, 1\}^4$. Define the number of transitions incompatible with the register by

$$d(c) = \sum_{\chi \in \{C, P, A\}} \sum_{j=0}^3 \mathbf{1}\{b_j^\chi L_{c_j} \neq b_{j+1}^\chi\}. \quad (\text{C.9})$$

Multiplying the preceding sequences by (C.8) gives

$$d(c) = 3 d_H(c, (0, 0, 1, 1)),$$

where d_H is Hamming distance: at each of the four positions, the alternative operator fails to satisfy all three regional transitions. The complete census is

c	$d(c)$	c	$d(c)$	c	$d(c)$	c	$d(c)$
0000	6	0100	9	1000	9	1100	12
0001	3	0101	6	1001	6	1101	9
0010	3	0110	6	1010	6	1110	9
0011	0	0111	3	1011	3	1111	6

Consequently, 0011 is the unique calendar reproducing the three complete sequences.

Sensitivity to matrix entries likewise admits an exhaustive proof without approximations. A single-entry modification replaces L_a by $L_a + \lambda E_{ij}$, with $a \in \{0, 1\}$, $1 \leq i, j \leq 6$ and $\lambda \in \mathbb{F}_3 \setminus \{0\}$. There are $2 \cdot 6 \cdot 6 \cdot 2 = 144$ such modifications. Since X_a is invertible,

$$X_a(L_a + \lambda E_{ij}) - Y_a = \lambda X_a E_{ij} \neq 0.$$

Every modification destroys at least one transition of the corresponding register. The executable certificate also enumerates the 144 modifications and preserves, for each, the number of violated transitions.

The proved property is rigidity of the operators with respect to the fixed regional register. Reproducing a prior construction of the register itself requires the coordinatewise action of its selection, transport and update operators. That dependency is kept separate from the rigidity theorem, the finite emission catalogue and the subsequent analytic realizations.

C.5 Reproduction certificate and location of the data

The program `verificar_nucleo_finito.py` traverses the 324 conditions of each cursor and forms the product of the resulting classes. The documentary catalogue is loaded after this construction to compare emissions and their multiplicities. In the second part, the four transition matrices are extracted from the verbatim copy of the source; solutions are computed by exact elimination modulo 3 before comparison with the printed matrices. The receipt `NUCLEO_FINITO.json` preserves the domains, preimages, emissions, 16 calendars and 144 modifications.

Verification uses exact integers and residues. It receives no metrological values or real expansions of the constants. Its scope is that of the finite operations and starting data specified in this appendix. The proofs of compatibility at unbounded depth and physical identifications retain their respective domains and expository obligations.

D Sources, preservation, and reproduction

The digital supplement preserves the complete cumulative document, its thirty-nine sources, the verification programs, and the corresponding records. The approved narrative and its continuation are retained verbatim in that archive. This edition recasts their conversational form as an academic exposition and records the correspondence between sections, while preserving the originals.

The antecedents originate in the series sources of 21 September 2026. The formal kernel is incorporated from its full documentary dependency closure in the constitutive article; the developments on action, incidence, mass, and temperature are preserved together with their proofs. The manifest identifies each file, extracted range, and hash. These data document provenance and preservation, while the proofs in the body establish the mathematical scope of the results.

The programs for composition of constants, hybridization, and compatibility check exact identities and finite examples. Numerical diagnostics are distinguished from rational checks. The proofs of monotonicity, convergence, inversion, and passage to the limit remain developed in the text: the executable examples complement those proofs.

Compilation uses LuaLaTeX. The included sources and figures allow the PDF to be reproduced independently of the author's original directories. The package instructions specify the compilation sequence and how to execute the checks.

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